Complete Zenodo Preservation

Part A - Papers, figures and publication history

Three-file edition | 396-page reading document

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Version 3 | Snapshot: 11 November 2026

Scope	Four projects / ten published releases
Original deposited downloads preserved	45
Original PDFs reproduced	44 distinct PDFs / 346 original pages
Raster figures reproduced	13
Text files moved to Part C	47,400
Full decimal expansions in Part B	6 files / 6,043,218 digits
Pages in this PDF	396

Keep all three files

This PDF contains the papers, figures, release histories, public descriptions, file manifests and binary catalogue. The 23,101-page printed software appendix has been moved into a separate text file. The large original-file recovery archive has also moved out of the PDF.

Part B - Full decimal expansions

v3_Part_B_Million_Digit_Markov_Numbers.txt

Part C - Software and textual supplements

v3_Part_C_Software_and_Textual_Supplements.txt

The text files retain all source content. Part C uses lossless text encoding for its compressed transcript and original-file recovery archive. Together, Parts B and C restore all 45 deposited downloads exactly.

Reading and exact recovery

Part B: the six full numbers

The six decimal expansions total 6,043,218 digits. Their original bytes and final newlines are unchanged. The final base64 blocks preserve the compressed numeric ZIP streams needed for exact reconstruction of the two numeric-supplement archives.

These expansions first appeared in record 24093257, whose publication-date field is 9 November 2026. Final rank confirmation completed 10 November at 04:18:33.296330 UTC (9 November at 22:18:33.296330 CST) and is documented in record 24096109. The later confirmation is not backdated.

https://zenodo.org/records/24093257

https://zenodo.org/records/24096109

Part C: encoded text and original-file archive

Part C uses ASCII Base85 to carry the complete compressed software transcript and recovery archive. Decode it with the included instructions to obtain all 47,400 exact UTF-8 file blocks. BEGIN / END markers, byte counts, hashes, original paths and release associations identify each file. Unicode, tabs and line endings remain unchanged.

A separate encoded payload contains the original-file recovery archive. It preserves binary certificates, executables, original PDF and figure files, and ZIP metadata. Lossless compression and text encoding keep Part C manageable and the reading PDF small.

Follow the decoder instructions at the beginning of Part C to obtain the readable transcript and v3_Originals_and_Recovery.tar.xz. Unpack that archive, then run its restore_exact_originals.py with the Part B TXT path and a destination directory. The restorer verifies byte counts, SHA-256 hashes and Zenodo MD5 checksums.

Part B SHA-256: 765d1bb69afbf2eb501a83c0be6643fecbff28133203e59b0e5f17f0de12fb55

Part C SHA-256: 8d3aac1651cb54e2947eb864919dfbac75418bc0923d6175b46bf70cc8ee619e

Decoded recovery archive SHA-256: 60a7312e156ca5626e40cd4efa8640d8dbe00654c4db91bec6872e7c9e6e09d4

The recovery payload is unchanged from the verified v2 archive. Historical filenames inside it remain unchanged; the v3 instructions identify the correct companion files. Recovery does not execute the archived research programs.

Release navigation

Each section begins with completion and publication history, public URLs, the Zenodo description and a deposited-file manifest. Page numbers below refer to this PDF.

Release / project	Zenodo record	Publication date field	PDF page
01. Computational supplements 1.0	23874075	2026-11-02	5
02. Meridian manuscript v4_ps	24043052	2026-11-07	58
03. Meridian manuscript v5_ps	24046012	2026-11-07	112
04. Aigner monotonicities v3	24053747	2026-11-08	147
05. Meridian manuscript v7_ps	24053720	2026-11-08	154
06. Meridian manuscript v9_ps	24084246	2026-11-09	189
07. Meridian manuscript v11	24089717	2026-11-09	228
08. Hyperbolic collision tiers v6	24091908	2026-11-09	272
09. Meridian manuscript v12-supplement-only (manuscript v11)	24093257	2026-11-09	311
10. Meridian manuscript v21	24096109	2026-11-09	314

Binary catalogue

The catalogue begins on global PDF page 386. Binary content itself is preserved in the recovery archive encoded in Part C.

The 44 distinct original PDFs and 13 raster figures retain their original mathematical content. Identical files are represented once, and all release associations remain in the manifests. The chronological notes distinguish deposited drafts from later computations.

Text supplement guide

After decoding Part C, search v3_Exact_Software_Transcript.txt for a release number, original path, hash or stable T-number. The embedded v3_Text_Index.json gives byte offsets for every original text block in that decoded UTF-8 transcript. These are byte offsets, not character positions.

First published in record	Text files	Release header byte offset in decoded transcript
23874075	26,127	773
24043052	37	150,258,787
24046012	13	151,285,100
24053747	7	151,374,783
24053720	7	151,408,257
24084246	25	151,456,461
24089717	14,667	151,604,875
24091908	35	229,291,008
24093257	62	229,592,495
24096109	6,420	230,079,851

History evidence and its limits

Publication-date fields, manuscript dates, completed computations, repository timestamps and separately retained public observations are distinguished. The June preprint date is source-reported; its DOI could not be independently retrieved. ResearchGate verifies the August upload. October X and LinkedIn dates come from retained public-post metadata. Additional November LinkedIn announcements and the older Meridian ResearchGate v13 URL could not be established. Unknown completion times remain unknown.

Small PDF attachments

v3_Text_Index.json
v3_Split_Manifest.json
v3_Verification.json
v3_Compilation_Source.zip
v3_Public_Metadata.zip

These attachments provide navigation and reproducibility metadata. The large software, original-file archive and full decimal expansions are in the two companion TXT files. Download all three originals; converted previews may omit PDF attachments.

Computational supplements for Meridians on the Markov Surface

Completion and release history

50-billion verification: final invocation and production audit ended 28 October 2026 at 05:38:33.644612263 CDT (10:38:33.644612263 UTC). The retained two-invocation execution time is 2 h 04 min 52.103 s; the wall span is 2 h 05 min 36.548 s. Source: the published version-4 computational chronology, pp. 4-5.

Degree-10,000 reproducibility package: dated 31 October 2026 in the published manuscript bibliography. A precise completion timestamp is not established here. Supplemental controls belong to manuscript version 26; their individual completion times are not independently established by the release metadata.

Zenodo publication-date field: 2026-11-02

Zenodo record created: 2026-11-01 20:16:18 CST / 2026-11-02 02:16:18 UTC

Zenodo record last modified: 2026-11-01 20:16:18 CST / 2026-11-02 02:16:18 UTC

The publication-date field is the date supplied in the deposit. The API created/modified timestamps are repository metadata, not a separately attested first-view time. Full fractional timestamps are retained in the metadata archived in Part C.

https://zenodo.org/records/23874075

https://doi.org/10.5281/zenodo.23874075

Earlier public announcements: 21 October 2026, 10:18:25 CDT / 15:18:25 UTC on X (more than a billion values in under 40 minutes); 23 October 2026, 09:54:57.771 CDT / 14:54:57.771 UTC on LinkedIn (9.3 billion in 24 minutes). These concern earlier computations, not the final trillion certificate.

https://x.com/webystar/status/2112926487091282350

https://www.linkedin.com/posts/elleard-felix-webster-heffern-joseph-by-confirmation-2953a3104_markovnumbers-activity-7519411048115003392-4NaJ

The dates and URLs are retained in the 7 November public-post evidence packet; the LinkedIn URL is also embedded in the published computational supplement. Additional November LinkedIn announcements and the older Meridian ResearchGate v13 URL could not be established in this compilation.

Complete public Zenodo description

Reproducibility materials accompanying Meridians on the Markov Surface: Geometry, Residue Structure, and Unicity Verification. The first archive is a public derivative of the sealed 50-billion verification package and supports the deterministic numerical-order verification of uniqueness of the first 50,000,000,000 distinct Markov values, including the original ten-billion-value anchor, frozen code and executables, compact retained evidence, timing receipts, and separate algorithmic audit programs. The second archive supports all-phase cross-Meridian separation within the complete family of 5,067,077 canonical Meridians of Laurent degree at most 10,000. Additional controls cover the manuscript’s structural identities and uniform unequal-degree collision-window bounds. These are separate finite computational results: the degree-bounded theorem does not exclude higher-degree competitors, and neither result establishes global Markov unicity. Independent audit denotes a separate algorithm or implementation within the project, not external peer review or proof-assistant certification. Some large intermediate streams are omitted from the compact evidence and must be regenerated using the supplied procedures. The degree-bounded archive is preserved byte-for-byte. The 50-billion public derivative removes only two nonresearch administrative receipts; its retained original members are byte-identical to the sealed original. Provenance identifies both archive hashes and the exact removals. Read the deposit README for file roles, dependencies, checksums and reproduction commands.

Licensing: Original software is licensed under MIT; original documentation and data under CC BY 4.0. Third-party materials and external dependencies retain their own terms. These scope-specific licences supersede historical 'licence approval pending' wording retained in the archives. Copyright (c) 2026 Elleard Felix Webster Heffern, to the extent copyright applies.

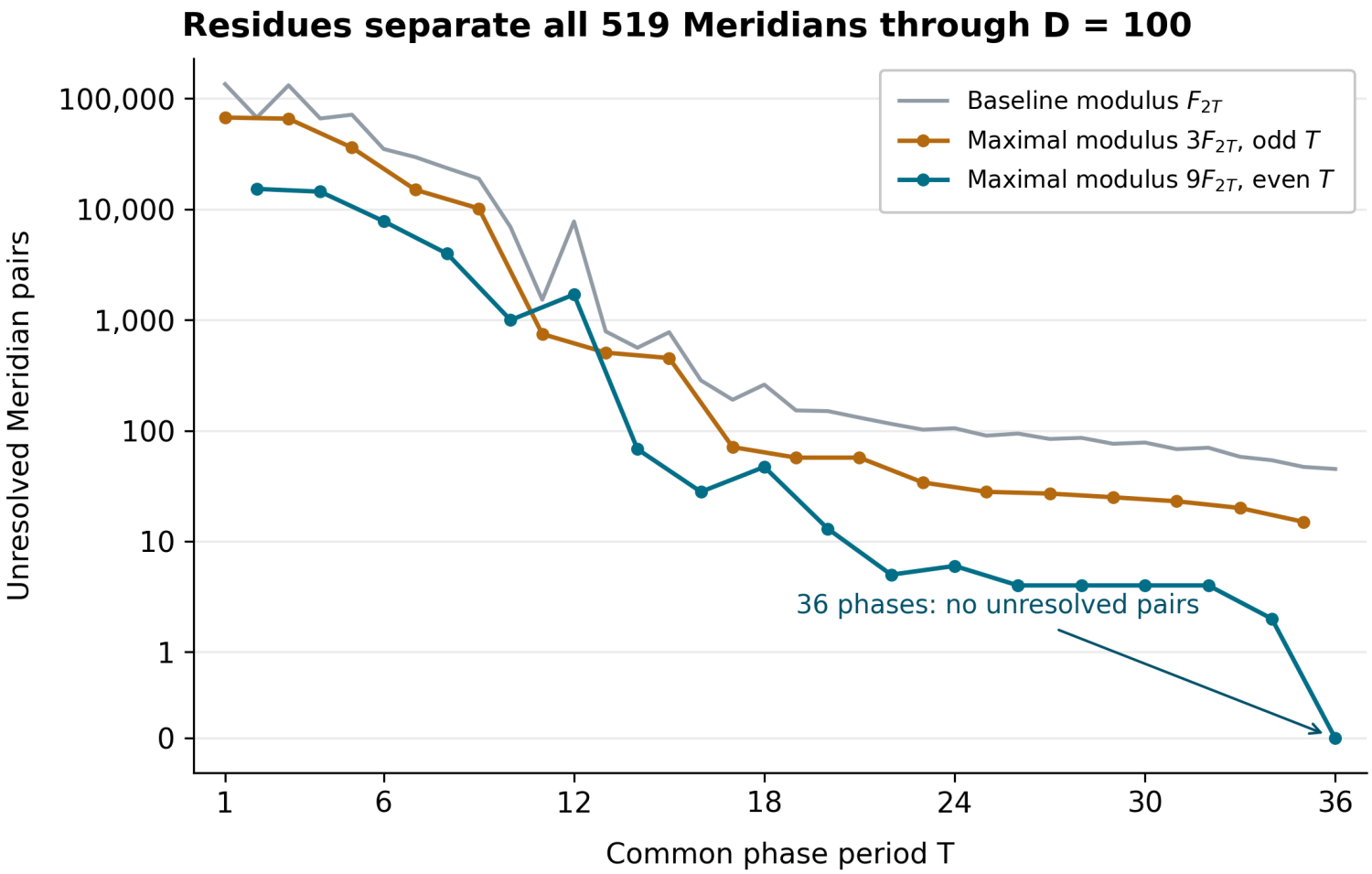
Funding: No funding.

Published files

Original filename / checksum	Bytes
Markov_Meridians_Supplemental_Controls_v26.zip MD5 43ad1a949e6a2f6f1d8905b6dddbc6e3 SHA256 8f11595a6cd15a8a2a545c702e5eaceea490de8ef479b103f218a863492b7db0	1,385,090
Markov_50_Billion_Verification_Public_v1_1.zip MD5 1e73f8f6588bb49c5fbac05330828bd5 SHA256 523fc2daa3b2d73e91aa5b625f7c8a9c4a2fa4f884eb013dfd516487d99032aa	43,018,800
Meridian_Degree_10000_Reproducibility.zip MD5 2c9a00bd81599c5021bc8d26fd758daf SHA256 e8197e663bc0569df46b0865a9a66f16d7f2293fd6f160a33bcf4c0e3a1f5d95	137,565,753

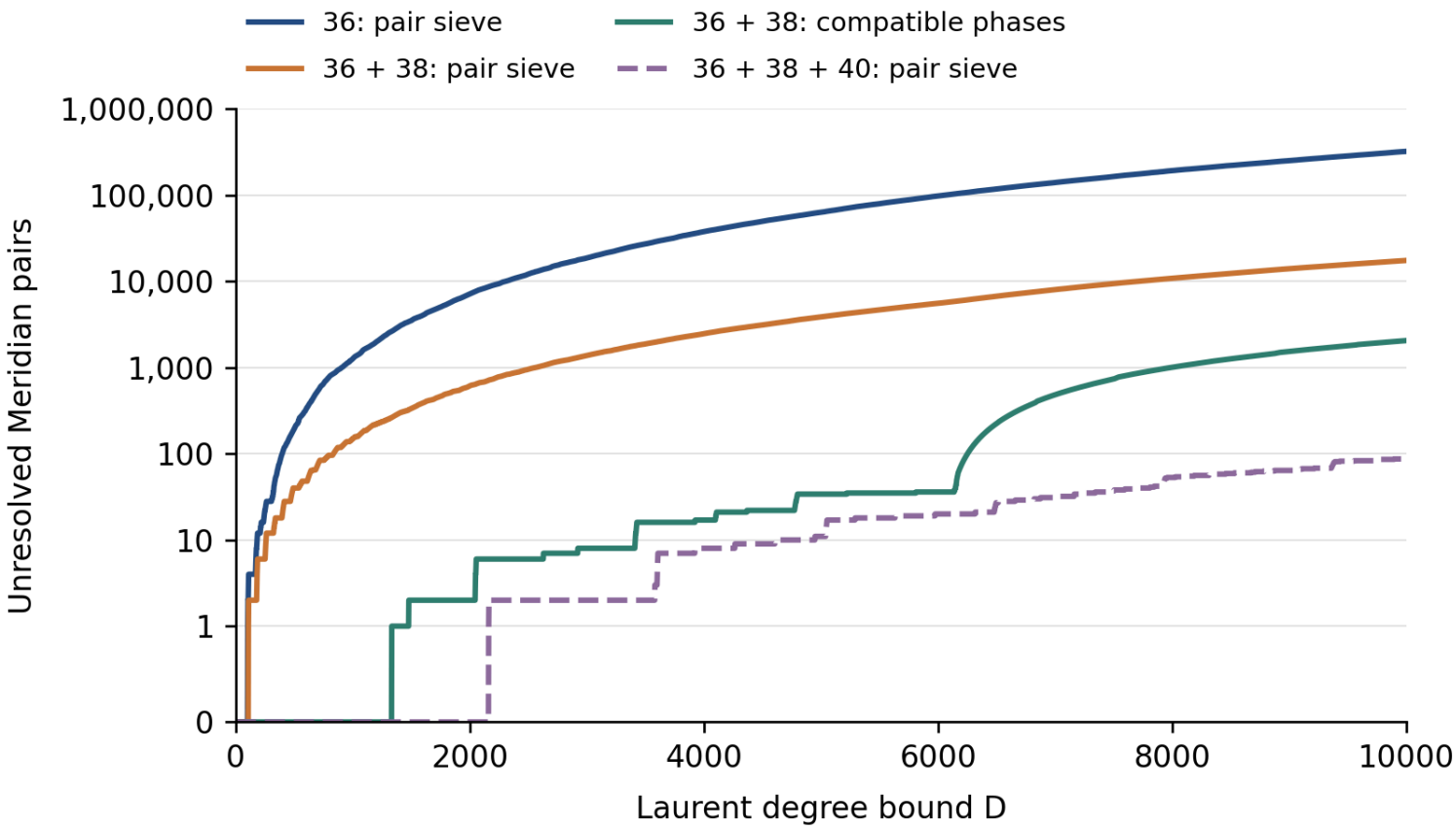
All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

Original figure: retained_v20/meridian_degree_100/astra/figures/Meridian_Residue_Separation_v1.png



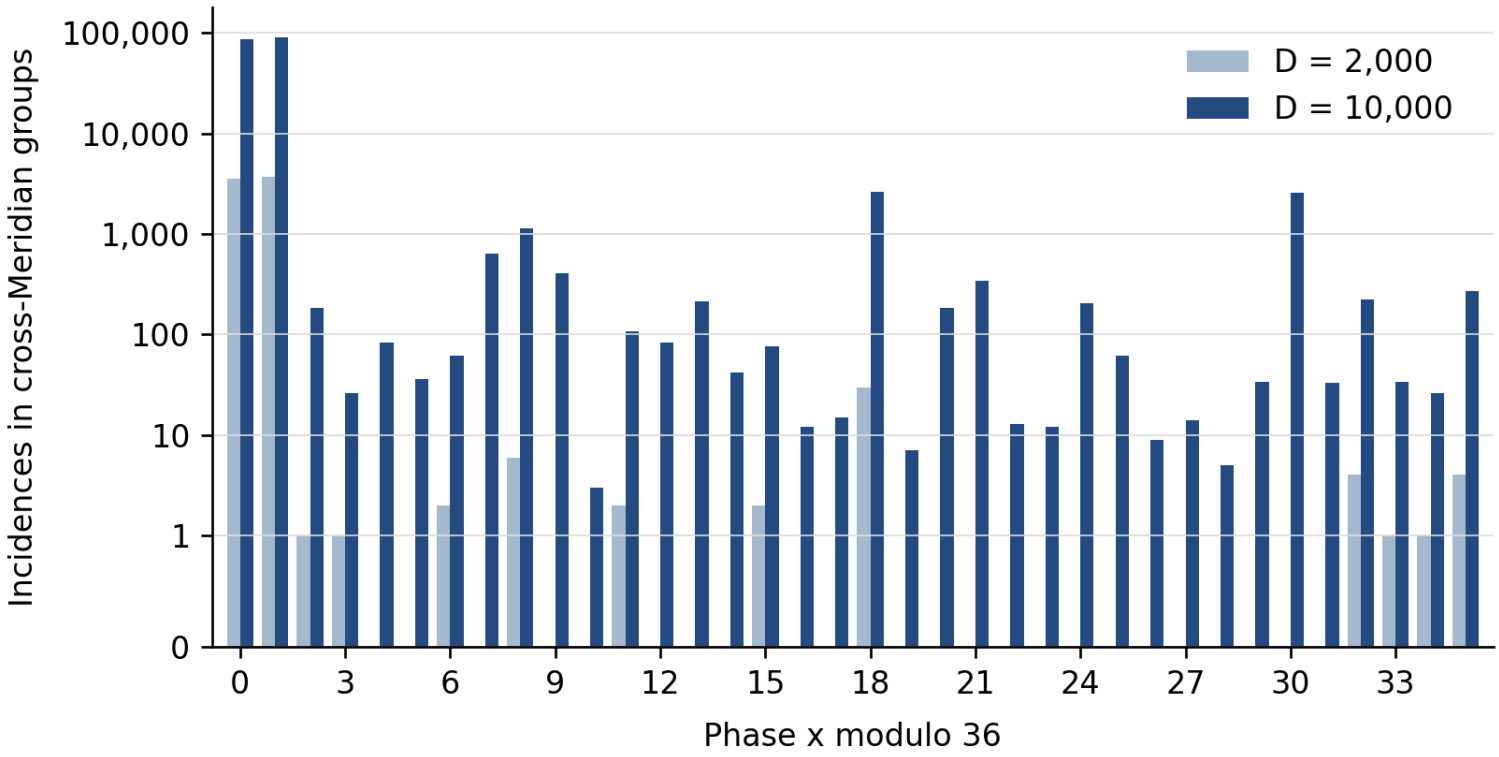
First in Zenodo record 23874075 | SHA256 906635ae9110820b510dab99

Original figure: Meridian_Degree_10000_Reproducibility/report/degree_survivors.png



First in Zenodo record 23874075 | SHA256 b76ddd7d06849479ef5145ec

Original figure: Meridian_Degree_10000_Reproducibility/report/phase_concentration.png



First in Zenodo record 23874075 | SHA256 adb7e1cc8cad8db3c7b3511d

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_50_Billion_Preflight_Audit_v1.pdf

First 50 Billion Markov Values

Coverage, pruning, storage, and reproducibility audit

28 October 2026 Version 1

Decision: ready to attempt the frozen run. The first-50-billion uniqueness comparison has **not** been completed. The full-domain count and storage preflight passed; the final launcher passed the complete 32-pass small trials and adversarial recovery tests.

What this review changed

The mathematical coverage and permitted pruning survived this review. The operational protocol needed additional safeguards, implemented in a separate launcher while preserving the arithmetic kernels and the validated 10-billion anchor byte for byte.

1. **Coverage.** An exact numerical cutoff contains at least 50 billion configurations; a proved height envelope contains every competitor below that cutoff. Distinctness is concluded only after collision verification succeeds. The disputed restriction on the relative word lengths of colliding Meridians is excluded from pruning.
2. **Anchor.** The original 10-billion cutoff certificate is required by exact identity. The 20-billion incremental certificate is rejected as an anchor by the current loader. No recursive anchor implementation was added.
3. **Storage and recovery.** The full-sized histogram validates 32 prefixes. The new launcher removes uncommitted merged-prefix copies before retry, keeps the run locked while orphaned native workers remain alive, and adds an exact-candidate audit memory guard.
4. **Reproducibility.** Sources and executables are hash-bound. Individual atomic timing receipts replace a prototype shared append journal that lost records in testing. A final independent evidence audit is mandatory.

The exact proposed search

Requested distinct prefix N	$50,000,000,000$
Inner height K	$573,574$
$C(K-1)$ / $C(K)$	$49,999,964,521$ / $50,000,097,577$
Numerical cutoff B	$F_{1,147,147}$
Cutoff bit length	$796,397$
Outer height L	$685,979$
Outer configurations $C(L)$	$71,517,775,875$

The completed historical extension certifies the first **20 billion** distinct values. Its $28,606,957,721$ configurations and roughly 34-minute combined execution must not be described as a first-28-billion result.

The following propositions supply the proof chain used by this release. They organize and specialize earlier manuscript and verification work; no claim of novelty for standard Vieta descent, mediants, or elementary Fibonacci bounds is intended. The package preserves the preceding proof documents.

Zenodo 23874075 | original page 1/7 | SHA256 8e8caff6b934b69d1019

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_50_Billion_Preflight_Audit_v1.pdf

1. Configuration coverage and Meridian ownership

Normalize a positive integral Markov triple as $a \le b \le c$, with $a^2 + b^2 + c^2 = 3abc$. A configuration means the normalized triple, before any assertion that its maximum uniquely identifies it.

Proposition 1. Include the separate root $(1, 1, 1)$ at coordinate $(0, 1)$ and the seed $(1, 1, 2)$ at $(1, 1)$. Ordinary mediants between $0/1$ and $1/1$, with maximum update $c = 3ab - w$, enumerate all other configurations once. Here w is the opposite value at the parent edge. Their height is $H = p + q$.

Proof. The Euclidean algorithm gives each reduced $0 < p < q$ exactly once in this mediant tree. For a sorted Markov triple, the other root $c' = 3ab - c = (a^2 + b^2)/c$ is positive. The quadratic evaluated at b is $a^2 + (2 - 3a)b^2 \le 0$, so $c' \le b$; outside the two initial exceptions the largest-coordinate mutation descends strictly. Mutating either smaller coordinate increases the maximum. Descent therefore reaches the root and gives a unique parent, without assuming maximum unicity. The increasing configuration tree agrees with the mediant update and its two children.

Endpoint heights add at a mediant. Every descendant has greater height, so a branch beyond L can be stopped. The native adaptive frontier either emits a node and splits both children, or assigns its entire remaining interval to one job. Those intervals are disjoint; its atomic job allocator processes each exactly once. Root and seed are emitted separately once. □

Canonical Meridian dictionary. Let a nonroot helm have coprime odd counts $1 \le r \le s$, and set $D = r + s$. Here D is the number of letters in the actual positive matrix word, not the number of entries in a Meridian address. The coordinates of phase x are

x	(p, q)	H
0	(r, s)	D
1	(r, D)	$D + r$
-1	(s, D)	$D + s$
$n \ge 2$	$(D, (n-1)D + r)$	$nD + r$
$-n \le -2$	$(D, (n-1)D + s)$	$nD + s.$

For the seed $(r, s) = (1, 1)$ retain only $x \ge 0$. Thus $D \le L$, with phase limits $n \le \lfloor (L - r)/D \rfloor$ and $n \le \lfloor (L - s)/D \rfloor$, covers exactly the height-bounded Meridian blocks.

Justification and inverse. The manuscript's ordered endpoint-word updates are linear in their two letter-count vectors. Substituting the seed bases at phases $0, 1, n \ge 2$, respectively $((1, 0), (0, 1))$, $((1, 1), (0, 1))$, and $((1, n), (1, n - 1))$, gives the table; negative phases exchange the endpoints. Conversely: odd/odd coordinates give $(r, s, x) = (p, q, 0)$; even q gives the sorted pair $p, q - p$ and phase $+1$ or -1; even p gives $q = tp + u$, the sorted pair $u, p - u$, and phase $\pm(t + 1)$. The sign is positive when the first listed entry is smaller. A tie forces the seed and its positive representative. This proves unique ownership.

Exact count, for $H \ge 2$. At height $h \ge 3$, the choices $0 < p < h/2$, $\gcd(p, h) = 1$, number $\varphi(h)/2$. Including root and seed,

$$C(H) = 1 + \frac{1}{2} \sum_{h=1}^{H} \varphi(h) = \frac{5 + \sum_{d=1}^{H} \mu(d) \lfloor H/d \rfloor^2}{4}.$$

The second identity counts ordered coprime pairs in a square and provides an independent Möbius implementation. These count configurations, not yet distinct maxima.

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2. The numerical-order guarantee and anchored extension

Proposition 2 (uniform bounds). For $H = p + q \geq 2$,

$$2^{H-1} \leq M(p/q) \leq F_{2H-1}, \qquad M(p/q)^2 \geq 4 \cdot 5^{H-2}.$$

Proof. For an outward mediant, the larger-root formula gives

$$c = \frac{3ab + \sqrt{9a^2b^2 - 4a^2 - 4b^2}}{2}.$$

Since $2a^2b^2 \geq a^2 + b^2$, we have $c \geq 2ab$. Above the initial endpoints, $\max(a, b) \geq 2$, so $5a^2b^2 \geq 4a^2 + 4b^2$ and $c \geq (5/2)ab$. The endpoints of heights one and two satisfy $M \geq (2/5)(\sqrt{5})^H$; this bound propagates because $(5/2)(2/5)^2 = 2/5$. Squaring proves the stronger lower bound; the power-of-two bound propagates similarly.

On the boundary $p = 1$, the recurrence $u_{j+1} = 3u_j - u_{j-1}$ gives $M(1/q) = F_{2q+1}$. For $p \geq 2$ both parent heights H_1, H_2 are at least two. Put $m = 2H_1 - 1$, $n = 2H_2 - 1$. Then

$$F_{m+n+1} = F_{m+1}F_{n+1} + F_mF_n \geq \tfrac{13}{4} F_mF_n > 3F_mF_n.$$

Using $c = 3ab - w < 3ab$ and induction proves the upper bound. □

Proposition 3 (first N distinct values). Choose the least K with $C(K) \geq N$, and set

$$B = F_{2K-1}, \qquad L = \max\{h \geq 2 : 4 \cdot 5^{h-2} \leq B^2\}.$$

If the maximum is injective on a retained set containing every configuration with $M \leq B$, then the first N distinct Markov values have unique normalized triples.

Proof. Every inner configuration $H \leq K$ has $M \leq B$. Every possible configuration with $M \leq B$ has $H \leq L$ by Proposition 2, with the root handled separately. After injectivity is established, the $C(K) \geq N$ inner configurations yield at least N distinct values below B, and all possible competitors for them have been considered. No count of configurations is interpreted as a distinct-value count beforehand. □

For the numbers on page 1, both count formulas agree. Exact integer arithmetic also checks $4 \cdot 5^{L-2} \leq B^2 < 4 \cdot 5^{L-1}$; logarithmic estimates alone do not choose the final boundary.

Proposition 4 (safe use of the 10-billion anchor). Let $K_0 = 256510$, $B_0 = F_{513019}$. The validated original run contains every configuration with $M \leq B_0$ and proves uniqueness throughout that numerical cutoff. The extension may omit emissions with $H \leq K_0$ while still traversing their descendants.

Proof. Every omitted value is at most B_0 by Proposition 2. A collision with common value at most B_0 is already impossible by the anchor, including competitors outside its inner region. If the common value exceeds B_0, neither configuration has $H \leq K_0$, so both remain eligible for the new comparison. This covers collisions across the old/new division. □

Only $10,000,015,326$ inner configurations receive this skip. The old outer height, all old retained records, and whole Meridians cannot be skipped by this argument. Reusing the 20-billion incremental result would require a different validated anchor implementation; it is not done here.

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Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_50_Billion_Preflight_Audit_v1.pdf

3. Permitted logarithmic pruning

The implemented exclusions use the maximum of the two generic lower bounds in Proposition 2 and the following positive-word lower bound. They do not assume a relative word-length window for potential collisions.

Proposition 5 (leading positive term). Put

$$\rho = \frac{3+\sqrt{5}}{2}, \quad \alpha_0 = \frac{15+3\sqrt{5}}{10}, \quad \beta_0 = \frac{15-3\sqrt{5}}{10}, \quad \gamma_0 = \frac{2}{\sqrt{5}}.$$

The manuscript's actual ordered positive endpoint words use

$$P(\xi) = \begin{pmatrix} \beta_0/\xi & \gamma_0 \\ \gamma_0 & \alpha_0\xi \end{pmatrix}, \qquad Q(\xi) = \begin{pmatrix} \alpha_0/\xi & \rho\gamma_0 \\ \gamma_0/\rho & \beta_0\xi \end{pmatrix}, \quad \xi = \rho^x.$$

Their node word satisfies $3M_\alpha(x) = \operatorname{tr}(UV)$. The endpoint updates $(U,V) \mapsto ((UV)^{m-1}V, (UV)^m V)$ and $((VU)^m U, (VU)^{m-1}U)$ preserve actual positive products.

Proof of the bound. In the trace expansion every term is nonnegative. For a word with r letters P and s letters Q, the path staying in the lower diagonal contributes $\alpha_0^r \beta_0^s \xi^D$, where $D = r + s$. The upper-diagonal path contributes $\beta_0^r \alpha_0^s \xi^{-D}$. Since $\alpha_0\beta_0 = 9/5$ and $\alpha_0/\beta_0 = \rho$, for $n \geq 0$ and $\sigma = \pm 1$,

$$\log M_\alpha(\sigma n) \geq \left(Dn + \sigma \frac{r-s}{2} \right) \log \rho + \frac{D}{2} \log(9/5) - \log 3. \tag{*}$$

Dropping all other positive terms proves the claim, including helm and near-helm phases. This uses the actual positive product representation, not positivity inferred from a scalar recurrence with subtraction. □

Substituting the inverse Meridian dictionary yields the native formula $\theta = E \log \rho + (D/2)\log(9/5) - \log 3$, with

coordinate parity	D	E
p, q odd	$p+q$	$(p-q)/2$
q even	q	$p + q/2$
p even	p	$q + p/2$.

Thus the fast path has no per-node Meridian division or reconstruction while retaining exactly the same proven lower bound.

Rounding obligation. Every logarithmic endpoint is a dyadic rational scaled by 2^{64}. Constants come from an integer square-root bracket and rational atanh series with a proved tail bound and 32 guard bits. A separate Fraction-based auditor checks the constants and cutoff. For negative E, the formula uses the *upper* endpoint of $\log \rho$; positive coefficients use lower endpoints, and the subtracted $\log 3$ uses its upper endpoint. The lower $\log 5 = 2\log 3 - \log(9/5)$ endpoint is nonnegative, so the nonnegative division by two rounds safely down in the native code.

An outer candidate is excluded only when its certified lower endpoint is **strictly greater** than the certified upper endpoint of $\log B$. Equality, overlapping windows, and every inner node remain retained. The root is handled separately. Generation and replay use the same predicate. For the reviewed coordinate limits, 32-bit coordinates, 64-bit counters, and signed 128-bit fixed-point intermediates have ample range; unsigned 64-bit multiplication and subtraction implement exact residues modulo 2^{64}.

Zenodo 23874075 | original page 4/7 | SHA256 8e8caff6b934b69d1019

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_50_Billion_Preflight_Audit_v1.pdf

4. Complete comparisons with bounded working storage

Proposition 6 (partition and deletion rule). Assign each retained configuration its exact key $M \bmod 2^{64}$. Equal maxima necessarily have equal keys, hence the same five-bit prefix and following eight-bit radix bucket. Comparing complete buckets over all worker shards therefore covers every possible equality among retained configurations.

Proof. A complete sorted bucket identifies every key with multiplicity at least two. Each singleton is unequal to all other retained values because its exact residue differs. Preserve all repeated keys with their multiplicities; then deterministic replay must recover exactly those occurrences and distinct coordinates. Independently reconstructed residues modulo 2^{64} and 5^{27} can prove inequality. Remaining shared signatures require disjoint certified windows or exact integer comparison. Unresolved groups cannot yield success. Distinct prefixes cannot share a key, so no cross-prefix comparison was lost. Combine this with the anchor argument for the omitted configurations. □

Generation receipts bind counts and shard hashes. Partition receipts bind their source shards and outputs. A bucket receipt and its repeated-key output are committed and synced before raw bucket inputs are deleted. A complete prefix receipt is committed before its individual bucket outputs are removed. Sorting one worker's shard in isolation would not justify deletion.

Two-copy bound, including recovery. Let T be the bytes of raw keys in one prefix. Generation uses at most T. Partitioning can temporarily retain an input and its output; across disjoint shards total data is at most $2T$, even with unbalanced workers. A repeated-key record is 16 bytes and represents at least two 8-byte input keys, so total scan output is no larger than its input. Concurrent scans and the final merge likewise need at most two copies.

This argument requires stale temporary outputs to be removed before retry. In particular, if interruption occurs after the merged `repeated.bin` is renamed but before `pass.json` commits, that uncommitted merged copy must be removed; the durable bucket outputs still own its evidence. Otherwise a retry could temporarily create a third copy. The new cleanup rule and its fault-injection test close this gap. Committed-prefix data is never discarded under this rule.

Measured or configured quantity	Bytes / records
Largest prefix, records	1,716,982,381
Largest raw prefix, T	13,735,859,048
Conservative active-prefix bound, $2T$	27,471,718,096
Launch requirement, $2T + 1$ GiB	28,545,459,920
Largest complete radix bucket, records	6,715,646
Configured bucket cap, records	25,000,000
Eight sort arrays at configured cap	3,200,000,000
Eight largest measured buckets, sort arrays	859,536,144

These byte bounds exclude small receipts and candidate evidence; the 1 GiB reserve accommodates them. After each prefix, candidate caps are checked. A separate 2 GiB planning guard bounds retained exact maxima using $M \le F_{2H-1} < 2^{2H}$, CPython integer storage, per-record metadata allowance, and 128 MiB fixed allowance. This prevents the auditor from accepting a replay size whose full integers exceed its planned memory. Array bounds and this planning estimate are not a measured whole-process RSS or an OS quota. The host limit is 8 GiB; any cap or I/O failure leaves verification uncertified.

Zenodo 23874075 | original page 5/7 | SHA256 8e8caff6b934b69d1019

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_50_Billion_Preflight_Audit_v1.pdf

5. Interruption protocol and recorded timing

Process ownership. A nonblocking lifetime lock is held by the launcher and inherited by every native subprocess. Killing only the parent therefore does not allow a new launcher to erase or race an old worker's outputs. A restart refuses until all owners exit. The normal Python error path kills and waits for its own child. The original bare CLI lacks this added protection.

Recovery. First verify source and configuration identity, then validate committed prefix evidence and finish any interrupted cleanup. Remove only known uncommitted generation, partition, scan, and merge outputs. Resume at the first missing durable stage. Changed committed hashes, missing inputs, incomplete coverage, or mismatched replay multiplicities stop the run; they are not waived. Actual parent-only process death was tested. Hardware power-loss recovery was not physically tested; durable writes use file and directory synchronization.

Timing. Each attempt has an atomic JSON receipt containing its start and, once complete, its finish, status, elapsed monotonic time, and native metrics. Generation, partition, sorting, replay, comparison, anchor audit, final audit, and whole invocations are recorded separately. Sorting also records its read, sort, and write durations.

For concurrent tasks the summary records both the sum of task durations and the union of occupied wall intervals. Nested stages overlap and must not be added. Whole-invocation time includes failed and repeated work, excluding idle gaps between invocations. The anchor's recorded time is shown separately and also added to the extension path. A hard kill can leave an unclosed attempt: its duration is unknown, and closed totals are explicitly lower bounds.

A prototype shared append journal lost some stage records during a concurrent small trial. The mathematical core completed, but the timing summary was rejected. The replacement uses one atomic file per attempt and reconciles receipt counts against executed starts before accepting a completed invocation. The diagnostic and compressed prototype journal are preserved; no unexplained loss was silently counted as a valid timed result.

Executed validation of the final implementation

Check	Outcome
Core regressions (61 executions) and new protocol tests (10 methods)	Passed
Interrupt at all commit boundaries and before prefix-merge receipt	Passed
Actual parent-only kill with a surviving stopped native worker	Passed
Disk/configuration/anchor and exact-audit memory guards	Passed
Corrupted committed data; missing timing receipt	Rejected
Complete 8-worker / 32-prefix / 256-bucket trial, 64-bit keys	Passed
Same layout with forced 13-bit primary collisions	Passed
Relocated package: anchor plus both trial audits and bindings	Passed

Both trials target 10,000 distinct values and retain 12,903 new records. The 13-bit trial replays all **10,265** repeated-key configurations. Independent literal-integer enumeration matches the complete repeated-key multiset and replayed coordinates; a separate value-bounded Vieta enumeration matches all 11,041 triples below its numerical cutoff. Production retains full 64-bit keys. The two final trials have exactly 8,480 and 8,481 native calls, each reconciled to durable timing receipts. These tests exercise the chosen layout; they are not the 50-billion comparison.

Zenodo 23874075 | original page 6/7 | SHA256 8e8caff6b934b69d1019

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_50_Billion_Preflight_Audit_v1.pdf

6. Measured preflight and frozen execution contract

The count instrument adds per-worker 8192-bin counters to the frozen generator, without changing traversal, modular arithmetic, or pruning. Its five changes are recorded. Two independent full-integer histogram checks through height 512 passed, each covering 39,927 configurations.

The full-size count visits **71,517,775,875** configurations, retaining **54,940,210,994**, anchor-skipping **10,000,015,326**, and excluding **6,577,549,555**. Counts add exactly. All 32 prefix and 8192 bucket counts are saved and required to match during production. Native wall time was **123.469610 s**; driver wall time was **123.474575 s**, with **979.647817 s** child CPU. Compilation and preparation are excluded. This is a counting/storage measurement, not a uniqueness runtime.

Final small trial: occupied wall seconds	64-bit	13-bit forced
Generation	0.164738	0.174984
Partition	3.390892	3.009971
Sorting, including native read/write	29.364955	27.699635
Native replay	—	0.004694
Comparison pipeline (includes replay)	0.004702	0.344841
Final independent audit	0.239358	0.375150
Complete launcher invocation	37.993934	36.401859

Stage rows are not additive. The retained anchor records 838.877233 s including its original build/startup; its internal timer was 838.847691 s. This review's read-only re-audit of the anchor took 1.490864 s and reconstructed all 38 repeated-key candidates as full integers, confirming 19 groups with distinct values. It did not regenerate the deleted singleton streams.

Frozen identities (SHA-256). Full source and executable hashes are in `release_binding.json`; every delivered package member is listed in `SHA256SUMS.json`. Selected executable/launcher identities are:

```
native_incremental:
5587ab32108fdce365eae968725bb2f7a3f729e3fdbeba873075ed5153fd4564
key_tools:
44066f9d950bdede3909452b049cee1508ecc4353fcf9e31dc978f5f1633dbcf
execute.py:
6f07a57ec94acae475045fac995931a45a8c26f3db4bd30be6c07af399606a3c
```

The environment is Linux x86-64, Python 3.12, GCC 13.3.0, eight CPU equivalents, and 8 GiB RAM. Build details are preserved. The launcher does not rebuild binaries.

Execution. From the package root, run or resume:

```
python3 step6_preflight/execute.py --work step6_preflight/runs/verified_50b_p5
```

Optionally append `--max-passes 1` for a clean first-prefix checkpoint. Acceptance requires all 32 prefixes, target 50 billion, core status `complete_unique`, and a passed final audit. The control summary must read `complete_unique_and_audited`. Any unclosed timing must be disclosed when reporting cumulative runtime.

Scope and provenance. The independent checks were implemented within this work; no external human review or proof-assistant certification is claimed. Inputs are manuscript v35, equations (15)–(18), the included Step 1/2 proofs, and Step 5 anchor arguments. Independently rechecking deleted singleton streams requires regeneration. The 50-billion uniqueness comparison remains unrun.

Zenodo 23874075 | original page 7/7 | SHA256 8e8caff6b934b69d1019

The First 50 Billion Markov Values

Finite verification results and reproducible execution record

28 October 2026 Version 1

Result. The completed computation and independent final audit certify that each of the first **50,000,000,000 distinct Markov values in numerical order** is the maximum of exactly one normalized positive Markov triple.

The verified domain

The numerical-order guarantee uses a proved finite envelope, rather than materializing a sorted list of enormous integers. With $C(H) = 1 + \frac{1}{2}\sum_{h=1}^{H} \varphi(h)$ for $H \geq 2$, the exact parameters are

Inner height K	$573,574$
$C(K-1)$ / $C(K)$	$49,999,964,521$ / $50,000,097,577$
Numerical cutoff B	$F_{1,147,147}$ ($796,397$ bits)
Outer height L / configurations $C(L)$	$685,979$ / $71,517,775,875$
Anchored inner configurations skipped	$10,000,015,326$
Configurations safely excluded above B	$6,577,549,555$
New retained configurations compared	$54,940,210,994$

All 32 high-bit prefix passes and all 8192 radix buckets completed. Their counts match the full-domain preflight histogram exactly. The pass sequence performs $2,288,568,828,000$ configuration visits before candidate replay; these repeated visits are not a count of distinct Markov values.

Measured execution time

The execution used the frozen Linux x86-64 binaries, eight workers and eight CPU equivalents, with an 8 GiB memory limit. No recompilation occurred.

Stage	Occupied wall seconds	CPU seconds
Generation: 32 passes	5,001.065659	37,211.908780
Partitioning	734.828642	1,260.188133
Sorting, including native read/write	484.227185	2,870.982905
Candidate replay	464.279119	3,633.077531
Final independent audit	107.371720	107.350548

Current verification: 2 h 04 min 52.103 s (7,492.102807 s). Including the original 10-billion anchor (838.877233 s), the cumulative execution path is **2 h 18 min 50.980 s** (8,330.980040 s). There were 0 failed native attempts and 0 unclosed timing receipts. Total launcher-parent and native-child CPU time was 46,788.073242 s.

Wall time sums two launcher invocations, excluding idle gaps. Concurrent task times use interval unions; hashing and orchestration account for the balance. CPU rows use native counters or the auditor's process timer. The comparison pipeline, including replay, took 464.343548 s. Checkpointing and maintenance ran concurrently. Address export, document production and earlier preflight/20-billion work are outside this timer.

Zenodo 23874075 | original page 1/2 | SHA256 b2c4683229716cddf2fe

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_50_Billion_Verification_Results_v1.pdf

Collision resolution and coverage proof

The complete buckets found 346 repeated residues modulo 2^{64}, covering 692 configurations. Replay recovered every recorded multiplicity and distinct coordinate. Independent residues modulo 5^{27} and the exact comparison machinery resolved the candidates. The final auditor reconstructed the full integer triple for every candidate, checked its Markov equation and recorded residues, and confirmed distinct full maxima. There were 0 actual equal-value collisions.

The production comparison recorded 0 repeated secondary-signature groups and 0 exact reconstructions. Those production counters are separate from the final auditor's full-integer reconstructions. Canonical Meridian addresses and phases for every candidate are archived.

The included six-proposition preflight audit proves that every inner configuration has $M \leq B$ and every competitor with $M \leq B$ has $H \leq L$. The bound $M^2 \geq 4 \cdot 5^{H-2}$ and exact check $4 \cdot 5^{L-2} \leq B^2 < 4 \cdot 5^{L-1}$ establish the outer limit; the root is included separately. After injectivity is verified, the $C(K) \geq 50$ billion inner configurations give that many distinct values below B, with all competitors covered. No distinct-value count is assumed beforehand.

Canonical Meridian ownership and the proved phase dictionary cover every normalized triple. Pruning uses only the height envelope, strict certified log exclusions, and the validated original 10-billion numerical-cutoff anchor. The uncertain restriction on relative Meridian word lengths is not used. The 20-billion incremental run was not reused as an anchor.

Storage, recovery, and reproducibility

Peak sampled verifier working storage across both invocations was **24.078 GB** (22.425 GiB). The largest complete bucket contained $6,715,646$ records, below the $25,000,000$ cap. The maximum observed sum of coordinator/native process RSS was 0.954 GiB, sampled every 15 seconds; this can double-count shared pages and is not a continuous RSS peak. The shared cgroup recorded one OOM-kill event. All 8481 native verifier tasks returned exit code zero, both coordinators completed, and the final audit passed. Kernel-log access was unavailable, so the victim was not identified. The RSS samples describe the tracked verifier processes; they do not certify an OOM-free workspace.

Raw inputs were removed only after complete-bucket and prefix receipts committed. A separate maintenance helper removed late disposable copies from already announced completed prefixes, checking the compact hashes before and after. It removed 15.788 GB of logical file data, including temporary copies. Its three deletion-guard tests passed. The live run retained the frozen arithmetic, pruning, sorting and comparison kernels. Recoverable checkpoints were saved after 8, 16, 24 and 32 completed comparison prefixes.

The release preserves the original anchor, sources, executables, build records, compact evidence, independent audits, Meridian addresses and atomic stage timings. Every archive member has a SHA-256 checksum. Selected identities are:

```
native_incremental:
5587ab32108fdce365eae968725bb2f7a3f729e3fdbeba873075ed5153fd4564
Core result:
652c169981af67fb51396f03ce23da5a2dc0ea15eb0f8f045535e7bd5f8524a2
```

The packaged read-only auditor can reconstruct the retained candidate integers and recheck the compact certificate. Independently rechecking deleted singleton streams requires rerunning the frozen enumeration. These are independent algorithms within this work, not an external human review or a proof-assistant certificate. This is a finite verification; no general Markov unicity theorem or trillion-value verification is asserted.

Zenodo 23874075 | original page 2/2 | SHA256 b2c4683229716cddf2fe

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_Disk_Saving_Verifier_Benchmarks_v1.pdf

Markov Verification with Reusable Coverage and Disposable Comparison Records

Benchmarks and proof addendum for Elleard Felix Webster Heffern

27 October 2026

Completed result. The largest completed run in this report verifies uniqueness of the first **20,000,000,000 distinct Markov values in numerical order**. It uses the finite Meridian coverage and certified logarithmic bounds established in Steps 1–2. The verifier took **1179.036 seconds wall time**, with **6608.473 aggregate CPU seconds** for the extension. Its measured peak working data was **20.621 GB**; the completed evidence occupies about **7.992 MB**. These are fresh finite computations, not a trillion-value verification.

First-N target	Prefix passes	Wall (s)	CPU (s)	Peak disk (GB)
100,000,000	1	13.47	34.06	1.27
1,000,000,000	2	67.12	256.44	6.67
10,000,000,000	8	838.85	4199.58	16.55
20,000,000,000†	8	1179.04	6608.47	20.62

All cases use eight workers. CPU is the coordinator plus all native children. GB is decimal. † The final row extends the completed 10,000,000,000-value cutoff. Its time and CPU are additional to the anchor; together they took 2017.883 verifier wall seconds. Retained evidence includes both runs. The corollary below states the reuse condition. The 100-million pilot used an earlier revision; the billion-scale results bind their exact source and executable hashes. Internal wall time includes planning, generation, hashing, partitioning, sorting, comparison and deletion. It excludes compilation, imports, final report production and separate validation. A separate invocation ledger includes build time and preserves failed or resumed attempts. The shared host is not an isolated benchmarking machine.

Largest-run accounting. The complete outer region contains 28,606,957,721 configurations. Every prefix pass enumerates that same region: 10,000,015,326 previously covered nodes, 2,631,005,156 strict logarithmic exclusions and 15,975,937,239 retained configurations. Across all prefixes, every retained occurrence is scanned exactly once. There are 29 repeated primary keys, representing 58 candidate configurations. The final second-signature and exact/log comparisons report no collision. Completed groups retain their counts, hashes, repeated keys and multiplicities; singleton keys and their full integers are not kept.

Zenodo 23874075 | original page 1/5 | SHA256 497c44a62ccbdd0e4beb

Why this verifies a numerical prefix

Let a configuration have reduced ordinary coordinate (p, q), $0 \le p \le q$, and height $H = p + q$. The separate root is $(0, 1)$ with maximum 1. Every other configuration has a unique canonical Meridian helm (r, s), with coprime positive odd $r \le s$, and signed phase x; only the seed omits its negative duplicate phases. Writing $D = r + s$,

$$(p, q) = \begin{cases} (r, s), & x = 0, \\ (r, D), & x = 1, \\ (s, D), & x = -1, \\ (D, (n-1)D + r), & x = n \ge 2, \\ (D, (n-1)D + s), & x = -n \le -2. \end{cases}$$

The earlier bijection proof does not assume maximum unicity. It therefore covers hypothetical equal maxima on different triples. The exact domain count is

$$C(L) = 1 + \frac{1}{2} \sum_{h=1}^{L} \varphi(h) \quad (L \ge 2).$$

Choose K with $C(K) \ge N$, put $B = F_{2K-1}$, and choose the largest integer L satisfying $4 \cdot 5^{L-2} \le B^2$. The proved inequalities

$$M(p, q) \le F_{2H-1}, \qquad M(p, q)^2 \ge 4 \cdot 5^{H-2}$$

place every competitor with maximum at most B inside height L, while all $C(K)$ inner configurations have maximum at most B. Original runs discard a point only after a strict certified lower-log excess over the upper-log enclosure of B. The corollary below also permits omission of previously covered key records in an extension. After the complete comparison, including any proved cutoff reuse, the inner configurations give at least N distinct values at most B. Thus every one of the first N distinct values, including its possible competing triples, is covered.

For the largest run,

$$K = 362{,}759, \quad C(K) = 20{,}000{,}010{,}785, \quad B = F_{725{,}517}, \quad L = 433{,}850.$$

These counts refer to configurations until the comparison succeeds. They are never used as distinct-value counts before injectivity is verified.

Fast propagation, same Meridian ownership. A disjoint frontier of ordinary mediant intervals carries its endpoint values and opposite value modulo 2^{64}; the next pole is $3ab - w$ using defined unsigned wraparound. This auxiliary traversal changes the order of visits, not the canonical Meridian owners. The exact dictionary above, full domain counts, inner counts, independent small-domain comparisons, and complete million-domain comparison check this interface. No floating-point operation decides coverage, pruning or equality. Signed 128-bit dyadic arithmetic implements the logarithmic bound at precision 64; overlapping candidate comparisons use precision 128 or full exact integers.

Zenodo 23874075 | original page 2/5 | SHA256 497c44a62ccbdd0e4beb

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_Disk_Saving_Verifier_Benchmarks_v1.pdf

Two concise additions suitable for the manuscript

The following are the two implementation-facing additions to the eight result groups in the earlier foundations document. The positive Laurent representation and finite coverage remain their explicit mathematical inputs.

Lemma 9 (the leading logarithm from ordinary coordinates). Let (p, q) be the ordinary coordinates of a nonroot configuration, with $1 \le p \le q$ and $\gcd(p, q) = 1$. Put $\rho = (3 + \sqrt{5})/2$ and define

$$(D, E) = \begin{cases} (p + q, (p - q)/2), & p, q \text{ odd}, \\ (q, p + q/2), & q \text{ even}, \\ (p, q + p/2), & p \text{ even}. \end{cases}$$

Then the existing Laurent leading-term lower bound for this configuration is

$$\theta = E \log \rho + \frac{D}{2} \log(9/5) - \log 3 \ \le \ \log M(p, q).$$

Thus its evaluation needs only the parity of p, q and integer arithmetic; neither the inverse Euclidean algorithm nor division to recover the phase is required. The root $(p, q) = (0, 1)$ is separate, with $\log M = 0$.

Proof. Let (r, s) be the coprime odd helm counts, $D = r + s$, and $x = \sigma n$, where $n \ge 0$ and $\sigma \in \{1, -1\}$; take $\sigma = 1$ at $x = 0$. The previously certified leading term has logarithm

$$\theta_\sigma(n) = \left(Dn + \sigma \frac{r - s}{2} \right) \log \rho + \frac{D}{2} \log(9/5) - \log 3.$$

At phase zero the dictionary gives $(p, q) = (r, s)$, proving the first case. At phases $1, -1$ it gives $(p, q) = (r, D), (s, D)$ respectively. In either case $Dn + \sigma(r - s)/2 = p + q/2$. At phases $n, -n$, $n \ge 2$, it gives

$$(p, q) = (D, (n - 1)D + r) \quad \text{or} \quad (p, q) = (D, (n - 1)D + s),$$

and the same coefficient equals $q + p/2$. Since r, s are odd and D is even, these are exactly the three displayed parity cases. Substitution therefore preserves the established lower bound. □

Directed rounding. If $[\lambda_-, \lambda_+]$, $[\nu_-, \nu_+]$, and $[\gamma_-, \gamma_+]$ enclose $\log \rho$, $\log(9/5)$, and $\log 3$, respectively, the certified lower endpoint remains

$$\theta_- = \begin{cases} E\lambda_-, & E \ge 0, \\ E\lambda_+, & E < 0 \end{cases} + \frac{D}{2} \nu_- - \gamma_+.$$

In particular, a negative coefficient uses the *upper* endpoint for $\log \rho$. Here E and $D/2$ are integers, so a shared fixed-point scale requires only exact signed integer operations. Existing outward rounding, range checks, and intersections with other certified lower bounds remain unchanged.

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Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_Disk_Saving_Verifier_Benchmarks_v1.pdf

Lemma 10 (safe deletion after a complete key prefix). Let Ω be a completely enumerated finite domain of canonical Meridian configurations, with each configuration emitted once. Retain the root and every configuration whose certified logarithmic lower endpoint is at most a certified upper endpoint $U_B \geq \log B$; exclusion is permitted only for the strict inequality $\ell(z) > U_B$. Denote the retained set by $\mathcal{R}$. Fix the deterministic exact key

$$\mathsf{K}(z) = M(z) \bmod 2^{64}, \qquad \mathsf{K}_b(z) = 2^{64-b} \left\lfloor \frac{\mathsf{K}(z)}{2^{64-b}} \right\rfloor, \quad 1 \leq b \leq 64.$$

The case $b = 64$ retains the full key. Partition $\mathcal{R}$ by disjoint high-bit prefixes of K_b. For each prefix, examine *all* configurations of $\mathcal{R}$ assigned to it, sort the keys exactly, and retain every repeated key together with its complete multiplicity. After that prefix's exact scan and repeated-key list have been durably committed, its original key records may be deleted. Doing so loses no possible equality of maxima.

Proof. An equality $M(z) = M(z')$ implies $\mathsf{K}(z) = \mathsf{K}(z')$, hence $\mathsf{K}_b(z) = \mathsf{K}_b(z')$, and therefore assigns both configurations to the same prefix. A key occurring once in a *complete* prefix scan cannot participate in a collision anywhere in $\mathcal{R}$. A repeated key retains every possible collision through its recorded multiplicity; coarsening the key only enlarges these candidate groups.

Replay the same finite canonical domain, source version, and retention predicate, emitting all configurations whose keys appear in the retained lists. The replay must recover precisely the committed multiplicity of each key, with no duplicated configuration. Canonical ownership reconstructs the configurations and their addresses, so storing an address record for every original key is unnecessary. Within each recovered group, different exact secondary signatures or disjoint certified log intervals prove inequality. Every remaining comparison is resolved by exact evaluation; an unresolved comparison prevents a successful certificate. Thus every possible equality survives until it is either disproved or reported. □

Completeness here is by *key prefix*, not by an arbitrary batch of Meridians: different batches can contain the same key, and a potential equal-maximum pair can cross their boundary. Such batches cannot discard their singleton keys independently. Committed counts and file checksums support restart and integrity checks; they do not replace the complete-domain argument or the exact key scan.

Corollary (reusing a completed numerical cutoff). Suppose a completed anchor computation covers every configuration with maximum at most $B_0 = F_{2K_0-1}$ and proves injectivity of its entire retained set. In a larger envelope, omit a node's key record when its ordinary height $H \leq K_0$, while continuing to traverse its descendants. Apply the identical omission during candidate replay.

Proof. Such a node has $M \leq F_{2H-1} \leq B_0$. A putative collision with value at most B_0 is already ruled out by the complete anchor. A collision with value above B_0 cannot have either configuration at height at most K_0, so both remain in the new comparison. Thus a successful new comparison and the anchor together prove injectivity through the new cutoff. A bare first-N_0 assertion is insufficient: the anchor must cover every competing configuration through B_0. Record the $C(K_0)$ omitted nodes separately and bind the anchor's complete configuration, result and replay evidence. □

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Validation, storage and reproducibility

Independent comparisons. The new traversal was compared against full exact-integer small domains, canonical Meridian blocks and the earlier modular evaluator. A separate audit compared every one of the 1,298,407 retained primary keys against the previously audited million-target run. Forced one-bit replay also recovered every retained coordinate exactly once, with its correct clipped key. That full comparison took 8.611 seconds wall time and deleted its temporary streams. It is separate from the verifier benchmark. Further independent audits reconstructed all 38 anchor candidates and all 58 extension candidates as full integers, checked their Markov identities and both residues, and reconciled every prefix and bucket certificate. They did not reconstruct every billion-scale configuration.

The native, key-tool and independent coordinator suites executed 61 tests, including forced repeated keys, partition edges, corrupted evidence, source binding, collision reporting, resource refusals and resume. A real process-group SIGKILL before generation commit was tested; resume regenerated the unfinished output and reported the lost timing interval explicitly. Hardware power-loss recovery was not tested. The large runs were not reconstructed a second time in full using huge integers, and no new external human or formal proof-checker certification is claimed.

Deletion and checkpoints. Complete buckets are compared before deletion. Data commits use fsync and atomic rename; replay must match each repeated key's multiplicity. A lock prevents overlapping writers. Executable hashes are checked before every launch, and all bound sources before accepting a result. Hashes support integrity, not a standalone proof of execution. Deleted singleton streams require regeneration for a complete reread.

Resources. Disk samples logical file sizes every 0.2 seconds; brief peaks may be missed. Its budget is monitored rather than a hard quota. Native RSS is per process. Sort arrays are capped at 6 GiB, and replay at 250,000 keys and 1,000,000 occurrences; cap failure never passes. The host provides eight CPU equivalents, 8 GiB RAM, and about 30 GB working disk after old reproducible streams were cleared. Source and historical evidence were preserved.

Reproduction. The accompanying archive includes source, tests, proof notes, benchmark JSON, every large-run prefix and bucket certificate, compact repeated keys, replay evidence and invocation ledgers. From its root:

```
cd step5_stream
python3 streaming.py build
python3 -m unittest discover -p 'test_*.py' -v
python3 streaming.py run --target 10000000000 --workers 8 \
  --pass-bits 3 --radix-bits 8 --work new_ten_billion
python3 incremental.py run --target 20000000000 --workers 8 \
  --anchor new_ten_billion --pass-bits 3 --radix-bits 8 \
  --work new_twenty_billion
```

The trillion target still has a sufficient domain of 1,430,355,600,632 configurations. Fixed disk capacity can be respected by more prefixes, but that increases complete-domain traversal work. The measurements here establish the completed targets above; they do not establish a trillion-value completion time. Further reductions in repeated traversal or candidate volume remain valuable.

Research basis. Manuscript v35, canonical Meridian bijection and positive-word representation; *Markov Meridian Verification Foundations*, Results 1–8; *Step 2: Envelope and Log Windows*; audited Step 3 Python reference and Step 4 compact backend. No lower word-length exclusion was added from the ambiguous helm-window claim. Shorter Meridians remain fully covered.

Zenodo 23874075 | original page 5/5 | SHA256 497c44a62ccbdd0e4beb

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_Meridian_Verification_Foundations_v1.pdf

Markov Meridian Verification Foundations

Selected findings and proofs for Elleard Felix Webster Heffern

27 October 2026 Version 1

Purpose and conclusion. Eight connected results support an exact finite verifier organized by canonical Meridians. They combine the current manuscript's coverage and cone machinery with selected results from Cone Bounds, Irrational Approximation, Markov Tier Separation Proof, Polynomial, and Markov Snake Collision Counting. Each result below has a proof, a stated domain, and an explicit computational role. Narrow special families and numerical observations that do not improve the general verification design are omitted.

The primary exact Meridian prototype has verified the first **100,000 distinct Markov maxima**, with complete normalized-triple agreement against a separately authored mutation-tree implementation. An additional reference run verified the first **1,000,000 distinct maxima**. Neither computation reaches the eventual target of 10^{12} values. The proofs establish coverage and the soundness of the stated filters; they do not establish global maximum unicity or the feasibility of the trillion-value production run.

Meaning of the target. The target is the smallest 10^{12} Markov values in increasing order, with every triple at each value inspected. It is not the much smaller set with maximum at most 10^{12}. A repeated maximum belonging to distinct normalized triples must be retained and reported. Even at the final rank, all configurations sharing that maximum must be processed.

Source convention. The authoritative manuscript is version 35, dated 26 October 2026 [S0]. Its address convention is used throughout. In particular $M_1(0), M_1(1), M_1(2), \ldots = 2, 5, 29, \ldots$. The symbol D below means full Laurent exponent or node-word length where specified; it is not an assumed sum of address subscripts. Source records and exact audited versions are listed at the end.

	Retained result	Role in verification
1	Canonical Meridian coverage and finite stopping	Complete enumeration and rank boundary
2	Exact residue transport	Safe comparison partitions
3	Cone enclosures and spectral refinement	Certified magnitude windows
4	Exact outward rounding	Independent arithmetic certificates
5	Coprime square allocation	Finite companion search for small cofactors
6	Coefficient norms and phase localization	Finite degree and signed-phase bounds
7	Full-return growth and a single crossing	One candidate on each applicable ray
8	Common-tail quadratic and ternary obstruction	Whole-prefix family rejection

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Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_Meridian_Verification_Foundations_v1.pdf

1 Canonical Meridian coverage and finite stopping

Foundation: manuscript [S0], equations (25)–(30), Section 5.12, Proposition 6.1.

For a positive integral Markov state,

$$a^2 + b^2 + H^2 = 3abH,$$

retain the companion order. Let F_j denote Fibonacci numbers, extended by $F_{-j} = (-1)^{j+1}F_j$. Define

$$V_1(x) = (F_{2x+1}, F_{2x-1}, 3F_{2x+1}F_{2x-1} - 1).$$

From $V_\alpha(x) = (a, b, H)$, set

$$u_0 = a, \quad u_1 = b, \quad u_{j+1} = 3Hu_j - u_{j-1}, \qquad V_{\alpha,k}(x) = (u_k, u_{k+1}, 3u_ku_{k+1} - H).$$

The canonical labels are $(1; x)$, $x \geq 0$, and

$$(1, k_1, \ldots, k_m; x), \quad m \geq 1, \quad k_1 > 0, \quad k_i \neq 0 \ (i \geq 2), \quad x \in \mathbb{Z}.$$

Proposition 1. These labels and the separate root $(1, 1, 1)$ name every sorted positive integral Markov triple exactly once. Every nonroot node has its maximum in the last coordinate. Enumeration by these labels through an integer cutoff terminates. A lazy merge of their outward rays gives a complete numerical ordering, including ties.

Proof. First, for $1 \leq a \leq b \leq c$, the other root of $q(z) = z^2 - 3abz + a^2 + b^2$ is $c' = 3ab - c = (a^2 + b^2)/c > 0$. Handle $a = b = 1$ first: its two possibilities are $c = 1, 2$, and the latter reflects to 1. Otherwise $q(b) = a^2 + (2 - 3a)b^2 < 0$, so $0 < c' < b < c$. Thus the largest coordinate decreases away from $(1, 1, 1)$. Two equal largest coordinates force that root: if $b = c$, then $a^2 = (3a - 2)b^2$, and $a \leq b$ forces $a = b = c = 1$. Repeated descent thus reaches $(1, 1, 1)$. It also proves pairwise coprimality, since mutation preserves the pairwise gcds.

For an oriented nonroot state (a, b, H), reflect its maximum to $P = 3ab - H < H$. If $P \geq 2$, reduce the oriented orbit about P by replacing its larger endpoint until both endpoints are below P. Each such step is a largest-coordinate descent, so it terminates. All orbit entries stay positive integers, and

$$u_{j+1} - 2u_j + u_{j-1} = (3P - 2)u_j > 0.$$

Consequently the indices with $u_j < P$ form an interval. Two edges wholly below P would provide three consecutive entries below P, contradicting $u_{j-1} + u_{j+1} = 3Pu_j \geq 3P$. The reduced edge is unique. Its signed displacement recovers the last nonzero index, and the recovered source has smaller pole P. Repeat. At $P = 1$, endpoint descent reaches $(1, 1)$ and determines the signed seed phase. Thus every oriented state has a unique raw address.

Exchanging the companions reverses the phase and every appended index simultaneously. The rule $k_1 > 0$, or $x \geq 0$ when there is no appended index, chooses exactly one representative. The only fixed state is $(1, 1, 2)$. This proves the bijection of configurations without assuming that their maxima are distinct.

For finite stopping, let $T = 3ab - H$, so $0 < T < H$. The first nodes on the two outward rays are

$$N_1 = (9b^2 - 2)H - T > 6H, \qquad N_{-1} = (9a^2 - 2)H - T > 6H.$$

The new larger endpoint exceeds $2H$, and the recurrence makes both outward sequences strictly increasing thereafter. Hence N_k increases with $|k|$ on each sign ray. A cutoff B permits only finitely many seed

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phases, finitely many indices per ray, and finite appended depth because each extension multiplies the maximum by more than six.

For numerical ordering, keep the next unused seed and next unused member of every active ray in an exact min-heap. Removing a seed activates its allowed signed-phase children; removing a nonseed activates its next sibling and its two child rays. Every unseen node lies on or below a pending ray whose first maximum is no greater. The least pending value therefore bounds every unseen maximum. Process a whole equal-value group before incrementing the rank; a strictly greater lookahead certifies completion of the final group. □

Implementation. The primary code uses both signed seed phases internally, emits the seed itself only for $x \geq 0$, appends only positive first indices, and allows both signs thereafter. The inverse decoder implements the proof. The independent mutation-tree oracle is a cross-check on the same configuration set. The two trees have different edges.

2 Exact residue transport

Foundation: [S0], Proposition 5.1; recent refinement: Cone Bounds [S1], response 45.

Proposition 2. For an actual source with preceding and current poles (T, H), appending $k \neq 0$ gives

$$(T,H) \longmapsto \begin{cases} (H,T) \pmod{27}, & k \text{ even}, \\ (H,7H-T) \pmod{27}, & k \text{ odd}. \end{cases}$$

The seed current residue is 2 at even phase and 5 at odd phase; its preceding pole is 1. Along actual scalar reflections $X_{j+1} = 3a_jb_j - X_j$,

$$X_k = (-1)^k X_0 + 3\sum_{j=0}^{k-1}(-1)^{k-1-j}a_jb_j, \qquad X_k \equiv (-1)^k X_0 \pmod 3.$$

Proof. No Markov coordinate is divisible by 3: descent reaches the root, and a mutation changes the mutated coordinate to its negative modulo 3. Thus each orbit square is 1 modulo 3. Direct expansion gives

$$N_{r+2} - N_r = 9H(u_{r+2}^2 - u_{r+1}^2) \equiv 0 \pmod{27H}.$$

Also $N_0 = T$ and $N_1 = (9b^2 - 2)H - T \equiv 7H - T \pmod{27H}$. These identities prove both parity classes in both directions. The seed recurrence $h(x+1) = 7h(x) - h(x-1) - 4$, starting at 2 and 5, gives the asserted phase residues. Induction on the actual scalar reflection identity proves the final formulas. □

Use. Different residues certify different maxima. Equal residues remain candidates. In particular, two histories with the same terminal value and opposite scalar-reflection parities cannot have equal starting maxima. Reflection count here is the number of the displayed scalar operations; it cannot be replaced by Meridian address length, phase, or an arbitrary tree depth. The stronger modulo $27H$ identity is local to its supplied source.

3 Cone enclosures and spectral refinement

Foundation: [S0], equation (60a); refinement: Irrational Approximation [S2], SC1–SC4.

Cone coordinates in this section are ordinary Farey coordinates. They are auxiliary coordinates for certified magnitude scheduling; canonical Meridian addresses remain the coverage and identity labels.

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Proposition 3. Consider a stopped Farey cone with endpoint values $1 \le h < b$, opposite value $1 \le z \le b$, and $h^2 + b^2 + z^2 = 3hbz$. Put

$$\Delta_h = \sqrt{9h^2 - 4}, \quad Q_h = (3h + \Delta_h)/2, \quad L_h = \log Q_h,$$
$$\gamma = \frac{bQ_h - z}{\Delta_h}, \quad \beta = \frac{z - b/Q_h}{\Delta_h}, \quad C_0 = \log(3\gamma).$$

Every actual node with primitive positive cone coordinates (m, n) satisfies

$$-\frac{n}{2b^2} \le \log(3M(m,n)) - (mL_h + nC_0) \le \frac{n}{b^2}. \tag{1}$$

Proof. The fixed-h orbit $b_{-1} = z, b_0 = b$ is $b_j = \gamma Q_h^j + \beta Q_h^{-j}$, with $\gamma\beta = h^2/(9h^2 - 4) > 0$ and $2\gamma = b + (3hb - 2z)/\Delta_h \ge b$. Hence, for $j \ge 0$,

$$0 \le \log(3b_j) - (jL_h + C_0) \le \beta/\gamma \le 4/(5b^2) < 1/b^2.$$

For $n = 1$ this is sufficient. Otherwise put $j = \lfloor m/n \rfloor$, $r = (j+1)n - m$, $s = m - jn$. Primitivity gives $r, s > 0$, $r + s = n$, placing the node between b_j, b_{j+1}. In every descendant interval the endpoints $u \le v$ are at least b, and its opposite value $w \le v$ obeys

$$3u(w/v) = 1 + (w/v)^2 + (u/v)^2 \le 3.$$

Thus $w/(3uv) \le 1/(3b^2)$. A mediant $t = 3uv - w$ subtracts from the sum of endpoint logarithms at most

$$-\log(1 - w/(3uv)) \le \frac{1}{3b^2 - 1} \le \frac{1}{2b^2}.$$

Induction bounds the accumulated subtraction by $(r + s - 1)/(2b^2)$. The two endpoint errors contribute between zero and n/b^2, and their affine parts sum to $mL_h + nC_0$. This proves (1), with the slightly stronger lower error $-(n-1)/(2b^2)$. □

Spectral refinement. Write $L(t) = \log((3t + \sqrt{9t^2 - 4})/2)$. For an outward mediant with endpoint values a, b, opposite value w, and larger pole $c = 3ab - w > w$, let

$$P = Q_a Q_b, \quad S = \sqrt{9a^2 - 4}\sqrt{9b^2 - 4}, \quad d = \frac{8}{S + 3(c - w)}.$$

Then

$$0 < \frac{d}{P - P^{-1}} \le L(a) + L(b) - L(c) \le \frac{d}{\sqrt{9c^2 - 4}} \le \frac{1}{4\sqrt{2}\, a^2 b^2}. \tag{2}$$

Indeed $S^2 - 9(c - w)^2 = 16$ and $P + P^{-1} - 3c = d$. Therefore the spectral defect is

$$\int_{3c}^{3c+d} \frac{dt}{\sqrt{t^2 - 4}}.$$

The integrand decreases, giving the two endpoint bounds. The larger-root formula gives $c \ge 2ab$, $c - w \ge ab$, while $S \ge 5ab$, $d \le 1/(ab)$, and $9c^2 - 4 \ge 32a^2b^2$; these give the final bound.

For a primitive descendant $mA + nB$ between adjacent Farey vectors carrying a, b, induction now yields

$$0 < mL(a) + nL(b) - L(M(m,n)) \le \frac{m + n - 1}{4\sqrt{2}\, a^2 b^2}. \tag{3}$$

At each mediant the two prior errors and the local defect add, and all later endpoint products are at least ab. The coefficients satisfy $(m_1 + n_1 - 1) + (m_2 + n_2 - 1) + 1 = m + n - 1$. This proves (3). Applied in the orbit gap, it replaces the error by $(n-1)/(4\sqrt{2}\, b_j^2 b_{j+1}^2)$. □

Use. Outward rational enclosures of L, together with the rational upper error $(m + n - 1)/(5a^2b^2)$, give checkable magnitude intervals. Only disjoint intervals certify inequality. Overlap calls for refinement or exact reconstruction. Neither (1) nor (3) says that different cones have disjoint height ranges.

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4 Exact outward rounding

Sources: [S2], SC57–SC60; Tier Separation [S3], Sections 1–2 and 6.

Proposition 4. Let (a, b, c) be a sorted nonroot Markov triple. For either $(h, k) = (a, b)$ or (b, a), set $E = 3hc - k$. Then

$$\frac{h}{3c^2} < Q_h - \frac{E}{c} < \frac{h}{2c^2}, \qquad 0 < cQ_h - E < \frac{1}{4}, \qquad E = \lfloor cQ_h \rfloor. \tag{4}$$

Proof. The larger-root formula gives $c \geq 2ab$, so $h, k \leq c/2$, and $c \geq 2$. The Markov equation gives $E^2 - 3hcE + c^2 = -h^2$. The larger root applies because $E/c = 3h - k/c > 3h/2$. Subtracting from Q_h and rationalizing yields

$$Q_h - E/c = \frac{2h^2}{c^2\left(\sqrt{9h^2 - 4} + \sqrt{9h^2 - 4 - 4h^2/c^2}\right)}.$$

Both radicals are below $3h$; the first exceeds $2h$, and the second is at least $2h$. This gives the strict bounds, and $h \leq c/2$ gives the floor conclusion. □

Exact certificate. For any positive integer scale B, put

$$v = \mathrm{isqrt}((9h^2 - 4)B^2), \quad L = \frac{3hB + v}{2B}, \quad U = \frac{3hB + v + 1}{2B}.$$

Since $L < Q_h < U$,

$$cL - \frac{h}{2c} < E < cU - \frac{h}{3c}. \tag{5}$$

Choosing $c(U - L) < 1/4$ makes this interval shorter than $1/3$, so it contains at most one integer. Equivalently, an entirely integral evaluation is

$$E = \left\lfloor \frac{3hc + \mathrm{isqrt}(c^2(9h^2 - 4))}{2} \right\rfloor.$$

These independently check outward recurrence values. Four accurate approximations sharing a denominator do not by themselves contradict a collision: their target Q_h's can differ. The error bound also does not supply a fixed positive approximation exponent uniformly for both companions.

5 Coprime square allocation and finite companion search

Source: Tier Separation [S4], Sections 1–7.

Proposition 5. Suppose two different normalized triples share maximum $c > 2$. Order them as $a < a' < b' < b < c$, and set

$$\Delta = ab - a'b', \quad C = bb' - aa', \quad D = a'b - ab'.$$

Then $\Delta > 0$, $C > D > 0$, and $CD = c^2\Delta$. Put $\ell = 1$ for odd c and $\ell = 2$ for even c, and $m = c/\ell$. There is a proper coprime factorization

$$m = uv, \quad u, v > 1, \quad \gcd(u, v) = 1, \quad u^2 \mid C/\ell, \quad v^2 \mid D/\ell. \tag{6}$$

If $q = p^j \| m$ is any full prime-power factor and $k = m/q$, then

$$aa' < \frac{\ell k^2}{4}. \tag{7}$$

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Proof. For fixed c, the reduced root b strictly decreases as a increases: implicit differentiation has positive numerator $3cb - 2a$ and negative denominator $2b - 3ca$. Shared companions identify the same reduced root; repeated companions outside the seeds are ruled out by coprimality. Hence the stated nesting holds. With $t = b/a > t' = b'/a' > 1$, the identity $t + 1/t = 3c - c^2/(ab)$ gives $\Delta > 0$. Also $C - D = (a + b)(b' - a') > 0$. Direct expansion gives

$$CD = (a^2 + b^2)a'b' - ab(a'^2 + b'^2) = c^2\Delta.$$

If c is even, all four companions are odd and reduction modulo 4 forces $c \equiv 2 \pmod 4$; both C, D are even and m is odd.

For an odd prime $p \mid m$, both normalized factors cannot be divisible by p. Otherwise

$$aC + bD = (b^2 - a^2)a', \qquad bC + aD = (b^2 - a^2)b'$$

force $p \mid b^2 - a^2$. Combining this with $a^2 + b^2 \equiv 0 \pmod p$ contradicts $\gcd(a, c) = 1$. Thus each p^{2j} from m^2 lies wholly in one normalized factor. Taking $u = \gcd(m, C/\ell)$, $v = \gcd(m, D/\ell)$ gives the coprime allocation. Furthermore

$$0 < D/\ell < C/\ell < bb'/\ell \le \frac{c^2}{4\ell aa'} = \frac{\ell m^2}{4aa'} < m^2.$$

Neither factor can receive all of m^2, proving properness. A selected q^2 divides one normalized factor, so $q^2 \le C/\ell < c^2/(4\ell aa')$; cancellation proves (7). □

Finite certificate. Given one known triple (a_0, b_0, c), every alternative smaller companion $h \ne a_0$ lies in

$$1 \le h \le \min\left\{\operatorname{isqrt}(\lfloor c/2 \rfloor), \left\lfloor \frac{\ell k^2 - 1}{4a_0} \right\rfloor\right\}. \tag{8}$$

Test every integer there. Put $R_h = 9h^2c^2 - 4(h^2 + c^2)$; an alternative requires a nonnegative exact square, even numerator in $b_h = (3hc - \sqrt{R_h})/2$, and $h \le b_h \le c$. Verify the original equation too. The prime p and its exact valuation must be certified. A probable-prime label is insufficient.

This gives a compact exclusion when a large prime-power factor leaves a small cofactor; it does not guarantee that every maximum has that property. The prime-power and twice-prime-power consequence is classical, and no novelty claim is made for it. In the new exact check, $(5, 13, 194)$ has $\ell = 2, q = 97, k = 1$, leaving no positive alternative h in (8).

6 Coefficient norms and signed phase localization

Source: Polynomial [S5], pass 1; positive harmonics and endpoints from [S0].

Let $\rho = (3 + \sqrt{5})/2$. The actual Meridian expansion has positive coefficients

$$M(x) = c_0 + \sum_{j>0}(c_j\rho^{jx} + c_{-j}\rho^{-jx}), \qquad c_{-j} = \overline{c_j},$$

where the sums run over positive even indices, the bar conjugates $\sqrt{5}$, and odd coefficients vanish. Its largest exponent D is even, and

$$c_D = C_D\rho^t, \quad c_{-D} = C_D\rho^{-t}, \quad C_D = \frac{1}{3}(9/5)^{D/2}, \quad |t| \le D/2 - 1.$$

Here t is an integer. Define

$$B_{\text{int}} = c_0 + 2\sum_{0<j<D}\sqrt{c_jc_{-j}}.$$

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Proposition 6. For every real x, $M(x) \geq B_{\text{int}} + 2C_D > 2C_D$. For integers $q \geq 1$ and $|x| \geq q$,

$$M(x) \geq B_{\text{int}} + C_D(\rho^{Dq-|t|} + \rho^{-Dq+|t|}). \tag{9}$$

Proof. AM–GM on each positive coefficient pair gives $c_j\rho^{jx} + c_{-j}\rho^{-jx} \geq 2\sqrt{c_j c_{-j}}$. Retain the endpoint pair exactly. Its exponent is $Dx + t$, whose absolute value is at least $Dq - |t| > 0$. The function $\rho^s + \rho^{-s}$ increases for $s \geq 0$. Summing proves (9); $c_0 > 0$ gives the strict endpoint-only lower bound. □

Manuscript input justification. Diagonalize the seed matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ with eigenvalues ρ, ρ^{-1}. In the manuscript's trace representation the two letter matrices have positive entries of the form constant times $\rho^x, 1, \rho^{-x}$. A closed-index expansion of the node-word trace therefore has positive harmonic coefficients. The all-upper and all-lower index sequences give $c_{\pm D} = \frac{1}{3}(9/5)^{D/2}\rho^{\pm(r-s)/2}$, where the positive odd helm counts satisfy $r + s = D$; thus $t = (r - s)/2$ and the stated bound follows. Conjugate reciprocity also follows directly from the seed. With $X = \rho^x$ and $\phi = (1 + \sqrt{5})/2$,

$$a_1(X) = \frac{\phi X + \phi^{-1}X^{-1}}{\sqrt{5}}, \qquad b_1(X) = \frac{\phi^{-1}X + \phi X^{-1}}{\sqrt{5}}.$$

Both expressions are fixed by $(X, \sqrt{5}) \mapsto (X^{-1}, -\sqrt{5})$. The integer-polynomial companion recurrence preserves this invariance, so its coefficient comparison gives $c_{-j} = \overline{c_j}$. This is the algebra behind [S0], Proposition 5.5 and Lemma 7.3.

Exact use. A candidate value H cannot occur at degree D when $3H \leq 2(9/5)^{D/2}$. Rational cross-multiplication tests this without logarithms. Define integers $L_0 = 2, L_1 = 3, L_{e+1} = 3L_e - L_{e-1}$; then $L_e = \rho^e + \rho^{-e}$. Use $C_D L_{Dq-|t|}$ and a rational lower bound for B_{int}, even just c_0, to exclude both signed tails. Increasing q terminates for finite H.

Normalization. Opposite coefficients are conjugates, generally not equal. Also c_0 need not be an integer: $M_{1,1}$ has $c_0 = 88/25$, $c_4 = (81 - 27\sqrt{5})/50$, and $c_2 = (78 - 18\sqrt{5})/25$. Consequently a small conjugate tail alone does not justify rounding an irrational head difference to an integer. The primary verifier evaluates the actual integer recurrence. Polynomial certificates must retain all denominators and conjugations.

7 Full return growth and a single crossing

Source: Polynomial [S5], pass 20.

Start from an actual oriented source (a, b, A), with $A > \max(a, b)$. Fix signs $\epsilon, \sigma \in \{-1, 1\}$. Let $V_A(k)$ be the maximum obtained by appending ϵk, then σ, at the unchanged source phase.

Proposition 7. For integer $k \geq 2$,

$$(3A - 1)^4 < \frac{V_A(k + 1)}{V_A(k)} < (3A + 1)^4. \tag{10}$$

For two such sources with maxima $A > B$, fixed independent signs and source phases, the ratio $V_A(k)/V_B(k)$ strictly increases and grows by a factor greater than $((3A - 1)/(3B + 1))^4 > 1$ per step. Equality has at most one integer location on the shared magnitude ray.

Proof. Write one outward arm as $w_0 = s, w_1 = 3As - o$, where s, o are its selected and other original companions. Then $w_1 > 2As \geq 2A$ and $(3A - 1)w_i < w_{i+1} < 3Aw_i$ for $i \geq 1$. At magnitude k, put $x = w_{k-1}, y = w_k, T = 3xy - A$. Both $x, y > 2A$. At the next magnitude, $z = 3Ay - x$ and $T' = 3yz - A$. With $\lambda = 3A - 1$, product bounds give $T' > \lambda^2 T$. The source conic gives the exact upper identity

$$T' - 9A^2T = -9Ax^2 - 3xy - A < 0.$$

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The final unit append selects one of x, y, denoted s_k, consistently along the fixed-sign ray. The complete returned value is

$$V_A(k) = (9s_k^2 - 2)T - A.$$

Its selected companion grows by a factor between λ and $3A$. Thus the next value is greater than $\lambda^4 V_A(k)$. For the other inequality, $T > A$ gives $V_A(k) > (9s_k^2 - 3)T$, and therefore

$$\frac{V_A(k+1)}{V_A(k)} < (3A)^4 \left(1 + \frac{1}{3s_k^2 - 1}\right) < (3A)^4 \left(1 + \frac{1}{12A^2 - 1}\right) < (3A+1)^4.$$

The last step follows from $1/(12A^2 - 1) < 4/(3A)$ and the binomial expansion. Dividing the lower bound for A by the upper bound for B, and using $3A - 1 > 3B + 1$, proves the ratio claim. □

Finite search. At the first allowed $k \geq 2$, a positive difference excludes all later indices. A zero requires comparison of the full triples. A negative difference can be bracketed by doubling and then exact bisection: the ratio tends to infinity by its fixed growth factor. Only the first nonnegative candidate can be equal. The same argument applies on a fixed arithmetic subsequence.

The two sides must use the same varying magnitude and the specified two-append form. Equal source maxima, unrelated suffix words, and independently varying indices remain outside the theorem. A monotone ratio is not itself a proof that its one candidate misses 1.

8 Common tail quadratics and a ternary obstruction

Source: Markov Snake Collision Counting [S6], common-tail analysis and manual extra pass 1.

Use the deterministic lower-left graph chop and its auxiliary transfer letters

$$E = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad F = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \qquad e = (1, 0).$$

For a certified actual folded half U, the helm count is $eUEU^T e^T$; the nonhelm count is $eUU^T e^T$. These formulas are input certificates from the actual graph transfer construction. An arbitrary word with the right letter counts is not automatically an actual Markov graph.

Suppose the compared halves are PS, PT. Put $r = eP = (x, y)$, and let

$$A = SES^T, \quad B = TET^T \text{ (helm)}, \quad \text{or } B = TT^T \text{ (nonhelm)}.$$

Then their difference is exactly

$$rDr^T = \alpha x^2 + 2\beta xy + \gamma y^2, \qquad D = A - B = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}. \tag{11}$$

Proposition 8. A nonempty canonical common prefix gives a positive primitive row (x, y). If $D \neq 0$, equality at that row is impossible unless $\alpha\gamma < 0$ and $\delta = \beta^2 - \alpha\gamma$ is an integer square. When those conditions hold, there is at most one primitive positive candidate row. It must still decode to the prescribed canonical prefix and actual graph.

Proof. Both A, B are positive definite with determinant one and nonnegative off-diagonal entries. For distinct such matrices,

$$\delta = \operatorname{tr}(AB^{-1}) - 2 > 0,$$

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because AB^{-1} is similar to a positive definite determinant-one matrix, with reciprocal positive eigenvalues, and trace 2 occurs only when $A = B$. If the two diagonal differences have the same weak sign, the off-diagonal difference has that sign too: from determinant one, the nonnegative off-diagonal entry is $\sqrt{A_{11}A_{22} - 1}$. At least one difference is strict, so the quadratic cannot vanish on a positive row. This includes a zero diagonal difference. Otherwise $\alpha\gamma < 0$, and the quadratic in y/x has exactly one positive real root. At an integer zero,

$$(\gamma y + \beta x)^2 = \delta x^2.$$

A rational square root of the integer δ is integral. Its positive root fixes a rational slope, which has at most one primitive positive representative. Unimodularity of the prefix product guarantees primitivity. □

For completeness, $D = 0$ is an identical-form case and certifies no exclusion. The empty prefix has $r = (1, 0)$ and is tested directly. Word reconstruction must keep the canonical initial letter: for a positive row, a last E gives output ratio $y/x > 1/2$, a last F gives $y/x < 1/2$, allowing inverse decoding until the boundary. At that boundary $eE = eF$; the prescribed initial E removes this ambiguity. These conditions prevent a square discriminant from being mistaken for a realized graph.

Ternary corollary for two helms. For distinct tails, $D = SES^T - TET^T$. If $|S|, |T|$ have opposite parity, then $rDr^T \not\equiv 0 \pmod 3$ for every primitive integer row. If their parities agree, let $h = \min_{i,j} v_3(D_{ij}) \geq 1$ and

$$C = 3^{-h}D = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}.$$

Then $\gamma \equiv \alpha + \beta \pmod 3$, and $\beta^2 - \alpha\gamma$ is 1 or 2 modulo 3. When it is 2,

$$v_3(rDr^T) = h$$

for every primitive row, excluding every such common prefix at once.

Proof. Direct multiplication gives $XEX^T \equiv -E \pmod 3$ for $X = E, F$. Opposite parity therefore leaves a signed E, whose binary quadratic discriminant is the nonsquare 2 modulo 3. For equal parity, write $K = TET^T$. Expanding $\det(K + 3^h C) = \det K = 1$, dividing by 3^h, and reducing modulo 3 gives $\alpha + \beta - \gamma = 0$. Since C is not zero modulo 3, its discriminant is $\beta^2 - \alpha^2 - \alpha\beta$, which takes only 1 or 2 on the eight nonzero pairs. In the nonsquare case the form vanishes on no nonzero projective row; primitivity makes r nonzero modulo 3. This proves the exact valuation. □

Distinct same-cap tails do give $D \neq 0$: test matrix words on the positive row $(1, 1)$. The last-letter ratio test uniquely decodes the word and the first coordinate strictly increases. Thus words are free; equality of the palindromic words SES^T, TET^T forces $S = T$. In the mixed-cap case their lengths have opposite parity. This argument uses matrix-word freeness, not the ambiguous boundary row e.

Actual example. The current canonical helms are $610 = M_{1,1,-1,1}(0)$ with triple $(1, 233, 610)$, and $433 = M_{1,2}(0)$ with triple $(5, 29, 433)$. Their halves are EEE, EFF. Removing the common E leaves

$$D = \begin{pmatrix} 15 & 24 \\ 24 & 21 \end{pmatrix} = 3\begin{pmatrix} 5 & 8 \\ 8 & 7 \end{pmatrix}.$$

The normalized discriminant is $29 \equiv 2 \pmod 3$; at $r = (2, 1)$ the gap is 177 and has valuation one. The theorem handles every primitive common prefix with these tails. It does not bound the number of tail pairs requiring analysis.

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9 Verification design and independent audit

Trusted core. The complete organizer is Proposition 1. Records retain the canonical address, signed phase, and ordered companions until normalization for comparison. Equal maxima are never deduplicated before their triples are compared. The root is separate; negative seed phases create canonical nonseed descendants but are not emitted as duplicate seed nodes. Inclusive cutoffs and whole final rank groups are part of the specification.

Production design. A future large run should replace memory-resident full integers with certified magnitude intervals and exact modular signatures, preserving reconstruction addresses. Each active Meridian ray or address subtree has a recorded frontier and a proved lower bound. A magnitude window is complete only when every remaining frontier is certified above its upper boundary, or when a validated cone scheduler accounts for every remaining canonical address. Cone coordinates must be translated with an audited bijection; they cannot silently replace the Meridian coverage proof.

All pruning is one-sided: a strict interval separation, different exact residue, or applicable theorem certificate rejects equality; every undecided comparison continues to tighter arithmetic or full exact reconstruction. Modular signatures can be sorted in external buckets without probabilistic assumptions: equal integers always have equal signatures. Every bucket must be processed, and every repeated signature retains all contributing addresses. Hashes authenticate artifacts but do not prove multiset equality. A short fingerprint is safe only as a preliminary filter that preserves every equal-signature group.

For a count target N, either consume N complete value groups from the certified ordered stream, or verify every triple through a certified cutoff containing at least N distinct values. A successful cutoff run proves more than the requested prefix. The accepted count must come from the complete enumeration, not an asymptotic estimate. Restart checkpoints must bind the code hash, cutoff/window, canonical frontier, processed counts, unresolved candidates, and output digests; partition accounting must show no omitted or duplicated configuration.

Implemented now. The delivered reference package contains the primary exact Meridian enumerator, reverse decoder, residue transport, independent Vieta oracle, independent small-bound discriminant enumeration, boundary fixtures, and exact identity checks for the selected research. It does not yet contain a production external-sort engine, certified logarithm library, all optional filter dispatchers, or a complete graph-to-Meridian certificate interface. Those remain implementation and audit tasks before a trillion-value run. The generic exact fallback already provides the correctness specification against which optimizations can be checked.

Measured runs. Times use monotonic wall clocks and process CPU clocks. These single-process measurements were made in a shared Linux workspace, Python 3.12.14, AMD EPYC 9V74, nine available logical CPUs, an eight-CPU quota, and an 8 GiB memory limit. Each JSON record states its timer boundaries; source retrieval, proof development, and document production are excluded.

Completed operation	Wall seconds	CPU seconds
Primary Meridian first 100,000 value groups	2.883506	2.880788
Meridian bounded rerun at its exact cutoff	0.945661	0.944657
Independent Vieta at the same cutoff	0.169888	0.169748
Set comparison and all address round trips	4.601196	4.597185
Independent Vieta first 1,000,000 value groups	29.797529	29.787335

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At the 100,000-value cutoff, all 100,000 full normalized triples agree between the two enumerators; no maximum repeats. Every address re-evaluates exactly, decodes from both companion orders, and matches the independent modulo-27 transport. The cutoff has 323 decimal digits and address

$$(1, 1, -1, 1, -2, 1, -1, -1, -1;\ x = 2).$$

The exact cutoff and first larger value are saved in the run-record JSON. The combined comparison process peaked at 248,128 KiB. The separate million-value Vieta run found no repeated maximum, had a 1,022-digit cutoff, and peaked at 1,065,742,336 bytes. The timings and memory growth do not justify a linear trillion-value extrapolation.

An independent reviewer additionally checked exact triple-set equality through 10^{100}, covering 9,670 triples, including 2,473 negative-phase nodes and 8,216 nodes with negative appended indices. Checks included 600 inclusive-cutoff cases, direct discriminant enumeration through 1,000, 250 random signed-address round trips, the first 10,000 complete groups, and an injected collision at the terminal rank. The reviewer found an invalid noninteger-rank input that could fail to terminate; both interfaces now reject it, and regression tests pass. No unresolved correctness defect was found in the audited exact kernels. These are software checks of the proved algorithms, not substitutes for their coverage proofs.

Independent research checks also covered 2,048 exact outward rounding cases, five finite cofactor certificates, 840 full-return growth inequalities, 6,336 ratio inequalities, 1,965 actual snake counts against a separate Farey construction, and 3,906 formal tail comparisons. Their exact scripts and outputs are included. The current manuscript's older billion-scale runs were not reproduced in this step and are not counted as newly verified coverage.

Gate for the next stage. Retain Meridian organization, implement certified magnitudes and deterministic bucket or window accounting, then benchmark successive scales against these exact kernels. Enable an optional research filter only after its hypotheses and rejection certificate have independent tests. The next stage can then measure the cost of reaching 10^{12} ranks with the actual storage and arithmetic strategy.

10 Source record and incorporation notes

The new proofs above are self-contained except for the stated actual-graph folding input in Result 8 and the explicitly explained manuscript trace representation in Result 6. The universal scopes come from their proofs; finite checks test code and examples.

S0 *Geometric Descent on the Markov Surface*, version 35, 26 October 2026. Sections 5–7; canonical definitions (25)–(30), Proposition 6.1, residue Proposition 5.1, cone (60a), harmonics Proposition 5.5 and Lemma 7.3. Snapshot SHA-256 is recorded in the package manifest.

S1 *Markov Cone Descent 100h Research Record*, version 48, updated 27 October 2026 at 16:57 UTC; response 45. Actual-reflection residue transport and its domain correction.

S2 *Markov Cone Irrational Approximation 100 Pass Record*, version 17, updated 27 October 2026 at 16:56 UTC. SC1–SC4 and SC57–SC60. Stable-norm and deep-sector dependencies are omitted from the trusted core here.

S3 *Markov Rational Relation Induction Approximation Pass*, version 1, 27 October 2026; Sections 1–2 and 6. Shared-denominator approximation bounds and limits.

S4 *Markov Rational Relation Induction Factor Splits*, version 2, 27 October 2026; Sections 1–7. Square allocation, strict cofactor bound, and finite alternative-companion test.

S5 *Meridian Polynomial Unicity 100 Pass Record*, version 32, updated 27 October 2026 at 17:18 UTC; passes 1 and 20. Passes 30–31 supply the cautions against continuous-sign and equality-preserving reverse-descent shortcuts. The audited cutoff is pass 31.

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S6 *Markov Snake Helm Nonhelm 100 Pass Record*, version 2, updated 27 October 2026 at 17:40 UTC; common-tail analysis and manual extra pass 1. Folding inputs also appear in *Cross Meridian Helm Nonhelm Tail Comparison*. All examples here have been reconciled with the current manuscript addresses.

For manuscript incorporation, Results 1–2 supply the complete organizer and exact arithmetic invariant; Results 3–8 can enter as separate lemmas with their domains intact. The verified numerical claims belong in a computational section with the accompanying source hashes and certificates. No general maximum-unicity conclusion follows from combining these results without resolving all remaining canonical configurations.

Original: Markov_50_Billion_Verification_Package_v1/documents/Markov_Step2_Envelope_and_Log_Windows_v1.pdf

A Certified Meridian Search Envelope and Logarithmic Windows

Step 2 addendum for Elleard Felix Webster Heffern

27 October 2026 Version 1

Resolved step. The missing interface was a finite, exactly counted search region that preserves canonical Meridian coverage while avoiding a numerical-order heap of enormous Markov integers. The result below supplies that region, exact phase bounds on each Meridian, and outward-rounded windows for $\log M_\alpha(x)$ that use only small coordinate data. It is one integrated addition to the eight findings in the Step 1 foundations [1].

The complete outer region for the proposed first-10^{12} verification has

$$\boxed{1,430,355,600,632 \text{ configurations.}}$$

This is a *workload and coverage certificate*, not a collision-free result. No trillion-scale uniqueness check was run. The log windows can discard provably irrelevant values and separate comparisons; overlapping windows remain unresolved.

1 The phase-dependent coordinate map

Use ordinary node coordinates with $M(0/1) = 1$, $M(1/1) = 2$, and $M(1/2) = 5$. These are auxiliary navigation coordinates; canonical Meridian addresses remain the organizing labels. A nonroot Meridian α has coprime odd helm counts (r, s), with $1 \le r \le s$, and $D = r + s$. The seed has $(r, s) = (1, 1)$. Its canonical phases are $x \ge 0$; every other Meridian has $x \in \mathbb{Z}$.

Lemma 1 (exact phase dictionary). The ordinary coordinate (p, q) of the configuration $M_\alpha(x)$ is

phase	(p, q)
0	(r, s)
1	(r, D)
-1	(s, D)
$n \ge 2$	$(D, (n-1)D + r)$
$-n \le -2$	$(D, (n-1)D + s)$.

(1)

Consequently its coordinate height is exactly

$$H_\alpha(0) = D, \qquad H_\alpha(n) = nD + r, \qquad H_\alpha(-n) = nD + s \quad (n \ge 1). \tag{2}$$

For a height cutoff L, a Meridian occurs only if $D \le L$, and its allowed phases are

$$0; \quad 1 \le x \le \left\lfloor \frac{L - r}{D} \right\rfloor; \quad 1 \le -x \le \left\lfloor \frac{L - s}{D} \right\rfloor. \tag{3}$$

Omit the negative range for the seed. These are integer limits; no approximate size test is needed to establish coverage.

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Proof of the dictionary. Retain the companion order in a source (a, b, H). If U, V are their adjacent coordinate vectors and $W = U + V$, one outward orbit step replaces the farther vector by its sum with W. Thus an append has exactly the manuscript's linear count rule

$$(U', V') = \begin{cases} ((k-1)W + V,\ kW + V), & k \geq 1, \\ (|k|W + U,\ (|k|-1)W + U), & k \leq -1. \end{cases} \tag{4}$$

The two matrices have determinant -1. This is an identity of actual neighboring configurations, not a replacement of a Meridian edge by one ordinary tree edge.

At phase zero the seed companion vectors are $(1, 0), (0, 1)$. At phase one they are $(1, 1), (0, 1)$. At phase $n \geq 2$ they are $(1, n), (1, n-1)$. Indeed the boundary values are $M(1/j) = F_{2j+1}$, obtained from $1, 2, 5, 13, \ldots$ by $f_{j+1} = 3f_j - f_{j-1}$; the seed companions are F_{2n+1}, F_{2n-1}. Negative phases exchange the two companion vectors.

Because (4) is linear, replacing the seed basis replaces every node vector by the same linear map. Applied to the helm vector (r, s) these basis changes give (1), hence (2)–(3). At the exceptional zero-phase seed, only positive first appends are allowed; they enter the half-tree $0 < p < q$. All subsequent node vectors remain there; their endpoints lie in $0 \leq p \leq q$.

For completeness, the ordinary mediant tree between $0/1$ and $1/1$ gives every reduced $0 < p < q$ once, by the Euclidean algorithm. Its configuration tree is the increasing Vieta tree above $(1, 2, 5)$; the separate configurations $(1, 1, 1)$ and $(1, 1, 2)$ supply the two initial exceptions. Largest-coordinate descent proves configuration coverage and uniqueness independently of maximum unicity [1, Result 1]. Hence the dictionary is compatible with that full configuration bijection.

There is also an explicit inverse, useful for audit. If p, q are odd, take $(r, s, x) = (p, q, 0)$. If q is even, sort $p, q - p$ into $r \leq s$ and choose $x = 1$ when $p \leq q - p$, otherwise $x = -1$. If p is even, write $q = tp + u$, $0 < u < p$; sort $u, p - u$ into $r \leq s$, taking $x = t + 1$ when $u \leq p - u$, otherwise $x = -t - 1$. The two sorted entries are odd and coprime. A tie forces the seed and selects its positive phase. Thus every coordinate has exactly one canonical Meridian and phase. In particular, phase-zero coordinates are precisely the coprime odd/odd pairs. □

Lemma 2 (exact configuration count). Including the separate root, let $C(L)$ count all configurations of coordinate height at most L. For $L \geq 2$,

$$C(L) = 2 + \frac{1}{2}\sum_{h=3}^{L} \varphi(h) = 1 + \frac{1}{2}\sum_{h=1}^{L} \varphi(h). \tag{5}$$

For fixed $h \geq 3$, the admissible nodes are $(p, h - p)$ with $0 < p < h/2$ and $\gcd(p, h) = 1$. Pairing p with $h - p$ gives $\varphi(h)/2$. The root and height-two seed add two. This counts *configurations*, even if two maxima coincide.

The count was checked with a separate Möbius sieve using

$$C(L) = \frac{5 + \sum_{d=1}^{L} \mu(d)\lfloor L/d \rfloor^2}{4}.$$

The sum counts ordered coprime pairs in $[1, L]^2$, which also number $2\sum_{h=1}^{L} \varphi(h) - 1$. This proves the independent counting identity.

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2 A finite envelope for the first trillion values

Lemma 3 (uniform size bounds). A node with coordinate height $h = p + q \geq 2$ satisfies

$$2^{h-1} \leq M(p/q) \leq F_{2h-1}, \qquad \boxed{M(p/q)^2 \geq 4 \cdot 5^{h-2}.} \tag{6}$$

The upper bound is attained on $p = 1$; it is strict for $p \geq 2$.

Proof. If an outward mediant has companion values a, b and maximum c, the larger-root formula gives

$$c = \frac{3ab + \sqrt{9a^2b^2 - 4a^2 - 4b^2}}{2} \geq 2ab,$$

because $2a^2b^2 \geq a^2 + b^2$ for $a, b \geq 1$. Also $c < 3ab$ since its opposite value is positive. The initial vectors $(0, 1), (1, 1)$ have values $1, 2$. Their heights add at each mediant. Induction with $c \geq 2ab$ proves the power-of-two bound.

For every mediant above these endpoints, $\max(a, b) \geq 2$, so $5a^2b^2 \geq 4a^2 + 4b^2$. The same larger-root formula now gives $c \geq \frac{5}{2}ab$. Both endpoint values satisfy $M \geq \frac{2}{5}(\sqrt{5})^h$. This inequality propagates because $\frac{5}{2}(\frac{2}{5})^2 = \frac{2}{5}$. Squaring gives the stronger lower bound in (6).

The boundary $p = 1$ has $M(1/q) = F_{2q+1} = F_{2h-1}$. If $p \geq 2$, neither parent is $(0, 1)$, so their heights h_1, h_2 are both at least two. Set $m = 2h_1 - 1$ and $n = 2h_2 - 1$. The Fibonacci addition identity and $F_{j+1} \geq \frac{3}{2}F_j$ for $j \geq 2$ give

$$F_{m+n+1} = F_{m+1}F_{n+1} + F_mF_n \geq \frac{13}{4}F_mF_n > 3F_mF_n.$$

Induction therefore yields $c < 3ab \leq 3F_mF_n < F_{2(h_1+h_2)-1}$. □

Theorem (complete prefix certificate). Choose $K \geq 2$ with $C(K) \geq N$, set

$$B = F_{2K-1}, \qquad L = \max\{h \geq 2 : 4 \cdot 5^{h-2} \leq B^2\}.$$

If maximum unicity is verified for every configuration of height at most L, then it holds for the first N distinct Markov values in numerical order.

Proof. All $C(K)$ inner configurations have maximum at most B. Every nonroot configuration with maximum at most B has $4 \cdot 5^{h-2} \leq B^2$, hence $h \leq L$; the root is included separately. Once the outer collision check succeeds, the inner configurations give at least N *distinct* values at most B, and every possible competing configuration at any of those values has been checked. Thus the first N values are covered. No step converts a configuration count into a distinct-value count before verifying injectivity. □

Exact specialization. Two integer sieves and one fast-doubling Fibonacci evaluation give

$$\begin{aligned} K &= 2,565,100, & C(K-1) &= 999,999,913,445, \\ B &= F_{5,130,199}, & C(K) &= 1,000,000,419,685, \\ L &= 3,067,794, & C(L) &= 1,430,355,600,632. \end{aligned} \tag{7}$$

The outer region is about $1.4304N$ configurations. It is sufficient, not asserted minimal. The single cutoff integer has 3,561,599 bits and occupies 445,200 unsigned bytes. A rational log interval locates L, and the separate exact check $4 \cdot 5^{L-2} \leq B^2 < 4 \cdot 5^{L-1}$ certifies it. Individual candidates need not be constructed as full integers to establish coverage.

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3 Certified log windows from Meridian data

Logarithmic evaluation fits this envelope directly. Let

$$\rho = \frac{3+\sqrt{5}}{2}, \quad \lambda = \log \rho, \quad t = \frac{r-s}{2}, \quad A_\sigma = \frac{1}{3}(9/5)^{D/2}\rho^{\sigma t}, \qquad \sigma \in \{1, -1\}.$$

The positive Laurent expansion and its leading coefficient are the stated input from [1, Result 6] and the manuscript's trace representation:

$$M_\alpha(x) = \sum_{j=-D,-D+2,\ldots,D} c_j \rho^{jx}, \quad c_j \geq 0, \quad c_{\sigma D} = A_\sigma. \tag{8}$$

The finite-envelope theorem above does not depend on (8); only the optional sharper log window does.

Corollary (no full-value evaluation). For $n \geq 0$, define

$$\begin{aligned} b_\sigma &= \frac{D}{2}\log(9/5) + \sigma t\lambda - \log 3, \\ \theta_\sigma(n) &= Dn\lambda + b_\sigma, \\ \eta_\sigma(n) &= (D-1-2n)\lambda - b_\sigma. \end{aligned}$$

Then

$$\theta_\sigma(n) \leq \log M_\alpha(\sigma n) \leq \theta_\sigma(n) + \log(1 + e^{\eta_\sigma(n)}). \tag{9}$$

For the actual coordinate height $h = H_\alpha(\sigma n)$, intersect this with

$$\max\{(h-1)\log 2,\ \log 2 + \tfrac{h-2}{2}\log 5\} \leq \log M_\alpha(\sigma n) < (h-1)\lambda. \tag{10}$$

Proof. Every nonleading exponent in (8), viewed in direction σ, is at most $D-2$. Therefore

$$A_\sigma \rho^{Dn} \leq M_\alpha(\sigma n) \leq A_\sigma \rho^{Dn} + (M_\alpha(0) - A_\sigma)\rho^{(D-2)n}.$$

By (6), $M_\alpha(0) \leq F_{2D-1} < \rho^{D-1}$ for $D \geq 2$. The strict last inequality follows immediately from Binet's formula. Enlarging the remainder to $\rho^{D-1+(D-2)n}$ and taking logarithms gives (9). Applying (6) and the same Fibonacci bound at height h gives (10). □

Arithmetic implementation. The delivered code encloses the four constants in rational intervals with denominator 2^{128} and performs signed integer interval operations. The constants are certified from

$$\log z = 2\sum_{j=0}^{J-1} \frac{v^{2j+1}}{2j+1} + R_J, \quad v = \frac{z-1}{z+1}, \quad 0 \leq R_J \leq \frac{9}{4(2J+1)3^{2J+1}}$$

after reducing $1 \leq z \leq 2$. An integer square root encloses $\sqrt{5}$. Every multiply and divide rounds outwards; 32 guard bits are retained before the final rounding. No ordinary floating-point logarithm decides inclusion or exclusion.

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The last term of (9) also needs no floating-point exponential. For a certified upper bound u on η, use

$$\log(1+e^{\eta}) \le \begin{cases} u+\log 2, & u \ge 0, \\ 2^{-k}, & u < 0, \quad k = \left\lfloor -u/\overline{\log 2} \right\rfloor, \end{cases}$$

where $\overline{\log 2}$ is a rational upper endpoint. In the second case $u \le -k\log 2$ and $\log(1+z) \le z$ prove the claim. Positive sub-ulp errors are rounded up, never to zero.

For $M_{1,1}(x)$, with helm counts $(1,3)$, the implemented natural-log window widths illustrate the phase contraction:

$x=0$	$x=10$	$x=20$	$x=40$
0.58469	2.3842×10^{-7}	8.8818×10^{-16}	2.4653×10^{-32}

The stored endpoints are exact dyadic rationals; these displayed widths are rounded for reading. These examples do not imply a uniform separation of different Meridians. Near a helm, this inexpensive bound can remain broad; the cone refinements and exact residue tests from Step 1 still matter.

Safe use in the verifier. Enumerate complete Meridian blocks (3). A candidate may be discarded from the cutoff task only when its log lower endpoint is strictly above a certified upper endpoint for $\log B$. Retain every boundary-straddling candidate. The retained set then contains all configurations with $M \le B$, including the $C(K)$ inner configurations. Proving injectivity on this retained set is sufficient by the theorem.

Disjoint log intervals certify different values. Overlap certifies neither equality nor inequality. Exact modular signatures may reject equality without full integer reconstruction; repeated signatures must retain all configurations for further checks. Assigning a configuration to a single window from a rounded log midpoint is unsafe at boundaries. Use certified membership with overlap handling, or retain a complete exact-residue partition. A detected collision outside B requires its numerical rank to be determined before calling it a counterexample within the requested prefix.

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4 Executed checks, timing, and scope

The exact configuration counts in (7) agree between the totient sieve and the independently implemented Möbius identity. The full target-certificate driver, including discovery of K, both count checks, the Fibonacci boundary, and its exact power comparison, took **1.978731 seconds wall time**. Driver CPU time was 1.794689 seconds; its child processes used 0.174154 seconds. The Fibonacci step alone took 0.744819 seconds wall time. Compilation and document production are excluded. This is *not* the runtime of a trillion-value collision check.

The coordinate audit took **8.048358 seconds wall / 8.039116 seconds CPU**. It checked 53,196 coordinate/value instances across eleven height cutoffs, all 39,927 configurations through height 512, sixteen inclusive maximum-cutoff cases, and the first 100,000 complete groups from the Step 1 Meridian implementation. Independent mediant enumeration agreed with every coordinate and triple. All phase/address round trips and all bounds in (6) passed. The first 100,000 values had largest coordinate height 843.

The log-window audit checked **3,240 cases** in **0.559921 seconds wall / 0.559868 seconds CPU**. Every window contained the separately computed integer-series log enclosure and agreed with a 100-digit Decimal check. The universal claim follows from the proof and input (8), not from this finite test. Additional boundary checks covered five precisions from 32 to 512 bits and small brute-force configuration counts. These are checks by separate algorithms within this work, not a new external or human review.

The measurements use Python 3.12.14 and optimized C++17 on the same shared Linux workspace as Step 1. Source hashes, exact endpoints, machine information, and timer boundaries are included in the accompanying package. No runtime extrapolation to 10^{12} is justified by these small computations.

What remains. The mathematical ownership, phase limits, and sufficient finite cutoff are now explicit. A production implementation still needs scalable modular evaluation, complete comparison buckets or windows, persistent checkpoints, and measured storage/runtime performance. The log-window code is an exact reference for such an implementation; it does not guarantee that all candidates can be separated at 128-bit precision. The original Step 1 maximum-unicity results remain 100,000 values by Meridian cross-check and 1,000,000 by the independent Vieta reference. This addendum establishes no larger collision-free prefix.

Sources and attribution. [1] *Markov Meridian Verification Foundations*, version 1, 27 October 2026; especially Results 1, 3, and 6. [2] *Geometric Descent on the Markov Surface*, version 35, equations (25)–(27), (56)–(58), and Section 6.2. The ordinary Farey-tree indexing is standard; see Lagisquet, Pelantová, Tavenas, and Vuillon, *On the Markov numbers: fixed numerator, denominator, and sum conjectures* (2020). The arguments above prove the specific bounds used here without importing a general ordering theorem or claiming these elementary size bounds are new. The contribution of this step is their explicit phase-dependent Meridian interface and the resulting executable verification certificate.

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Original: retained_v20/meridian_degree_100/astra/Astra_Residue_Proof_Report_v1.pdf

Residues, Trend Explanations, and Proofs

An all-phase residue certificate for the degree-100 family
Prepared for Elleard Felix Webster Heffern
31 October 2026

The residue investigation gives a new certificate for the existing all-phase result. For all 519 canonical Meridians with $D = r + s \leq 100$, their residue sets modulo

$$\boxed{3F_{72} = 1{,}495{,}362{,}035{,}637{,}792}$$

are pairwise disjoint. Every Meridian has phase period 36 modulo this number. Consequently no two distinct Meridians in this family have equal values at any canonical integer phases. The check uses 18,684 phase entries and does not depend on tail bounds or a finite numerical ceiling.

Finding	Status and meaning
Exact common-period moduli	For every degree, period T works modulo m for all Meridians iff $m \mid q(T)$, where $q(T) = 3F_{2T}$ for odd T and $9F_{2T}$ for even T. Proof on page 2.
Ten classes modulo 27	All canonical Meridians belong to ten disjoint phase-residue classes. Different classes cannot collide. In this family they exclude 119,206 of 134,421 pairs (88.68%).
Minimal period for this test	Checking $q(T)$ for every $1 \leq T < 36$ leaves an overlap; $T = 36$ separates every pair. Minimality is for one modulus with one common period, including the seed.
Strip-cutoff ordering	Sol's fixed-D ordering is a corollary of fixed-sum monotonicity. It holds at every degree and for every fixed positive cutoff threshold.
Closest finite pair	Its two residue classes are $\{2, 5\}$ and $\{14, 20\}$. This proves the entire two sequences are disjoint, despite their small observed relative gap.

What is new within this pass. The maximal modulus formula sharpens the manuscript's inherited-period results; the ten-class classification completes its modulo-27 transport picture. The complete separation and period minimality are finite computational results supported by a standalone checker. These statements make no claim of priority in the literature.

Scope. D is the extreme Laurent exponent in ρ^x; the polynomial degree in ρ^{2x} is $D/2$. The separation theorem concerns distinct Meridians with both degrees at most 100. It does not exclude competitors of larger degree, prove global Markov unicity, enlarge the numerical-order prefix, or establish injectivity among phases of a single Meridian.

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Original: retained_v20/meridian_degree_100/astra/Astra_Residue_Proof_Report_v1.pdf

1. Exact classification of common-period moduli

Let $F_0 = 0, F_1 = 1$, with signed Fibonacci extension to negative indices. Put $h(x) = M_1(x) = 2 + 3F_{2x}^2$.

Theorem. For integers $T, m \geq 1$, the following are equivalent: (i) every canonical Meridian has phase period T modulo m; (ii) the seed has this period; (iii) $m \mid q(T)$, where

$$q(T) = 3F_{2T}F_{\gcd(2T,4)} = \begin{cases} 3F_{2T}, & T \text{ odd}, \\ 9F_{2T}, & T \text{ even}. \end{cases}$$

A period here need not be the least period of an individual Meridian.

Proof: the seed's exact modulus. The Fibonacci addition identity gives

$$h(x+T) - h(x) = 3\big(F_{2x+2T}^2 - F_{2x}^2\big) = 3F_{2T}F_{4x+2T}.$$

Strong divisibility, $\gcd(F_a, F_b) = F_{\gcd(a,b)}$, follows by the Euclidean algorithm from the Fibonacci addition formula and coprimality of consecutive Fibonacci numbers. Thus every F_{4x+2T} is divisible by $F_{\gcd(2T,4)}$. Conversely the two differences at $x = 0, 1$ already have gcd

$$3F_{2T}\gcd(F_{2T}, F_{2T+4}) = q(T).$$

Hence the gcd of *all* seed differences is exactly $q(T)$, proving (ii) equivalent to (iii), including necessity on the canonical seed domain $x \geq 0$.

Proof: inheritance to every address. This uses the polynomial record already present in the manuscript. For a state (a, b, H), retain $(A, B, C, H) = (9a^2, 9b^2, 3ab, H)$. One forward or backward orbit step, and the pole switch, are respectively

$$\begin{aligned} \mathcal{R}(A, B, C, H) &= (B, A + 9H^2B - 18HC, HB - C, H), \\ \mathcal{R}^{-1}(A, B, C, H) &= (B + 9H^2A - 18HC, A, HA - C, H), \\ \mathcal{J}(A, B, C, H) &= (A, B, C, C - H). \end{aligned}$$

The seed record is

$$\big(8h(x) - h(x-1) - 2,\ h(x) + h(x-1) + 2,\ h(x) + 1,\ h(x)\big).$$

A seed period preserves this whole record. Every address is a composition of these integer polynomial maps, so it preserves congruences modulo m. This proves (ii)⇒(i); (i)⇒(ii) includes the seed. □

A compact certificate of any proposed period. The seed recurrence $h(x+1) = 7h(x) - h(x-1) - 4$ is reversible. Starting with $(h(-1), h(0)) = (5, 2)$, it suffices to check the same pair after T steps modulo m. This proves a two-sided period and then transfers through the record. The standalone checker uses this exact recurrence return as an additional period check.

A useful ordering principle. If $T \mid T'$, then $q(T) \mid q(T')$, either directly from the formula or because a period T is also a period T'. Separation therefore persists along divisibility chains. There is no comparable monotonicity merely from $T < T'$: the tested moduli need not divide one another.

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2. Ten disjoint classes modulo 27

Write (t, h) for the preceding and current poles. The manuscript's local transport becomes

$$S(t,h) = (h,t) \quad (k \text{ even}), \qquad R(t,h) = (h, 7h-t) \quad (k \text{ odd}) \pmod{27}.$$

Indeed, for $u_{j+1} = 3hu_j - u_{j-1}$ and $N_j = 3u_ju_{j+1} - h$,

$$N_0 = t, \quad N_1 = (9u_1^2 - 2)h - t, \quad N_{j+2} - N_j = 9h(u_{j+2}^2 - u_{j+1}^2).$$

Every Markov entry is a unit modulo 3, so its square is 1 modulo 3. Therefore $N_{j+2} \equiv N_j$ and $N_1 \equiv 7h - t$ modulo 27. The argument covers negative indices as well. Only the parity of each appended index matters; its sign and absolute size otherwise disappear.

Track even and odd phases together, starting with $(1, 2, 1, 5)$. The simultaneous maps S, R satisfy $S^2 = I$ and $SRS = R^{-1}$. The orbit under R returns after 18 steps. Its ordered current-pole pairs, in order, are

$$\begin{array}{cccccc} (2,5) & (13,7) & (8,17) & (16,4) & (23,11) & (10,19) \\ (20,14) & (22,25) & (26,26) & (25,22) & (14,20) & (19,10) \\ (11,23) & (4,16) & (17,8) & (7,13) & (5,2) & (1,1). \end{array}$$

The preceding pair is the previous pair cyclically. This list can be checked by $b_{j+1} = 7b_j - b_{j-1}$ in each coordinate. Applying S supplies the other 18 states. The relations above show these 36 states are closed under both maps, so induction covers every finite address. Their unordered current-pole sets are exactly the following ten classes, which are pairwise disjoint.

Residue set	Count at $D \le 100$	Residue set	Count at $D \le 100$
$\{1\}$	42	$\{10, 19\}$	47
$\{2, 5\}$	88	$\{11, 23\}$	47
$\{4, 16\}$	61	$\{14, 20\}$	38
$\{7, 13\}$	78	$\{22, 25\}$	35
$\{8, 17\}$	65	$\{26\}$	18

Corollary. Meridians in different classes cannot share an integer value at any phases, at any degrees. For the 519-family, only $\sum_c \binom{n_c}{2} = 15{,}215$ pairs share a class; the other 119,206 are excluded. The classification is universal, while these counts are specific to $D \le 100$.

This also explains a distinction in Sol's leading-gap data: 595 of the 752 pairs sharing the weakest leading-log gap already lie in different modulo-27 classes. The remaining 157 are resolved by the larger-modulus certificate. Weakness of a leading-term bound need not reflect arithmetic difficulty of the full values.

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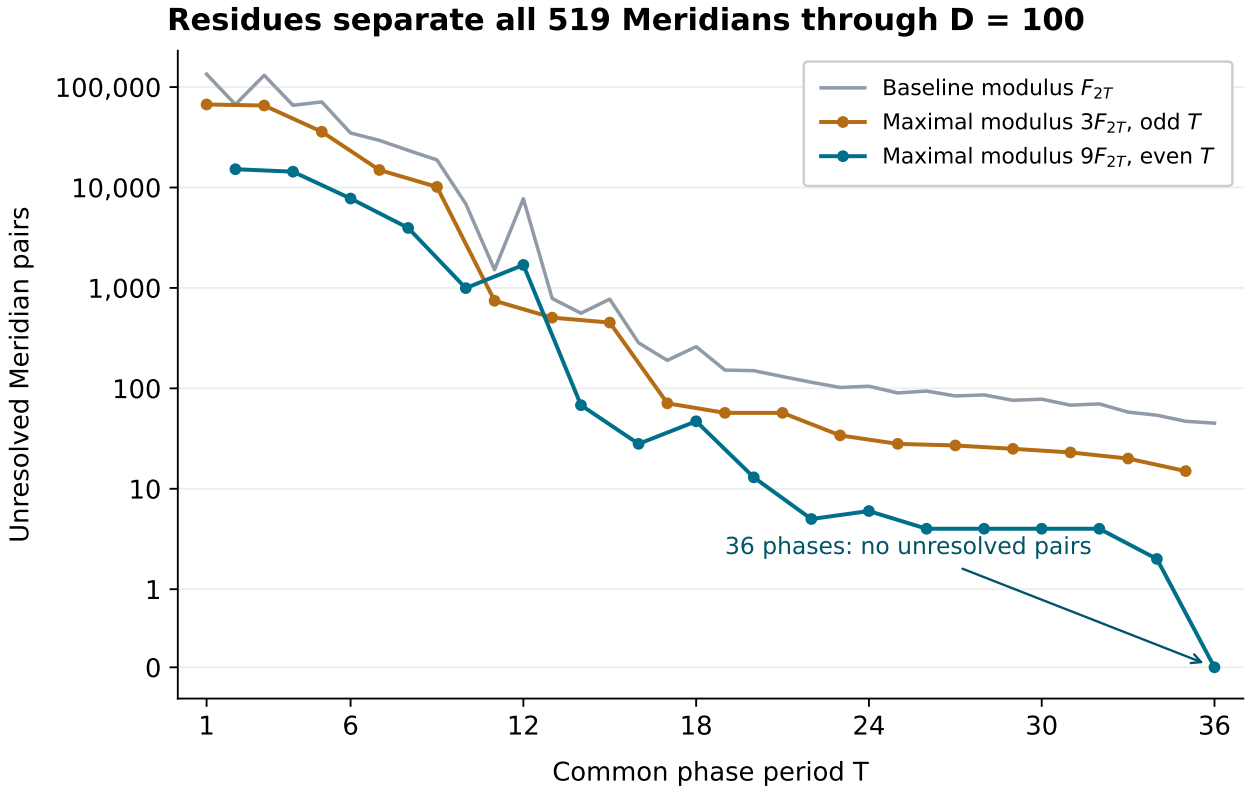
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3. The finite computation proves an all-phase result

The independent checker rebuilds the canonical address inventory from endpoint count vectors and compares it with all coprime positive odd pairs $r \le s$, $r + s \le 100$. It finds exactly 519 labels. Child degrees strictly increase; the bound $D' \ge 2|k|D - ||v| - |u||$, with $|u|, |v|$ the endpoint word lengths, makes its index enumeration exhaustive.

For each address it evaluates phases $0, \ldots, 35$ modulo $3F_{72}$ in three ways: the signed orbit definition; the polynomial record; and the palindromic Fibonacci-word half-row sum of squares. All three agree entry by entry.

The resulting 18,684 entries contain 18,667 distinct residues. Every repeated residue belongs to the same Meridian: there are **zero cross-Meridian overlaps**. Since every integer phase is congruent to one of the checked phases, any equality between different Meridians would contradict this finite table. The seed uses its nonnegative canonical phases; negative phases of other Meridians are covered by the same periodicity proof.



Exact unresolved-pair counts for each individual modulus, not a cumulative filter. The vertical scale is logarithmic above 1 and linear around zero. Lines join discrete tests. Baseline F_{2T} leaves 45 pairs at $T = 36$; the maximal modulus $9F_{72}$ leaves none. The intermediate $3F_{72}$ already leaves none and is the modulus used in the delivered certificate.

Minimality for this certificate format. An independent scan checks $q(T)$ for every $1 \le T \le 36$, saving an overlap witness for each $T < 36$. At $T = 34, 35, 36$ the unresolved counts are respectively $2, 15, 0$. Any modulus giving a common period T for this family must divide $q(T)$, because the family includes the seed. An overlap modulo $q(T)$ remains an overlap modulo every divisor. Therefore no smaller T works in this format. This is not a claim of optimal total work among multi-modulus, pair-specific, or other certificate methods.

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4. Proofs explaining Sol's window trends

Let $\rho = (3+\sqrt{5})/2$, fix an even D, and let r range over admissible odd integers with $s = D - r \geq r$ and $\gcd(r, D) = 1$. Write

$$H(r) = M_\alpha(0) = m_{r/(D-r)}, \qquad A_+(r) = \frac{1}{3}(9/5)^{D/2}\rho^{r-D/2}, \qquad R_+(r) = \frac{H(r)}{A_+(r)} - 1.$$

For any fixed $\eta > 0$, define the strip cutoff used in this analysis by

$$Q_\eta(r) = \min\{q \in \mathbb{Z}_{\geq 0} : R_+(r)\rho^{-2(q+1)} < \eta\}.$$

Because $r \leq s$, $A_+ \leq A_-$ and hence $R_+ \geq R_-$. This is the larger of the two tail remainders, so it controls both arm signs.

Corollary of fixed-sum monotonicity. $Q_\eta(r)$ is nonincreasing in r.

Proof. For two admissible $r_1 < r_2$, Aigner's fixed-sum theorem, as established in the manuscript, gives $H(r_1) > H(r_2)$. Meanwhile $A_+(r_1) < A_+(r_2)$. Positivity gives

$$\frac{H(r_1)}{A_+(r_1)} > \frac{H(r_2)}{A_+(r_2)}.$$

Thus R_+ strictly decreases. Every integer q satisfying the cutoff inequality at r_1 also satisfies it at r_2, proving the claim. The degree-2 seed is the sole admissible label and is trivial. □

This proves the trend at every degree, not just the 50 degrees sampled by Sol. The maximum occurs at $r = 1$ and a minimum at the most balanced admissible pair, with possible plateaus from integer rounding. The largest admissible r is $D/2 - 1$ for $4 \mid D$, and $D/2 - 2$ for $D \equiv 2 \pmod 4$, $D > 2$: these numbers are odd and their gcd with D is 1. This recovers Sol's degree-100 extremes 1/99 and 49/51, where Q is respectively 61 and 33. The proof works for $\eta = 13/1000$ without asserting that this number is a uniform collision gap at higher degrees.

The arm-window difference is a floor effect. For a common ceiling $C \geq H$, the exact leading-term limits are

$$N_\pm = \left\lfloor \frac{\log C - \log A_\pm}{D \log \rho} \right\rfloor.$$

Before taking floors the positive-arm limit exceeds the negative-arm limit by

$$\frac{\log A_- - \log A_+}{D \log \rho} = \frac{s - r}{D} \in [0, 1).$$

Hence $N_+ - N_- \in \{0, 1\}$. Of the 518 nonseed rows, 250 have difference zero and 268 have difference one, exactly as this predicts. The seed's negative arm is omitted by its canonical convention. Likewise the approximately inverse-D window sizes follow from the common-ceiling formula itself; they do not furnish an independent growth law for the Markov numbers.

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5. The closest finite pair and reproducibility

Sol's closest relative pair within the certified windows is

$$A = M_{[1,2,-3]}(0) = 274{,}928{,}228{,}722{,}804{,}802,$$
$$B = M_{[1,1,1,-1,1,-1,1]}(1) = 274{,}931{,}806{,}748{,}021{,}285.$$

Their Fibonacci half rows are $(302045899, 428598301)$ and $(452244079, 265343362)$, respectively. The sum-of-squares representation gives

$$\begin{aligned} B - A &= (452244079^2 - 302045899^2) - (428598301^2 - 265343362^2) \\ &= 113292981887840040 - 113289403862623557 \\ &= 3578025216483 > 0. \end{aligned}$$

Thus the small relative gap, approximately 1.30144×10^{-5}, comes from near cancellation between two much larger square differences. This explains the observed arithmetic and its sign exactly; it does not establish a general family of small gaps or a global closest pair.

The modular explanation is stronger for non-equality. Starting with the seed's even/odd pair $(2, 5)$, append even then odd for $[1, 2, -3]$ to obtain $(5, 2)$. For the second address, all six appended indices are odd, giving $(20, 14)$. Their unordered phase-residue sets are disjoint. Consequently the *entire* two sequences are disjoint; no special treatment of phases 0 and 1 is needed.

What this pass settles and leaves open. The fixed-degree cutoff ordering, arm asymmetry, exact common moduli, and ten-class classification now have proofs. The degree-100 residue separation and its minimal common period have exact computational certificates. Pair-specific tail ceilings were not recomputed: the residue certificate removes the need for them for this family. Sol's 10,516 improved exponent lattices still should not be described as a measured runtime improvement. No independent address-prefix effect, general near-helm concentration, or all-degree separation theorem has been proved here.

Reproduce the main checks. From the delivered package root:

```
python3 astra/code/residue_certificate.py --out rerun
python3 astra/code/period_optimality.py --out rerun
```

Both use only the Python standard library. The first regenerates the inventory, three-evaluator residue table, 36-state closure, class counts, and negative controls. The second regenerates the full $T = 1, \ldots, 36$ scan with modular overlap witnesses. The figure script requires matplotlib. The table digest is

```
00a12cea468e54be2439ea985f2d530af
980bf354af9e4f2cff810dfb9892327
```

Source dependence. Mathematical definitions and the fixed-sum theorem come from manuscript v19, Sections 3–4 and its Fibonacci appendix. Sol's statistics and the Step 3 inventory are retained for comparison, but the main residue checker imports neither their code nor their data. Supporting source excerpts, exact outputs, scripts, and input/output hashes are included. The manuscript itself has not been edited.

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Meridians through degree 10,000

Independent certification, phase compatibility,
and Fibonacci residue obstructions

Research continuation of the version-20 manuscript

Theorem 1 (Certified computation)**.** *Let $M_{r,s}$ denote the canonical Meridian with coprime odd helm counts $1 \le r \le s$ and Laurent degree $D = r + s$. Among all such Meridians with $D \le 10,000$, no two distinct Meridians take the same value at any canonical integer phases. The inventory contains exactly $5,067,077$ Meridians, comprising $12,837,632,128,426$ distinct unordered pairs.*

The degree-2,000 subfamily was run and independently certified first. It contains $202,862$ Meridians and $20,576,394,091$ distinct pairs. The two independently generated runs agree exactly on their common pair sets after every shared filtering stage.

Here "certified" means a deterministic exhaustive computation checked by a separate mathematical evaluator, complete independent residue grouping, and deliberately failing controls. It is not a proof-assistant formalization. The degree is the extreme Laurent exponent in ρ^x, as in version 20; it is not an address-index sum. The conclusion concerns *different* Meridians. It neither asserts injectivity within one Meridian nor extends the separate numerical-order verification of fifty billion values.

The main new result is a stronger sieve. Retaining the compatibility of the two phases across moduli proves Theorem 1 using only periods $36, 38, 40$. A less informative sieve, which retains only pairs of Meridian names, required $36, 38, 40, 42, 44$ in this run. Both calculations were completed; the stronger result was checked in two different ways.

Cumulative tests	Degree 2,000	Degree 10,000
36: unresolved pairs	7,121	320,755
36, 38: unresolved pairs	612	17,427
36, 38: compatible phases	2	2,052
36, 38, 40: unresolved pairs	0	87
36, 38, 40: compatible phases	**0**	**0**
36, 38, 40, 42: unresolved pairs	—	1
36, 38, 40, 42, 44: unresolved pairs	—	0

The dashes mean that the degree-2,000 calculation had already terminated. A surviving pair in this table is only a modular coincidence, not an equality of Markov values.

1 Specification and certification chain

The seed is $a = F_{2x+1}$, $b = F_{2x-1}$, $H = 3ab - 1 = 2 + 3F_{2x}^2$. An appended nonzero signed address index iterates

$$(a, b) \longmapsto (b, 3Hb - a) \quad \text{or} \quad (3Ha - b, a)$$

with H fixed, and then replaces the pole by $3ab - H$. The first appended index is positive. Subsequent indices may have either sign. The seed uses $x \ge 0$; all other canonical Meridians use integer x.

Version 20 proves that the largest modulus admitting common phase period T is

$$q(T) = \begin{cases} 3F_{2T}, & T \text{ odd}, \\ 9F_{2T}, & T \text{ even}. \end{cases}$$

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For clarity, the transfer does not assume that the seed companions individually have period T. The seed difference is

$$H(x+T) - H(x) = 3F_{2T}F_{4x+2T},$$

and the integer polynomial record $(9a^2, 9b^2, 3ab, H)$ transfers its period through every signed address. Reversibility of $H_{x+1} = 7H_x - H_{x-1} - 4$ extends the residue period to negative phases. Each run checks the seed return modulo every modulus it uses.

T	$q(T)$
36	4,486,086,106,913,376
38	30,748,091,606,160,363
40	210,750,555,136,209,165
42	1,444,505,794,347,303,792
44	9,900,790,005,294,917,379

Inventory. Production enumerates the canonical address tree with increasing degree and reuses one orbit step between successive siblings. An independent checker ignores those address links and verifies equality with the entire set of coprime odd pairs $r \le s$, $r + s \le D$. Thus completeness is checked by membership, not only by the count. The exact count also satisfies

$$N(2n) = 1 + \frac{1}{2}\sum_{k=2}^{n} \varphi(2k),$$

because degree $2k > 2$ contributes $\varphi(2k)/2$ complementary coprime odd pairs.

Independent values. The checker uses the Fibonacci matrices

$$C(z) = \begin{pmatrix} F_{2z+3} & F_{2z+1} \\ F_{2z+1} & F_{2z-1} \end{pmatrix}, \qquad A = C(x), \quad B = C(x-1),$$

and the version-20 word specification. Set $d = r + s$ and

$$\epsilon_j = \left\lceil \frac{jr}{d} \right\rceil - \left\lceil \frac{(j-1)r}{d} \right\rceil.$$

Let $U_{r,s}$ have jth factor A for $\epsilon_j = 1$, and B otherwise. Its first factor is A. Then

$$M_{r,s}(x) = [BA^{-1}U_{r,s}(A, B)]_{11}. \tag{1}$$

The prefix changes that first factor to B. The identity holds at every integer phase, by the Laurent-polynomial argument already proved in version 20.

The checker constructs these words by a separate staircase traversal and Euclidean batching. It imports neither the production evaluator nor the address recurrence. For adjacent slopes $r_1/d_1 < r_2/d_2$ with $r_2d_1 - r_1d_2 = 1$, the upper staircase word at the mediant is the right word followed by the left word. One can check this directly: on the first d_2 positions the difference between the two linear heights is less than $1/d_2$, and on the remaining d_1 positions it is less than $1/d_1$. Coprimality keeps their ceilings unchanged, including the endpoint cases. This supplies an exact fast construction of the specified word, rather than a second version of the address algorithm.

Complete grouping. At period 36, production writes every tagged pair (Meridian, phase) to one of 64 unsorted residue buckets. The independent evaluator verifies a bijection between those records and the complete phase table: no missing or repeated tags, correct bucket, correct value. A separate NumPy implementation sorts each original bucket and independently reconstructs every cross-Meridian group and every resulting unordered pair. It compares complete sets with production, not just counts or hashes. Later tests evaluate every phase of each active Meridian and independently intersect its finite image with the image of each remaining partner.

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Audit quantity	Degree 2,000	Degree 10,000
Baseline phase records	7,303,032	182,414,772
All independently compared values	7,525,814	188,369,412
Cross-Meridian residue groups at 36	2,675	70,009

These are values actually checked, not a sample. The later production stages traverse the full address tree to obtain active entries; the table counts list only entries retained and independently compared. The degree-100 control additionally checks all 18,684 entries against a literal word evaluator and the polynomial record. All 88 period-44 entries are also checked with literal words and Python unbounded integers, covering the modulus above 2^{63}. Five negative controls reject an altered residue, altered modulus, duplicate phase tag, omitted phase record, and invented surviving pair.

All residue arithmetic is exact. Products of 64-bit residues are reduced using 128-bit intermediates before multiplying by small coefficients. The checker uses signed 128-bit Fibonacci and seed calculations only within the explicitly enforced period range $1 \le T \le 44$; its largest relevant Fibonacci indices and moduli fit safely. Floating point is used only for elapsed times and plots.

2 A proved phase-compatibility refinement

For distinct Meridians a, b, define their complete witness relation

$$W_T(a,b) = \{(u,v) \in (\mathbb{Z}/T\mathbb{Z})^2 : M_a(u) \equiv M_b(v) \pmod{q(T)}\}.$$

The pair-only sieve asks whether every W_T is nonempty. An actual equality requires *one common pair of integer phases* to give witnesses in all of them.

Proposition 2 (Compatible-phase sieve)**.** *For periods $T_1, \dots, T_k$, form the joint witness relation by joining witnesses whose first coordinates agree modulo each pairwise gcd, and whose second coordinates do likewise. If this joint relation is empty, then $M_a(x) \neq M_b(y)$ for all integer phases x, y.*

Proof. An equality supplies the residues of its actual x, y in every witness relation, so they are compatible. Conversely, the generalized Chinese remainder theorem says that a compatible choice of witness phases lifts to a pair modulo $\operatorname{lcm}(T_1, \dots, T_k)$. This converse concerns simultaneous modular witnesses only; it does not assert an equality of integer values. □

The implementation joins exact phase pairs modulo successive least common multiples. A second checker avoids this algorithm entirely for $36, 38, 40$: periods 36 and 40 need only be compared modulo 4, while period 38 imposes parity. For each pair, it intersects three 16-bit masks representing $(x \bmod 4, y \bmod 4)$. A period-38 witness is lifted to all mod-4 pairs with its parity. An intersection is nonempty exactly when the three witness relations have a compatible choice. The two checkers agree on the complete set of 2,052 survivors after period 38 and both find none after period 40.

This gives a useful strengthening of the degree-2,000 calculation as well: the first two periods leave only two phase-compatible pairs, with helm counts

$$(35, 1223) \text{ versus } (37, 1297), \qquad (37, 1371) \text{ versus } (39, 1441).$$

Their compatible phases are respectively $(1, 1)$ and $(0, 0)$ modulo 684. Period 40 eliminates both.

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Cumulative sieve	First unresolved degree	Certified below it
36, pairs only	104	102
36, 38, pairs only	110	108
36, 38, compatible phases	1,334	1,332
36, 38, 40, pairs only	2,162	2,160
36, 38, 40, 42, pairs only	9,108	9,106
36, 38, 40, compatible phases	none through 10,000	10,000

These are exact thresholds for these specified sieves, obtained from the fully checked degree-10,000 inventory. They are not claims of optimal periods or minimal computation among all possible modular tests.

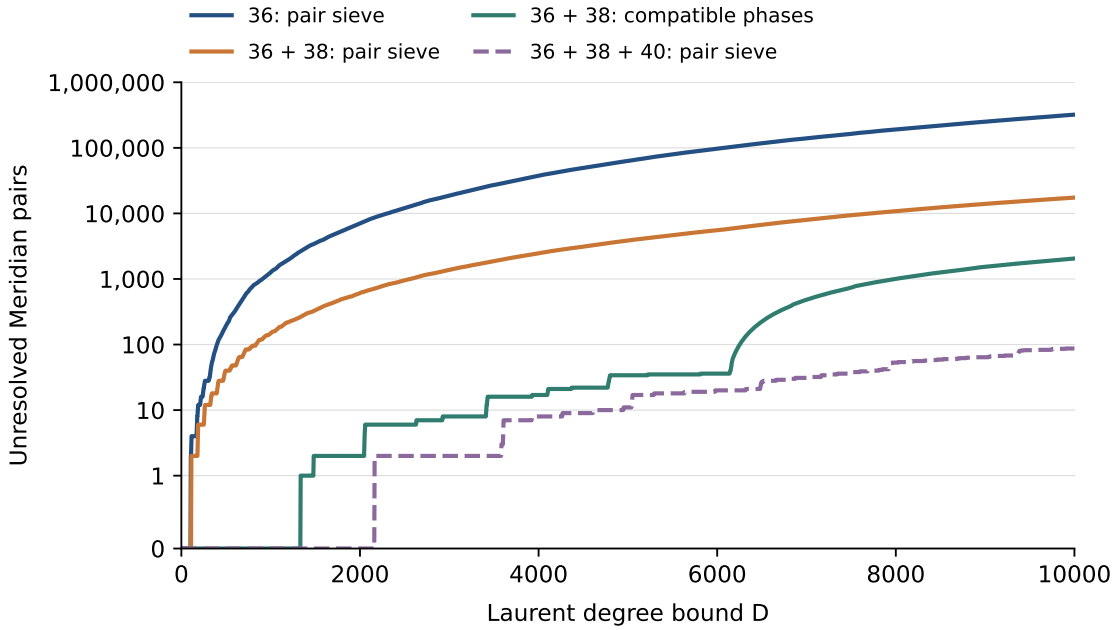


Figure 1: Exact survivor counts over all even degree bounds up to 10,000. The ordinate is linear from zero to one and logarithmic above one. The compatible-phase curve after period 40 is identically zero and is omitted. The large separation between the two period-38 curves measures information lost by discarding phase compatibility.

3 Where the modular coincidences occur

The baseline coincidences are concentrated near phases 0 and 1. At degree 10,000, those phases contribute $177,898$ of $187,655$ tagged incidences in cross-Meridian residue groups, or 94.80%. Furthermore, $312,994$ of the $320,755$ unresolved pairs (97.58%) have a witness with *both* phases in $\{0,1\}$. Only 7,761 pairs lack such a witness. These counts use complete residue groups and count a tagged incidence once per group, not once for every partner in the group.

The concentration weakens with degree: the corresponding incidence fraction at degree 2,000 is $7,234/7,288 = 99.26\%$. At degree 10,000, phases 18 and 30 are the largest remaining peaks, with 2,618 and 2,552 incidences. These are observations for $q(36)$, not a universal distribution law. The

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phase concentration and large groups (up to 60 different Meridians sharing one baseline residue) make a model of independent uniform random residues inappropriate here.

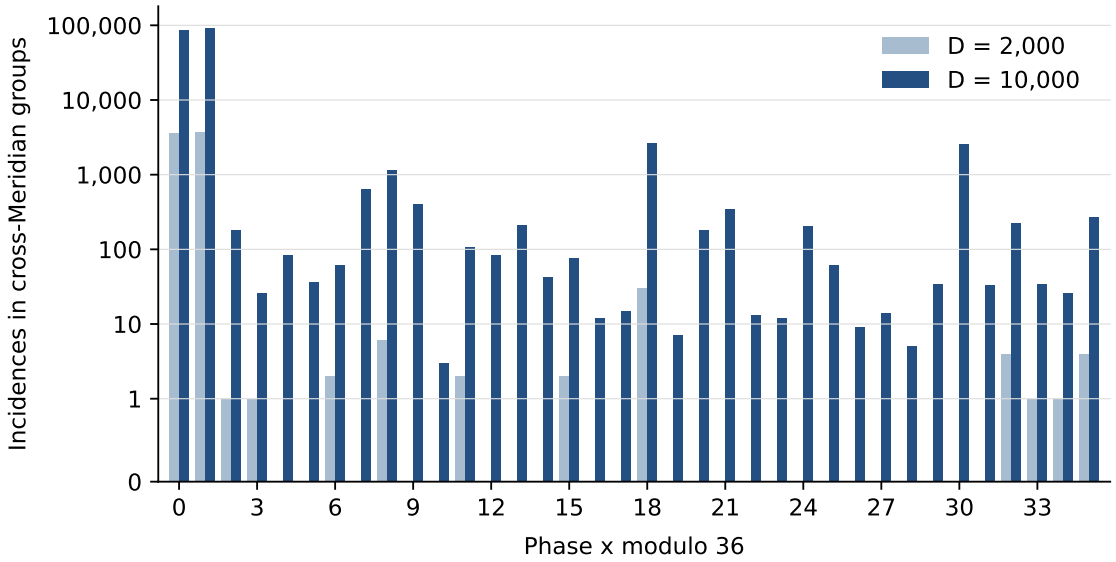


Figure 2: Complete phase-incidence counts in cross-Meridian groups modulo $q(36)$. The ordinate is linear from zero to one and logarithmic above one. The central peaks are structural; the secondary peaks have not been classified. No limiting proportion is claimed.

The Fibonacci boundary $r = 1$ is especially prominent among late survivors. It accounts for 74,445 of the baseline pairs, 2,321 of the 17,427 pairs after period 38, and 47 of the 87 pairs after period 40. Among the 2,052 *phase-compatible* survivors after period 38, 1,922 have $r = 1$ on both sides. Thus the phase refinement removes most other apparent coincidences and makes a concrete arithmetic source visible.

4 An exact Fibonacci explanation

Lemma 3 (Boundary values)**.** *For every positive odd s,*

$$M_{1,s}(0) = F_{2s+1}, \qquad M_{1,s}(1) = F_{2s+3}.$$

Proof. For $r = 1$, the corrected word in (1) consists of $s + 1$ copies of B. At phases 0 and 1 these matrices are respectively

$$\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Their dth powers have upper-left entries F_{2d-1} and F_{2d+1}. Put $d = s + 1$. □

Theorem 4 (Exact Fibonacci matrix period)**.** *For every even $T \geq 2$, the matrix $Q = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has exact order $36T$ modulo $q(T) = 9F_{2T}$.*

Proof. Put $n = 2T$ and $f = F_n$. Then $4 \mid n$ and $3 \mid f$. First, Q has order $2n$ modulo f. Indeed, $Q^j = I$ implies $f \mid F_j$, which by strong divisibility and strict growth forces $n \mid j$. But $Q^n \equiv F_{n-1}I \not\equiv I \pmod{f}$, whereas Cassini's identity gives $Q^{2n} \equiv I \pmod{f}$.

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Write $Q^{2n} = I + fE$. Its off-diagonal entry satisfies $E_{12} = F_{2n}/F_n = F_{n-1} + F_{n+1} \not\equiv 0 \pmod 3$, since consecutive Fibonacci numbers are coprime. The binomial expansion gives

$$(I + fE)^3 = I + 3fE_1, \qquad E_1 \equiv E \pmod 3,$$

because $3 \mid f$. Cubing again gives $(I + fE)^9 \equiv I \pmod{9f}$. Conversely, if $(I + fE)^k \equiv I \pmod{9f}$, reducing first modulo $3f$ and looking at the off-diagonal entry forces $3 \mid k$. Write $k = 3\ell$ and apply the same argument to $I + 3fE_1$ modulo $9f$; then $3 \mid \ell$. Thus its order is exactly 9. Every order of Q modulo $9f$ is a multiple of its order $2n$ modulo f, so the exact order is $9(2n) = 36T$. □

Combining the two results with $F_{-j} = F_j$ for odd j gives explicit sufficient conditions for a central-phase coincidence modulo $q(T)$, where $u, v \in \{0, 1\}$:

$$s - s' + u - v \equiv 0 \pmod{18T} \qquad \text{(translation)}, \tag{2}$$

$$s + s' + 1 + u + v \equiv 0 \pmod{18T} \qquad \text{(reflection)}. \tag{3}$$

Because s, s' are odd and T is even, translation forces $u = v$, while reflection forces $u \neq v$. These are sufficient mechanisms, not a classification of every Fibonacci residue coincidence.

The last pair in the less informative sieve

The only pair surviving periods $36, 38, 40, 42$ without phase information is

$$M_{1,4571} \quad \text{and} \quad M_{1,9107},$$

of degrees 4,572 and 9,108. Their *complete* phase witness relations in the checked tables are

Period	Witness phases (u, v)	Explanation
36	$(0,0), (1,1)$	translation
38	$(0,1), (1,0)$	reflection
40	$(0,1), (1,0)$	reflection
42	$(0,0), (1,1)$	translation
44	none	separated

Here $9107 - 4571 = 4536 = 7(18{\cdot}36) = 6(18{\cdot}42)$, while $4571 + 9107 + 2 = 13680 = 20(18{\cdot}38) = 19(18{\cdot}40)$. Thus the displayed congruences explain every surviving witness. The witnesses for periods 36 and 38 already demand contradictory phase parities, so they cannot arise from one equality. This explains why the phase refinement rejects the pair long before period 44.

5 A quantitative barrier and the next computations

Proposition 5 (Unavoidable boundary alias)**.** *Let $\mathcal{T}$ be any nonempty finite set of even periods and let $L = \operatorname{lcm}\{T : T \in \mathcal{T}\}$. The two distinct Meridians*

$$M_{1,9L-3}(0), \qquad M_{1,9L+1}(1)$$

are congruent modulo every $q(T)$, $T \in \mathcal{T}$. Therefore those modular tests alone, even with full phase compatibility, cannot certify separation for the entire family through degree $9L + 2$.

Zenodo 23874075 | original page 6/7 | SHA256 08d045813336c4dd1f6e

Original: Meridian_Degree_10000_Reproducibility/report/Meridian_Degree_10000_Certification_and_Trends.pdf

Proof. The two values are F_{18L-5} and F_{18L+5}. Their indices are odd and sum to $36L$, a multiple of every Fibonacci matrix period $36T$ in Theorem 4. They are therefore congruent modulo every modulus. The same actual phases $(0, 1)$ work in every test. Both labels are positive, odd and coprime, and their degrees are $9L - 2$ and $9L + 2$. The integer values themselves are unequal by strict growth of Fibonacci numbers. □

For the newly sufficient periods $36, 38, 40$, one has $L = 6840$. Consequently the fixed scheme is guaranteed to fail by degree **61,562**, on the explicit labels $(1, 61557)$ and $(1, 61561)$. This is an upper bound on how far this scheme can go; it may fail earlier on other pairs. It is not a counterexample to Meridian disjointness.

A necessary condition for any such even-period scheme to certify every Meridian through degree D is

$$\boxed{L > \frac{D-2}{9}}.$$

This gives a concrete design rule: increase the least common multiple of the periods, in addition to increasing the moduli. Adding period 42 raises the displayed obstruction to degree 430,922; adding both 42 and 44 raises it to 4,740,122. Passing this necessary test is not sufficient for separation.

Practical next scale. The new native producer took 39.05 seconds for degree 10,000. The independent value audit took 46.93 seconds and independent grouping 48.97 seconds; those two audits ran concurrently, so their times should not be added as elapsed wall time. Degree 2,000 production took 1.01 seconds. These are measurements of this environment and implementation, not complexity guarantees.

The family count grows quadratically in the degree bound. In this implementation, the baseline phase table and unsorted bucket payload together occupy exactly $720N(D)$ bytes, before inventory and later stages. At degree 10,000 that is about 3.65 GB. Quadratic planning estimates put that payload near 14.6 GB at degree 20,000, 32.8 GB at 30,000, and 91.2 GB at 50,000. A 20,000–30,000 run is a sensible next use of the present design; 50,000 or 100,000 calls for streaming the independent comparison and reorganizing disk storage. The arithmetic is already fast enough to make these targets worth pursuing, but no separation result above 10,000 is claimed here.

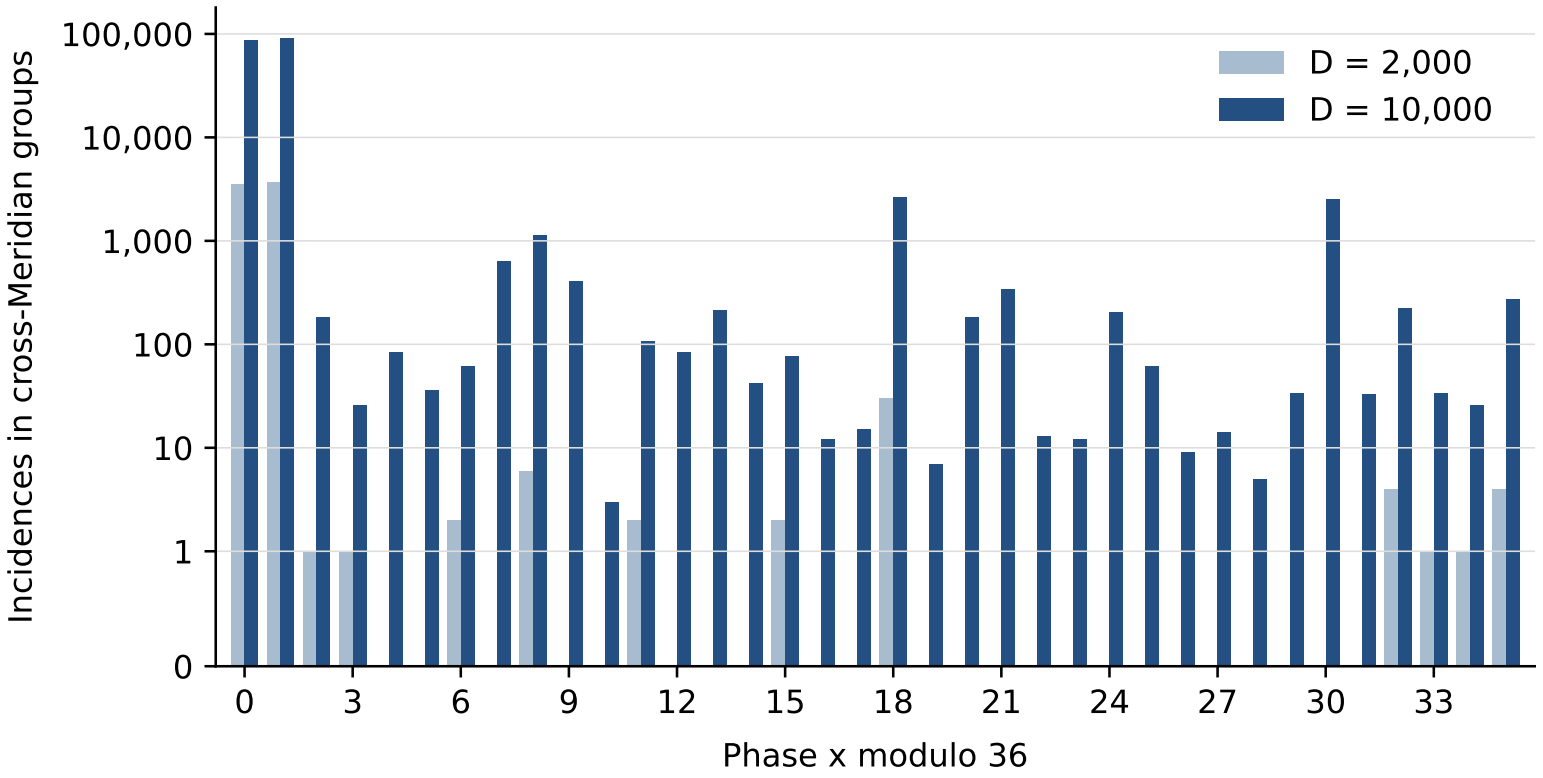
There is no need to promote the observed phase peaks to a new conjecture. The proved results are the compatibility criterion, the exact boundary matrix period, and the explicit obstruction. The secondary phase peaks and the degree at which the three-period compatible sieve first fails above 10,000 remain open computational questions. The previous manuscript's modular-separability conjecture remains a conjecture.

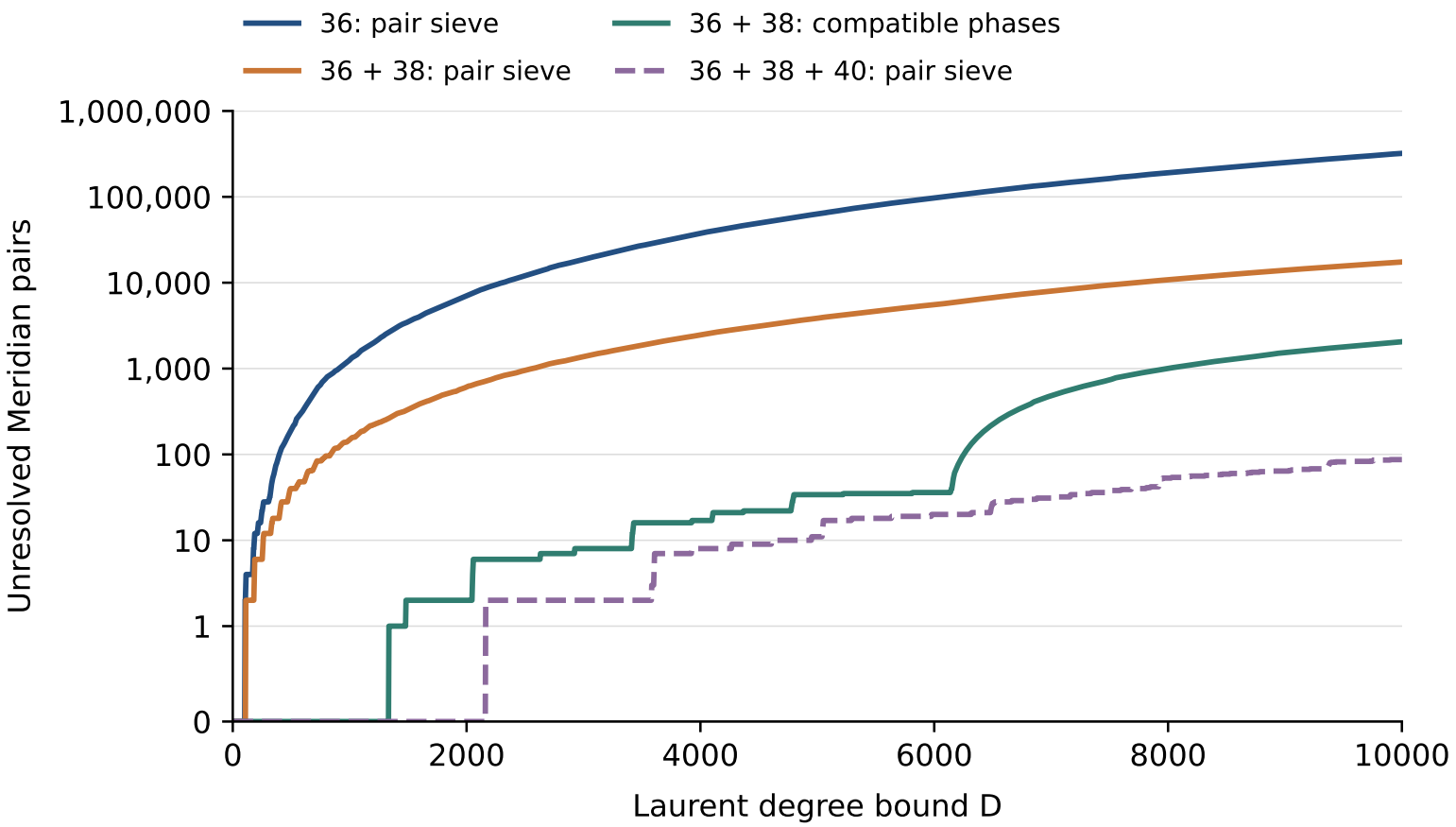
6 Reproducible evidence

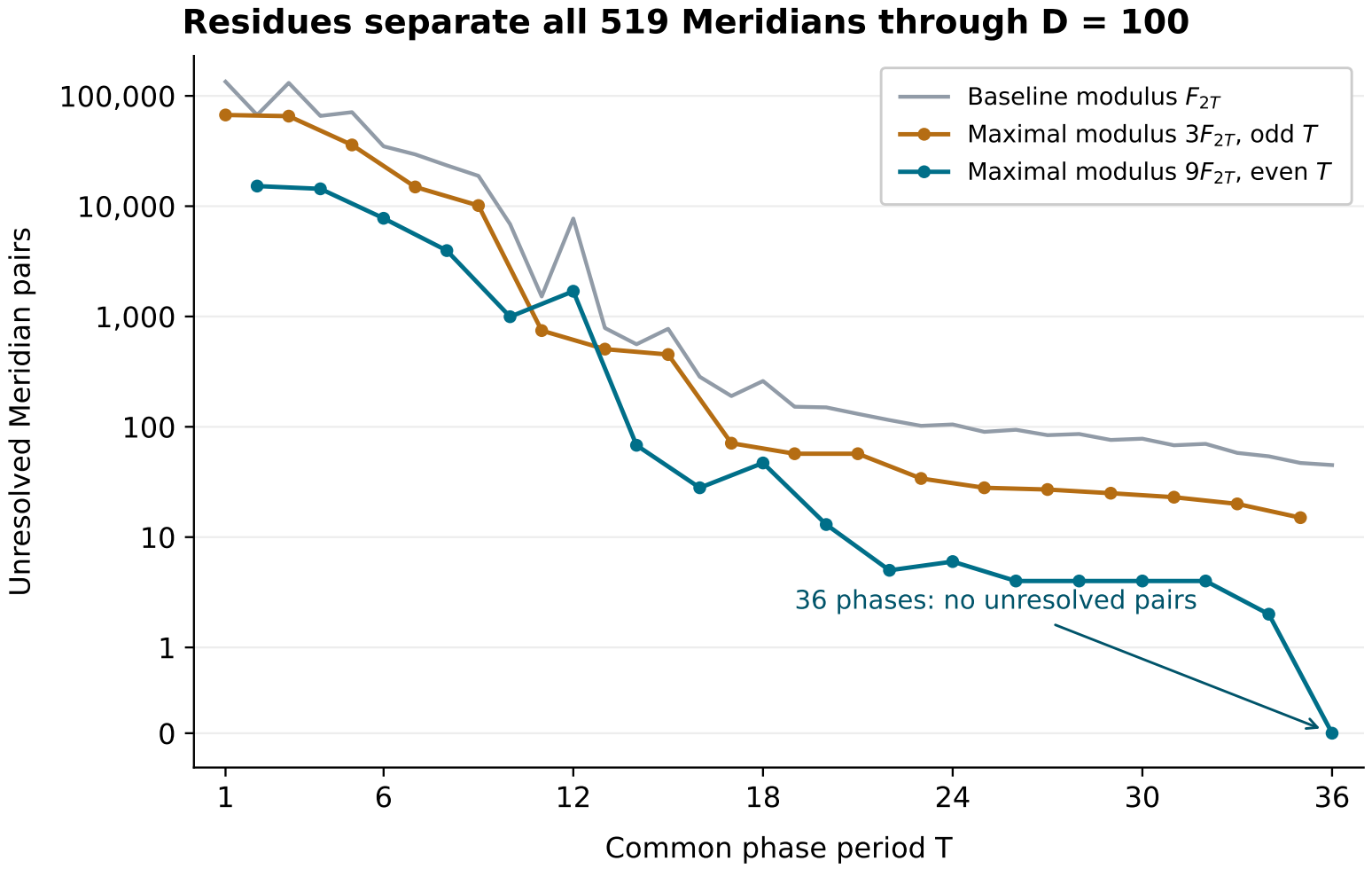
The accompanying archive contains source code, compact evidence for both degree bounds, all later active-phase tables, the exact degree profile, phase masks, execution receipts, a hash manifest, this report source, and an insertable manuscript statement. The multi-gigabyte baseline phase table and raw buckets are regenerated by `python3 code/reproduce.py`; they are not included in the compact archive. The manifest records their exact byte sizes and hashes from the audited runs. Hashes identify artifacts; they were never used as a substitute for equality, complete coverage, or independent evaluation.

Reproduction builds two C++20 programs, certifies the degree-100 control and the two requested bounds in order, then repeats the phase checks, analyses and seal. It creates fresh runs without altering the archived evidence. Requirements are NumPy, a suitable C++20 compiler, a little-endian Linux host and several gigabytes of working space. Runtime fields need not reproduce bit for bit.

Zenodo 23874075 | original page 7/7 | SHA256 08d045813336c4dd1f6e


Incidences in cross-Meridian groups
100,000
10,000
1,000
100
10
1
0
0
3
6
9
12
15
18
21
24
27
30
33
Phase x modulo 36
D = 2,000
D = 10,000


36: pair sieve
36 + 38: pair sieve
36 + 38: compatible phases
36 + 38 + 40: pair sieve
Unresolved Meridian pairs
1,000,000
100,000
10,000
1,000
100
10
1
0
0
2000
4000
6000
8000
10000
Laurent degree bound D


Residues separate all 519 Meridians through D = 100
Baseline modulus F_{2T}
Maximal modulus $3F_{2T}$, odd T
Maximal modulus $9F_{2T}$, even T
36 phases: no unresolved pairs
Unresolved Meridian pairs
100,000
10,000
1,000
100
10
1
0
1
6
12
18
24
30
36
Common phase period T

Meridians on the Markov Surface: Geometry, Residues, and Collision Windows

Completion and release history

Manuscript and computational chronology dated 7 November 2026. The historical 50-billion computation was completed 28 October; this release did not perform a new enumeration. A separate manuscript-completion time is not supplied.

Zenodo publication-date field: 2026-11-07

Zenodo record created: 2026-11-07 13:37:47 CST / 2026-11-07 19:37:47 UTC

Zenodo record last modified: 2026-11-07 13:37:48 CST / 2026-11-07 19:37:48 UTC

The publication-date field is the date supplied in the deposit. The API created/modified timestamps are repository metadata, not a separately attested first-view time. Full fractional timestamps are retained in the metadata archived in Part C.

https://zenodo.org/records/24043052

https://doi.org/10.5281/zenodo.24043052

This is the first public Zenodo version located in the Meridian manuscript family.

Earlier public announcements: 21 October 2026, 10:18:25 CDT / 15:18:25 UTC on X (more than a billion values in under 40 minutes); 23 October 2026, 09:54:57.771 CDT / 14:54:57.771 UTC on LinkedIn (9.3 billion in 24 minutes). These concern earlier computations, not the final trillion certificate.

https://x.com/webystar/status/2112926487091282350

https://www.linkedin.com/posts/elleard-felix-webster-heffern-joseph-by-confirmation-2953a3104_markovnumbers-activity-7519411048115003392-4NaJ

The dates and URLs are retained in the 7 November public-post evidence packet; the LinkedIn URL is also embedded in the published computational supplement. Additional November LinkedIn announcements and the older Meridian ResearchGate v13 URL could not be established in this compilation.

Complete public Zenodo description

This edition organizes Markov configurations by reversed Meridian addresses, geometric descent, Fibonacci-modulus and modulo-27 behavior, equal-degree separation, finite unequal-degree collision windows, corridor recurrences and a numerator-denominator collision window. It retains the snake-graph arguments and the archived degree-10,000 certificate. The unrestricted unicity problem remains open within this framework.

The revised computational chronology includes the earlier Claude-assisted historical search and the separately archived fifty-billion numerical-order verification. The working draft reports 9,581,528,580 triples against a cutoff of 10^100000 in 24 minutes on two cores with 7 GB memory. A public LinkedIn announcement is timestamped 23 October 2026, 14:54:57.771 UTC (09:54:57.771 CDT), with caption wording of 9.3 billion in 24 minutes. This is the announcement timestamp, not a recovered execution start or finish. The earlier draft's qualification about uncertified historical logs and cutoff error bounds is retained. This search is not the later ten-billion anchor.

The fifty-billion verifier completed on 28 October 2026. Its recorded start was 08:32:57.096455198 UTC and its audit-inclusive finish was 10:38:33.644612263 UTC (03:32:57-05:38:34 CDT, rounded). Two closed invocations total 7,492.102807105 seconds (2 h 04 min 52.103 s). The calendar span is 7,536.548157065 seconds, including the invocation pause. Including the separate original ten-billion anchor gives 8,330.980040248007 seconds (2 h 18 min 50.980 s). These historical execution dates are distinct from this edition's release date.

The exact numerical envelope covers 71,517,775,875 configurations. After anchored omissions and certified lower-bound exclusions, 54,940,210,994 new configurations were compared. All 346 repeated primary-residue pairs separated; the final auditor reconstructed all 692 candidate triples and rechecked the 38 anchor candidates. The protocol establishes uniqueness for the first 50,000,000,000 distinct Markov values in numerical order. It used eight workers, an eight-CPU-equivalent quota and an 8-GiB memory limit.

Included are the 31-page main paper, six-page computational proof and timing supplement, editable source, all figure assets, selected historical evidence and receipts, and a new read-only reconciliation of 26,151 archive member checksums and 8,522 closed stage receipts. The full frozen historical verifier archive is available separately at https://doi.org/10.5281/zenodo.23874075. This release is an editorial revision, not a new enumeration; no completed 500-billion result is claimed.

Licensing: Original software is licensed under MIT; original documentation and data under CC BY 4.0. Third-party materials and external dependencies retain their own terms. Copyright (c) 2026 Elleard Felix Webster Heffern, to the extent copyright applies.

Funding: No funding. Substantial generative-AI assistance is disclosed in the manuscript; the author directed the research and is responsible for the work.

Published files

Original filename / checksum	Bytes
v4_ps_Markov_Meridians_TLMS.pdf MD5 e8fd2ce9060d15e97a23246bb2dcf934 SHA256 2384d0ac3d797c993132978c81520c0d2409b9c246bd658c0dbc0346a3c0b72e	7,051,356
v4_ps_Markov_Meridians_TLMS_editable_source.zip MD5 a5a51a46e0cd3922ce3aa1fbb4b0a1cf SHA256 e1acba7686d0506b108a3af3da3cdfe71802cc51869236fd182f25c69a53473d	26,203,004
v4_50_Billion_Verification_Supplement.pdf MD5 6c966b341c1ec55b6b7bdc3a9dc50aaf SHA256 b477d344aff0971fee600ce1d087fc05f15509582bb43b22eb337180d6d3c446	310,901

All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

Original: v4_ps_Markov_Meridians_TLMS.pdf

MERIDIANS ON THE MARKOV SURFACE: GEOMETRY, RESIDUES, AND COLLISION WINDOWS

ELLEARD FELIX WEBSTER HEFFERN

Abstract. We organize positive integral Markov triples by Meridians: real curves obtained by reversing a geometric descent, with a signed integer address for the curve and an integer phase for each arithmetic configuration. This gives a bijection with normalized triples without assuming uniqueness of their largest entries. We prove a common Fibonacci-modulus period law and a classification into ten disjoint residue classes modulo 27. Established Markov orderings imply unicity throughout each fixed Meridian degree; positive matrix words place every unequal-degree collision in an explicit finite phase window. A second organization, by corridor walls, gives exact recurrences across both Meridian and phasor steps and a complementary bound on denominators in terms of numerators and their difference. We give snake-graph proofs of the three established Aigner monotonicities and a folded-piece comparison criterion. Finally, complete residue and phase certificates separate all 5,067,077 Meridians of degree at most 10,000 at every integer phase. A separate archived computation, completed on 28 October 2026, verifies the first 50 billion distinct Markov values in numerical order. The unrestricted unicity problem remains open within this framework.

Post-submission draft. This edition incorporates changes requiring a new submission. The prefix `ps` retains that distinction.

1. Introduction

A normalized Markov configuration is a positive integral triple satisfying

$$a^2 + b^2 + M^2 = 3abM, \qquad 1 \le a \le b \le M. \tag{1}$$

The unicity question asks whether the largest entry M determines a, b. The classical Vieta tree enumerates configurations by replacing one coordinate with the other root of its quadratic. A tree of configurations is available independently of unicity; treating its vertices as distinct numerical maxima would assume the conclusion. Aigner's monograph gives the historical and arithmetic setting [5]. Continued fractions, snake graphs and Christoffel words supply several established orderings [12, 9, 10, 11], which will be used with their precise hypotheses.

This paper develops two ways to organize that same set of configurations. A *Meridian* holds fixed an itinerary of geometric descent and varies a real phase. A *corridor wall* follows a different route through the alternative tree, eventually holding a Farey numerator fixed while advancing the denominator. These organizations answer different comparison questions: Meridian degree controls a phase window, while corridor numerator controls a denominator window.

The geometric construction comes first in Section 2. Section 3 gives the canonical addresses, their inverse, and the exact conversion to Farey labels. Section 4 proves the modulo-27 classification and the common period s modulo F_{2s}. Sections 5 and 6 establish the two collision windows. Section 7 supplies the matching-count proofs used in the ordering arguments. Section 8 presents a finite certificate with an infinite phase range:

Date: v4_ps — Post-submission edition — 7 November 2026.
2020 *Mathematics Subject Classification.* Primary 11D25; Secondary 11B39, 05C70, 11Y50.
Key words and phrases. Markov numbers, unicity, geometric descent, Meridians, corridors, snake graphs.

Zenodo 24043052 | original page 1/31 | SHA256 2384d0ac3d797c993132

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Theorem 1.1 (Degree-10,000 separation)**.** *Among canonical Meridians of Laurent degree at most* 10,000*, distinct canonical integer-phase marks have distinct maxima. There are exactly* 5,067,077 *Meridians in this family.*

The computational part separates different Meridians at all phases. The fixed-degree theorem also separates marks on a single Meridian, so the two conclusions combine as stated. The bound is on *degree*, not on the size of a Markov number, and it does not exclude a competitor of higher degree. Section 9 records the separate fifty-billion numerical-order verification and its dated execution history.

Throughout, $M_\alpha(x)$ denotes the maximum at Meridian address α and phase x. The notation $m_{p/q}$ denotes the maximum attached to the standard reduced Farey label p/q; it labels a configuration before any assertion of unicity. These are two names for values of the same configuration map. The seed formula $M_1(x) = 2 + 3F_{2x}^2$ should not be confused with the boundary family $m_{1/q} = F_{2q+1}$. Analytical logarithms are natural; plotted horizontal axes use $\log_{10} M$.

2. Geometric descent and arithmetic closure

2.1. **A unit edge on a calibrated hyperbola.** For a real host $h > 2/3$, define

$$L_h = \operatorname{arcosh}(3h/2), \qquad \alpha_h = \frac{2h}{\sqrt{9h^2-4}}, \tag{2}$$
$$f_h(t) = \alpha_h \cosh(L_h t), \qquad g_h(t) = \alpha_h \sinh(L_h t).$$

The Cartesian curve $(g_h(t), f_h(t))$ is a hyperbola. Its height profile $t \mapsto f_h(t)$ is the curve used in the figures. Hyperbolic addition gives

$$f_h(t+1) + f_h(t-1) = 3h f_h(t), \tag{3}$$

$$f_h(t)^2 + f_h(t-1)^2 + h^2 = 3h f_h(t) f_h(t-1). \tag{4}$$

Thus two contacts one horizontal unit apart, with heights $A = f_h(c)$ and $B = f_h(c-1)$, form a real Markov state. The two compatible hosts are exactly

$$h, \qquad K = 3AB - h = \frac{A^2 + B^2}{h}. \tag{5}$$

Begin with an edge straddling the vertex, $0 \le c < 1$. Seek a wider calibrated profile through the same two ordered contacts, allowing horizontal translation. Smaller h means a wider profile. If the other host satisfies $K < h$, fit that profile and walk the unit lattice inward until the selected edge again straddles its vertex. Otherwise stop. Figure 1 shows one complete step. The fit is explicitly

$$v = \frac{1}{2} + \frac{1}{L_K} \operatorname{arsinh}\left(\frac{(A-B)\sqrt{3K+2}}{2K} \right), \qquad f_K(v) = A, \quad f_K(v-1) = B. \tag{6}$$

The walk replaces v by $v - \lfloor v \rfloor$. Its orientation is retained: reflection exchanges the contacts and sends phase to its negative modulo one.

Proposition 2.1 (Uniqueness of the fit)**.** *A marked edge has at most one distinct compatible profile, hence at most one wider fit. For fixed* $0 < a < b$*, a same-arm fit* $f_h(u-1) = a$*,* $f_h(u) = b$ *with* $u > 1$ *is unique if it exists.*

Proof. The hosts are the two roots of $X^2 - 3ABX + A^2 + B^2$. The other root belongs to the geometric domain because $(3K-2)AB = K^2 + (A-B)^2 > 0$. For a same-arm fit the horizontal distance between heights a, b is

$$\frac{1}{L_h} \int_a^b \frac{dy}{\sqrt{y^2 - \alpha_h^2}}.$$

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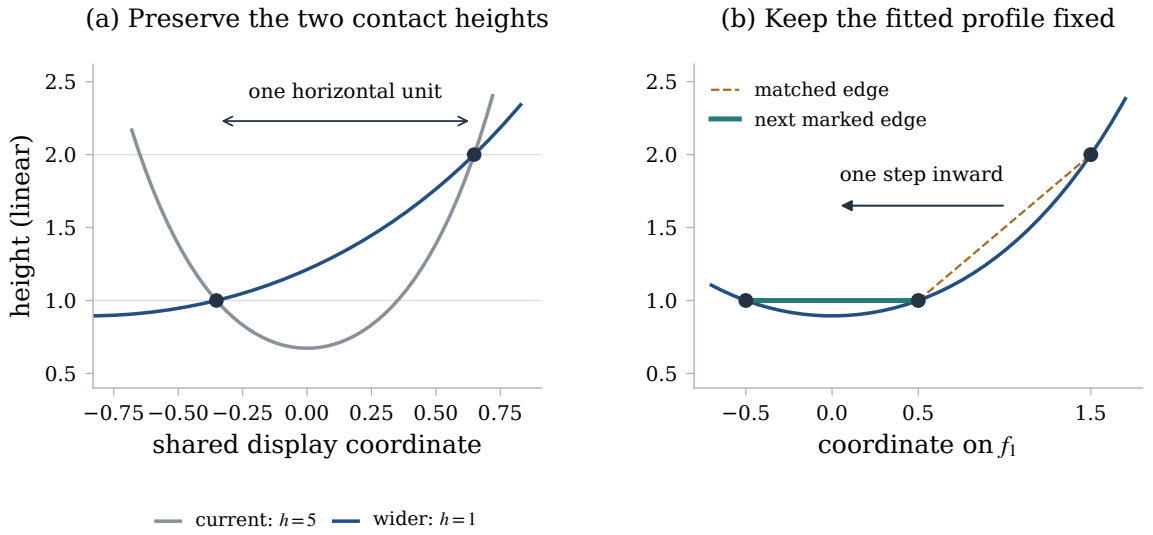


Figure 1. Geometric descent from $(1, 2, 5)$. The wider profile passes through the same unit-spaced contacts. On f_1 their coordinates are $1/2, 3/2$; one inward step selects $-1/2, 1/2$, with heights $1, 1$. The curve is fixed during the walk.

When $a > 2/3$ and the lower height is in range, both factors strictly decrease with h. The distance can therefore equal one at most once. The profile is strictly increasing on that arm, which also determines u. If $a \le 2/3$ there is no fit. □

2.2. **Termination.** Write $\mathcal{D}(M, c) = (H, \phi)$ for the terminal host and oriented phase.

Theorem 2.2 (Finite descent)**.** *Every straddling input with $M \ge 1$ terminates. On $1 \le M \le R$ the number of host changes is uniformly bounded. The terminal map is continuous into $(2/3, \infty) \times \mathbb{R}/\mathbb{Z}$ and analytic on cells with a fixed strict itinerary.*

Proof. Temporarily sort the endpoints as $a \le b$. Inward walking does not increase either endpoint. Every current host is at most R, and the discriminant in (4) gives

$$(9h^2 - 4)a^2 - 4h^2 = (2b - 3ha)^2 \ge 0.$$

Consequently the endpoints are at least $a_R = 2R/\sqrt{9R^2 - 4} > 2/3$. At an accepted change, $a^2 + b^2 < h^2$, so the endpoints are also below $h \le R$. Using either endpoint as the host in the same discriminant identity bounds the new host below by a_R. The initial host is at least $1 > a_R$.

If the new host K is at least b, its edge already straddles the vertex, because both endpoints are below $f_K(1)$; the next candidate is the old, larger host, so descent stops. Otherwise

$$\frac{h}{K} > \frac{h}{b} = 3a - \frac{K}{b} > 3a - 1 \ge 3a_R - 1 > 1.$$

All changes except possibly the last therefore contract the host by a fixed factor. An upper bound is $1 + \lceil \log(R/a_R)/\log(3a_R - 1) \rceil$.

At a boundary $K = h$, the fitted coordinate in (6) is the original c, so the accepted and stopped maps agree. At a phase cut, $K(h, 0) = h(9h^2 + 4)/(9h^2 - 4) > h$, and a neighborhood is stopped. The uniform bound on compositions then gives continuity, with phase taken modulo one. The strict formulas are analytic. □

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2.3. **Which inputs close arithmetically?** The two reversible state operations are

$$T(a,b,h) = (3ha - b, a, h), \qquad J(a,b,h) = (a, b, 3ab - h). \tag{7}$$

They and their inverses are integer polynomial maps. A terminal state has the nonnegative defect

$$\begin{aligned} H f_H(\phi) f_H(\phi - 1) - 1 = \frac{(H-1)^2(H+2)}{3H-2} \\ + \frac{4H^3}{9H^2-4} \sinh^2\Big(L_H(\phi - \tfrac{1}{2})\Big). \end{aligned} \tag{8}$$

For $0 \le \phi < 1$ it vanishes exactly at $(H, \phi) = (1, 1/2)$.

Proposition 2.3 (Arithmetic closure)**.** *A straddling input with $M \ge 1$ reaches $(1, 1/2)$ if and only if its three entries are positive integers. Modulo companion exchange, these inputs are exactly the normalized configurations* (1)*.*

Proof. The product-to-sum identity gives (8). Reversing a finite closure itinerary from $(1,1,1)$ by (7) preserves integrality. Conversely, integral descent stays integral. At an integral terminal state order the entries as $u \le v \le w$. There is no coordinate-decreasing mutation. But for $Q(X) = X^2 - 3uvX + u^2 + v^2$,

$$Q(v) = u^2 + (2 - 3u)v^2 \le 3u^2(1 - u) \le 0.$$

If neither root is below $w \ge v$, equality is forced, giving $u = v = 1$; terminality then gives $w = 1$. Finally, a straddling edge has no decreasing endpoint mutation, so its host is a largest entry. Its ordered split is uniquely recovered by (6). □

Both terminal coordinates matter. Figure 2 shows inputs with terminal host one but different terminal phases. A crossing of the height level $H = 1$ alone does not certify integrality. The marked configurations follow the host itineraries $37666 \to 5 \to 1$, $7561 \to 5 \to 1$, and $14701 \to 2 \to 1$, all ending at phase $1/2$.

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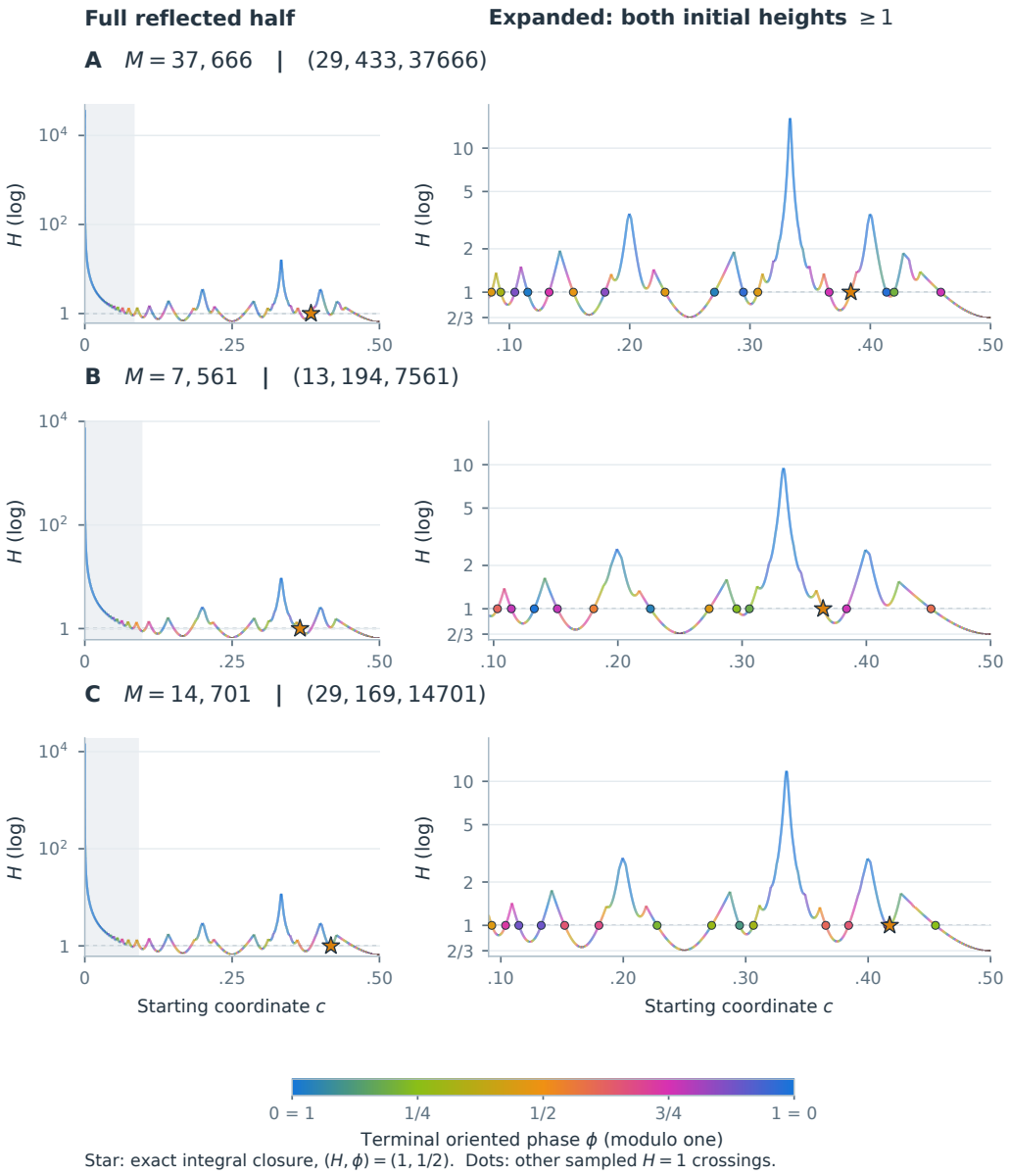


Figure 2. Terminal host and phase for $M = 37666, 7561, 14701$. Height is H on a logarithmic scale and hue is oriented phase. Left: $0 \le c \le 1/2$, with gray where an initial endpoint is below one. Right: the admissible portion enlarged. Stars mark the exact integral starts; dots are other numerically located $H = 1$ crossings. These numerical plots illustrate the closure criterion.

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3. Meridians, addresses, and the alternative tree

3.1. **The landscape and its two coordinates.** On the landscape $(\log_{10} M, c)$, the point (M, c) represents the contacts $f_M(c), f_M(c-1)$. We draw the reflected half $0 < c \le 1/2$; exchanging the contacts sends c to $1-c$. A *Vieta arc* keeps one initial contact height fixed. For example, the branch with height a has

$$c = \frac{\operatorname{arcosh}(a/\alpha_M)}{L_M}.$$

A *Vieta trek* follows such arcs, changing the retained contact when necessary. A *Meridian* is a connected component of terminal host $H(M, c) = 1$. A *phasor* holds the terminal phase ϕ fixed. The alternative tree uses phasors at $\phi = 1/2$ to join successive Meridians.

The address phase x on a Meridian is real and unrolled. Its terminal circle phase is $\phi \equiv x + 1/2 \pmod 1$. Every integer mark therefore closes arithmetically; the distinguished arrival mark $x = 0$ is called the *helm*. It need not be the minimum of the real profile.

Figure 3 keeps the same configurations in three views. The alternative tree is in the middle panel. The bottom panel uses smooth schematic connectors for the Vieta parent relation: node coordinates and adjacency are exact, but those connectors are not literal level curves. The two trees each have 55 edges. An unmarked crossing is not a vertex.

For reproducibility, the field in panel (a) uses the fixed cyclic phase palette with HSL saturation and lightness

$$w(H) = \frac{1}{1 + (\ln H / \ln 1.02)^2}, \qquad S = w(H), \qquad L = 1 - 23w(H)/40.$$

The background is a numerical illustration, independent of all exact enumerations and exclusions below.

3.2. **Evaluating an address.** The seed Meridian is

$$\begin{aligned} a_1(x) = f_1(x + \tfrac{1}{2}), \qquad b_1(x) = f_1(x - \tfrac{1}{2}), \\ M_1(x) = 3a_1(x)b_1(x) - 1. \end{aligned} \tag{9}$$

For integer x, its companions are F_{2x+1}, F_{2x-1}, where $F_{-j} = (-1)^{j+1}F_j$. Hence

$$M_1(x) = 2 + 3F_{2x}^2, \qquad M_1(0), M_1(1), M_1(2), \ldots = 2, 5, 29, 194, \ldots. \tag{10}$$

From any ordered state (a, b, H) form the two-sided orbit

$$u_0 = a, \quad u_1 = b, \qquad u_{j+1} = 3Hu_j - u_{j-1}.$$

A nonzero index k replaces the state by

$$(u_k, u_{k+1}, 3u_k u_{k+1} - H). \tag{11}$$

The address is written

$$\alpha = (1, k_m, \ldots, k_1): \quad k_1 \text{ is applied first and written at the right.}$$

This is the reversed indexing convention: the former $M_{1,a,b,c}(x)$ is now $M_{1,c,b,a}(x)$. Neither the signs nor x are reversed. In terms of (7), with $S = T^{-1}$ and $\mathcal{W}_k = JS^k$,

$$(a_\alpha(x), b_\alpha(x), M_\alpha(x)) = \mathcal{W}_{k_m} \cdots \mathcal{W}_{k_1} JT^x(1, 1, 1). \tag{12}$$

Composition acts from right to left. A recursive extension inserts its index immediately after the leading 1. For example, $M_{1,1}(0) = 13$. Index zero is forbidden because (11) would return the preceding pole $3ab - H$, rather than create a new component.

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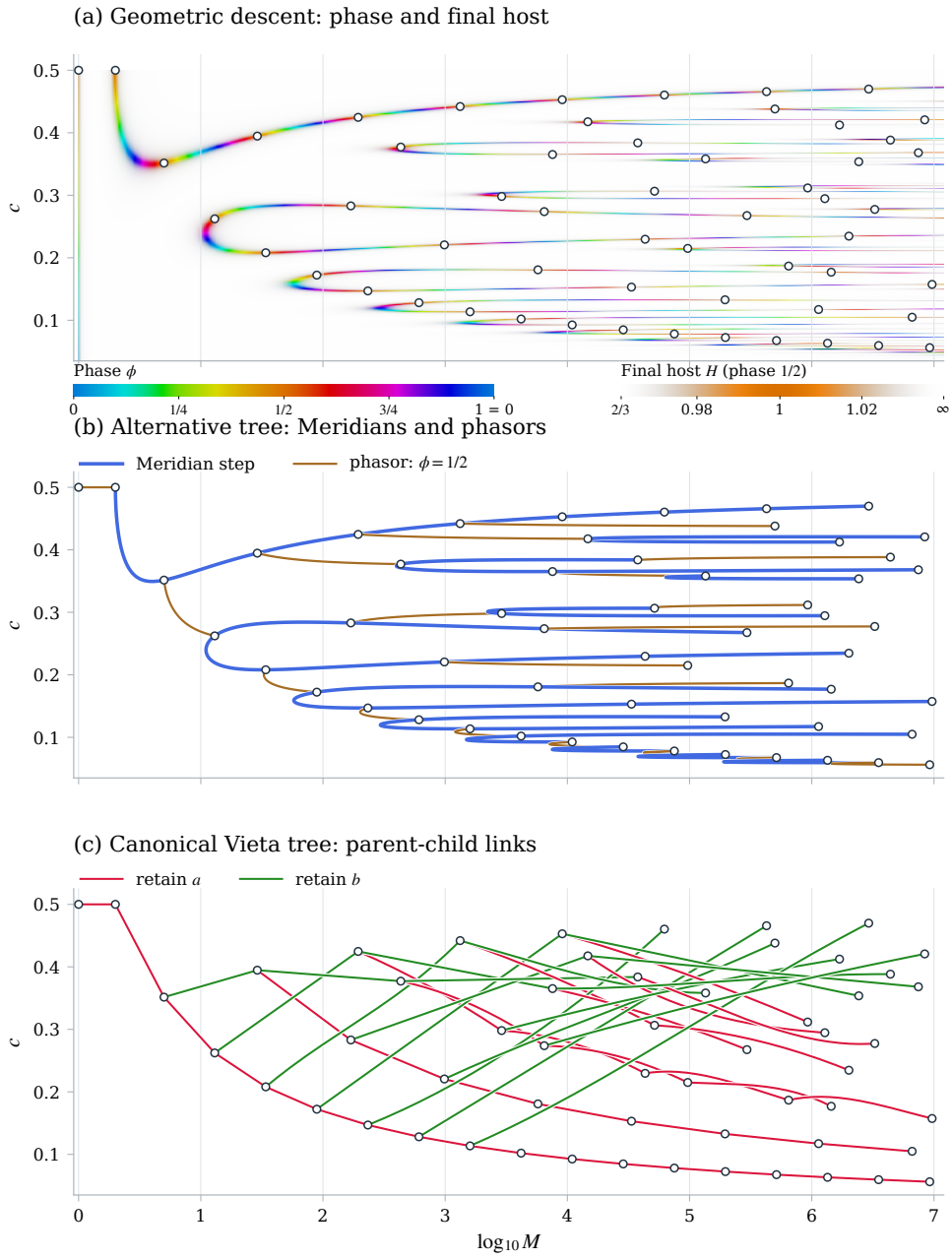


Figure 3. The same 56 configurations with $M \leq 10^7$. (a) Numerical geometric descent, with phase encoded by hue and host by color intensity. (b) The alternative tree, with Meridian and phasor edges. (c) The canonical Vieta tree, with simplified connectors retaining exact endpoints. Red retains the smaller companion, green the larger. Every edge runs toward larger M. Crossings without a node mark are not vertices.

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Definition 3.1 (Canonical marks)**.** The root $(1,1,1)$ is separate. For the seed (1) use integer phases $x \geq 0$. For every other address require $k_1 > 0$, $k_i \neq 0$ for $i > 1$, and $x \in \mathbb{Z}$. Evaluation is followed by sorting the two companions.

Theorem 3.2 (Configuration bijection)**.** *Canonical marks are in bijection with all normalized positive Markov triples except the root. The displayed value $M_\alpha(x)$ is the largest entry.*

Proof. The quadratic argument in Proposition 2.3 shows that replacing the largest coordinate by its other pole gives a smaller maximum, apart from the initial repeated entries. This is the forced Vieta reduction of a configuration and does not assume scalar unicity.

To recover a word, reflect the current maximum H to $P = 3ab - H$. When $P \geq 2$, walk its oriented endpoint orbit inward until both endpoints are below P. This edge is unique. Indeed $u_{j+1} - 2u_j + u_{j-1} = (3P-2)u_j > 0$, so the entries below P form an interval; three consecutive entries below P would contradict $u_{j-1} + u_{j+1} = 3Pu_j \geq 3P$. Endpoint descent supplies the edge, and its nonzero signed displacement recovers the last-applied, leftmost index. Repeat at the smaller pole. At pole one, the orbit reaches $(1,1)$ and fixes the seed phase.

Conversely, all outward blocks are positive and reduced. If the preceding pole is $P < H$, the first reflected poles on the two sides are $(9b^2-2)H - P > 6H$ and $(9a^2-2)H - P > 6H$. Their farther endpoints exceed $2H$, and the recurrence makes subsequent outward entries increase. The reduction therefore undoes every prescribed block. Finally, companion exchange negates all indices and x simultaneously. The canonical conditions select one of the two orientations; only $(1,1,2)$ is fixed. □

3.3. **The two parent rules.** At a nonzero phase the alternative parent is the adjacent inward mark, $(\alpha; x) \mapsto (\alpha; x - \operatorname{sgn} x)$. At a nonseed helm it is

$$(1, k_m, \ldots, k_2, k_1; 0) \longmapsto (1, k_m, \ldots, k_2; -k_1), \tag{13}$$

followed by simultaneous sign reflection if the remaining rightmost index is negative. An empty tail uses the seed's nonnegative phase. The seed helm with maximum two has parent $(1,1,1)$.

The alternative parent deletes the *rightmost* index at a helm. Recursive evaluation removes the *leftmost* index after 1. These operations serve different purposes. The depth

$$d(\alpha; x) = 1 + \sum_{i=1}^{m} (|k_i| + 1) + |x| \tag{14}$$

decreases by one under either parent edge, with root depth zero. Thus the parent rule defines a rooted tree on the same configurations. The ordinary outward Vieta moves instead are

$$(a, b, M) \longmapsto (a, M, 3aM - b), \qquad (a, b, M) \longmapsto (b, M, 3bM - a). \tag{15}$$

For example, $29 \to 194$ is a Meridian edge, whereas $29 \to 169$ is a Vieta edge. Figure 4 compares the two sets of links.

3.4. **Real components and phasor edges.**

Proposition 3.3. *The canonical real families in* (12)*, modulo companion reflection, are the nonterminal connected components of $H(M, c) = 1$. Their terminal phase is $x + 1/2$ modulo one. The separate terminal component is $M = 1$.*

Proof. The seed companions are at least $2/\sqrt{5}$, and $M_1 \geq 2$ is strictly larger than either. More generally, let $h \geq 1$ and $2/\sqrt{5} \leq a, b \leq h$. The first outward endpoint satisfies $3hb - a \geq (6/\sqrt{5} - 1)h > h$; its reflected pole is larger still. The reversed orbit gives the same conclusion on the other side. Every outward block thus has a strict inverse descent to its source and preserves the terminal phase.

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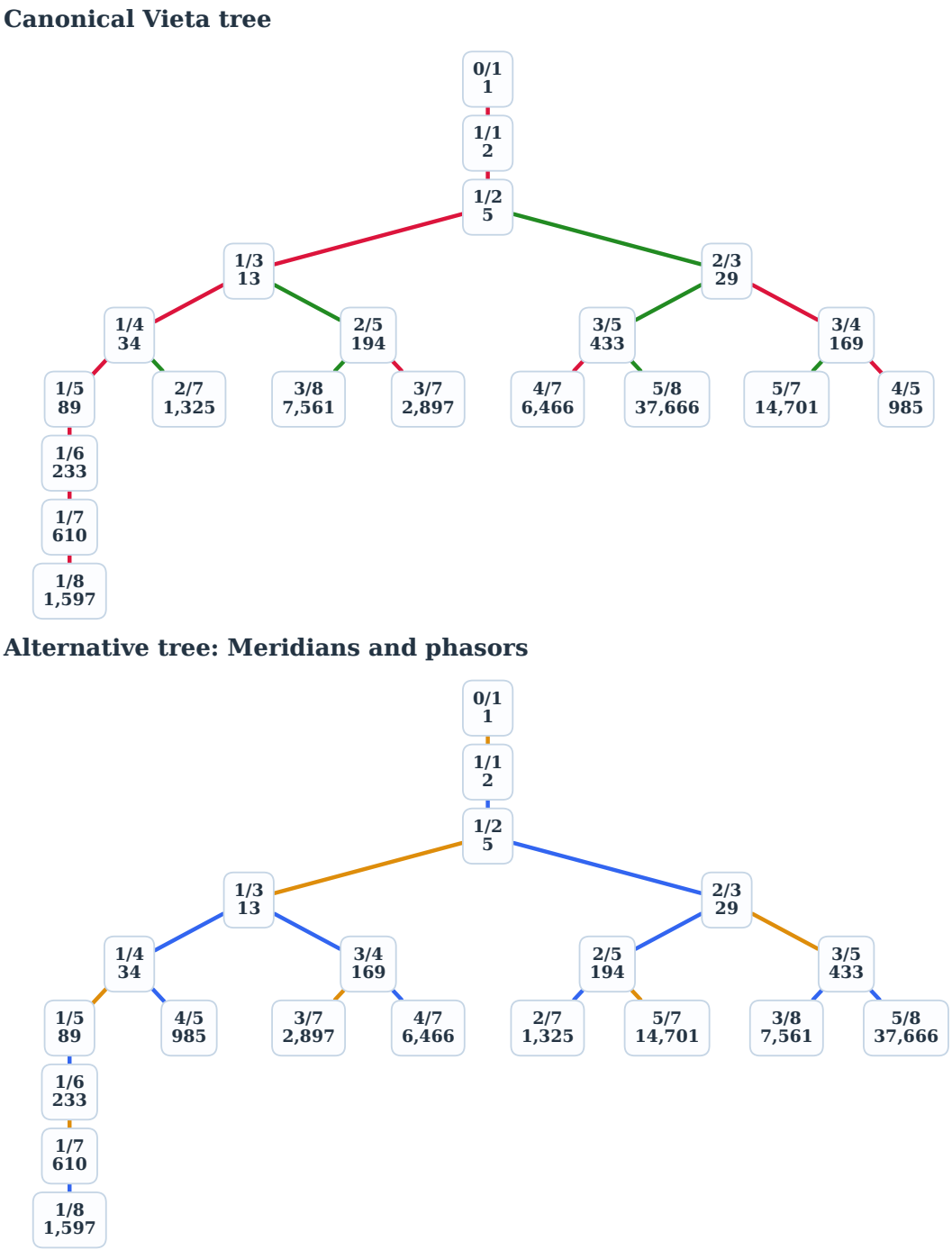


Figure 4. Two trees on the same 20 configurations: the 16 with $M \leq 3000$ and the four additional values $6466, 7561, 14701, 37666$. Each node displays its reduced standard Farey label and maximum. Red and green denote the Vieta moves; blue and gold denote Meridian and phasor steps.

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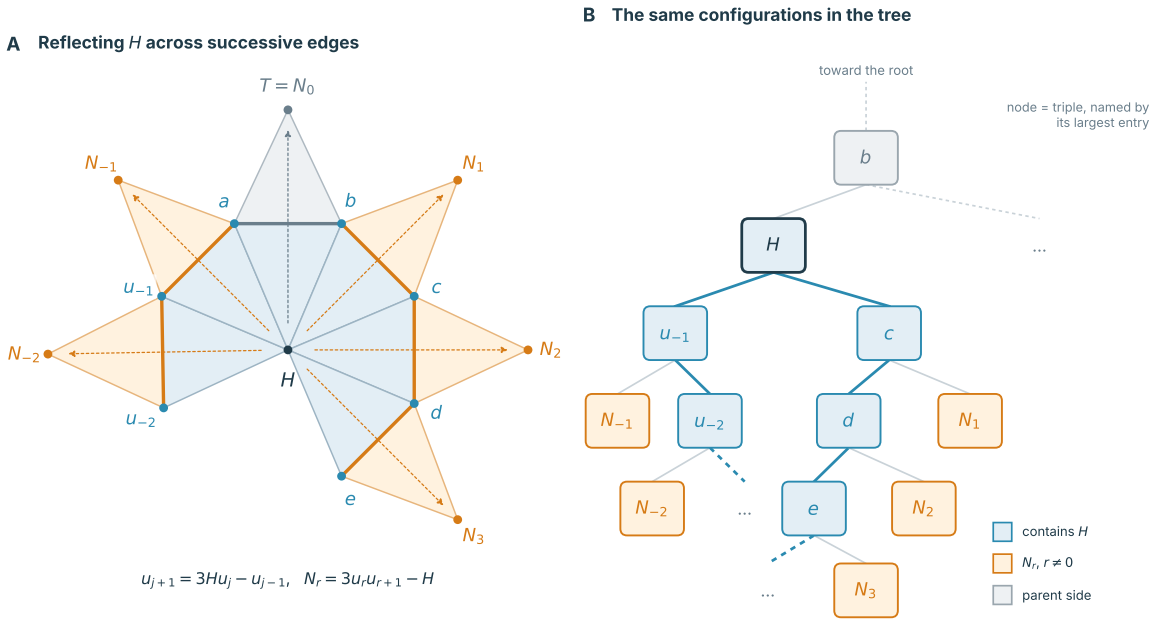


Figure 5. The orbit underlying one recursive extension. Reflecting H across (u_j, u_{j+1}) gives $N_j = 3u_j u_{j+1} - H$; N_0 is the preceding pole. For source $(a, b, H) = (1, 5, 13)$ at $M_{1,1}(0)$, the preceding pole is two, $(u_{-2}, u_{-1}, u_2, u_3, u_4) = (1325, 34, 194, 7561, 294685)$, and $(N_{-2}, N_{-1}, N_1, N_2, N_3) = (135137, 89, 2897, 4400489, 6684339842)$. The right panel locates these reflections in the Vieta tree.

Conversely, reverse any descent ending at host one. Its last host change produces the seed with $x = v - 1/2$, and every preceding walk is nonzero: a zero walk would simply have stopped at the smaller host. These strict blocks uniquely recover a locally constant word. The unrolled x joins the final phase charts continuously. The seed is injective as an ordered curve, as its endpoint difference is strictly monotone; each block is invertible. The positive Laurent expansion proved in Proposition 5.2 makes every family proper. The inverse charts are relatively open in the level set, and properness makes the families closed. Each connected family therefore fills a component. Reflection gives exactly the canonical identifications. □

The phasor edges are obtained by applying the same outward blocks to

$$\left(\frac{h}{\sqrt{3h-2}}, \frac{h}{\sqrt{3h-2}}, h\right), \qquad 1 \le h \le 2. \tag{16}$$

This symmetric arc joins $(1, 1, 1)$ to $(1, 1, 2)$. Its other host is $2h/(3h-2)$, and both possible terminal fits have phase $1/2$. Its endpoints satisfy the bounds used in the proposition, so the outward blocks preserve that phase. The displayed phasors consequently arise from the descent itself, rather than interpolation between integer nodes.

3.5. **Degree and the exact Farey dictionary.** Attach endpoint count vectors $u = (1, 0)$ and $v = (0, 1)$ to the seed. Under an index k update them by

$$(u, v) \longmapsto \begin{cases} ((k-1)(u+v)+v,\ k(u+v)+v), & k > 0, \\ ((|k|(u+v)+u,\ (|k|-1)(u+v)+u), & k < 0. \end{cases} \tag{17}$$

Write their sum as (r, s). Their determinant remains ± 1 and their unordered residues modulo two remain $\{(1, 0), (0, 1)\}$. Thus r, s are positive coprime odd integers. Canonical orientation gives $r \le s$.

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The *Meridian degree* is $D = r + s$. It is the extreme Laurent exponent in ρ^x, where $\rho = (3 + \sqrt{5})/2$; the degree in ρ^{2x} is $D/2$.

Proposition 3.4 (Helm classification)**.** *Canonical Meridians correspond bijectively to coprime positive odd pairs $r \le s$. Their helm labels are the standard Farey fractions r/s.*

Proof. The endpoint vectors are the two Stern–Brocot parents: their sum is (r, s) and their determinant has absolute value one. Conversely, the parents of an odd/odd pair have opposite coordinate parities. Their componentwise absolute difference is a smaller positive primitive odd/odd pair, except at $(1, 1)$. The difference has one componentwise sign; an opposite sign would give the initial pair. Iteration reaches $(1, 1)$.

To recover the signed indices, put $n_\alpha = (r, s)$ and let α^- remove the leftmost index after 1. Consecutive difference vectors have determinant of absolute value two, so three consecutive vectors satisfy $n = 2c\, n^- + \varepsilon n^{--}$ with $c \ge 1$ and $\varepsilon = \pm 1$. Determinant division gives an integer coefficient, parity makes it even, and decreasing coordinate sums fix its magnitude and sign. At the bottom use $(1, 1), (-1, 1)$; canonical orientation fixes the first index positive. The magnitude is fixed by the ratio of successive degrees, which lies in $[2c, 2c+1)$ or $(2c-1, 2c]$ according to the sign; equality occurs only at the first applied index. Reversing the process gives precisely $n_{1,k,k_m,\ldots,k_1} = 2|k| n_\alpha + \operatorname{sgn}(kk_m) n_{\alpha^-}$. For the seed use $k_0 = 1$ and $n_\varnothing = (-1, 1)$, so the first extension has $(r, s) = (2k_1 - 1, 2k_1 + 1)$. This recovers the unique address. Mediant addition identifies the label. □

Lemma 3.5 (Phase dictionary)**.** *For helm (r, s) and degree $D = r + s$, the standard Farey label at phase x is*

x	(p, q)	$p + q$
0	(r, s)	D
1	(r, D)	$D + r$
-1	(s, D)	$D + s$
$n \ge 2$	$(D, (n-1)D + r)$	$nD + r$
$-n \le -2$	$(D, (n-1)D + s)$	$nD + s.$

The negative seed phases are omitted.

Proof. At seed phases $0, 1, n \ge 2$, the endpoint bases are respectively $((1, 0), (0, 1))$, $((1, 1), (0, 1))$, and $((1, n), (1, n-1))$. The count updates are linear, so substitution gives the table; negative phase exchanges the bases. For the inverse, odd/odd p, q give $(r, s, x) = (p, q, 0)$. If q is even, sort $p, q - p$ and choose $+1$ when p is the smaller entry, otherwise -1. If p is even, write $q = tp + u$ with $0 < u < p$; sort $u, p - u$ and choose $+(t + 1)$ or $-(t + 1)$ in the same way. A tie is the seed and uses the nonnegative representative. These formulas recover every mark. □

In particular, degree can be read directly from a reduced label:

$$D = \begin{cases} p + q, & p, q \text{ odd}, \\ q, & q \text{ even}, \\ p, & p \text{ even}. \end{cases} \tag{18}$$

This dictionary connects the Meridian and corridor collision windows below.

4. Meridian residues: modulo 27 and Fibonacci moduli

4.1. **Transferring a seed congruence to every address.** The recurrence for the companions contains no division, but a seed period of the *maximum* need not visibly preserve both companions.

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The following four-entry record resolves that issue:

$$\mathcal{S} = (A, B, C, H) = (9a^2, 9b^2, 3ab, H).$$

Forward and backward orbit steps and pole reflection are the polynomial maps

$$\begin{aligned} \mathcal{R}(A, B, C, H) &= (B, A + 9H^2B - 18HC, HB - C, H), \\ \mathcal{R}^{-1}(A, B, C, H) &= (B + 9H^2A - 18HC, A, HA - C, H), \\ \mathcal{J}(A, B, C, H) &= (A, B, C, C - H). \end{aligned} \tag{19}$$

For $h(x) = M_1(x)$ the seed record is

$$\mathcal{S}_1(x) = (8h(x) - h(x-1) - 2,\ h(x) + h(x-1) + 2,\ h(x) + 1,\ h(x)).$$

Every congruence preserving two consecutive seed maxima therefore propagates along every signed address.

Theorem 4.1 (Largest modulus with a common phase period)**.** *Every canonical Meridian has phase period T modulo m if and only if*

$$m \mid q(T), \qquad q(T) = 3F_{2T}F_{\gcd(2T,4)} = \begin{cases} 3F_{2T}, & T \text{ odd}, \\ 9F_{2T}, & T \text{ even}. \end{cases} \tag{20}$$

In particular, s is a common period modulo F_{2s}.

Proof. Fibonacci addition and (10) give

$$h(x + T) - h(x) = 3F_{2T}F_{4x+2T}.$$

Fibonacci strong divisibility, $\gcd(F_a, F_b) = F_{\gcd(a,b)}$, follows by the Euclidean algorithm from the addition formula. The gcd of all displayed differences is already the gcd of those at $x = 0, 1$, namely $q(T)$. Thus the seed has period T modulo m exactly when $m \mid q(T)$. The record (19) proves sufficiency at every address; the seed itself proves necessity. □

The theorem specifies a period common to all Meridians. It does not assert that the period is the smallest for each individual sequence. Nor does a large period exclude a scalar residue match at two earlier phases. For powers of three it gives

$$M_\alpha(x + 2 \cdot 3^n) \equiv M_\alpha(x) \pmod{3^{n+3}}, \qquad n \geq 0. \tag{21}$$

Indeed, $F_{3j} = F_j(5F_j^2 + 3)$ for even j, beginning with $F_4 = 3$, gives $v_3(F_{4\cdot 3^n}) = n + 1$. Figure 6 shows the common periods at the requested even-indexed Fibonacci moduli.

4.2. **Why modulo 27 gives only ten classes.** No positive Markov entry is divisible by three: mutation negates the changed coordinate modulo three and the integral tree starts at $(1, 1, 1)$. Let t, h be preceding and current poles. An extension by a nonzero index has the particularly simple action

$$S(t, h) = (h, t) \pmod{k \text{ even}}, \qquad R(t, h) = (h, 7h - t) \pmod{k \text{ odd}} \pmod{27}. \tag{22}$$

To see this, put $N_j = 3u_ju_{j+1} - h$. Direct use of the orbit recurrence gives

$$N_0 = t, \quad N_1 = (9u_1^2 - 2)h - t, \quad N_{j+2} - N_j = 9h(u_{j+2}^2 - u_{j+1}^2).$$

All orbit squares are one modulo three, so $N_{j+2} \equiv N_j$ and $N_1 \equiv 7h - t \pmod{27}$. This works for negative indices as well. Only the parity of each recursively applied index survives.

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x	M1							M1.1							M1.2							M1.1.1							M1.-1.1						
s	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8
F_{2s}	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987
-12								1	5	13	31	13	231	181	1	1	13	50	1	88	937	2	1	20	50	17	328	944	2	1	5	9	89	285	215
-11								1	2	13	4	34	283	559	1	1	1	46	73	21	715	2	1	2	13	89	94	170	2	1	2	21	89	333	107
-10								1	1	19	13	121	248	544	1	2	16	48	82	270	541	2	2	20	37	98	29	482	2	5	8	34	53	272	869
-9								1	5	1	34	61	57	169	1	1	13	26	1	291	160	2	1	2	51	89	343	653	2	1	8	13	17	241	806
-8								1	2	13	50	130	169	13	1	1	13	53	73	343	433	2	1	20	31	17	34	923	2	1	5	21	17	86	89
-7								1	1	13	31	25	13	34	1	2	1	50	82	56	652	2	2	2	50	98	258	317	2	5	2	9	125	89	233
-6								1	5	19	4	13	34	985	1	1	16	46	1	21	541	2	1	20	13	17	369	104	2	1	8	21	89	233	890
-5								1	2	1	13	34	231	820	1	1	13	48	73	88	580	2	1	2	37	89	328	758	2	1	8	34	89	285	71
-4								1	1	13	34	121	283	181	1	2	13	26	82	21	937	2	2	20	51	98	94	944	2	5	5	13	53	333	215
-3								1	5	13	50	61	248	559	1	1	1	53	1	270	715	2	1	2	31	89	29	170	2	1	2	21	17	272	107
-2								1	2	19	31	130	57	544	1	1	16	50	73	291	541	2	1	20	50	17	343	482	2	1	8	9	17	241	869
-1								1	1	1	4	25	169	169	1	2	13	46	82	343	160	2	2	2	13	98	34	653	2	5	8	21	125	86	806
0	2	2	2	2	2	2	2	1	5	13	13	13	13	13	1	1	13	48	1	56	433	2	1	20	37	17	258	923	2	1	5	34	89	89	89
1	2	5	5	5	5	5	5	1	2	13	34	34	34	34	1	1	1	26	73	21	652	2	1	2	51	89	369	317	2	1	2	13	89	233	233
2	2	5	8	29	29	29	29	1	1	19	50	121	231	985	1	2	16	53	82	88	541	2	2	20	31	98	328	104	2	5	8	21	53	285	890
3	2	2	5	29	50	194	194	1	5	1	31	61	283	820	1	1	13	50	1	21	580	2	1	2	50	89	94	758	2	1	8	9	17	333	71
4	2	5	2	5	29	194	338	1	2	13	4	130	248	181	1	1	13	46	73	270	937	2	1	20	13	17	29	944	2	1	5	21	17	272	215
5	2	5	5	2	5	29	194	1	1	13	13	25	57	559	1	2	1	48	82	291	715	2	2	2	37	98	343	170	2	5	2	34	125	241	107
6	2	2	8	5	2	5	29	1	5	19	34	13	169	544	1	1	16	26	1	343	541	2	1	20	51	17	34	482	2	1	8	13	89	86	869
7	2	5	5	29	5	2	5	1	2	1	50	34	13	169	1	1	13	53	73	56	160	2	1	2	31	89	258	653	2	1	8	21	89	89	806
8	2	5	2	29	29	5	2	1	1	13	31	121	34	13	1	2	13	50	82	21	433	2	2	20	50	98	369	923	2	5	5	9	53	233	89
9	2	2	5	5	50	29	5	1	5	13	4	61	231	34	1	1	1	46	1	88	652	2	1	2	13	89	328	317	2	1	2	21	17	285	233
10	2	5	8	2	29	194	29	1	2	19	13	130	283	985	1	1	16	48	73	21	541	2	1	20	37	17	94	104	2	1	8	34	17	333	890
11	2	5	5	5	5	194	194	1	1	1	34	25	248	820	1	2	13	26	82	270	580	2	2	2	51	98	29	758	2	5	8	13	125	272	71
12	2	2	2	29	2	29	338	1	5	13	50	13	57	181	1	1	13	53	1	291	937	2	1	20	31	17	343	944	2	1	5	21	89	241	215

Figure 6. Five Meridians modulo F_{2s} for $s = 2, \ldots, 8$: the moduli are $3, 8, 21, 55, 144, 377, 987$. Each column has common phase period s, possibly a multiple of an individual least period. The shaded row is the helm; blank cells omit the seed's negative phases. The displayed residues use the reversed Meridian indexing of this edition.

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Proposition 4.2 (Universal modulo-27 classes)**.** *At integer phases every Meridian alternates between the two residues of one of the following sets, allowing the two singleton cases:*

$$\{1\},\ \{2,5\},\ \{4,16\},\ \{7,13\},\ \{8,17\},\ \{10,19\},\ \{11,23\},\ \{14,20\},\ \{22,25\},\ \{26\}.$$

The ten sets are pairwise disjoint. Meridians in different classes cannot share an integer value, regardless of degree or phase.

Proof. The common phase period is two because $q(2) = 27$. Track preceding and current poles at both phase parities, starting from $(1, 2, 1, 5)$. The simultaneous maps satisfy $S^2 = I$ and $SRS = R^{-1}$. Successive current-pole pairs on the R orbit are

$$\begin{array}{cccccc} (2,5) & (13,7) & (8,17) & (16,4) & (23,11) & (10,19) \\ (20,14) & (22,25) & (26,26) & (25,22) & (14,20) & (19,10) \\ (11,23) & (4,16) & (17,8) & (7,13) & (5,2) & (1,1). \end{array}$$

The preceding pair is the previous pair cyclically, and $b_{j+1} = 7b_j - b_{j-1}$ checks the return in 18 steps. These states and their S images form a closed set of 36 states. Their current-pole images are exactly the ten listed sets. Congruence of integer values would require membership in the same set. □

Table 1. The ten universal classes and their populations among the 519 Meridians of degree at most 100. Only the population column is bounded-degree data.

Residue set	Population	Residue set	Population
$\{1\}$	42	$\{10, 19\}$	47
$\{2, 5\}$	88	$\{11, 23\}$	47
$\{4, 16\}$	61	$\{14, 20\}$	38
$\{7, 13\}$	78	$\{22, 25\}$	35
$\{8, 17\}$	65	$\{26\}$	18

At that degree bound, the classification leaves 15,215 of the 134,421 pairs for further comparison. Larger Fibonacci moduli and compatible phases, rather than extrapolation from this table, supply the degree-10,000 certificate in Section 8.

5. Equal-degree separation and unequal-degree windows

5.1. **A complete result at each fixed degree.** We first use the established fixed-sum, fixed-numerator and fixed-denominator orderings of Markov numbers. Section 7 proves them by matching counts and records their earlier sources.

Theorem 5.1 (Equal-degree unicity)**.** *Distinct canonical integer-phase marks of the same Meridian degree have different maxima. The marks may belong to the same or to different Meridians.*

Proof. For $D \geq 4$, Lemma 3.5 partitions the labels into three classes. Helms r/s have fixed sum D; near-helms $r/D, s/D$ have fixed denominator D; outer phases have fixed numerator D. Each class is strictly ordered by its corresponding monotonicity, and the dictionary ensures that different marks have different labels.

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The largest helm is $m_{1/(D-1)}$. The near-helm extrema are $m_{1/D}, m_{(D-1)/D}$; the smallest outer value is $m_{D/(D+1)}$. They are separated by

$$m_{1/(D-1)} < m_{1/D}, \qquad m_{(D-1)/D} < m_{(D-1)/(D+1)} < m_{D/(D+1)}.$$

The intermediate fraction is reduced because D is even. The first two inequalities hold at fixed numerator and the last at fixed denominator. For $D = 2$, the unique seed gives $2, 5$ and then the increasing family $m_{2/(2n-1)}$ for $n \geq 2$, starting at 29. These values are also distinct. □

Thus the remaining Meridian comparison has unequal degrees. To bound it, we need exact leading terms and an explicit positive remainder.

5.2. **Positive words and a uniform tail bound.** Put $\rho = (3 + \sqrt{5})/2$, $\lambda = \log \rho$, and $\xi = \rho^x$. The endpoint matrices are

$$P(\xi) = \begin{pmatrix} B_0/\xi & C_0 \\ C_0 & A_0\xi \end{pmatrix}, \qquad Q(\xi) = \begin{pmatrix} A_0/\xi & \rho C_0 \\ C_0/\rho & B_0\xi \end{pmatrix}, \tag{23}$$

$$A_0 = \frac{15 + 3\sqrt{5}}{10}, \quad B_0 = \frac{15 - 3\sqrt{5}}{10}, \quad C_0 = \frac{2}{\sqrt{5}}.$$

Both have determinant one and positive entries. With seed endpoint words $U = P, V = Q$, the scalar state is

$$a_\alpha = \operatorname{tr} U/3, \quad b_\alpha = \operatorname{tr} V/3, \quad M_\alpha = \operatorname{tr}(UV)/3.$$

Its extension is represented by

$$(U, V) \longmapsto \begin{cases} ((UV)^{k-1}V, (UV)^k V), & k > 0, \\ ((VU)^{|k|}U, (VU)^{|k|-1}U), & k < 0. \end{cases} \tag{24}$$

For verification, $\operatorname{tr}((UV)^{j-1}V)/3$ begins at a, b and obeys the endpoint recurrence by Cayley–Hamilton. The determinant-one trace identity $\operatorname{tr}(XY) = \operatorname{tr} X \operatorname{tr} Y - \operatorname{tr}(XY^{-1})$ then gives the reflected pole. Negative indices use the reversed orbit. The count updates are exactly (17).

Proposition 5.2 (Laurent profile)**.** *For helm counts (r, s), degree $D = r + s$, and $t = (r - s)/2$,*

$$M_\alpha(x) = \sum_{j=-D,-D+2,\ldots,D} c_j \rho^{jx}, \quad c_j \geq 0, \quad c_{\sigma D} = \frac{1}{3}(9/5)^{D/2}\rho^{\sigma t}. \tag{25}$$

The logarithm is strictly convex, and the real profile has one minimum.

Proof. Expand the trace into closed paths on the two matrix states. All weights are nonnegative. A closed path has an even number of off-diagonal choices, which fixes the parity of its exponent. The two constant-state paths give the extreme coefficients because $A_0B_0 = 9/5$ and $A_0/B_0 = \rho$. The second derivative of the logarithm is λ^2 times the variance of the exponents under their positive normalized weights. Both extremes occur, so the variance is positive. They also force growth at both ends. □

Lemma 5.3 (Uniform positive-word remainder)**.** *Let W have r letters P and s letters Q, of total length $g = r + s > 0$. For $n \geq 0$ and $\sigma = \pm 1$,*

$$\frac{\operatorname{tr} W(\rho^{\sigma n})}{3} = \frac{1}{3}(9/5)^{g/2}\rho^{\sigma(r-s)/2}\rho^{gn}(1 + \epsilon_{W,\sigma}(n)), \qquad 0 \leq \epsilon_{W,\sigma}(n) \leq (1 + \rho^{1-2n})^g - 1. \tag{26}$$

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Proof. Write $z = \rho^{-n}$, $\varphi = \sqrt{\rho}$, and $L = 2\varphi/3$. On the positive arm, divide the P, Q factors by $A_0\rho^n, B_0\rho^n$ respectively and conjugate by $\mathrm{diag}(\rho, 1)$. The normalized factors are

$$\begin{pmatrix} \rho^{-1}z^2 & L\rho^{-2}z \\ Lz & 1 \end{pmatrix}, \qquad \begin{pmatrix} \rho z^2 & Lz \\ Lz & 1 \end{pmatrix}.$$

The negative arm gives the same pair in reverse roles after exchanging the basis states. Both are bounded entrywise by the rank-one matrix

$$K = \begin{pmatrix} \rho z^2 & \varphi z \\ \varphi z & 1 \end{pmatrix}.$$

Products preserve the entrywise bound, and $\operatorname{tr} K^g = (1+\rho z^2)^g$. The constant dominant-state path contributes one to the normalized trace. Restoring the removed factors proves the bounds. □

5.3. **A finite rectangle depending only on the degrees.**

Theorem 5.4 (Unequal-degree collision window)**.** *Let $D_\alpha \neq D_\beta$, and define*

$$m = \frac{|D_\alpha - D_\beta|}{2}, \qquad E = \max(D_\alpha, D_\beta),$$
$$h = \min\{j \in \mathbb{Z}_{\geq 0} : \rho^j \geq E\}, \qquad N = 3m + h + 3, \qquad T = E(N+2).$$

An integer equality $M_\alpha(x) = M_\beta(y)$ requires

$$|x| \leq \left\lceil \frac{T}{D_\alpha} \right\rceil, \qquad |y| \leq \left\lceil \frac{T}{D_\beta} \right\rceil, \qquad M_\alpha(x) < \rho^T/3. \tag{27}$$

The seed still uses only nonnegative phases.

Proof. The degrees are even, so m is a positive integer. First,

$$|m\log(9/5) - j\log\rho| > \frac{1}{81^m + 1} \qquad (j \in \mathbb{Z}). \tag{28}$$

Choose $j_0 \geq 0$ with $1 < (9/5)^m\rho^{-j_0} < \rho$; equality at an endpoint is impossible by taking the field norm. The positive algebraic integers $9^m - 5^m\rho^{j_0}$ and $5^m\rho^{j_0+1} - 9^m$ have nonzero integer norms and conjugates of absolute value below 9^m. Each is therefore greater than 9^{-m}. Dividing by the appropriate positive denominators makes both adjacent multiplicative gaps greater than 81^{-m}. The inequality $\log(1+u) > u/(1+u)$ gives (28); all other integers j are farther away.

At phases $x = \sigma n$, $y = \tau n'$, the leading logarithms differ by

$$\frac{D_\alpha - D_\beta}{2}\log(9/5) + (D_\alpha n - D_\beta n' + \sigma t_\alpha - \tau t_\beta)\log\rho.$$

The coefficient of $\log\rho$ is integral. Put $r_m = (81^m + 1)^{-1}$. Since $E \leq \rho^h$, $\rho^6 > 81$ and $\rho^5 > 5$,

$$E\rho^{1-2N} \leq \rho^{-6m-5} < \frac{1}{2\cdot 81^m + 3} = \frac{r_m}{r_m + 2}.$$

For $gu < 1$, the elementary bound $(1+u)^g \leq \sum_{j\geq 0}(gu)^j = (1-gu)^{-1}$ makes each relative remainder in Lemma 5.3 smaller than $r_m/2$ when its phase magnitude is at least N. Their logarithmic corrections cannot cancel (28). At least one collision phase therefore has magnitude less than N.

The leading coefficient of a length-g word lies strictly between $\rho^{-g}/3$ and $\rho^g/3$, because $\rho^{-1} < B_0 < A_0 < \rho$. Lemma 5.3, with $1 + \rho^{1-2N} < \rho$, gives $\operatorname{tr} W(\rho^{\pm N})/3 < \rho^{g(N+2)}/3 \leq \rho^T/3$. Convexity bounds a value inside either finite phase strip by its endpoints, so the common value is below $\rho^T/3$. Positivity also gives the lower bound $\operatorname{tr} W(\rho^{\sigma n})/3 > \rho^{g(n-1)}/3$. Comparing the two bounds yields $n < 1 + T/g$, hence the integer ceilings in (27). □

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This theorem bounds the remaining candidates; it does not assert that any candidate collides. It also does not bound the degrees of a hypothetical collision. The next section supplies a different finite reduction, expressed in Farey numerators and denominators rather than Meridian phases.

6. Corridor walls and numerator–denominator windows

6.1. **A second way to travel through the same tree.** The corridor observation groups the walls between neighboring phasors. After an initial entrance, a wall follows a fixed-numerator family

$$m_{p/(q_0+np)}, \qquad n \geq 0.$$

Both Meridian and phasor edges occur in such a route. Their common law is the advance $q \mapsto q+p$, not a rule that distinguishes the two edge colors. For the blue corridor the fixed-numerator portions are $3/(4+3n)$ and $3/(5+3n)$, tending to the historical coordinate $c = 1/3$. The first wall begins

$$169 \xrightarrow{\text{phasor}} 2897 \xrightarrow{\text{Meridian}} 51641 \xrightarrow{\text{phasor}} 925765 \longrightarrow \cdots,$$

and the other begins $433, 7561, 135137, 2423525, \ldots$ with the opposite edge alternation.

Corridor role and edge color must be kept separate. With corridor-wall travel designated *standard* and a new entrance *exotic*, the two entry routes here are

$$5 \xrightarrow{\text{exotic}} 29 \xrightarrow{\text{standard}} 433 \xrightarrow{\text{standard}} 7561 \longrightarrow \cdots, \qquad 13 \xrightarrow{\text{exotic}} 169 \xrightarrow{\text{standard}} 2897 \longrightarrow \cdots.$$

Both exotic entrances are Meridian edges. In the first route the initial standard step $29 \to 433$ changes the numerator from two to three; the fixed-numerator formula starts at 433. Thus excluding the concave ceiling arc from a wall's recurrence does not move the exotic entrance to that arc. The graph and its actual depth (14) are unchanged.

6.2. **The exact recurrence behind the observed regularity.** Use the standard Christoffel matrices

$$D_a = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad D_b = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{29}$$

Their trace encoding is

$$3m_{p/q} = \operatorname{tr}\left[\prod_{j=1}^{p} D_a^{\lfloor jq/p \rfloor - \lfloor (j-1)q/p \rfloor} D_b\right]. \tag{30}$$

The word is a cyclic conjugate of the lower Christoffel word. Trace is unchanged by that rotation. The normalization is fixed by $\operatorname{tr} D_a = 3$, $\operatorname{tr}(D_a D_b) = 6$ and the Farey trace identity, which give the Markov recurrence with the initial values one and two. This is the classical matrix-word framework [9]; Section 7 also gives its matching interpretation.

Replacing q by $q_0 + np$ adds n to every exponent in (30). The eigenvalues of D_a are ρ, ρ^{-1}. Choose its orthogonal eigenbasis so that D_b becomes

$$\begin{pmatrix} \frac{3}{2} + \frac{3}{2\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{3}{2} - \frac{3}{2\sqrt{5}} \end{pmatrix}.$$

All four entries are positive. Expanding the trace over cyclic choices of the two eigenstates gives the exact identity

$$m_{p/(q_0+np)} = \sum_{j=0}^{p} c_{p,q_0,j}\, \rho^{(p-2j)n}, \qquad c_{p,q_0,j} > 0. \tag{31}$$

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Every mode occurs. The least order of a constant-coefficient recurrence is therefore $p+1$, with characteristic polynomial

$$\prod_{j=0}^{p}(X-\rho^{p-2j}).$$

Pairing conjugate roots shows that its coefficients are integers. For $p=3$ the polynomial is $(X^2-18X+1)(X^2-3X+1)$, so either blue wall satisfies

$$m_{3/(q+12)} = 21m_{3/(q+9)} - 56m_{3/(q+6)} + 21m_{3/(q+3)} - m_{3/q}. \tag{32}$$

This equality holds at every step, for both colors of edge.

The exact regularity is a shared recurrence, not a constant output ratio. The positive spectral sum gives strict log-convexity, hence

$$\log_{10}\frac{m_{p/(q_0+(n+1)p)}}{m_{p/(q_0+np)}} \nearrow p\log_{10}\rho.$$

The small deviation is retained in the collision argument, rather than discarded as an asymptotic error. Figure 7 shows both the geometric walls and that deviation.

6.3. **A positive correction with a uniform bound.** The dominant eigenstate in (30) yields

$$3m_{p/q} = \left(\frac{3}{\sqrt{5}}\right)^{p}\left(\frac{1+\sqrt{5}}{2}\right)^{2q+p}(1+\varepsilon_{p,q}), \tag{33}$$

where

$$0 < \varepsilon_{p,q} \le \left(1+\rho^{-2\lfloor q/p\rfloor}\right)^{p} - 1. \tag{34}$$

To prove the bound, divide the transformed D_b by its upper-left entry; every resulting entry is at most one. Choosing the contracting eigenstate at a factor costs at least $\rho^{-2\lfloor q/p\rfloor}$. Summing over all nonempty subsets of contracting choices gives (34). This bound is independent of the remainder class of q modulo p.

Along a fixed wall the correction decreases strictly to zero. Consequently its logarithmic deviations telescope exactly:

$$\sum_{k=0}^{n-1}\left[p\log\rho - \log\frac{m_{p/(q+(k+1)p)}}{m_{p/(q+kp)}}\right] = \log\frac{1+\varepsilon_{p,q}}{1+\varepsilon_{p,q+np}}. \tag{35}$$

Thus there is no accumulated error proportional to the path length. For a route changing corridors, subtraction of the endpoint formula still telescopes, but monotone decay is claimed only within a fixed wall.

Theorem 6.1 (Necessary correction gap)**.** *If $p \neq p'$ and $m_{p/q} = m_{p'/q'}$ for reduced labels, then*

$$\frac{|\varepsilon_{p,q} - \varepsilon_{p',q'}|}{1+\min(\varepsilon_{p,q},\varepsilon_{p',q'})} > 9^{-|p-p'|}. \tag{36}$$

Proof. Assume $p > p'$. The ratio of the leading terms is

$$R = \left(\frac{3(1+\sqrt{5})}{2\sqrt{5}}\right)^{p-p'}\rho^{q-q'}.$$

Its quadratic-field norm is $(9/5)^{p-p'}$, so $R \neq 1$. The number $(\sqrt{5})^{p-p'}(R-1)$ is an algebraic integer, because the powers of $(1+\sqrt{5})/2$ and ρ are units. Its nonzero norm gives

$$|(R-1)(\overline{R}-1)| \ge 5^{-(p-p')}, \qquad \overline{R} = (9/5)^{p-p'}/R > 0.$$

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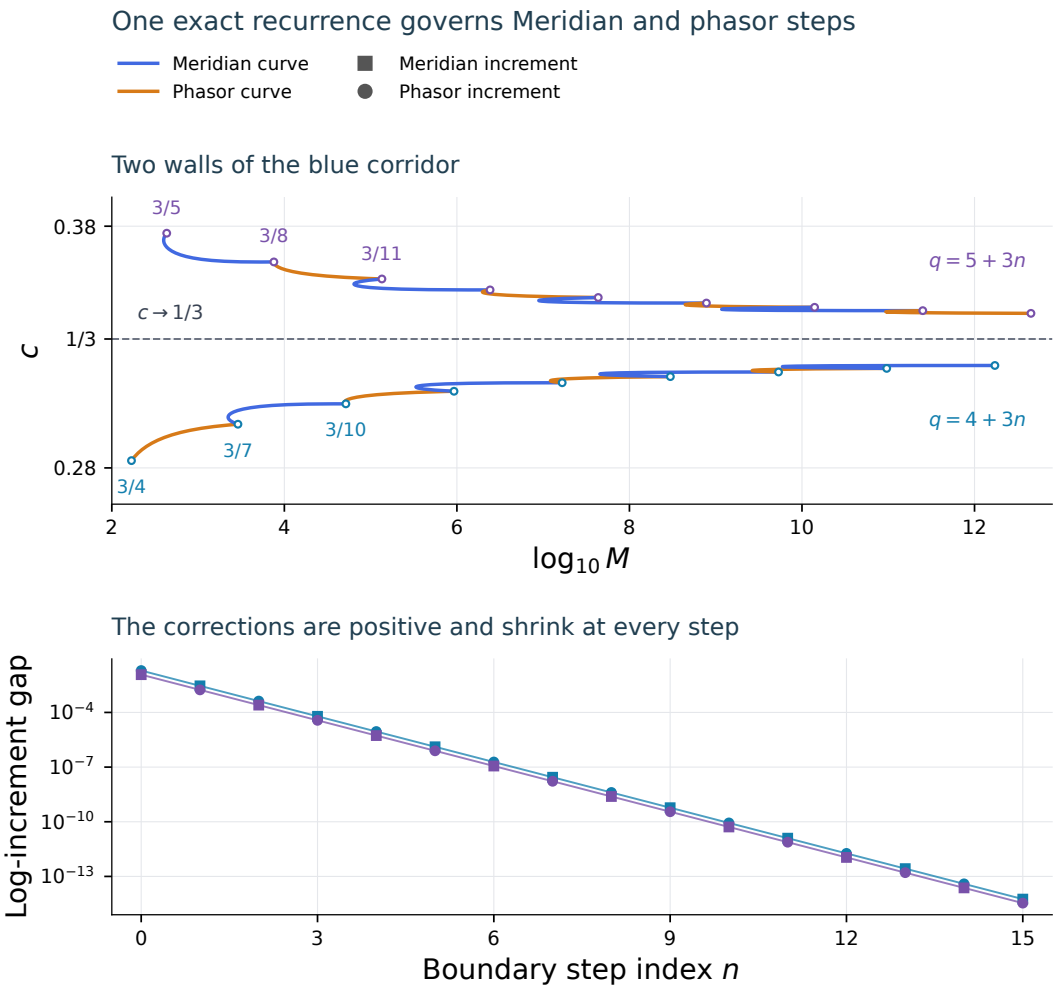


Figure 7. The two fixed-numerator walls of the blue corridor. The concave ceiling arc is omitted from this recurrence illustration. Both walls obey (32). In the lower panel, the positive difference between $3\log_{10}\rho$ and the actual logarithmic increment decreases at every step. Squares denote Meridian increments and circles phasor increments. The values are exact integers; only the plotted curves and logarithms are numerical.

Put $\eta = \max(R, R^{-1}) - 1$. If $R \geq 1$, then $|R - 1| = \eta$ and $|\overline{R} - 1| < (9/5)^{p-p'}$. If $R < 1$, their product is $\eta((9/5)^{p-p'} - 1/(1+\eta)) < \eta(9/5)^{p-p'}$. Either case forces $\eta > 9^{-(p-p')}$. At equality of the Markov values, $R = (1+\varepsilon_{p',q'})/(1+\varepsilon_{p,q})$, so η is precisely the left side of (36). □

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6.4. **The corridor collision window.**

Theorem 6.2 (Numerator–denominator window)**.** *Let $p \neq p'$, $P = \max(p, p')$, and let K be the least nonnegative integer satisfying*

$$\rho^{2K} > P(9^{|p-p'|} + 1). \tag{37}$$

Every equality $m_{p/q} = m_{p'/q'}$ requires

$$\max(q, q') \leq P(K + 1) - 1. \tag{38}$$

For bounded numerators a simpler, weaker consequence is $\max(q, q') \leq 2P^2 + P - 1$.

Proof. If both quotient floors are at least K, then $P\rho^{-2K} < 1/(9^{|p-p'|} + 1)$. The geometric-series estimate used in Lemma 5.3 makes both corrections strictly smaller than $9^{-|p-p'|}$, contradicting Theorem 6.1. Thus one denominator, say q, is at most $Kp - 1$.

Both matrices in (29) have operator norm ρ. The trace bound and the positive leading term imply

$$\frac{1}{3}\rho^{q'} < m_{p'/q'} = m_{p/q} \leq \frac{2}{3}\rho^{q+p}.$$

Since $\log_\rho 2 < 1$ and the exponents are integers, $q' \leq q + p$. It follows that both denominators are at most $P(K + 1) - 1$. For the simpler bound it suffices to use $K = 2P$: $\rho^{4P} > 36^P > P(9^{P-1} + 1)$ for $P \geq 2$, and $|p - p'| \leq P - 1$. □

For example, numerators at most 64 permit $K = 75$, so every remaining comparison lies at denominators at most 4863. With $P = 64$ and numerator difference five, $K = 8$ gives 575. These are bounds on candidates, not new numerical verification claims.

Theorem 5.1 first discards equal Meridian degrees. For the rest, the phase rectangle (27) and denominator cap (38) can be imposed together through the dictionary (18). Fixed-numerator ordering already separates all labels with $p = p'$. What remains is a comparison of different numerators and degrees inside their joint windows. Neither numerator nor degree is globally bounded by these theorems, so this is not yet a finite reduction of the entire unicity conjecture.

7. Snake-graph counts and ordering proofs

7.1. **From a Farey label to an actual matching count.** A snake graph is a chain of square tiles, each attached to the right of or above the preceding tile. Shared edges are counted once. A perfect matching covers every vertex exactly once. The Markov snake associated with a reduced $0 < p/q < 1$ is built as follows. Take the lower staircase from $(0, 0)$ to (q, p): at each horizontal coordinate j, step right and then up to height $\lfloor pj/q \rfloor$. Write right steps as a and up steps as b. The resulting lower Christoffel word has the form $w_{p/q} = azb$. Remove its first and last letters; start with one square and replace each a of z by two added squares to the right and each b by two upward. The resulting snake has $2(p + q) - 3$ tiles.

Write $\mu(w)$ for its number of perfect matchings. The established Markov matching correspondence gives

$$\mu(w_{p/q}) = m_{p/q}. \tag{39}$$

See [12, Section 4.1, Theorem 4.2] and [6]. For example, $w_{2/5} = a^3ba^2b$, its trimmed word is $aabaa$, and the eleven-tile snake has 194 matchings. Auxiliary rearranged words also specify snakes, counted by μ; they need not be canonical Markov snakes.

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7.2. **Six pieces and two connector states.** Figure 8 fixes the five three-tile pieces and the three one-square variants of type 6. In a cut between pieces, a matching uses both yellow connector edges or neither: the cut leaves an even number of vertices on either side. Denote these states by 1 and 0. At a selected port remove the two vertices already covered by its connector pair. Direct counting within the pieces gives

(40)

piece	00	01	10	11
1	5	2	2	1
2	5	3	3	2
3	5	2	3	1
4	5	3	2	1
5	5	2	2	1
6i	2	1	1	1

The state order is incoming, outgoing. The one-port end squares 6ii and 6iii have counts $(2, 1)$.

For the ordering surgeries, use a fixed one-ended chop: retain tile triples $(1, 2, 3), (5, 6, 7), \ldots$, with tiles $4, 8, \ldots$ providing paired connectors. A one-square remainder occurs exactly when $p + q$ is even and is counted with the closed end piece. For every connector sequence σ, multiply the applicable row entries in (40); then sum over σ, with outer ports closed. Each matching determines one sequence and one internal completion per piece, and gluing reverses the construction. This proves the counting rule, without an independence assumption. Equivalently it is a sum of terms $5^{k(\sigma)}3^{j(\sigma)}2^{\ell(\sigma)}$, where the exponents count factors of the indicated size. For the $2/5$ snake the piece list $2, 1, 2$ gives

$$5^3 + 5 \cdot 2 \cdot 3 + 3 \cdot 2 \cdot 5 + 3^2 = 194.$$

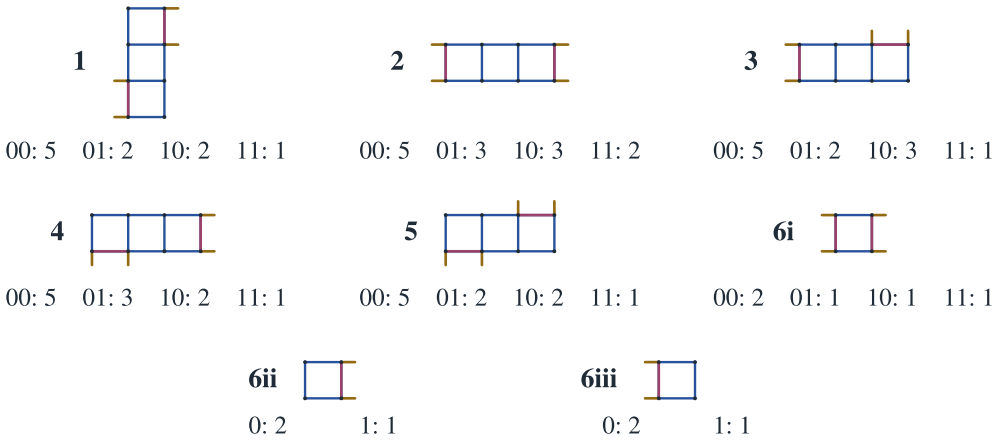


Figure 8. The six physical pieces. Gold edges are paired connectors, magenta marks ports, and dots mark tile vertices. Two-port counts use the order $00, 01, 10, 11$. The central square 6i has counts $(2, 1, 1, 1)$; the one-port end squares have $(2, 1)$. Types 1 and 5 have equal count rows but different physical orientations.

7.3. **Fibonacci run counts and normalization.** With $F_0 = 0, F_1 = 1$ and $F_{-j} = (-1)^{j+1}F_j$, put

(41)
$$C(t) = \begin{pmatrix} F_{2t+3} & F_{2t+1} \\ F_{2t+1} & F_{2t-1} \end{pmatrix}, \qquad t \geq 0.$$

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If $w = a^{h_1}b\cdots a^{h_p}b$, with $h_1 \geq 1$ and subsequent runs possibly zero, the boundary convention is

$$\mu(w) = [C(h_1 - 1)C(h_2)\cdots C(h_p)]_{11}. \tag{42}$$

The canonical run lengths are $h_j = \lceil jq/p\rceil - \lceil (j-1)q/p\rceil$. For $k = 1 + \lfloor (q-1)/p\rfloor$, the factors in (42) are the two Fibonacci matrices

$$\begin{aligned} A = C(k) &= \begin{pmatrix} F_{2k+3} & F_{2k+1} \\ F_{2k+1} & F_{2k-1} \end{pmatrix}, \\ B = C(k-1) &= \begin{pmatrix} F_{2k+1} & F_{2k-1} \\ F_{2k-1} & F_{2k-3} \end{pmatrix}. \end{aligned} \tag{43}$$

The end factors are B; for example $m_{3/8} = [BAB]_{11}$ with $k = 3$.

For completeness, the single-letter boundary matrices are the D_a, D_b of (29). With $P = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$ and $P^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, Fibonacci addition gives $P^{-1}D_a^t D_b P = C(t)$. Counting at the outermost corner shows that $\mu(w) = (D_w)_{12}$. After conjugation this is the lower-left entry of the product of $C(h_j)$; the lower row of $C(h_1)$ is the upper row of $C(h_1 - 1)$, proving (42).

An exact run-balancing identity is

$$C(r)C(s) - C(r-1)C(s+1) = 2\begin{pmatrix} F_{2(r-s)-2} & F_{2(r-s)} \\ F_{2(r-s)-4} & F_{2(r-s)-2} \end{pmatrix}. \tag{44}$$

It follows from Fibonacci addition, for $r \geq 1, s \geq 0$. When $r \geq s+2$ every entry is nonnegative. At $r = s+1$ the off-diagonal signs differ, so normalization must also account for the surrounding word.

Lemma 7.1 (Normalization)**.** *For coprime positive p, q, every word $w = azb$ with q letters a and p letters b satisfies $\mu(w_{p/q}) \leq \mu(w)$.*

Proof. At an interchange $uabv \leftrightarrow ubav$, put $(x, y) = (1, 0)D_u$ and $(s, t) = (0, 1)D_{v^{\mathrm{R}}}$. The letter matrices are symmetric, so direct multiplication gives

$$\mu(uabv) - \mu(ubav) = 2(ys - xt). \tag{45}$$

Compare u^{R} and v from the interchange outward. A common letter cancels from this determinant because its matrix has determinant one. At first unequal letters a, b, one has $y \geq x$, $s \geq t$, so the difference is nonnegative. At b, a, one has $x > y$, $t > s$, so it is negative. Exhaustion of either word leaves $(1, 0)$ or $(0, 1)$ and the same strict negative sign. This supplies the missing exterior test.

Here is a terminating choice of interchanges, the extremal-path normalization of LPTV [9, Section 3] in these boundary counts. Let $N = p + q$ and measure staircase height by $\Delta(x, y) = qy - px$. If there is a positive peak, lower the rightmost highest one by $ba \to ab$. An obstructing exterior order would be a, b; after canceling its common outward word, the two exterior heights sum to twice that peak. Neither can exceed the peak, so the right one would be another highest point farther right, a contradiction. The count therefore strictly decreases. The inversion count decreases too, so the process reaches $\Delta \leq 0$.

If the minimum h is below $-N$, raise the leftmost lowest trough by $ab \to ba$. An obstructing order b, a would similarly force an earlier minimum. If a comparison word is exhausted, the exterior heights sum to $2h + N < h$; one is the endpoint zero, forcing the other below the minimum. Hence this move does not increase the count. Its new height $h + N$ is still negative, and the bounded

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inversion count increases, so repetition ends with all heights in $[-N, 0]$. The endpoint letters stay unchanged: boundary troughs have height $-p$ or $-q$ and cannot be selected.

At step j the height is congruent to $-pj \pmod N$. Coprimality gives a unique representative in $[-N, 0]$ at each interior step. These are exactly the lower Christoffel staircase. Thus the terminating path is $w_{p/q}$, and no operation has increased its matching count. □

7.4. **The three Aigner orderings.** Fixed-numerator monotonicity was proved by Rabideau and Schiffler [12]; all three orderings were proved by LPTV, Lee–Li–Rabideau–Schiffler, and McShane [9, 10, 11]. The proofs below use the author's physical pieces for the two surgeries and Fibonacci runs for insertion.

Theorem 7.2 (Established monotonicities)**.** *For $i > 0$, whenever both fractions are reduced and lie in $(0, 1)$,*

$$\begin{aligned} m_{p/q} &> m_{(p+i)/(q-i)} && \textit{(fixed sum)},\\ m_{p/q} &< m_{p/(q+i)} && \textit{(fixed numerator)},\\ m_{p/q} &< m_{(p+i)/q} && \textit{(fixed denominator)}. \end{aligned}$$

Fixed sum. At the same physical cuts, the changes $2 \to 3$ or 4, then 3 or $4 \to 5$, increase no entry of (40); the changed factors are $3 \to 2$ or $2 \to 1$. Rotation $5 \leftrightarrow 1$ preserves the row. Selecting the changed connector and no others exhibits a strictly smaller positive summand whenever an equal-direction join becomes unequal.

The trimmed canonical word begins and ends with a, contains no bb, and has $q - p - 1$ equal joins, all aa. At its rightmost equal join, interchange a, b strictly to the right. That suffix is alternating, so only the chosen join changes. Exactly one prescribed piece changes by the stated rules, while any rotated pieces keep their rows. Each surgery strictly decreases the count. Odd-numbered operations replace one horizontal step by a vertical step in net; even-numbered operations keep the totals. After $2i$ operations the endpoint is $(q - i, p + i)$. There are enough joins because $p + i < q - i$. For the resulting auxiliary word w_*, normalization gives $m_{(p+i)/(q-i)} \le \mu(w_*) < m_{p/q}$. This is the piece-count realization of the tail surgery in [9, Proposition 22]. □

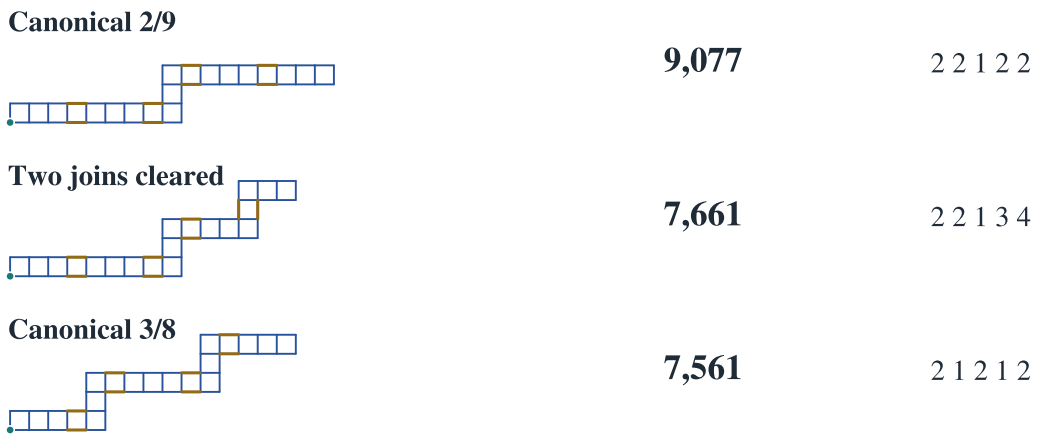


Figure 9. Fixed-sum surgery on the same five pieces. The canonical 2/9 snake has 9077 matchings; after two changes the auxiliary snake has 7661; normalization gives the 3/8 snake with 7561. The intermediate first change, not shown, has 8263. Gold pairs mark cuts and teal dots the starting corners.

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Fixed numerator. Fibonacci recurrence gives, for $t \geq 1$,

$$C(t) - C(t-1) = \begin{pmatrix} F_{2t+2} & F_{2t} \\ F_{2t} & F_{2t-2} \end{pmatrix}.$$

All entries are nonnegative and the upper-left is positive. Start with $w_{p/(q+i)}$ and delete i internal letters a, retaining the first. Each deletion changes one factor $C(t)$ to $C(t-1)$ with $t \geq 1$. Surrounding factors are positive, so the matching count strictly falls. The resulting word ends at (q, p); normalization gives $m_{p/q} \leq \mu(w_*) < m_{p/(q+i)}$. □

Fixed denominator. Refine the cuts to retain individual odd-numbered tiles and use the even tiles as paired connectors. A retained square has counts $(2, 1, 1, 1)$ for opposite ports and $(2, 1, 1, 0)$ for adjacent ports. The zero forbids covering their common corner twice. Summing an intervening state regroups these squares into the original three-tile pieces.

Insert one internal b, adding two vertical squares. The affected local counts are the following complete table, with outside ports fixed:

	before				after			
directions	00	01	10	11	00	01	10	11
a, a	2	1	1	1	5	2	2	1
b, b	2	1	1	1	5	3	3	2
a, b	2	1	1	0	5	3	2	1
b, a	2	1	1	0	5	2	3	1

.

The endpoint caps have the corresponding rows with an outer port closed; the empty trimmed word has the increase $2 \to 5$. No outside count falls, and the all-zero choice witnesses a strict increase. Thus the two added squares compensate for every changed direction.

Starting from $w_{(p+i)/q}$, delete i internal b's and reverse this comparison. At least i are available. Normalizing the resulting word to (q, p) proves $m_{p/q} \leq \mu(w_*) < m_{(p+i)/q}$. □

7.5. **Folding an actual snake at its center.** The same local counts also compare families sharing a long initial sequence of pieces. Represent the four counts by

$$R_1 = R_5 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}, \quad R_4 = \begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}.$$

Attaching piece i to an open-count row (x, y) replaces it by $(x, y)R_i$. This is the sum over the common connector state, not an additional assumption about matchings.

For a helm, use the symmetric central chop: retain the middle square as 6i and group retained squares in pairs outward on each side. For odd/odd p, q, each half has $(p+q-2)/2$ retained squares. If this number is odd, one square remains at each outer end, giving pieces 6ii and 6iii with initial counts $(2, 1)$. Otherwise the half starts at $(1, 0)$ and consists of complete three-tile pieces. This repartition requires rebuilding the piece list; one cannot move a cut and keep its old boundary counts.

Proposition 7.3 (Folded matching counts)**.** *For a reduced $0 < p < q$, let (x, y) be the open counts of one reflected half under the central chop. If p, q are odd, then*

$$m_{p/q} = 2x^2 + 2xy + y^2 = (x+y)^2 + x^2. \tag{46}$$

If $p + q$ is odd, then $m_{p/q} = x^2 + y^2$. In either case $\gcd(x, y) = 1$; a nonempty half has positive counts.

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Proof. The trimmed Christoffel word is a palindrome: for $0 < j < q$, $\lfloor p(q-j)/q \rfloor = p - 1 - \lfloor pj/q \rfloor$. For odd/odd labels the two halves meet at an opposite-port central square, whose four counts are $2, 1, 1, 1$. The reflected halves contribute x^2, xy, yx, y^2, giving (46). For opposite parities they meet across a single connector and must use its same state, giving $x^2 + y^2$. Every full piece table and every exposed-square table has determinant ± 1, so its integral inverse preserves the gcd of an open row. Starting at $(1, 0)$ proves primitivity. The first square gives $(2, 1)$, and every subsequent permitted table preserves positivity. □

The seed $1/1$ consists only of its central square. The two examples in Figure 10 give $m_{1/3} = 2 \cdot 2^2 + 2 \cdot 2 \cdot 1 + 1^2 = 13$ and $m_{1/5} = 2 \cdot 5^2 + 2 \cdot 5 \cdot 3 + 3^2 = 89$.

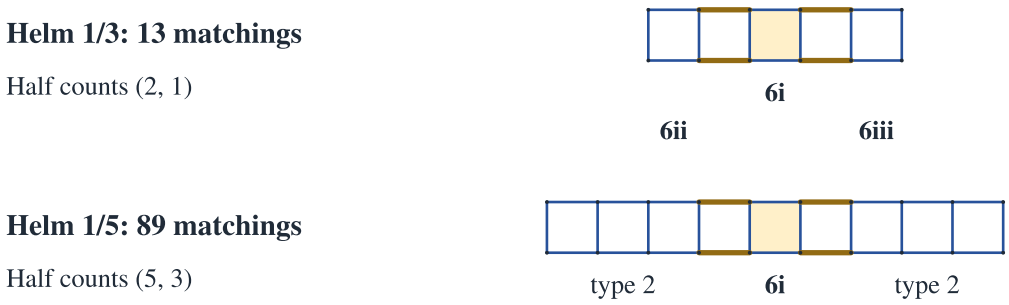


Figure 10. Central and end pieces in actual helms. At $1/3$, central square 6i has the end squares 6ii and 6iii as its halves, with counts $(2, 1)$. At $1/5$, each half is a type-2 triple with counts $(5, 3)$. Folding produces 13 and 89.

7.6. **A common prefix leaves one quadratic comparison.** Two folded graphs with the same outer boundary convention may share a prefix with open counts (x, y). Sum over all remaining states, leaving that cut open. The two centers give positive definite symmetric tables $G, \widetilde{G}$ with determinant one and nonnegative off-diagonal entries. For a helm they have the form SQS^{T} with $Q = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$; for opposite-parity labels they have form SS^{T}. Thus

$$m - \widetilde{m} = \alpha x^2 + 2\beta xy + \gamma y^2, \qquad G - \widetilde{G} = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}. \tag{47}$$

Proposition 7.4 (At most one primitive common prefix)**.** *For a nonempty common part and $G \neq \widetilde{G}$, equality requires $\alpha\gamma < 0$ and $\beta^2 - \alpha\gamma$ to be an integer square. There is then at most one primitive positive row (x, y) giving equality. That row must still arise from both actual prescribed snakes.*

Proof. A center table $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ has $ac - b^2 = 1$ and $b \geq 0$. If its two diagonal differences from another table have the same weak sign, then the difference of $b = \sqrt{ac - 1}$ has that sign too. A nonzero difference cannot vanish at a positive row. Opposite diagonal signs are therefore necessary. The resulting quadratic in y/x has one positive root. An integer zero obeys $(\gamma y + \beta x)^2 = (\beta^2 - \alpha\gamma)x^2$; a

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rational square root of an integer is integral. A positive rational root has only one primitive positive representative. □

For an empty prefix, test its actual boundary row. If $G = \widetilde{G}$, the criterion supplies no exclusion. Equal count tables do not certify equal physical graphs: types 1 and 5 already show why orientations and the prescribed Christoffel word must be retained.

A useful uniform exclusion follows directly modulo three. The physical piece tables satisfy $R_i Q R_i^{\mathsf{T}} \equiv Q \pmod 3$, whereas a single retained square gives $-Q$. A helm tail with d retained squares therefore has center table $(-1)^d Q \pmod 3$. Opposite parities give a nonzero difference at every primitive row because the discriminant of Q is -1, a nonsquare modulo three. For equal parity and nonzero difference, let $h = \min_{i,j} v_3((G - \widetilde{G})_{ij})$ and write $3^{-h}(G - \widetilde{G}) = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$. Then $h \geq 1$. Expanding the equal determinants and reducing modulo three gives $\gamma \equiv \alpha + \beta$; the normalized discriminant is one or two modulo three. If it is two, every primitive shared row has count difference of valuation exactly h. These are exclusions for specified tails with arbitrary actual common prefixes, not a classification of all possible snake-graph collisions.

8. All-phase verification through degree 10,000

8.1. **Why a finite residue test covers infinitely many phases.** For a period T, define the complete image

$$\mathcal{R}_\alpha(T) = \{M_\alpha(x) \bmod q(T) : 0 \leq x < T\}.$$

Theorem 4.1 makes this the image at *every* integer phase. Disjoint images therefore separate two entire Meridians. If images overlap for several periods, a true equality must use compatible phases in all of them. Define the witness relation

$$W_T(\alpha, \beta) = \{(u, v) \in (\mathbb{Z}/T\mathbb{Z})^2 : M_\alpha(u) \equiv M_\beta(v) \pmod{q(T)}\}.$$

Proposition 8.1 (Compatible witnesses)**.** *Equality of two integer values requires one witness in each selected W_T, with first coordinates agreeing modulo every pairwise gcd of the periods, and second coordinates agreeing in the same way. If no compatible choice exists, the two Meridians are disjoint at all integer phases.*

Proof. A true equality uses the same integers x, y in every modulus and hence supplies the required residues. Conversely, the generalized Chinese remainder theorem lifts compatible coordinates to a phase pair modulo the least common multiple. This converse concerns modular witnesses only; it does not imply equality of the full values. □

The moduli need not be independent. In particular, the periods $36, 38, 40$ are all even, so contradictory parity requirements alone can exclude a pair that survives each separate image test.

8.2. **Complete inventory and the final exclusion.**

Proof of Theorem 1.1. Proposition 3.4 identifies the inventory with coprime positive odd pairs $r \leq s$, $r + s \leq 10{,}000$. It contains 5,067,077 Meridians and 12,837,632,128,426 unordered pairs. The producer enumerates the count-vector address tree; an independent checker enumerates the coprime pairs directly and compares the inventories. The child degree is $2|k|D \pm (|v| - |u|) > D$, where $|u|, |v|$ are the endpoint word lengths. This gives finite exhaustive index ranges.

Every one of the 36 phase residues of every Meridian is evaluated modulo $q(36)$. Equal-residue grouping lists all pairs whose images intersect; its complement is excluded at once. Periods 38 and 40 are then evaluated on every endpoint of the preceding survivor sets. The complete image tests leave 87 pairs. For each of them, project every full witness relation onto the four parity pairs

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$(u \bmod 2, v \bmod 2)$. The three projected sets have empty intersection in all 87 cases. Proposition 8.1 therefore excludes every pair of different Meridians at every integer phase. Theorem 5.1 also excludes different marks within each individual Meridian, completing the statement. □

Table 2. Complete cumulative survivor counts from the archived certificates. "Images" forgets meeting phases; "phases" also imposes compatibility. Survivors are modular coincidences, not integer collisions.

Tests retained	$D \le 2{,}000$	$D \le 10{,}000$
36: images	7,121	320,755
36, 38: images	612	17,427
36, 38: phases	2	2,052
36, 38, 40: images	0	87
36, 38, 40: phases	**0**	**0**
36, 38, 40, 42: images	—	1
36, 38, 40, 42, 44: images	—	0

8.3. **What was checked independently.** The degree-10,000 producer uses signed endpoint-orbit updates. The separate evaluator uses Fibonacci staircase words with Euclidean matrix reduction. They agree on all 188,369,412 retained phase evaluations. The checker verifies the 182,414,772 baseline phase tags bijectively, so an omitted or repeated phase record cannot pass unnoticed. Independent sorting of the original unsorted buckets reconstructs all 70,009 cross-Meridian residue groups and the subsequent complete pair sets. Coverage is checked by exact records, not inferred merely from their counts or from a matching digest.

The phase result is checked three ways: full generalized-CRT joins, an independent 16-bit mask computation on $(x \bmod 4, y \bmod 4)$, and the weaker four-state parity projection, already sufficient for the final 87. The degree-2,000 certificate, containing 202,862 Meridians, was completed first; its common survivor sets agree exactly with the larger inventory when identified by helm labels.

Five deliberate corruptions—an altered value, an incorrect modulus, a duplicate tag, an omitted record and an invented pair—were all rejected. The degree-100 control compares 18,684 residues with literal Fibonacci words and the polynomial record of (19). The optional image-only extension through periods 42 and 44 also ends with zero pairs; all 88 period-44 values were separately checked using arbitrary-size Python integers because $q(44) > 2^{63}$.

8.4. **Reproducibility and scope.** The degree reproducibility package [7] contains the inventory, compact residue groups, phase witnesses, producer and checker sources, audit receipts, and a regeneration driver. Large baseline tables are regenerated; their recorded sizes and hashes identify the original run. The reindexed edition retains the original computational kernels with an explicit address-convention adapter. Reversing the written address tail does not change a helm label, degree, phase, or certified numerical value.

This is a deterministic computer-assisted certificate, not an external referee report or proof-assistant formalization. Its scope is a bounded set of Meridians with unbounded phase. The numerical-order verification in Section 9 is a separate computation with its own coverage proof and execution record.

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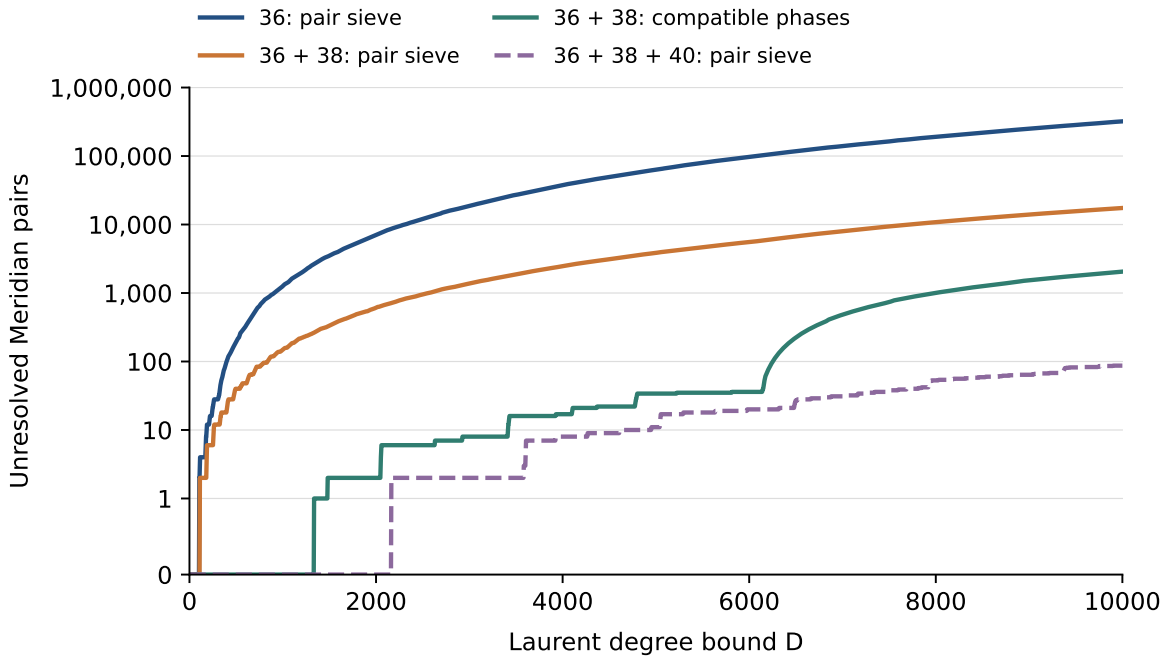


Figure 11. Exact unresolved-pair profiles in the degree-10,000 inventory. The phase-compatible sieve through period 40 has no survivors in the entire displayed range. Lines join discrete even-degree bounds. The figure is a record of the finite certificate, with no extrapolation to larger degrees.

9. Computational chronology and numerical-order verification

9.1. **The earlier Claude-assisted search.** The historical working draft [1], pp. 38–40, reports a Claude-assisted search of **9,581,528,580 sorted Markov triples** against the cutoff $10^{100,000}$, taking **24 minutes** on a two-core machine with 7 GB of memory. It compared maxima modulo 3^{39} in 16 passes and 4,096 buckets; all 21 repeated-residue pairs separated modulo $2^{61} - 1$. No equal maxima were reported.

An earlier public announcement of a 24-minute search is timestamped **23 October 2026, 09:54:57.771 CDT** (14:54:57.771 UTC) [2]. Its caption says “9.3 billion”; the later draft gives the exact count above. This dates the *announcement*, not the beginning or end of execution. Original execution timestamps have not been recovered. The draft itself labels these as historical computational reports whose original logs and floating-point cutoff bounds had not been independently certified. We preserve that status here. This search is separate from the later ten-billion anchor, and its reported 24 minutes are not included in the cumulative timing below.

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9.2. **The completed fifty-billion verification.** The archived computation [3] establishes that each of the **first 50,000,000,000 distinct Markov values in numerical order** is the maximum of exactly one normalized positive Markov triple. This complements the degree-10,000 result: the latter covers infinitely many phases on a bounded set of Meridians, whereas this computation certifies a numerical prefix. Its coverage and pruning proofs are collected in the separate verification supplement [4].

Run date and duration. The recorded execution began on **28 October 2026 at 03:32:57 CDT** and finished, including its final audit, at **05:38:34 CDT** (UTC−05:00). The underlying timestamps are 08:32:57.096455–10:38:33.644612 UTC. Two completed coordinator invocations occupied **7,492.102807 seconds: 2 h 04 min 52.103 s**. Including the original ten-billion anchor, the cumulative recorded execution path is **8,330.980040 seconds: 2 h 18 min 50.980 s**. The calendar span is 7,536.548157 seconds; it additionally includes the approximately 44-second pause between invocations. These are historical execution dates and measured durations, distinct from this edition's publication date.

Table 3. The completed fifty-billion run. Counts are configurations unless explicitly identified as distinct values.

Certified distinct-value prefix	50,000,000,000
Complete outer domain	71,517,775,875
New retained configurations compared	54,940,210,994
Repeated primary residues / candidate configurations	346 / 692
Actual equal-value collisions	0

The numerical cutoff was $B = F_{1,147,147}$, with 796,397 bits. The inner height $K = 573{,}574$ contains 50,000,097,577 configurations at or below B; the proved outer height $L = 685{,}979$ contains every competitor below that cutoff. The original ten-billion cutoff certificate justifies 10,000,015,326 inner omissions. Strict certified lower bounds exclude a further 6,577,549,555 configurations above B.

All 32 primary prefix passes and all 8,192 complete buckets finished. Every repeated residue modulo 2^{64} was replayed; all 346 candidate pairs separated modulo 5^{27}. The final auditor independently reconstructed their 692 full integer triples and checked the Markov equation, residues and unequal maxima, as well as the 38 anchor candidates. No probabilistic hash-collision assumption enters this comparison.

The run used eight workers on Linux x86-64, an eight-CPU-equivalent quota and an 8-GiB memory limit, with frozen GCC 13.3.0 executables. The timing includes generation, partitioning, sorting, replay, orchestration and the 107.371720-second final audit; earlier preflight and later archival checks are excluded. The supplement gives the stage timings, retained receipts, source identities, audit scope and reproducibility limitations. This edition reports the completed run; it does not announce a result for the planned 500-billion verification.

10. What remains to prove

The organization now separates three questions. The address bijection settles coverage of configurations. Fixed-degree ordering and the two collision windows restrict equality of maxima. Complete residue witnesses settle the remaining comparisons for the degree-bounded family in Theorem 1.1. The fifty-billion computation additionally excludes every competitor for a finite prefix in numerical order.

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For unrestricted unicity the unresolved pair has different Meridian degrees, the same modulo-27 class, and different Farey numerators. It must satisfy both its phase rectangle and its numerator–denominator window. A proof must exclude those candidates uniformly as the degrees and numerators grow. Enlarging a finite certificate alone does not supply that step. Likewise, an invertible corridor recurrence or a long modular period does not by itself prevent an earlier match of one scalar residue.

Corridors offer a way to formulate the next comparison with explicit entrance states and independently chosen wall lengths. Their positive corrections are exact and telescope; the common-prefix snake calculation offers a different reduction to a quadratic in primitive counts. Neither mechanism has yet been shown to cover every unrestricted pair. The purpose of retaining their precise hypotheses is to make the remaining uniform statement identifiable, rather than hide it inside an induction.

Data and code. The editable source accompanying this edition includes all figure assets, the revised Figure 3 generator, a source and figure inventory, and checks used for this rewrite. The separate degree reproducibility package contains the full computational evidence cited in Section 8. The fifty-billion proof and timing supplement accompanies this edition; its full frozen computational archive is cited separately [3, 4]. The corridor derivations are developed further in the supplementary research record [8]. Numerical drawings are explanatory; the arithmetic claims use the displayed identities or complete certificates.

Author contributions and AI assistance. The author conceived and directed the research program, including the hyperbolic representation and geometric descent, the two-dimensional host–phase landscape, the Meridian indexing and alternative tree, the six-piece snake-graph decomposition, and the corridor observation. The author proposed using these structures for coverage, comparisons and verification, and specified the questions pursued at each stage. The Aigner monotonicities are established results; their prior proofs are credited in Section 7.

Generative AI provided substantial mathematical and computational assistance, not only language editing. The development record identifies ChatGPT (OpenAI), with the recorded GPT-6.0 Astra and GPT-6.1 Sol model labels and reasoning settings, and Claude 5.5 Opus Extra (Anthropic) in the preceding geometric-descent work; exact backend revision identifiers were not retained. Within the author's framework, these tools assisted with detailed derivations, examples and counterexamples, proof construction and revision, verifier and independent-checker code, figures, and manuscript drafting. ChatGPT also assisted with this reorganization and Figure 3 revision. The project's separate computational checks are algorithmic audits, not external peer review. The author is responsible for the claims, references and final submission.

Funding and competing interests. No funding was received. The author declares no competing interests.

References

[1] E. F. W. Heffern, *Geometric Descent on the Markov Surface*, working draft v32, archived October 2026, pp. 38–40, historical computational reports. The earlier search was assisted by Claude.

[2] E. F. W. Heffern, public LinkedIn announcement of a 24-minute Markov-number search, 23 October 2026, 14:54:57.771 UTC, activity 7519411048115003392. Platform metadata and the source URL are retained in the verification supplement's evidence packet; see the public announcement.

[3] E. F. W. Heffern, *Markov 50 Billion Verification Package*, version 1, completed and sealed 28 October 2026; public derivative version 1.1 in *Computational supplements for Meridians on the Markov Surface*, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

[4] E. F. W. Heffern, *Computational chronology and proofs for the fifty-billion verification*, supplement to the present manuscript, version 4, 7 November 2026.

[5] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013. doi:10.1007/978-3-319-00888-2.

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Original: v4_ps_Markov_Meridians_TLMS.pdf

[6] İ. Çanakçı and R. Schiffler, *Snake graphs and continued fractions*, European J. Combin. **86** (2020), 103081. doi:10.1016/j.ejc.2020.103081.
[7] E. F. W. Heffern, *Meridian Degree 10,000 Reproducibility Package*, computational supplement, 31 October 2026; address-convention revision, 7 November 2026. Original computational release in *Computational supplements for Meridians on the Markov Surface*, version 1.0, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.
[8] E. F. W. Heffern, *Markov Meridian Residue Supplement*, version 1.18, and *Corridor-to-Corridor Induction*, version 1.3, unpublished supplementary research records, 7 November 2026. The corridor results used here are proved in Section 6.
[9] C. Lagisquet, E. Pelantová, S. Tavenas and L. Vuillon, *On the Markov numbers: fixed numerator, denominator, and sum conjectures*, Adv. Appl. Math. **130** (2021), 102227. doi:10.1016/j.aam.2021.102227. Preprint: arXiv:2010.10335.
[10] K. Lee, L. Li, M. Rabideau and R. Schiffler, *On the ordering of the Markov numbers*, Adv. Appl. Math. **143** (2023), 102453. doi:10.1016/j.aam.2022.102453.
[11] G. McShane, *Convexity and Aigner's conjectures*, preprint (2021), arXiv:2101.03316.
[12] M. Rabideau and R. Schiffler, *Continued fractions and orderings on the Markov numbers*, Adv. Math. **370** (2020), 107231. doi:10.1016/j.aim.2020.107231.

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COMPUTATIONAL CHRONOLOGY AND PROOFS FOR THE FIFTY-BILLION VERIFICATION

ELLEARD FELIX WEBSTER HEFFERN

Abstract. This supplement records the earlier search assisted by Claude and the separately archived fifty-billion verification for *Meridians on the Markov Surface*. It distinguishes announcement timestamps from execution receipts, proves the finite numerical envelope and safe omissions, and states the deterministic comparison and timing conventions. The fifty-billion run completed on 28 October 2026; this edition does not constitute a new enumeration or an unrestricted unicity proof.

1. Chronology and the earlier Claude-assisted search

The archived v32 working draft [2], pp. 38–40, reports 9,581,528,580 sorted triples in a search against the bound $10^{100,000}$, taking 24 minutes on a two-core machine with 7 GB of memory. The author identifies this earlier computation as Claude-assisted. Its reported implementation walked the outward Vieta tree, carried residues modulo 3^{39} in Montgomery form, and used 16 passes and 4,096 buckets. Exactly 21 primary residues occurred twice; all 21 pairs separated modulo $2^{61} - 1$. The report records no equal maxima.

The source states that double-precision logarithms controlled the stopping rule, with a relative allowance of 10^{-9} on $10^5 \log 10$. It explicitly classifies this account as a historical computational report whose original logs and cutoff error bounds had not been independently certified in that revision. We preserve that qualification. The archived count is not silently promoted to an exact numerical-prefix certificate.

The public LinkedIn post [3] has platform `datePublished` equal to **2026-10-23T14:54:57.771Z**, or **23 October 2026, 09:54:57.771 CDT**. Its caption describes a 24-minute search as "9.3 billion"; the later working draft gives 9,581,528,580. These are the respective source wordings, not equal numerical counts or a rounding identity. No original run receipt reconciling them was recovered for this edition. The timestamp dates the public announcement; it does not recover an execution start or finish. A current public snapshot also does not establish the absence of later edits to a post.

The source packet retains the platform metadata, the archived page snapshot, the relevant three pages of v32, and a provenance record with SHA-256 identities. No execution timestamp or more precise duration is invented. This historical search is distinct from the original ten-billion cutoff certificate reused in the fifty-billion computation. Its 24-minute reported duration is not added to the latter's cumulative time.

2. Exact numerical coverage

We use the main paper's standard mediant label p/q, with $0 < p \leq q$, $\gcd(p, q) = 1$, height $h = p + q$, and maximum $M(p/q)$. The boundary $0/1$ represents the root of value 1, and $1/1$ has value 2. These labels are configuration coordinates. Their bijection with normalized triples, and the reversed Meridian address dictionary, do not assume that maxima are distinct.

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Lemma 2.1 (Configuration count)**.** *Including the root, for $L \geq 2$ the number of configurations of height at most L is*

$$C(L) = 1 + \frac{1}{2}\sum_{h=1}^{L} \varphi(h) = \frac{5 + \sum_{d=1}^{L} \mu(d)\lfloor L/d \rfloor^2}{4}.$$

Here φ is Euler's totient and μ is the Möbius function.

Proof. For $h \geq 3$, the labels are $(p, h-p)$ with $0 < p < h/2$ and $\gcd(p, h) = 1$, giving $\varphi(h)/2$ labels. Add the root and seed. The Möbius sum counts ordered coprime pairs in $[1, L]^2$, whose number is also $2\sum_{h=1}^{L} \varphi(h) - 1$. □

Lemma 2.2 (Uniform magnitude bounds)**.** *For every nonroot configuration of height $h \geq 2$,*

$$2^{h-1} \leq M(p/q) \leq F_{2h-1}, \qquad M(p/q)^2 \geq 4 \cdot 5^{h-2}.$$

The Fibonacci upper bound is attained for $p = 1$ and is strict for $p \geq 2$.

Proof. The outward mediant of parents with values a, b has value

$$c = \frac{3ab + \sqrt{9a^2b^2 - 4a^2 - 4b^2}}{2}.$$

The inequality $2a^2b^2 \geq a^2 + b^2$ gives $c \geq 2ab$. Since heights add at a mediant, induction from the two boundary values gives the first lower bound. For the outward nodes above those endpoints, $\max(a, b) \geq 2$ and $5a^2b^2 \geq 4a^2 + 4b^2$, so $c \geq (5/2)ab$. The endpoint values satisfy $M \geq (2/5)(\sqrt{5})^h$; the last inequality propagates this bound. Squaring yields $4 \cdot 5^{h-2}$.

The boundary recurrence gives $M(1/q) = F_{2q+1}$. For $p \geq 2$ neither parent is $0/1$, and both parent heights h_1, h_2 are at least two. Put $m = 2h_1 - 1, n = 2h_2 - 1$. The Fibonacci addition formula and $F_{j+1}/F_j \geq 3/2$ for odd $j \geq 3$ give

$$F_{m+n+1} = F_{m+1}F_{n+1} + F_mF_n \geq \frac{13}{4}F_mF_n > 3F_mF_n.$$

The opposite value is positive, so the Vieta update has $c < 3ab$. Induction proves the strict upper bound. □

Theorem 2.3 (Numerical-order envelope)**.** *Choose $K \geq 2$ with $C(K) \geq N$ and define*

$$B = F_{2K-1}, \qquad L = \max\{h \geq 2 : 4 \cdot 5^{h-2} \leq B^2\}.$$

Every configuration with maximum at most B has height at most L, apart from the separately included root. If maximum unicity is established for all configurations with maximum at most B, it holds for the first N distinct values in numerical order.

Proof. The lower bound in Lemma 2.2 contains every competitor in the outer domain. Its upper bound places all $C(K)$ inner configurations at or below B. Once injectivity has been established, these provide at least N distinct values. Distinctness is not assumed in counting the domain or choosing K. □

For the completed run the exact parameters are

$$\begin{aligned} N &= 50{,}000{,}000{,}000, & K &= 573{,}574, \\ C(K-1) &= 49{,}999{,}964{,}521, & C(K) &= 50{,}000{,}097{,}577, \\ B &= F_{1{,}147{,}147}, & L &= 685{,}979, \\ C(L) &= 71{,}517{,}775{,}875. \end{aligned}$$

The cutoff has 796,397 bits. Integer arithmetic verifies $4 \cdot 5^{685{,}977} \leq B^2 < 4 \cdot 5^{685{,}978}$; thus no floating-point rounding selects the outer boundary. Totient and Möbius evaluations provide independent exact count formulas.

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3. Safe omissions and conservative pruning

The reusable anchor [4] is the later original ten-billion cutoff certificate, with $K_0 = 256{,}510$, $B_0 = F_{513{,}019}$ and $C(K_0) = 10{,}000{,}015{,}326$. It covers every competitor for a value at most B_0, including configurations outside its inner height range.

Lemma 3.1 (Anchored omission)**.** *The extension may omit emissions of height $h \le K_0$ while continuing to traverse their descendants.*

Proof. Every omitted value is at most B_0. Equality at or below B_0 has already been excluded by the anchor, including comparisons with outside competitors. Equality above B_0 cannot involve an omitted configuration, so both members remain eligible for the new comparison. □

This permits the specified inner omissions only; it does not permit skipping the old outer region, entire Meridians, or descendants of an omitted node. No twenty-billion intermediate extension is used as an anchor.

The native scheduler traverses the complete mediant coordinate domain, which the main paper's dictionary assigns uniquely to Meridian addresses and phases. Disjoint frontier intervals are assigned once; a worker either visits a node and its two child intervals or passes an entire remaining interval to another job. Height increases strictly down a branch, so a branch can stop when its height exceeds L.

In addition to Lemma 2.2, positivity of the main paper's Meridian Laurent expansion gives a lower bound on $\log M$:

$$E \log \rho + \frac{D}{2} \log(9/5) - \log 3, \qquad \rho = \frac{3+\sqrt{5}}{2},$$

coordinate parity	D	E
p, q odd	$p+q$	$(p-q)/2$
q even	q	$p+q/2$
p even	p	$q+p/2$

This is the leading positive monomial after substitution of the inverse dictionary. The negative coefficient in the first row is intentional; the claim is a lower bound, not an approximation used as an equality.

Constants are enclosed by integer square-root brackets and rational logarithm series. For $z = (x-1)/(x+1)$ and $|z| < 1$,

$$\log x = 2 \sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1} + R_J, \qquad |R_J| \le \frac{2|z|^{2J+1}}{(2J+1)(1-z^2)}.$$

The bound follows by bounding the remaining denominators below by $2J+1$ and summing a geometric series in z^2. The implementation uses dyadic endpoints scaled by 2^{64}, with 32 guard bits; a separate rational implementation checks the enclosures. A negative coefficient uses the upper constant endpoint to form a lower bound, as does a subtracted constant. The maximum of the valid lower endpoints remains a lower endpoint.

A configuration is excluded only if its lower endpoint is strictly greater than an upper endpoint for $\log B$. All overlapping intervals and every new inner configuration are retained. Generation and replay apply the same predicate. The exact accounting is

$$71{,}517{,}775{,}875 = 10{,}000{,}015{,}326 + 6{,}577{,}549{,}555 + 54{,}940{,}210{,}994,$$

respectively anchored omissions, certified exclusions above B, and new retained configurations. No relative word-length restriction, near-helm restriction, optional cone estimate, or cofactor factorization is used.

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Coordinates fit in 32 bits: trial endpoint-height sums are at most $2L < 2^{21}$. Counts fit in 64 bits; the repeated visits before replay are below 2^{42}. Fixed-point intermediates have absolute value below 2^{90} and fit in signed 128-bit arithmetic. Unsigned residue multiplication deliberately takes place modulo 2^{64}.

4. Deterministic comparison and its result

Each retained configuration emits its exact residue $M \bmod 2^{64}$. The high five bits select one of 32 passes and the next eight bits one of 256 buckets. Each of the 8,192 complete buckets merges all worker shards and sorts the full retained key. No further shortening of the key is used.

Proposition 4.1 (Complete-bucket exclusion)**.** *If every complete bucket is processed and every repeated key is resolved, every possible equality among retained maxima is detected.*

Proof. Equal integers have the same exact residue, prefix and bucket. Sorting the complete bucket identifies every residue of multiplicity at least two. A singleton therefore cannot share its integer value with another retained configuration. Preserve repeated keys and multiplicities; replay recovers all contributing coordinates. Different residues modulo 5^{27} prove inequality within a primary group. A surviving signature must be resolved by further certified comparison or full integers; an unresolved group cannot yield a successful certificate. □

The completed run has 346 repeated primary keys, each of multiplicity two, giving 692 candidate configurations. Replay recovered all candidates and multiplicities. Every pair separated modulo 5^{27}, leaving zero repeated secondary groups and zero equal-value collisions. The production path therefore required no full-integer candidate reconstructions. Separately, the final auditor reconstructed all 692 complete triples, checked the Markov equation and both residues, and confirmed unequal maxima. It also rechecked the 38 anchor candidates.

Theorem 4.2 (Archived finite verification)**.** *The completed execution and audit recorded in* [4]*, with the coverage and omission rules above, establish that every one of the first 50,000,000,000 distinct Markov values in numerical order belongs to exactly one normalized positive Markov triple.*

Proof. The canonical traversal covers the height envelope, hence all competitors with maximum at most B. The anchor excludes equality at or below B_0. A larger equal pair at or below B would have both configurations retained, because neither anchor omission nor strict cutoff pruning could remove it. Proposition 4.1 and the completed residue comparisons exclude that pair. Thus all values at or below B are unique. Theorem 2.3 and $C(K) \geq 50{,}000{,}000{,}000$ give the claim. The execution of the finite comparison is the computational part of this argument; the archive supplies its program and recorded evidence. □

The full outer domain is traversed once per prefix: there were 2,288,568,828,000 configuration visits before replay. This workload is not a count of distinct values. Cryptographic hashes authenticate retained files; they do not replace the deterministic equality argument or by themselves establish enumeration coverage.

5. Execution timestamps and measured time

The first invocation started at **28 October 2026, 08:32:57.096455198 UTC**; the final invocation finished, including the production audit, at **10:38:33.644612263 UTC**. These are respectively **03:32:57.096455198** and **05:38:33.644612263 CDT** (UTC−05:00), on the same date. The retained integer Unix-epoch timestamps are 1793176377096455198 and 1793183913644612263 nanoseconds. Precision is that of the recorded clock fields, not a claim of externally calibrated nanosecond accuracy.

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The two closed invocations total **7,492.102807105 seconds**, or **2 h 04 min 52.103 s**. The wall-clock span between the endpoints is 7,536.548157065 seconds (2 h 05 min 36.548 s). Their difference, 44.445349960 seconds, reflects the pause between invocations and any small difference between wall and monotonic clocks. The first invocation ended at a checkpoint; the second completed the run.

The original ten-billion anchor recorded 838.8772331430082 seconds including its build. Adding that once gives the retained cumulative execution path of **8,330.980040248007 seconds**, or **2 h 18 min 50.980 s**. The historical 24-minute Claude report, preflight work, intermediate extensions and later archival checks are excluded from this sum. Timer boundaries of the earlier anchor differ from the later audit-inclusive protocol; the sum describes the archived execution path rather than a single uninterrupted benchmark.

TABLE 1. Recorded fifty-billion execution stages. Occupied wall time is the union of each stage's intervals, not the sum of concurrent workers.

Stage	Occupied wall (s)	CPU (s)
Generation, 32 passes	5,001.065659	37,211.908780
Partitioning	734.828642	1,260.188133
Sorting, including native I/O	484.227185	2,870.982905
Candidate replay	464.279119	3,633.077531
Final audit	107.371720	107.350548

The comparison pipeline including replay occupied 464.343548 seconds. Total coordinator-parent and native-child CPU time was 46,788.073242 seconds. Nested stages may overlap; stage rows should not be added to obtain overall elapsed time. Orchestration, hashing and synchronization also contribute to the invocation total. A later fresh-archive audit took 105.898774 seconds and is outside that total.

The execution environment was Linux x86-64, Python 3.12.14 and frozen GCC 13.3.0 executables, using eight workers, an eight-CPU-equivalent quota and an 8-GiB memory limit. All 8,481 tracked native tasks returned zero: 32 generation tasks, 256 partitions, 8,192 sorts and one replay. Both coordinators closed normally, with no failed native attempts or unclosed timing receipts.

Sampled working storage reached 24,078,417,344 bytes (22.425 GiB). The largest sampled sum of coordinator/native resident memory was 1,024,335,872 bytes (0.954 GiB), sampled every 15 seconds. Shared pages can be counted twice, and sampling does not establish a continuous peak. The shared cgroup recorded one out-of-memory kill whose victim could not be identified; all tracked verifier tasks nevertheless completed normally and the final audit passed. The measurements do not establish an out-of-memory-free workspace or a speed guarantee on other hardware.

6. Retained evidence and reproducibility

The frozen archive retains sources, executables, build records, the anchor, configuration, complete-bucket receipts, candidate coordinates and Meridian addresses, atomic stage timings, and the auditors. Compact receipts were committed before disposable raw records were removed. The final archive does not retain every singleton key stream. Independently rechecking those deleted streams requires rerunning the frozen enumeration.

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Selected SHA-256 identities are:

```
Frozen native_incremental executable:
5587ab32108fdce365eae968725bb2f7a3f729e3fdbeba873075ed5153fd4564

Core result:
652c169981af67fb51396f03ce23da5a2dc0ea15eb0f8f045535e7bd5f8524a2

Sealed original verification archive:
3652546ff076573f196682787bc493f1dd7b36c4bbb0d718b67b1c825f9cb1e2
```

The existing public derivative, version 1.1 [4], is documented as omitting only two noncomputational private storage-save receipts. Its computational content remains the separately frozen historical archive; this edition's manuscript source packet is not a replacement for that full verifier archive.

For this edition, a new read-only reconciliation verified all 26,151 member checksums in the sealed original and reconciled 8,522 closed stage receipts with the recorded invocation and occupied-stage times. It also checked exact coverage arithmetic and the domain partition. Its script and machine-readable result accompany the editable source. This editorial check did not rerun the fifty-billion enumeration or the full-integer candidate auditor. The original successful candidate audits are reported from their retained records.

Here an independent auditor means a separate algorithm or implementation within the project. It does not mean external human review or a proof-assistant formalization. The earlier Claude report, the bounded-degree certificate in the main paper, and the fifty-billion numerical prefix have different coverage and evidence; their counts and runtimes are not interchangeable. No completed 500-billion result is claimed in this release.

References

[1] E. F. W. Heffern, *Meridians on the Markov Surface: Geometry, Residues, and Collision Windows*, v4_ps, 7 November 2026.

[2] E. F. W. Heffern, *Geometric Descent on the Markov Surface*, working draft v32, archived October 2026, pp. 38–40. Historical computational reports; relevant pages retained with this source packet.

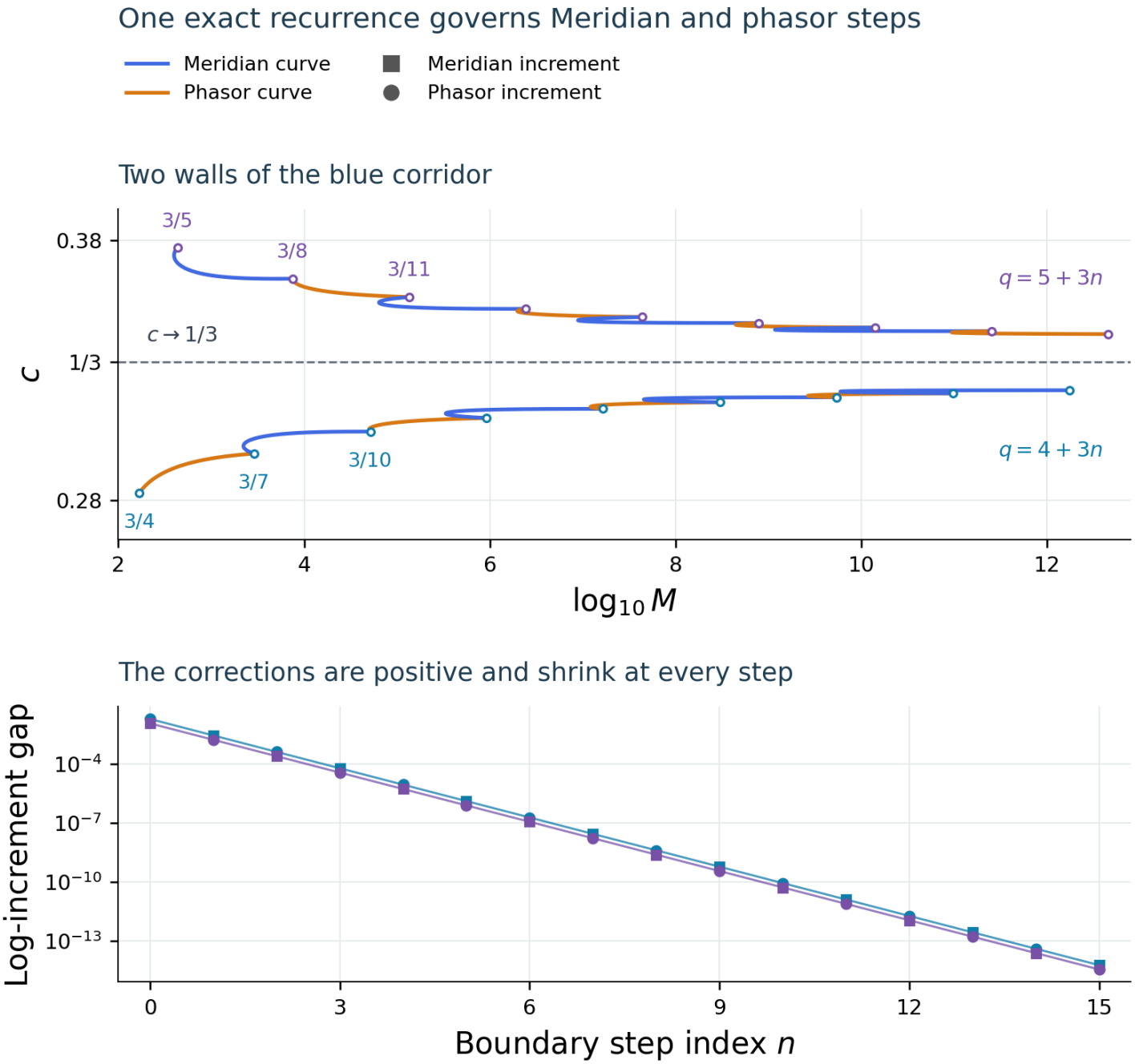
[3] E. F. W. Heffern, public LinkedIn announcement, activity 7519411048115003392, platform publication timestamp 23 October 2026, 14:54:57.771 UTC. Public post. The archived metadata and page snapshot accompany this supplement.

[4] E. F. W. Heffern, *Markov 50 Billion Verification Package*, v1, completed and sealed 28 October 2026; public derivative v1.1 in *Computational supplements for Meridians on the Markov Surface*, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

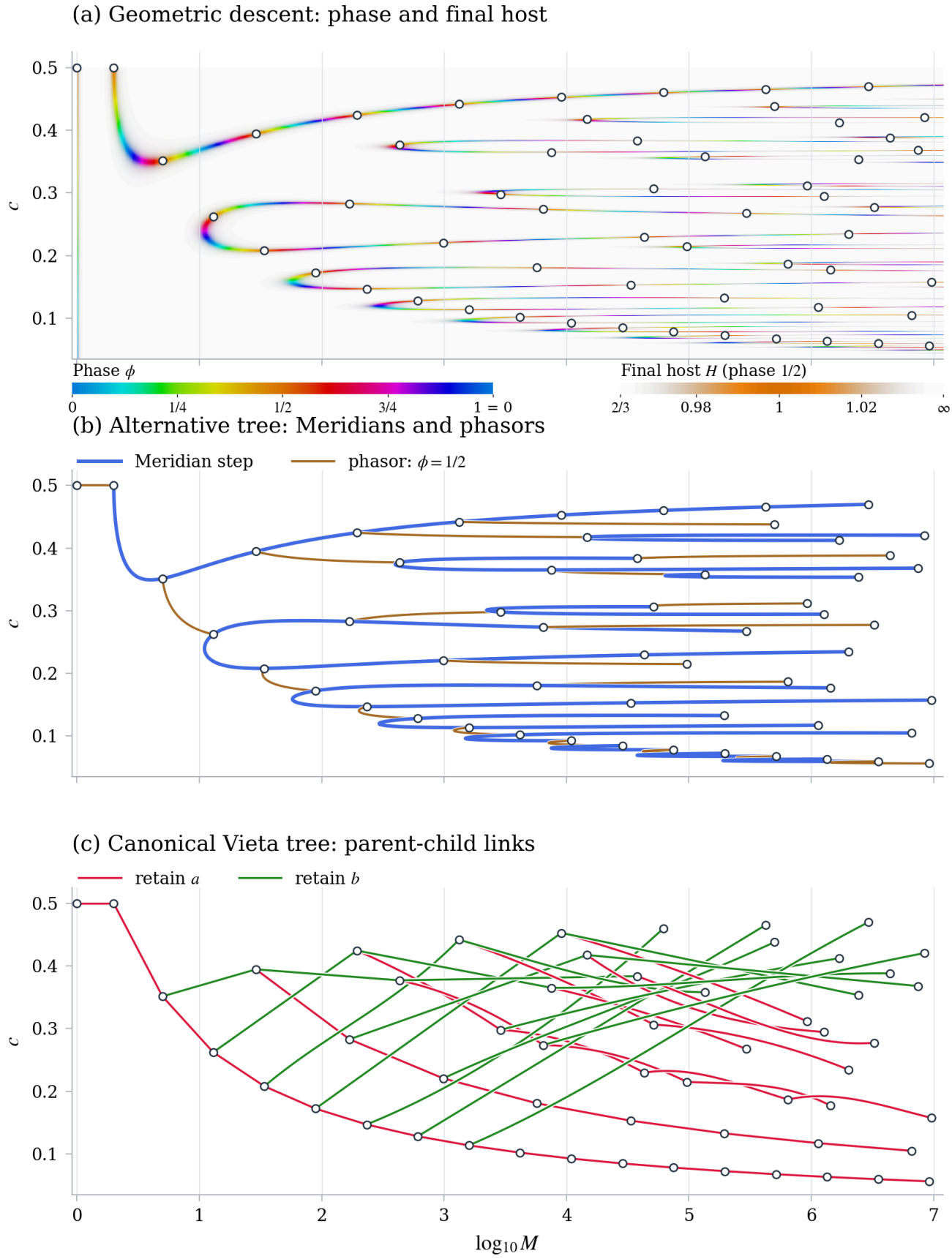
Unaffiliated, St. Louis, Missouri, USA

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Original figure: v11_Markov_Meridians_source/figures/corridor_growth.png



Original figure: v11_Markov_Meridians_source/figures/fig2A_two_trees_landscape.png



Original: v11_Markov_Meridians_source/checks/historical_claude/v32_historical_report_pages_38_40.pdf

7. Historical computational reports

The following records preserve earlier claims of a search below $10^{100,000}$ and separation for word degree at most 16,384. Their original search logs and error bounds have not been independently certified in this revision. They retain their historical conventions; Section 6 supplies the current definitions and exact results. Figure 10 and Section 7.4 belong to those earlier reports.

7.1. Construction checks and conventions. The current Section 6 recursion gives $M_{1,1,1}(0) = 2897$ and $M_{1,1,-1}(0) = 89$. The earlier 7,029-site comparison used older address labels; those labels do not define the present construction. The new exact companion, affine-row and digit checks are documented in Sections 6.13–6.22. The degree bounds and searches below retain their original conventions and reported validation; they have not been rerun or newly certified in this revision.

7.2. **A complete search below $10^{100,000}$ by 3-adic residues.** Write the Vieta descent of a triple with maximum M as a path of local triples (a_i, b_i, m_i), $0 \le i \le n$, with $m_{i-1} = 3b_i m_i - a_i$, $m_{-1} = M$ and $(a_n, b_n, m_n) = (1, 1, 1)$, as in [5]. With

$$B_i = \prod_{k=0}^{i} (3b_k), \qquad T_i = \sum_{k=0}^{i} B_{k-1} a_k \qquad (B_{-1} = 1),$$

one has

$$M = m_i B_i - T_i \qquad (0 \le i \le n). \tag{53}$$

No coordinate of a Markov triple is divisible by 3, so $v_3(B_i) = i + 1$, and the first $i + 1$ steps of a path determine M mod 3^{i+1}. A collision $M = M'$ between two paths therefore forces agreement of these residues at every precision. The 3-adic lifting conditions of [5], such as the exact valuation $v_3(T_i - T'_{i+l}) = i + 1$ when the second path is longer by $l > 0$, are consequences of this agreement. Instead of testing pairs of paths one at a time, we reduce every maximum modulo 3^{39} and compare all residues at once.

Every sorted triple $a \le b \le c$ occurs exactly once in the tree rooted at $(1, 2, 5)$ in which (a, b, c) has the children $(a, c, 3ac - b)$ and $(b, c, 3bc - a)$. A C++ program walks this whole tree below the bound. The entries of each node are carried modulo 3^{39} in Montgomery form [6], together with

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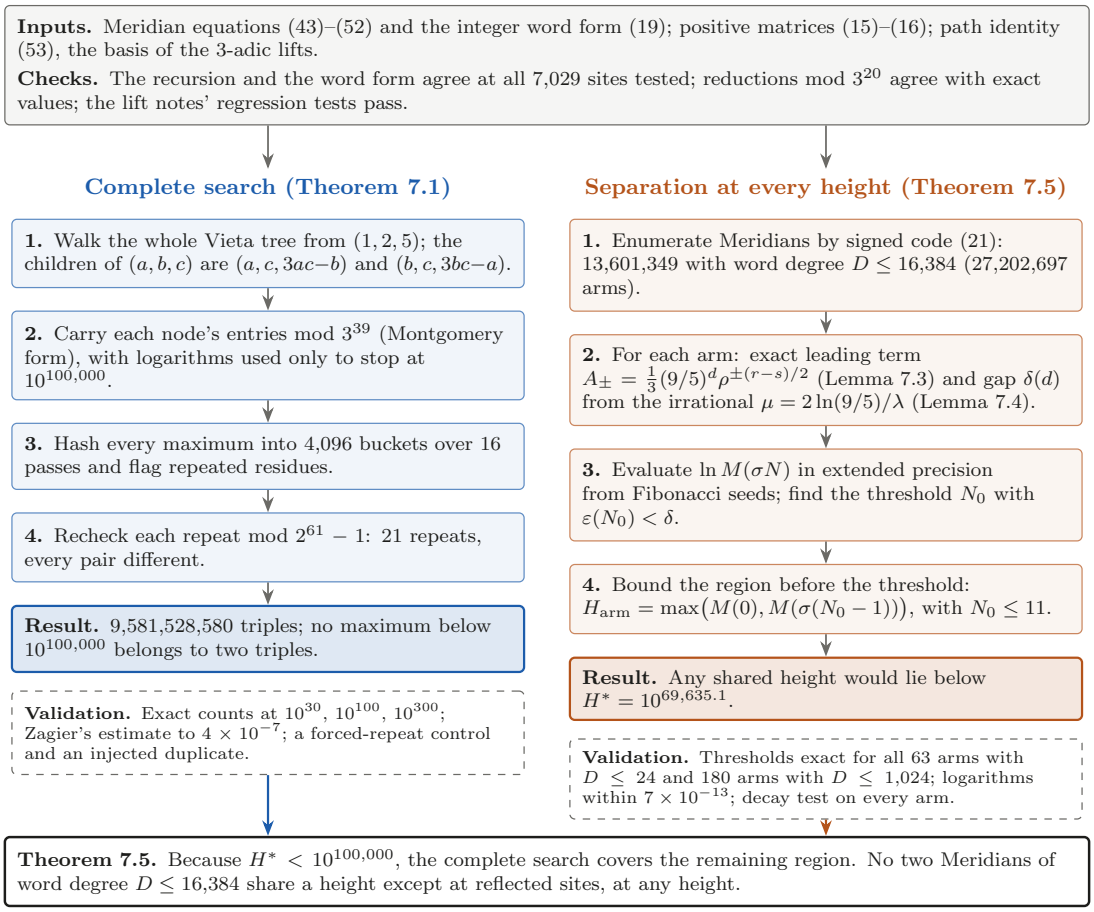


FIGURE 10. The computational flow behind Theorems 7.1 and 7.5. Left: the complete search, which reduces every maximum below $10^{100,000}$ modulo 3^{39} and compares all residues at once. Right: the separation argument, which uses the exact leading terms and the irrational gap to confine any shared height between Meridians of word degree at most 16,384 to a finite region below $10^{69,635.1}$. The complete search covers that region, which gives Theorem 7.5. Dashed boxes list the independent checks.

double-precision logarithms that serve only to stop the walk. The logarithm of a child is computed as $\ln(x^2 + c^2) - \ln o$ from the Vieta product, so no cancellation occurs, and a child is kept when its logarithm is at most $(1 + 10^{-9}) \cdot 10^5 \ln 10$; this margin is far larger than the accumulated rounding error. The residues of the maxima are distributed by a hash into 4,096 buckets over 16 passes and checked for repeats. A repeated residue triggers a second walk that compares the corresponding maxima modulo the prime $2^{61} - 1$.

Theorem 7.1. *The walk visits* 9,581,528,580 *sorted Markov triples, a set that contains every triple with maximum at most* $10^{100,000}$*. Their maxima are pairwise distinct. Consequently no Markov number below* $10^{100,000}$ *is the maximum of two different Markov triples.*

Proof. Distinct residues modulo 3^{39} are distinct integers. Exactly 21 residues occur twice, and none occurs more often; for each of these 21 pairs the two maxima differ modulo $2^{61} - 1$. □

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TABLE 1. Complete searches with 3-adic residues. The estimate is Zagier's asymptotic count $C(\ln 3x)^2$, $C = 0.18071704711507\ldots$ [9]. Times are on a two-core machine with 7 GB of memory.

Bound x	Triples walked	Repeated residues mod 3^{39}	Zagier estimate	Time
$10^{10,000}$	95,823,446	0	95,823,480	2 s
$10^{30,000}$	862,356,775	0	862,356,463	28 s
$10^{100,000}$	9,581,528,580	21	9,581,525,137	24 min

The counts in Table 1 agree with exact enumeration at 10^{30}, 10^{100} and 10^{300} (893, 9,670 and 86,518 triples), and an independent implementation working modulo $2^{61}-1$ reproduced them at $10^{10,000}$ and $10^{30,000}$. They also agree with Zagier's estimate to within four parts in 10^7, consistent with a complete walk. Two controls exercise the repeat-handling path: with the modulus 3^9 the program reports 12,728 false repeats at 10^{300} and clears all of them, and an artificially duplicated maximum is reported as a candidate collision.

Remark 7.2. The 21 repeats are pairs of different Markov numbers that agree in their last 39 base-3 digits, close to the 17 expected if the residues of the $N = 9.58 \times 10^9$ maxima were uniformly distributed among the $2 \cdot 3^{38}$ units modulo 3^{39}. Each such pair passes 39 levels of 3-adic lifting and is rejected only by the second modulus: a residue pass, however deep, is not a collision. Earlier computer verifications, by Borosh [3] and Baragar [2], are cited in recent work as reaching 10^{140} [7].

7.3. **Separation of Meridians at every height.** The positive matrices of Section 4.2 give each Meridian an explicit leading term.

Lemma 7.3 (Leading terms)**.** *Let a nonseed Meridian have helm letter counts* (r, s) *as in* (21)*, and put* $D = r + s = 2d$*. Then*

$$M_\alpha(x) = \sum_{j=-d}^{d} c_j\, e^{2j\lambda x}, \qquad c_j \geq 0, \qquad c_{\pm d} = \frac{1}{3}\Big(\frac{9}{5}\Big)^d \rho^{\pm(r-s)/2}.$$

Consequently, for $\sigma \in \{1, -1\}$ *and integers* $N \geq 0$*,*

$$M_\alpha(\sigma N) = A_\sigma\, \rho^{2dN}\big(1 + \varepsilon_\sigma(N)\big), \qquad A_\sigma = \frac{1}{3}\Big(\frac{9}{5}\Big)^d \rho^{\sigma(r-s)/2}, \tag{54}$$

where $0 \leq \varepsilon_\sigma(N+1) \leq \rho^{-2}\varepsilon_\sigma(N)$*.*

Proof. By (16), $3M_\alpha(x) = \operatorname{tr} W(\xi)$ for the node word W, a product of r factors $P(\xi)$ and s factors $Q(\xi)$ from (15). Every entry of P and Q is a positive constant times ξ^{-1} (upper left), ξ^0 (off the diagonal) or ξ^1 (lower right). Expanding the trace as a sum over closed index paths gives nonnegative coefficients, and the exponent of each term is congruent to D modulo 2. The coefficient of ξ^D comes only from the lower-right entries, giving $\alpha_0^r\beta_0^s$; that of ξ^{-D} comes only from the upper-left entries, giving $\beta_0^r\alpha_0^s$. Since $\alpha_0 = 3\rho^{1/2}/\sqrt{5}$ and $\beta_0 = 3\rho^{-1/2}/\sqrt{5}$, these equal $(9/5)^d\rho^{\pm(r-s)/2}$, and $c_{\pm d}$ is one third of them. With $\xi = e^{\lambda x}$ and $e^\lambda = \rho$, the remaining terms on the arm $x = \sigma N$ are nonnegative multiples of ρ^{2jN} with $j \leq d-1$, relative to ρ^{2dN}, which gives the decay of ε_σ. □

The extreme coefficients were also confirmed exactly, together with the nonnegativity of all coefficients, for all 63 Meridians with $D \leq 24$, from exact rational fits of the harmonic expansion.

Put

$$\mu = \frac{2\ln(9/5)}{\lambda} = 1.2214717808094493\ldots.$$

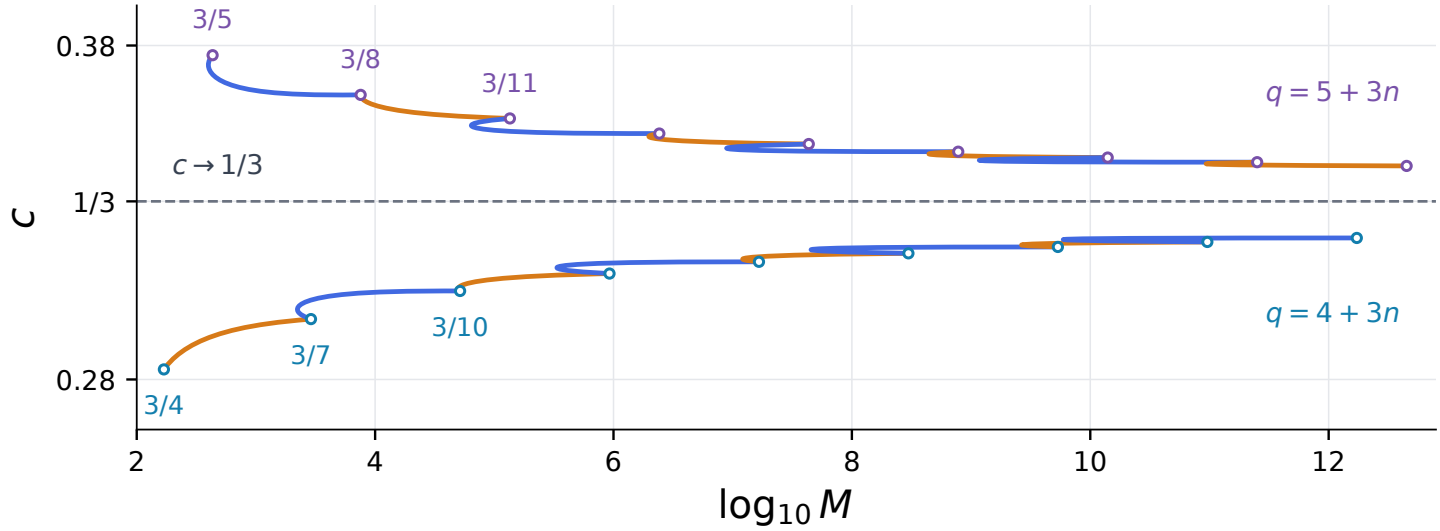
The number μ is irrational: $\mu = p/q$ would give $(9/5)^{2q} = \rho^p$, which is impossible for $p \neq 0$ because ρ^p is irrational.

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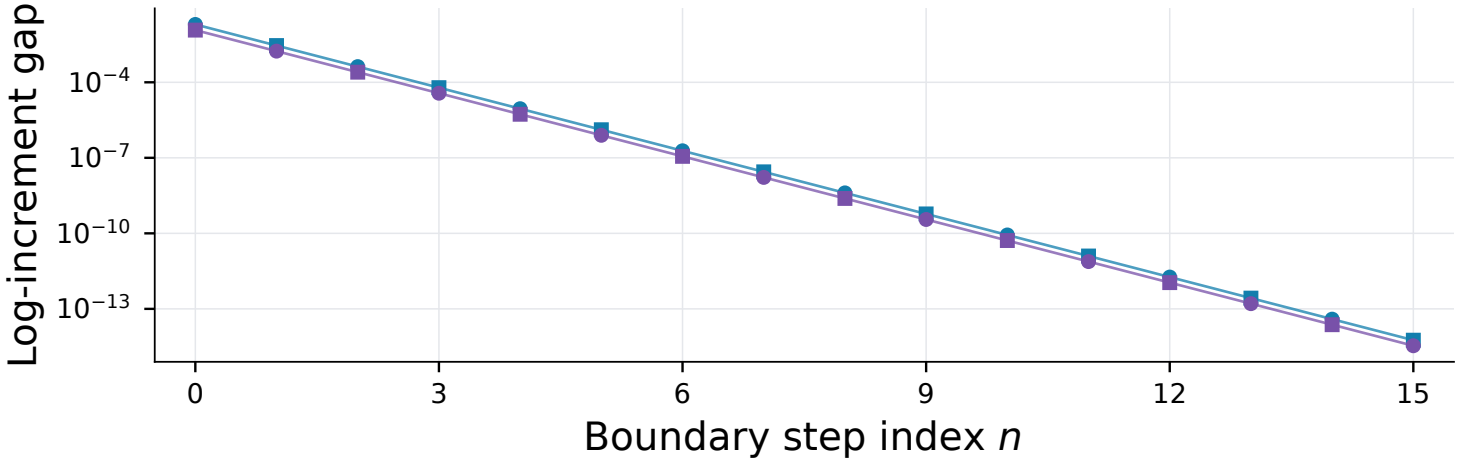
One exact recurrence governs Meridian and phasor steps

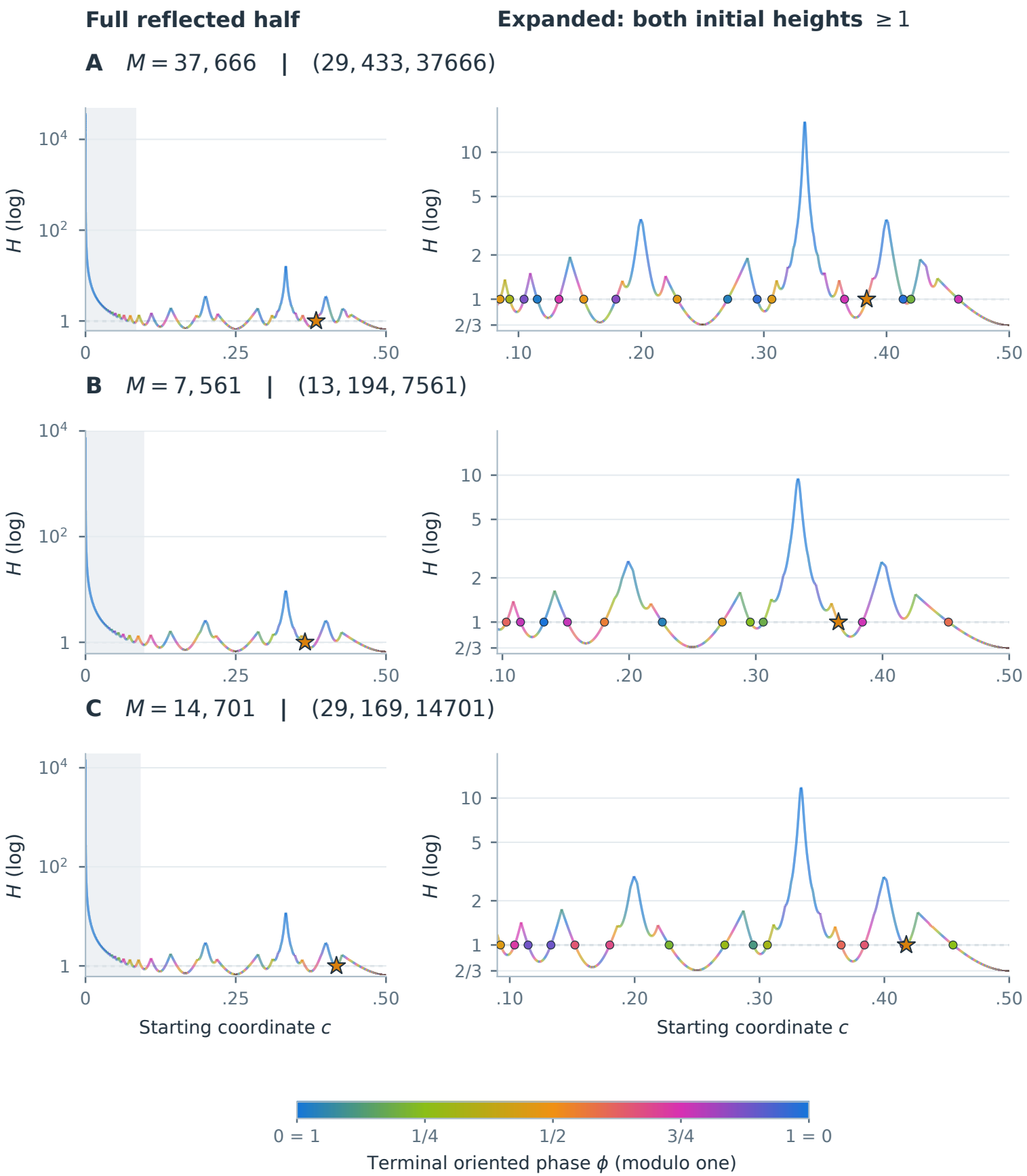
Meridian curve
Phasor curve
Meridian increment
Phasor increment

Two walls of the blue corridor



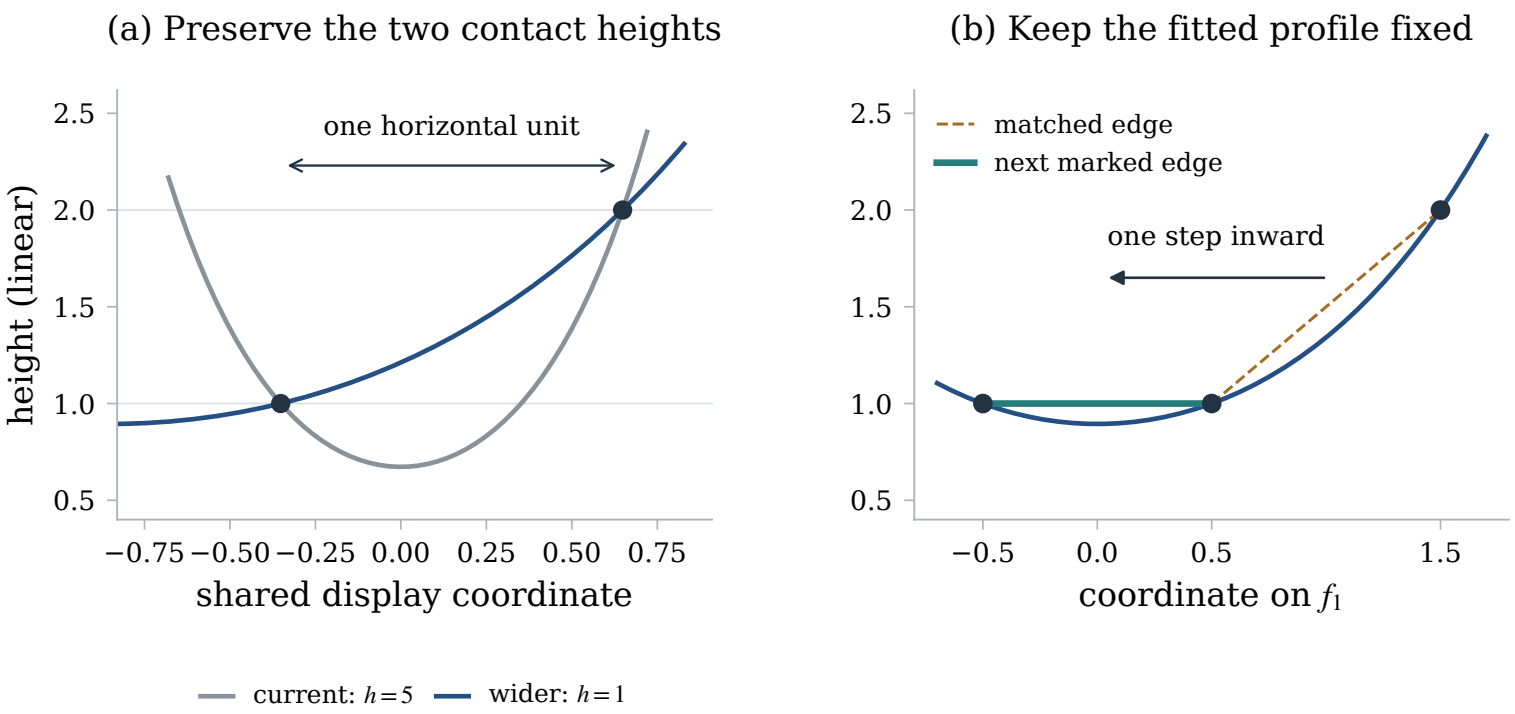
The corrections are positive and shrink at every step



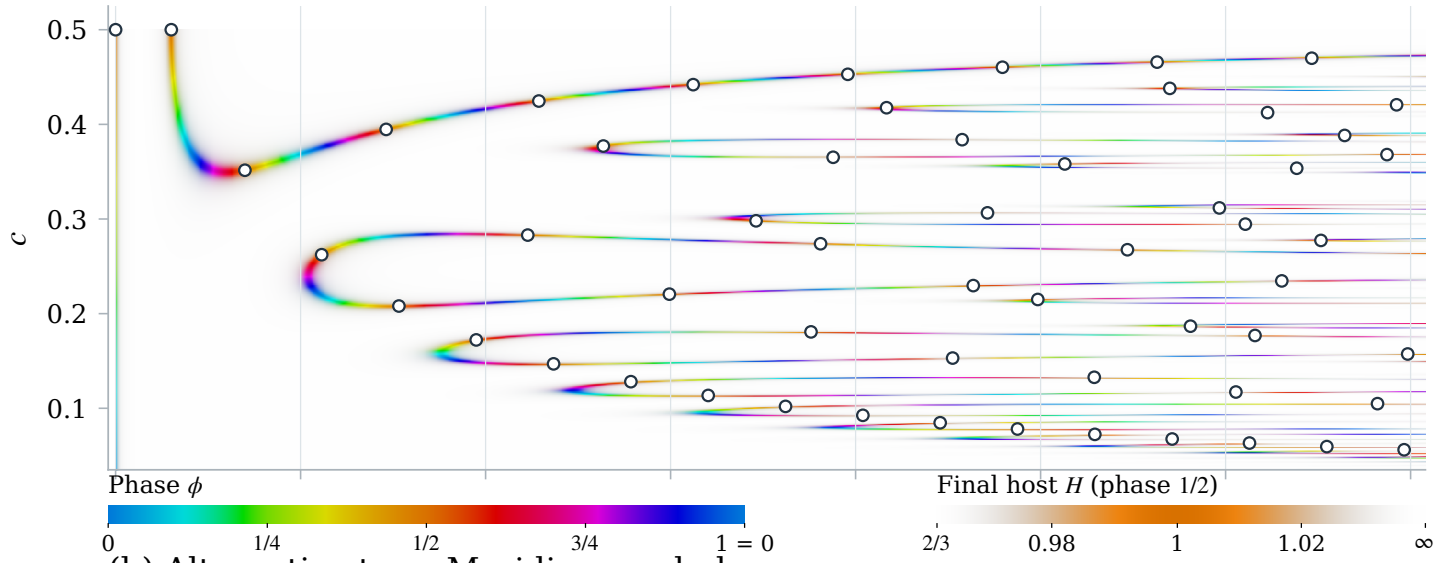


Star: exact integral closure, $(H, \phi) = (1, 1/2)$. Dots: other sampled $H = 1$ crossings.

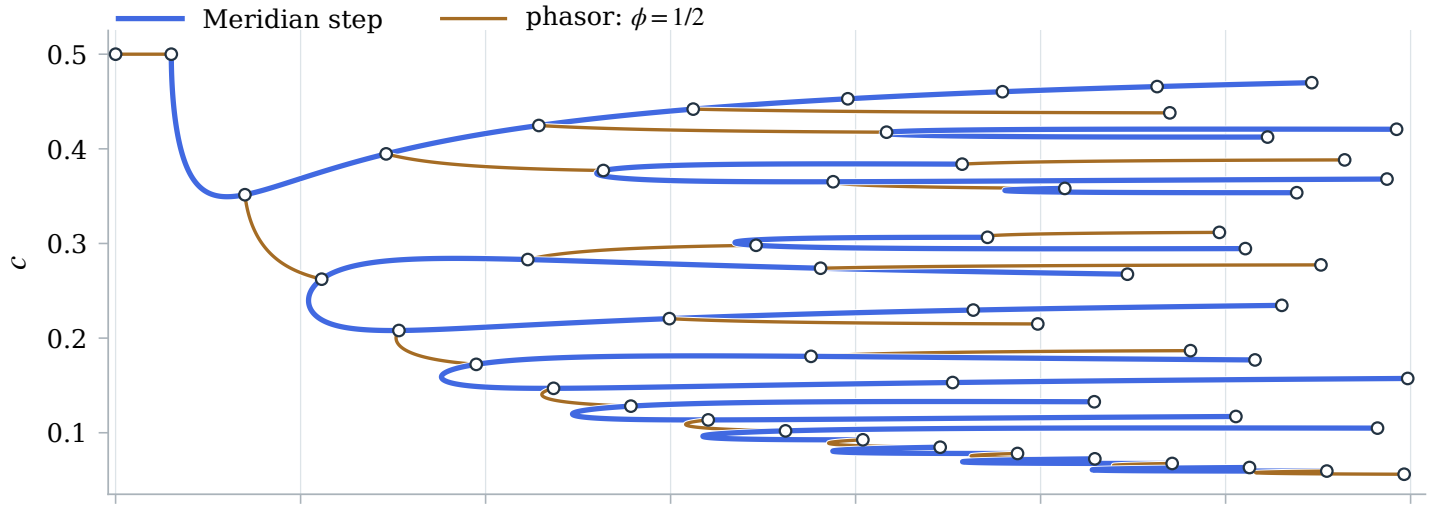
Gray: an initial endpoint is below 1. All displayed starts undergo the same real descent.


(a) Preserve the two contact heights
one horizontal unit
height (linear)
2.5
2.0
1.5
1.0
0.5
−0.75 −0.50 −0.25 0.00 0.25 0.50 0.75
shared display coordinate
current: $h = 5$
wider: $h = 1$
(b) Keep the fitted profile fixed
matched edge
next marked edge
one step inward
2.5
2.0
1.5
1.0
0.5
−0.5 0.0 0.5 1.5
coordinate on f_1

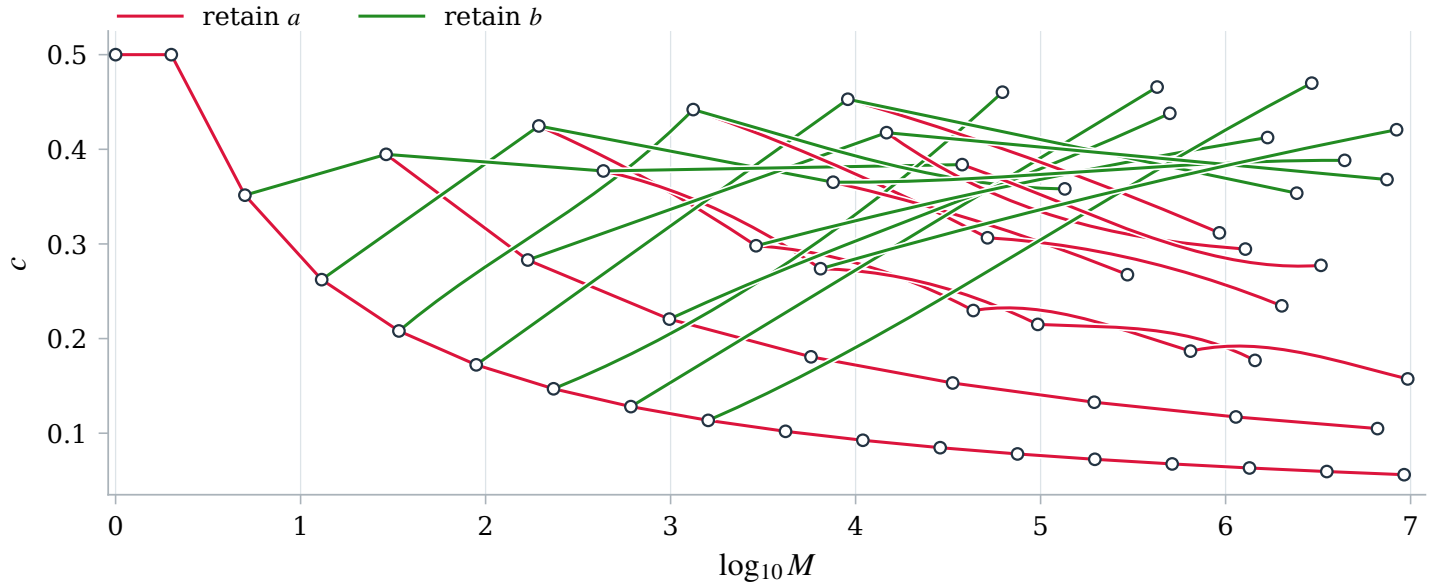
(a) Geometric descent: phase and final host



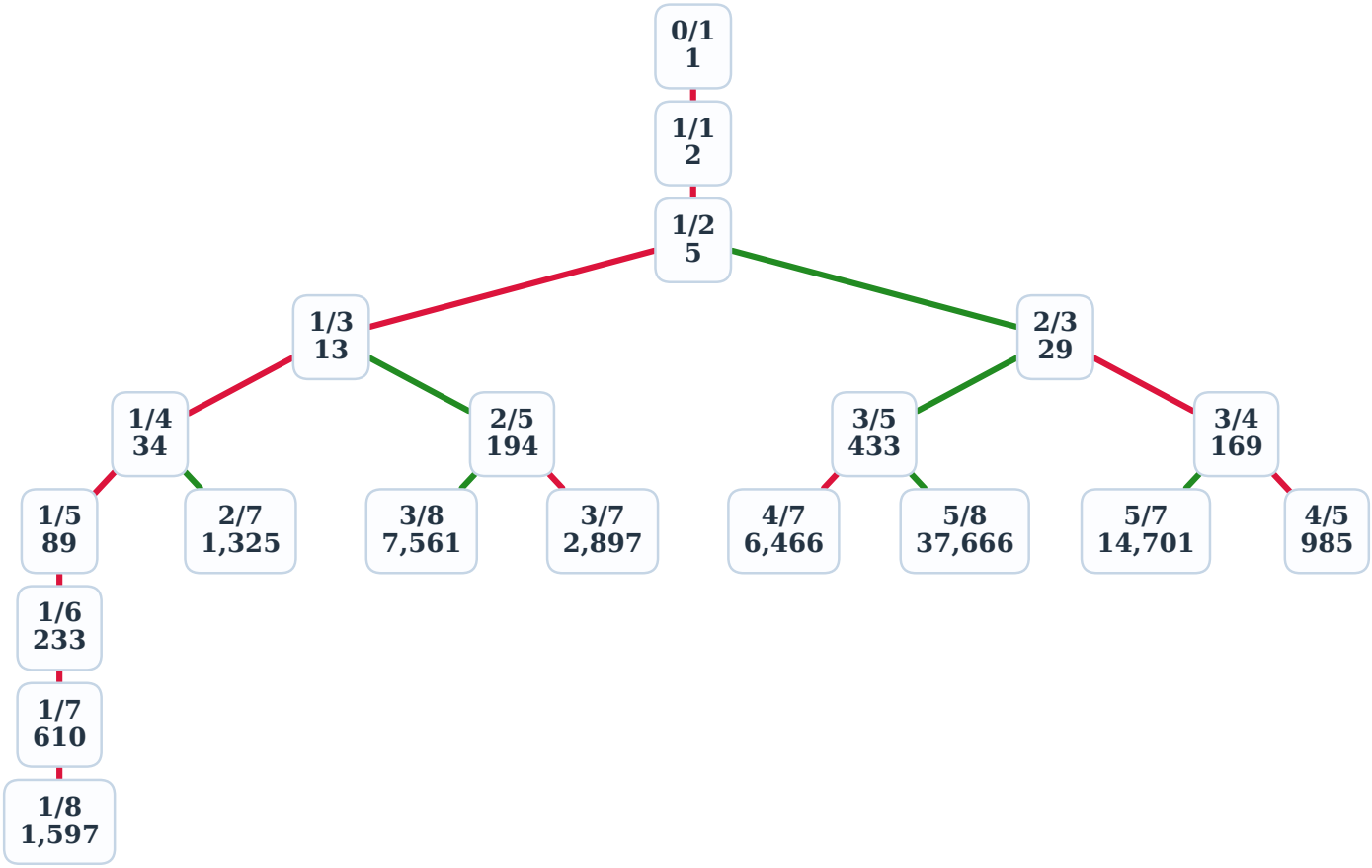
(b) Alternative tree: Meridians and phasors



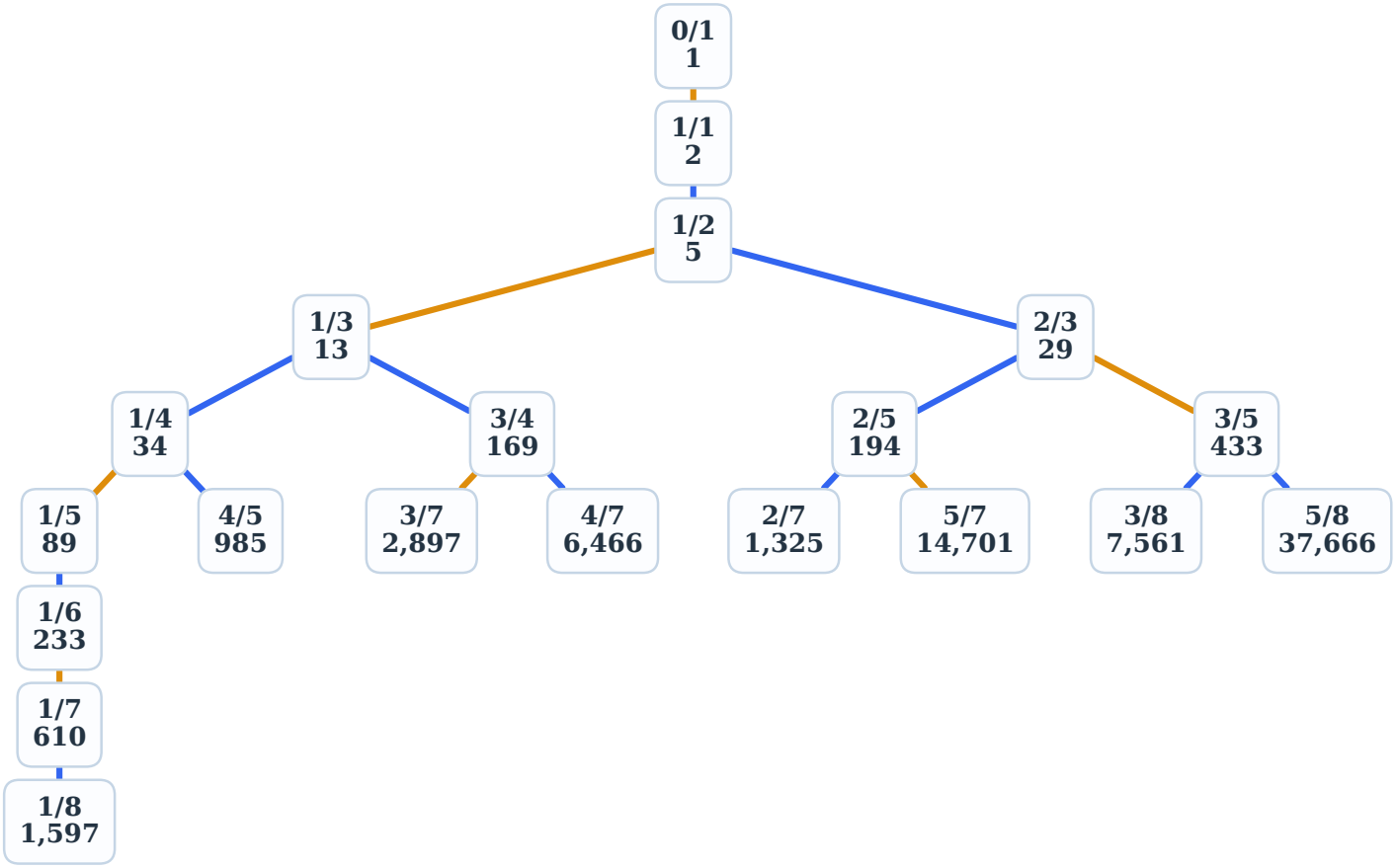
(c) Canonical Vieta tree: parent-child links



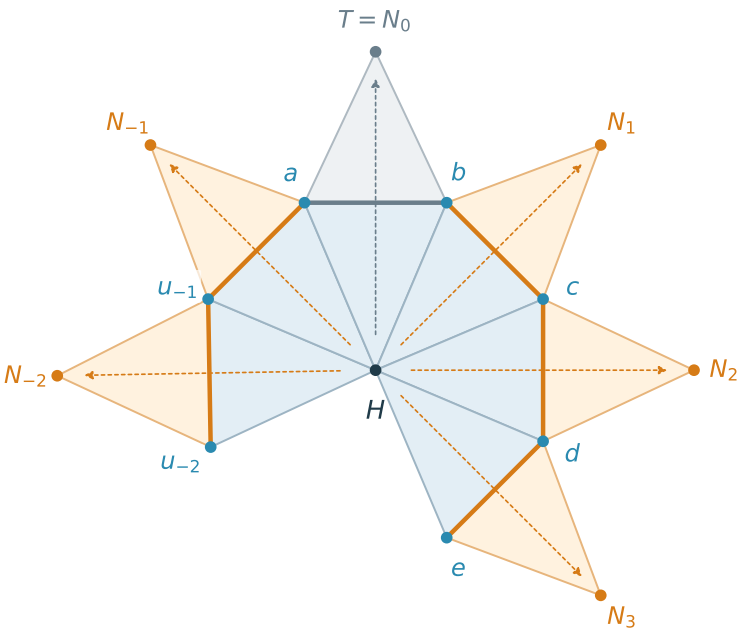
Canonical Vieta tree



Alternative tree: Meridians and phasors

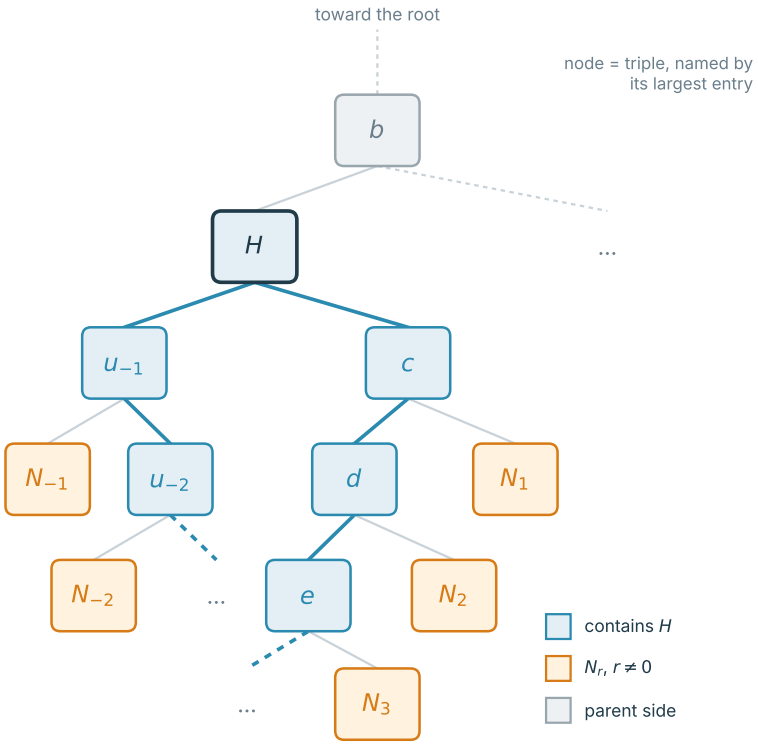


A Reflecting H across successive edges



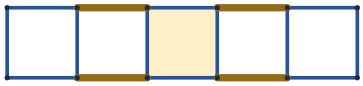
$$u_{j+1} = 3Hu_j - u_{j-1}, \quad N_r = 3u_r u_{r+1} - H$$

B The same configurations in the tree



Helm 1/3: 13 matchings

Half counts (2, 1)

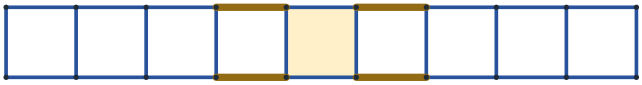


6i

6ii **6iii**

Helm 1/5: 89 matchings

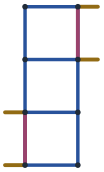
Half counts (5, 3)



type 2 **6i** type 2

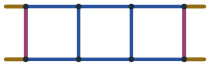
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F_{2s}	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987
-12								1	5	13	31	13	231	181	1	1	13	50	1	88	937	2	1	20	50	17	328	944	2	1	5	9	89	285	215
-11								1	2	13	4	34	283	559	1	1	1	46	73	21	715	2	1	2	13	89	94	170	2	1	2	21	89	333	107
-10								1	1	19	13	121	248	544	1	2	16	48	82	270	541	2	2	20	37	98	29	482	2	5	8	34	53	272	869
-9								1	5	1	34	61	57	169	1	1	13	26	1	291	160	2	1	2	51	89	343	653	2	1	8	13	17	241	806
-8								1	2	13	50	130	169	13	1	1	13	53	73	343	433	2	1	20	31	17	34	923	2	1	5	21	17	86	89
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-6								1	5	19	4	13	34	985	1	1	16	46	1	21	541	2	1	20	13	17	369	104	2	1	8	21	89	233	890
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-3								1	5	13	50	61	248	559	1	1	1	53	1	270	715	2	1	2	31	89	29	170	2	1	2	21	17	272	107
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12	2	2	2	29	2	29	338	1	5	13	50	13	57	181	1	1	13	53	1	291	937	2	1	20	31	17	343	944	2	1	5	21	89	241	215

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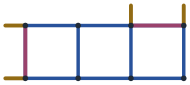
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2



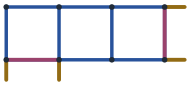
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3



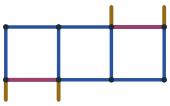
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4



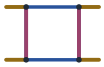
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5



00: 5 01: 2 10: 2 11: 1

6i



00: 2 01: 1 10: 1 11: 1

6ii



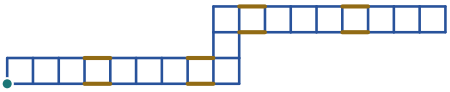
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6iii



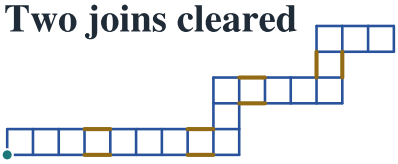
0: 2 1: 1

Canonical 2/9



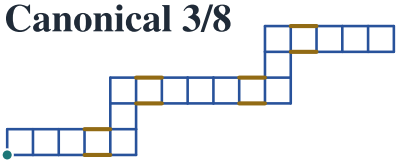
9,077 2 2 1 2 2

Two joins cleared



7,661 2 2 1 3 4

Canonical 3/8



7,561 2 1 2 1 2

Meridians on the Markov Surface: Geometry, Residues, and Collision Windows

Completion and release history

Manuscript dated 7 November 2026 (v5_ps). No separate completion time is established. The UTC upload date falls on 8 November; the local Central posting date is 7 November.

Zenodo publication-date field: 2026-11-07

Zenodo record created: 2026-11-07 18:35:22 CST / 2026-11-08 00:35:22 UTC

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The publication-date field is the date supplied in the deposit. The API created/modified timestamps are repository metadata, not a separately attested first-view time. Full fractional timestamps are retained in the metadata archived in Part C.

https://zenodo.org/records/24046012

https://doi.org/10.5281/zenodo.24046012

Earlier published Zenodo drafts in this version family:

v4_ps | 2026-11-07 13:37:47 CST / 2026-11-07 19:37:47 UTC | https://zenodo.org/records/24043052

Earlier public announcements: 21 October 2026, 10:18:25 CDT / 15:18:25 UTC on X (more than a billion values in under 40 minutes); 23 October 2026, 09:54:57.771 CDT / 14:54:57.771 UTC on LinkedIn (9.3 billion in 24 minutes). These concern earlier computations, not the final trillion certificate.

https://x.com/webystar/status/2112926487091282350

https://www.linkedin.com/posts/elleard-felix-webster-heffern-joseph-by-confirmation-2953a3104_markovnumbers-activity-7519411048115003392-4NaJ

The dates and URLs are retained in the 7 November public-post evidence packet; the LinkedIn URL is also embedded in the published computational supplement. Additional November LinkedIn announcements and the older Meridian ResearchGate v13 URL could not be established in this compilation.

Complete public Zenodo description

This v5_ps revision presents the author's Fibonacci-matrix proofs of normalization and fixed-numerator monotonicity, and his six-piece proofs of the remaining fixed-sum comparison and fixed-denominator monotonicity. The normalization starts from (AC)_11 - (B^2)_11 = 2 and proves the local comparisons and termination in the same matrices. Related earlier normalization is cited as similar work, while the author's argument retains its explicit attribution. The matching interpretation is credited to Musiker's 27 June 2002 memorandum and the Itsara-Musiker-Propp-Viana manuscript of 10 January 2003.

The paper organizes Markov configurations by reversed Meridian addresses, geometric descent, Fibonacci-modulus and modulo-27 behavior, equal-degree separation, finite unequal-degree collision windows, corridor recurrences and a numerator-denominator collision window. The archived degree-10,000 certificate is retained. The unrestricted unicity problem remains open within this framework.

The revised computational chronology includes the earlier Claude-assisted historical search and the separately archived fifty-billion numerical-order verification. The working draft reports 9,581,528,580 triples against a cutoff of 10^100000 in 24 minutes on two cores with 7 GB memory. A public LinkedIn announcement is timestamped 23 October 2026, 14:54:57.771 UTC (09:54:57.771 CDT), with caption wording of 9.3 billion in 24 minutes. This is the announcement timestamp, not a recovered execution start or finish. The earlier draft's qualification about uncertified historical logs and cutoff error bounds is retained. This search is not the later ten-billion anchor.

The fifty-billion verifier completed on 28 October 2026. Its recorded start was 08:32:57.096455198 UTC and its audit-inclusive finish was 10:38:33.644612263 UTC (03:32:57-05:38:34 CDT, rounded). Two closed invocations total 7,492.102807105 seconds (2 h 04 min 52.103 s). The calendar span is 7,536.548157065 seconds, including the invocation pause. Including the separate original ten-billion anchor gives 8,330.980040248007 seconds (2 h 18 min 50.980 s). These historical execution dates are distinct from this edition's release date.

The exact numerical envelope covers 71,517,775,875 configurations. After anchored omissions and certified lower-bound exclusions, 54,940,210,994 new configurations were compared. All 346 repeated primary-residue pairs separated; the final auditor reconstructed all 692 candidate triples and rechecked the 38 anchor candidates. The protocol establishes uniqueness for the first

50,000,000,000 distinct Markov values in numerical order. It used eight workers, an eight-CPU-equivalent quota and an 8-GiB memory limit.

Included are the 33-page main paper, six-page computational proof and timing supplement, editable source, all figure assets, selected historical evidence and receipts, and a new read-only reconciliation of 26,151 archive member checksums and 8,522 closed stage receipts. The full frozen historical verifier archive is available separately at https://doi.org/10.5281/zenodo.23874075. This release is an editorial revision, not a new enumeration; no completed 500-billion result is claimed.

Licensing: Original software is licensed under MIT; original documentation and data under CC BY 4.0. Third-party materials and external dependencies retain their own terms. Copyright (c) 2026 Elleard Felix Webster Heffern, to the extent copyright applies.

Funding: No funding. Substantial generative-AI assistance is disclosed in the manuscript; the author directed the research and is responsible for the work.

Published files

Original filename / checksum	Bytes
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All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

Original: v5_ps_Markov_Meridians_TLMS.pdf

MERIDIANS ON THE MARKOV SURFACE: GEOMETRY, RESIDUES, AND COLLISION WINDOWS

ELLEARD FELIX WEBSTER HEFFERN

Abstract. We organize positive integral Markov triples by Meridians: real curves obtained by reversing a geometric descent, with a signed integer address for the curve and an integer phase for each arithmetic configuration. This gives a bijection with normalized triples without assuming uniqueness of their largest entries. We prove a common Fibonacci-modulus period law and a classification into ten disjoint residue classes modulo 27. Established Markov orderings imply unicity throughout each fixed Meridian degree; positive matrix words place every unequal-degree collision in an explicit finite phase window. A second organization, by corridor walls, gives exact recurrences across both Meridian and phasor steps and a complementary bound on denominators in terms of numerators and their difference. The author's Fibonacci matrices prove normalization and fixed-numerator monotonicity; his six pieces prove the remaining fixed-sum comparison and fixed-denominator monotonicity. We also give a folded-piece comparison criterion. Finally, complete residue and phase certificates separate all 5,067,077 Meridians of degree at most 10,000 at every integer phase. A separate archived computation, completed on 28 October 2026, verifies the first 50 billion distinct Markov values in numerical order. The unrestricted unicity problem remains open within this framework.

Post-submission draft. This edition incorporates changes requiring a new submission. The prefix `ps` retains that distinction.

1. Introduction

A normalized Markov configuration is a positive integral triple satisfying

$$a^2 + b^2 + M^2 = 3abM, \qquad 1 \le a \le b \le M. \tag{1}$$

The unicity question asks whether the largest entry M determines a, b. The classical Vieta tree enumerates configurations by replacing one coordinate with the other root of its quadratic. A tree of configurations is available independently of unicity; treating its vertices as distinct numerical maxima would assume the conclusion. Aigner's monograph gives the historical and arithmetic setting [5]. Continued fractions, snake graphs and Christoffel words supply several established orderings [12, 9, 10, 11], which will be used with their precise hypotheses.

This paper develops two ways to organize that same set of configurations. A *Meridian* holds fixed an itinerary of geometric descent and varies a real phase. A *corridor wall* follows a different route through the alternative tree, eventually holding a Farey numerator fixed while advancing the denominator. These organizations answer different comparison questions: Meridian degree controls a phase window, while corridor numerator controls a denominator window.

The geometric construction comes first in Section 2. Section 3 gives the canonical addresses, their inverse, and the exact conversion to Farey labels. Section 4 proves the modulo-27 classification and the common period s modulo F_{2s}. Sections 5 and 6 establish the two collision windows. Section 7 supplies the matching-count proofs used in the ordering arguments. Section 8 presents a finite certificate with an infinite phase range:

Date: v5_ps — Post-submission edition — 7 November 2026.
2020 *Mathematics Subject Classification.* Primary 11D25; Secondary 11B39, 05C70, 11Y50.
Key words and phrases. Markov numbers, unicity, geometric descent, Meridians, corridors, snake graphs.

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Theorem 1.1 (Degree-10,000 separation)**.** *Among canonical Meridians of Laurent degree at most* 10,000*, distinct canonical integer-phase marks have distinct maxima. There are exactly* 5,067,077 *Meridians in this family.*

The computational part separates different Meridians at all phases. The fixed-degree theorem also separates marks on a single Meridian, so the two conclusions combine as stated. The bound is on *degree*, not on the size of a Markov number, and it does not exclude a competitor of higher degree. Section 9 records the separate fifty-billion numerical-order verification and its dated execution history.

Throughout, $M_\alpha(x)$ denotes the maximum at Meridian address α and phase x. The notation $m_{p/q}$ denotes the maximum attached to the standard reduced Farey label p/q; it labels a configuration before any assertion of unicity. These are two names for values of the same configuration map. The seed formula $M_1(x) = 2 + 3F_{2x}^2$ should not be confused with the boundary family $m_{1/q} = F_{2q+1}$. Analytical logarithms are natural; plotted horizontal axes use $\log_{10} M$.

2. Geometric descent and arithmetic closure

2.1. **A unit edge on a calibrated hyperbola.** For a real host $h > 2/3$, define

$$L_h = \operatorname{arcosh}(3h/2), \qquad \alpha_h = \frac{2h}{\sqrt{9h^2-4}}, \tag{2}$$
$$f_h(t) = \alpha_h \cosh(L_h t), \qquad g_h(t) = \alpha_h \sinh(L_h t).$$

The Cartesian curve $(g_h(t), f_h(t))$ is a hyperbola. Its height profile $t \mapsto f_h(t)$ is the curve used in the figures. Hyperbolic addition gives

$$f_h(t+1) + f_h(t-1) = 3h f_h(t), \tag{3}$$

$$f_h(t)^2 + f_h(t-1)^2 + h^2 = 3h f_h(t) f_h(t-1). \tag{4}$$

Thus two contacts one horizontal unit apart, with heights $A = f_h(c)$ and $B = f_h(c-1)$, form a real Markov state. The two compatible hosts are exactly

$$h, \qquad K = 3AB - h = \frac{A^2 + B^2}{h}. \tag{5}$$

Begin with an edge straddling the vertex, $0 \le c < 1$. Seek a wider calibrated profile through the same two ordered contacts, allowing horizontal translation. Smaller h means a wider profile. If the other host satisfies $K < h$, fit that profile and walk the unit lattice inward until the selected edge again straddles its vertex. Otherwise stop. Figure 1 shows one complete step. The fit is explicitly

$$v = \frac{1}{2} + \frac{1}{L_K} \operatorname{arsinh}\left(\frac{(A-B)\sqrt{3K+2}}{2K} \right), \qquad f_K(v) = A, \quad f_K(v-1) = B. \tag{6}$$

The walk replaces v by $v - \lfloor v \rfloor$. Its orientation is retained: reflection exchanges the contacts and sends phase to its negative modulo one.

Proposition 2.1 (Uniqueness of the fit)**.** *A marked edge has at most one distinct compatible profile, hence at most one wider fit. For fixed* $0 < a < b$*, a same-arm fit* $f_h(u-1) = a$*,* $f_h(u) = b$ *with* $u > 1$ *is unique if it exists.*

Proof. The hosts are the two roots of $X^2 - 3ABX + A^2 + B^2$. The other root belongs to the geometric domain because $(3K-2)AB = K^2 + (A-B)^2 > 0$. For a same-arm fit the horizontal distance between heights a, b is

$$\frac{1}{L_h} \int_a^b \frac{dy}{\sqrt{y^2 - \alpha_h^2}}.$$

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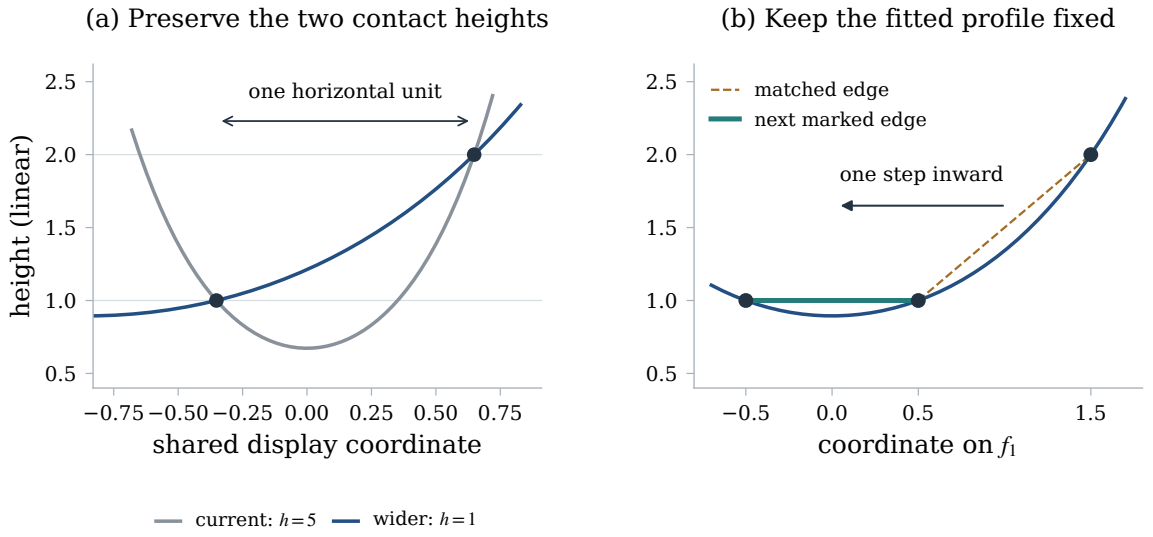


Figure 1. Geometric descent from $(1, 2, 5)$. The wider profile passes through the same unit-spaced contacts. On f_1 their coordinates are $1/2, 3/2$; one inward step selects $-1/2, 1/2$, with heights $1, 1$. The curve is fixed during the walk.

When $a > 2/3$ and the lower height is in range, both factors strictly decrease with h. The distance can therefore equal one at most once. The profile is strictly increasing on that arm, which also determines u. If $a \le 2/3$ there is no fit. □

2.2. **Termination.** Write $\mathcal{D}(M, c) = (H, \phi)$ for the terminal host and oriented phase.

Theorem 2.2 (Finite descent)**.** *Every straddling input with $M \ge 1$ terminates. On $1 \le M \le R$ the number of host changes is uniformly bounded. The terminal map is continuous into $(2/3, \infty) \times \mathbb{R}/\mathbb{Z}$ and analytic on cells with a fixed strict itinerary.*

Proof. Temporarily sort the endpoints as $a \le b$. Inward walking does not increase either endpoint. Every current host is at most R, and the discriminant in (4) gives

$$(9h^2 - 4)a^2 - 4h^2 = (2b - 3ha)^2 \ge 0.$$

Consequently the endpoints are at least $a_R = 2R/\sqrt{9R^2 - 4} > 2/3$. At an accepted change, $a^2 + b^2 < h^2$, so the endpoints are also below $h \le R$. Using either endpoint as the host in the same discriminant identity bounds the new host below by a_R. The initial host is at least $1 > a_R$.

If the new host K is at least b, its edge already straddles the vertex, because both endpoints are below $f_K(1)$; the next candidate is the old, larger host, so descent stops. Otherwise

$$\frac{h}{K} > \frac{h}{b} = 3a - \frac{K}{b} > 3a - 1 \ge 3a_R - 1 > 1.$$

All changes except possibly the last therefore contract the host by a fixed factor. An upper bound is $1 + \lceil \log(R/a_R) / \log(3a_R - 1) \rceil$.

At a boundary $K = h$, the fitted coordinate in (6) is the original c, so the accepted and stopped maps agree. At a phase cut, $K(h, 0) = h(9h^2 + 4)/(9h^2 - 4) > h$, and a neighborhood is stopped. The uniform bound on compositions then gives continuity, with phase taken modulo one. The strict formulas are analytic. □

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2.3. **Which inputs close arithmetically?** The two reversible state operations are

$$T(a,b,h) = (3ha - b, a, h), \qquad J(a,b,h) = (a, b, 3ab - h). \tag{7}$$

They and their inverses are integer polynomial maps. A terminal state has the nonnegative defect

$$\begin{aligned} H f_H(\phi) f_H(\phi - 1) - 1 = {} & \frac{(H-1)^2(H+2)}{3H-2} \\ & + \frac{4H^3}{9H^2-4} \sinh^2\Big(L_H(\phi - \tfrac{1}{2})\Big). \end{aligned} \tag{8}$$

For $0 \le \phi < 1$ it vanishes exactly at $(H, \phi) = (1, 1/2)$.

Proposition 2.3 (Arithmetic closure)**.** *A straddling input with $M \ge 1$ reaches $(1, 1/2)$ if and only if its three entries are positive integers. Modulo companion exchange, these inputs are exactly the normalized configurations* (1)*.*

Proof. The product-to-sum identity gives (8). Reversing a finite closure itinerary from $(1,1,1)$ by (7) preserves integrality. Conversely, integral descent stays integral. At an integral terminal state order the entries as $u \le v \le w$. There is no coordinate-decreasing mutation. But for $Q(X) = X^2 - 3uvX + u^2 + v^2$,

$$Q(v) = u^2 + (2 - 3u)v^2 \le 3u^2(1 - u) \le 0.$$

If neither root is below $w \ge v$, equality is forced, giving $u = v = 1$; terminality then gives $w = 1$. Finally, a straddling edge has no decreasing endpoint mutation, so its host is a largest entry. Its ordered split is uniquely recovered by (6). □

Both terminal coordinates matter. Figure 2 shows inputs with terminal host one but different terminal phases. A crossing of the height level $H = 1$ alone does not certify integrality. The marked configurations follow the host itineraries $37666 \to 5 \to 1$, $7561 \to 5 \to 1$, and $14701 \to 2 \to 1$, all ending at phase $1/2$.

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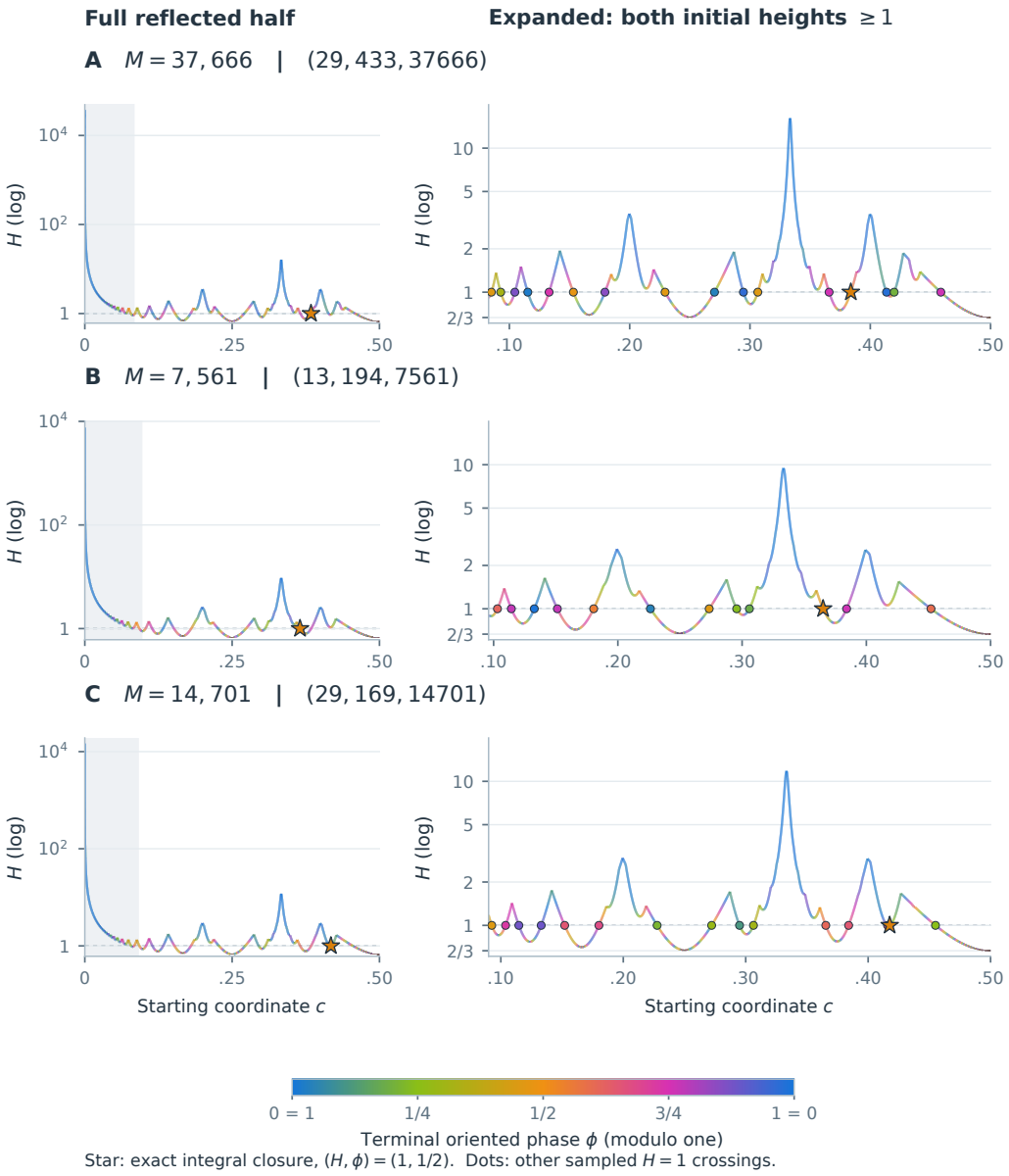


Figure 2. Terminal host and phase for $M = 37666, 7561, 14701$. Height is H on a logarithmic scale and hue is oriented phase. Left: $0 \le c \le 1/2$, with gray where an initial endpoint is below one. Right: the admissible portion enlarged. Stars mark the exact integral starts; dots are other numerically located $H = 1$ crossings. These numerical plots illustrate the closure criterion.

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3. Meridians, addresses, and the alternative tree

3.1. **The landscape and its two coordinates.** On the landscape $(\log_{10} M, c)$, the point (M, c) represents the contacts $f_M(c), f_M(c-1)$. We draw the reflected half $0 < c \le 1/2$; exchanging the contacts sends c to $1-c$. A *Vieta arc* keeps one initial contact height fixed. For example, the branch with height a has

$$c = \frac{\operatorname{arcosh}(a/\alpha_M)}{L_M}.$$

A *Vieta trek* follows such arcs, changing the retained contact when necessary. A *Meridian* is a connected component of terminal host $H(M, c) = 1$. A *phasor* holds the terminal phase ϕ fixed. The alternative tree uses phasors at $\phi = 1/2$ to join successive Meridians.

The address phase x on a Meridian is real and unrolled. Its terminal circle phase is $\phi \equiv x + 1/2 \pmod 1$. Every integer mark therefore closes arithmetically; the distinguished arrival mark $x = 0$ is called the *helm*. It need not be the minimum of the real profile.

Figure 3 keeps the same configurations in three views. The alternative tree is in the middle panel. The bottom panel uses smooth schematic connectors for the Vieta parent relation: node coordinates and adjacency are exact, but those connectors are not literal level curves. The two trees each have 55 edges. An unmarked crossing is not a vertex.

For reproducibility, the field in panel (a) uses the fixed cyclic phase palette with HSL saturation and lightness

$$w(H) = \frac{1}{1 + (\ln H / \ln 1.02)^2}, \qquad S = w(H), \qquad L = 1 - 23w(H)/40.$$

The background is a numerical illustration, independent of all exact enumerations and exclusions below.

3.2. **Evaluating an address.** The seed Meridian is

$$\begin{aligned} a_1(x) = f_1(x + \tfrac{1}{2}), \qquad b_1(x) = f_1(x - \tfrac{1}{2}), \\ M_1(x) = 3a_1(x)b_1(x) - 1. \end{aligned} \tag{9}$$

For integer x, its companions are F_{2x+1}, F_{2x-1}, where $F_{-j} = (-1)^{j+1}F_j$. Hence

$$M_1(x) = 2 + 3F_{2x}^2, \qquad M_1(0), M_1(1), M_1(2), \ldots = 2, 5, 29, 194, \ldots. \tag{10}$$

From any ordered state (a, b, H) form the two-sided orbit

$$u_0 = a, \quad u_1 = b, \qquad u_{j+1} = 3Hu_j - u_{j-1}.$$

A nonzero index k replaces the state by

$$(u_k, u_{k+1}, 3u_k u_{k+1} - H). \tag{11}$$

The address is written

$$\alpha = (1, k_m, \ldots, k_1): \quad k_1 \text{ is applied first and written at the right.}$$

This is the reversed indexing convention: the former $M_{1,a,b,c}(x)$ is now $M_{1,c,b,a}(x)$. Neither the signs nor x are reversed. In terms of (7), with $S = T^{-1}$ and $\mathcal{W}_k = JS^k$,

$$(a_\alpha(x), b_\alpha(x), M_\alpha(x)) = \mathcal{W}_{k_m} \cdots \mathcal{W}_{k_1} JT^x(1, 1, 1). \tag{12}$$

Composition acts from right to left. A recursive extension inserts its index immediately after the leading 1. For example, $M_{1,1}(0) = 13$. Index zero is forbidden because (11) would return the preceding pole $3ab - H$, rather than create a new component.

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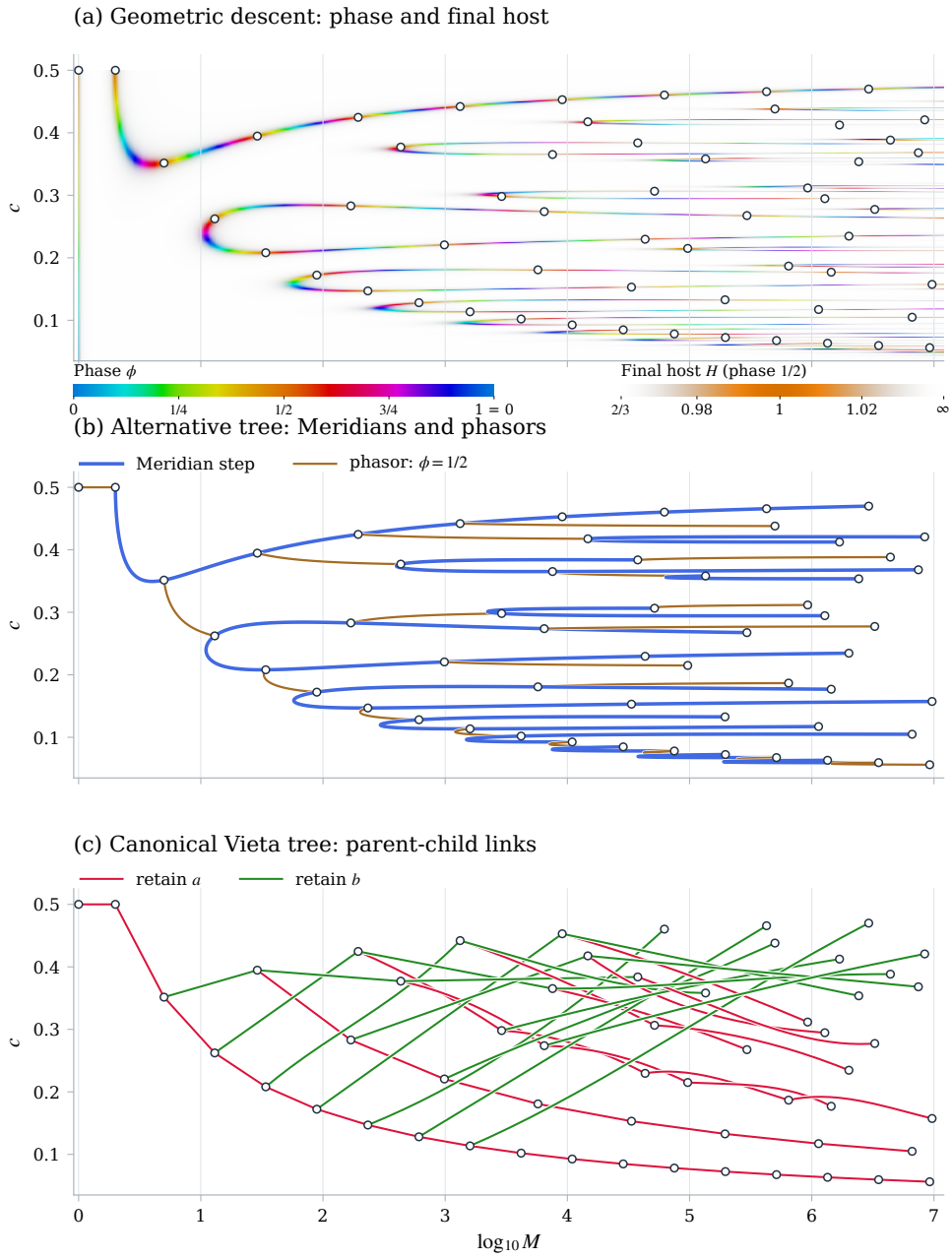


Figure 3. The same 56 configurations with $M \leq 10^7$. (a) Numerical geometric descent, with phase encoded by hue and host by color intensity. (b) The alternative tree, with Meridian and phasor edges. (c) The canonical Vieta tree, with simplified connectors retaining exact endpoints. Red retains the smaller companion, green the larger. Every edge runs toward larger M. Crossings without a node mark are not vertices.

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Definition 3.1 (Canonical marks)**.** The root $(1,1,1)$ is separate. For the seed (1) use integer phases $x \geq 0$. For every other address require $k_1 > 0$, $k_i \neq 0$ for $i > 1$, and $x \in \mathbb{Z}$. Evaluation is followed by sorting the two companions.

Theorem 3.2 (Configuration bijection)**.** *Canonical marks are in bijection with all normalized positive Markov triples except the root. The displayed value $M_\alpha(x)$ is the largest entry.*

Proof. The quadratic argument in Proposition 2.3 shows that replacing the largest coordinate by its other pole gives a smaller maximum, apart from the initial repeated entries. This is the forced Vieta reduction of a configuration and does not assume scalar unicity.

To recover a word, reflect the current maximum H to $P = 3ab - H$. When $P \geq 2$, walk its oriented endpoint orbit inward until both endpoints are below P. This edge is unique. Indeed $u_{j+1} - 2u_j + u_{j-1} = (3P-2)u_j > 0$, so the entries below P form an interval; three consecutive entries below P would contradict $u_{j-1} + u_{j+1} = 3Pu_j \geq 3P$. Endpoint descent supplies the edge, and its nonzero signed displacement recovers the last-applied, leftmost index. Repeat at the smaller pole. At pole one, the orbit reaches $(1,1)$ and fixes the seed phase.

Conversely, all outward blocks are positive and reduced. If the preceding pole is $P < H$, the first reflected poles on the two sides are $(9b^2-2)H - P > 6H$ and $(9a^2-2)H - P > 6H$. Their farther endpoints exceed $2H$, and the recurrence makes subsequent outward entries increase. The reduction therefore undoes every prescribed block. Finally, companion exchange negates all indices and x simultaneously. The canonical conditions select one of the two orientations; only $(1,1,2)$ is fixed. □

3.3. **The two parent rules.** At a nonzero phase the alternative parent is the adjacent inward mark, $(\alpha; x) \mapsto (\alpha; x - \operatorname{sgn} x)$. At a nonseed helm it is

$$(1, k_m, \ldots, k_2, k_1; 0) \longmapsto (1, k_m, \ldots, k_2; -k_1), \tag{13}$$

followed by simultaneous sign reflection if the remaining rightmost index is negative. An empty tail uses the seed's nonnegative phase. The seed helm with maximum two has parent $(1,1,1)$.

The alternative parent deletes the *rightmost* index at a helm. Recursive evaluation removes the *leftmost* index after 1. These operations serve different purposes. The depth

$$d(\alpha; x) = 1 + \sum_{i=1}^{m} (|k_i| + 1) + |x| \tag{14}$$

decreases by one under either parent edge, with root depth zero. Thus the parent rule defines a rooted tree on the same configurations. The ordinary outward Vieta moves instead are

$$(a, b, M) \longmapsto (a, M, 3aM - b), \qquad (a, b, M) \longmapsto (b, M, 3bM - a). \tag{15}$$

For example, $29 \to 194$ is a Meridian edge, whereas $29 \to 169$ is a Vieta edge. Figure 4 compares the two sets of links.

3.4. **Real components and phasor edges.**

Proposition 3.3. *The canonical real families in* (12)*, modulo companion reflection, are the nonterminal connected components of $H(M, c) = 1$. Their terminal phase is $x + 1/2$ modulo one. The separate terminal component is $M = 1$.*

Proof. The seed companions are at least $2/\sqrt{5}$, and $M_1 \geq 2$ is strictly larger than either. More generally, let $h \geq 1$ and $2/\sqrt{5} \leq a, b \leq h$. The first outward endpoint satisfies $3hb - a \geq (6/\sqrt{5} - 1)h > h$; its reflected pole is larger still. The reversed orbit gives the same conclusion on the other side. Every outward block thus has a strict inverse descent to its source and preserves the terminal phase.

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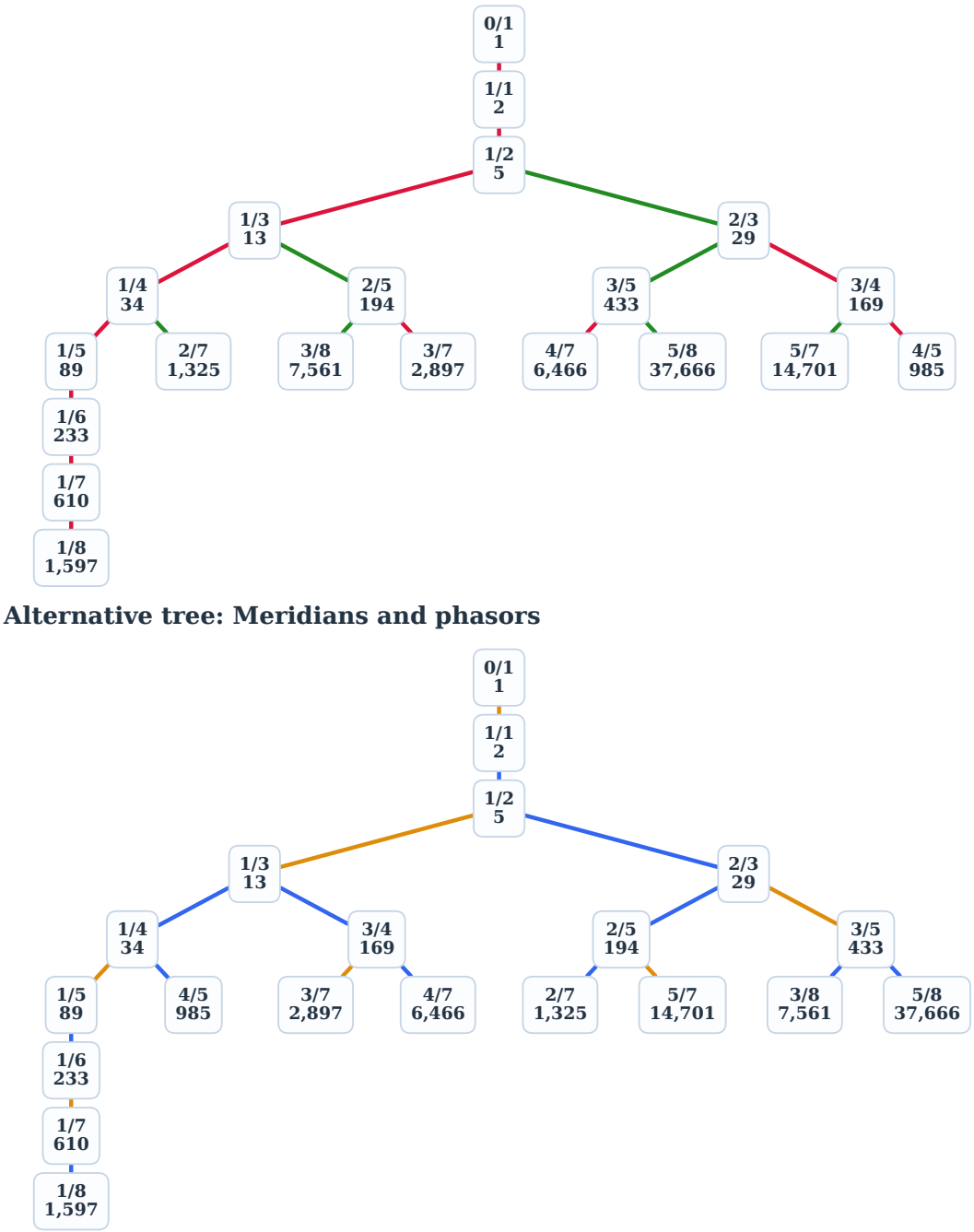


Figure 4. Two trees on the same 20 configurations: the 16 with $M \leq 3000$ and the four additional values $6466, 7561, 14701, 37666$. Each node displays its reduced standard Farey label and maximum. Red and green denote the Vieta moves; blue and gold denote Meridian and phasor steps.

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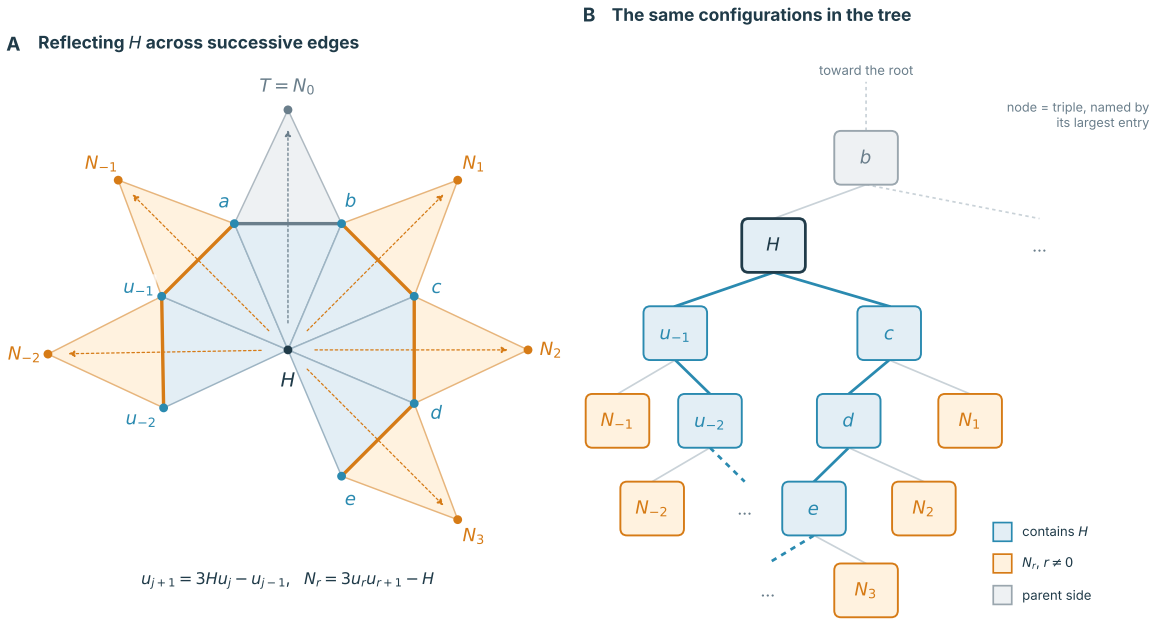


Figure 5. The orbit underlying one recursive extension. Reflecting H across (u_j, u_{j+1}) gives $N_j = 3u_j u_{j+1} - H$; N_0 is the preceding pole. For source $(a, b, H) = (1, 5, 13)$ at $M_{1,1}(0)$, the preceding pole is two, $(u_{-2}, u_{-1}, u_2, u_3, u_4) = (1325, 34, 194, 7561, 294685)$, and $(N_{-2}, N_{-1}, N_1, N_2, N_3) = (135137, 89, 2897, 4400489, 6684339842)$. The right panel locates these reflections in the Vieta tree.

Conversely, reverse any descent ending at host one. Its last host change produces the seed with $x = v - 1/2$, and every preceding walk is nonzero: a zero walk would simply have stopped at the smaller host. These strict blocks uniquely recover a locally constant word. The unrolled x joins the final phase charts continuously. The seed is injective as an ordered curve, as its endpoint difference is strictly monotone; each block is invertible. The positive Laurent expansion proved in Proposition 5.2 makes every family proper. The inverse charts are relatively open in the level set, and properness makes the families closed. Each connected family therefore fills a component. Reflection gives exactly the canonical identifications. □

The phasor edges are obtained by applying the same outward blocks to

$$\left(\frac{h}{\sqrt{3h-2}}, \frac{h}{\sqrt{3h-2}}, h\right), \qquad 1 \le h \le 2. \tag{16}$$

This symmetric arc joins $(1, 1, 1)$ to $(1, 1, 2)$. Its other host is $2h/(3h-2)$, and both possible terminal fits have phase $1/2$. Its endpoints satisfy the bounds used in the proposition, so the outward blocks preserve that phase. The displayed phasors consequently arise from the descent itself, rather than interpolation between integer nodes.

3.5. **Degree and the exact Farey dictionary.** Attach endpoint count vectors $u = (1, 0)$ and $v = (0, 1)$ to the seed. Under an index k update them by

$$(u, v) \longmapsto \begin{cases} ((k-1)(u+v)+v,\ k(u+v)+v), & k > 0, \\ ((|k|(u+v)+u,\ (|k|-1)(u+v)+u), & k < 0. \end{cases} \tag{17}$$

Write their sum as (r, s). Their determinant remains ± 1 and their unordered residues modulo two remain $\{(1, 0), (0, 1)\}$. Thus r, s are positive coprime odd integers. Canonical orientation gives $r \le s$.

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The *Meridian degree* is $D = r + s$. It is the extreme Laurent exponent in ρ^x, where $\rho = (3 + \sqrt{5})/2$; the degree in ρ^{2x} is $D/2$.

Proposition 3.4 (Helm classification)**.** *Canonical Meridians correspond bijectively to coprime positive odd pairs $r \le s$. Their helm labels are the standard Farey fractions r/s.*

Proof. The endpoint vectors are the two Stern–Brocot parents: their sum is (r, s) and their determinant has absolute value one. Conversely, the parents of an odd/odd pair have opposite coordinate parities. Their componentwise absolute difference is a smaller positive primitive odd/odd pair, except at $(1, 1)$. The difference has one componentwise sign; an opposite sign would give the initial pair. Iteration reaches $(1, 1)$.

To recover the signed indices, put $n_\alpha = (r, s)$ and let α^- remove the leftmost index after 1. Consecutive difference vectors have determinant of absolute value two, so three consecutive vectors satisfy $n = 2c\, n^- + \varepsilon n^{--}$ with $c \ge 1$ and $\varepsilon = \pm 1$. Determinant division gives an integer coefficient, parity makes it even, and decreasing coordinate sums fix its magnitude and sign. At the bottom use $(1, 1), (-1, 1)$; canonical orientation fixes the first index positive. The magnitude is fixed by the ratio of successive degrees, which lies in $[2c, 2c+1)$ or $(2c-1, 2c]$ according to the sign; equality occurs only at the first applied index. Reversing the process gives precisely $n_{1,k,k_m,\ldots,k_1} = 2|k| n_\alpha + \operatorname{sgn}(kk_m) n_{\alpha^-}$. For the seed use $k_0 = 1$ and $n_\varnothing = (-1, 1)$, so the first extension has $(r, s) = (2k_1 - 1, 2k_1 + 1)$. This recovers the unique address. Mediant addition identifies the label. □

Lemma 3.5 (Phase dictionary)**.** *For helm (r, s) and degree $D = r + s$, the standard Farey label at phase x is*

x	(p, q)	$p + q$
0	(r, s)	D
1	(r, D)	$D + r$
-1	(s, D)	$D + s$
$n \ge 2$	$(D, (n-1)D + r)$	$nD + r$
$-n \le -2$	$(D, (n-1)D + s)$	$nD + s.$

The negative seed phases are omitted.

Proof. At seed phases $0, 1, n \ge 2$, the endpoint bases are respectively $((1, 0), (0, 1))$, $((1, 1), (0, 1))$, and $((1, n), (1, n-1))$. The count updates are linear, so substitution gives the table; negative phase exchanges the bases. For the inverse, odd/odd p, q give $(r, s, x) = (p, q, 0)$. If q is even, sort $p, q - p$ and choose $+1$ when p is the smaller entry, otherwise -1. If p is even, write $q = tp + u$ with $0 < u < p$; sort $u, p - u$ and choose $+(t + 1)$ or $-(t + 1)$ in the same way. A tie is the seed and uses the nonnegative representative. These formulas recover every mark. □

In particular, degree can be read directly from a reduced label:

$$D = \begin{cases} p + q, & p, q \text{ odd}, \\ q, & q \text{ even}, \\ p, & p \text{ even}. \end{cases} \tag{18}$$

This dictionary connects the Meridian and corridor collision windows below.

4. Meridian residues: modulo 27 and Fibonacci moduli

4.1. **Transferring a seed congruence to every address.** The recurrence for the companions contains no division, but a seed period of the *maximum* need not visibly preserve both companions.

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The following four-entry record resolves that issue:

$$\mathcal{S} = (A, B, C, H) = (9a^2, 9b^2, 3ab, H).$$

Forward and backward orbit steps and pole reflection are the polynomial maps

$$\begin{aligned} \mathcal{R}(A, B, C, H) &= (B, A + 9H^2B - 18HC, HB - C, H), \\ \mathcal{R}^{-1}(A, B, C, H) &= (B + 9H^2A - 18HC, A, HA - C, H), \\ \mathcal{J}(A, B, C, H) &= (A, B, C, C - H). \end{aligned} \tag{19}$$

For $h(x) = M_1(x)$ the seed record is

$$\mathcal{S}_1(x) = (8h(x) - h(x-1) - 2,\ h(x) + h(x-1) + 2,\ h(x) + 1,\ h(x)).$$

Every congruence preserving two consecutive seed maxima therefore propagates along every signed address.

Theorem 4.1 (Largest modulus with a common phase period)**.** *Every canonical Meridian has phase period T modulo m if and only if*

$$m \mid q(T), \qquad q(T) = 3F_{2T}F_{\gcd(2T,4)} = \begin{cases} 3F_{2T}, & T \text{ odd}, \\ 9F_{2T}, & T \text{ even}. \end{cases} \tag{20}$$

In particular, s is a common period modulo F_{2s}.

Proof. Fibonacci addition and (10) give

$$h(x + T) - h(x) = 3F_{2T}F_{4x+2T}.$$

Fibonacci strong divisibility, $\gcd(F_a, F_b) = F_{\gcd(a,b)}$, follows by the Euclidean algorithm from the addition formula. The gcd of all displayed differences is already the gcd of those at $x = 0, 1$, namely $q(T)$. Thus the seed has period T modulo m exactly when $m \mid q(T)$. The record (19) proves sufficiency at every address; the seed itself proves necessity. □

The theorem specifies a period common to all Meridians. It does not assert that the period is the smallest for each individual sequence. Nor does a large period exclude a scalar residue match at two earlier phases. For powers of three it gives

$$M_\alpha(x + 2 \cdot 3^n) \equiv M_\alpha(x) \pmod{3^{n+3}}, \qquad n \geq 0. \tag{21}$$

Indeed, $F_{3j} = F_j(5F_j^2 + 3)$ for even j, beginning with $F_4 = 3$, gives $v_3(F_{4\cdot 3^n}) = n + 1$. Figure 6 shows the common periods at the requested even-indexed Fibonacci moduli.

4.2. **Why modulo 27 gives only ten classes.** No positive Markov entry is divisible by three: mutation negates the changed coordinate modulo three and the integral tree starts at $(1, 1, 1)$. Let t, h be preceding and current poles. An extension by a nonzero index has the particularly simple action

$$S(t, h) = (h, t) \quad (k \text{ even}), \qquad R(t, h) = (h, 7h - t) \quad (k \text{ odd}) \pmod{27}. \tag{22}$$

To see this, put $N_j = 3u_ju_{j+1} - h$. Direct use of the orbit recurrence gives

$$N_0 = t, \quad N_1 = (9u_1^2 - 2)h - t, \quad N_{j+2} - N_j = 9h(u_{j+2}^2 - u_{j+1}^2).$$

All orbit squares are one modulo three, so $N_{j+2} \equiv N_j$ and $N_1 \equiv 7h - t \pmod{27}$. This works for negative indices as well. Only the parity of each recursively applied index survives.

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x	M1							M1.1							M1.2							M1.1.1							M1.-1.1						
s	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8
F_{2s}	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987
-12								1	5	13	31	13	231	181	1	1	13	50	1	88	937	2	1	20	50	17	328	944	2	1	5	9	89	285	215
-11								1	2	13	4	34	283	559	1	1	1	46	73	21	715	2	1	2	13	89	94	170	2	1	2	21	89	333	107
-10								1	1	19	13	121	248	544	1	2	16	48	82	270	541	2	2	20	37	98	29	482	2	5	8	34	53	272	869
-9								1	5	1	34	61	57	169	1	1	13	26	1	291	160	2	1	2	51	89	343	653	2	1	8	13	17	241	806
-8								1	2	13	50	130	169	13	1	1	13	53	73	343	433	2	1	20	31	17	34	923	2	1	5	21	17	86	89
-7								1	1	13	31	25	13	34	1	2	1	50	82	56	652	2	2	2	50	98	258	317	2	5	2	9	125	89	233
-6								1	5	19	4	13	34	985	1	1	16	46	1	21	541	2	1	20	13	17	369	104	2	1	8	21	89	233	890
-5								1	2	1	13	34	231	820	1	1	13	48	73	88	580	2	1	2	37	89	328	758	2	1	8	34	89	285	71
-4								1	1	13	34	121	283	181	1	2	13	26	82	21	937	2	2	20	51	98	94	944	2	5	5	13	53	333	215
-3								1	5	13	50	61	248	559	1	1	1	53	1	270	715	2	1	2	31	89	29	170	2	1	2	21	17	272	107
-2								1	2	19	31	130	57	544	1	1	16	50	73	291	541	2	1	20	50	17	343	482	2	1	8	9	17	241	869
-1								1	1	1	4	25	169	169	1	2	13	46	82	343	160	2	2	2	13	98	34	653	2	5	8	21	125	86	806
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12	2	2	2	29	2	29	338	1	5	13	50	13	57	181	1	1	13	53	1	291	937	2	1	20	31	17	343	944	2	1	5	21	89	241	215

Figure 6. Five Meridians modulo F_{2s} for $s = 2, \ldots, 8$: the moduli are $3, 8, 21, 55, 144, 377, 987$. Each column has common phase period s, possibly a multiple of an individual least period. The shaded row is the helm; blank cells omit the seed's negative phases. The displayed residues use the reversed Meridian indexing of this edition.

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Proposition 4.2 (Universal modulo-27 classes)**.** *At integer phases every Meridian alternates between the two residues of one of the following sets, allowing the two singleton cases:*

$$\{1\},\ \{2,5\},\ \{4,16\},\ \{7,13\},\ \{8,17\},\ \{10,19\},\ \{11,23\},\ \{14,20\},\ \{22,25\},\ \{26\}.$$

The ten sets are pairwise disjoint. Meridians in different classes cannot share an integer value, regardless of degree or phase.

Proof. The common phase period is two because $q(2) = 27$. Track preceding and current poles at both phase parities, starting from $(1, 2, 1, 5)$. The simultaneous maps satisfy $S^2 = I$ and $SRS = R^{-1}$. Successive current-pole pairs on the R orbit are

$$\begin{array}{cccccc} (2,5) & (13,7) & (8,17) & (16,4) & (23,11) & (10,19) \\ (20,14) & (22,25) & (26,26) & (25,22) & (14,20) & (19,10) \\ (11,23) & (4,16) & (17,8) & (7,13) & (5,2) & (1,1). \end{array}$$

The preceding pair is the previous pair cyclically, and $b_{j+1} = 7b_j - b_{j-1}$ checks the return in 18 steps. These states and their S images form a closed set of 36 states. Their current-pole images are exactly the ten listed sets. Congruence of integer values would require membership in the same set. □

Table 1. The ten universal classes and their populations among the 519 Meridians of degree at most 100. Only the population column is bounded-degree data.

Residue set	Population	Residue set	Population
$\{1\}$	42	$\{10, 19\}$	47
$\{2, 5\}$	88	$\{11, 23\}$	47
$\{4, 16\}$	61	$\{14, 20\}$	38
$\{7, 13\}$	78	$\{22, 25\}$	35
$\{8, 17\}$	65	$\{26\}$	18

At that degree bound, the classification leaves 15,215 of the 134,421 pairs for further comparison. Larger Fibonacci moduli and compatible phases, rather than extrapolation from this table, supply the degree-10,000 certificate in Section 8.

5. Equal-degree separation and unequal-degree windows

5.1. **A complete result at each fixed degree.** We first use the established fixed-sum, fixed-numerator and fixed-denominator orderings of Markov numbers. Section 7 proves them by matching counts and records their earlier sources.

Theorem 5.1 (Equal-degree unicity)**.** *Distinct canonical integer-phase marks of the same Meridian degree have different maxima. The marks may belong to the same or to different Meridians.*

Proof. For $D \geq 4$, Lemma 3.5 partitions the labels into three classes. Helms r/s have fixed sum D; near-helms $r/D, s/D$ have fixed denominator D; outer phases have fixed numerator D. Each class is strictly ordered by its corresponding monotonicity, and the dictionary ensures that different marks have different labels.

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The largest helm is $m_{1/(D-1)}$. The near-helm extrema are $m_{1/D}, m_{(D-1)/D}$; the smallest outer value is $m_{D/(D+1)}$. They are separated by

$$m_{1/(D-1)} < m_{1/D}, \qquad m_{(D-1)/D} < m_{(D-1)/(D+1)} < m_{D/(D+1)}.$$

The intermediate fraction is reduced because D is even. The first two inequalities hold at fixed numerator and the last at fixed denominator. For $D = 2$, the unique seed gives $2, 5$ and then the increasing family $m_{2/(2n-1)}$ for $n \geq 2$, starting at 29. These values are also distinct. □

Thus the remaining Meridian comparison has unequal degrees. To bound it, we need exact leading terms and an explicit positive remainder.

5.2. **Positive words and a uniform tail bound.** Put $\rho = (3 + \sqrt{5})/2$, $\lambda = \log \rho$, and $\xi = \rho^x$. The endpoint matrices are

$$P(\xi) = \begin{pmatrix} B_0/\xi & C_0 \\ C_0 & A_0\xi \end{pmatrix}, \qquad Q(\xi) = \begin{pmatrix} A_0/\xi & \rho C_0 \\ C_0/\rho & B_0\xi \end{pmatrix}, \tag{23}$$
$$A_0 = \frac{15 + 3\sqrt{5}}{10}, \quad B_0 = \frac{15 - 3\sqrt{5}}{10}, \quad C_0 = \frac{2}{\sqrt{5}}.$$

Both have determinant one and positive entries. With seed endpoint words $U = P, V = Q$, the scalar state is

$$a_\alpha = \operatorname{tr} U/3, \quad b_\alpha = \operatorname{tr} V/3, \quad M_\alpha = \operatorname{tr}(UV)/3.$$

Its extension is represented by

$$(U, V) \longmapsto \begin{cases} ((UV)^{k-1}V, (UV)^k V), & k > 0, \\ ((VU)^{|k|}U, (VU)^{|k|-1}U), & k < 0. \end{cases} \tag{24}$$

For verification, $\operatorname{tr}((UV)^{j-1}V)/3$ begins at a, b and obeys the endpoint recurrence by Cayley–Hamilton. The determinant-one trace identity $\operatorname{tr}(XY) = \operatorname{tr} X \operatorname{tr} Y - \operatorname{tr}(XY^{-1})$ then gives the reflected pole. Negative indices use the reversed orbit. The count updates are exactly (17).

Proposition 5.2 (Laurent profile)**.** *For helm counts (r, s), degree $D = r + s$, and $t = (r - s)/2$,*

$$M_\alpha(x) = \sum_{j=-D,-D+2,\ldots,D} c_j \rho^{jx}, \quad c_j \geq 0, \quad c_{\sigma D} = \frac{1}{3}(9/5)^{D/2}\rho^{\sigma t}. \tag{25}$$

The logarithm is strictly convex, and the real profile has one minimum.

Proof. Expand the trace into closed paths on the two matrix states. All weights are nonnegative. A closed path has an even number of off-diagonal choices, which fixes the parity of its exponent. The two constant-state paths give the extreme coefficients because $A_0B_0 = 9/5$ and $A_0/B_0 = \rho$. The second derivative of the logarithm is λ^2 times the variance of the exponents under their positive normalized weights. Both extremes occur, so the variance is positive. They also force growth at both ends. □

Lemma 5.3 (Uniform positive-word remainder)**.** *Let W have r letters P and s letters Q, of total length $g = r + s > 0$. For $n \geq 0$ and $\sigma = \pm 1$,*

$$\frac{\operatorname{tr} W(\rho^{\sigma n})}{3} = \frac{1}{3}(9/5)^{g/2}\rho^{\sigma(r-s)/2}\rho^{gn}(1 + \epsilon_{W,\sigma}(n)), \qquad 0 \leq \epsilon_{W,\sigma}(n) \leq (1 + \rho^{1-2n})^g - 1. \tag{26}$$

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Proof. Write $z = \rho^{-n}$, $\varphi = \sqrt{\rho}$, and $L = 2\varphi/3$. On the positive arm, divide the P, Q factors by $A_0\rho^n, B_0\rho^n$ respectively and conjugate by $\mathrm{diag}(\rho, 1)$. The normalized factors are

$$\begin{pmatrix} \rho^{-1}z^2 & L\rho^{-2}z \\ Lz & 1 \end{pmatrix}, \qquad \begin{pmatrix} \rho z^2 & Lz \\ Lz & 1 \end{pmatrix}.$$

The negative arm gives the same pair in reverse roles after exchanging the basis states. Both are bounded entrywise by the rank-one matrix

$$K = \begin{pmatrix} \rho z^2 & \varphi z \\ \varphi z & 1 \end{pmatrix}.$$

Products preserve the entrywise bound, and $\operatorname{tr} K^g = (1 + \rho z^2)^g$. The constant dominant-state path contributes one to the normalized trace. Restoring the removed factors proves the bounds. □

5.3. **A finite rectangle depending only on the degrees.**

Theorem 5.4 (Unequal-degree collision window)**.** *Let $D_\alpha \neq D_\beta$, and define*

$$m = \frac{|D_\alpha - D_\beta|}{2}, \qquad E = \max(D_\alpha, D_\beta),$$
$$h = \min\{j \in \mathbb{Z}_{\geq 0} : \rho^j \geq E\}, \qquad N = 3m + h + 3, \qquad T = E(N + 2).$$

An integer equality $M_\alpha(x) = M_\beta(y)$ requires

$$|x| \leq \left\lceil \frac{T}{D_\alpha} \right\rceil, \qquad |y| \leq \left\lceil \frac{T}{D_\beta} \right\rceil, \qquad M_\alpha(x) < \rho^T/3. \tag{27}$$

The seed still uses only nonnegative phases.

Proof. The degrees are even, so m is a positive integer. First,

$$|m \log(9/5) - j \log \rho| > \frac{1}{81^m + 1} \qquad (j \in \mathbb{Z}). \tag{28}$$

Choose $j_0 \geq 0$ with $1 < (9/5)^m \rho^{-j_0} < \rho$; equality at an endpoint is impossible by taking the field norm. The positive algebraic integers $9^m - 5^m\rho^{j_0}$ and $5^m\rho^{j_0+1} - 9^m$ have nonzero integer norms and conjugates of absolute value below 9^m. Each is therefore greater than 9^{-m}. Dividing by the appropriate positive denominators makes both adjacent multiplicative gaps greater than 81^{-m}. The inequality $\log(1 + u) > u/(1 + u)$ gives (28); all other integers j are farther away.

At phases $x = \sigma n$, $y = \tau n'$, the leading logarithms differ by

$$\frac{D_\alpha - D_\beta}{2} \log(9/5) + (D_\alpha n - D_\beta n' + \sigma t_\alpha - \tau t_\beta) \log \rho.$$

The coefficient of $\log \rho$ is integral. Put $r_m = (81^m + 1)^{-1}$. Since $E \leq \rho^h$, $\rho^6 > 81$ and $\rho^5 > 5$,

$$E\rho^{1-2N} \leq \rho^{-6m-5} < \frac{1}{2 \cdot 81^m + 3} = \frac{r_m}{r_m + 2}.$$

For $gu < 1$, the elementary bound $(1+u)^g \leq \sum_{j\geq 0}(gu)^j = (1-gu)^{-1}$ makes each relative remainder in Lemma 5.3 smaller than $r_m/2$ when its phase magnitude is at least N. Their logarithmic corrections cannot cancel (28). At least one collision phase therefore has magnitude less than N.

The leading coefficient of a length-g word lies strictly between $\rho^{-g}/3$ and $\rho^g/3$, because $\rho^{-1} < B_0 < A_0 < \rho$. Lemma 5.3, with $1 + \rho^{1-2N} < \rho$, gives $\operatorname{tr} W(\rho^{\pm N})/3 < \rho^{g(N+2)}/3 \leq \rho^T/3$. Convexity bounds a value inside either finite phase strip by its endpoints, so the common value is below $\rho^T/3$. Positivity also gives the lower bound $\operatorname{tr} W(\rho^{\sigma n})/3 > \rho^{g(n-1)}/3$. Comparing the two bounds yields $n < 1 + T/g$, hence the integer ceilings in (27). □

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This theorem bounds the remaining candidates; it does not assert that any candidate collides. It also does not bound the degrees of a hypothetical collision. The next section supplies a different finite reduction, expressed in Farey numerators and denominators rather than Meridian phases.

6. Corridor walls and numerator–denominator windows

6.1. **A second way to travel through the same tree.** The corridor observation groups the walls between neighboring phasors. After an initial entrance, a wall follows a fixed-numerator family

$$m_{p/(q_0+np)}, \qquad n \geq 0.$$

Both Meridian and phasor edges occur in such a route. Their common law is the advance $q \mapsto q+p$, not a rule that distinguishes the two edge colors. For the blue corridor the fixed-numerator portions are $3/(4+3n)$ and $3/(5+3n)$, tending to the historical coordinate $c = 1/3$. The first wall begins

$$169 \xrightarrow{\text{phasor}} 2897 \xrightarrow{\text{Meridian}} 51641 \xrightarrow{\text{phasor}} 925765 \longrightarrow \cdots,$$

and the other begins $433, 7561, 135137, 2423525, \ldots$ with the opposite edge alternation.

Corridor role and edge color must be kept separate. With corridor-wall travel designated *standard* and a new entrance *exotic*, the two entry routes here are

$$5 \xrightarrow{\text{exotic}} 29 \xrightarrow{\text{standard}} 433 \xrightarrow{\text{standard}} 7561 \longrightarrow \cdots, \qquad 13 \xrightarrow{\text{exotic}} 169 \xrightarrow{\text{standard}} 2897 \longrightarrow \cdots.$$

Both exotic entrances are Meridian edges. In the first route the initial standard step $29 \to 433$ changes the numerator from two to three; the fixed-numerator formula starts at 433. Thus excluding the concave ceiling arc from a wall's recurrence does not move the exotic entrance to that arc. The graph and its actual depth (14) are unchanged.

6.2. **The exact recurrence behind the observed regularity.** Use the standard Christoffel matrices

$$D_a = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad D_b = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{29}$$

Their trace encoding is

$$3m_{p/q} = \operatorname{tr}\left[\prod_{j=1}^{p} D_a^{\lfloor jq/p \rfloor - \lfloor (j-1)q/p \rfloor} D_b\right]. \tag{30}$$

The word is a cyclic conjugate of the lower Christoffel word. Trace is unchanged by that rotation. The normalization is fixed by $\operatorname{tr} D_a = 3$, $\operatorname{tr}(D_a D_b) = 6$ and the Farey trace identity, which give the Markov recurrence with the initial values one and two. This is the classical matrix-word framework [9]; Section 7 also gives its matching interpretation.

Replacing q by $q_0 + np$ adds n to every exponent in (30). The eigenvalues of D_a are ρ, ρ^{-1}. Choose its orthogonal eigenbasis so that D_b becomes

$$\begin{pmatrix} \frac{3}{2} + \frac{3}{2\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{3}{2} - \frac{3}{2\sqrt{5}} \end{pmatrix}.$$

All four entries are positive. Expanding the trace over cyclic choices of the two eigenstates gives the exact identity

$$m_{p/(q_0+np)} = \sum_{j=0}^{p} c_{p,q_0,j}\, \rho^{(p-2j)n}, \qquad c_{p,q_0,j} > 0. \tag{31}$$

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Every mode occurs. The least order of a constant-coefficient recurrence is therefore $p+1$, with characteristic polynomial

$$\prod_{j=0}^{p}(X-\rho^{p-2j}).$$

Pairing conjugate roots shows that its coefficients are integers. For $p=3$ the polynomial is $(X^2-18X+1)(X^2-3X+1)$, so either blue wall satisfies

$$m_{3/(q+12)} = 21m_{3/(q+9)} - 56m_{3/(q+6)} + 21m_{3/(q+3)} - m_{3/q}. \tag{32}$$

This equality holds at every step, for both colors of edge.

The exact regularity is a shared recurrence, not a constant output ratio. The positive spectral sum gives strict log-convexity, hence

$$\log_{10}\frac{m_{p/(q_0+(n+1)p)}}{m_{p/(q_0+np)}} \nearrow p\log_{10}\rho.$$

The small deviation is retained in the collision argument, rather than discarded as an asymptotic error. Figure 7 shows both the geometric walls and that deviation.

6.3. **A positive correction with a uniform bound.** The dominant eigenstate in (30) yields

$$3m_{p/q} = \left(\frac{3}{\sqrt{5}}\right)^{p}\left(\frac{1+\sqrt{5}}{2}\right)^{2q+p}(1+\varepsilon_{p,q}), \tag{33}$$

where

$$0 < \varepsilon_{p,q} \le \left(1+\rho^{-2\lfloor q/p\rfloor}\right)^{p} - 1. \tag{34}$$

To prove the bound, divide the transformed D_b by its upper-left entry; every resulting entry is at most one. Choosing the contracting eigenstate at a factor costs at least $\rho^{-2\lfloor q/p\rfloor}$. Summing over all nonempty subsets of contracting choices gives (34). This bound is independent of the remainder class of q modulo p.

Along a fixed wall the correction decreases strictly to zero. Consequently its logarithmic deviations telescope exactly:

$$\sum_{k=0}^{n-1}\left[p\log\rho - \log\frac{m_{p/(q+(k+1)p)}}{m_{p/(q+kp)}}\right] = \log\frac{1+\varepsilon_{p,q}}{1+\varepsilon_{p,q+np}}. \tag{35}$$

Thus there is no accumulated error proportional to the path length. For a route changing corridors, subtraction of the endpoint formula still telescopes, but monotone decay is claimed only within a fixed wall.

Theorem 6.1 (Necessary correction gap)**.** *If $p \neq p'$ and $m_{p/q} = m_{p'/q'}$ for reduced labels, then*

$$\frac{|\varepsilon_{p,q} - \varepsilon_{p',q'}|}{1+\min(\varepsilon_{p,q},\varepsilon_{p',q'})} > 9^{-|p-p'|}. \tag{36}$$

Proof. Assume $p > p'$. The ratio of the leading terms is

$$R = \left(\frac{3(1+\sqrt{5})}{2\sqrt{5}}\right)^{p-p'}\rho^{q-q'}.$$

Its quadratic-field norm is $(9/5)^{p-p'}$, so $R \neq 1$. The number $(\sqrt{5})^{p-p'}(R-1)$ is an algebraic integer, because the powers of $(1+\sqrt{5})/2$ and ρ are units. Its nonzero norm gives

$$|(R-1)(\overline{R}-1)| \ge 5^{-(p-p')}, \qquad \overline{R} = (9/5)^{p-p'}/R > 0.$$

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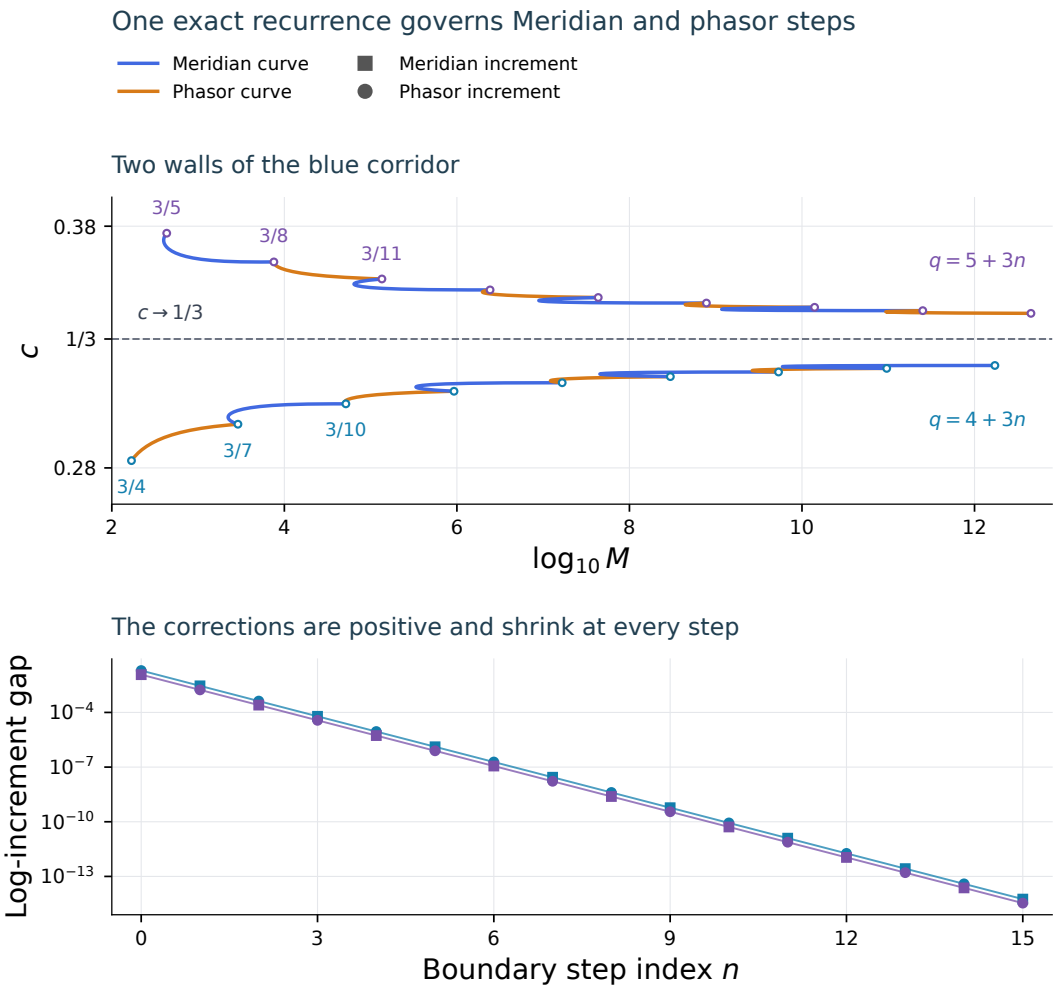


Figure 7. The two fixed-numerator walls of the blue corridor. The concave ceiling arc is omitted from this recurrence illustration. Both walls obey (32). In the lower panel, the positive difference between $3\log_{10}\rho$ and the actual logarithmic increment decreases at every step. Squares denote Meridian increments and circles phasor increments. The values are exact integers; only the plotted curves and logarithms are numerical.

Put $\eta = \max(R, R^{-1}) - 1$. If $R \geq 1$, then $|R - 1| = \eta$ and $|\overline{R} - 1| < (9/5)^{p-p'}$. If $R < 1$, their product is $\eta((9/5)^{p-p'} - 1/(1+\eta)) < \eta(9/5)^{p-p'}$. Either case forces $\eta > 9^{-(p-p')}$. At equality of the Markov values, $R = (1+\varepsilon_{p',q'})/(1+\varepsilon_{p,q})$, so η is precisely the left side of (36). □

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6.4. **The corridor collision window.**

Theorem 6.2 (Numerator–denominator window)**.** *Let $p \neq p'$, $P = \max(p, p')$, and let K be the least nonnegative integer satisfying*

$$\rho^{2K} > P(9^{|p-p'|} + 1). \tag{37}$$

Every equality $m_{p/q} = m_{p'/q'}$ requires

$$\max(q, q') \leq P(K + 1) - 1. \tag{38}$$

For bounded numerators a simpler, weaker consequence is $\max(q, q') \leq 2P^2 + P - 1$.

Proof. If both quotient floors are at least K, then $P\rho^{-2K} < 1/(9^{|p-p'|} + 1)$. The geometric-series estimate used in Lemma 5.3 makes both corrections strictly smaller than $9^{-|p-p'|}$, contradicting Theorem 6.1. Thus one denominator, say q, is at most $Kp - 1$.

Both matrices in (29) have operator norm ρ. The trace bound and the positive leading term imply

$$\frac{1}{3}\rho^{q'} < m_{p'/q'} = m_{p/q} \leq \frac{2}{3}\rho^{q+p}.$$

Since $\log_\rho 2 < 1$ and the exponents are integers, $q' \leq q + p$. It follows that both denominators are at most $P(K + 1) - 1$. For the simpler bound it suffices to use $K = 2P$: $\rho^{4P} > 36^P > P(9^{P-1} + 1)$ for $P \geq 2$, and $|p - p'| \leq P - 1$. □

For example, numerators at most 64 permit $K = 75$, so every remaining comparison lies at denominators at most 4863. With $P = 64$ and numerator difference five, $K = 8$ gives 575. These are bounds on candidates, not new numerical verification claims.

Theorem 5.1 first discards equal Meridian degrees. For the rest, the phase rectangle (27) and denominator cap (38) can be imposed together through the dictionary (18). Fixed-numerator ordering already separates all labels with $p = p'$. What remains is a comparison of different numerators and degrees inside their joint windows. Neither numerator nor degree is globally bounded by these theorems, so this is not yet a finite reduction of the entire unicity conjecture.

7. Snake-graph counts and ordering proofs

7.1. **From a Farey label to an actual matching count.** A snake graph is a chain of square tiles, each attached to the right of or above the preceding tile. Shared edges are counted once. A perfect matching covers every vertex exactly once. The Markov snake associated with a reduced $0 < p/q < 1$ is built as follows. Take the lower staircase from $(0, 0)$ to (q, p): at each horizontal coordinate j, step right and then up to height $\lfloor pj/q \rfloor$. Write right steps as a and up steps as b. The resulting lower Christoffel word has the form $w_{p/q} = azb$. Remove its first and last letters; start with one square and replace each a of z by two added squares to the right and each b by two upward. The resulting snake has $2(p + q) - 3$ tiles.

Write $\mu(w)$ for its number of perfect matchings. The established Markov matching correspondence gives

$$\mu(w_{p/q}) = m_{p/q}. \tag{39}$$

The matching interpretation originated in the REACH work of 2002–2003. Musiker's memorandum, dated 27 June 2002, proposed the recursive graph model [13]; Itsara, Musiker, Propp, and Viana's manuscript of 10 January 2003 stated and proved the snake-matching correspondence [14, Theorem 3.1]. Setting all edge weights to one gives the ordinary perfect-matching counts. Propp's later exposition, first posted in 2005, records the broader collaborative development [15]. These original contributions precede the later proofs of Aigner's orderings.

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For a modern treatment and the Farey-indexed construction, see [12, Section 4.1, Theorem 4.2] and [6]. For example, $w_{2/5} = a^3ba^2b$, its trimmed word is $aabaa$, and the eleven-tile snake has 194 matchings. Auxiliary rearranged words also specify snakes, counted by μ; they need not be canonical Markov snakes.

7.2. **Six pieces and two connector states.** Figure 8 fixes the five three-tile pieces and the three one-square variants of type 6. In a cut between pieces, a matching uses both yellow connector edges or neither: the cut leaves an even number of vertices on either side. Denote these states by 1 and 0. At a selected port remove the two vertices already covered by its connector pair. Direct counting within the pieces gives

(40)

piece	00	01	10	11
1	5	2	2	1
2	5	3	3	2
3	5	2	3	1
4	5	3	2	1
5	5	2	2	1
6i	2	1	1	1

The state order is incoming, outgoing. The one-port end squares 6ii and 6iii have counts $(2, 1)$.

For the ordering surgeries, use a fixed one-ended chop: retain tile triples $(1, 2, 3), (5, 6, 7), \ldots$, with tiles $4, 8, \ldots$ providing paired connectors. A one-square remainder occurs exactly when $p + q$ is even and is counted with the closed end piece. For every connector sequence σ, multiply the applicable row entries in (40); then sum over σ, with outer ports closed. Each matching determines one sequence and one internal completion per piece, and gluing reverses the construction. This proves the counting rule, without an independence assumption. Equivalently it is a sum of terms $5^{k(\sigma)}3^{j(\sigma)}2^{\ell(\sigma)}$, where the exponents count factors of the indicated size. For the $2/5$ snake the piece list $2, 1, 2$ gives

$$5^3 + 5 \cdot 2 \cdot 3 + 3 \cdot 2 \cdot 5 + 3^2 = 194.$$

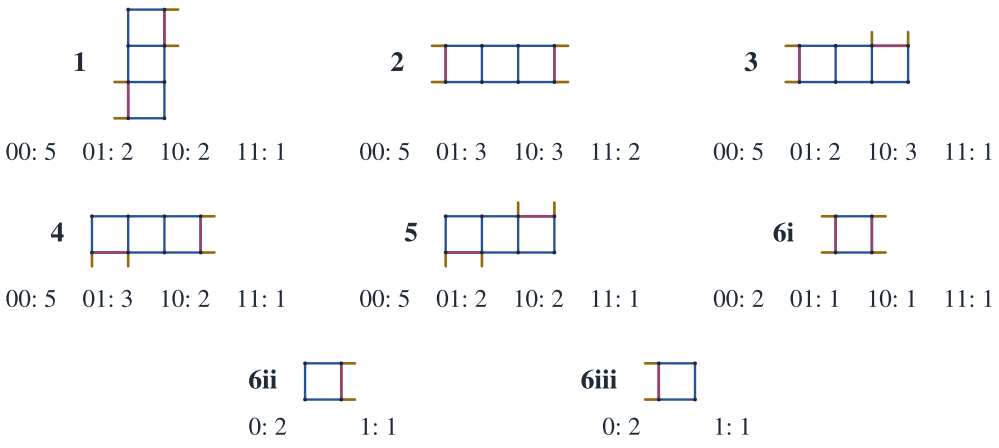


Figure 8. The six physical pieces. Gold edges are paired connectors, magenta marks ports, and dots mark tile vertices. Two-port counts use the order $00, 01, 10, 11$. The central square 6i has counts $(2, 1, 1, 1)$; the one-port end squares have $(2, 1)$. Types 1 and 5 have equal count rows but different physical orientations.

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7.3. **Fibonacci-matrix normalization.** The Fibonacci-matrix formulation and normalization argument below are the author's. We begin with the comparison of three consecutive blocks and then apply it to complete words.

With $F_0 = 0, F_1 = 1$ and $F_{-j} = (-1)^{j+1}F_j$, the basic odd-index identity is

$$F_{2k-1}F_{2k+3} - F_{2k+1}^2 = 1. \tag{41}$$

For example, $5 \cdot 34 - 13^2 = 1$. Consider the author's three consecutive Fibonacci blocks:

$$\begin{aligned} A &= \begin{pmatrix} F_{2k+3} & F_{2k+1} \\ F_{2k+1} & F_{2k-1} \end{pmatrix}, \\ B &= \begin{pmatrix} F_{2k+1} & F_{2k-1} \\ F_{2k-1} & F_{2k-3} \end{pmatrix}, \\ C &= \begin{pmatrix} F_{2k-1} & F_{2k-3} \\ F_{2k-3} & F_{2k-5} \end{pmatrix}, \qquad k \geq 2. \end{aligned} \tag{42}$$

Applying (41) twice gives precisely the upper-left comparison used in normalization:

$$\begin{aligned} (AC)_{11} - (B^2)_{11} &= (F_{2k+3}F_{2k-1} - F_{2k+1}^2) \\ &\quad + (F_{2k+1}F_{2k-3} - F_{2k-1}^2) = 2. \end{aligned} \tag{43}$$

The same identities in the other entries give

$$AC - B^2 = \begin{pmatrix} 2 & 6 \\ 0 & 2 \end{pmatrix}, \qquad CA - B^2 = \begin{pmatrix} 2 & 0 \\ 6 & 2 \end{pmatrix}.$$

Thus replacing AC or CA by B^2 decreases the upper-left entry even inside a longer product: the surrounding Fibonacci blocks have nonnegative entries and positive upper-left entries. For $k = 3$ this reads $(AC)_{11} = 34 \cdot 5 + 13 \cdot 2 = 196 > 194 = 13^2 + 5^2 = (B^2)_{11}$.

7.3.1. *Run counts and general index shifts.* To apply the same comparison at arbitrary run lengths, index the blocks by

$$\mathcal{A}_t = \begin{pmatrix} F_{2t+3} & F_{2t+1} \\ F_{2t+1} & F_{2t-1} \end{pmatrix}, \qquad t \geq 0. \tag{44}$$

Thus $A = \mathcal{A}_k$, $B = \mathcal{A}_{k-1}$ and $C = \mathcal{A}_{k-2}$ in (42). The four boundary counts of a straight block of $2t + 1$ tiles are $F_{2t+3}, F_{2t+1}, F_{2t+1}, F_{2t-1}$: these follow by conditioning on the first edge of a ladder, giving the Fibonacci recurrence; at $t = 0$ they are $(2, 1, 1, 1)$. Gluing successive runs with their paired connector states therefore gives, for $w = a^{h_1}b \cdots a^{h_p}b$ with $h_1 \geq 1$ and $h_j \geq 0$,

$$\mu(w) = [\mathcal{A}_{h_1-1}\mathcal{A}_{h_2} \cdots \mathcal{A}_{h_p}]_{11}. \tag{45}$$

The first run is shortened by the removed initial a; the final b closes the other end. Equivalently, a block-index list $(t_1, \ldots, t_p)$ represents $w = a^{t_1+1}ba^{t_2}b \cdots a^{t_p}b$, with $t_1 + \cdots + t_p = q - 1$.

Fibonacci addition gives the scalar index-shifting identity

$$F_{2r+1}F_{2s+1} - F_{2r-1}F_{2s+3} = F_{2(r-s)-2}.$$

Apply it to the two summands of each product entry to obtain

$$\mathcal{A}_r\mathcal{A}_s - \mathcal{A}_{r-1}\mathcal{A}_{s+1} = 2\begin{pmatrix} F_{2(r-s)-2} & F_{2(r-s)} \\ F_{2(r-s)-4} & F_{2(r-s)-2} \end{pmatrix}, \quad r \geq 1,\ s \geq 0. \tag{46}$$

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Consequently, whenever two adjacent indices differ by at least two, moving one unit from the larger to the smaller strictly decreases the whole product's upper-left entry. The case $r = s + 2$ is exactly (43). The only remaining local comparison has adjacent indices differing by one; it is an interchange of the two blocks.

7.3.2. *Comparing the surrounding Fibonacci blocks.* For a finite list of block indices, define the ratio of its two open counts by

$$\rho(u_1, \dots, u_m) = \frac{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{21}}{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{11}}, \qquad \rho(\varnothing) = 0.$$

All these ratios lie in $[0, 1)$. Direct multiplication gives

$$\rho(t, U) = \frac{F_{2t+1} + F_{2t-1}\rho(U)}{F_{2t+3} + F_{2t+1}\rho(U)}. \tag{47}$$

Its right-hand side increases strictly with $\rho(U)$, since (41) gives $\det \mathcal{A}_t = 1$. Moreover, another application of that identity gives

$$\frac{F_{2t+1}}{F_{2t+3}} - \frac{F_{2t+3} + F_{2t+1}}{F_{2t+5} + F_{2t+3}} = \frac{1}{F_{2t+3}(F_{2t+5} + F_{2t+3})} > 0.$$

Hence $\rho(t + 1, U) < \rho(t, V)$ for any two finite tails U, V. Two lists can therefore be compared by cancelling equal initial indices: at the first unequal pair, the larger index gives the smaller ratio; an exhausted list has the smaller ratio than any nonempty remainder.

For adjacent blocks in positions $j, j + 1$, the outward lists are $(t_{j-1}, \dots, t_1)$ on the left and $(t_{j+2}, \dots, t_p)$ on the right. At $r = s + 1$, (46) becomes

$$\mathcal{A}_{s+1}\mathcal{A}_s - \mathcal{A}_s\mathcal{A}_{s+1} = 2 \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

After multiplication by the surrounding blocks, its upper-left entry has the sign of

$$\rho(t_{j+2}, \dots, t_p) - \rho(t_{j-1}, \dots, t_1). \tag{48}$$

Indeed, it equals this difference times twice the positive upper-left entries of the two surrounding products. This proves the needed surrounding-block test entirely in the same Fibonacci matrices.

Lemma 7.1 (Fibonacci-matrix normalization)**.** *For coprime positive p, q, every word $w = azb$ with q letters a and p letters b satisfies $\mu(w_{p/q}) \le \mu(w)$.*

Proof. For $p = 1$ there is only the word $a^q b$. Assume $p > 1$ and use the block-index list in (45). Its prefix discrepancies are

$$\delta_j = qj - p(1 + t_1 + \cdots + t_j) \quad (1 \le j \le p), \qquad \delta_0 = \delta_p = 0.$$

Changing (t_j, t_{j+1}) to $(t_j + 1, t_{j+1} - 1)$ decreases only δ_j, by p; the reverse change increases only δ_j, by p. We show that an appropriate change always decreases the matching count until $-p \le \delta_j \le 0$ for every j.

The sign checks use just a prefix-sum identity. Put $h_1 = t_1 + 1$ and $h_i = t_i$ for $i > 1$, the actual run lengths. Whenever $j - m \ge 0$ and $j + m \le p$,

$$\delta_{j-m} + \delta_{j+m} = 2\delta_j + p \left(\sum_{i=j-m+1}^{j} h_i - \sum_{i=j+1}^{j+m} h_i \right). \tag{49}$$

In the block indices, the left sum has the extra 1 precisely when it includes t_1.

First remove positive discrepancies. Choose the rightmost largest $\delta_j > 0$ and transfer one unit from t_{j+1} to t_j. Comparison with its two neighboring discrepancies gives $h_j < q/p < h_{j+1}$; coprimality

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makes these inequalities strict. Thus the transfer is allowed and $t_{j+1} \geq t_j + 1$; for $j = 1$ the gap is at least two. Gaps of at least two are handled by (46).

If $t_{j+1} = t_j + 1$, compare the outward lists. A first unequal pair with the left index larger would make the parenthesis in (49) nonnegative: the central difference is -1, the equal outward indices cancel, and the first unequal difference is at least 1. This contradicts maximality and the choice of the rightmost maximum. If the left list were exhausted first, its extra initial 1 would instead give $\delta_{2j} = 2\delta_j > \delta_j$, again impossible (including simultaneous exhaustion, when $2j = p$). The left outward ratio is therefore strictly larger than the right one. Reversing the interchange in (48) shows that the selected transfer strictly decreases the count. It also decreases $\sum_{j=1}^{p-1} \delta_j$. There are only finitely many nonnegative block-index lists of sum $q - 1$, so this phase ends with all $\delta_j \leq 0$.

Then remove discrepancies below $-p$. Choose the leftmost smallest $\delta_j < -p$ and transfer one unit from t_j to t_{j+1}. For $j > 1$, the neighboring comparisons give $h_j > q/p > h_{j+1}$; for $j = 1$, the inequality $\delta_1 < -p$ gives $t_1 > q/p$, while $t_2 < q/p$. Thus $t_j \geq t_{j+1} + 1$ and the transfer is allowed. Again a gap of at least two is settled by (46).

In the gap-one case, a first unequal outward pair with the left index smaller would make the parenthesis in (49) nonpositive, contradicting minimality and the leftmost choice. If that pair reaches t_1, its extra 1 instead gives $\delta_{2j} \leq 2\delta_j + p < \delta_j$, the same contradiction. If the right list were exhausted first, the identity would give $\delta_{2j-p} = 2\delta_j + p < \delta_j$. Simultaneous exhaustion would give $\delta_j = -p$. Thus the left outward ratio is strictly smaller than the right one, and (48) proves a strict decrease in count. The changed discrepancy is $\delta_j + p < 0$, so positivity is not reintroduced. This phase increases the bounded sum of discrepancies by p each time and terminates with $-p \leq \delta_j \leq 0$.

Finally, $\delta_j \equiv qj \pmod{p}$. For $1 \leq j < p$, coprimality leaves exactly one possible value in $[-p, 0]$, forcing

$$1 + t_1 + \cdots + t_j = \left\lceil \frac{jq}{p} \right\rceil .$$

These are precisely the canonical run lengths. Every transfer has decreased the upper-left product entry, proving the assertion. □

For the canonical word, $h_j = \lceil jq/p \rceil - \lceil (j-1)q/p \rceil$. With $k = 1 + \lfloor (q-1)/p \rfloor$, the factors are the author's A, B from (42), and the end factors are B. For example, $m_{3/8} = [BAB]_{11}$ with $k = 3$. A similar normalization result appears in [9, Proposition 21]; the proof above is self-contained in the author's Fibonacci matrices.

7.4. **The three Aigner orderings.** Fixed-numerator monotonicity was proved by Rabideau and Schiffler [12]; all three orderings were proved by LPTV, Lee–Li–Rabideau–Schiffler, and McShane [9, 10, 11]. The author's Fibonacci matrices prove normalization and the fixed-numerator ordering. His six pieces prove the remaining fixed-sum comparison and the fixed-denominator ordering, using normalization to return to a canonical Markov snake.

Theorem 7.2 (Established monotonicities)**.** *For $i > 0$, whenever both fractions are reduced and lie in $(0, 1)$,*

$$\begin{aligned} &m_{p/q} > m_{(p+i)/(q-i)} && \textit{(fixed sum)},\\ &m_{p/q} < m_{p/(q+i)} && \textit{(fixed numerator)},\\ &m_{p/q} < m_{(p+i)/q} && \textit{(fixed denominator)}. \end{aligned}$$

Fixed numerator. Fibonacci recurrence gives, for $t \geq 1$,

$$\mathcal{A}_t - \mathcal{A}_{t-1} = \begin{pmatrix} F_{2t+2} & F_{2t} \\ F_{2t} & F_{2t-2} \end{pmatrix} .$$

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All entries are nonnegative and the upper-left is positive. Start with $w_{p/(q+i)}$ and delete i internal letters a, retaining the first. Each deletion changes one factor $\mathcal{A}_t$ to $\mathcal{A}_{t-1}$ with $t \geq 1$. Surrounding factors are positive, so the matching count strictly falls. The resulting word ends at (q, p); normalization gives $m_{p/q} \leq \mu(w_*) < m_{p/(q+i)}$. □

Fixed sum. At the same physical cuts, the changes $2 \to 3$ or 4, then 3 or $4 \to 5$, increase no entry of (40); the changed factors are $3 \to 2$ or $2 \to 1$. Rotation $5 \leftrightarrow 1$ preserves the row. Selecting the changed connector and no others exhibits a strictly smaller positive summand whenever an equal-direction join becomes unequal.

The trimmed canonical word begins and ends with a, contains no bb, and has $q - p - 1$ equal joins, all aa. At its rightmost equal join, interchange a, b strictly to the right. That suffix is alternating, so only the chosen join changes. Exactly one prescribed piece changes by the stated rules, while any rotated pieces keep their rows. Each surgery strictly decreases the count. Odd-numbered operations replace one horizontal step by a vertical step in net; even-numbered operations keep the totals. After $2i$ operations the endpoint is $(q - i, p + i)$. There are enough joins because $p + i < q - i$. For the resulting auxiliary word w_*, normalization gives $m_{(p+i)/(q-i)} \leq \mu(w_*) < m_{p/q}$. A similar tail comparison appears in [9, Proposition 22]; here the comparison is proved using the author's six physical pieces. □

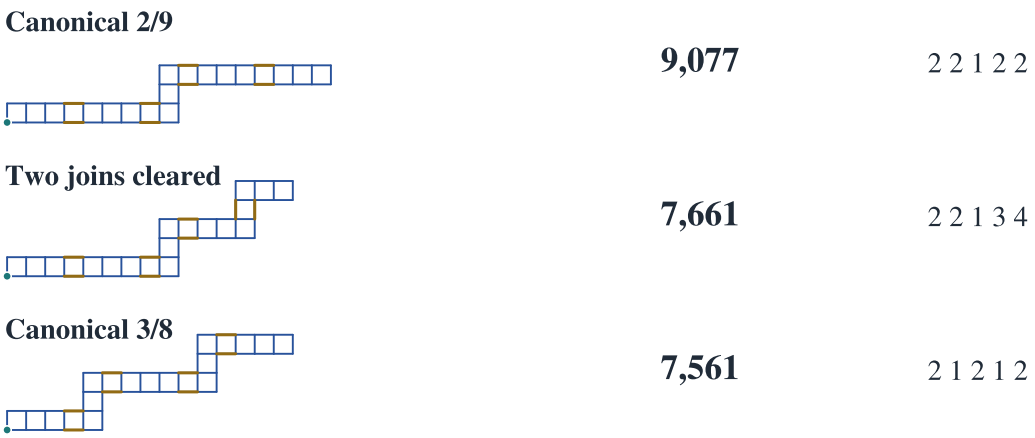


Figure 9. Fixed-sum surgery on the same five pieces. The canonical 2/9 snake has 9077 matchings; after two changes the auxiliary snake has 7661; normalization gives the 3/8 snake with 7561. The intermediate first change, not shown, has 8263. Gold pairs mark cuts and teal dots the starting corners.

Fixed denominator. For the fixed-denominator comparison, refine the same six-piece chop into single-square pieces of type 6, retaining odd-numbered tiles and using the even tiles as paired connectors. A retained square has counts $(2, 1, 1, 1)$ for opposite ports and $(2, 1, 1, 0)$ for adjacent ports. The zero forbids covering their common corner twice. Summing an intervening state regroups these squares into the original three-tile pieces.

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Insert one internal b, adding two vertical squares. The affected local counts are the following complete table, with outside ports fixed:

	before				after			
directions	00	01	10	11	00	01	10	11
a, a	2	1	1	1	5	2	2	1
b, b	2	1	1	1	5	3	3	2
a, b	2	1	1	0	5	3	2	1
b, a	2	1	1	0	5	2	3	1

.

The endpoint caps have the corresponding rows with an outer port closed; the empty trimmed word has the increase $2 \to 5$. No outside count falls, and the all-zero choice witnesses a strict increase. Thus the two added squares compensate for every changed direction.

Starting from $w_{(p+i)/q}$, delete i internal b's and reverse this comparison. At least i are available. Normalizing the resulting word to (q, p) proves $m_{p/q} \le \mu(w_*) < m_{(p+i)/q}$. □

7.5. **Folding an actual snake at its center.** The same local counts also compare families sharing a long initial sequence of pieces. Represent the four counts by

$$R_1 = R_5 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}, \quad R_4 = \begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}.$$

Attaching piece i to an open-count row (x, y) replaces it by $(x, y)R_i$. This is the sum over the common connector state, not an additional assumption about matchings.

For a helm, use the symmetric central chop: retain the middle square as 6i and group retained squares in pairs outward on each side. For odd/odd p, q, each half has $(p+q-2)/2$ retained squares. If this number is odd, one square remains at each outer end, giving pieces 6ii and 6iii with initial counts $(2, 1)$. Otherwise the half starts at $(1, 0)$ and consists of complete three-tile pieces. This repartition requires rebuilding the piece list; one cannot move a cut and keep its old boundary counts.

Proposition 7.3 (Folded matching counts)**.** *For a reduced $0 < p < q$, let (x, y) be the open counts of one reflected half under the central chop. If p, q are odd, then*

$$m_{p/q} = 2x^2 + 2xy + y^2 = (x + y)^2 + x^2. \tag{50}$$

If $p + q$ is odd, then $m_{p/q} = x^2 + y^2$. In either case $\gcd(x, y) = 1$; a nonempty half has positive counts.

Proof. The trimmed Christoffel word is a palindrome: for $0 < j < q$, $\lfloor p(q-j)/q \rfloor = p - 1 - \lfloor pj/q \rfloor$. For odd/odd labels the two halves meet at an opposite-port central square, whose four counts are $2, 1, 1, 1$. The reflected halves contribute x^2, xy, yx, y^2, giving (50). For opposite parities they meet across a single connector and must use its same state, giving $x^2 + y^2$. Every full piece table and every exposed-square table has determinant ± 1, so its integral inverse preserves the gcd of an open row. Starting at $(1, 0)$ proves primitivity. The first square gives $(2, 1)$, and every subsequent permitted table preserves positivity. □

The seed $1/1$ consists only of its central square. The two examples in Figure 10 give $m_{1/3} = 2 \cdot 2^2 + 2 \cdot 2 \cdot 1 + 1^2 = 13$ and $m_{1/5} = 2 \cdot 5^2 + 2 \cdot 5 \cdot 3 + 3^2 = 89$.

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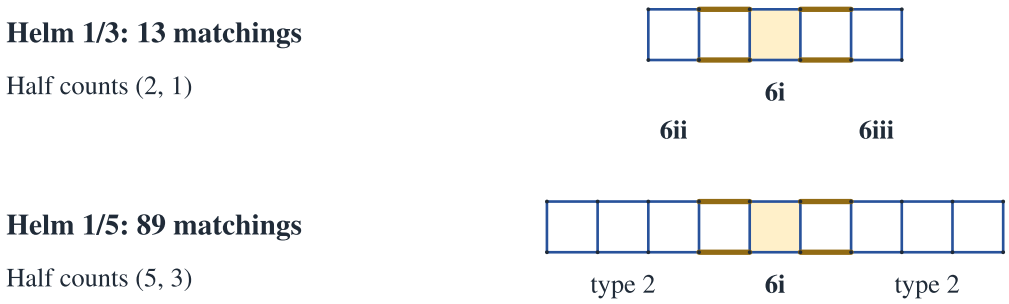


Figure 10. Central and end pieces in actual helms. At 1/3, central square 6i has the end squares 6ii and 6iii as its halves, with counts $(2,1)$. At 1/5, each half is a type-2 triple with counts $(5,3)$. Folding produces 13 and 89.

7.6. **A common prefix leaves one quadratic comparison.** Two folded graphs with the same outer boundary convention may share a prefix with open counts (x,y). Sum over all remaining states, leaving that cut open. The two centers give positive definite symmetric tables $G, \widetilde{G}$ with determinant one and nonnegative off-diagonal entries. For a helm they have the form SQS^{T} with $Q = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$; for opposite-parity labels they have form SS^{T}. Thus

$$m - \widetilde{m} = \alpha x^2 + 2\beta xy + \gamma y^2, \qquad G - \widetilde{G} = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}. \tag{51}$$

Proposition 7.4 (At most one primitive common prefix)**.** *For a nonempty common part and $G \neq \widetilde{G}$, equality requires $\alpha\gamma < 0$ and $\beta^2 - \alpha\gamma$ to be an integer square. There is then at most one primitive positive row (x,y) giving equality. That row must still arise from both actual prescribed snakes.*

Proof. A center table $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ has $ac - b^2 = 1$ and $b \geq 0$. If its two diagonal differences from another table have the same weak sign, then the difference of $b = \sqrt{ac-1}$ has that sign too. A nonzero difference cannot vanish at a positive row. Opposite diagonal signs are therefore necessary. The resulting quadratic in y/x has one positive root. An integer zero obeys $(\gamma y + \beta x)^2 = (\beta^2 - \alpha\gamma)x^2$; a rational square root of an integer is integral. A positive rational root has only one primitive positive representative. □

For an empty prefix, test its actual boundary row. If $G = \widetilde{G}$, the criterion supplies no exclusion. Equal count tables do not certify equal physical graphs: types 1 and 5 already show why orientations and the prescribed Christoffel word must be retained.

A useful uniform exclusion follows directly modulo three. The physical piece tables satisfy $R_i Q R_i^{\mathsf{T}} \equiv Q \pmod 3$, whereas a single retained square gives $-Q$. A helm tail with d retained squares therefore has center table $(-1)^d Q \pmod 3$. Opposite parities give a nonzero difference at every primitive row because the discriminant of Q is -1, a nonsquare modulo three. For equal parity

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and nonzero difference, let $h = \min_{i,j} v_3((G - \widetilde{G})_{ij})$ and write $3^{-h}(G - \widetilde{G}) = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$. Then $h \geq 1$. Expanding the equal determinants and reducing modulo three gives $\gamma \equiv \alpha + \beta$; the normalized discriminant is one or two modulo three. If it is two, every primitive shared row has count difference of valuation exactly h. These are exclusions for specified tails with arbitrary actual common prefixes, not a classification of all possible snake-graph collisions.

8. All-phase verification through degree 10,000

8.1. **Why a finite residue test covers infinitely many phases.** For a period T, define the complete image

$$\mathcal{R}_\alpha(T) = \{M_\alpha(x) \bmod q(T) : 0 \leq x < T\}.$$

Theorem 4.1 makes this the image at *every* integer phase. Disjoint images therefore separate two entire Meridians. If images overlap for several periods, a true equality must use compatible phases in all of them. Define the witness relation

$$W_T(\alpha, \beta) = \{(u, v) \in (\mathbb{Z}/T\mathbb{Z})^2 : M_\alpha(u) \equiv M_\beta(v) \pmod{q(T)}\}.$$

Proposition 8.1 (Compatible witnesses)**.** *Equality of two integer values requires one witness in each selected W_T, with first coordinates agreeing modulo every pairwise gcd of the periods, and second coordinates agreeing in the same way. If no compatible choice exists, the two Meridians are disjoint at all integer phases.*

Proof. A true equality uses the same integers x, y in every modulus and hence supplies the required residues. Conversely, the generalized Chinese remainder theorem lifts compatible coordinates to a phase pair modulo the least common multiple. This converse concerns modular witnesses only; it does not imply equality of the full values. □

The moduli need not be independent. In particular, the periods $36, 38, 40$ are all even, so contradictory parity requirements alone can exclude a pair that survives each separate image test.

8.2. **Complete inventory and the final exclusion.**

Proof of Theorem 1.1. Proposition 3.4 identifies the inventory with coprime positive odd pairs $r \leq s$, $r + s \leq 10{,}000$. It contains 5,067,077 Meridians and 12,837,632,128,426 unordered pairs. The producer enumerates the count-vector address tree; an independent checker enumerates the coprime pairs directly and compares the inventories. The child degree is $2|k|D \pm (|v| - |u|) > D$, where $|u|, |v|$ are the endpoint word lengths. This gives finite exhaustive index ranges.

Every one of the 36 phase residues of every Meridian is evaluated modulo $q(36)$. Equal-residue grouping lists all pairs whose images intersect; its complement is excluded at once. Periods 38 and 40 are then evaluated on every endpoint of the preceding survivor sets. The complete image tests leave 87 pairs. For each of them, project every full witness relation onto the four parity pairs $(u \bmod 2, v \bmod 2)$. The three projected sets have empty intersection in all 87 cases. Proposition 8.1 therefore excludes every pair of different Meridians at every integer phase. Theorem 5.1 also excludes different marks within each individual Meridian, completing the statement. □

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Table 2. Complete cumulative survivor counts from the archived certificates. "Images" forgets meeting phases; "phases" also imposes compatibility. Survivors are modular coincidences, not integer collisions.

Tests retained	$D \le 2{,}000$	$D \le 10{,}000$
36: images	7,121	320,755
36, 38: images	612	17,427
36, 38: phases	2	2,052
36, 38, 40: images	0	87
36, 38, 40: phases	**0**	**0**
36, 38, 40, 42: images	—	1
36, 38, 40, 42, 44: images	—	0

8.3. **What was checked independently.** The degree-10,000 producer uses signed endpoint-orbit updates. The separate evaluator uses Fibonacci staircase words with Euclidean matrix reduction. They agree on all 188,369,412 retained phase evaluations. The checker verifies the 182,414,772 baseline phase tags bijectively, so an omitted or repeated phase record cannot pass unnoticed. Independent sorting of the original unsorted buckets reconstructs all 70,009 cross-Meridian residue groups and the subsequent complete pair sets. Coverage is checked by exact records, not inferred merely from their counts or from a matching digest.

The phase result is checked three ways: full generalized-CRT joins, an independent 16-bit mask computation on $(x \bmod 4, y \bmod 4)$, and the weaker four-state parity projection, already sufficient for the final 87. The degree-2,000 certificate, containing 202,862 Meridians, was completed first; its common survivor sets agree exactly with the larger inventory when identified by helm labels.

Five deliberate corruptions—an altered value, an incorrect modulus, a duplicate tag, an omitted record and an invented pair—were all rejected. The degree-100 control compares 18,684 residues with literal Fibonacci words and the polynomial record of (19). The optional image-only extension through periods 42 and 44 also ends with zero pairs; all 88 period-44 values were separately checked using arbitrary-size Python integers because $q(44) > 2^{63}$.

8.4. **Reproducibility and scope.** The degree reproducibility package [7] contains the inventory, compact residue groups, phase witnesses, producer and checker sources, audit receipts, and a regeneration driver. Large baseline tables are regenerated; their recorded sizes and hashes identify the original run. The reindexed edition retains the original computational kernels with an explicit address-convention adapter. Reversing the written address tail does not change a helm label, degree, phase, or certified numerical value.

This is a deterministic computer-assisted certificate, not an external referee report or proof-assistant formalization. Its scope is a bounded set of Meridians with unbounded phase. The numerical-order verification in Section 9 is a separate computation with its own coverage proof and execution record.

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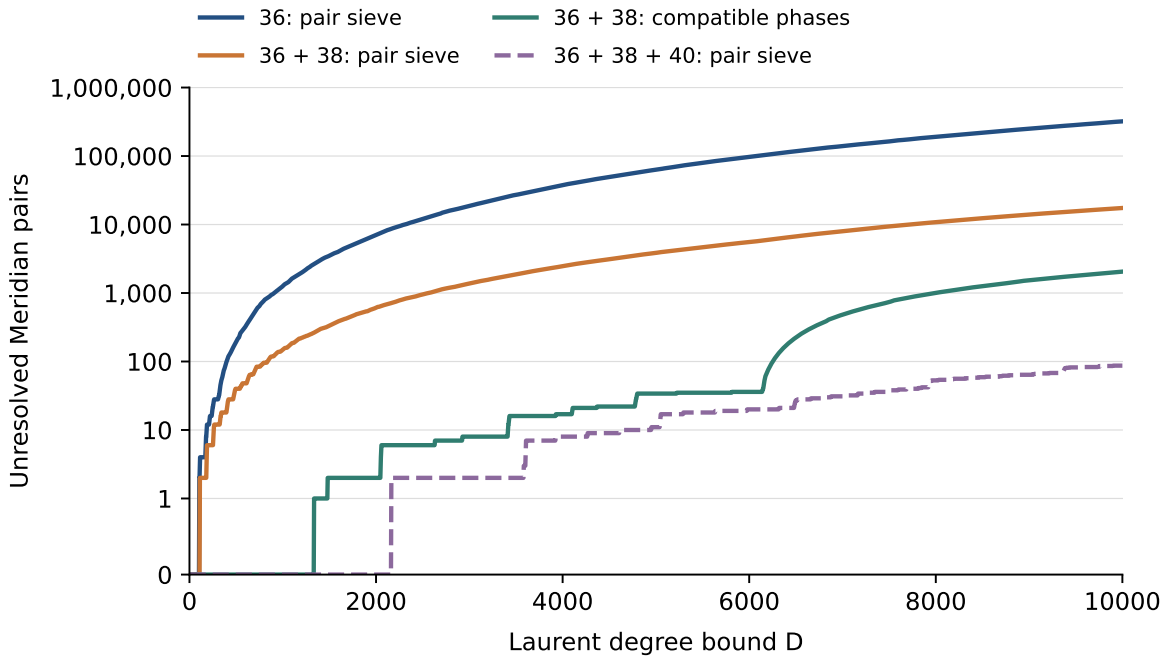


Figure 11. Exact unresolved-pair profiles in the degree-10,000 inventory. The phase-compatible sieve through period 40 has no survivors in the entire displayed range. Lines join discrete even-degree bounds. The figure is a record of the finite certificate, with no extrapolation to larger degrees.

9. Computational chronology and numerical-order verification

9.1. **The earlier Claude-assisted search.** The historical working draft [1], pp. 38–40, reports a Claude-assisted search of **9,581,528,580 sorted Markov triples** against the cutoff $10^{100,000}$, taking **24 minutes** on a two-core machine with 7 GB of memory. It compared maxima modulo 3^{39} in 16 passes and 4,096 buckets; all 21 repeated-residue pairs separated modulo $2^{61}-1$. No equal maxima were reported.

An earlier public announcement of a 24-minute search is timestamped **23 October 2026, 09:54:57.771 CDT** (14:54:57.771 UTC) [2]. Its caption says “9.3 billion”; the later draft gives the exact count above. This dates the *announcement*, not the beginning or end of execution. Original execution timestamps have not been recovered. The draft itself labels these as historical computational reports whose original logs and floating-point cutoff bounds had not been independently certified. We preserve that status here. This search is separate from the later ten-billion anchor, and its reported 24 minutes are not included in the cumulative timing below.

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9.2. **The completed fifty-billion verification.** The archived computation [3] establishes that each of the **first 50,000,000,000 distinct Markov values in numerical order** is the maximum of exactly one normalized positive Markov triple. This complements the degree-10,000 result: the latter covers infinitely many phases on a bounded set of Meridians, whereas this computation certifies a numerical prefix. Its coverage and pruning proofs are collected in the separate verification supplement [4].

Run date and duration. The recorded execution began on **28 October 2026 at 03:32:57 CDT** and finished, including its final audit, at **05:38:34 CDT** (UTC−05:00). The underlying timestamps are 08:32:57.096455–10:38:33.644612 UTC. Two completed coordinator invocations occupied **7,492.102807 seconds: 2 h 04 min 52.103 s**. Including the original ten-billion anchor, the cumulative recorded execution path is **8,330.980040 seconds: 2 h 18 min 50.980 s**. The calendar span is 7,536.548157 seconds; it additionally includes the approximately 44-second pause between invocations. These are historical execution dates and measured durations, distinct from this edition's publication date.

Table 3. The completed fifty-billion run. Counts are configurations unless explicitly identified as distinct values.

Certified distinct-value prefix	50,000,000,000
Complete outer domain	71,517,775,875
New retained configurations compared	54,940,210,994
Repeated primary residues / candidate configurations	346 / 692
Actual equal-value collisions	0

The numerical cutoff was $B = F_{1,147,147}$, with 796,397 bits. The inner height $K = 573{,}574$ contains 50,000,097,577 configurations at or below B; the proved outer height $L = 685{,}979$ contains every competitor below that cutoff. The original ten-billion cutoff certificate justifies 10,000,015,326 inner omissions. Strict certified lower bounds exclude a further 6,577,549,555 configurations above B.

All 32 primary prefix passes and all 8,192 complete buckets finished. Every repeated residue modulo 2^{64} was replayed; all 346 candidate pairs separated modulo 5^{27}. The final auditor independently reconstructed their 692 full integer triples and checked the Markov equation, residues and unequal maxima, as well as the 38 anchor candidates. No probabilistic hash-collision assumption enters this comparison.

The run used eight workers on Linux x86-64, an eight-CPU-equivalent quota and an 8-GiB memory limit, with frozen GCC 13.3.0 executables. The timing includes generation, partitioning, sorting, replay, orchestration and the 107.371720-second final audit; earlier preflight and later archival checks are excluded. The supplement gives the stage timings, retained receipts, source identities, audit scope and reproducibility limitations. This edition reports the completed run; it does not announce a result for the planned 500-billion verification.

10. What remains to prove

The organization now separates three questions. The address bijection settles coverage of configurations. Fixed-degree ordering and the two collision windows restrict equality of maxima. Complete residue witnesses settle the remaining comparisons for the degree-bounded family in Theorem 1.1. The fifty-billion computation additionally excludes every competitor for a finite prefix in numerical order.

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For unrestricted unicity the unresolved pair has different Meridian degrees, the same modulo-27 class, and different Farey numerators. It must satisfy both its phase rectangle and its numerator–denominator window. A proof must exclude those candidates uniformly as the degrees and numerators grow. Enlarging a finite certificate alone does not supply that step. Likewise, an invertible corridor recurrence or a long modular period does not by itself prevent an earlier match of one scalar residue.

Corridors offer a way to formulate the next comparison with explicit entrance states and independently chosen wall lengths. Their positive corrections are exact and telescope; the common-prefix snake calculation offers a different reduction to a quadratic in primitive counts. Neither mechanism has yet been shown to cover every unrestricted pair. The purpose of retaining their precise hypotheses is to make the remaining uniform statement identifiable, rather than hide it inside an induction.

Data and code. The editable source accompanying this edition includes all figure assets, the revised Figure 3 generator, a source and figure inventory, and checks used for this rewrite. The separate degree reproducibility package contains the full computational evidence cited in Section 8. The fifty-billion proof and timing supplement accompanies this edition; its full frozen computational archive is cited separately [3, 4]. The corridor derivations are developed further in the supplementary research record [8]. Numerical drawings are explanatory; the arithmetic claims use the displayed identities or complete certificates.

Author contributions and AI assistance. The author conceived and directed the research program, including the hyperbolic representation and geometric descent, the two-dimensional host–phase landscape, the Meridian indexing and alternative tree, the six-piece snake-graph decomposition, the Fibonacci-matrix formulation and normalization argument, and the corridor observation. The author proposed using these structures for coverage, comparisons and verification, and specified the questions pursued at each stage. The Aigner monotonicities are established results; their prior proofs are credited in Section 7. The Fibonacci-matrix normalization argument presented there is the author's; the related earlier normalization result is cited for comparison, and the proof is given in full from Fibonacci identities.

Generative AI provided substantial mathematical and computational assistance, not only language editing. The development record identifies ChatGPT (OpenAI), with the recorded GPT-6.0 Astra and GPT-6.1 Sol model labels and reasoning settings, and Claude 5.5 Opus Extra (Anthropic) in the preceding geometric-descent work; exact backend revision identifiers were not retained. Within the author's framework, these tools assisted with detailed derivations, examples and counterexamples, proof construction and revision, verifier and independent-checker code, figures, and manuscript drafting. ChatGPT also assisted with this reorganization and Figure 3 revision. The project's separate computational checks are algorithmic audits, not external peer review. The author is responsible for the claims, references and final submission.

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References

[1] E. F. W. Heffern, *Geometric Descent on the Markov Surface*, working draft v32, archived October 2026, pp. 38–40, historical computational reports. The earlier search was assisted by Claude.

[2] E. F. W. Heffern, public LinkedIn announcement of a 24-minute Markov-number search, 23 October 2026, 14:54:57.771 UTC, activity 7519411048115003392. Platform metadata and the source URL are retained in the verification supplement's evidence packet; see the public announcement.

[3] E. F. W. Heffern, *Markov 50 Billion Verification Package*, version 1, completed and sealed 28 October 2026; public derivative version 1.1 in *Computational supplements for Meridians on the Markov Surface*, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

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Original: v5_ps_Markov_Meridians_TLMS.pdf

[4] E. F. W. Heffern, *Computational chronology and proofs for the fifty-billion verification*, supplement to the present manuscript, version 4, 7 November 2026.
[5] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013. doi:10.1007/978-3-319-00888-2.
[6] İ. Çanakçı and R. Schiffler, *Snake graphs and continued fractions*, European J. Combin. **86** (2020), 103081. doi:10.1016/j.ejc.2020.103081.
[7] E. F. W. Heffern, *Meridian Degree 10,000 Reproducibility Package*, computational supplement, 31 October 2026; address-convention revision, 7 November 2026. Original computational release in *Computational supplements for Meridians on the Markov Surface*, version 1.0, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.
[8] E. F. W. Heffern, *Markov Meridian Residue Supplement*, version 1.18, and *Corridor-to-Corridor Induction*, version 1.3, unpublished supplementary research records, 7 November 2026. The corridor results used here are proved in Section 6.
[9] C. Lagisquet, E. Pelantová, S. Tavenas and L. Vuillon, *On the Markov numbers: fixed numerator, denominator, and sum conjectures*, Adv. Appl. Math. **130** (2021), 102227. doi:10.1016/j.aam.2021.102227. Preprint: arXiv:2010.10335.
[10] K. Lee, L. Li, M. Rabideau and R. Schiffler, *On the ordering of the Markov numbers*, Adv. Appl. Math. **143** (2023), 102453. doi:10.1016/j.aam.2022.102453.
[11] G. McShane, *Convexity and Aigner's conjectures*, preprint (2021), arXiv:2101.03316.
[12] M. Rabideau and R. Schiffler, *Continued fractions and orderings on the Markov numbers*, Adv. Math. **370** (2020), 107231. doi:10.1016/j.aim.2020.107231.
[13] G. Musiker, *A conjectured combinatorial interpretation for Markov numbers*, REACH memorandum, 27 June 2002. Original memorandum; REACH archive.
[14] A. Itsara, G. Musiker, J. Propp, and R. Viana, *Combinatorial interpretations for the Markov numbers*, REACH manuscript, 10 January 2003, Theorem 3.1. Original manuscript.
[15] J. Propp, *The combinatorics of frieze patterns and Markoff numbers*, arXiv:math/0511633, first posted 25 November 2005; published in *Integers* **20** (2020), A12.

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Aigner's Three Monotonicities from Fibonacci Matrices and Six Snake Pieces

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Related earlier presentations of the Fibonacci-matrix and six-piece arguments appear in the Meridian manuscript editions v4_ps and v5_ps. No earlier standalone Zenodo draft or separate LinkedIn announcement was verified.

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Complete public Zenodo description

Starting from the established fact that Markov numbers count perfect matchings of their Markov snake graphs, this six-page paper gives a self-contained proof of Aigner's fixed-numerator, fixed-sum and fixed-denominator orderings. The author's Fibonacci matrices establish normalization and fixed-numerator monotonicity; his six-piece decomposition establishes the remaining fixed-sum comparison and fixed-denominator monotonicity. The normalization begins with (AC)_11 - (B^2)_11 = 2 for consecutive odd-indexed Fibonacci blocks. The fixed-sum example is illustrated in Figure 2, and the three ordering proofs have parallel subsection headings.

The matching correspondence is credited to Musiker's 27 June 2002 memorandum and the Itsara-Musiker-Propp-Viana manuscript of 10 January 2003; earlier proofs of Aigner's orderings are cited separately.

Included are the six-page PDF and editable LaTeX source, figures, and finite-check scripts and reports.

Funding: No funding. Generative-AI assistance in proof development, exact checking, reference retrieval and typesetting is disclosed in the manuscript; the author directed the research and is responsible for the work.

Published files

Original filename / checksum	Bytes
v3_Aigner_Fibonacci_Six_Piece_Source.zip MD5 16b3be89a69d75cbe7caf1589228d0b4 SHA256 2ba24326784f097128a7d668db387bcc766b4f5d638dcd58480ab1cd5e1cfe88	42,272
v3_Aigner_Fibonacci_Six_Piece_Proof.pdf MD5 105e8b0c34ccb7400cad68a7570c4e9e SHA256 2f0dc276bd85e415456565b6ca90465fd2627269c5e72c8984ef6f46a4715067	443,270

All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

Original: v3_Aigner_Fibonacci_Six_Piece_Proof.pdf

AIGNER'S THREE MONOTONICITIES FROM FIBONACCI MATRICES AND SIX SNAKE PIECES

ELLEARD FELIX WEBSTER HEFFERN

Abstract. Starting from the established fact that Markov numbers count perfect matchings of their Markov snake graphs, we give a self-contained proof of the fixed-sum, fixed-numerator, and fixed-denominator orderings proposed by Aigner. The author's Fibonacci-matrix normalization begins with $(AC)_{11}-(B^2)_{11}=2$ for consecutive odd-indexed Fibonacci blocks. Fibonacci matrices prove normalization and fixed-numerator monotonicity. The author's six-piece decomposition proves the remaining fixed-sum comparison and fixed-denominator monotonicity. We credit the matching correspondence to its original 2002–2003 REACH development and distinguish that history from the later ordering theorems.

1. The starting correspondence and the three orderings

Write $m_{p/q}$ for the Markov number with the Farey label used in Aigner's conjectures. A snake graph is a chain of square tiles, each attached to the right of or above the preceding tile. Shared edges are counted once. A perfect matching covers every vertex exactly once. The Markov snake associated with a reduced $0<p/q<1$ is built as follows. Take the lower staircase from $(0,0)$ to (q,p): at each horizontal coordinate j, step right and then up to height $\lfloor pj/q\rfloor$. Write right steps as a and up steps as b. The resulting lower Christoffel word has the form $w_{p/q}=azb$. Remove its first and last letters; start with one square and replace each a of z by two added squares to the right and each b by two upward. The resulting snake has $2(p+q)-3$ tiles.

Write $\mu(w)$ for its number of perfect matchings. Our starting precept is the established Markov matching correspondence

$$\mu(w_{p/q})=m_{p/q}. \tag{1}$$

For example, $w_{2/5}=a^3ba^2b$, its trimmed word is $aabaa$, and the eleven-tile snake has 194 matchings. Auxiliary rearranged words also specify snakes, counted by μ; they need not be canonical Markov snakes.

The matching interpretation originated in the REACH work of 2002–2003. Musiker's memorandum, dated 27 June 2002, proposed the recursive graph model [1]; Itsara, Musiker, Propp, and Viana's manuscript of 10 January 2003 stated and proved the snake-matching correspondence [2, Theorem 3.1]. Setting all edge weights to one gives the ordinary perfect-matching counts. Propp's later exposition, first posted in 2005, records the broader collaborative development [3]. These original contributions precede the later proofs of Aigner's orderings.

Aigner formulated these orderings in [4, Chapter 10]. Fixed-numerator monotonicity was proved by Rabideau and Schiffler [5]; all three orderings were proved by LPTV, Lee–Li–Rabideau–Schiffler, and McShane [6, 7, 8]. The author's Fibonacci matrices prove normalization and the fixed-numerator ordering; his six pieces prove the remaining fixed-sum comparison and the fixed-denominator ordering. Every argument after the matching correspondence is proved below.

Theorem 1 (Aigner's three monotonicities)**.** *For $i>0$, whenever both fractions are reduced and lie in $(0,1)$,*

$$\begin{aligned} &m_{p/q}>m_{(p+i)/(q-i)} && \textit{(fixed sum)},\\ &m_{p/q}<m_{p/(q+i)} && \textit{(fixed numerator)},\\ &m_{p/q}<m_{(p+i)/q} && \textit{(fixed denominator)}. \end{aligned}$$

2. Six pieces and two connector states

Figure 1 fixes the five three-tile pieces and the three one-square variants of type 6. In a cut between pieces, a matching uses both yellow connector edges or neither: the cut leaves an even number of vertices on either side.

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Denote these states by 1 and 0. At a selected port remove the two vertices already covered by its connector pair. Direct counting within the pieces gives

(2)

piece	00	01	10	11
1	5	2	2	1
2	5	3	3	2
3	5	2	3	1
4	5	3	2	1
5	5	2	2	1
6i	2	1	1	1

The state order is incoming, outgoing. The one-port end squares 6ii and 6iii have counts $(2, 1)$.

For the ordering surgeries, use a fixed one-ended chop: retain tile triples $(1, 2, 3), (5, 6, 7), \ldots$, with tiles $4, 8, \ldots$ providing paired connectors. A one-square remainder occurs exactly when $p + q$ is even and is counted with the closed end piece. For every connector sequence σ, multiply the applicable row entries in (2); then sum over σ, with outer ports closed. Each matching determines one sequence and one internal completion per piece, and gluing reverses the construction. This proves the counting rule, without an independence assumption. Equivalently it is a sum of terms $5^{k(\sigma)}3^{j(\sigma)}2^{\ell(\sigma)}$, where the exponents count factors of the indicated size. For the 2/5 snake the piece list $2, 1, 2$ gives

$$5^3 + 5 \cdot 2 \cdot 3 + 3 \cdot 2 \cdot 5 + 3^2 = 194.$$

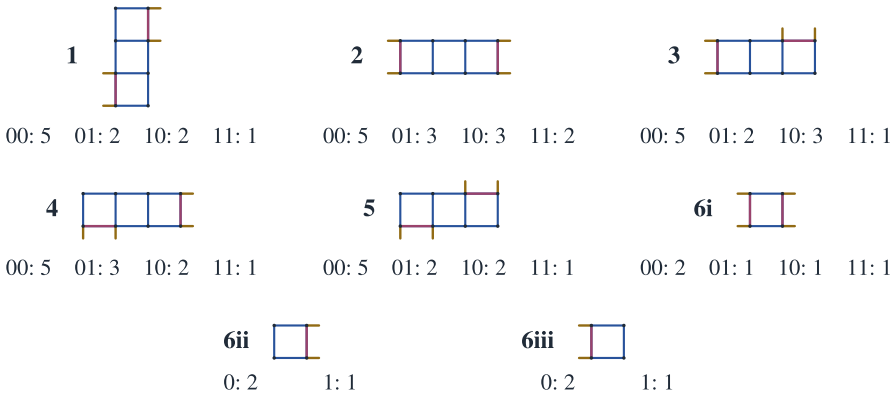


Figure 1. The six physical pieces. Gold edges are paired connectors, magenta marks ports, and dots mark tile vertices. Two-port counts use the order $00, 01, 10, 11$. The central square 6i has counts $(2, 1, 1, 1)$; the one-port end squares have $(2, 1)$. Types 1 and 5 have equal count rows but different physical orientations.

3. Normalization with Fibonacci matrices

The Fibonacci-matrix formulation and normalization argument below are the author's. We begin with the comparison of three consecutive blocks and then apply it to complete words.

With $F_0 = 0, F_1 = 1$ and $F_{-j} = (-1)^{j+1}F_j$, the basic odd-index identity is

$$F_{2k-1}F_{2k+3} - F_{2k+1}^2 = 1. \tag{3}$$

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For example, $5 \cdot 34 - 13^2 = 1$. Consider the author's three consecutive Fibonacci blocks:

$$A = \begin{pmatrix} F_{2k+3} & F_{2k+1} \\ F_{2k+1} & F_{2k-1} \end{pmatrix},$$
$$B = \begin{pmatrix} F_{2k+1} & F_{2k-1} \\ F_{2k-1} & F_{2k-3} \end{pmatrix}, \tag{4}$$
$$C = \begin{pmatrix} F_{2k-1} & F_{2k-3} \\ F_{2k-3} & F_{2k-5} \end{pmatrix}, \qquad k \geq 2.$$

Applying (3) twice gives precisely the upper-left comparison used in normalization:

$$\begin{aligned}(AC)_{11} - (B^2)_{11} = &\left(F_{2k+3}F_{2k-1} - F_{2k+1}^2\right) \\ &+ \left(F_{2k+1}F_{2k-3} - F_{2k-1}^2\right) = 2.\end{aligned} \tag{5}$$

The same identities in the other entries give

$$AC - B^2 = \begin{pmatrix} 2 & 6 \\ 0 & 2 \end{pmatrix}, \qquad CA - B^2 = \begin{pmatrix} 2 & 0 \\ 6 & 2 \end{pmatrix}.$$

Thus replacing AC or CA by B^2 decreases the upper-left entry even inside a longer product: the surrounding Fibonacci blocks have nonnegative entries and positive upper-left entries. For $k = 3$ this reads $(AC)_{11} = 34 \cdot 5 + 13 \cdot 2 = 196 > 194 = 13^2 + 5^2 = (B^2)_{11}$.

3.1. **Run counts and general index shifts.** To apply the same comparison at arbitrary run lengths, index the blocks by

$$\mathcal{A}_t = \begin{pmatrix} F_{2t+3} & F_{2t+1} \\ F_{2t+1} & F_{2t-1} \end{pmatrix}, \qquad t \geq 0. \tag{6}$$

Thus $A = \mathcal{A}_k$, $B = \mathcal{A}_{k-1}$ and $C = \mathcal{A}_{k-2}$ in (4). The four boundary counts of a straight block of $2t+1$ tiles are $F_{2t+3}, F_{2t+1}, F_{2t+1}, F_{2t-1}$: these follow by conditioning on the first edge of a ladder, giving the Fibonacci recurrence; at $t = 0$ they are $(2, 1, 1, 1)$. Gluing successive runs with their paired connector states therefore gives, for $w = a^{h_1}b \cdots a^{h_p}b$ with $h_1 \geq 1$ and $h_j \geq 0$,

$$\mu(w) = [\mathcal{A}_{h_1-1}\mathcal{A}_{h_2} \cdots \mathcal{A}_{h_p}]_{11}. \tag{7}$$

The first run is shortened by the removed initial a; the final b closes the other end. Equivalently, a block-index list $(t_1, \ldots, t_p)$ represents $w = a^{t_1+1}ba^{t_2}b \cdots a^{t_p}b$, with $t_1 + \cdots + t_p = q - 1$.

Fibonacci addition gives the scalar index-shifting identity

$$F_{2r+1}F_{2s+1} - F_{2r-1}F_{2s+3} = F_{2(r-s)-2}.$$

Apply it to the two summands of each product entry to obtain

$$\mathcal{A}_r\mathcal{A}_s - \mathcal{A}_{r-1}\mathcal{A}_{s+1} = 2\begin{pmatrix} F_{2(r-s)-2} & F_{2(r-s)} \\ F_{2(r-s)-4} & F_{2(r-s)-2} \end{pmatrix}, \quad r \geq 1,\ s \geq 0. \tag{8}$$

Consequently, whenever two adjacent indices differ by at least two, moving one unit from the larger to the smaller strictly decreases the whole product's upper-left entry. The case $r = s + 2$ is exactly (5). The only remaining local comparison has adjacent indices differing by one; it is an interchange of the two blocks.

3.2. **Comparing the surrounding Fibonacci blocks.** For a finite list of block indices, define the ratio of its two open counts by

$$\rho(u_1, \ldots, u_m) = \frac{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{21}}{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{11}}, \qquad \rho(\varnothing) = 0.$$

All these ratios lie in $[0, 1)$. Direct multiplication gives

$$\rho(t, U) = \frac{F_{2t+1} + F_{2t-1}\rho(U)}{F_{2t+3} + F_{2t+1}\rho(U)}. \tag{9}$$

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Its right-hand side increases strictly with $\rho(U)$, since (3) gives $\det \mathcal{A}_t = 1$. Moreover, another application of that identity gives

$$\frac{F_{2t+1}}{F_{2t+3}} - \frac{F_{2t+3} + F_{2t+1}}{F_{2t+5} + F_{2t+3}} = \frac{1}{F_{2t+3}(F_{2t+5} + F_{2t+3})} > 0.$$

Hence $\rho(t+1, U) < \rho(t, V)$ for any two finite tails U, V. Two lists can therefore be compared by cancelling equal initial indices: at the first unequal pair, the larger index gives the smaller ratio; an exhausted list has the smaller ratio than any nonempty remainder.

For adjacent blocks in positions $j, j+1$, the outward lists are $(t_{j-1}, \dots, t_1)$ on the left and $(t_{j+2}, \dots, t_p)$ on the right. At $r = s+1$, (8) becomes

$$\mathcal{A}_{s+1}\mathcal{A}_s - \mathcal{A}_s\mathcal{A}_{s+1} = 2\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

After multiplication by the surrounding blocks, its upper-left entry has the sign of

$$\rho(t_{j+2}, \dots, t_p) - \rho(t_{j-1}, \dots, t_1). \tag{10}$$

Indeed, it equals this difference times twice the positive upper-left entries of the two surrounding products. This proves the needed surrounding-block test entirely in the same Fibonacci matrices.

Lemma 1 (Fibonacci-matrix normalization)**.** *For coprime positive p, q, every word $w = azb$ with q letters a and p letters b satisfies $\mu(w_{p/q}) \le \mu(w)$.*

Proof. For $p = 1$ there is only the word $a^q b$. Assume $p > 1$ and use the block-index list in (7). Its prefix discrepancies are

$$\delta_j = qj - p(1 + t_1 + \cdots + t_j) \quad (1 \le j \le p), \qquad \delta_0 = \delta_p = 0.$$

Changing (t_j, t_{j+1}) to $(t_j + 1, t_{j+1} - 1)$ decreases only δ_j, by p; the reverse change increases only δ_j, by p. We show that an appropriate change always decreases the matching count until $-p \le \delta_j \le 0$ for every j.

The sign checks use just a prefix-sum identity. Put $h_1 = t_1 + 1$ and $h_i = t_i$ for $i > 1$, the actual run lengths. Whenever $j - m \ge 0$ and $j + m \le p$,

$$\delta_{j-m} + \delta_{j+m} = 2\delta_j + p\left(\sum_{i=j-m+1}^{j} h_i - \sum_{i=j+1}^{j+m} h_i\right). \tag{11}$$

In the block indices, the left sum has the extra 1 precisely when it includes t_1.

First remove positive discrepancies. Choose the rightmost largest $\delta_j > 0$ and transfer one unit from t_{j+1} to t_j. Comparison with its two neighboring discrepancies gives $h_j < q/p < h_{j+1}$; coprimality makes these inequalities strict. Thus the transfer is allowed and $t_{j+1} \ge t_j + 1$; for $j = 1$ the gap is at least two. Gaps of at least two are handled by (8).

If $t_{j+1} = t_j + 1$, compare the outward lists. A first unequal pair with the left index larger would make the parenthesis in (11) nonnegative: the central difference is -1, the equal outward indices cancel, and the first unequal difference is at least 1. This contradicts maximality and the choice of the rightmost maximum. If the left list were exhausted first, its extra initial 1 would instead give $\delta_{2j} = 2\delta_j > \delta_j$, again impossible (including simultaneous exhaustion, when $2j = p$). The left outward ratio is therefore strictly larger than the right one. Reversing the interchange in (10) shows that the selected transfer strictly decreases the count. It also decreases $\sum_{j=1}^{p-1} \delta_j$. There are only finitely many nonnegative block-index lists of sum $q - 1$, so this phase ends with all $\delta_j \le 0$.

Then remove discrepancies below $-p$. Choose the leftmost smallest $\delta_j < -p$ and transfer one unit from t_j to t_{j+1}. For $j > 1$, the neighboring comparisons give $h_j > q/p > h_{j+1}$; for $j = 1$, the inequality $\delta_1 < -p$ gives $t_1 > q/p$, while $t_2 < q/p$. Thus $t_j \ge t_{j+1} + 1$ and the transfer is allowed. Again a gap of at least two is settled by (8).

In the gap-one case, a first unequal outward pair with the left index smaller would make the parenthesis in (11) nonpositive, contradicting minimality and the leftmost choice. If that pair reaches t_1, its extra 1 instead gives $\delta_{2j} \le 2\delta_j + p < \delta_j$, the same contradiction. If the right list were exhausted first, the identity would give $\delta_{2j-p} = 2\delta_j + p < \delta_j$. Simultaneous exhaustion would give $\delta_j = -p$. Thus the left outward ratio is strictly smaller than the right one, and (10) proves a strict decrease in count. The changed discrepancy is $\delta_j + p < 0$, so positivity is not reintroduced. This phase increases the bounded sum of discrepancies by p each time and terminates with $-p \le \delta_j \le 0$.

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Finally, $\delta_j \equiv qj \pmod{p}$. For $1 \le j < p$, coprimality leaves exactly one possible value in $[-p, 0]$, forcing

$$1 + t_1 + \cdots + t_j = \left\lceil \frac{jq}{p} \right\rceil .$$

These are precisely the canonical run lengths. Every transfer has decreased the upper-left product entry, proving the assertion. □

For the canonical word, $h_j = \lceil jq/p \rceil - \lceil (j-1)q/p \rceil$. With $k = 1 + \lfloor (q-1)/p \rfloor$, the factors are the author's A, B from (4), and the end factors are B. For example, $m_{3/8} = [BAB]_{11}$ with $k = 3$. A similar normalization result appears in [6, Proposition 21]; the proof above is self-contained in the author's Fibonacci matrices.

4. The three ordering proofs

4.1. **Fixed numerator: Fibonacci matrices.**

Proof. Fibonacci recurrence gives, for $t \ge 1$,

$$\mathcal{A}_t - \mathcal{A}_{t-1} = \begin{pmatrix} F_{2t+2} & F_{2t} \\ F_{2t} & F_{2t-2} \end{pmatrix}.$$

All entries are nonnegative and the upper-left is positive. Start with $w_{p/(q+i)}$ and delete i internal letters a, retaining the first. Each deletion changes one factor $\mathcal{A}_t$ to $\mathcal{A}_{t-1}$ with $t \ge 1$. Surrounding factors are positive, so the matching count strictly falls. The resulting word ends at (q, p); normalization gives $m_{p/q} \le \mu(w_*) < m_{p/(q+i)}$. □

4.2. **Fixed sum: six pieces.**

Proof. At the same physical cuts, the changes $2 \to 3$ or 4, then 3 or $4 \to 5$, increase no entry of (2); the changed factors are $3 \to 2$ or $2 \to 1$. Rotation $5 \leftrightarrow 1$ preserves the row. Selecting the changed connector and no others exhibits a strictly smaller positive summand whenever an equal-direction join becomes unequal.

The trimmed canonical word begins and ends with a, contains no bb, and has $q - p - 1$ equal joins, all aa. At its rightmost equal join, interchange a, b strictly to the right. That suffix is alternating, so only the chosen join changes. Exactly one prescribed piece changes by the stated rules, while any rotated pieces keep their rows. Each surgery strictly decreases the count. Odd-numbered operations replace one horizontal step by a vertical step in net; even-numbered operations keep the totals. After $2i$ operations the endpoint is $(q - i, p + i)$. There are enough joins because $p + i < q - i$. For the resulting auxiliary word w_*, normalization gives $m_{(p+i)/(q-i)} \le \mu(w_*) < m_{p/q}$. A similar tail comparison appears in [6, Proposition 22]; here the comparison is proved using the author's six physical pieces. □

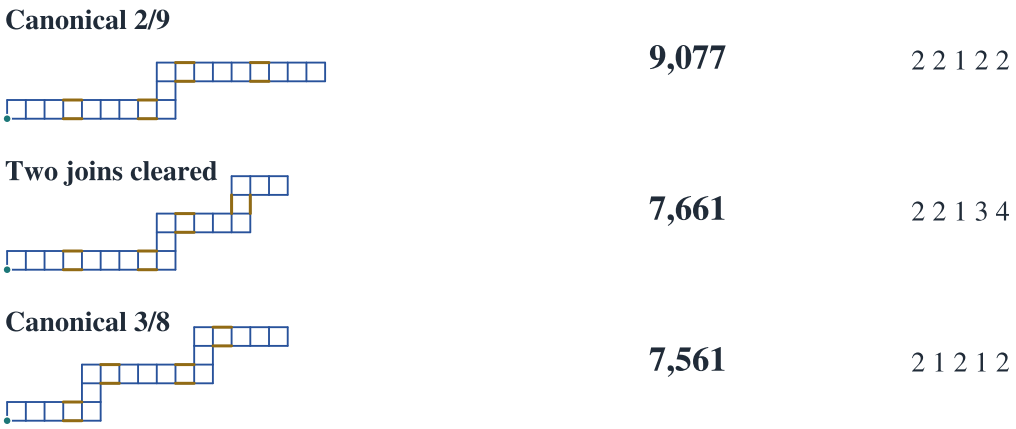


Figure 2. Fixed-sum surgery on five retained pieces. The canonical 2/9 snake has 9077 matchings; after two changes the auxiliary snake has 7661; Fibonacci-matrix normalization gives the canonical 3/8 snake with 7561. The intermediate first change, not shown, has 8263. Gold pairs mark cuts and teal dots the starting corners.

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4.3. **Fixed denominator: six pieces.**

Proof. For the fixed-denominator comparison, refine the same six-piece chop into single-square pieces of type 6, retaining odd-numbered tiles and using the even tiles as paired connectors. A retained square has counts $(2, 1, 1, 1)$ for opposite ports and $(2, 1, 1, 0)$ for adjacent ports. The zero forbids covering their common corner twice. Summing an intervening state regroups these squares into the original three-tile pieces.

Insert one internal b, adding two vertical squares. The affected local counts are the following complete table, with outside ports fixed:

	before				after			
directions	00	01	10	11	00	01	10	11
a, a	2	1	1	1	5	2	2	1
b, b	2	1	1	1	5	3	3	2
a, b	2	1	1	0	5	3	2	1
b, a	2	1	1	0	5	2	3	1

.

The endpoint caps have the corresponding rows with an outer port closed; the empty trimmed word has the increase $2 \to 5$. No outside count falls, and the all-zero choice witnesses a strict increase. Thus the two added squares compensate for every changed direction.

Starting from $w_{(p+i)/q}$, delete i internal b's and reverse this comparison. At least i are available. Normalizing the resulting word to (q, p) proves $m_{p/q} \leq \mu(w_*) < m_{(p+i)/q}$. □

Author contributions and assistance. The author devised the Fibonacci-matrix formulation and six-piece approach. Generative AI assisted with proof development, exact checking, reference retrieval, and typesetting. The author is responsible for the final claims. No funding was received; the author declares no competing interests.

References

[1] G. Musiker, *A conjectured combinatorial interpretation for Markov numbers*, REACH memorandum, 27 June 2002. Original memorandum; REACH archive.

[2] A. Itsara, G. Musiker, J. Propp, and R. Viana, *Combinatorial interpretations for the Markov numbers*, REACH manuscript, 10 January 2003, Theorem 3.1. Original manuscript.

[3] J. Propp, *The combinatorics of frieze patterns and Markoff numbers*, arXiv:math/0511633, first posted 25 November 2005; published in *Integers* **20** (2020), A12.

[4] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture*, Springer, 2013. doi:10.1007/978-3-319-00888-2.

[5] M. Rabideau and R. Schiffler, *Continued fractions and orderings on the Markov numbers*, Adv. Math. **370** (2020), 107231. doi:10.1016/j.aim.2020.107231.

[6] C. Lagisquet, E. Pelantová, S. Tavenas, and L. Vuillon, *On the Markov numbers: fixed numerator, denominator, and sum conjectures*, Adv. Appl. Math. **130** (2021), 102227. arXiv:2010.10335.

[7] K. Lee, L. Li, M. Rabideau, and R. Schiffler, *On the ordering of the Markov numbers*, Adv. Appl. Math. **143** (2023), 102453. arXiv:2010.13010.

[8] G. McShane, *Convexity and Aigner's conjectures*, preprint (2021), arXiv:2101.03316.

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Meridians on the Markov Surface: Geometry, Residues, and Collision Windows

Completion and release history

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v5_ps | 2026-11-07 18:35:22 CST / 2026-11-08 00:35:22 UTC | https://zenodo.org/records/24046012

Earlier public announcements: 21 October 2026, 10:18:25 CDT / 15:18:25 UTC on X (more than a billion values in under 40 minutes); 23 October 2026, 09:54:57.771 CDT / 14:54:57.771 UTC on LinkedIn (9.3 billion in 24 minutes). These concern earlier computations, not the final trillion certificate.

https://x.com/webystar/status/2112926487091282350

https://www.linkedin.com/posts/elleard-felix-webster-heffern-joseph-by-confirmation-2953a3104_markovnumbers-activity-7519411048115003392-4NaJ

The dates and URLs are retained in the 7 November public-post evidence packet; the LinkedIn URL is also embedded in the published computational supplement. Additional November LinkedIn announcements and the older Meridian ResearchGate v13 URL could not be established in this compilation.

Complete public Zenodo description

This v7_ps revision presents the author's Fibonacci-matrix proofs of normalization and fixed-numerator monotonicity, and his six-piece proofs of the remaining fixed-sum comparison and fixed-denominator monotonicity. The normalization starts from (AC)_11 - (B^2)_11 = 2 and proves the local comparisons and termination in the same matrices. Related earlier normalization is cited as similar work, while the author's argument retains its explicit attribution. The matching interpretation is credited to Musiker's 27 June 2002 memorandum and the Itsara-Musiker-Propp-Viana manuscript of 10 January 2003.

The fixed-numerator, fixed-sum and fixed-denominator proofs now have parallel subsection headings.

The paper organizes Markov configurations by reversed Meridian addresses, geometric descent, Fibonacci-modulus and modulo-27 behavior, equal-degree separation, finite unequal-degree collision windows, corridor recurrences and a numerator-denominator collision window. The archived degree-10,000 certificate is retained. The unrestricted unicity problem remains open within this framework.

The revised computational chronology includes the earlier Claude-assisted historical search and the separately archived fifty-billion numerical-order verification. The working draft reports 9,581,528,580 triples against a cutoff of 10^100000 in 24 minutes on two cores with 7 GB memory. A public LinkedIn announcement is timestamped 23 October 2026, 14:54:57.771 UTC (09:54:57.771 CDT), with caption wording of 9.3 billion in 24 minutes. This is the announcement timestamp, not a recovered execution start or finish. The earlier draft's qualification about uncertified historical logs and cutoff error bounds is retained. This search is not the later ten-billion anchor.

The fifty-billion verifier completed on 28 October 2026. Its recorded start was 08:32:57.096455198 UTC and its audit-inclusive finish was 10:38:33.644612263 UTC (03:32:57-05:38:34 CDT, rounded). Two closed invocations total 7,492.102807105 seconds (2 h 04 min 52.103 s). The calendar span is 7,536.548157065 seconds, including the invocation pause. Including the separate original ten-billion anchor gives 8,330.980040248007 seconds (2 h 18 min 50.980 s). These historical execution dates are distinct from this edition's release date.

The exact numerical envelope covers 71,517,775,875 configurations. After anchored omissions and certified lower-bound exclusions, 54,940,210,994 new configurations were compared. All 346 repeated primary-residue pairs separated; the final auditor

reconstructed all 692 candidate triples and rechecked the 38 anchor candidates. The protocol establishes uniqueness for the first 50,000,000,000 distinct Markov values in numerical order. It used eight workers, an eight-CPU-equivalent quota and an 8-GiB memory limit.

Included are the 33-page main paper, six-page computational proof and timing supplement, editable source, all figure assets, selected historical evidence and receipts, and a new read-only reconciliation of 26,151 archive member checksums and 8,522 closed stage receipts. The full frozen historical verifier archive is available separately at https://doi.org/10.5281/zenodo.23874075. This release is an editorial revision, not a new enumeration; no completed 500-billion result is claimed.

Licensing: Original software is licensed under MIT; original documentation and data under CC BY 4.0. Third-party materials and external dependencies retain their own terms. Copyright (c) 2026 Elleard Felix Webster Heffern, to the extent copyright applies.

Funding: No funding. Substantial generative-AI assistance is disclosed in the manuscript; the author directed the research and is responsible for the work.

Published files

Original filename / checksum	Bytes
v4_50_Billion_Verification_Supplement.pdf MD5 6c966b341c1ec55b6b7bdc3a9dc50aaf SHA256 b477d344aff0971fee600ce1d087fc05f15509582bb43b22eb337180d6d3c446	310,901
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All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

Original: v7_ps_Markov_Meridians_TLMS.pdf

MERIDIANS ON THE MARKOV SURFACE: GEOMETRY, RESIDUES, AND COLLISION WINDOWS

ELLEARD FELIX WEBSTER HEFFERN

Abstract. We organize positive integral Markov triples by Meridians: real curves obtained by reversing a geometric descent, with a signed integer address for the curve and an integer phase for each arithmetic configuration. This gives a bijection with normalized triples without assuming uniqueness of their largest entries. We prove a common Fibonacci-modulus period law and a classification into ten disjoint residue classes modulo 27. Established Markov orderings imply unicity throughout each fixed Meridian degree; positive matrix words place every unequal-degree collision in an explicit finite phase window. A second organization, by corridor walls, gives exact recurrences across both Meridian and phasor steps and a complementary bound on denominators in terms of numerators and their difference. The author's Fibonacci matrices prove normalization and fixed-numerator monotonicity; his six pieces prove the remaining fixed-sum comparison and fixed-denominator monotonicity. We also give a folded-piece comparison criterion. Finally, complete residue and phase certificates separate all 5,067,077 Meridians of degree at most 10,000 at every integer phase. A separate archived computation, completed on 28 October 2026, verifies the first 50 billion distinct Markov values in numerical order. The unrestricted unicity problem remains open within this framework.

Post-submission draft. This edition incorporates changes requiring a new submission. The prefix `ps` retains that distinction.

1. Introduction

A normalized Markov configuration is a positive integral triple satisfying

$$a^2 + b^2 + M^2 = 3abM, \qquad 1 \le a \le b \le M. \tag{1}$$

The unicity question asks whether the largest entry M determines a, b. The classical Vieta tree enumerates configurations by replacing one coordinate with the other root of its quadratic. A tree of configurations is available independently of unicity; treating its vertices as distinct numerical maxima would assume the conclusion. Aigner's monograph gives the historical and arithmetic setting [5]. Continued fractions, snake graphs and Christoffel words supply several established orderings [12, 9, 10, 11], which will be used with their precise hypotheses.

This paper develops two ways to organize that same set of configurations. A *Meridian* holds fixed an itinerary of geometric descent and varies a real phase. A *corridor wall* follows a different route through the alternative tree, eventually holding a Farey numerator fixed while advancing the denominator. These organizations answer different comparison questions: Meridian degree controls a phase window, while corridor numerator controls a denominator window.

The geometric construction comes first in Section 2. Section 3 gives the canonical addresses, their inverse, and the exact conversion to Farey labels. Section 4 proves the modulo-27 classification and the common period s modulo F_{2s}. Sections 5 and 6 establish the two collision windows. Section 7 supplies the matching-count proofs used in the ordering arguments. Section 8 presents a finite certificate with an infinite phase range:

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2020 *Mathematics Subject Classification.* Primary 11D25; Secondary 11B39, 05C70, 11Y50.
Key words and phrases. Markov numbers, unicity, geometric descent, Meridians, corridors, snake graphs.

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Theorem 1.1 (Degree-10,000 separation)**.** *Among canonical Meridians of Laurent degree at most* 10,000*, distinct canonical integer-phase marks have distinct maxima. There are exactly* 5,067,077 *Meridians in this family.*

The computational part separates different Meridians at all phases. The fixed-degree theorem also separates marks on a single Meridian, so the two conclusions combine as stated. The bound is on *degree*, not on the size of a Markov number, and it does not exclude a competitor of higher degree. Section 9 records the separate fifty-billion numerical-order verification and its dated execution history.

Throughout, $M_\alpha(x)$ denotes the maximum at Meridian address α and phase x. The notation $m_{p/q}$ denotes the maximum attached to the standard reduced Farey label p/q; it labels a configuration before any assertion of unicity. These are two names for values of the same configuration map. The seed formula $M_1(x) = 2 + 3F_{2x}^2$ should not be confused with the boundary family $m_{1/q} = F_{2q+1}$. Analytical logarithms are natural; plotted horizontal axes use $\log_{10} M$.

2. Geometric descent and arithmetic closure

2.1. **A unit edge on a calibrated hyperbola.** For a real host $h > 2/3$, define

$$L_h = \operatorname{arcosh}(3h/2), \qquad \alpha_h = \frac{2h}{\sqrt{9h^2-4}}, \tag{2}$$
$$f_h(t) = \alpha_h \cosh(L_h t), \qquad g_h(t) = \alpha_h \sinh(L_h t).$$

The Cartesian curve $(g_h(t), f_h(t))$ is a hyperbola. Its height profile $t \mapsto f_h(t)$ is the curve used in the figures. Hyperbolic addition gives

$$f_h(t+1) + f_h(t-1) = 3h f_h(t), \tag{3}$$

$$f_h(t)^2 + f_h(t-1)^2 + h^2 = 3h f_h(t) f_h(t-1). \tag{4}$$

Thus two contacts one horizontal unit apart, with heights $A = f_h(c)$ and $B = f_h(c-1)$, form a real Markov state. The two compatible hosts are exactly

$$h, \qquad K = 3AB - h = \frac{A^2 + B^2}{h}. \tag{5}$$

Begin with an edge straddling the vertex, $0 \le c < 1$. Seek a wider calibrated profile through the same two ordered contacts, allowing horizontal translation. Smaller h means a wider profile. If the other host satisfies $K < h$, fit that profile and walk the unit lattice inward until the selected edge again straddles its vertex. Otherwise stop. Figure 1 shows one complete step. The fit is explicitly

$$v = \frac{1}{2} + \frac{1}{L_K} \operatorname{arsinh}\left(\frac{(A-B)\sqrt{3K+2}}{2K} \right), \qquad f_K(v) = A, \quad f_K(v-1) = B. \tag{6}$$

The walk replaces v by $v - \lfloor v \rfloor$. Its orientation is retained: reflection exchanges the contacts and sends phase to its negative modulo one.

Proposition 2.1 (Uniqueness of the fit)**.** *A marked edge has at most one distinct compatible profile, hence at most one wider fit. For fixed* $0 < a < b$*, a same-arm fit* $f_h(u-1) = a$*,* $f_h(u) = b$ *with* $u > 1$ *is unique if it exists.*

Proof. The hosts are the two roots of $X^2 - 3ABX + A^2 + B^2$. The other root belongs to the geometric domain because $(3K-2)AB = K^2 + (A-B)^2 > 0$. For a same-arm fit the horizontal distance between heights a, b is

$$\frac{1}{L_h} \int_a^b \frac{dy}{\sqrt{y^2 - \alpha_h^2}}.$$

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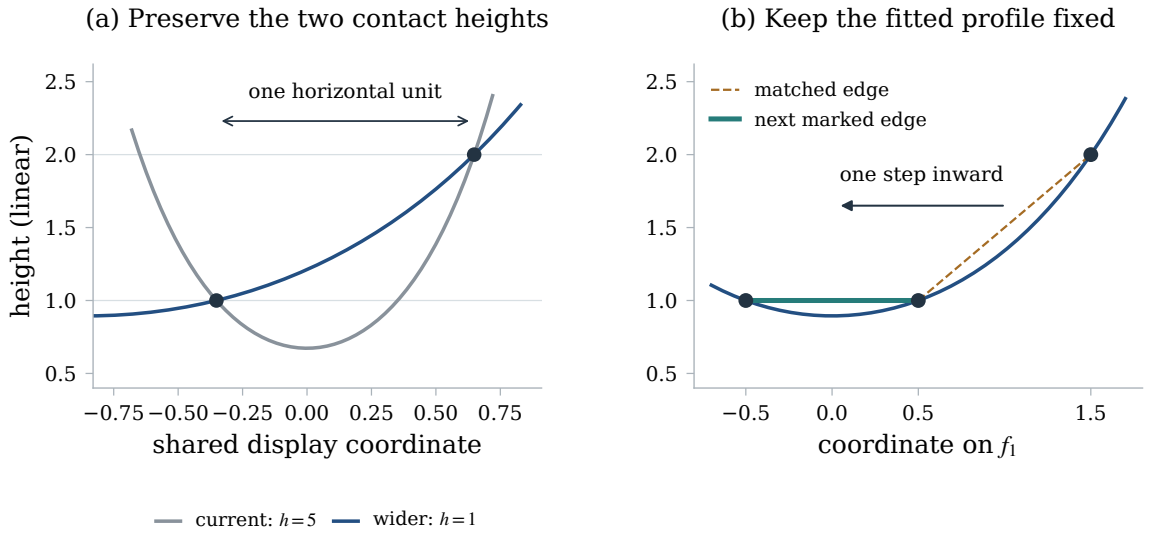


Figure 1. Geometric descent from $(1, 2, 5)$. The wider profile passes through the same unit-spaced contacts. On f_1 their coordinates are $1/2, 3/2$; one inward step selects $-1/2, 1/2$, with heights $1, 1$. The curve is fixed during the walk.

When $a > 2/3$ and the lower height is in range, both factors strictly decrease with h. The distance can therefore equal one at most once. The profile is strictly increasing on that arm, which also determines u. If $a \le 2/3$ there is no fit. □

2.2. **Termination.** Write $\mathcal{D}(M, c) = (H, \phi)$ for the terminal host and oriented phase.

Theorem 2.2 (Finite descent)**.** *Every straddling input with $M \ge 1$ terminates. On $1 \le M \le R$ the number of host changes is uniformly bounded. The terminal map is continuous into $(2/3, \infty) \times \mathbb{R}/\mathbb{Z}$ and analytic on cells with a fixed strict itinerary.*

Proof. Temporarily sort the endpoints as $a \le b$. Inward walking does not increase either endpoint. Every current host is at most R, and the discriminant in (4) gives

$$(9h^2 - 4)a^2 - 4h^2 = (2b - 3ha)^2 \ge 0.$$

Consequently the endpoints are at least $a_R = 2R/\sqrt{9R^2 - 4} > 2/3$. At an accepted change, $a^2 + b^2 < h^2$, so the endpoints are also below $h \le R$. Using either endpoint as the host in the same discriminant identity bounds the new host below by a_R. The initial host is at least $1 > a_R$.

If the new host K is at least b, its edge already straddles the vertex, because both endpoints are below $f_K(1)$; the next candidate is the old, larger host, so descent stops. Otherwise

$$\frac{h}{K} > \frac{h}{b} = 3a - \frac{K}{b} > 3a - 1 \ge 3a_R - 1 > 1.$$

All changes except possibly the last therefore contract the host by a fixed factor. An upper bound is $1 + \lceil \log(R/a_R) / \log(3a_R - 1) \rceil$.

At a boundary $K = h$, the fitted coordinate in (6) is the original c, so the accepted and stopped maps agree. At a phase cut, $K(h, 0) = h(9h^2 + 4)/(9h^2 - 4) > h$, and a neighborhood is stopped. The uniform bound on compositions then gives continuity, with phase taken modulo one. The strict formulas are analytic. □

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2.3. **Which inputs close arithmetically?** The two reversible state operations are

$$T(a,b,h) = (3ha - b, a, h), \qquad J(a,b,h) = (a, b, 3ab - h). \tag{7}$$

They and their inverses are integer polynomial maps. A terminal state has the nonnegative defect

$$\begin{aligned} H f_H(\phi) f_H(\phi - 1) - 1 = {} & \frac{(H-1)^2(H+2)}{3H-2} \\ & + \frac{4H^3}{9H^2-4} \sinh^2\Big(L_H(\phi - \tfrac{1}{2})\Big). \end{aligned} \tag{8}$$

For $0 \le \phi < 1$ it vanishes exactly at $(H, \phi) = (1, 1/2)$.

Proposition 2.3 (Arithmetic closure)**.** *A straddling input with $M \ge 1$ reaches $(1, 1/2)$ if and only if its three entries are positive integers. Modulo companion exchange, these inputs are exactly the normalized configurations* (1)*.*

Proof. The product-to-sum identity gives (8). Reversing a finite closure itinerary from $(1,1,1)$ by (7) preserves integrality. Conversely, integral descent stays integral. At an integral terminal state order the entries as $u \le v \le w$. There is no coordinate-decreasing mutation. But for $Q(X) = X^2 - 3uvX + u^2 + v^2$,

$$Q(v) = u^2 + (2 - 3u)v^2 \le 3u^2(1 - u) \le 0.$$

If neither root is below $w \ge v$, equality is forced, giving $u = v = 1$; terminality then gives $w = 1$. Finally, a straddling edge has no decreasing endpoint mutation, so its host is a largest entry. Its ordered split is uniquely recovered by (6). □

Both terminal coordinates matter. Figure 2 shows inputs with terminal host one but different terminal phases. A crossing of the height level $H = 1$ alone does not certify integrality. The marked configurations follow the host itineraries $37666 \to 5 \to 1$, $7561 \to 5 \to 1$, and $14701 \to 2 \to 1$, all ending at phase $1/2$.

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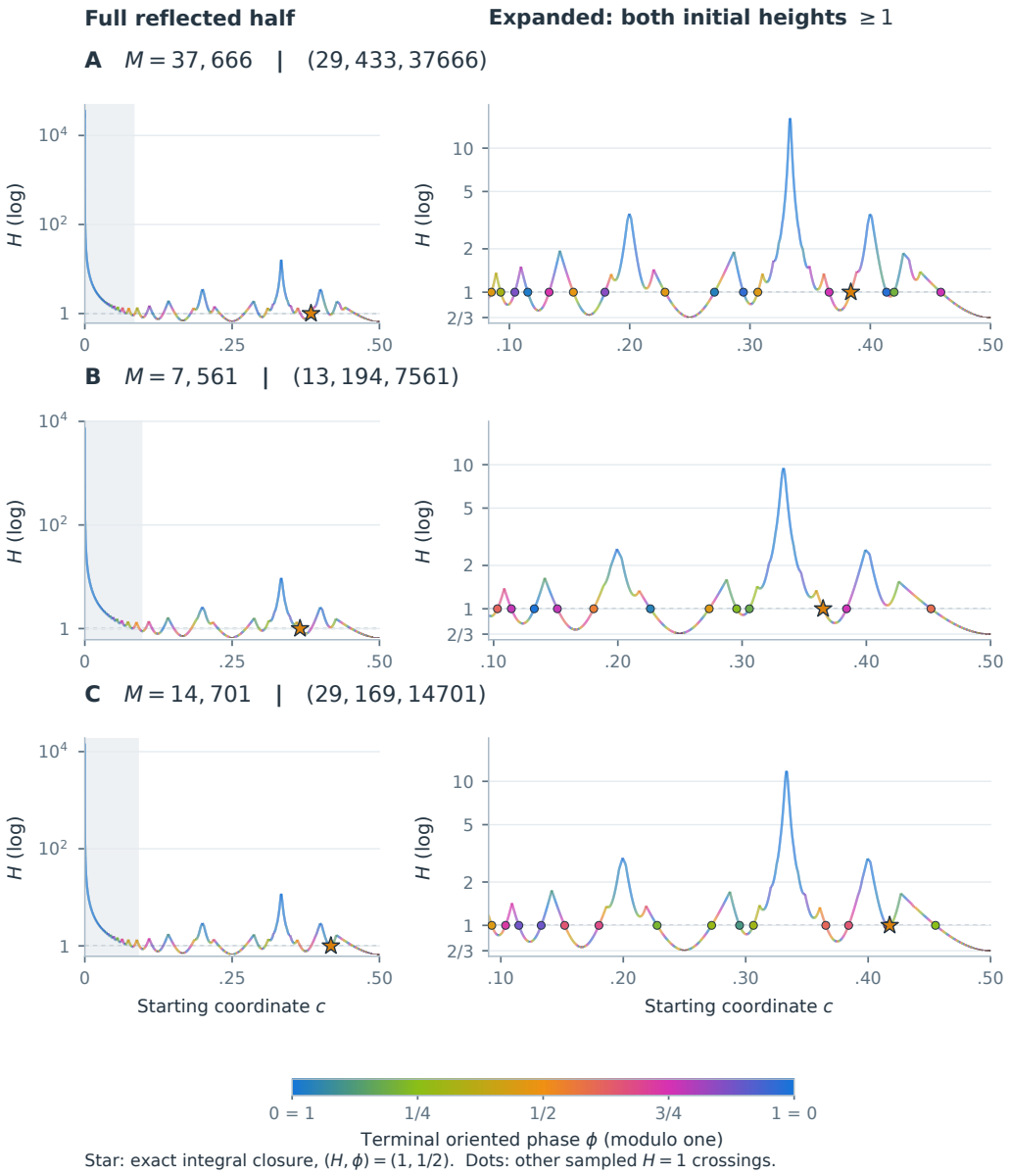


Figure 2. Terminal host and phase for $M = 37666, 7561, 14701$. Height is H on a logarithmic scale and hue is oriented phase. Left: $0 \le c \le 1/2$, with gray where an initial endpoint is below one. Right: the admissible portion enlarged. Stars mark the exact integral starts; dots are other numerically located $H = 1$ crossings. These numerical plots illustrate the closure criterion.

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3. Meridians, addresses, and the alternative tree

3.1. **The landscape and its two coordinates.** On the landscape $(\log_{10} M, c)$, the point (M, c) represents the contacts $f_M(c), f_M(c-1)$. We draw the reflected half $0 < c \le 1/2$; exchanging the contacts sends c to $1-c$. A *Vieta arc* keeps one initial contact height fixed. For example, the branch with height a has

$$c = \frac{\operatorname{arcosh}(a/\alpha_M)}{L_M}.$$

A *Vieta trek* follows such arcs, changing the retained contact when necessary. A *Meridian* is a connected component of terminal host $H(M, c) = 1$. A *phasor* holds the terminal phase ϕ fixed. The alternative tree uses phasors at $\phi = 1/2$ to join successive Meridians.

The address phase x on a Meridian is real and unrolled. Its terminal circle phase is $\phi \equiv x + 1/2 \pmod 1$. Every integer mark therefore closes arithmetically; the distinguished arrival mark $x = 0$ is called the *helm*. It need not be the minimum of the real profile.

Figure 3 keeps the same configurations in three views. The alternative tree is in the middle panel. The bottom panel uses smooth schematic connectors for the Vieta parent relation: node coordinates and adjacency are exact, but those connectors are not literal level curves. The two trees each have 55 edges. An unmarked crossing is not a vertex.

For reproducibility, the field in panel (a) uses the fixed cyclic phase palette with HSL saturation and lightness

$$w(H) = \frac{1}{1 + (\ln H / \ln 1.02)^2}, \qquad S = w(H), \qquad L = 1 - 23w(H)/40.$$

The background is a numerical illustration, independent of all exact enumerations and exclusions below.

3.2. **Evaluating an address.** The seed Meridian is

$$\begin{aligned} a_1(x) = f_1(x + \tfrac{1}{2}), \qquad b_1(x) = f_1(x - \tfrac{1}{2}), \\ M_1(x) = 3a_1(x)b_1(x) - 1. \end{aligned} \tag{9}$$

For integer x, its companions are F_{2x+1}, F_{2x-1}, where $F_{-j} = (-1)^{j+1}F_j$. Hence

$$M_1(x) = 2 + 3F_{2x}^2, \qquad M_1(0), M_1(1), M_1(2), \ldots = 2, 5, 29, 194, \ldots. \tag{10}$$

From any ordered state (a, b, H) form the two-sided orbit

$$u_0 = a, \quad u_1 = b, \qquad u_{j+1} = 3Hu_j - u_{j-1}.$$

A nonzero index k replaces the state by

$$(u_k, u_{k+1}, 3u_k u_{k+1} - H). \tag{11}$$

The address is written

$$\alpha = (1, k_m, \ldots, k_1): \quad k_1 \text{ is applied first and written at the right.}$$

This is the reversed indexing convention: the former $M_{1,a,b,c}(x)$ is now $M_{1,c,b,a}(x)$. Neither the signs nor x are reversed. In terms of (7), with $S = T^{-1}$ and $\mathcal{W}_k = JS^k$,

$$(a_\alpha(x), b_\alpha(x), M_\alpha(x)) = \mathcal{W}_{k_m} \cdots \mathcal{W}_{k_1} JT^x(1, 1, 1). \tag{12}$$

Composition acts from right to left. A recursive extension inserts its index immediately after the leading 1. For example, $M_{1,1}(0) = 13$. Index zero is forbidden because (11) would return the preceding pole $3ab - H$, rather than create a new component.

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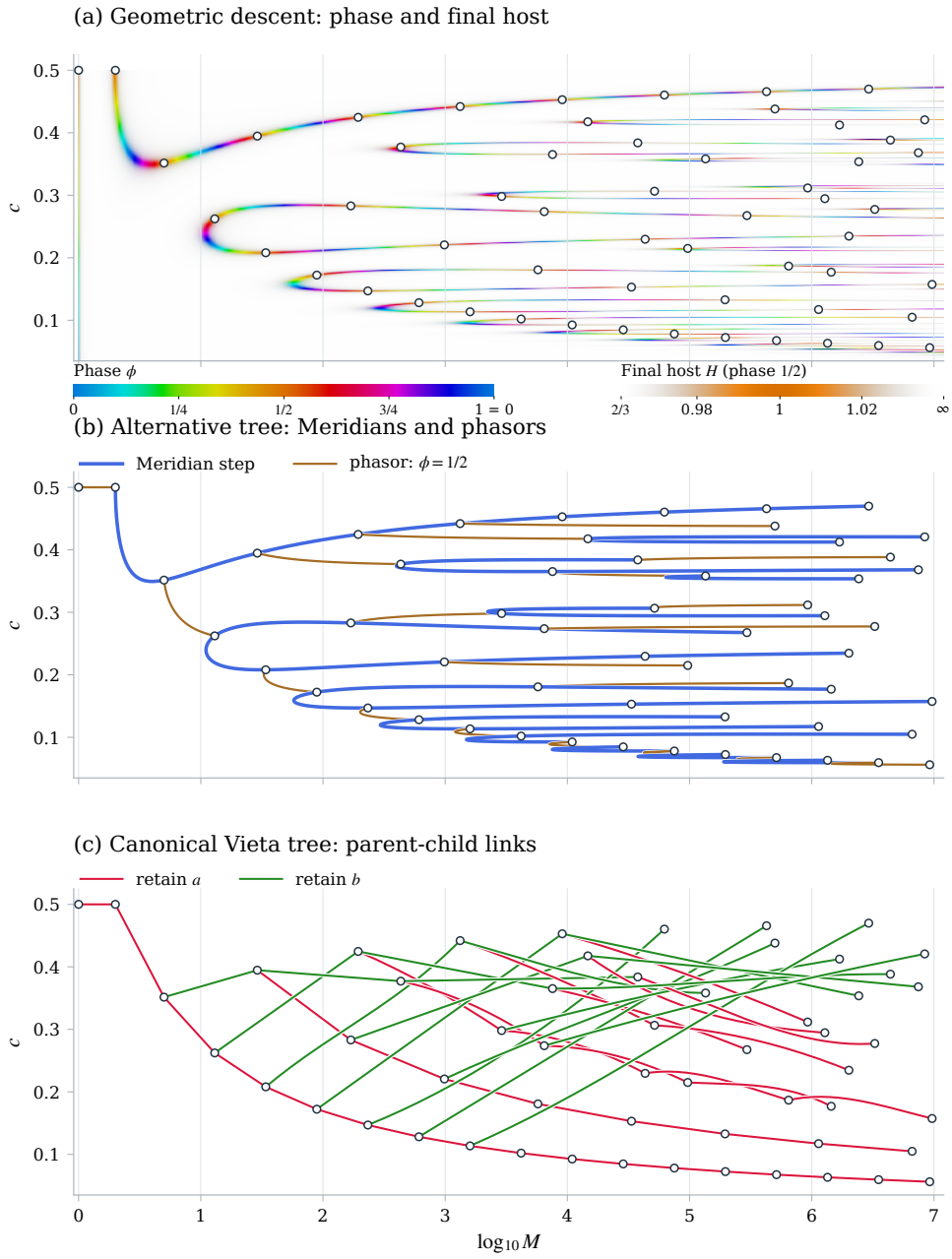


Figure 3. The same 56 configurations with $M \leq 10^7$. (a) Numerical geometric descent, with phase encoded by hue and host by color intensity. (b) The alternative tree, with Meridian and phasor edges. (c) The canonical Vieta tree, with simplified connectors retaining exact endpoints. Red retains the smaller companion, green the larger. Every edge runs toward larger M. Crossings without a node mark are not vertices.

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Definition 3.1 (Canonical marks)**.** The root $(1,1,1)$ is separate. For the seed (1) use integer phases $x \geq 0$. For every other address require $k_1 > 0$, $k_i \neq 0$ for $i > 1$, and $x \in \mathbb{Z}$. Evaluation is followed by sorting the two companions.

Theorem 3.2 (Configuration bijection)**.** *Canonical marks are in bijection with all normalized positive Markov triples except the root. The displayed value $M_\alpha(x)$ is the largest entry.*

Proof. The quadratic argument in Proposition 2.3 shows that replacing the largest coordinate by its other pole gives a smaller maximum, apart from the initial repeated entries. This is the forced Vieta reduction of a configuration and does not assume scalar unicity.

To recover a word, reflect the current maximum H to $P = 3ab - H$. When $P \geq 2$, walk its oriented endpoint orbit inward until both endpoints are below P. This edge is unique. Indeed $u_{j+1} - 2u_j + u_{j-1} = (3P-2)u_j > 0$, so the entries below P form an interval; three consecutive entries below P would contradict $u_{j-1} + u_{j+1} = 3Pu_j \geq 3P$. Endpoint descent supplies the edge, and its nonzero signed displacement recovers the last-applied, leftmost index. Repeat at the smaller pole. At pole one, the orbit reaches $(1,1)$ and fixes the seed phase.

Conversely, all outward blocks are positive and reduced. If the preceding pole is $P < H$, the first reflected poles on the two sides are $(9b^2-2)H - P > 6H$ and $(9a^2-2)H - P > 6H$. Their farther endpoints exceed $2H$, and the recurrence makes subsequent outward entries increase. The reduction therefore undoes every prescribed block. Finally, companion exchange negates all indices and x simultaneously. The canonical conditions select one of the two orientations; only $(1,1,2)$ is fixed. □

3.3. **The two parent rules.** At a nonzero phase the alternative parent is the adjacent inward mark, $(\alpha; x) \mapsto (\alpha; x - \operatorname{sgn} x)$. At a nonseed helm it is

$$(1, k_m, \ldots, k_2, k_1; 0) \longmapsto (1, k_m, \ldots, k_2; -k_1), \tag{13}$$

followed by simultaneous sign reflection if the remaining rightmost index is negative. An empty tail uses the seed's nonnegative phase. The seed helm with maximum two has parent $(1,1,1)$.

The alternative parent deletes the *rightmost* index at a helm. Recursive evaluation removes the *leftmost* index after 1. These operations serve different purposes. The depth

$$d(\alpha; x) = 1 + \sum_{i=1}^{m} (|k_i| + 1) + |x| \tag{14}$$

decreases by one under either parent edge, with root depth zero. Thus the parent rule defines a rooted tree on the same configurations. The ordinary outward Vieta moves instead are

$$(a, b, M) \longmapsto (a, M, 3aM - b), \qquad (a, b, M) \longmapsto (b, M, 3bM - a). \tag{15}$$

For example, $29 \to 194$ is a Meridian edge, whereas $29 \to 169$ is a Vieta edge. Figure 4 compares the two sets of links.

3.4. **Real components and phasor edges.**

Proposition 3.3. *The canonical real families in* (12)*, modulo companion reflection, are the nonterminal connected components of $H(M, c) = 1$. Their terminal phase is $x + 1/2$ modulo one. The separate terminal component is $M = 1$.*

Proof. The seed companions are at least $2/\sqrt{5}$, and $M_1 \geq 2$ is strictly larger than either. More generally, let $h \geq 1$ and $2/\sqrt{5} \leq a, b \leq h$. The first outward endpoint satisfies $3hb - a \geq (6/\sqrt{5} - 1)h > h$; its reflected pole is larger still. The reversed orbit gives the same conclusion on the other side. Every outward block thus has a strict inverse descent to its source and preserves the terminal phase.

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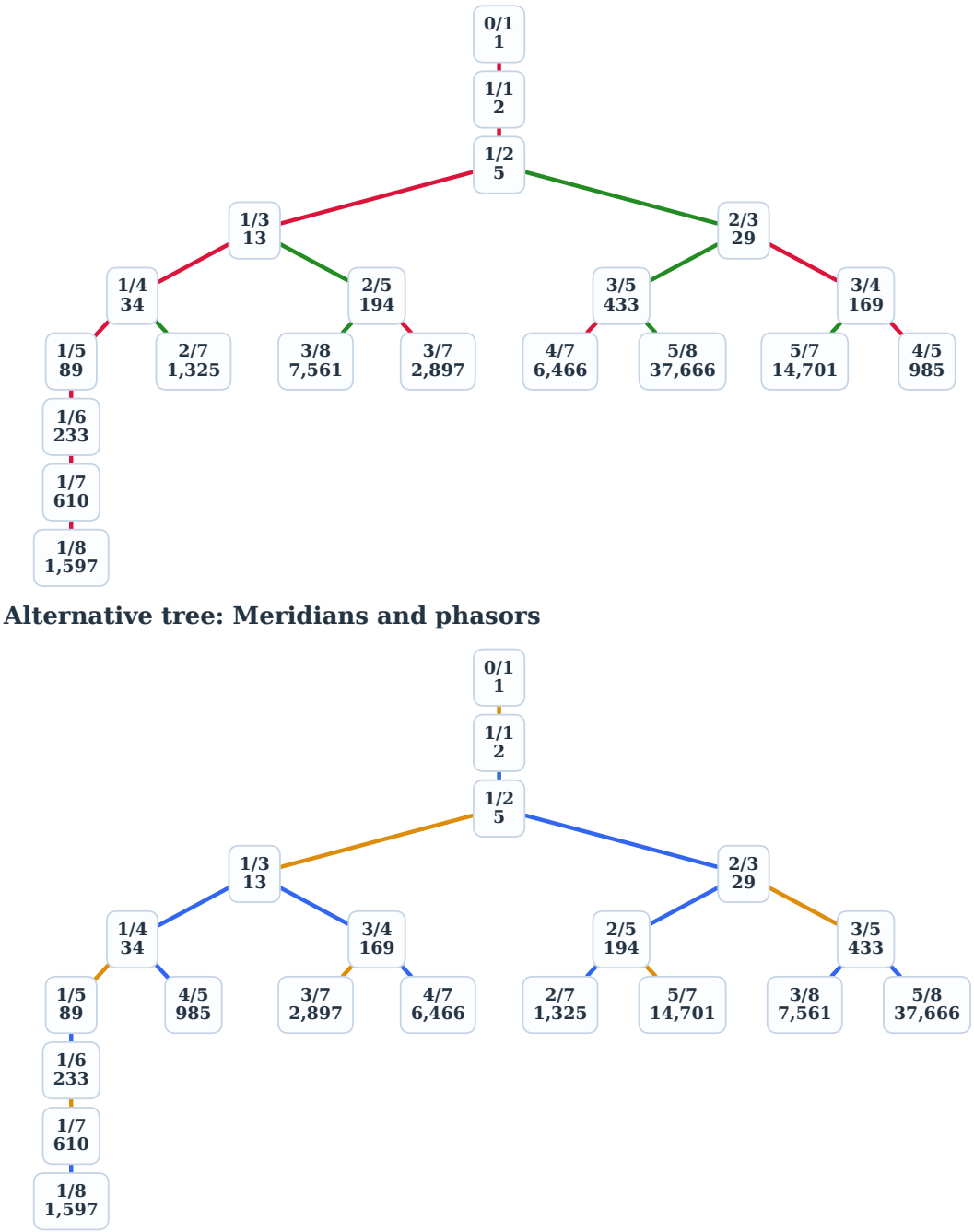


Figure 4. Two trees on the same 20 configurations: the 16 with $M \leq 3000$ and the four additional values $6466, 7561, 14701, 37666$. Each node displays its reduced standard Farey label and maximum. Red and green denote the Vieta moves; blue and gold denote Meridian and phasor steps.

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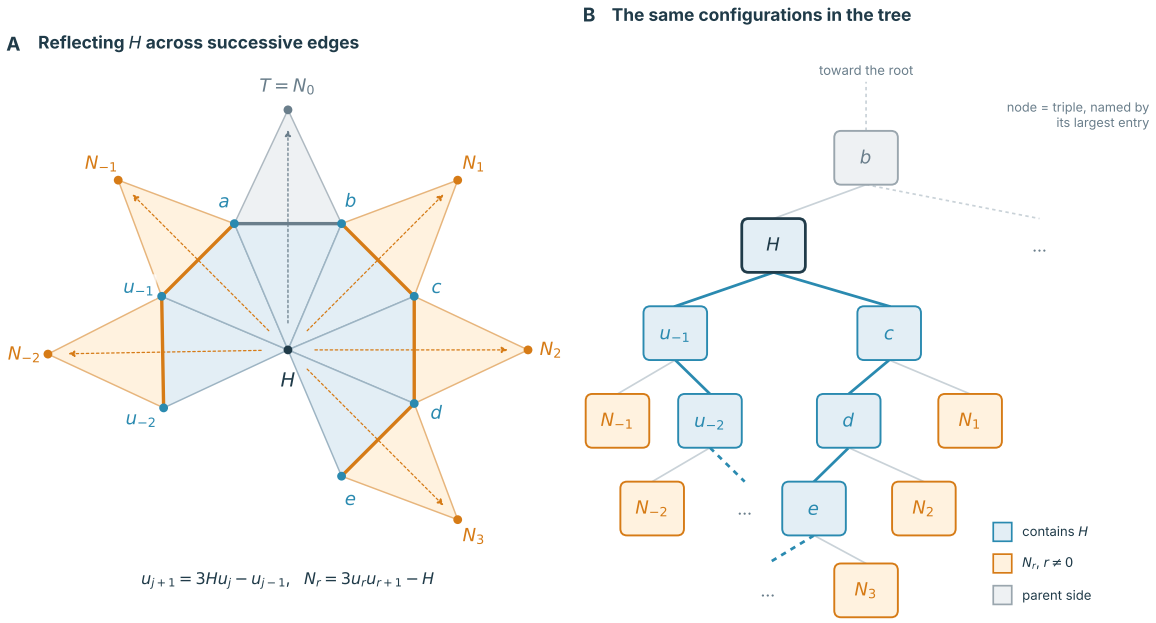


Figure 5. The orbit underlying one recursive extension. Reflecting H across (u_j, u_{j+1}) gives $N_j = 3u_j u_{j+1} - H$; N_0 is the preceding pole. For source $(a, b, H) = (1, 5, 13)$ at $M_{1,1}(0)$, the preceding pole is two, $(u_{-2}, u_{-1}, u_2, u_3, u_4) = (1325, 34, 194, 7561, 294685)$, and $(N_{-2}, N_{-1}, N_1, N_2, N_3) = (135137, 89, 2897, 4400489, 6684339842)$. The right panel locates these reflections in the Vieta tree.

Conversely, reverse any descent ending at host one. Its last host change produces the seed with $x = v - 1/2$, and every preceding walk is nonzero: a zero walk would simply have stopped at the smaller host. These strict blocks uniquely recover a locally constant word. The unrolled x joins the final phase charts continuously. The seed is injective as an ordered curve, as its endpoint difference is strictly monotone; each block is invertible. The positive Laurent expansion proved in Proposition 5.2 makes every family proper. The inverse charts are relatively open in the level set, and properness makes the families closed. Each connected family therefore fills a component. Reflection gives exactly the canonical identifications. □

The phasor edges are obtained by applying the same outward blocks to

$$\left(\frac{h}{\sqrt{3h-2}}, \frac{h}{\sqrt{3h-2}}, h\right), \qquad 1 \le h \le 2. \tag{16}$$

This symmetric arc joins $(1, 1, 1)$ to $(1, 1, 2)$. Its other host is $2h/(3h-2)$, and both possible terminal fits have phase $1/2$. Its endpoints satisfy the bounds used in the proposition, so the outward blocks preserve that phase. The displayed phasors consequently arise from the descent itself, rather than interpolation between integer nodes.

3.5. **Degree and the exact Farey dictionary.** Attach endpoint count vectors $u = (1, 0)$ and $v = (0, 1)$ to the seed. Under an index k update them by

$$(u, v) \longmapsto \begin{cases} ((k-1)(u+v)+v,\ k(u+v)+v), & k > 0, \\ ((|k|(u+v)+u,\ (|k|-1)(u+v)+u), & k < 0. \end{cases} \tag{17}$$

Write their sum as (r, s). Their determinant remains ± 1 and their unordered residues modulo two remain $\{(1, 0), (0, 1)\}$. Thus r, s are positive coprime odd integers. Canonical orientation gives $r \le s$.

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The *Meridian degree* is $D = r + s$. It is the extreme Laurent exponent in ρ^x, where $\rho = (3 + \sqrt{5})/2$; the degree in ρ^{2x} is $D/2$.

Proposition 3.4 (Helm classification)**.** *Canonical Meridians correspond bijectively to coprime positive odd pairs $r \le s$. Their helm labels are the standard Farey fractions r/s.*

Proof. The endpoint vectors are the two Stern–Brocot parents: their sum is (r, s) and their determinant has absolute value one. Conversely, the parents of an odd/odd pair have opposite coordinate parities. Their componentwise absolute difference is a smaller positive primitive odd/odd pair, except at $(1, 1)$. The difference has one componentwise sign; an opposite sign would give the initial pair. Iteration reaches $(1, 1)$.

To recover the signed indices, put $n_\alpha = (r, s)$ and let α^- remove the leftmost index after 1. Consecutive difference vectors have determinant of absolute value two, so three consecutive vectors satisfy $n = 2c\, n^- + \varepsilon n^{--}$ with $c \ge 1$ and $\varepsilon = \pm 1$. Determinant division gives an integer coefficient, parity makes it even, and decreasing coordinate sums fix its magnitude and sign. At the bottom use $(1, 1), (-1, 1)$; canonical orientation fixes the first index positive. The magnitude is fixed by the ratio of successive degrees, which lies in $[2c, 2c+1)$ or $(2c-1, 2c]$ according to the sign; equality occurs only at the first applied index. Reversing the process gives precisely $n_{1,k,k_m,\ldots,k_1} = 2|k| n_\alpha + \operatorname{sgn}(kk_m) n_{\alpha^-}$. For the seed use $k_0 = 1$ and $n_\varnothing = (-1, 1)$, so the first extension has $(r, s) = (2k_1 - 1, 2k_1 + 1)$. This recovers the unique address. Mediant addition identifies the label. □

Lemma 3.5 (Phase dictionary)**.** *For helm (r, s) and degree $D = r + s$, the standard Farey label at phase x is*

x	(p, q)	$p + q$
0	(r, s)	D
1	(r, D)	$D + r$
-1	(s, D)	$D + s$
$n \ge 2$	$(D, (n-1)D + r)$	$nD + r$
$-n \le -2$	$(D, (n-1)D + s)$	$nD + s.$

The negative seed phases are omitted.

Proof. At seed phases $0, 1, n \ge 2$, the endpoint bases are respectively $((1, 0), (0, 1))$, $((1, 1), (0, 1))$, and $((1, n), (1, n-1))$. The count updates are linear, so substitution gives the table; negative phase exchanges the bases. For the inverse, odd/odd p, q give $(r, s, x) = (p, q, 0)$. If q is even, sort $p, q - p$ and choose $+1$ when p is the smaller entry, otherwise -1. If p is even, write $q = tp + u$ with $0 < u < p$; sort $u, p - u$ and choose $+(t + 1)$ or $-(t + 1)$ in the same way. A tie is the seed and uses the nonnegative representative. These formulas recover every mark. □

In particular, degree can be read directly from a reduced label:

$$D = \begin{cases} p + q, & p, q \text{ odd}, \\ q, & q \text{ even}, \\ p, & p \text{ even}. \end{cases} \tag{18}$$

This dictionary connects the Meridian and corridor collision windows below.

4. Meridian residues: modulo 27 and Fibonacci moduli

4.1. **Transferring a seed congruence to every address.** The recurrence for the companions contains no division, but a seed period of the *maximum* need not visibly preserve both companions.

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The following four-entry record resolves that issue:

$$\mathcal{S} = (A, B, C, H) = (9a^2, 9b^2, 3ab, H).$$

Forward and backward orbit steps and pole reflection are the polynomial maps

$$\begin{aligned} \mathcal{R}(A, B, C, H) &= (B, A + 9H^2B - 18HC, HB - C, H), \\ \mathcal{R}^{-1}(A, B, C, H) &= (B + 9H^2A - 18HC, A, HA - C, H), \\ \mathcal{J}(A, B, C, H) &= (A, B, C, C - H). \end{aligned} \tag{19}$$

For $h(x) = M_1(x)$ the seed record is

$$\mathcal{S}_1(x) = (8h(x) - h(x-1) - 2,\ h(x) + h(x-1) + 2,\ h(x) + 1,\ h(x)).$$

Every congruence preserving two consecutive seed maxima therefore propagates along every signed address.

Theorem 4.1 (Largest modulus with a common phase period)**.** *Every canonical Meridian has phase period T modulo m if and only if*

$$m \mid q(T), \qquad q(T) = 3F_{2T}F_{\gcd(2T,4)} = \begin{cases} 3F_{2T}, & T \text{ odd}, \\ 9F_{2T}, & T \text{ even}. \end{cases} \tag{20}$$

In particular, s is a common period modulo F_{2s}.

Proof. Fibonacci addition and (10) give

$$h(x + T) - h(x) = 3F_{2T}F_{4x+2T}.$$

Fibonacci strong divisibility, $\gcd(F_a, F_b) = F_{\gcd(a,b)}$, follows by the Euclidean algorithm from the addition formula. The gcd of all displayed differences is already the gcd of those at $x = 0, 1$, namely $q(T)$. Thus the seed has period T modulo m exactly when $m \mid q(T)$. The record (19) proves sufficiency at every address; the seed itself proves necessity. □

The theorem specifies a period common to all Meridians. It does not assert that the period is the smallest for each individual sequence. Nor does a large period exclude a scalar residue match at two earlier phases. For powers of three it gives

$$M_\alpha(x + 2 \cdot 3^n) \equiv M_\alpha(x) \pmod{3^{n+3}}, \qquad n \geq 0. \tag{21}$$

Indeed, $F_{3j} = F_j(5F_j^2 + 3)$ for even j, beginning with $F_4 = 3$, gives $v_3(F_{4\cdot 3^n}) = n + 1$. Figure 6 shows the common periods at the requested even-indexed Fibonacci moduli.

4.2. **Why modulo 27 gives only ten classes.** No positive Markov entry is divisible by three: mutation negates the changed coordinate modulo three and the integral tree starts at $(1, 1, 1)$. Let t, h be preceding and current poles. An extension by a nonzero index has the particularly simple action

$$S(t, h) = (h, t) \quad (k \text{ even}), \qquad R(t, h) = (h, 7h - t) \quad (k \text{ odd}) \pmod{27}. \tag{22}$$

To see this, put $N_j = 3u_ju_{j+1} - h$. Direct use of the orbit recurrence gives

$$N_0 = t, \quad N_1 = (9u_1^2 - 2)h - t, \quad N_{j+2} - N_j = 9h(u_{j+2}^2 - u_{j+1}^2).$$

All orbit squares are one modulo three, so $N_{j+2} \equiv N_j$ and $N_1 \equiv 7h - t \pmod{27}$. This works for negative indices as well. Only the parity of each recursively applied index survives.

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x	M1							M1.1							M1.2							M1.1.1							M1.-1.1						
s	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8
F_{2s}	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987
-12								1	5	13	31	13	231	181	1	1	13	50	1	88	937	2	1	20	50	17	328	944	2	1	5	9	89	285	215
-11								1	2	13	4	34	283	559	1	1	1	46	73	21	715	2	1	2	13	89	94	170	2	1	2	21	89	333	107
-10								1	1	19	13	121	248	544	1	2	16	48	82	270	541	2	2	20	37	98	29	482	2	5	8	34	53	272	869
-9								1	5	1	34	61	57	169	1	1	13	26	1	291	160	2	1	2	51	89	343	653	2	1	8	13	17	241	806
-8								1	2	13	50	130	169	13	1	1	13	53	73	343	433	2	1	20	31	17	34	923	2	1	5	21	17	86	89
-7								1	1	13	31	25	13	34	1	2	1	50	82	56	652	2	2	2	50	98	258	317	2	5	2	9	125	89	233
-6								1	5	19	4	13	34	985	1	1	16	46	1	21	541	2	1	20	13	17	369	104	2	1	8	21	89	233	890
-5								1	2	1	13	34	231	820	1	1	13	48	73	88	580	2	1	2	37	89	328	758	2	1	8	34	89	285	71
-4								1	1	13	34	121	283	181	1	2	13	26	82	21	937	2	2	20	51	98	94	944	2	5	5	13	53	333	215
-3								1	5	13	50	61	248	559	1	1	1	53	1	270	715	2	1	2	31	89	29	170	2	1	2	21	17	272	107
-2								1	2	19	31	130	57	544	1	1	16	50	73	291	541	2	1	20	50	17	343	482	2	1	8	9	17	241	869
-1								1	1	1	4	25	169	169	1	2	13	46	82	343	160	2	2	2	13	98	34	653	2	5	8	21	125	86	806
0	2	2	2	2	2	2	2	1	5	13	13	13	13	13	1	1	13	48	1	56	433	2	1	20	37	17	258	923	2	1	5	34	89	89	89
1	2	5	5	5	5	5	5	1	2	13	34	34	34	34	1	1	1	26	73	21	652	2	1	2	51	89	369	317	2	1	2	13	89	233	233
2	2	5	8	29	29	29	29	1	1	19	50	121	231	985	1	2	16	53	82	88	541	2	2	20	31	98	328	104	2	5	8	21	53	285	890
3	2	2	5	29	50	194	194	1	5	1	31	61	283	820	1	1	13	50	1	21	580	2	1	2	50	89	94	758	2	1	8	9	17	333	71
4	2	5	2	5	29	194	338	1	2	13	4	130	248	181	1	1	13	46	73	270	937	2	1	20	13	17	29	944	2	1	5	21	17	272	215
5	2	5	5	2	5	29	194	1	1	13	13	25	57	559	1	2	1	48	82	291	715	2	2	2	37	98	343	170	2	5	2	34	125	241	107
6	2	2	8	5	2	5	29	1	5	19	34	13	169	544	1	1	16	26	1	343	541	2	1	20	51	17	34	482	2	1	8	13	89	86	869
7	2	5	5	29	5	2	5	1	2	1	50	34	13	169	1	1	13	53	73	56	160	2	1	2	31	89	258	653	2	1	8	21	89	89	806
8	2	5	2	29	29	5	2	1	1	13	31	121	34	13	1	2	13	50	82	21	433	2	2	20	50	98	369	923	2	5	5	9	53	233	89
9	2	2	5	5	50	29	5	1	5	13	4	61	231	34	1	1	1	46	1	88	652	2	1	2	13	89	328	317	2	1	2	21	17	285	233
10	2	5	8	2	29	194	29	1	2	19	13	130	283	985	1	1	16	48	73	21	541	2	1	20	37	17	94	104	2	1	8	34	17	333	890
11	2	5	5	5	5	194	194	1	1	1	34	25	248	820	1	2	13	26	82	270	580	2	2	2	51	98	29	758	2	5	8	13	125	272	71
12	2	2	2	29	2	29	338	1	5	13	50	13	57	181	1	1	13	53	1	291	937	2	1	20	31	17	343	944	2	1	5	21	89	241	215

Figure 6. Five Meridians modulo F_{2s} for $s = 2, \ldots, 8$: the moduli are $3, 8, 21, 55, 144, 377, 987$. Each column has common phase period s, possibly a multiple of an individual least period. The shaded row is the helm; blank cells omit the seed's negative phases. The displayed residues use the reversed Meridian indexing of this edition.

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Proposition 4.2 (Universal modulo-27 classes)**.** *At integer phases every Meridian alternates between the two residues of one of the following sets, allowing the two singleton cases:*

$$\{1\},\ \{2,5\},\ \{4,16\},\ \{7,13\},\ \{8,17\},\ \{10,19\},\ \{11,23\},\ \{14,20\},\ \{22,25\},\ \{26\}.$$

The ten sets are pairwise disjoint. Meridians in different classes cannot share an integer value, regardless of degree or phase.

Proof. The common phase period is two because $q(2) = 27$. Track preceding and current poles at both phase parities, starting from $(1, 2, 1, 5)$. The simultaneous maps satisfy $S^2 = I$ and $SRS = R^{-1}$. Successive current-pole pairs on the R orbit are

$$\begin{array}{cccccc} (2,5) & (13,7) & (8,17) & (16,4) & (23,11) & (10,19) \\ (20,14) & (22,25) & (26,26) & (25,22) & (14,20) & (19,10) \\ (11,23) & (4,16) & (17,8) & (7,13) & (5,2) & (1,1). \end{array}$$

The preceding pair is the previous pair cyclically, and $b_{j+1} = 7b_j - b_{j-1}$ checks the return in 18 steps. These states and their S images form a closed set of 36 states. Their current-pole images are exactly the ten listed sets. Congruence of integer values would require membership in the same set. □

Table 1. The ten universal classes and their populations among the 519 Meridians of degree at most 100. Only the population column is bounded-degree data.

Residue set	Population	Residue set	Population
$\{1\}$	42	$\{10, 19\}$	47
$\{2, 5\}$	88	$\{11, 23\}$	47
$\{4, 16\}$	61	$\{14, 20\}$	38
$\{7, 13\}$	78	$\{22, 25\}$	35
$\{8, 17\}$	65	$\{26\}$	18

At that degree bound, the classification leaves 15,215 of the 134,421 pairs for further comparison. Larger Fibonacci moduli and compatible phases, rather than extrapolation from this table, supply the degree-10,000 certificate in Section 8.

5. Equal-degree separation and unequal-degree windows

5.1. **A complete result at each fixed degree.** We first use the established fixed-sum, fixed-numerator and fixed-denominator orderings of Markov numbers. Section 7 proves them by matching counts and records their earlier sources.

Theorem 5.1 (Equal-degree unicity)**.** *Distinct canonical integer-phase marks of the same Meridian degree have different maxima. The marks may belong to the same or to different Meridians.*

Proof. For $D \geq 4$, Lemma 3.5 partitions the labels into three classes. Helms r/s have fixed sum D; near-helms $r/D, s/D$ have fixed denominator D; outer phases have fixed numerator D. Each class is strictly ordered by its corresponding monotonicity, and the dictionary ensures that different marks have different labels.

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The largest helm is $m_{1/(D-1)}$. The near-helm extrema are $m_{1/D}, m_{(D-1)/D}$; the smallest outer value is $m_{D/(D+1)}$. They are separated by

$$m_{1/(D-1)} < m_{1/D}, \qquad m_{(D-1)/D} < m_{(D-1)/(D+1)} < m_{D/(D+1)}.$$

The intermediate fraction is reduced because D is even. The first two inequalities hold at fixed numerator and the last at fixed denominator. For $D = 2$, the unique seed gives $2, 5$ and then the increasing family $m_{2/(2n-1)}$ for $n \geq 2$, starting at 29. These values are also distinct. □

Thus the remaining Meridian comparison has unequal degrees. To bound it, we need exact leading terms and an explicit positive remainder.

5.2. **Positive words and a uniform tail bound.** Put $\rho = (3 + \sqrt{5})/2$, $\lambda = \log \rho$, and $\xi = \rho^x$. The endpoint matrices are

$$P(\xi) = \begin{pmatrix} B_0/\xi & C_0 \\ C_0 & A_0\xi \end{pmatrix}, \qquad Q(\xi) = \begin{pmatrix} A_0/\xi & \rho C_0 \\ C_0/\rho & B_0\xi \end{pmatrix}, \tag{23}$$
$$A_0 = \frac{15 + 3\sqrt{5}}{10}, \quad B_0 = \frac{15 - 3\sqrt{5}}{10}, \quad C_0 = \frac{2}{\sqrt{5}}.$$

Both have determinant one and positive entries. With seed endpoint words $U = P, V = Q$, the scalar state is

$$a_\alpha = \operatorname{tr} U/3, \quad b_\alpha = \operatorname{tr} V/3, \quad M_\alpha = \operatorname{tr}(UV)/3.$$

Its extension is represented by

$$(U, V) \longmapsto \begin{cases} ((UV)^{k-1}V, (UV)^k V), & k > 0, \\ ((VU)^{|k|}U, (VU)^{|k|-1}U), & k < 0. \end{cases} \tag{24}$$

For verification, $\operatorname{tr}((UV)^{j-1}V)/3$ begins at a, b and obeys the endpoint recurrence by Cayley–Hamilton. The determinant-one trace identity $\operatorname{tr}(XY) = \operatorname{tr} X \operatorname{tr} Y - \operatorname{tr}(XY^{-1})$ then gives the reflected pole. Negative indices use the reversed orbit. The count updates are exactly (17).

Proposition 5.2 (Laurent profile)**.** *For helm counts (r, s), degree $D = r + s$, and $t = (r - s)/2$,*

$$M_\alpha(x) = \sum_{j=-D,-D+2,\ldots,D} c_j \rho^{jx}, \quad c_j \geq 0, \quad c_{\sigma D} = \frac{1}{3}(9/5)^{D/2}\rho^{\sigma t}. \tag{25}$$

The logarithm is strictly convex, and the real profile has one minimum.

Proof. Expand the trace into closed paths on the two matrix states. All weights are nonnegative. A closed path has an even number of off-diagonal choices, which fixes the parity of its exponent. The two constant-state paths give the extreme coefficients because $A_0B_0 = 9/5$ and $A_0/B_0 = \rho$. The second derivative of the logarithm is λ^2 times the variance of the exponents under their positive normalized weights. Both extremes occur, so the variance is positive. They also force growth at both ends. □

Lemma 5.3 (Uniform positive-word remainder)**.** *Let W have r letters P and s letters Q, of total length $g = r + s > 0$. For $n \geq 0$ and $\sigma = \pm 1$,*

$$\frac{\operatorname{tr} W(\rho^{\sigma n})}{3} = \frac{1}{3}(9/5)^{g/2}\rho^{\sigma(r-s)/2}\rho^{gn}(1 + \epsilon_{W,\sigma}(n)), \qquad 0 \leq \epsilon_{W,\sigma}(n) \leq (1 + \rho^{1-2n})^g - 1. \tag{26}$$

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Proof. Write $z = \rho^{-n}$, $\varphi = \sqrt{\rho}$, and $L = 2\varphi/3$. On the positive arm, divide the P, Q factors by $A_0\rho^n, B_0\rho^n$ respectively and conjugate by $\mathrm{diag}(\rho, 1)$. The normalized factors are

$$\begin{pmatrix} \rho^{-1}z^2 & L\rho^{-2}z \\ Lz & 1 \end{pmatrix}, \qquad \begin{pmatrix} \rho z^2 & Lz \\ Lz & 1 \end{pmatrix}.$$

The negative arm gives the same pair in reverse roles after exchanging the basis states. Both are bounded entrywise by the rank-one matrix

$$K = \begin{pmatrix} \rho z^2 & \varphi z \\ \varphi z & 1 \end{pmatrix}.$$

Products preserve the entrywise bound, and $\operatorname{tr} K^g = (1 + \rho z^2)^g$. The constant dominant-state path contributes one to the normalized trace. Restoring the removed factors proves the bounds. □

5.3. **A finite rectangle depending only on the degrees.**

Theorem 5.4 (Unequal-degree collision window)**.** *Let $D_\alpha \neq D_\beta$, and define*

$$m = \frac{|D_\alpha - D_\beta|}{2}, \qquad E = \max(D_\alpha, D_\beta),$$
$$h = \min\{j \in \mathbb{Z}_{\geq 0} : \rho^j \geq E\}, \qquad N = 3m + h + 3, \qquad T = E(N + 2).$$

An integer equality $M_\alpha(x) = M_\beta(y)$ requires

$$|x| \leq \left\lceil \frac{T}{D_\alpha} \right\rceil, \qquad |y| \leq \left\lceil \frac{T}{D_\beta} \right\rceil, \qquad M_\alpha(x) < \rho^T/3. \tag{27}$$

The seed still uses only nonnegative phases.

Proof. The degrees are even, so m is a positive integer. First,

$$|m \log(9/5) - j \log \rho| > \frac{1}{81^m + 1} \qquad (j \in \mathbb{Z}). \tag{28}$$

Choose $j_0 \geq 0$ with $1 < (9/5)^m \rho^{-j_0} < \rho$; equality at an endpoint is impossible by taking the field norm. The positive algebraic integers $9^m - 5^m\rho^{j_0}$ and $5^m\rho^{j_0+1} - 9^m$ have nonzero integer norms and conjugates of absolute value below 9^m. Each is therefore greater than 9^{-m}. Dividing by the appropriate positive denominators makes both adjacent multiplicative gaps greater than 81^{-m}. The inequality $\log(1 + u) > u/(1 + u)$ gives (28); all other integers j are farther away.

At phases $x = \sigma n$, $y = \tau n'$, the leading logarithms differ by

$$\frac{D_\alpha - D_\beta}{2} \log(9/5) + (D_\alpha n - D_\beta n' + \sigma t_\alpha - \tau t_\beta) \log \rho.$$

The coefficient of $\log \rho$ is integral. Put $r_m = (81^m + 1)^{-1}$. Since $E \leq \rho^h$, $\rho^6 > 81$ and $\rho^5 > 5$,

$$E\rho^{1-2N} \leq \rho^{-6m-5} < \frac{1}{2 \cdot 81^m + 3} = \frac{r_m}{r_m + 2}.$$

For $gu < 1$, the elementary bound $(1+u)^g \leq \sum_{j\geq 0}(gu)^j = (1-gu)^{-1}$ makes each relative remainder in Lemma 5.3 smaller than $r_m/2$ when its phase magnitude is at least N. Their logarithmic corrections cannot cancel (28). At least one collision phase therefore has magnitude less than N.

The leading coefficient of a length-g word lies strictly between $\rho^{-g}/3$ and $\rho^g/3$, because $\rho^{-1} < B_0 < A_0 < \rho$. Lemma 5.3, with $1 + \rho^{1-2N} < \rho$, gives $\operatorname{tr} W(\rho^{\pm N})/3 < \rho^{g(N+2)}/3 \leq \rho^T/3$. Convexity bounds a value inside either finite phase strip by its endpoints, so the common value is below $\rho^T/3$. Positivity also gives the lower bound $\operatorname{tr} W(\rho^{\sigma n})/3 > \rho^{g(n-1)}/3$. Comparing the two bounds yields $n < 1 + T/g$, hence the integer ceilings in (27). □

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This theorem bounds the remaining candidates; it does not assert that any candidate collides. It also does not bound the degrees of a hypothetical collision. The next section supplies a different finite reduction, expressed in Farey numerators and denominators rather than Meridian phases.

6. Corridor walls and numerator–denominator windows

6.1. **A second way to travel through the same tree.** The corridor observation groups the walls between neighboring phasors. After an initial entrance, a wall follows a fixed-numerator family

$$m_{p/(q_0+np)}, \qquad n \geq 0.$$

Both Meridian and phasor edges occur in such a route. Their common law is the advance $q \mapsto q+p$, not a rule that distinguishes the two edge colors. For the blue corridor the fixed-numerator portions are $3/(4+3n)$ and $3/(5+3n)$, tending to the historical coordinate $c = 1/3$. The first wall begins

$$169 \xrightarrow{\text{phasor}} 2897 \xrightarrow{\text{Meridian}} 51641 \xrightarrow{\text{phasor}} 925765 \longrightarrow \cdots,$$

and the other begins $433, 7561, 135137, 2423525, \ldots$ with the opposite edge alternation.

Corridor role and edge color must be kept separate. With corridor-wall travel designated *standard* and a new entrance *exotic*, the two entry routes here are

$$5 \xrightarrow{\text{exotic}} 29 \xrightarrow{\text{standard}} 433 \xrightarrow{\text{standard}} 7561 \longrightarrow \cdots, \qquad 13 \xrightarrow{\text{exotic}} 169 \xrightarrow{\text{standard}} 2897 \longrightarrow \cdots.$$

Both exotic entrances are Meridian edges. In the first route the initial standard step $29 \to 433$ changes the numerator from two to three; the fixed-numerator formula starts at 433. Thus excluding the concave ceiling arc from a wall's recurrence does not move the exotic entrance to that arc. The graph and its actual depth (14) are unchanged.

6.2. **The exact recurrence behind the observed regularity.** Use the standard Christoffel matrices

$$D_a = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad D_b = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{29}$$

Their trace encoding is

$$3m_{p/q} = \operatorname{tr}\left[\prod_{j=1}^{p} D_a^{\lfloor jq/p \rfloor - \lfloor (j-1)q/p \rfloor} D_b\right]. \tag{30}$$

The word is a cyclic conjugate of the lower Christoffel word. Trace is unchanged by that rotation. The normalization is fixed by $\operatorname{tr} D_a = 3$, $\operatorname{tr}(D_a D_b) = 6$ and the Farey trace identity, which give the Markov recurrence with the initial values one and two. This is the classical matrix-word framework [9]; Section 7 also gives its matching interpretation.

Replacing q by $q_0 + np$ adds n to every exponent in (30). The eigenvalues of D_a are ρ, ρ^{-1}. Choose its orthogonal eigenbasis so that D_b becomes

$$\begin{pmatrix} \frac{3}{2} + \frac{3}{2\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{3}{2} - \frac{3}{2\sqrt{5}} \end{pmatrix}.$$

All four entries are positive. Expanding the trace over cyclic choices of the two eigenstates gives the exact identity

$$m_{p/(q_0+np)} = \sum_{j=0}^{p} c_{p,q_0,j}\, \rho^{(p-2j)n}, \qquad c_{p,q_0,j} > 0. \tag{31}$$

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Every mode occurs. The least order of a constant-coefficient recurrence is therefore $p+1$, with characteristic polynomial

$$\prod_{j=0}^{p}(X-\rho^{p-2j}).$$

Pairing conjugate roots shows that its coefficients are integers. For $p=3$ the polynomial is $(X^2-18X+1)(X^2-3X+1)$, so either blue wall satisfies

$$m_{3/(q+12)} = 21m_{3/(q+9)} - 56m_{3/(q+6)} + 21m_{3/(q+3)} - m_{3/q}. \tag{32}$$

This equality holds at every step, for both colors of edge.

The exact regularity is a shared recurrence, not a constant output ratio. The positive spectral sum gives strict log-convexity, hence

$$\log_{10}\frac{m_{p/(q_0+(n+1)p)}}{m_{p/(q_0+np)}} \nearrow p\log_{10}\rho.$$

The small deviation is retained in the collision argument, rather than discarded as an asymptotic error. Figure 7 shows both the geometric walls and that deviation.

6.3. **A positive correction with a uniform bound.** The dominant eigenstate in (30) yields

$$3m_{p/q} = \left(\frac{3}{\sqrt{5}}\right)^{p}\left(\frac{1+\sqrt{5}}{2}\right)^{2q+p}(1+\varepsilon_{p,q}), \tag{33}$$

where

$$0 < \varepsilon_{p,q} \le \left(1+\rho^{-2\lfloor q/p\rfloor}\right)^{p} - 1. \tag{34}$$

To prove the bound, divide the transformed D_b by its upper-left entry; every resulting entry is at most one. Choosing the contracting eigenstate at a factor costs at least $\rho^{-2\lfloor q/p\rfloor}$. Summing over all nonempty subsets of contracting choices gives (34). This bound is independent of the remainder class of q modulo p.

Along a fixed wall the correction decreases strictly to zero. Consequently its logarithmic deviations telescope exactly:

$$\sum_{k=0}^{n-1}\left[p\log\rho - \log\frac{m_{p/(q+(k+1)p)}}{m_{p/(q+kp)}}\right] = \log\frac{1+\varepsilon_{p,q}}{1+\varepsilon_{p,q+np}}. \tag{35}$$

Thus there is no accumulated error proportional to the path length. For a route changing corridors, subtraction of the endpoint formula still telescopes, but monotone decay is claimed only within a fixed wall.

Theorem 6.1 (Necessary correction gap)**.** *If $p \neq p'$ and $m_{p/q} = m_{p'/q'}$ for reduced labels, then*

$$\frac{|\varepsilon_{p,q} - \varepsilon_{p',q'}|}{1+\min(\varepsilon_{p,q},\varepsilon_{p',q'})} > 9^{-|p-p'|}. \tag{36}$$

Proof. Assume $p > p'$. The ratio of the leading terms is

$$R = \left(\frac{3(1+\sqrt{5})}{2\sqrt{5}}\right)^{p-p'}\rho^{q-q'}.$$

Its quadratic-field norm is $(9/5)^{p-p'}$, so $R \neq 1$. The number $(\sqrt{5})^{p-p'}(R-1)$ is an algebraic integer, because the powers of $(1+\sqrt{5})/2$ and ρ are units. Its nonzero norm gives

$$|(R-1)(\overline{R}-1)| \ge 5^{-(p-p')}, \qquad \overline{R} = (9/5)^{p-p'}/R > 0.$$

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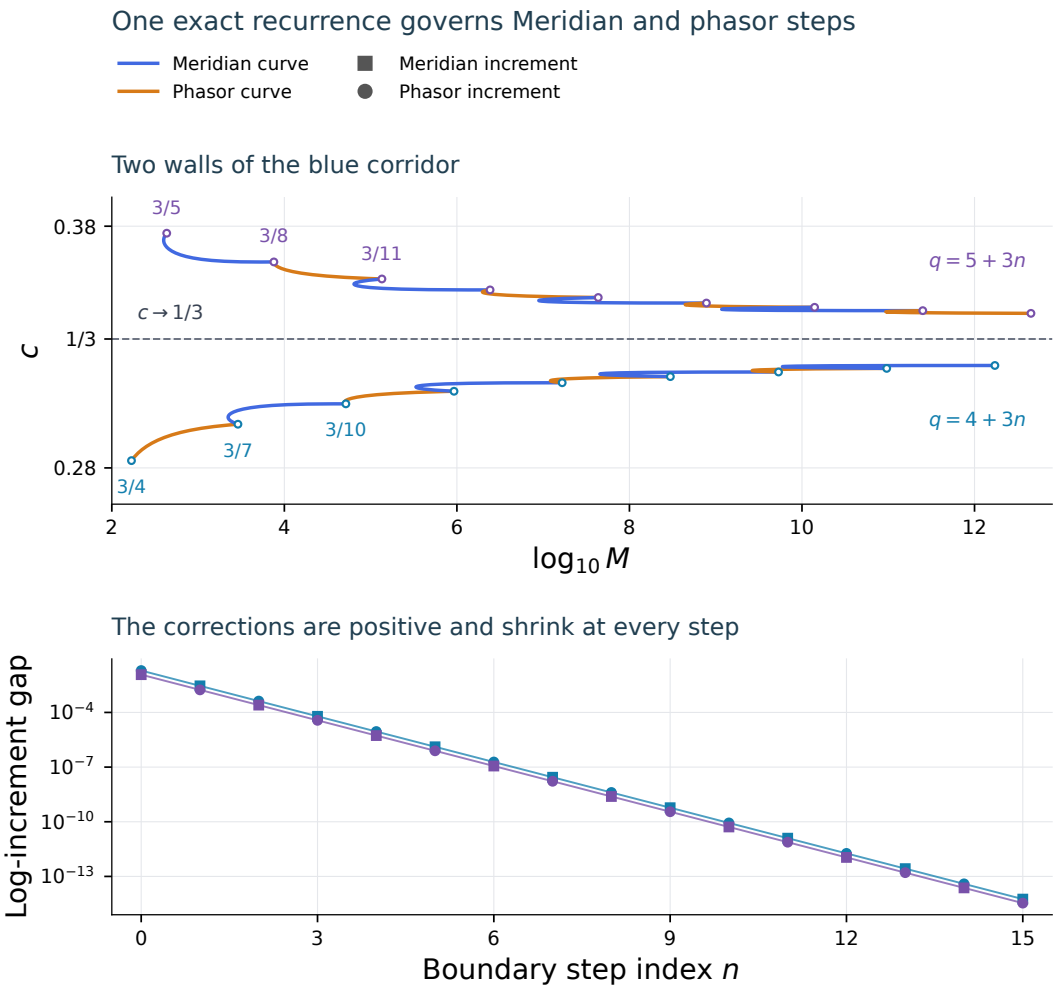


Figure 7. The two fixed-numerator walls of the blue corridor. The concave ceiling arc is omitted from this recurrence illustration. Both walls obey (32). In the lower panel, the positive difference between $3\log_{10}\rho$ and the actual logarithmic increment decreases at every step. Squares denote Meridian increments and circles phasor increments. The values are exact integers; only the plotted curves and logarithms are numerical.

Put $\eta = \max(R, R^{-1}) - 1$. If $R \geq 1$, then $|R - 1| = \eta$ and $|\overline{R} - 1| < (9/5)^{p-p'}$. If $R < 1$, their product is $\eta((9/5)^{p-p'} - 1/(1+\eta)) < \eta(9/5)^{p-p'}$. Either case forces $\eta > 9^{-(p-p')}$. At equality of the Markov values, $R = (1 + \varepsilon_{p',q'})/(1 + \varepsilon_{p,q})$, so η is precisely the left side of (36). □

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6.4. **The corridor collision window.**

Theorem 6.2 (Numerator–denominator window)**.** *Let $p \neq p'$, $P = \max(p, p')$, and let K be the least nonnegative integer satisfying*

$$\rho^{2K} > P(9^{|p-p'|} + 1). \tag{37}$$

Every equality $m_{p/q} = m_{p'/q'}$ requires

$$\max(q, q') \leq P(K + 1) - 1. \tag{38}$$

For bounded numerators a simpler, weaker consequence is $\max(q, q') \leq 2P^2 + P - 1$.

Proof. If both quotient floors are at least K, then $P\rho^{-2K} < 1/(9^{|p-p'|} + 1)$. The geometric-series estimate used in Lemma 5.3 makes both corrections strictly smaller than $9^{-|p-p'|}$, contradicting Theorem 6.1. Thus one denominator, say q, is at most $Kp - 1$.

Both matrices in (29) have operator norm ρ. The trace bound and the positive leading term imply

$$\frac{1}{3}\rho^{q'} < m_{p'/q'} = m_{p/q} \leq \frac{2}{3}\rho^{q+p}.$$

Since $\log_\rho 2 < 1$ and the exponents are integers, $q' \leq q + p$. It follows that both denominators are at most $P(K + 1) - 1$. For the simpler bound it suffices to use $K = 2P$: $\rho^{4P} > 36^P > P(9^{P-1} + 1)$ for $P \geq 2$, and $|p - p'| \leq P - 1$. □

For example, numerators at most 64 permit $K = 75$, so every remaining comparison lies at denominators at most 4863. With $P = 64$ and numerator difference five, $K = 8$ gives 575. These are bounds on candidates, not new numerical verification claims.

Theorem 5.1 first discards equal Meridian degrees. For the rest, the phase rectangle (27) and denominator cap (38) can be imposed together through the dictionary (18). Fixed-numerator ordering already separates all labels with $p = p'$. What remains is a comparison of different numerators and degrees inside their joint windows. Neither numerator nor degree is globally bounded by these theorems, so this is not yet a finite reduction of the entire unicity conjecture.

7. Snake-graph counts and ordering proofs

7.1. **From a Farey label to an actual matching count.** A snake graph is a chain of square tiles, each attached to the right of or above the preceding tile. Shared edges are counted once. A perfect matching covers every vertex exactly once. The Markov snake associated with a reduced $0 < p/q < 1$ is built as follows. Take the lower staircase from $(0, 0)$ to (q, p): at each horizontal coordinate j, step right and then up to height $\lfloor pj/q \rfloor$. Write right steps as a and up steps as b. The resulting lower Christoffel word has the form $w_{p/q} = azb$. Remove its first and last letters; start with one square and replace each a of z by two added squares to the right and each b by two upward. The resulting snake has $2(p + q) - 3$ tiles.

Write $\mu(w)$ for its number of perfect matchings. The established Markov matching correspondence gives

$$\mu(w_{p/q}) = m_{p/q}. \tag{39}$$

The matching interpretation originated in the REACH work of 2002–2003. Musiker's memorandum, dated 27 June 2002, proposed the recursive graph model [13]; Itsara, Musiker, Propp, and Viana's manuscript of 10 January 2003 stated and proved the snake-matching correspondence [14, Theorem 3.1]. Setting all edge weights to one gives the ordinary perfect-matching counts. Propp's later exposition, first posted in 2005, records the broader collaborative development [15]. These original contributions precede the later proofs of Aigner's orderings.

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For a modern treatment and the Farey-indexed construction, see [12, Section 4.1, Theorem 4.2] and [6]. For example, $w_{2/5} = a^3ba^2b$, its trimmed word is $aabaa$, and the eleven-tile snake has 194 matchings. Auxiliary rearranged words also specify snakes, counted by μ; they need not be canonical Markov snakes.

7.2. **Six pieces and two connector states.** Figure 8 fixes the five three-tile pieces and the three one-square variants of type 6. In a cut between pieces, a matching uses both yellow connector edges or neither: the cut leaves an even number of vertices on either side. Denote these states by 1 and 0. At a selected port remove the two vertices already covered by its connector pair. Direct counting within the pieces gives

(40)

piece	00	01	10	11
1	5	2	2	1
2	5	3	3	2
3	5	2	3	1
4	5	3	2	1
5	5	2	2	1
6i	2	1	1	1

The state order is incoming, outgoing. The one-port end squares 6ii and 6iii have counts $(2, 1)$.

For the ordering surgeries, use a fixed one-ended chop: retain tile triples $(1, 2, 3), (5, 6, 7), \ldots$, with tiles $4, 8, \ldots$ providing paired connectors. A one-square remainder occurs exactly when $p + q$ is even and is counted with the closed end piece. For every connector sequence σ, multiply the applicable row entries in (40); then sum over σ, with outer ports closed. Each matching determines one sequence and one internal completion per piece, and gluing reverses the construction. This proves the counting rule, without an independence assumption. Equivalently it is a sum of terms $5^{k(\sigma)}3^{j(\sigma)}2^{\ell(\sigma)}$, where the exponents count factors of the indicated size. For the 2/5 snake the piece list $2, 1, 2$ gives

$$5^3 + 5 \cdot 2 \cdot 3 + 3 \cdot 2 \cdot 5 + 3^2 = 194.$$

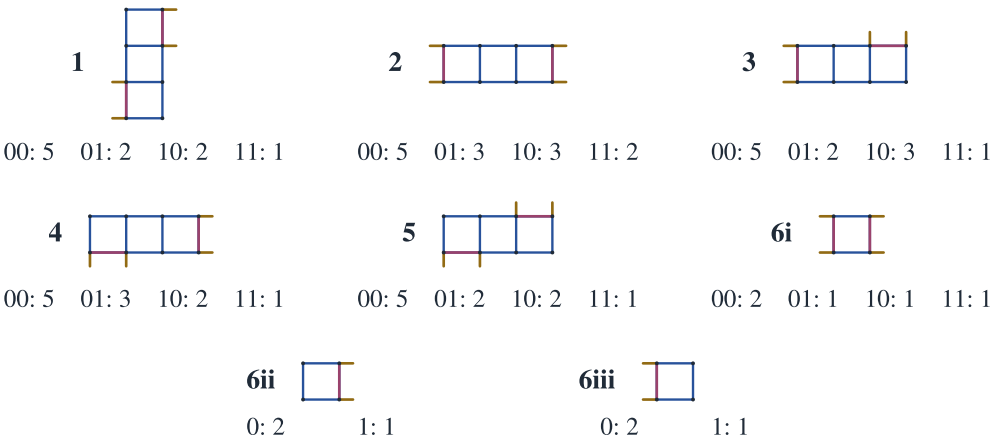


Figure 8. The six physical pieces. Gold edges are paired connectors, magenta marks ports, and dots mark tile vertices. Two-port counts use the order $00, 01, 10, 11$. The central square 6i has counts $(2, 1, 1, 1)$; the one-port end squares have $(2, 1)$. Types 1 and 5 have equal count rows but different physical orientations.

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7.3. **Fibonacci-matrix normalization.** The Fibonacci-matrix formulation and normalization argument below are the author's. We begin with the comparison of three consecutive blocks and then apply it to complete words.

With $F_0 = 0, F_1 = 1$ and $F_{-j} = (-1)^{j+1}F_j$, the basic odd-index identity is

$$F_{2k-1}F_{2k+3} - F_{2k+1}^2 = 1. \tag{41}$$

For example, $5 \cdot 34 - 13^2 = 1$. Consider the author's three consecutive Fibonacci blocks:

$$\begin{aligned} A &= \begin{pmatrix} F_{2k+3} & F_{2k+1} \\ F_{2k+1} & F_{2k-1} \end{pmatrix}, \\ B &= \begin{pmatrix} F_{2k+1} & F_{2k-1} \\ F_{2k-1} & F_{2k-3} \end{pmatrix}, \\ C &= \begin{pmatrix} F_{2k-1} & F_{2k-3} \\ F_{2k-3} & F_{2k-5} \end{pmatrix}, \qquad k \geq 2. \end{aligned} \tag{42}$$

Applying (41) twice gives precisely the upper-left comparison used in normalization:

$$\begin{aligned} (AC)_{11} - (B^2)_{11} &= (F_{2k+3}F_{2k-1} - F_{2k+1}^2) \\ &\quad + (F_{2k+1}F_{2k-3} - F_{2k-1}^2) = 2. \end{aligned} \tag{43}$$

The same identities in the other entries give

$$AC - B^2 = \begin{pmatrix} 2 & 6 \\ 0 & 2 \end{pmatrix}, \qquad CA - B^2 = \begin{pmatrix} 2 & 0 \\ 6 & 2 \end{pmatrix}.$$

Thus replacing AC or CA by B^2 decreases the upper-left entry even inside a longer product: the surrounding Fibonacci blocks have nonnegative entries and positive upper-left entries. For $k = 3$ this reads $(AC)_{11} = 34 \cdot 5 + 13 \cdot 2 = 196 > 194 = 13^2 + 5^2 = (B^2)_{11}$.

7.3.1. *Run counts and general index shifts.* To apply the same comparison at arbitrary run lengths, index the blocks by

$$\mathcal{A}_t = \begin{pmatrix} F_{2t+3} & F_{2t+1} \\ F_{2t+1} & F_{2t-1} \end{pmatrix}, \qquad t \geq 0. \tag{44}$$

Thus $A = \mathcal{A}_k$, $B = \mathcal{A}_{k-1}$ and $C = \mathcal{A}_{k-2}$ in (42). The four boundary counts of a straight block of $2t + 1$ tiles are $F_{2t+3}, F_{2t+1}, F_{2t+1}, F_{2t-1}$: these follow by conditioning on the first edge of a ladder, giving the Fibonacci recurrence; at $t = 0$ they are $(2, 1, 1, 1)$. Gluing successive runs with their paired connector states therefore gives, for $w = a^{h_1}b \cdots a^{h_p}b$ with $h_1 \geq 1$ and $h_j \geq 0$,

$$\mu(w) = [\mathcal{A}_{h_1-1}\mathcal{A}_{h_2} \cdots \mathcal{A}_{h_p}]_{11}. \tag{45}$$

The first run is shortened by the removed initial a; the final b closes the other end. Equivalently, a block-index list $(t_1, \ldots, t_p)$ represents $w = a^{t_1+1}ba^{t_2}b \cdots a^{t_p}b$, with $t_1 + \cdots + t_p = q - 1$.

Fibonacci addition gives the scalar index-shifting identity

$$F_{2r+1}F_{2s+1} - F_{2r-1}F_{2s+3} = F_{2(r-s)-2}.$$

Apply it to the two summands of each product entry to obtain

$$\mathcal{A}_r\mathcal{A}_s - \mathcal{A}_{r-1}\mathcal{A}_{s+1} = 2\begin{pmatrix} F_{2(r-s)-2} & F_{2(r-s)} \\ F_{2(r-s)-4} & F_{2(r-s)-2} \end{pmatrix}, \quad r \geq 1,\ s \geq 0. \tag{46}$$

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Consequently, whenever two adjacent indices differ by at least two, moving one unit from the larger to the smaller strictly decreases the whole product's upper-left entry. The case $r = s + 2$ is exactly (43). The only remaining local comparison has adjacent indices differing by one; it is an interchange of the two blocks.

7.3.2. *Comparing the surrounding Fibonacci blocks.* For a finite list of block indices, define the ratio of its two open counts by

$$\rho(u_1, \dots, u_m) = \frac{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{21}}{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{11}}, \qquad \rho(\varnothing) = 0.$$

All these ratios lie in $[0, 1)$. Direct multiplication gives

$$\rho(t, U) = \frac{F_{2t+1} + F_{2t-1}\rho(U)}{F_{2t+3} + F_{2t+1}\rho(U)}. \tag{47}$$

Its right-hand side increases strictly with $\rho(U)$, since (41) gives $\det \mathcal{A}_t = 1$. Moreover, another application of that identity gives

$$\frac{F_{2t+1}}{F_{2t+3}} - \frac{F_{2t+3} + F_{2t+1}}{F_{2t+5} + F_{2t+3}} = \frac{1}{F_{2t+3}(F_{2t+5} + F_{2t+3})} > 0.$$

Hence $\rho(t + 1, U) < \rho(t, V)$ for any two finite tails U, V. Two lists can therefore be compared by cancelling equal initial indices: at the first unequal pair, the larger index gives the smaller ratio; an exhausted list has the smaller ratio than any nonempty remainder.

For adjacent blocks in positions $j, j + 1$, the outward lists are $(t_{j-1}, \dots, t_1)$ on the left and $(t_{j+2}, \dots, t_p)$ on the right. At $r = s + 1$, (46) becomes

$$\mathcal{A}_{s+1}\mathcal{A}_s - \mathcal{A}_s\mathcal{A}_{s+1} = 2 \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

After multiplication by the surrounding blocks, its upper-left entry has the sign of

$$\rho(t_{j+2}, \dots, t_p) - \rho(t_{j-1}, \dots, t_1). \tag{48}$$

Indeed, it equals this difference times twice the positive upper-left entries of the two surrounding products. This proves the needed surrounding-block test entirely in the same Fibonacci matrices.

Lemma 7.1 (Fibonacci-matrix normalization)**.** *For coprime positive p, q, every word $w = azb$ with q letters a and p letters b satisfies $\mu(w_{p/q}) \le \mu(w)$.*

Proof. For $p = 1$ there is only the word $a^q b$. Assume $p > 1$ and use the block-index list in (45). Its prefix discrepancies are

$$\delta_j = qj - p(1 + t_1 + \cdots + t_j) \quad (1 \le j \le p), \qquad \delta_0 = \delta_p = 0.$$

Changing (t_j, t_{j+1}) to $(t_j + 1, t_{j+1} - 1)$ decreases only δ_j, by p; the reverse change increases only δ_j, by p. We show that an appropriate change always decreases the matching count until $-p \le \delta_j \le 0$ for every j.

The sign checks use just a prefix-sum identity. Put $h_1 = t_1 + 1$ and $h_i = t_i$ for $i > 1$, the actual run lengths. Whenever $j - m \ge 0$ and $j + m \le p$,

$$\delta_{j-m} + \delta_{j+m} = 2\delta_j + p \left(\sum_{i=j-m+1}^{j} h_i - \sum_{i=j+1}^{j+m} h_i \right). \tag{49}$$

In the block indices, the left sum has the extra 1 precisely when it includes t_1.

First remove positive discrepancies. Choose the rightmost largest $\delta_j > 0$ and transfer one unit from t_{j+1} to t_j. Comparison with its two neighboring discrepancies gives $h_j < q/p < h_{j+1}$; coprimality

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makes these inequalities strict. Thus the transfer is allowed and $t_{j+1} \geq t_j + 1$; for $j = 1$ the gap is at least two. Gaps of at least two are handled by (46).

If $t_{j+1} = t_j + 1$, compare the outward lists. A first unequal pair with the left index larger would make the parenthesis in (49) nonnegative: the central difference is -1, the equal outward indices cancel, and the first unequal difference is at least 1. This contradicts maximality and the choice of the rightmost maximum. If the left list were exhausted first, its extra initial 1 would instead give $\delta_{2j} = 2\delta_j > \delta_j$, again impossible (including simultaneous exhaustion, when $2j = p$). The left outward ratio is therefore strictly larger than the right one. Reversing the interchange in (48) shows that the selected transfer strictly decreases the count. It also decreases $\sum_{j=1}^{p-1} \delta_j$. There are only finitely many nonnegative block-index lists of sum $q - 1$, so this phase ends with all $\delta_j \leq 0$.

Then remove discrepancies below $-p$. Choose the leftmost smallest $\delta_j < -p$ and transfer one unit from t_j to t_{j+1}. For $j > 1$, the neighboring comparisons give $h_j > q/p > h_{j+1}$; for $j = 1$, the inequality $\delta_1 < -p$ gives $t_1 > q/p$, while $t_2 < q/p$. Thus $t_j \geq t_{j+1} + 1$ and the transfer is allowed. Again a gap of at least two is settled by (46).

In the gap-one case, a first unequal outward pair with the left index smaller would make the parenthesis in (49) nonpositive, contradicting minimality and the leftmost choice. If that pair reaches t_1, its extra 1 instead gives $\delta_{2j} \leq 2\delta_j + p < \delta_j$, the same contradiction. If the right list were exhausted first, the identity would give $\delta_{2j-p} = 2\delta_j + p < \delta_j$. Simultaneous exhaustion would give $\delta_j = -p$. Thus the left outward ratio is strictly smaller than the right one, and (48) proves a strict decrease in count. The changed discrepancy is $\delta_j + p < 0$, so positivity is not reintroduced. This phase increases the bounded sum of discrepancies by p each time and terminates with $-p \leq \delta_j \leq 0$.

Finally, $\delta_j \equiv qj \pmod{p}$. For $1 \leq j < p$, coprimality leaves exactly one possible value in $[-p, 0]$, forcing

$$1 + t_1 + \cdots + t_j = \left\lceil \frac{jq}{p} \right\rceil .$$

These are precisely the canonical run lengths. Every transfer has decreased the upper-left product entry, proving the assertion. □

For the canonical word, $h_j = \lceil jq/p \rceil - \lceil (j-1)q/p \rceil$. With $k = 1 + \lfloor (q-1)/p \rfloor$, the factors are the author's A, B from (42), and the end factors are B. For example, $m_{3/8} = [BAB]_{11}$ with $k = 3$. A similar normalization result appears in [9, Proposition 21]; the proof above is self-contained in the author's Fibonacci matrices.

7.4. **The three Aigner orderings.** Fixed-numerator monotonicity was proved by Rabideau and Schiffler [12]; all three orderings were proved by LPTV, Lee–Li–Rabideau–Schiffler, and McShane [9, 10, 11]. The author's Fibonacci matrices prove normalization and the fixed-numerator ordering. His six pieces prove the remaining fixed-sum comparison and the fixed-denominator ordering, using normalization to return to a canonical Markov snake.

Theorem 7.2 (Established monotonicities)**.** *For $i > 0$, whenever both fractions are reduced and lie in $(0, 1)$,*

$$\begin{aligned} m_{p/q} &> m_{(p+i)/(q-i)} && \textit{(fixed sum)}, \\ m_{p/q} &< m_{p/(q+i)} && \textit{(fixed numerator)}, \\ m_{p/q} &< m_{(p+i)/q} && \textit{(fixed denominator)}. \end{aligned}$$

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7.4.1. *Fixed numerator: Fibonacci matrices.*

Proof. Fibonacci recurrence gives, for $t \geq 1$,

$$\mathcal{A}_t - \mathcal{A}_{t-1} = \begin{pmatrix} F_{2t+2} & F_{2t} \\ F_{2t} & F_{2t-2} \end{pmatrix}.$$

All entries are nonnegative and the upper-left is positive. Start with $w_{p/(q+i)}$ and delete i internal letters a, retaining the first. Each deletion changes one factor $\mathcal{A}_t$ to $\mathcal{A}_{t-1}$ with $t \geq 1$. Surrounding factors are positive, so the matching count strictly falls. The resulting word ends at (q, p); normalization gives $m_{p/q} \leq \mu(w_*) < m_{p/(q+i)}$. □

7.4.2. *Fixed sum: six pieces.*

Proof. At the same physical cuts, the changes $2 \to 3$ or 4, then 3 or $4 \to 5$, increase no entry of (40); the changed factors are $3 \to 2$ or $2 \to 1$. Rotation $5 \leftrightarrow 1$ preserves the row. Selecting the changed connector and no others exhibits a strictly smaller positive summand whenever an equal-direction join becomes unequal.

The trimmed canonical word begins and ends with a, contains no bb, and has $q-p-1$ equal joins, all aa. At its rightmost equal join, interchange a, b strictly to the right. That suffix is alternating, so only the chosen join changes. Exactly one prescribed piece changes by the stated rules, while any rotated pieces keep their rows. Each surgery strictly decreases the count. Odd-numbered operations replace one horizontal step by a vertical step in net; even-numbered operations keep the totals. After $2i$ operations the endpoint is $(q-i, p+i)$. There are enough joins because $p+i < q-i$. For the resulting auxiliary word w_*, normalization gives $m_{(p+i)/(q-i)} \leq \mu(w_*) < m_{p/q}$. A similar tail comparison appears in [9, Proposition 22]; here the comparison is proved using the author's six physical pieces. □

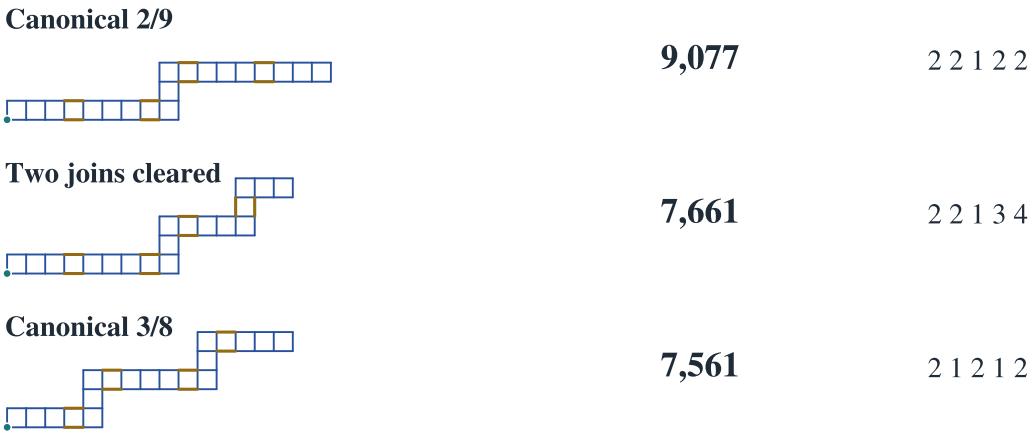


Figure 9. Fixed-sum surgery on the same five pieces. The canonical 2/9 snake has 9077 matchings; after two changes the auxiliary snake has 7661; normalization gives the 3/8 snake with 7561. The intermediate first change, not shown, has 8263. Gold pairs mark cuts and teal dots the starting corners.

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7.4.3. *Fixed denominator: six pieces.*

Proof. For the fixed-denominator comparison, refine the same six-piece chop into single-square pieces of type 6, retaining odd-numbered tiles and using the even tiles as paired connectors. A retained square has counts $(2,1,1,1)$ for opposite ports and $(2,1,1,0)$ for adjacent ports. The zero forbids covering their common corner twice. Summing an intervening state regroups these squares into the original three-tile pieces.

Insert one internal b, adding two vertical squares. The affected local counts are the following complete table, with outside ports fixed:

	before				after			
directions	00	01	10	11	00	01	10	11
a,a	2	1	1	1	5	2	2	1
b,b	2	1	1	1	5	3	3	2
a,b	2	1	1	0	5	3	2	1
b,a	2	1	1	0	5	2	3	1

.

The endpoint caps have the corresponding rows with an outer port closed; the empty trimmed word has the increase $2 \to 5$. No outside count falls, and the all-zero choice witnesses a strict increase. Thus the two added squares compensate for every changed direction.

Starting from $w_{(p+i)/q}$, delete i internal b's and reverse this comparison. At least i are available. Normalizing the resulting word to (q,p) proves $m_{p/q} \le \mu(w_*) < m_{(p+i)/q}$. □

7.5. **Folding an actual snake at its center.** The same local counts also compare families sharing a long initial sequence of pieces. Represent the four counts by

$$R_1 = R_5 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}, \quad R_4 = \begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}.$$

Attaching piece i to an open-count row (x,y) replaces it by $(x,y)R_i$. This is the sum over the common connector state, not an additional assumption about matchings.

For a helm, use the symmetric central chop: retain the middle square as 6i and group retained squares in pairs outward on each side. For odd/odd p,q, each half has $(p+q-2)/2$ retained squares. If this number is odd, one square remains at each outer end, giving pieces 6ii and 6iii with initial counts $(2,1)$. Otherwise the half starts at $(1,0)$ and consists of complete three-tile pieces. This repartition requires rebuilding the piece list; one cannot move a cut and keep its old boundary counts.

Proposition 7.3 (Folded matching counts)**.** *For a reduced $0<p<q$, let (x,y) be the open counts of one reflected half under the central chop. If p,q are odd, then*

$$m_{p/q} = 2x^2 + 2xy + y^2 = (x+y)^2 + x^2. \tag{50}$$

If $p+q$ is odd, then $m_{p/q} = x^2+y^2$. In either case $\gcd(x,y)=1$; a nonempty half has positive counts.

Proof. The trimmed Christoffel word is a palindrome: for $0<j<q$, $\lfloor p(q-j)/q \rfloor = p-1-\lfloor pj/q \rfloor$. For odd/odd labels the two halves meet at an opposite-port central square, whose four counts are $2,1,1,1$. The reflected halves contribute x^2, xy, yx, y^2, giving (50). For opposite parities they meet across a single connector and must use its same state, giving x^2+y^2. Every full piece table and every exposed-square table has determinant ± 1, so its integral inverse preserves the gcd of an

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open row. Starting at $(1,0)$ proves primitivity. The first square gives $(2,1)$, and every subsequent permitted table preserves positivity. □

The seed $1/1$ consists only of its central square. The two examples in Figure 10 give $m_{1/3} = 2\cdot 2^2 + 2\cdot 2\cdot 1 + 1^2 = 13$ and $m_{1/5} = 2\cdot 5^2 + 2\cdot 5\cdot 3 + 3^2 = 89$.

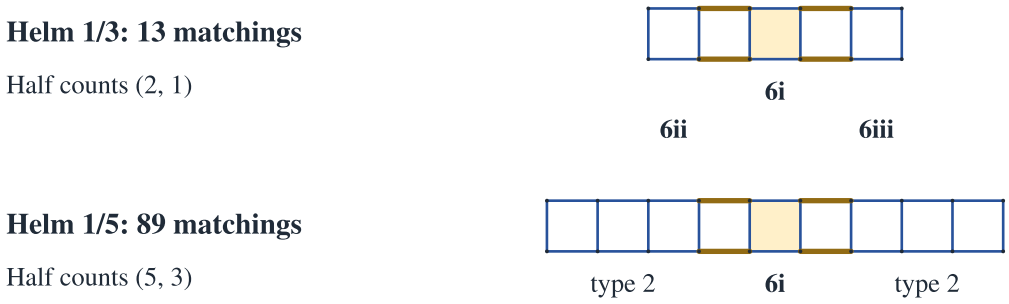


Figure 10. Central and end pieces in actual helms. At $1/3$, central square 6i has the end squares 6ii and 6iii as its halves, with counts $(2,1)$. At $1/5$, each half is a type-2 triple with counts $(5,3)$. Folding produces 13 and 89.

7.6. **A common prefix leaves one quadratic comparison.** Two folded graphs with the same outer boundary convention may share a prefix with open counts (x,y). Sum over all remaining states, leaving that cut open. The two centers give positive definite symmetric tables $G, \widetilde{G}$ with determinant one and nonnegative off-diagonal entries. For a helm they have the form SQS^{T} with $Q = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$; for opposite-parity labels they have form SS^{T}. Thus

$$m - \widetilde{m} = \alpha x^2 + 2\beta xy + \gamma y^2, \qquad G - \widetilde{G} = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}. \tag{51}$$

Proposition 7.4 (At most one primitive common prefix)**.** *For a nonempty common part and $G \neq \widetilde{G}$, equality requires $\alpha\gamma < 0$ and $\beta^2 - \alpha\gamma$ to be an integer square. There is then at most one primitive positive row (x,y) giving equality. That row must still arise from both actual prescribed snakes.*

Proof. A center table $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ has $ac - b^2 = 1$ and $b \geq 0$. If its two diagonal differences from another table have the same weak sign, then the difference of $b = \sqrt{ac-1}$ has that sign too. A nonzero difference cannot vanish at a positive row. Opposite diagonal signs are therefore necessary. The resulting quadratic in y/x has one positive root. An integer zero obeys $(\gamma y + \beta x)^2 = (\beta^2 - \alpha\gamma)x^2$; a rational square root of an integer is integral. A positive rational root has only one primitive positive representative. □

For an empty prefix, test its actual boundary row. If $G = \widetilde{G}$, the criterion supplies no exclusion. Equal count tables do not certify equal physical graphs: types 1 and 5 already show why orientations and the prescribed Christoffel word must be retained.

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A useful uniform exclusion follows directly modulo three. The physical piece tables satisfy $R_i Q R_i^{\mathsf{T}} \equiv Q \pmod 3$, whereas a single retained square gives $-Q$. A helm tail with d retained squares therefore has center table $(-1)^d Q \pmod 3$. Opposite parities give a nonzero difference at every primitive row because the discriminant of Q is -1, a nonsquare modulo three. For equal parity and nonzero difference, let $h = \min_{i,j} v_3((G - \widetilde{G})_{ij})$ and write $3^{-h}(G - \widetilde{G}) = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$. Then $h \geq 1$. Expanding the equal determinants and reducing modulo three gives $\gamma \equiv \alpha + \beta$; the normalized discriminant is one or two modulo three. If it is two, every primitive shared row has count difference of valuation exactly h. These are exclusions for specified tails with arbitrary actual common prefixes, not a classification of all possible snake-graph collisions.

8. All-phase verification through degree 10,000

8.1. **Why a finite residue test covers infinitely many phases.**

For a period T, define the complete image

$$\mathcal{R}_\alpha(T) = \{M_\alpha(x) \bmod q(T) : 0 \leq x < T\}.$$

Theorem 4.1 makes this the image at *every* integer phase. Disjoint images therefore separate two entire Meridians. If images overlap for several periods, a true equality must use compatible phases in all of them. Define the witness relation

$$W_T(\alpha, \beta) = \{(u, v) \in (\mathbb{Z}/T\mathbb{Z})^2 : M_\alpha(u) \equiv M_\beta(v) \pmod{q(T)}\}.$$

Proposition 8.1 (Compatible witnesses)**.** *Equality of two integer values requires one witness in each selected W_T, with first coordinates agreeing modulo every pairwise gcd of the periods, and second coordinates agreeing in the same way. If no compatible choice exists, the two Meridians are disjoint at all integer phases.*

Proof. A true equality uses the same integers x, y in every modulus and hence supplies the required residues. Conversely, the generalized Chinese remainder theorem lifts compatible coordinates to a phase pair modulo the least common multiple. This converse concerns modular witnesses only; it does not imply equality of the full values. □

The moduli need not be independent. In particular, the periods $36, 38, 40$ are all even, so contradictory parity requirements alone can exclude a pair that survives each separate image test.

8.2. **Complete inventory and the final exclusion.**

Proof of Theorem 1.1. Proposition 3.4 identifies the inventory with coprime positive odd pairs $r \leq s$, $r + s \leq 10{,}000$. It contains 5,067,077 Meridians and 12,837,632,128,426 unordered pairs. The producer enumerates the count-vector address tree; an independent checker enumerates the coprime pairs directly and compares the inventories. The child degree is $2|k|D \pm (|v| - |u|) > D$, where $|u|, |v|$ are the endpoint word lengths. This gives finite exhaustive index ranges.

Every one of the 36 phase residues of every Meridian is evaluated modulo $q(36)$. Equal-residue grouping lists all pairs whose images intersect; its complement is excluded at once. Periods 38 and 40 are then evaluated on every endpoint of the preceding survivor sets. The complete image tests leave 87 pairs. For each of them, project every full witness relation onto the four parity pairs $(u \bmod 2, v \bmod 2)$. The three projected sets have empty intersection in all 87 cases. Proposition 8.1 therefore excludes every pair of different Meridians at every integer phase. Theorem 5.1 also excludes different marks within each individual Meridian, completing the statement. □

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Table 2. Complete cumulative survivor counts from the archived certificates. "Images" forgets meeting phases; "phases" also imposes compatibility. Survivors are modular coincidences, not integer collisions.

Tests retained	$D \le 2{,}000$	$D \le 10{,}000$
36: images	7,121	320,755
36, 38: images	612	17,427
36, 38: phases	2	2,052
36, 38, 40: images	0	87
36, 38, 40: phases	**0**	**0**
36, 38, 40, 42: images	—	1
36, 38, 40, 42, 44: images	—	0

8.3. **What was checked independently.** The degree-10,000 producer uses signed endpoint-orbit updates. The separate evaluator uses Fibonacci staircase words with Euclidean matrix reduction. They agree on all 188,369,412 retained phase evaluations. The checker verifies the 182,414,772 baseline phase tags bijectively, so an omitted or repeated phase record cannot pass unnoticed. Independent sorting of the original unsorted buckets reconstructs all 70,009 cross-Meridian residue groups and the subsequent complete pair sets. Coverage is checked by exact records, not inferred merely from their counts or from a matching digest.

The phase result is checked three ways: full generalized-CRT joins, an independent 16-bit mask computation on $(x \bmod 4, y \bmod 4)$, and the weaker four-state parity projection, already sufficient for the final 87. The degree-2,000 certificate, containing 202,862 Meridians, was completed first; its common survivor sets agree exactly with the larger inventory when identified by helm labels.

Five deliberate corruptions—an altered value, an incorrect modulus, a duplicate tag, an omitted record and an invented pair—were all rejected. The degree-100 control compares 18,684 residues with literal Fibonacci words and the polynomial record of (19). The optional image-only extension through periods 42 and 44 also ends with zero pairs; all 88 period-44 values were separately checked using arbitrary-size Python integers because $q(44) > 2^{63}$.

8.4. **Reproducibility and scope.** The degree reproducibility package [7] contains the inventory, compact residue groups, phase witnesses, producer and checker sources, audit receipts, and a regeneration driver. Large baseline tables are regenerated; their recorded sizes and hashes identify the original run. The reindexed edition retains the original computational kernels with an explicit address-convention adapter. Reversing the written address tail does not change a helm label, degree, phase, or certified numerical value.

This is a deterministic computer-assisted certificate, not an external referee report or proof-assistant formalization. Its scope is a bounded set of Meridians with unbounded phase. The numerical-order verification in Section 9 is a separate computation with its own coverage proof and execution record.

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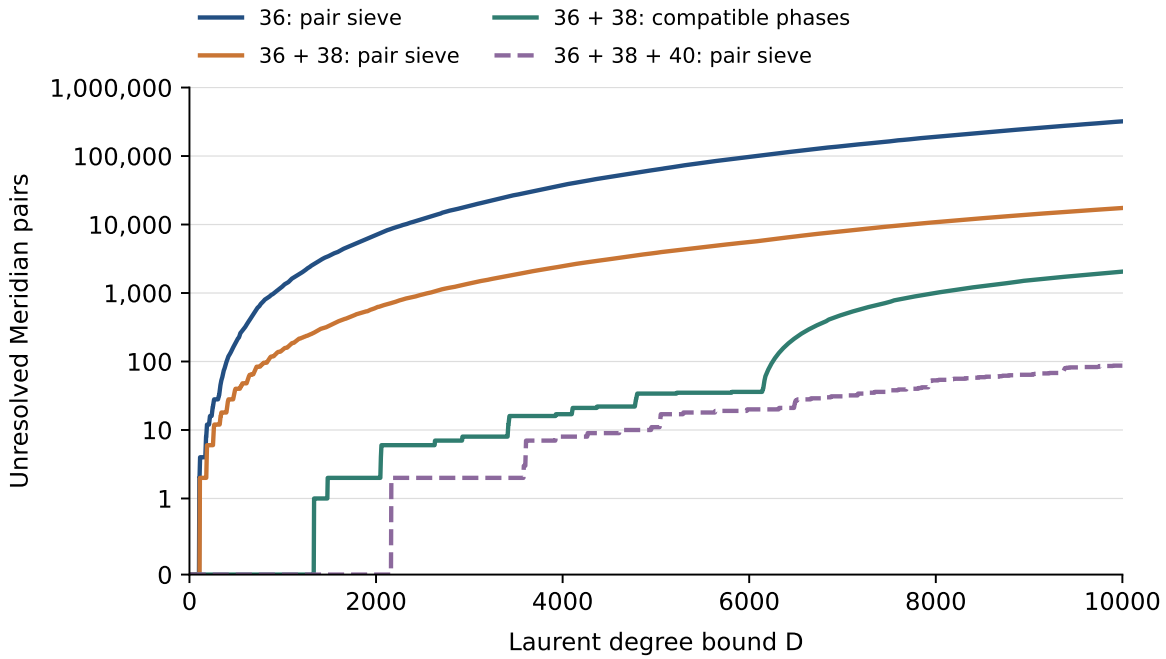


Figure 11. Exact unresolved-pair profiles in the degree-10,000 inventory. The phase-compatible sieve through period 40 has no survivors in the entire displayed range. Lines join discrete even-degree bounds. The figure is a record of the finite certificate, with no extrapolation to larger degrees.

9. Computational chronology and numerical-order verification

9.1. **The earlier Claude-assisted search.** The historical working draft [1], pp. 38–40, reports a Claude-assisted search of **9,581,528,580 sorted Markov triples** against the cutoff $10^{100,000}$, taking **24 minutes** on a two-core machine with 7 GB of memory. It compared maxima modulo 3^{39} in 16 passes and 4,096 buckets; all 21 repeated-residue pairs separated modulo $2^{61} - 1$. No equal maxima were reported.

An earlier public announcement of a 24-minute search is timestamped **23 October 2026, 09:54:57.771 CDT** (14:54:57.771 UTC) [2]. Its caption says “9.3 billion”; the later draft gives the exact count above. This dates the *announcement*, not the beginning or end of execution. Original execution timestamps have not been recovered. The draft itself labels these as historical computational reports whose original logs and floating-point cutoff bounds had not been independently certified. We preserve that status here. This search is separate from the later ten-billion anchor, and its reported 24 minutes are not included in the cumulative timing below.

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9.2. **The completed fifty-billion verification.** The archived computation [3] establishes that each of the **first 50,000,000,000 distinct Markov values in numerical order** is the maximum of exactly one normalized positive Markov triple. This complements the degree-10,000 result: the latter covers infinitely many phases on a bounded set of Meridians, whereas this computation certifies a numerical prefix. Its coverage and pruning proofs are collected in the separate verification supplement [4].

Run date and duration. The recorded execution began on **28 October 2026 at 03:32:57 CDT** and finished, including its final audit, at **05:38:34 CDT** (UTC−05:00). The underlying timestamps are 08:32:57.096455–10:38:33.644612 UTC. Two completed coordinator invocations occupied **7,492.102807 seconds: 2 h 04 min 52.103 s**. Including the original ten-billion anchor, the cumulative recorded execution path is **8,330.980040 seconds: 2 h 18 min 50.980 s**. The calendar span is 7,536.548157 seconds; it additionally includes the approximately 44-second pause between invocations. These are historical execution dates and measured durations, distinct from this edition's publication date.

Table 3. The completed fifty-billion run. Counts are configurations unless explicitly identified as distinct values.

Certified distinct-value prefix	50,000,000,000
Complete outer domain	71,517,775,875
New retained configurations compared	54,940,210,994
Repeated primary residues / candidate configurations	346 / 692
Actual equal-value collisions	0

The numerical cutoff was $B = F_{1,147,147}$, with 796,397 bits. The inner height $K = 573{,}574$ contains 50,000,097,577 configurations at or below B; the proved outer height $L = 685{,}979$ contains every competitor below that cutoff. The original ten-billion cutoff certificate justifies 10,000,015,326 inner omissions. Strict certified lower bounds exclude a further 6,577,549,555 configurations above B.

All 32 primary prefix passes and all 8,192 complete buckets finished. Every repeated residue modulo 2^{64} was replayed; all 346 candidate pairs separated modulo 5^{27}. The final auditor independently reconstructed their 692 full integer triples and checked the Markov equation, residues and unequal maxima, as well as the 38 anchor candidates. No probabilistic hash-collision assumption enters this comparison.

The run used eight workers on Linux x86-64, an eight-CPU-equivalent quota and an 8-GiB memory limit, with frozen GCC 13.3.0 executables. The timing includes generation, partitioning, sorting, replay, orchestration and the 107.371720-second final audit; earlier preflight and later archival checks are excluded. The supplement gives the stage timings, retained receipts, source identities, audit scope and reproducibility limitations. This edition reports the completed run; it does not announce a result for the planned 500-billion verification.

10. What remains to prove

The organization now separates three questions. The address bijection settles coverage of configurations. Fixed-degree ordering and the two collision windows restrict equality of maxima. Complete residue witnesses settle the remaining comparisons for the degree-bounded family in Theorem 1.1. The fifty-billion computation additionally excludes every competitor for a finite prefix in numerical order.

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For unrestricted unicity the unresolved pair has different Meridian degrees, the same modulo-27 class, and different Farey numerators. It must satisfy both its phase rectangle and its numerator–denominator window. A proof must exclude those candidates uniformly as the degrees and numerators grow. Enlarging a finite certificate alone does not supply that step. Likewise, an invertible corridor recurrence or a long modular period does not by itself prevent an earlier match of one scalar residue.

Corridors offer a way to formulate the next comparison with explicit entrance states and independently chosen wall lengths. Their positive corrections are exact and telescope; the common-prefix snake calculation offers a different reduction to a quadratic in primitive counts. Neither mechanism has yet been shown to cover every unrestricted pair. The purpose of retaining their precise hypotheses is to make the remaining uniform statement identifiable, rather than hide it inside an induction.

Data and code. The editable source accompanying this edition includes all figure assets, the revised Figure 3 generator, a source and figure inventory, and checks used for this rewrite. The separate degree reproducibility package contains the full computational evidence cited in Section 8. The fifty-billion proof and timing supplement accompanies this edition; its full frozen computational archive is cited separately [3, 4]. The corridor derivations are developed further in the supplementary research record [8]. Numerical drawings are explanatory; the arithmetic claims use the displayed identities or complete certificates.

Author contributions and AI assistance. The author conceived and directed the research program, including the hyperbolic representation and geometric descent, the two-dimensional host–phase landscape, the Meridian indexing and alternative tree, the six-piece snake-graph decomposition, the Fibonacci-matrix formulation and normalization argument, and the corridor observation. The author proposed using these structures for coverage, comparisons and verification, and specified the questions pursued at each stage. The Aigner monotonicities are established results; their prior proofs are credited in Section 7. The Fibonacci-matrix normalization argument presented there is the author's; the related earlier normalization result is cited for comparison, and the proof is given in full from Fibonacci identities.

Generative AI provided substantial mathematical and computational assistance, not only language editing. The development record identifies ChatGPT (OpenAI), with the recorded GPT-6.0 Astra and GPT-6.1 Sol model labels and reasoning settings, and Claude 5.5 Opus Extra (Anthropic) in the preceding geometric-descent work; exact backend revision identifiers were not retained. Within the author's framework, these tools assisted with detailed derivations, examples and counterexamples, proof construction and revision, verifier and independent-checker code, figures, and manuscript drafting. ChatGPT also assisted with this reorganization and Figure 3 revision. The project's separate computational checks are algorithmic audits, not external peer review. The author is responsible for the claims, references and final submission.

Funding and competing interests. No funding was received. The author declares no competing interests.

References

[1] E. F. W. Heffern, *Geometric Descent on the Markov Surface*, working draft v32, archived October 2026, pp. 38–40, historical computational reports. The earlier search was assisted by Claude.

[2] E. F. W. Heffern, public LinkedIn announcement of a 24-minute Markov-number search, 23 October 2026, 14:54:57.771 UTC, activity 7519411048115003392. Platform metadata and the source URL are retained in the verification supplement's evidence packet; see the public announcement.

[3] E. F. W. Heffern, *Markov 50 Billion Verification Package*, version 1, completed and sealed 28 October 2026; public derivative version 1.1 in *Computational supplements for Meridians on the Markov Surface*, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

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[4] E. F. W. Heffern, *Computational chronology and proofs for the fifty-billion verification*, supplement to the present manuscript, version 4, 7 November 2026.
[5] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013. doi:10.1007/978-3-319-00888-2.
[6] İ. Çanakçı and R. Schiffler, *Snake graphs and continued fractions*, European J. Combin. **86** (2020), 103081. doi:10.1016/j.ejc.2020.103081.
[7] E. F. W. Heffern, *Meridian Degree 10,000 Reproducibility Package*, computational supplement, 31 October 2026; address-convention revision, 7 November 2026. Original computational release in *Computational supplements for Meridians on the Markov Surface*, version 1.0, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.
[8] E. F. W. Heffern, *Markov Meridian Residue Supplement*, version 1.18, and *Corridor-to-Corridor Induction*, version 1.3, unpublished supplementary research records, 7 November 2026. The corridor results used here are proved in Section 6.
[9] C. Lagisquet, E. Pelantová, S. Tavenas and L. Vuillon, *On the Markov numbers: fixed numerator, denominator, and sum conjectures*, Adv. Appl. Math. **130** (2021), 102227. doi:10.1016/j.aam.2021.102227. Preprint: arXiv:2010.10335.
[10] K. Lee, L. Li, M. Rabideau and R. Schiffler, *On the ordering of the Markov numbers*, Adv. Appl. Math. **143** (2023), 102453. doi:10.1016/j.aam.2022.102453.
[11] G. McShane, *Convexity and Aigner's conjectures*, preprint (2021), arXiv:2101.03316.
[12] M. Rabideau and R. Schiffler, *Continued fractions and orderings on the Markov numbers*, Adv. Math. **370** (2020), 107231. doi:10.1016/j.aim.2020.107231.
[13] G. Musiker, *A conjectured combinatorial interpretation for Markov numbers*, REACH memorandum, 27 June 2002. Original memorandum; REACH archive.
[14] A. Itsara, G. Musiker, J. Propp, and R. Viana, *Combinatorial interpretations for the Markov numbers*, REACH manuscript, 10 January 2003, Theorem 3.1. Original manuscript.
[15] J. Propp, *The combinatorics of frieze patterns and Markoff numbers*, arXiv:math/0511633, first posted 25 November 2005; published in *Integers* **20** (2020), A12.

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Meridians on the Markov Surface: Geometry, Residues, and Collision Windows

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v5_ps | 2026-11-07 18:35:22 CST / 2026-11-08 00:35:22 UTC | https://zenodo.org/records/24046012

v7_ps | 2026-11-08 07:26:11 CST / 2026-11-08 13:26:11 UTC | https://zenodo.org/records/24053720

Earlier public announcements: 21 October 2026, 10:18:25 CDT / 15:18:25 UTC on X (more than a billion values in under 40 minutes); 23 October 2026, 09:54:57.771 CDT / 14:54:57.771 UTC on LinkedIn (9.3 billion in 24 minutes). These concern earlier computations, not the final trillion certificate.

https://x.com/webystar/status/2112926487091282350

https://www.linkedin.com/posts/elleard-felix-webster-heffern-joseph-by-confirmation-2953a3104_markovnumbers-activity-7519411048115003392-4NaJ

The dates and URLs are retained in the 7 November public-post evidence packet; the LinkedIn URL is also embedded in the published computational supplement. Additional November LinkedIn announcements and the older Meridian ResearchGate v13 URL could not be established in this compilation.

Complete public Zenodo description

This v9_ps revision rewrites the abstract and introduction around the catenary construction. It defines the host, derives the unit-step Markov identity through explicit cancellations and multiplication, and distinguishes a general position x from selected positions c and d. A new two-panel figure on the host-5 catenary shows one choice giving the integer triple (1,2,5) and another giving noninteger heights. The text explains when two distinct admissible integral choices would constitute a collision.

Profiles, phase recovery, and related derivations are written throughout in explicit powers, square roots, and logarithms. The manuscript retains the established theorem statements; the degree-10,000 separation theorem now appears in Section 8. The general unicity problem remains open.

The paper retains the author's Fibonacci-matrix proofs of normalization and fixed-numerator monotonicity, and his six-piece proofs of the remaining fixed-sum comparison and fixed-denominator monotonicity. The normalization starts from (AC)_11 - (B^2)_11 = 2 and proves the local comparisons and termination in the same matrices. Related earlier normalization is cited as similar work, while the author's argument retains its explicit attribution. The matching interpretation is credited to Musiker's 27 June 2002 memorandum and the Itsara-Musiker-Propp-Viana manuscript of 10 January 2003.

The fixed-numerator, fixed-sum and fixed-denominator proofs have parallel subsection headings.

The paper organizes Markov configurations by reversed Meridian addresses, geometric descent, Fibonacci-modulus and modulo-27 behavior, equal-degree separation, finite unequal-degree collision windows, corridor recurrences and a numerator-denominator collision window. The archived degree-10,000 certificate is retained. The unrestricted unicity problem remains open within this framework.

The revised computational chronology includes the earlier Claude-assisted historical search and the separately archived fifty-billion numerical-order verification. The working draft reports 9,581,528,580 triples against a cutoff of 10^100000 in 24 minutes on two cores with 7 GB memory. A public LinkedIn announcement is timestamped 23 October 2026, 14:54:57.771 UTC (09:54:57.771 CDT), with caption wording of 9.3 billion in 24 minutes. This is the announcement timestamp, not a recovered execution start or finish. The earlier draft's qualification about uncertified historical logs and cutoff error bounds is retained. This search is not the later

ten-billion anchor.

The fifty-billion verifier completed on 28 October 2026. Its recorded start was 08:32:57.096455198 UTC and its audit-inclusive finish was 10:38:33.644612263 UTC (03:32:57-05:38:34 CDT, rounded). Two closed invocations total 7,492.102807105 seconds (2 h 04 min 52.103 s). The calendar span is 7,536.548157065 seconds, including the invocation pause. Including the separate original ten-billion anchor gives 8,330.980040248007 seconds (2 h 18 min 50.980 s). These historical execution dates are distinct from this edition's release date.

The exact numerical envelope covers 71,517,775,875 configurations. After anchored omissions and certified lower-bound exclusions, 54,940,210,994 new configurations were compared. All 346 repeated primary-residue pairs separated; the final auditor reconstructed all 692 candidate triples and rechecked the 38 anchor candidates. The protocol establishes uniqueness for the first 50,000,000,000 distinct Markov values in numerical order. It used eight workers, an eight-CPU-equivalent quota and an 8-GiB memory limit.

Included are the 35-page main paper, six-page computational proof and timing supplement, editable source, all figure assets, a standalone PDF of the new catenary figure, selected historical evidence and receipts, and the retained read-only reconciliation of 26,151 archive member checksums and 8,522 closed stage receipts. The full frozen historical verifier archive is available separately at https://doi.org/10.5281/zenodo.23874075. This release is an editorial revision, not a new enumeration; no completed 500-billion result is claimed.

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Funding: No funding. Substantial generative-AI assistance is disclosed in the manuscript; the author directed the research and is responsible for the work.

Published files

Original filename / checksum	Bytes
v9_ps_Markov_Meridians_TLMS_editable_source.zip MD5 f423428243ac30aa57cd86bea2f7d4cd SHA256 7f922ad072b14b7bfe4686e28f061cfa4a6088e3cb678f0c24798ef440b0af7b	26,389,932
v4_50_Billion_Verification_Supplement.pdf MD5 6c966b341c1ec55b6b7bdc3a9dc50aaf SHA256 b477d344aff0971fee600ce1d087fc05f15509582bb43b22eb337180d6d3c446	310,901
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v9_ps_Markov_Catenary_Choices.pdf MD5 710cbea1d67e7d03e3112818469930a1 SHA256 957c018cc631ddf931a2ad232762309e0ed84e3fc732ae090ecabd3297ebb19d	20,426

All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

Original: v9_ps_Markov_Meridians_TLMS.pdf

MERIDIANS ON THE MARKOV SURFACE: GEOMETRY, RESIDUES, AND COLLISION WINDOWS

ELLEARD FELIX WEBSTER HEFFERN

Abstract. The Markov unicity problem asks whether the largest entry of a positive integer solution determines the other two. We approach it through catenary profiles. A fixed host sets the curve; two heights one unit apart solve the Markov equation with that host. Integer triples occur at selected positions. Reversing geometric descent groups the triples into families called Meridians. We give an exact indexing of these families and their integer triples, then use remainder patterns and growth bounds to restrict possible collisions. Corridor walls give a second way to compare the same triples. Fibonacci matrices and a six-piece graph decomposition supply the ordering proofs. A computer-checked certificate separates the integer configurations on all 5,067,077 Meridians of degree at most 10,000, at every integer phase. A separate archived computation, completed on 28 October 2026, verifies the first 50 billion distinct Markov values in numerical order. The general unicity problem remains open.

Post-submission draft. Changes in this edition require a new submission.

1. Introduction

A Markov triple consists of three positive integers satisfying

$$a^2 + b^2 + M^2 = 3abM, \qquad 1 \le a \le b \le M. \tag{1}$$

For example, $(1, 2, 5)$ and $(1, 5, 13)$ are Markov triples. The largest entry of such a triple is a Markov number. The *unicity problem* asks a simple question: does that largest entry determine the other two? A counterexample would consist of two different ordered triples with the same largest entry. We call such an equality a *collision*. For the history and wider setting of the problem, see Aigner [5].

Our starting point is to let the entries vary continuously. We build a curve on which the Markov equation holds automatically, then ask which positions give integer entries. This separates the equation itself from the requirement that its solutions be integers. It also gives a picture of what a collision would have to look like.

1.1. The host and its catenary profile.

The *host* h is the fixed third entry around which we build the curve. It determines a profile f_h: two heights on this profile, taken one horizontal unit apart, will satisfy the Markov equation together with h. We first allow positive real entries and call any such solution a *real Markov state*. The host need not yet be an integer or the largest entry; those requirements will be imposed when we return to the unicity question.

For $h > 2/3$, define

$$q_h = \frac{3h}{2}, \qquad R_h = \sqrt{q_h^2 - 1}, \qquad Q_h = q_h + R_h,$$
$$f_h(x) = \frac{h}{2R_h}(Q_h^x + Q_h^{-x}). \tag{2}$$

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2020 *Mathematics Subject Classification.* Primary 11D25; Secondary 11B39, 05C70, 11Y50.
Key words and phrases. Markov numbers, unicity, geometric descent, Meridians, corridors, snake graphs.

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The restriction $h > 2/3$ makes R_h positive and $Q_h > 1$; it includes every positive integer host. The graph of f_h is a rescaled catenary: a smooth, symmetric U-shaped curve, with its minimum at $x = 0$. In particular, $f_h(-x) = f_h(x)$. In (2), x is a free position on this profile. We use c below for a particular selected position.

The constants in (2) set the spacing and height scale so that one horizontal unit has the required arithmetic meaning. The calculation is short. First,

$$(q_h + R_h)(q_h - R_h) = q_h^2 - R_h^2 = 1,$$

so

$$Q_h^{-1} = q_h - R_h, \qquad Q_h + Q_h^{-1} = 3h, \qquad Q_h - Q_h^{-1} = 2R_h.$$

These three identities follow directly from the definitions.

1.2. **Why two heights one unit apart satisfy the equation.** Compare $f_h(x+1)$ with $Q_h f_h(x)$. The terms containing Q_h^{x+1} cancel, leaving

$$\begin{aligned} f_h(x+1) - Q_h f_h(x) &= \frac{h}{2R_h}(Q_h^{-x-1} - Q_h^{1-x}) \\ &= \frac{h}{2R_h} Q_h^{-x}(Q_h^{-1} - Q_h) = -hQ_h^{-x}. \end{aligned}$$

Multiplying $f_h(x)$ by Q_h^{-1} instead cancels the terms containing Q_h^{-x-1}:

$$\begin{aligned} f_h(x+1) - Q_h^{-1} f_h(x) &= \frac{h}{2R_h}(Q_h^{x+1} - Q_h^{x-1}) \\ &= \frac{h}{2R_h} Q_h^{x}(Q_h - Q_h^{-1}) = hQ_h^{x}. \end{aligned}$$

Now multiply these two equalities. The powers on the right cancel:

$$(f_h(x+1) - Q_h f_h(x))(f_h(x+1) - Q_h^{-1} f_h(x)) = -h^2.$$

Expanding the left side gives

$$f_h(x+1)^2 - (Q_h + Q_h^{-1}) f_h(x) f_h(x+1) + f_h(x)^2 = -h^2.$$

Finally, use $Q_h + Q_h^{-1} = 3h$ and move h^2 to the left:

$$\boxed{f_h(x)^2 + f_h(x+1)^2 + h^2 = 3f_h(x)f_h(x+1)h.} \tag{3}$$

Thus *every* real position x gives a real Markov state. No choice of an integer value or a special position was used in this calculation.

1.3. **Choosing an integral state: the role of c.** To look for an ordered triple with largest entry h, select $0 < c \le 1/2$ and take the unit interval $[c-1, c]$ across the bottom of the curve. Its two heights are $f_h(c-1)$ and $f_h(c)$. By symmetry, $f_h(c-1) = f_h(1-c)$, so the state, with its companions in increasing order, is

$$(f_h(c),\ f_h(1-c),\ h).$$

It solves the equation by (3), with $x = c - 1$. For it to be a normalized Markov triple, h and both heights must be positive integers, and $f_h(1-c) \le h$ must hold. The latter condition keeps the chosen host as the largest entry. Restricting c to this half of the unit interval avoids counting the reflected pair twice.

Figure 1 shows the distinction for $h = 5$. One choice, $c \approx 0.351515$, gives the heights 1 and 2, hence the integer triple $(1, 2, 5)$. A different choice, $d = 0.44$, gives heights approximately 1.207458 and 1.602614. It still solves the same equation with host 5, but its companion heights are not integers.

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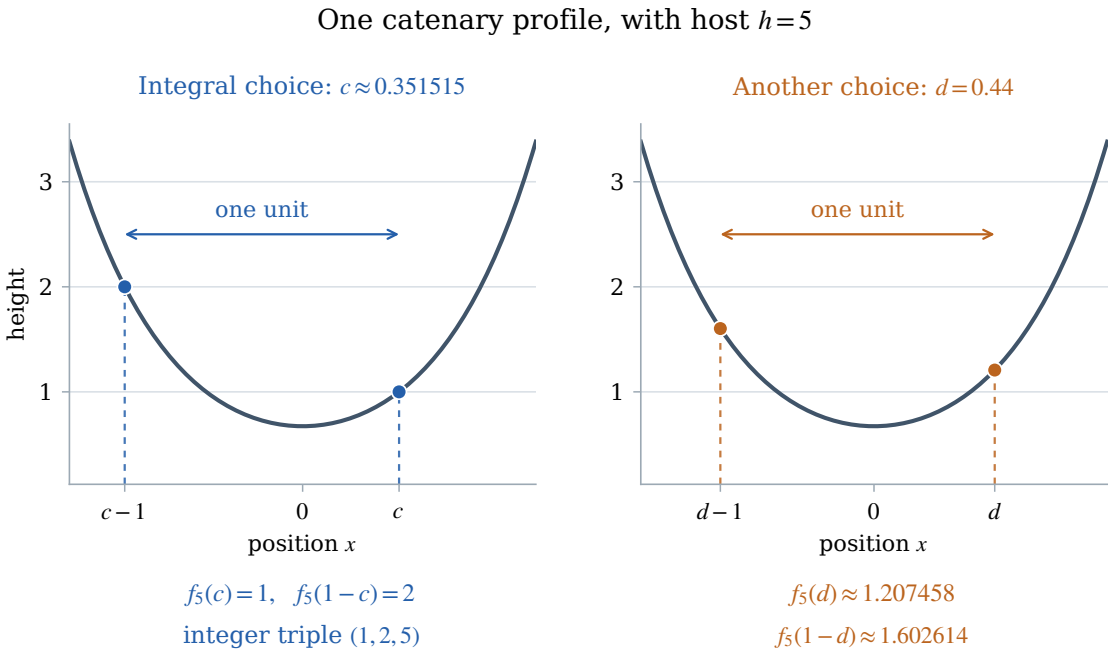


Figure 1. Two choices on the same catenary profile, with host $h = 5$. In each panel the marked contacts are exactly one horizontal unit apart. The left contact has height $f_5(1 - c)$ or $f_5(1 - d)$ by symmetry. The choice c gives an integer triple; the displayed choice d gives only a real state. A collision would require two distinct choices with both pairs of heights integral and with the same host as their largest entry. No collision is shown here.

This is the geometric form of the unicity question. For a fixed integer host h, could two distinct choices c and d in $(0, 1/2]$ both give integer companion heights no larger than h? If so, they would give different ordered triples with the same maximum. The curve makes these possible comparisons visible. To exclude them requires information about the integer values on the curve.

1.4. **How the paper organizes those comparisons.** The next step is *descent*. Starting from two contact heights, we look for a smaller host whose profile passes through the same heights one unit apart. We then move inward along that profile. Repeating these operations reduces a normalized integer Markov triple to $(1, 1, 1)$. Section 2 makes the procedure precise and explains how its final host and final position recognize the integer starting states.

Reversing the steps organizes the triples into families called *Meridians*. Each family has an address recording the steps used to construct it, and a phase specifying a position along it. This Meridian phase is a separate coordinate from the selected c on a fixed catenary. Integer Meridian phases pick out integer triples; the corresponding c need not be an integer, as Figure 1 shows. Section 3 proves that these addresses and integer phases recover every normalized triple exactly once. This is a statement about triples: whether two of them have the same largest entry remains the question to be answered. The same section introduces the alternative tree and the *phasors* that connect its Meridians.

Two kinds of information then restrict collisions. First, the remainders of the Markov values follow exact periodic patterns (Section 4). Second, their growth limits where two families could meet. A Meridian's *degree* measures the largest exponent in its formula. We prove that distinct

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integer marks of the same degree cannot collide. For different degrees, we bound the finite range of phases that needs comparison (Section 5). *Corridor walls* follow other paths through the same tree, using both Meridian and phasor steps. Their recurrence gives a second bound on possible collisions (Section 6).

Section 7 supplies the ordering proofs used in these comparisons. It represents Markov numbers by counts of perfect matchings on chains of square tiles, called snake graphs; a perfect matching pairs all the vertices using edges of the graph. The author's Fibonacci matrices and six-piece decomposition provide concrete ways to compare these counts. The graph construction and the needed notation are introduced in that section.

The resulting computer-checked certificate covers all 5,067,077 Meridians of degree at most 10,000, at every integer phase (Theorem 8.1). Distinct triples within that family have different maxima. This bounds the complexity of the families, while allowing their values to grow without limit; comparisons with higher-degree Meridians remain outside this certificate. Section 9 records a separate computation, completed on 28 October 2026, that verifies the first 50 billion distinct Markov values in numerical order. The general unicity problem remains open within this framework.

As the later sections introduce their coordinates, $M_\alpha(x)$ will denote the value at Meridian address α and phase x, and $m_{p/q}$ the value of the triple indexed by the reduced fraction p/q. Both label triples before any assertion that their maxima are distinct. Analytical logarithms are natural; plotted horizontal axes use $\log_{10} M$.

2. Geometric descent and arithmetic closure

2.1. **Moving along a profile and changing its host.** The catenary profile (2) turns two elementary operations on Markov states into movements of the contacts. The first operation is a unit step along a fixed profile. Directly from its powers,

$$\begin{aligned} f_h(x+1) + f_h(x-1) &= \frac{h}{2R_h}(Q_h + Q_h^{-1})(Q_h^x + Q_h^{-x}) \\ &= 3h f_h(x). \end{aligned} \tag{4}$$

Thus the height beyond a pair is obtained by multiplying the nearer height by $3h$ and subtracting the farther height. This is the usual Vieta replacement of a companion entry, expressed on the curve.

The second operation keeps the contact heights fixed and changes the host. For any real x, put $A = f_h(x)$ and $B = f_h(x+1)$. Their possible hosts are the roots of $X^2 - 3ABX + A^2 + B^2 = 0$. One root is h, so the other is

$$h, \qquad K = 3AB - h = \frac{A^2 + B^2}{h}. \tag{5}$$

For descent, select an edge straddling the minimum of the profile. We record its right endpoint as c, with $0 \le c < 1$, and its ordered contact heights as $A = f_h(c)$ and $B = f_h(c-1)$. This is the same pair used in the introduction, since $f_h(c-1) = f_h(1-c)$. We retain the full interval of c here to keep track of orientation. Seek a wider profile through these same two heights, still one unit apart, allowing a horizontal translation. Smaller h gives a wider profile. If the other host satisfies $K < h$, fit that profile and walk the unit lattice inward until the selected edge again straddles its vertex. Otherwise stop. Figure 2 shows one complete step. The fit is explicitly

$$v = \frac{\log\big((Q_K A - B)/K\big)}{\log Q_K}, \qquad f_K(v) = A, \quad f_K(v-1) = B. \tag{6}$$

Indeed, $Q_K f_K(v) - f_K(v-1) = K Q_K^v$, so taking logarithms recovers v directly. The Markov relation gives

$$(Q_K A - B)(B - Q_K^{-1} A) = K^2.$$

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Both factors are positive: they have the same sign, and their sum is $2R_K A > 0$. This also fixes the real branch of the logarithm and verifies the fitted contact heights by direct substitution. The walk replaces v by $v - \lfloor v \rfloor$. Its orientation is retained: reflection exchanges the contacts and sends phase to its negative modulo one.

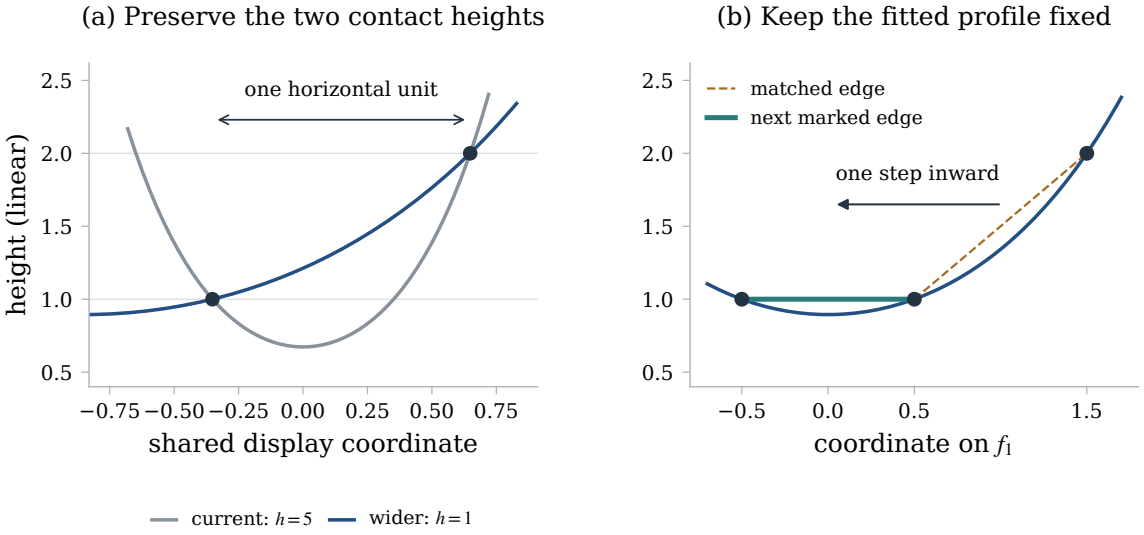


Figure 2. Geometric descent from $(1, 2, 5)$. The wider profile passes through the same unit-spaced contacts. On f_1 their coordinates are $1/2, 3/2$; one inward step selects $-1/2, 1/2$, with heights $1, 1$. The curve is fixed during the walk.

Proposition 2.1 (Uniqueness of the fit)**.** *A marked edge has at most one distinct compatible profile, hence at most one wider fit. For fixed $0 < a < b$, a same-arm fit $f_h(u-1) = a$, $f_h(u) = b$ with $u > 1$ is unique if it exists.*

Proof. The hosts are the two roots of $X^2 - 3ABX + A^2 + B^2$. The other root belongs to the geometric domain because $(3K-2)AB = K^2 + (A-B)^2 > 0$. For a same-arm fit the horizontal distance between heights a, b is

$$\frac{1}{\log Q_h} \int_a^b \frac{dy}{\sqrt{y^2 - h^2/R_h^2}}.$$

When $a > 2/3$ and the lower height is in range, Q_h strictly increases and $h/R_h = 2h/\sqrt{9h^2-4}$ strictly decreases with h. Thus the prefactor and the integrand strictly decrease with h. The distance can therefore equal one at most once. The profile is strictly increasing on that arm, which also determines u. If $a \le 2/3$ there is no fit. □

2.2. **Termination.** Write $\mathcal{D}(M, c) = (H, \phi)$ for the terminal host and oriented phase.

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Theorem 2.2 (Finite descent)**.** *Every straddling input with $M \geq 1$ terminates. On $1 \leq M \leq R$ the number of host changes is uniformly bounded. The terminal map is continuous into $(2/3, \infty) \times \mathbb{R}/\mathbb{Z}$ and analytic on cells with a fixed strict itinerary.*

Proof. Temporarily sort the endpoints as $a \leq b$. Inward walking does not increase either endpoint. Every current host is at most R, and the discriminant in (3) gives

$$(9h^2 - 4)a^2 - 4h^2 = (2b - 3ha)^2 \geq 0.$$

Consequently the endpoints are at least $a_R = 2R/\sqrt{9R^2 - 4} > 2/3$. At an accepted change, $a^2 + b^2 < h^2$, so the endpoints are also below $h \leq R$. Using either endpoint as the host in the same discriminant identity bounds the new host below by a_R. The initial host is at least $1 > a_R$.

If the new host K is at least b, its edge already straddles the vertex, because both endpoints are below $f_K(1)$; the next candidate is the old, larger host, so descent stops. Otherwise

$$\frac{h}{K} > \frac{h}{b} = 3a - \frac{K}{b} > 3a - 1 \geq 3a_R - 1 > 1.$$

All changes except possibly the last therefore contract the host by a fixed factor. An upper bound is $1 + \lceil \log(R/a_R)/\log(3a_R - 1) \rceil$.

At a boundary $K = h$, the fitted coordinate in (6) is the original c, so the accepted and stopped maps agree. At a phase cut, $K(h, 0) = h(9h^2 + 4)/(9h^2 - 4) > h$, and a neighborhood is stopped. The uniform bound on compositions then gives continuity, with phase taken modulo one. The strict formulas are analytic. □

2.3. **Which inputs close arithmetically?** The two reversible state operations are

$$T(a, b, h) = (3ha - b, a, h), \qquad J(a, b, h) = (a, b, 3ab - h). \tag{7}$$

They and their inverses are integer polynomial maps. A terminal state has the nonnegative defect

$$\begin{aligned} H f_H(\phi) f_H(\phi - 1) - 1 = {} & \frac{(H-1)^2(H+2)}{3H-2} \\ & + \frac{H^3}{4R_H^2}\big(Q_H^{\phi - 1/2} - Q_H^{1/2 - \phi}\big)^2. \end{aligned} \tag{8}$$

For $0 \leq \phi < 1$ it vanishes exactly at $(H, \phi) = (1, 1/2)$.

Proposition 2.3 (Arithmetic closure)**.** *A straddling input with $M \geq 1$ reaches $(1, 1/2)$ if and only if its three entries are positive integers. Modulo companion exchange, these inputs are exactly the normalized configurations* (1)*.*

Proof. Expanding the two profiles gives

$$H f_H(\phi) f_H(\phi - 1) - 1 = \frac{H^3}{4R_H^2}\big(Q_H^{2\phi - 1} + Q_H^{1 - 2\phi} + 3H\big) - 1.$$

Replace the first two powers by $\big(Q_H^{\phi - 1/2} - Q_H^{1/2 - \phi}\big)^2 + 2$. The constant term is $H^3/(3H - 2) - 1 = (H - 1)^2(H + 2)/(3H - 2)$, which proves (8). Reversing a finite closure itinerary from $(1, 1, 1)$ by (7) preserves integrality. Conversely, integral descent stays integral. At an integral terminal state order the entries as $u \leq v \leq w$. There is no coordinate-decreasing mutation. But for $Q(X) = X^2 - 3uvX + u^2 + v^2$,

$$Q(v) = u^2 + (2 - 3u)v^2 \leq 3u^2(1 - u) \leq 0.$$

If neither root is below $w \geq v$, equality is forced, giving $u = v = 1$; terminality then gives $w = 1$. Finally, a straddling edge has no decreasing endpoint mutation, so its host is a largest entry. Its ordered split is uniquely recovered by (6). □

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Both terminal coordinates matter. Figure 3 shows inputs with terminal host one but different terminal phases. A crossing of the height level $H = 1$ alone does not certify integrality. The marked configurations follow the host itineraries $37666 \to 5 \to 1$, $7561 \to 5 \to 1$, and $14701 \to 2 \to 1$, all ending at phase $1/2$.

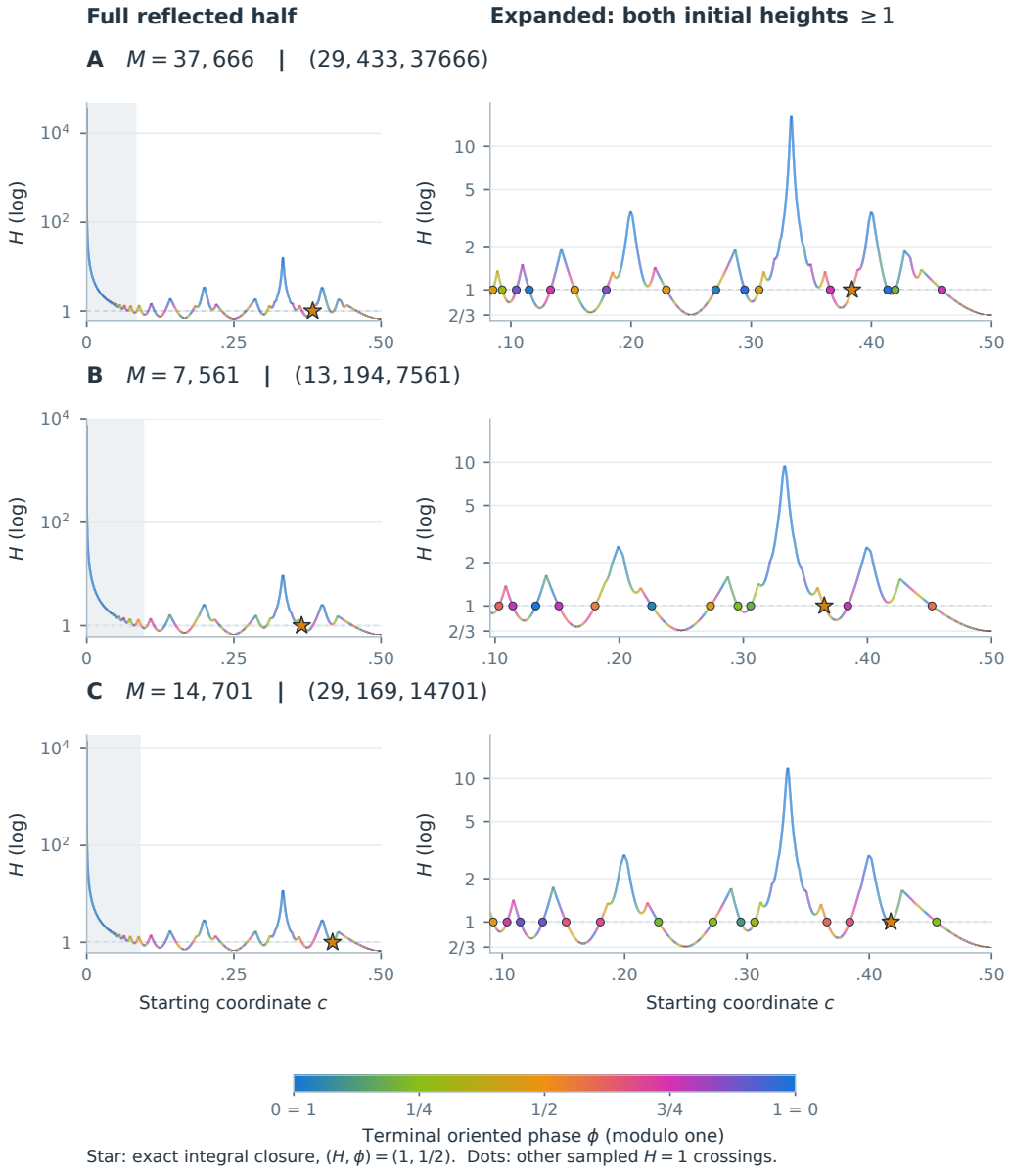


Figure 3. Terminal host and phase for $M = 37666, 7561, 14701$. Height is H on a logarithmic scale and hue is oriented phase. Left: $0 \le c \le 1/2$, with gray where an initial endpoint is below one. Right: the admissible portion enlarged. Stars mark the exact integral starts; dots are other numerically located $H = 1$ crossings. These numerical plots illustrate the closure criterion.

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3. Meridians, addresses, and the alternative tree

3.1. **The landscape and its two coordinates.** On the landscape $(\log_{10} M, c)$, the point (M, c) represents the contacts $f_M(c), f_M(c-1)$. We draw the reflected half $0 < c \le 1/2$; exchanging the contacts sends c to $1-c$. A *Vieta arc* keeps one initial contact height fixed. For example, the branch with height a has

$$c = \frac{\log\Big((R_M a + \sqrt{R_M^2 a^2 - M^2})/M\Big)}{\log Q_M}.$$

This follows by solving $Q_M^c + Q_M^{-c} = 2R_M a/M$ for $Q_M^c \ge 1$ and taking logarithms. The square root is real exactly when $a \ge M/R_M$, the minimum height of the profile; reflection supplies the other arm. A *Vieta trek* follows such arcs, changing the retained contact when necessary. A *Meridian* is a connected component of terminal host $H(M, c) = 1$. A *phasor* holds the terminal phase ϕ fixed. The alternative tree uses phasors at $\phi = 1/2$ to join successive Meridians.

The address phase x on a Meridian is real and unrolled. Its terminal circle phase is $\phi \equiv x + 1/2 \pmod 1$. Every integer mark therefore closes arithmetically; the distinguished arrival mark $x = 0$ is called the *helm*. It need not be the minimum of the real profile.

Figure 4 keeps the same configurations in three views. The alternative tree is in the middle panel. The bottom panel uses smooth schematic connectors for the Vieta parent relation: node coordinates and adjacency are exact, but those connectors are not literal level curves. The two trees each have 55 edges. An unmarked crossing is not a vertex.

For reproducibility, the field in panel (a) uses the fixed cyclic phase palette with HSL saturation and lightness

$$w(H) = \frac{1}{1 + (\ln H/\ln 1.02)^2}, \qquad S = w(H), \qquad L = 1 - 23w(H)/40.$$

The background is a numerical illustration, independent of all exact enumerations and exclusions below.

3.2. **Evaluating an address.** The seed Meridian is

$$\begin{aligned} a_1(x) = f_1(x + \tfrac{1}{2}), \qquad b_1(x) = f_1(x - \tfrac{1}{2}), \\ M_1(x) = 3a_1(x)b_1(x) - 1. \end{aligned} \tag{9}$$

For integer x, its companions are F_{2x+1}, F_{2x-1}, where $F_{-j} = (-1)^{j+1}F_j$. Hence

$$M_1(x) = 2 + 3F_{2x}^2, \qquad M_1(0), M_1(1), M_1(2), \ldots = 2, 5, 29, 194, \ldots. \tag{10}$$

From any ordered state (a, b, H) form the two-sided orbit

$$u_0 = a, \quad u_1 = b, \qquad u_{j+1} = 3Hu_j - u_{j-1}.$$

A nonzero index k replaces the state by

$$(u_k, u_{k+1}, 3u_k u_{k+1} - H). \tag{11}$$

The address is written

$$\alpha = (1, k_m, \ldots, k_1): \quad k_1 \text{ is applied first and written at the right.}$$

This is the reversed indexing convention: the former $M_{1,a,b,c}(x)$ is now $M_{1,c,b,a}(x)$. Neither the signs nor x are reversed. In terms of (7), with $S = T^{-1}$ and $\mathcal{W}_k = JS^k$,

$$(a_\alpha(x), b_\alpha(x), M_\alpha(x)) = \mathcal{W}_{k_m} \cdots \mathcal{W}_{k_1} JT^x(1, 1, 1). \tag{12}$$

Composition acts from right to left. A recursive extension inserts its index immediately after the leading 1. For example, $M_{1,1}(0) = 13$. Index zero is forbidden because (11) would return the preceding pole $3ab - H$, rather than create a new component.

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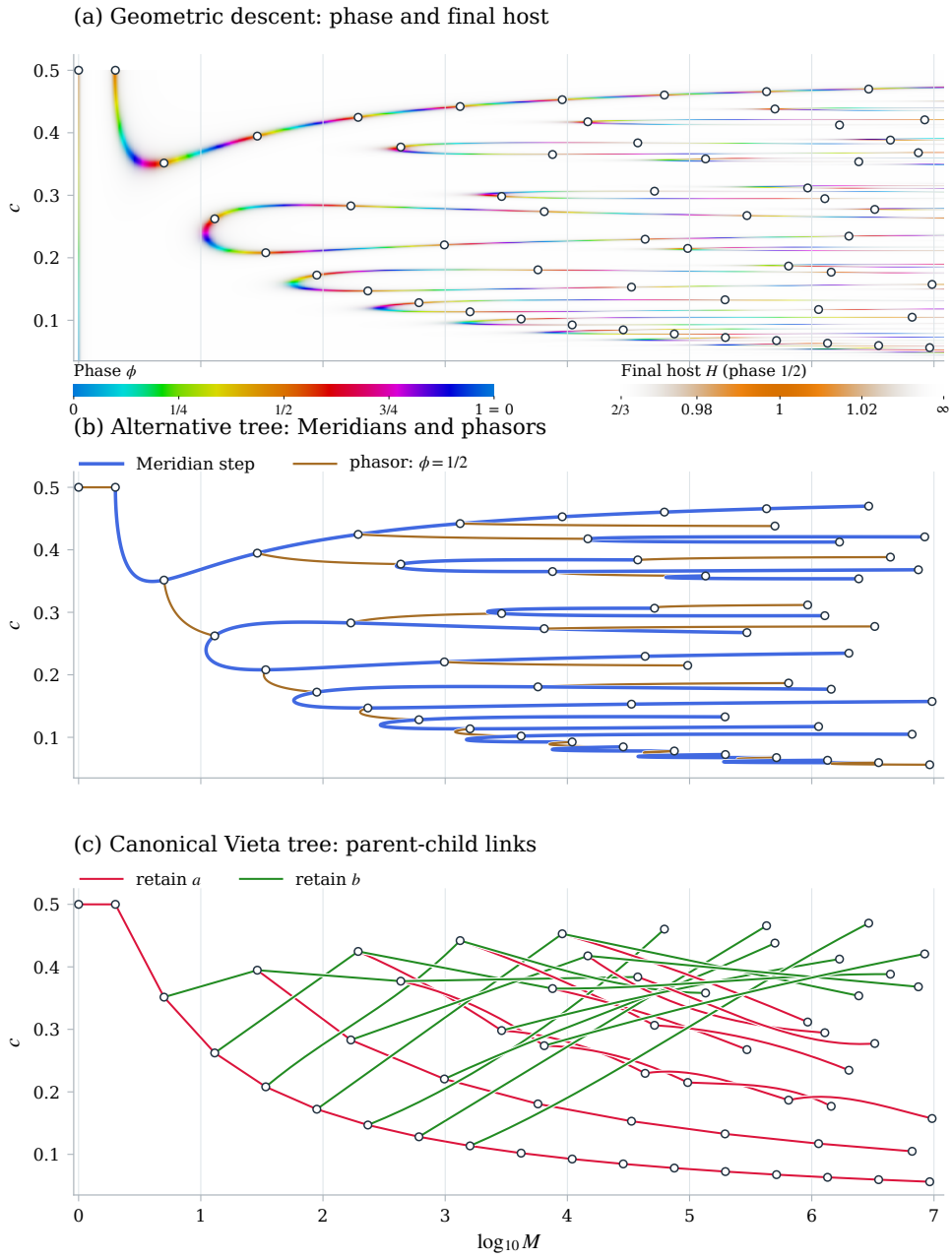


Figure 4. The same 56 configurations with $M \leq 10^7$. (a) Numerical geometric descent, with phase encoded by hue and host by color intensity. (b) The alternative tree, with Meridian and phasor edges. (c) The canonical Vieta tree, with simplified connectors retaining exact endpoints. Red retains the smaller companion, green the larger. Every edge runs toward larger M. Crossings without a node mark are not vertices.

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Definition 3.1 (Canonical marks)**.** The root $(1,1,1)$ is separate. For the seed (1) use integer phases $x \geq 0$. For every other address require $k_1 > 0$, $k_i \neq 0$ for $i > 1$, and $x \in \mathbb{Z}$. Evaluation is followed by sorting the two companions.

Theorem 3.2 (Configuration bijection)**.** *Canonical marks are in bijection with all normalized positive Markov triples except the root. The displayed value $M_\alpha(x)$ is the largest entry.*

Proof. The quadratic argument in Proposition 2.3 shows that replacing the largest coordinate by its other pole gives a smaller maximum, apart from the initial repeated entries. This is the forced Vieta reduction of a configuration and does not assume scalar unicity.

To recover a word, reflect the current maximum H to $P = 3ab - H$. When $P \geq 2$, walk its oriented endpoint orbit inward until both endpoints are below P. This edge is unique. Indeed $u_{j+1} - 2u_j + u_{j-1} = (3P-2)u_j > 0$, so the entries below P form an interval; three consecutive entries below P would contradict $u_{j-1} + u_{j+1} = 3Pu_j \geq 3P$. Endpoint descent supplies the edge, and its nonzero signed displacement recovers the last-applied, leftmost index. Repeat at the smaller pole. At pole one, the orbit reaches $(1,1)$ and fixes the seed phase.

Conversely, all outward blocks are positive and reduced. If the preceding pole is $P < H$, the first reflected poles on the two sides are $(9b^2-2)H - P > 6H$ and $(9a^2-2)H - P > 6H$. Their farther endpoints exceed $2H$, and the recurrence makes subsequent outward entries increase. The reduction therefore undoes every prescribed block. Finally, companion exchange negates all indices and x simultaneously. The canonical conditions select one of the two orientations; only $(1,1,2)$ is fixed. □

3.3. **The two parent rules.** At a nonzero phase the alternative parent is the adjacent inward mark, $(\alpha; x) \mapsto (\alpha; x - \operatorname{sgn} x)$. At a nonseed helm it is

$$(1, k_m, \ldots, k_2, k_1; 0) \longmapsto (1, k_m, \ldots, k_2; -k_1), \tag{13}$$

followed by simultaneous sign reflection if the remaining rightmost index is negative. An empty tail uses the seed's nonnegative phase. The seed helm with maximum two has parent $(1,1,1)$.

The alternative parent deletes the *rightmost* index at a helm. Recursive evaluation removes the *leftmost* index after 1. These operations serve different purposes. The depth

$$d(\alpha; x) = 1 + \sum_{i=1}^{m} (|k_i| + 1) + |x| \tag{14}$$

decreases by one under either parent edge, with root depth zero. Thus the parent rule defines a rooted tree on the same configurations. The ordinary outward Vieta moves instead are

$$(a, b, M) \longmapsto (a, M, 3aM - b), \qquad (a, b, M) \longmapsto (b, M, 3bM - a). \tag{15}$$

For example, $29 \to 194$ is a Meridian edge, whereas $29 \to 169$ is a Vieta edge. Figure 5 compares the two sets of links.

3.4. **Real components and phasor edges.**

Proposition 3.3. *The canonical real families in* (12)*, modulo companion reflection, are the nonterminal connected components of $H(M, c) = 1$. Their terminal phase is $x + 1/2$ modulo one. The separate terminal component is $M = 1$.*

Proof. The seed companions are at least $2/\sqrt{5}$, and $M_1 \geq 2$ is strictly larger than either. More generally, let $h \geq 1$ and $2/\sqrt{5} \leq a, b \leq h$. The first outward endpoint satisfies $3hb - a \geq (6/\sqrt{5} - 1)h > h$; its reflected pole is larger still. The reversed orbit gives the same conclusion on the other side. Every outward block thus has a strict inverse descent to its source and preserves the terminal phase.

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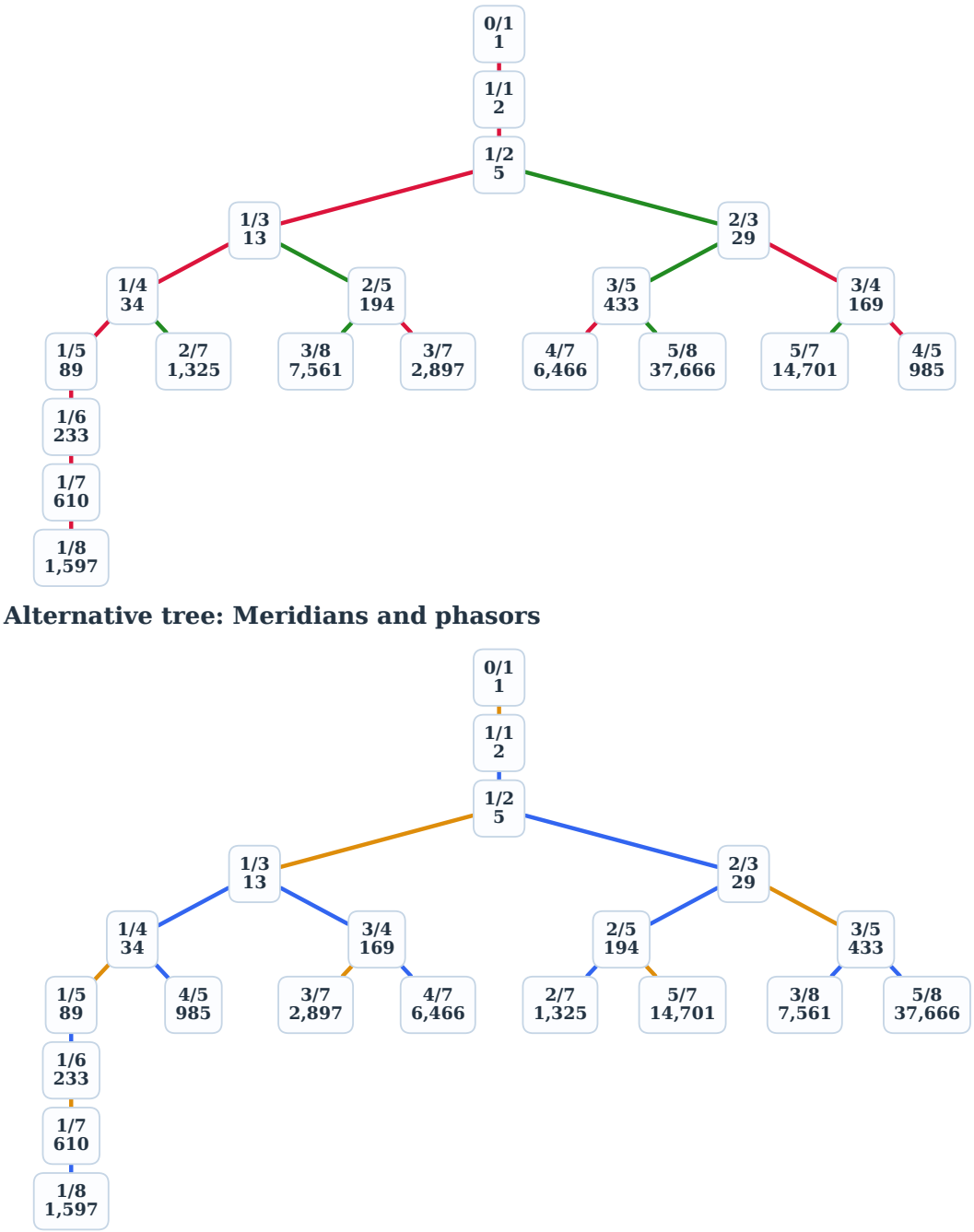


Figure 5. Two trees on the same 20 configurations: the 16 with $M \leq 3000$ and the four additional values $6466, 7561, 14701, 37666$. Each node displays its reduced standard Farey label and maximum. Red and green denote the Vieta moves; blue and gold denote Meridian and phasor steps.

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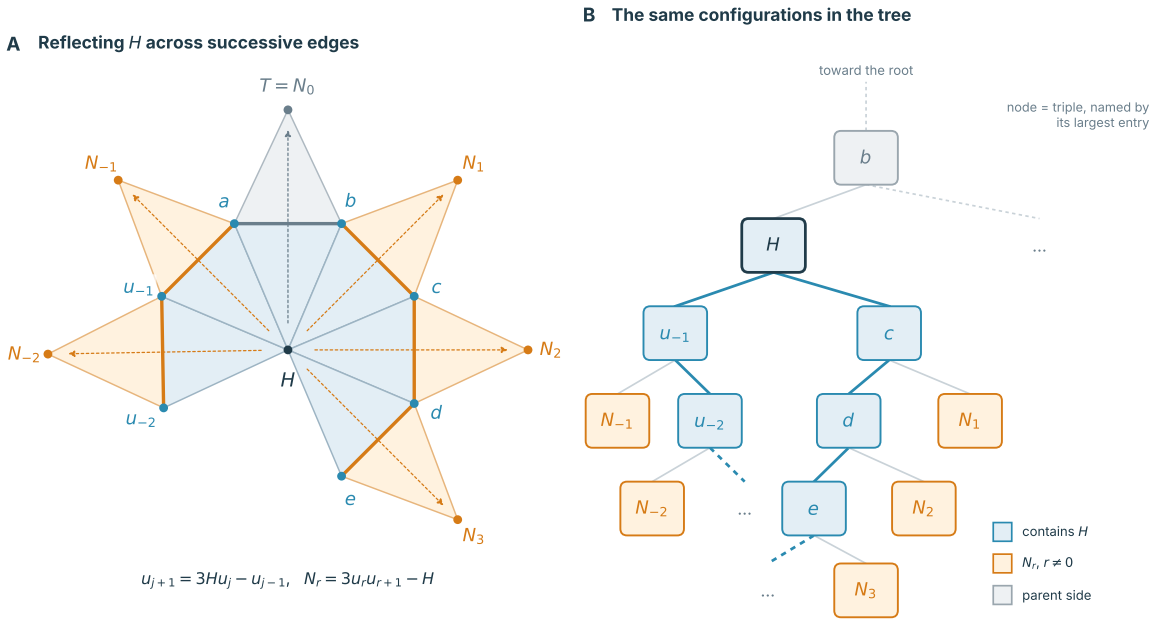


Figure 6. The orbit underlying one recursive extension. Reflecting H across (u_j, u_{j+1}) gives $N_j = 3u_j u_{j+1} - H$; N_0 is the preceding pole. For source $(a, b, H) = (1, 5, 13)$ at $M_{1,1}(0)$, the preceding pole is two, $(u_{-2}, u_{-1}, u_2, u_3, u_4) = (1325, 34, 194, 7561, 294685)$, and $(N_{-2}, N_{-1}, N_1, N_2, N_3) = (135137, 89, 2897, 4400489, 6684339842)$. The right panel locates these reflections in the Vieta tree.

Conversely, reverse any descent ending at host one. Its last host change produces the seed with $x = v - 1/2$, and every preceding walk is nonzero: a zero walk would simply have stopped at the smaller host. These strict blocks uniquely recover a locally constant word. The unrolled x joins the final phase charts continuously. The seed is injective as an ordered curve, as its endpoint difference is strictly monotone; each block is invertible. The positive Laurent expansion proved in Proposition 5.2 makes every family proper. The inverse charts are relatively open in the level set, and properness makes the families closed. Each connected family therefore fills a component. Reflection gives exactly the canonical identifications. □

The phasor edges are obtained by applying the same outward blocks to

$$\left(\frac{h}{\sqrt{3h-2}}, \frac{h}{\sqrt{3h-2}}, h\right), \qquad 1 \le h \le 2. \tag{16}$$

This symmetric arc joins $(1, 1, 1)$ to $(1, 1, 2)$. Its other host is $2h/(3h-2)$, and both possible terminal fits have phase $1/2$. Its endpoints satisfy the bounds used in the proposition, so the outward blocks preserve that phase. The displayed phasors consequently arise from the descent itself, rather than interpolation between integer nodes.

3.5. **Degree and the exact Farey dictionary.** Attach endpoint count vectors $u = (1, 0)$ and $v = (0, 1)$ to the seed. Under an index k update them by

$$(u, v) \longmapsto \begin{cases} ((k-1)(u+v)+v,\ k(u+v)+v), & k > 0, \\ ((|k|(u+v)+u,\ (|k|-1)(u+v)+u), & k < 0. \end{cases} \tag{17}$$

Write their sum as (r, s). Their determinant remains ± 1 and their unordered residues modulo two remain $\{(1, 0), (0, 1)\}$. Thus r, s are positive coprime odd integers. Canonical orientation gives $r \le s$.

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The *Meridian degree* is $D = r + s$. It is the extreme Laurent exponent in ρ^x, where $\rho = (3 + \sqrt{5})/2$; the degree in ρ^{2x} is $D/2$.

Proposition 3.4 (Helm classification)**.** *Canonical Meridians correspond bijectively to coprime positive odd pairs $r \le s$. Their helm labels are the standard Farey fractions r/s.*

Proof. The endpoint vectors are the two Stern–Brocot parents: their sum is (r, s) and their determinant has absolute value one. Conversely, the parents of an odd/odd pair have opposite coordinate parities. Their componentwise absolute difference is a smaller positive primitive odd/odd pair, except at $(1, 1)$. The difference has one componentwise sign; an opposite sign would give the initial pair. Iteration reaches $(1, 1)$.

To recover the signed indices, put $n_\alpha = (r, s)$ and let α^- remove the leftmost index after 1. Consecutive difference vectors have determinant of absolute value two, so three consecutive vectors satisfy $n = 2c\, n^- + \varepsilon n^{--}$ with $c \ge 1$ and $\varepsilon = \pm 1$. Determinant division gives an integer coefficient, parity makes it even, and decreasing coordinate sums fix its magnitude and sign. At the bottom use $(1, 1), (-1, 1)$; canonical orientation fixes the first index positive. The magnitude is fixed by the ratio of successive degrees, which lies in $[2c, 2c+1)$ or $(2c-1, 2c]$ according to the sign; equality occurs only at the first applied index. Reversing the process gives precisely $n_{1,k,k_m,\ldots,k_1} = 2|k| n_\alpha + \operatorname{sgn}(kk_m) n_{\alpha^-}$. For the seed use $k_0 = 1$ and $n_\varnothing = (-1, 1)$, so the first extension has $(r, s) = (2k_1 - 1, 2k_1 + 1)$. This recovers the unique address. Mediant addition identifies the label. □

Lemma 3.5 (Phase dictionary)**.** *For helm (r, s) and degree $D = r + s$, the standard Farey label at phase x is*

x	(p, q)	$p + q$
0	(r, s)	D
1	(r, D)	$D + r$
-1	(s, D)	$D + s$
$n \ge 2$	$(D, (n-1)D + r)$	$nD + r$
$-n \le -2$	$(D, (n-1)D + s)$	$nD + s.$

The negative seed phases are omitted.

Proof. At seed phases $0, 1, n \ge 2$, the endpoint bases are respectively $((1, 0), (0, 1))$, $((1, 1), (0, 1))$, and $((1, n), (1, n-1))$. The count updates are linear, so substitution gives the table; negative phase exchanges the bases. For the inverse, odd/odd p, q give $(r, s, x) = (p, q, 0)$. If q is even, sort $p, q - p$ and choose $+1$ when p is the smaller entry, otherwise -1. If p is even, write $q = tp + u$ with $0 < u < p$; sort $u, p - u$ and choose $+(t + 1)$ or $-(t + 1)$ in the same way. A tie is the seed and uses the nonnegative representative. These formulas recover every mark. □

In particular, degree can be read directly from a reduced label:

$$D = \begin{cases} p + q, & p, q \text{ odd}, \\ q, & q \text{ even}, \\ p, & p \text{ even}. \end{cases} \tag{18}$$

This dictionary connects the Meridian and corridor collision windows below.

4. Meridian residues: modulo 27 and Fibonacci moduli

4.1. **Transferring a seed congruence to every address.** The recurrence for the companions contains no division, but a seed period of the *maximum* need not visibly preserve both companions.

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The following four-entry record resolves that issue:

$$\mathcal{S} = (A, B, C, H) = (9a^2, 9b^2, 3ab, H).$$

Forward and backward orbit steps and pole reflection are the polynomial maps

$$\begin{aligned} \mathcal{R}(A, B, C, H) &= (B, A + 9H^2B - 18HC, HB - C, H), \\ \mathcal{R}^{-1}(A, B, C, H) &= (B + 9H^2A - 18HC, A, HA - C, H), \\ \mathcal{J}(A, B, C, H) &= (A, B, C, C - H). \end{aligned} \tag{19}$$

For $h(x) = M_1(x)$ the seed record is

$$\mathcal{S}_1(x) = (8h(x) - h(x-1) - 2,\ h(x) + h(x-1) + 2,\ h(x) + 1,\ h(x)).$$

Every congruence preserving two consecutive seed maxima therefore propagates along every signed address.

Theorem 4.1 (Largest modulus with a common phase period)**.** *Every canonical Meridian has phase period T modulo m if and only if*

$$m \mid q(T), \qquad q(T) = 3F_{2T}F_{\gcd(2T,4)} = \begin{cases} 3F_{2T}, & T \text{ odd}, \\ 9F_{2T}, & T \text{ even}. \end{cases} \tag{20}$$

In particular, s is a common period modulo F_{2s}.

Proof. Fibonacci addition and (10) give

$$h(x + T) - h(x) = 3F_{2T}F_{4x+2T}.$$

Fibonacci strong divisibility, $\gcd(F_a, F_b) = F_{\gcd(a,b)}$, follows by the Euclidean algorithm from the addition formula. The gcd of all displayed differences is already the gcd of those at $x = 0, 1$, namely $q(T)$. Thus the seed has period T modulo m exactly when $m \mid q(T)$. The record (19) proves sufficiency at every address; the seed itself proves necessity. □

The theorem specifies a period common to all Meridians. It does not assert that the period is the smallest for each individual sequence. Nor does a large period exclude a scalar residue match at two earlier phases. For powers of three it gives

$$M_\alpha(x + 2 \cdot 3^n) \equiv M_\alpha(x) \pmod{3^{n+3}}, \qquad n \geq 0. \tag{21}$$

Indeed, $F_{3j} = F_j(5F_j^2 + 3)$ for even j, beginning with $F_4 = 3$, gives $v_3(F_{4\cdot 3^n}) = n + 1$. Figure 7 shows the common periods at the requested even-indexed Fibonacci moduli.

4.2. **Why modulo 27 gives only ten classes.** No positive Markov entry is divisible by three: mutation negates the changed coordinate modulo three and the integral tree starts at $(1, 1, 1)$. Let t, h be preceding and current poles. An extension by a nonzero index has the particularly simple action

$$S(t, h) = (h, t) \pmod{k \text{ even}}, \qquad R(t, h) = (h, 7h - t) \pmod{k \text{ odd}} \pmod{27}. \tag{22}$$

To see this, put $N_j = 3u_ju_{j+1} - h$. Direct use of the orbit recurrence gives

$$N_0 = t, \quad N_1 = (9u_1^2 - 2)h - t, \quad N_{j+2} - N_j = 9h(u_{j+2}^2 - u_{j+1}^2).$$

All orbit squares are one modulo three, so $N_{j+2} \equiv N_j$ and $N_1 \equiv 7h - t \pmod{27}$. This works for negative indices as well. Only the parity of each recursively applied index survives.

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x	M1							M1.1							M1.2							M1.1.1							M1.-1.1						
s	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8
F_{2s}	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987
-12								1	5	13	31	13	231	181	1	1	13	50	1	88	937	2	1	20	50	17	328	944	2	1	5	9	89	285	215
-11								1	2	13	4	34	283	559	1	1	1	46	73	21	715	2	1	2	13	89	94	170	2	1	2	21	89	333	107
-10								1	1	19	13	121	248	544	1	2	16	48	82	270	541	2	2	20	37	98	29	482	2	5	8	34	53	272	869
-9								1	5	1	34	61	57	169	1	1	13	26	1	291	160	2	1	2	51	89	343	653	2	1	8	13	17	241	806
-8								1	2	13	50	130	169	13	1	1	13	53	73	343	433	2	1	20	31	17	34	923	2	1	5	21	17	86	89
-7								1	1	13	31	25	13	34	1	2	1	50	82	56	652	2	2	2	50	98	258	317	2	5	2	9	125	89	233
-6								1	5	19	4	13	34	985	1	1	16	46	1	21	541	2	1	20	13	17	369	104	2	1	8	21	89	233	890
-5								1	2	1	13	34	231	820	1	1	13	48	73	88	580	2	1	2	37	89	328	758	2	1	8	34	89	285	71
-4								1	1	13	34	121	283	181	1	2	13	26	82	21	937	2	2	20	51	98	94	944	2	5	5	13	53	333	215
-3								1	5	13	50	61	248	559	1	1	1	53	1	270	715	2	1	2	31	89	29	170	2	1	2	21	17	272	107
-2								1	2	19	31	130	57	544	1	1	16	50	73	291	541	2	1	20	50	17	343	482	2	1	8	9	17	241	869
-1								1	1	1	4	25	169	169	1	2	13	46	82	343	160	2	2	2	13	98	34	653	2	5	8	21	125	86	806
0	2	2	2	2	2	2	2	1	5	13	13	13	13	13	1	1	13	48	1	56	433	2	1	20	37	17	258	923	2	1	5	34	89	89	89
1	2	5	5	5	5	5	5	1	2	13	34	34	34	34	1	1	1	26	73	21	652	2	1	2	51	89	369	317	2	1	2	13	89	233	233
2	2	5	8	29	29	29	29	1	1	19	50	121	231	985	1	2	16	53	82	88	541	2	2	20	31	98	328	104	2	5	8	21	53	285	890
3	2	2	5	29	50	194	194	1	5	1	31	61	283	820	1	1	13	50	1	21	580	2	1	2	50	89	94	758	2	1	8	9	17	333	71
4	2	5	2	5	29	194	338	1	2	13	4	130	248	181	1	1	13	46	73	270	937	2	1	20	13	17	29	944	2	1	5	21	17	272	215
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6	2	2	8	5	2	5	29	1	5	19	34	13	169	544	1	1	16	26	1	343	541	2	1	20	51	17	34	482	2	1	8	13	89	86	869
7	2	5	5	29	5	2	5	1	2	1	50	34	13	169	1	1	13	53	73	56	160	2	1	2	31	89	258	653	2	1	8	21	89	89	806
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10	2	5	8	2	29	194	29	1	2	19	13	130	283	985	1	1	16	48	73	21	541	2	1	20	37	17	94	104	2	1	8	34	17	333	890
11	2	5	5	5	5	194	194	1	1	1	34	25	248	820	1	2	13	26	82	270	580	2	2	2	51	98	29	758	2	5	8	13	125	272	71
12	2	2	2	29	2	29	338	1	5	13	50	13	57	181	1	1	13	53	1	291	937	2	1	20	31	17	343	944	2	1	5	21	89	241	215

Figure 7. Five Meridians modulo F_{2s} for $s = 2, \ldots, 8$: the moduli are $3, 8, 21, 55, 144, 377, 987$. Each column has common phase period s, possibly a multiple of an individual least period. The shaded row is the helm; blank cells omit the seed's negative phases. The displayed residues use the reversed Meridian indexing of this edition.

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Proposition 4.2 (Universal modulo-27 classes)**.** *At integer phases every Meridian alternates between the two residues of one of the following sets, allowing the two singleton cases:*

$$\{1\},\ \{2,5\},\ \{4,16\},\ \{7,13\},\ \{8,17\},\ \{10,19\},\ \{11,23\},\ \{14,20\},\ \{22,25\},\ \{26\}.$$

The ten sets are pairwise disjoint. Meridians in different classes cannot share an integer value, regardless of degree or phase.

Proof. The common phase period is two because $q(2) = 27$. Track preceding and current poles at both phase parities, starting from $(1, 2, 1, 5)$. The simultaneous maps satisfy $S^2 = I$ and $SRS = R^{-1}$. Successive current-pole pairs on the R orbit are

$$\begin{array}{cccccc} (2,5) & (13,7) & (8,17) & (16,4) & (23,11) & (10,19) \\ (20,14) & (22,25) & (26,26) & (25,22) & (14,20) & (19,10) \\ (11,23) & (4,16) & (17,8) & (7,13) & (5,2) & (1,1). \end{array}$$

The preceding pair is the previous pair cyclically, and $b_{j+1} = 7b_j - b_{j-1}$ checks the return in 18 steps. These states and their S images form a closed set of 36 states. Their current-pole images are exactly the ten listed sets. Congruence of integer values would require membership in the same set. □

Table 1. The ten universal classes and their populations among the 519 Meridians of degree at most 100. Only the population column is bounded-degree data.

Residue set	Population	Residue set	Population
$\{1\}$	42	$\{10, 19\}$	47
$\{2, 5\}$	88	$\{11, 23\}$	47
$\{4, 16\}$	61	$\{14, 20\}$	38
$\{7, 13\}$	78	$\{22, 25\}$	35
$\{8, 17\}$	65	$\{26\}$	18

At that degree bound, the classification leaves 15,215 of the 134,421 pairs for further comparison. Larger Fibonacci moduli and compatible phases, rather than extrapolation from this table, supply the degree-10,000 certificate in Section 8.

5. Equal-degree separation and unequal-degree windows

5.1. **A complete result at each fixed degree.** We first use the established fixed-sum, fixed-numerator and fixed-denominator orderings of Markov numbers. Section 7 proves them by matching counts and records their earlier sources.

Theorem 5.1 (Equal-degree unicity)**.** *Distinct canonical integer-phase marks of the same Meridian degree have different maxima. The marks may belong to the same or to different Meridians.*

Proof. For $D \geq 4$, Lemma 3.5 partitions the labels into three classes. Helms r/s have fixed sum D; near-helms $r/D, s/D$ have fixed denominator D; outer phases have fixed numerator D. Each class is strictly ordered by its corresponding monotonicity, and the dictionary ensures that different marks have different labels.

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The largest helm is $m_{1/(D-1)}$. The near-helm extrema are $m_{1/D}, m_{(D-1)/D}$; the smallest outer value is $m_{D/(D+1)}$. They are separated by

$$m_{1/(D-1)} < m_{1/D}, \qquad m_{(D-1)/D} < m_{(D-1)/(D+1)} < m_{D/(D+1)}.$$

The intermediate fraction is reduced because D is even. The first two inequalities hold at fixed numerator and the last at fixed denominator. For $D = 2$, the unique seed gives $2, 5$ and then the increasing family $m_{2/(2n-1)}$ for $n \geq 2$, starting at 29. These values are also distinct. □

Thus the remaining Meridian comparison has unequal degrees. To bound it, we need exact leading terms and an explicit positive remainder.

5.2. **Positive words and a uniform tail bound.** Put $\rho = (3 + \sqrt{5})/2$, $\lambda = \log \rho$, and $\xi = \rho^x$. The endpoint matrices are

$$P(\xi) = \begin{pmatrix} B_0/\xi & C_0 \\ C_0 & A_0\xi \end{pmatrix}, \qquad Q(\xi) = \begin{pmatrix} A_0/\xi & \rho C_0 \\ C_0/\rho & B_0\xi \end{pmatrix}, \tag{23}$$
$$A_0 = \frac{15 + 3\sqrt{5}}{10}, \quad B_0 = \frac{15 - 3\sqrt{5}}{10}, \quad C_0 = \frac{2}{\sqrt{5}}.$$

Both have determinant one and positive entries. With seed endpoint words $U = P, V = Q$, the scalar state is

$$a_\alpha = \operatorname{tr} U/3, \quad b_\alpha = \operatorname{tr} V/3, \quad M_\alpha = \operatorname{tr}(UV)/3.$$

Its extension is represented by

$$(U, V) \longmapsto \begin{cases} ((UV)^{k-1}V, (UV)^k V), & k > 0, \\ ((VU)^{|k|}U, (VU)^{|k|-1}U), & k < 0. \end{cases} \tag{24}$$

For verification, $\operatorname{tr}((UV)^{j-1}V)/3$ begins at a, b and obeys the endpoint recurrence by Cayley–Hamilton. The determinant-one trace identity $\operatorname{tr}(XY) = \operatorname{tr} X \operatorname{tr} Y - \operatorname{tr}(XY^{-1})$ then gives the reflected pole. Negative indices use the reversed orbit. The count updates are exactly (17).

Proposition 5.2 (Laurent profile)**.** *For helm counts (r, s), degree $D = r + s$, and $t = (r - s)/2$,*

$$M_\alpha(x) = \sum_{j=-D,-D+2,\ldots,D} c_j \rho^{jx}, \quad c_j \geq 0, \quad c_{\sigma D} = \frac{1}{3}(9/5)^{D/2}\rho^{\sigma t}. \tag{25}$$

The logarithm is strictly convex, and the real profile has one minimum.

Proof. Expand the trace into closed paths on the two matrix states. All weights are nonnegative. A closed path has an even number of off-diagonal choices, which fixes the parity of its exponent. The two constant-state paths give the extreme coefficients because $A_0B_0 = 9/5$ and $A_0/B_0 = \rho$. The second derivative of the logarithm is λ^2 times the variance of the exponents under their positive normalized weights. Both extremes occur, so the variance is positive. They also force growth at both ends. □

Lemma 5.3 (Uniform positive-word remainder)**.** *Let W have r letters P and s letters Q, of total length $g = r + s > 0$. For $n \geq 0$ and $\sigma = \pm 1$,*

$$\frac{\operatorname{tr} W(\rho^{\sigma n})}{3} = \frac{1}{3}(9/5)^{g/2}\rho^{\sigma(r-s)/2}\rho^{gn}(1 + \epsilon_{W,\sigma}(n)), \qquad 0 \leq \epsilon_{W,\sigma}(n) \leq (1 + \rho^{1-2n})^g - 1. \tag{26}$$

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Proof. Write $z = \rho^{-n}$, $\varphi = \sqrt{\rho}$, and $L = 2\varphi/3$. On the positive arm, divide the P, Q factors by $A_0\rho^n, B_0\rho^n$ respectively and conjugate by $\mathrm{diag}(\rho, 1)$. The normalized factors are

$$\begin{pmatrix} \rho^{-1}z^2 & L\rho^{-2}z \\ Lz & 1 \end{pmatrix}, \qquad \begin{pmatrix} \rho z^2 & Lz \\ Lz & 1 \end{pmatrix}.$$

The negative arm gives the same pair in reverse roles after exchanging the basis states. Both are bounded entrywise by the rank-one matrix

$$K = \begin{pmatrix} \rho z^2 & \varphi z \\ \varphi z & 1 \end{pmatrix}.$$

Products preserve the entrywise bound, and $\operatorname{tr} K^g = (1 + \rho z^2)^g$. The constant dominant-state path contributes one to the normalized trace. Restoring the removed factors proves the bounds. □

5.3. **A finite rectangle depending only on the degrees.**

Theorem 5.4 (Unequal-degree collision window)**.** *Let $D_\alpha \neq D_\beta$, and define*

$$m = \frac{|D_\alpha - D_\beta|}{2}, \qquad E = \max(D_\alpha, D_\beta),$$
$$h = \min\{j \in \mathbb{Z}_{\geq 0} : \rho^j \geq E\}, \qquad N = 3m + h + 3, \qquad T = E(N + 2).$$

An integer equality $M_\alpha(x) = M_\beta(y)$ requires

$$|x| \leq \left\lceil \frac{T}{D_\alpha} \right\rceil, \qquad |y| \leq \left\lceil \frac{T}{D_\beta} \right\rceil, \qquad M_\alpha(x) < \rho^T/3. \tag{27}$$

The seed still uses only nonnegative phases.

Proof. The degrees are even, so m is a positive integer. First,

$$|m \log(9/5) - j \log \rho| > \frac{1}{81^m + 1} \qquad (j \in \mathbb{Z}). \tag{28}$$

Choose $j_0 \geq 0$ with $1 < (9/5)^m \rho^{-j_0} < \rho$; equality at an endpoint is impossible by taking the field norm. The positive algebraic integers $9^m - 5^m\rho^{j_0}$ and $5^m\rho^{j_0+1} - 9^m$ have nonzero integer norms and conjugates of absolute value below 9^m. Each is therefore greater than 9^{-m}. Dividing by the appropriate positive denominators makes both adjacent multiplicative gaps greater than 81^{-m}. The inequality $\log(1 + u) > u/(1 + u)$ gives (28); all other integers j are farther away.

At phases $x = \sigma n$, $y = \tau n'$, the leading logarithms differ by

$$\frac{D_\alpha - D_\beta}{2} \log(9/5) + (D_\alpha n - D_\beta n' + \sigma t_\alpha - \tau t_\beta) \log \rho.$$

The coefficient of $\log \rho$ is integral. Put $r_m = (81^m + 1)^{-1}$. Since $E \leq \rho^h$, $\rho^6 > 81$ and $\rho^5 > 5$,

$$E\rho^{1-2N} \leq \rho^{-6m-5} < \frac{1}{2 \cdot 81^m + 3} = \frac{r_m}{r_m + 2}.$$

For $gu < 1$, the elementary bound $(1+u)^g \leq \sum_{j\geq 0}(gu)^j = (1-gu)^{-1}$ makes each relative remainder in Lemma 5.3 smaller than $r_m/2$ when its phase magnitude is at least N. Their logarithmic corrections cannot cancel (28). At least one collision phase therefore has magnitude less than N.

The leading coefficient of a length-g word lies strictly between $\rho^{-g}/3$ and $\rho^g/3$, because $\rho^{-1} < B_0 < A_0 < \rho$. Lemma 5.3, with $1 + \rho^{1-2N} < \rho$, gives $\operatorname{tr} W(\rho^{\pm N})/3 < \rho^{g(N+2)}/3 \leq \rho^T/3$. Convexity bounds a value inside either finite phase strip by its endpoints, so the common value is below $\rho^T/3$. Positivity also gives the lower bound $\operatorname{tr} W(\rho^{\sigma n})/3 > \rho^{g(n-1)}/3$. Comparing the two bounds yields $n < 1 + T/g$, hence the integer ceilings in (27). □

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This theorem bounds the remaining candidates; it does not assert that any candidate collides. It also does not bound the degrees of a hypothetical collision. The next section supplies a different finite reduction, expressed in Farey numerators and denominators rather than Meridian phases.

6. Corridor walls and numerator–denominator windows

6.1. **A second way to travel through the same tree.** The corridor observation groups the walls between neighboring phasors. After an initial entrance, a wall follows a fixed-numerator family

$$m_{p/(q_0+np)}, \qquad n \geq 0.$$

Both Meridian and phasor edges occur in such a route. Their common law is the advance $q \mapsto q+p$, not a rule that distinguishes the two edge colors. For the blue corridor the fixed-numerator portions are $3/(4+3n)$ and $3/(5+3n)$, tending to the historical coordinate $c = 1/3$. The first wall begins

$$169 \xrightarrow{\text{phasor}} 2897 \xrightarrow{\text{Meridian}} 51641 \xrightarrow{\text{phasor}} 925765 \longrightarrow \cdots,$$

and the other begins $433, 7561, 135137, 2423525, \ldots$ with the opposite edge alternation.

Corridor role and edge color must be kept separate. With corridor-wall travel designated *standard* and a new entrance *exotic*, the two entry routes here are

$$5 \xrightarrow{\text{exotic}} 29 \xrightarrow{\text{standard}} 433 \xrightarrow{\text{standard}} 7561 \longrightarrow \cdots, \qquad 13 \xrightarrow{\text{exotic}} 169 \xrightarrow{\text{standard}} 2897 \longrightarrow \cdots.$$

Both exotic entrances are Meridian edges. In the first route the initial standard step $29 \to 433$ changes the numerator from two to three; the fixed-numerator formula starts at 433. Thus excluding the concave ceiling arc from a wall's recurrence does not move the exotic entrance to that arc. The graph and its actual depth (14) are unchanged.

6.2. **The exact recurrence behind the observed regularity.** Use the standard Christoffel matrices

$$D_a = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad D_b = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{29}$$

Their trace encoding is

$$3m_{p/q} = \operatorname{tr}\left[\prod_{j=1}^{p} D_a^{\lfloor jq/p \rfloor - \lfloor (j-1)q/p \rfloor} D_b\right]. \tag{30}$$

The word is a cyclic conjugate of the lower Christoffel word. Trace is unchanged by that rotation. The normalization is fixed by $\operatorname{tr} D_a = 3$, $\operatorname{tr}(D_a D_b) = 6$ and the Farey trace identity, which give the Markov recurrence with the initial values one and two. This is the classical matrix-word framework [9]; Section 7 also gives its matching interpretation.

Replacing q by $q_0 + np$ adds n to every exponent in (30). The eigenvalues of D_a are ρ, ρ^{-1}. Choose its orthogonal eigenbasis so that D_b becomes

$$\begin{pmatrix} \frac{3}{2} + \frac{3}{2\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{3}{2} - \frac{3}{2\sqrt{5}} \end{pmatrix}.$$

All four entries are positive. Expanding the trace over cyclic choices of the two eigenstates gives the exact identity

$$m_{p/(q_0+np)} = \sum_{j=0}^{p} c_{p,q_0,j}\, \rho^{(p-2j)n}, \qquad c_{p,q_0,j} > 0. \tag{31}$$

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Every mode occurs. The least order of a constant-coefficient recurrence is therefore $p+1$, with characteristic polynomial

$$\prod_{j=0}^{p}(X-\rho^{p-2j}).$$

Pairing conjugate roots shows that its coefficients are integers. For $p=3$ the polynomial is $(X^2-18X+1)(X^2-3X+1)$, so either blue wall satisfies

$$m_{3/(q+12)} = 21m_{3/(q+9)} - 56m_{3/(q+6)} + 21m_{3/(q+3)} - m_{3/q}. \tag{32}$$

This equality holds at every step, for both colors of edge.

The exact regularity is a shared recurrence, not a constant output ratio. The positive spectral sum gives strict log-convexity, hence

$$\log_{10}\frac{m_{p/(q_0+(n+1)p)}}{m_{p/(q_0+np)}} \nearrow p\log_{10}\rho.$$

The small deviation is retained in the collision argument, rather than discarded as an asymptotic error. Figure 8 shows both the geometric walls and that deviation.

6.3. **A positive correction with a uniform bound.** The dominant eigenstate in (30) yields

$$3m_{p/q} = \left(\frac{3}{\sqrt{5}}\right)^{p}\left(\frac{1+\sqrt{5}}{2}\right)^{2q+p}(1+\varepsilon_{p,q}), \tag{33}$$

where

$$0 < \varepsilon_{p,q} \le \left(1+\rho^{-2\lfloor q/p\rfloor}\right)^{p} - 1. \tag{34}$$

To prove the bound, divide the transformed D_b by its upper-left entry; every resulting entry is at most one. Choosing the contracting eigenstate at a factor costs at least $\rho^{-2\lfloor q/p\rfloor}$. Summing over all nonempty subsets of contracting choices gives (34). This bound is independent of the remainder class of q modulo p.

Along a fixed wall the correction decreases strictly to zero. Consequently its logarithmic deviations telescope exactly:

$$\sum_{k=0}^{n-1}\left[p\log\rho - \log\frac{m_{p/(q+(k+1)p)}}{m_{p/(q+kp)}}\right] = \log\frac{1+\varepsilon_{p,q}}{1+\varepsilon_{p,q+np}}. \tag{35}$$

Thus there is no accumulated error proportional to the path length. For a route changing corridors, subtraction of the endpoint formula still telescopes, but monotone decay is claimed only within a fixed wall.

Theorem 6.1 (Necessary correction gap)**.** *If $p \neq p'$ and $m_{p/q} = m_{p'/q'}$ for reduced labels, then*

$$\frac{|\varepsilon_{p,q} - \varepsilon_{p',q'}|}{1+\min(\varepsilon_{p,q},\varepsilon_{p',q'})} > 9^{-|p-p'|}. \tag{36}$$

Proof. Assume $p > p'$. The ratio of the leading terms is

$$R = \left(\frac{3(1+\sqrt{5})}{2\sqrt{5}}\right)^{p-p'}\rho^{q-q'}.$$

Its quadratic-field norm is $(9/5)^{p-p'}$, so $R \neq 1$. The number $(\sqrt{5})^{p-p'}(R-1)$ is an algebraic integer, because the powers of $(1+\sqrt{5})/2$ and ρ are units. Its nonzero norm gives

$$|(R-1)(\overline{R}-1)| \ge 5^{-(p-p')}, \qquad \overline{R} = (9/5)^{p-p'}/R > 0.$$

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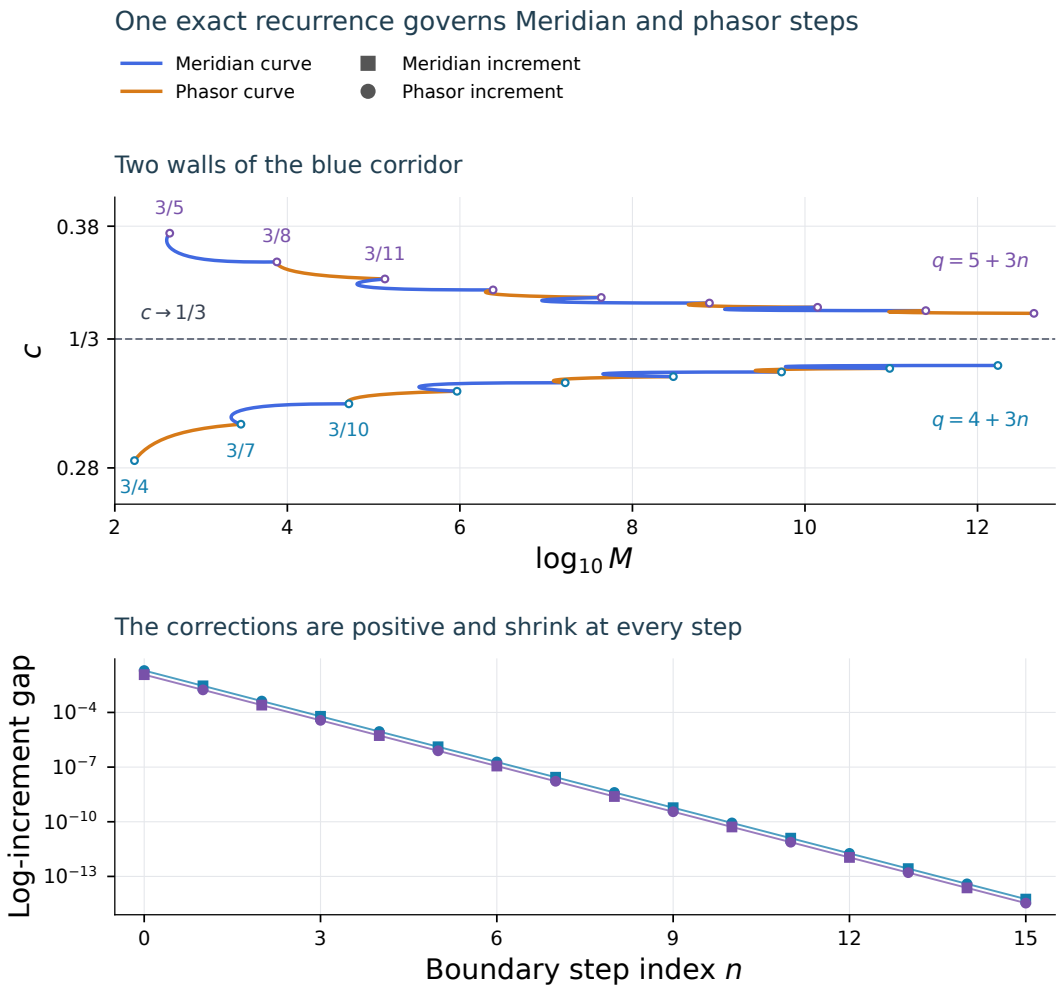


Figure 8. The two fixed-numerator walls of the blue corridor. The concave ceiling arc is omitted from this recurrence illustration. Both walls obey (32). In the lower panel, the positive difference between $3\log_{10}\rho$ and the actual logarithmic increment decreases at every step. Squares denote Meridian increments and circles phasor increments. The values are exact integers; only the plotted curves and logarithms are numerical.

Put $\eta = \max(R, R^{-1}) - 1$. If $R \geq 1$, then $|R - 1| = \eta$ and $|\overline{R} - 1| < (9/5)^{p-p'}$. If $R < 1$, their product is $\eta((9/5)^{p-p'} - 1/(1+\eta)) < \eta(9/5)^{p-p'}$. Either case forces $\eta > 9^{-(p-p')}$. At equality of the Markov values, $R = (1+\varepsilon_{p',q'})/(1+\varepsilon_{p,q})$, so η is precisely the left side of (36). □

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6.4. **The corridor collision window.**

Theorem 6.2 (Numerator–denominator window)**.** *Let $p \neq p'$, $P = \max(p, p')$, and let K be the least nonnegative integer satisfying*

$$\rho^{2K} > P(9^{|p-p'|} + 1). \tag{37}$$

Every equality $m_{p/q} = m_{p'/q'}$ requires

$$\max(q, q') \leq P(K + 1) - 1. \tag{38}$$

For bounded numerators a simpler, weaker consequence is $\max(q, q') \leq 2P^2 + P - 1$.

Proof. If both quotient floors are at least K, then $P\rho^{-2K} < 1/(9^{|p-p'|} + 1)$. The geometric-series estimate used in Lemma 5.3 makes both corrections strictly smaller than $9^{-|p-p'|}$, contradicting Theorem 6.1. Thus one denominator, say q, is at most $Kp - 1$.

Both matrices in (29) have operator norm ρ. The trace bound and the positive leading term imply

$$\frac{1}{3}\rho^{q'} < m_{p'/q'} = m_{p/q} \leq \frac{2}{3}\rho^{q+p}.$$

Since $\log_\rho 2 < 1$ and the exponents are integers, $q' \leq q + p$. It follows that both denominators are at most $P(K + 1) - 1$. For the simpler bound it suffices to use $K = 2P$: $\rho^{4P} > 36^P > P(9^{P-1} + 1)$ for $P \geq 2$, and $|p - p'| \leq P - 1$. □

For example, numerators at most 64 permit $K = 75$, so every remaining comparison lies at denominators at most 4863. With $P = 64$ and numerator difference five, $K = 8$ gives 575. These are bounds on candidates, not new numerical verification claims.

Theorem 5.1 first discards equal Meridian degrees. For the rest, the phase rectangle (27) and denominator cap (38) can be imposed together through the dictionary (18). Fixed-numerator ordering already separates all labels with $p = p'$. What remains is a comparison of different numerators and degrees inside their joint windows. Neither numerator nor degree is globally bounded by these theorems, so this is not yet a finite reduction of the entire unicity conjecture.

7. Snake-graph counts and ordering proofs

7.1. **From a Farey label to an actual matching count.** A snake graph is a chain of square tiles, each attached to the right of or above the preceding tile. Shared edges are counted once. A perfect matching covers every vertex exactly once. The Markov snake associated with a reduced $0 < p/q < 1$ is built as follows. Take the lower staircase from $(0, 0)$ to (q, p): at each horizontal coordinate j, step right and then up to height $\lfloor pj/q \rfloor$. Write right steps as a and up steps as b. The resulting lower Christoffel word has the form $w_{p/q} = azb$. Remove its first and last letters; start with one square and replace each a of z by two added squares to the right and each b by two upward. The resulting snake has $2(p + q) - 3$ tiles.

Write $\mu(w)$ for its number of perfect matchings. The established Markov matching correspondence gives

$$\mu(w_{p/q}) = m_{p/q}. \tag{39}$$

The matching interpretation originated in the REACH work of 2002–2003. Musiker's memorandum, dated 27 June 2002, proposed the recursive graph model [13]; Itsara, Musiker, Propp, and Viana's manuscript of 10 January 2003 stated and proved the snake-matching correspondence [14, Theorem 3.1]. Setting all edge weights to one gives the ordinary perfect-matching counts. Propp's later exposition, first posted in 2005, records the broader collaborative development [15]. These original contributions precede the later proofs of Aigner's orderings.

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For a modern treatment and the Farey-indexed construction, see [12, Section 4.1, Theorem 4.2] and [6]. For example, $w_{2/5} = a^3ba^2b$, its trimmed word is $aabaa$, and the eleven-tile snake has 194 matchings. Auxiliary rearranged words also specify snakes, counted by μ; they need not be canonical Markov snakes.

7.2. **Six pieces and two connector states.** Figure 9 fixes the five three-tile pieces and the three one-square variants of type 6. In a cut between pieces, a matching uses both yellow connector edges or neither: the cut leaves an even number of vertices on either side. Denote these states by 1 and 0. At a selected port remove the two vertices already covered by its connector pair. Direct counting within the pieces gives

(40)

piece	00	01	10	11
1	5	2	2	1
2	5	3	3	2
3	5	2	3	1
4	5	3	2	1
5	5	2	2	1
6i	2	1	1	1

The state order is incoming, outgoing. The one-port end squares 6ii and 6iii have counts $(2, 1)$.

For the ordering surgeries, use a fixed one-ended chop: retain tile triples $(1, 2, 3), (5, 6, 7), \ldots$, with tiles $4, 8, \ldots$ providing paired connectors. A one-square remainder occurs exactly when $p + q$ is even and is counted with the closed end piece. For every connector sequence σ, multiply the applicable row entries in (40); then sum over σ, with outer ports closed. Each matching determines one sequence and one internal completion per piece, and gluing reverses the construction. This proves the counting rule, without an independence assumption. Equivalently it is a sum of terms $5^{k(\sigma)}3^{j(\sigma)}2^{\ell(\sigma)}$, where the exponents count factors of the indicated size. For the $2/5$ snake the piece list $2, 1, 2$ gives

$$5^3 + 5 \cdot 2 \cdot 3 + 3 \cdot 2 \cdot 5 + 3^2 = 194.$$

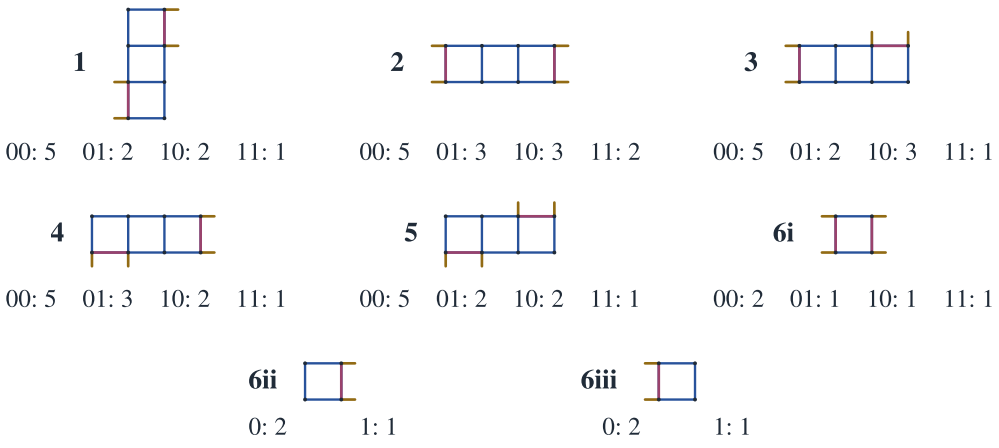


Figure 9. The six physical pieces. Gold edges are paired connectors, magenta marks ports, and dots mark tile vertices. Two-port counts use the order $00, 01, 10, 11$. The central square 6i has counts $(2, 1, 1, 1)$; the one-port end squares have $(2, 1)$. Types 1 and 5 have equal count rows but different physical orientations.

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7.3. **Fibonacci-matrix normalization.** The Fibonacci-matrix formulation and normalization argument below are the author's. We begin with the comparison of three consecutive blocks and then apply it to complete words.

With $F_0 = 0, F_1 = 1$ and $F_{-j} = (-1)^{j+1}F_j$, the basic odd-index identity is

$$F_{2k-1}F_{2k+3} - F_{2k+1}^2 = 1. \tag{41}$$

For example, $5 \cdot 34 - 13^2 = 1$. Consider the author's three consecutive Fibonacci blocks:

$$\begin{aligned} A &= \begin{pmatrix} F_{2k+3} & F_{2k+1} \\ F_{2k+1} & F_{2k-1} \end{pmatrix}, \\ B &= \begin{pmatrix} F_{2k+1} & F_{2k-1} \\ F_{2k-1} & F_{2k-3} \end{pmatrix}, \\ C &= \begin{pmatrix} F_{2k-1} & F_{2k-3} \\ F_{2k-3} & F_{2k-5} \end{pmatrix}, \qquad k \geq 2. \end{aligned} \tag{42}$$

Applying (41) twice gives precisely the upper-left comparison used in normalization:

$$\begin{aligned} (AC)_{11} - (B^2)_{11} &= (F_{2k+3}F_{2k-1} - F_{2k+1}^2) \\ &\quad + (F_{2k+1}F_{2k-3} - F_{2k-1}^2) = 2. \end{aligned} \tag{43}$$

The same identities in the other entries give

$$AC - B^2 = \begin{pmatrix} 2 & 6 \\ 0 & 2 \end{pmatrix}, \qquad CA - B^2 = \begin{pmatrix} 2 & 0 \\ 6 & 2 \end{pmatrix}.$$

Thus replacing AC or CA by B^2 decreases the upper-left entry even inside a longer product: the surrounding Fibonacci blocks have nonnegative entries and positive upper-left entries. For $k = 3$ this reads $(AC)_{11} = 34 \cdot 5 + 13 \cdot 2 = 196 > 194 = 13^2 + 5^2 = (B^2)_{11}$.

7.3.1. *Run counts and general index shifts.* To apply the same comparison at arbitrary run lengths, index the blocks by

$$\mathcal{A}_t = \begin{pmatrix} F_{2t+3} & F_{2t+1} \\ F_{2t+1} & F_{2t-1} \end{pmatrix}, \qquad t \geq 0. \tag{44}$$

Thus $A = \mathcal{A}_k$, $B = \mathcal{A}_{k-1}$ and $C = \mathcal{A}_{k-2}$ in (42). The four boundary counts of a straight block of $2t + 1$ tiles are $F_{2t+3}, F_{2t+1}, F_{2t+1}, F_{2t-1}$: these follow by conditioning on the first edge of a ladder, giving the Fibonacci recurrence; at $t = 0$ they are $(2, 1, 1, 1)$. Gluing successive runs with their paired connector states therefore gives, for $w = a^{h_1}b \cdots a^{h_p}b$ with $h_1 \geq 1$ and $h_j \geq 0$,

$$\mu(w) = [\mathcal{A}_{h_1-1}\mathcal{A}_{h_2} \cdots \mathcal{A}_{h_p}]_{11}. \tag{45}$$

The first run is shortened by the removed initial a; the final b closes the other end. Equivalently, a block-index list $(t_1, \ldots, t_p)$ represents $w = a^{t_1+1}ba^{t_2}b \cdots a^{t_p}b$, with $t_1 + \cdots + t_p = q - 1$.

Fibonacci addition gives the scalar index-shifting identity

$$F_{2r+1}F_{2s+1} - F_{2r-1}F_{2s+3} = F_{2(r-s)-2}.$$

Apply it to the two summands of each product entry to obtain

$$\mathcal{A}_r\mathcal{A}_s - \mathcal{A}_{r-1}\mathcal{A}_{s+1} = 2\begin{pmatrix} F_{2(r-s)-2} & F_{2(r-s)} \\ F_{2(r-s)-4} & F_{2(r-s)-2} \end{pmatrix}, \quad r \geq 1,\ s \geq 0. \tag{46}$$

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Consequently, whenever two adjacent indices differ by at least two, moving one unit from the larger to the smaller strictly decreases the whole product's upper-left entry. The case $r = s + 2$ is exactly (43). The only remaining local comparison has adjacent indices differing by one; it is an interchange of the two blocks.

7.3.2. *Comparing the surrounding Fibonacci blocks.* For a finite list of block indices, define the ratio of its two open counts by

$$\rho(u_1, \dots, u_m) = \frac{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{21}}{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{11}}, \qquad \rho(\varnothing) = 0.$$

All these ratios lie in $[0, 1)$. Direct multiplication gives

$$\rho(t, U) = \frac{F_{2t+1} + F_{2t-1}\rho(U)}{F_{2t+3} + F_{2t+1}\rho(U)}. \tag{47}$$

Its right-hand side increases strictly with $\rho(U)$, since (41) gives $\det \mathcal{A}_t = 1$. Moreover, another application of that identity gives

$$\frac{F_{2t+1}}{F_{2t+3}} - \frac{F_{2t+3} + F_{2t+1}}{F_{2t+5} + F_{2t+3}} = \frac{1}{F_{2t+3}(F_{2t+5} + F_{2t+3})} > 0.$$

Hence $\rho(t + 1, U) < \rho(t, V)$ for any two finite tails U, V. Two lists can therefore be compared by cancelling equal initial indices: at the first unequal pair, the larger index gives the smaller ratio; an exhausted list has the smaller ratio than any nonempty remainder.

For adjacent blocks in positions $j, j + 1$, the outward lists are $(t_{j-1}, \dots, t_1)$ on the left and $(t_{j+2}, \dots, t_p)$ on the right. At $r = s + 1$, (46) becomes

$$\mathcal{A}_{s+1}\mathcal{A}_s - \mathcal{A}_s\mathcal{A}_{s+1} = 2 \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

After multiplication by the surrounding blocks, its upper-left entry has the sign of

$$\rho(t_{j+2}, \dots, t_p) - \rho(t_{j-1}, \dots, t_1). \tag{48}$$

Indeed, it equals this difference times twice the positive upper-left entries of the two surrounding products. This proves the needed surrounding-block test entirely in the same Fibonacci matrices.

Lemma 7.1 (Fibonacci-matrix normalization)**.** *For coprime positive p, q, every word $w = azb$ with q letters a and p letters b satisfies $\mu(w_{p/q}) \le \mu(w)$.*

Proof. For $p = 1$ there is only the word $a^q b$. Assume $p > 1$ and use the block-index list in (45). Its prefix discrepancies are

$$\delta_j = qj - p(1 + t_1 + \cdots + t_j) \quad (1 \le j \le p), \qquad \delta_0 = \delta_p = 0.$$

Changing (t_j, t_{j+1}) to $(t_j + 1, t_{j+1} - 1)$ decreases only δ_j, by p; the reverse change increases only δ_j, by p. We show that an appropriate change always decreases the matching count until $-p \le \delta_j \le 0$ for every j.

The sign checks use just a prefix-sum identity. Put $h_1 = t_1 + 1$ and $h_i = t_i$ for $i > 1$, the actual run lengths. Whenever $j - m \ge 0$ and $j + m \le p$,

$$\delta_{j-m} + \delta_{j+m} = 2\delta_j + p \left(\sum_{i=j-m+1}^{j} h_i - \sum_{i=j+1}^{j+m} h_i \right). \tag{49}$$

In the block indices, the left sum has the extra 1 precisely when it includes t_1.

First remove positive discrepancies. Choose the rightmost largest $\delta_j > 0$ and transfer one unit from t_{j+1} to t_j. Comparison with its two neighboring discrepancies gives $h_j < q/p < h_{j+1}$; coprimality

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makes these inequalities strict. Thus the transfer is allowed and $t_{j+1} \geq t_j + 1$; for $j = 1$ the gap is at least two. Gaps of at least two are handled by (46).

If $t_{j+1} = t_j + 1$, compare the outward lists. A first unequal pair with the left index larger would make the parenthesis in (49) nonnegative: the central difference is -1, the equal outward indices cancel, and the first unequal difference is at least 1. This contradicts maximality and the choice of the rightmost maximum. If the left list were exhausted first, its extra initial 1 would instead give $\delta_{2j} = 2\delta_j > \delta_j$, again impossible (including simultaneous exhaustion, when $2j = p$). The left outward ratio is therefore strictly larger than the right one. Reversing the interchange in (48) shows that the selected transfer strictly decreases the count. It also decreases $\sum_{j=1}^{p-1} \delta_j$. There are only finitely many nonnegative block-index lists of sum $q - 1$, so this phase ends with all $\delta_j \leq 0$.

Then remove discrepancies below $-p$. Choose the leftmost smallest $\delta_j < -p$ and transfer one unit from t_j to t_{j+1}. For $j > 1$, the neighboring comparisons give $h_j > q/p > h_{j+1}$; for $j = 1$, the inequality $\delta_1 < -p$ gives $t_1 > q/p$, while $t_2 < q/p$. Thus $t_j \geq t_{j+1} + 1$ and the transfer is allowed. Again a gap of at least two is settled by (46).

In the gap-one case, a first unequal outward pair with the left index smaller would make the parenthesis in (49) nonpositive, contradicting minimality and the leftmost choice. If that pair reaches t_1, its extra 1 instead gives $\delta_{2j} \leq 2\delta_j + p < \delta_j$, the same contradiction. If the right list were exhausted first, the identity would give $\delta_{2j-p} = 2\delta_j + p < \delta_j$. Simultaneous exhaustion would give $\delta_j = -p$. Thus the left outward ratio is strictly smaller than the right one, and (48) proves a strict decrease in count. The changed discrepancy is $\delta_j + p < 0$, so positivity is not reintroduced. This phase increases the bounded sum of discrepancies by p each time and terminates with $-p \leq \delta_j \leq 0$.

Finally, $\delta_j \equiv qj \pmod{p}$. For $1 \leq j < p$, coprimality leaves exactly one possible value in $[-p, 0]$, forcing

$$1 + t_1 + \cdots + t_j = \left\lceil \frac{jq}{p} \right\rceil .$$

These are precisely the canonical run lengths. Every transfer has decreased the upper-left product entry, proving the assertion. □

For the canonical word, $h_j = \lceil jq/p \rceil - \lceil (j-1)q/p \rceil$. With $k = 1 + \lfloor (q-1)/p \rfloor$, the factors are the author's A, B from (42), and the end factors are B. For example, $m_{3/8} = [BAB]_{11}$ with $k = 3$. A similar normalization result appears in [9, Proposition 21]; the proof above is self-contained in the author's Fibonacci matrices.

7.4. **The three Aigner orderings.** Fixed-numerator monotonicity was proved by Rabideau and Schiffler [12]; all three orderings were proved by LPTV, Lee–Li–Rabideau–Schiffler, and McShane [9, 10, 11]. The author's Fibonacci matrices prove normalization and the fixed-numerator ordering. His six pieces prove the remaining fixed-sum comparison and the fixed-denominator ordering, using normalization to return to a canonical Markov snake.

Theorem 7.2 (Established monotonicities)**.** *For $i > 0$, whenever both fractions are reduced and lie in $(0, 1)$,*

$$\begin{aligned} m_{p/q} &> m_{(p+i)/(q-i)} && \textit{(fixed sum)}, \\ m_{p/q} &< m_{p/(q+i)} && \textit{(fixed numerator)}, \\ m_{p/q} &< m_{(p+i)/q} && \textit{(fixed denominator)}. \end{aligned}$$

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7.4.1. *Fixed numerator: Fibonacci matrices.*

Proof. Fibonacci recurrence gives, for $t \geq 1$,

$$\mathcal{A}_t - \mathcal{A}_{t-1} = \begin{pmatrix} F_{2t+2} & F_{2t} \\ F_{2t} & F_{2t-2} \end{pmatrix}.$$

All entries are nonnegative and the upper-left is positive. Start with $w_{p/(q+i)}$ and delete i internal letters a, retaining the first. Each deletion changes one factor $\mathcal{A}_t$ to $\mathcal{A}_{t-1}$ with $t \geq 1$. Surrounding factors are positive, so the matching count strictly falls. The resulting word ends at (q, p); normalization gives $m_{p/q} \leq \mu(w_*) < m_{p/(q+i)}$. □

7.4.2. *Fixed sum: six pieces.*

Proof. At the same physical cuts, the changes $2 \to 3$ or 4, then 3 or $4 \to 5$, increase no entry of (40); the changed factors are $3 \to 2$ or $2 \to 1$. Rotation $5 \leftrightarrow 1$ preserves the row. Selecting the changed connector and no others exhibits a strictly smaller positive summand whenever an equal-direction join becomes unequal.

The trimmed canonical word begins and ends with a, contains no bb, and has $q - p - 1$ equal joins, all aa. At its rightmost equal join, interchange a, b strictly to the right. That suffix is alternating, so only the chosen join changes. Exactly one prescribed piece changes by the stated rules, while any rotated pieces keep their rows. Each surgery strictly decreases the count. Odd-numbered operations replace one horizontal step by a vertical step in net; even-numbered operations keep the totals. After $2i$ operations the endpoint is $(q - i, p + i)$. There are enough joins because $p + i < q - i$. For the resulting auxiliary word w_*, normalization gives $m_{(p+i)/(q-i)} \leq \mu(w_*) < m_{p/q}$. A similar tail comparison appears in [9, Proposition 22]; here the comparison is proved using the author's six physical pieces. □

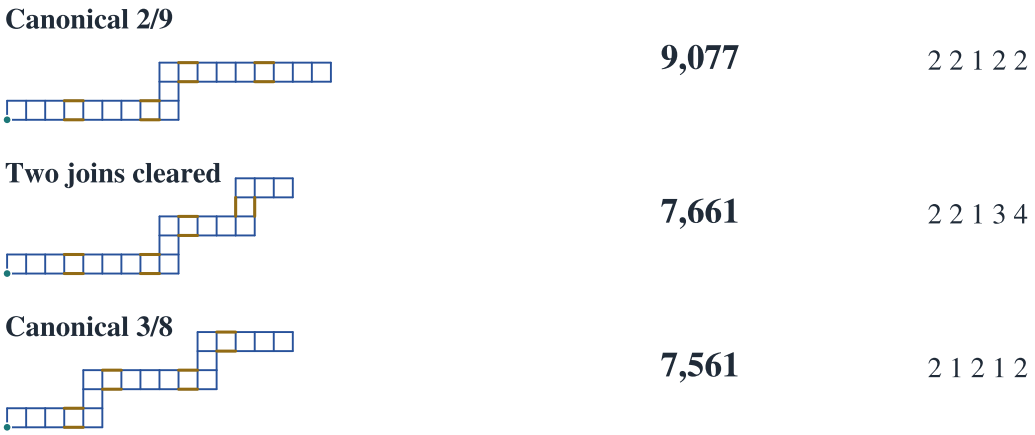


Figure 10. Fixed-sum surgery on the same five pieces. The canonical 2/9 snake has 9077 matchings; after two changes the auxiliary snake has 7661; normalization gives the 3/8 snake with 7561. The intermediate first change, not shown, has 8263. Gold pairs mark cuts and teal dots the starting corners.

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7.4.3. *Fixed denominator: six pieces.*

Proof. For the fixed-denominator comparison, refine the same six-piece chop into single-square pieces of type 6, retaining odd-numbered tiles and using the even tiles as paired connectors. A retained square has counts $(2,1,1,1)$ for opposite ports and $(2,1,1,0)$ for adjacent ports. The zero forbids covering their common corner twice. Summing an intervening state regroups these squares into the original three-tile pieces.

Insert one internal b, adding two vertical squares. The affected local counts are the following complete table, with outside ports fixed:

	before				after			
directions	00	01	10	11	00	01	10	11
a,a	2	1	1	1	5	2	2	1
b,b	2	1	1	1	5	3	3	2
a,b	2	1	1	0	5	3	2	1
b,a	2	1	1	0	5	2	3	1

The endpoint caps have the corresponding rows with an outer port closed; the empty trimmed word has the increase $2 \to 5$. No outside count falls, and the all-zero choice witnesses a strict increase. Thus the two added squares compensate for every changed direction.

Starting from $w_{(p+i)/q}$, delete i internal b's and reverse this comparison. At least i are available. Normalizing the resulting word to (q,p) proves $m_{p/q} \le \mu(w_*) < m_{(p+i)/q}$. □

7.5. **Folding an actual snake at its center.** The same local counts also compare families sharing a long initial sequence of pieces. Represent the four counts by

$$R_1 = R_5 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}, \quad R_4 = \begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}.$$

Attaching piece i to an open-count row (x,y) replaces it by $(x,y)R_i$. This is the sum over the common connector state, not an additional assumption about matchings.

For a helm, use the symmetric central chop: retain the middle square as 6i and group retained squares in pairs outward on each side. For odd/odd p,q, each half has $(p+q-2)/2$ retained squares. If this number is odd, one square remains at each outer end, giving pieces 6ii and 6iii with initial counts $(2,1)$. Otherwise the half starts at $(1,0)$ and consists of complete three-tile pieces. This repartition requires rebuilding the piece list; one cannot move a cut and keep its old boundary counts.

Proposition 7.3 (Folded matching counts)**.** *For a reduced $0<p<q$, let (x,y) be the open counts of one reflected half under the central chop. If p,q are odd, then*

$$m_{p/q} = 2x^2 + 2xy + y^2 = (x+y)^2 + x^2. \tag{50}$$

If $p+q$ is odd, then $m_{p/q} = x^2+y^2$. In either case $\gcd(x,y)=1$; a nonempty half has positive counts.

Proof. The trimmed Christoffel word is a palindrome: for $0<j<q$, $\lfloor p(q-j)/q \rfloor = p-1-\lfloor pj/q \rfloor$. For odd/odd labels the two halves meet at an opposite-port central square, whose four counts are $2,1,1,1$. The reflected halves contribute x^2, xy, yx, y^2, giving (50). For opposite parities they meet across a single connector and must use its same state, giving x^2+y^2. Every full piece table and every exposed-square table has determinant ± 1, so its integral inverse preserves the gcd of an

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open row. Starting at $(1,0)$ proves primitivity. The first square gives $(2,1)$, and every subsequent permitted table preserves positivity. □

The seed $1/1$ consists only of its central square. The two examples in Figure 11 give $m_{1/3} = 2\cdot 2^2 + 2\cdot 2\cdot 1 + 1^2 = 13$ and $m_{1/5} = 2\cdot 5^2 + 2\cdot 5\cdot 3 + 3^2 = 89$.

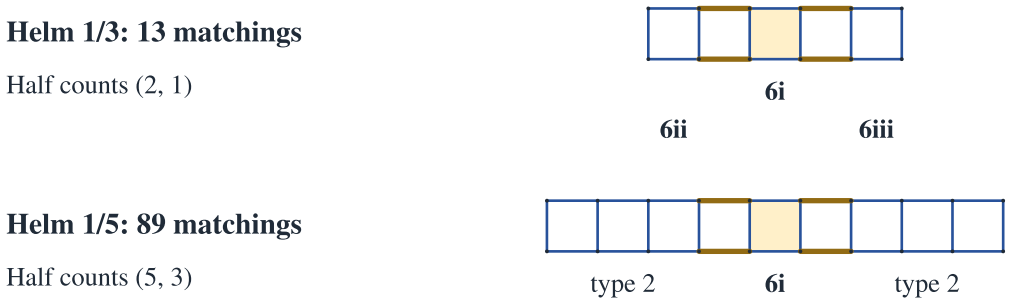


Figure 11. Central and end pieces in actual helms. At $1/3$, central square 6i has the end squares 6ii and 6iii as its halves, with counts $(2,1)$. At $1/5$, each half is a type-2 triple with counts $(5,3)$. Folding produces 13 and 89.

7.6. **A common prefix leaves one quadratic comparison.** Two folded graphs with the same outer boundary convention may share a prefix with open counts (x,y). Sum over all remaining states, leaving that cut open. The two centers give positive definite symmetric tables $G, \widetilde{G}$ with determinant one and nonnegative off-diagonal entries. For a helm they have the form SQS^{T} with $Q = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$; for opposite-parity labels they have form SS^{T}. Thus

$$m - \widetilde{m} = \alpha x^2 + 2\beta xy + \gamma y^2, \qquad G - \widetilde{G} = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}. \tag{51}$$

Proposition 7.4 (At most one primitive common prefix)**.** *For a nonempty common part and $G \neq \widetilde{G}$, equality requires $\alpha\gamma < 0$ and $\beta^2 - \alpha\gamma$ to be an integer square. There is then at most one primitive positive row (x,y) giving equality. That row must still arise from both actual prescribed snakes.*

Proof. A center table $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ has $ac - b^2 = 1$ and $b \geq 0$. If its two diagonal differences from another table have the same weak sign, then the difference of $b = \sqrt{ac-1}$ has that sign too. A nonzero difference cannot vanish at a positive row. Opposite diagonal signs are therefore necessary. The resulting quadratic in y/x has one positive root. An integer zero obeys $(\gamma y + \beta x)^2 = (\beta^2 - \alpha\gamma)x^2$; a rational square root of an integer is integral. A positive rational root has only one primitive positive representative. □

For an empty prefix, test its actual boundary row. If $G = \widetilde{G}$, the criterion supplies no exclusion. Equal count tables do not certify equal physical graphs: types 1 and 5 already show why orientations and the prescribed Christoffel word must be retained.

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A useful uniform exclusion follows directly modulo three. The physical piece tables satisfy $R_iQR_i^{\mathsf{T}} \equiv Q \pmod 3$, whereas a single retained square gives $-Q$. A helm tail with d retained squares therefore has center table $(-1)^dQ \pmod 3$. Opposite parities give a nonzero difference at every primitive row because the discriminant of Q is -1, a nonsquare modulo three. For equal parity and nonzero difference, let $h = \min_{i,j} v_3((G-\widetilde{G})_{ij})$ and write $3^{-h}(G-\widetilde{G}) = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$. Then $h \geq 1$. Expanding the equal determinants and reducing modulo three gives $\gamma \equiv \alpha + \beta$; the normalized discriminant is one or two modulo three. If it is two, every primitive shared row has count difference of valuation exactly h. These are exclusions for specified tails with arbitrary actual common prefixes, not a classification of all possible snake-graph collisions.

8. All-phase verification through degree 10,000

Theorem 8.1 (Degree-10,000 separation)**.** *Among canonical Meridians of Laurent degree at most* 10,000*, distinct canonical integer-phase marks have distinct maxima. There are exactly* 5,067,077 *Meridians in this family.*

The computational part separates different Meridians at all integer phases. The fixed-degree theorem also separates marks on a single Meridian, so the two conclusions combine as stated. The bound is on *degree*, not on the size of a Markov number, and it does not exclude a competitor of higher degree.

8.1. Why a finite residue test covers infinitely many phases.

For a period T, define the complete image

$$\mathcal{R}_\alpha(T) = \{M_\alpha(x) \bmod q(T) : 0 \leq x < T\}.$$

Theorem 4.1 makes this the image at *every* integer phase. Disjoint images therefore separate two entire Meridians. If images overlap for several periods, a true equality must use compatible phases in all of them. Define the witness relation

$$W_T(\alpha,\beta) = \{(u,v) \in (\mathbb{Z}/T\mathbb{Z})^2 : M_\alpha(u) \equiv M_\beta(v) \pmod{q(T)}\}.$$

Proposition 8.2 (Compatible witnesses)**.** *Equality of two integer values requires one witness in each selected W_T, with first coordinates agreeing modulo every pairwise gcd of the periods, and second coordinates agreeing in the same way. If no compatible choice exists, the two Meridians are disjoint at all integer phases.*

Proof. A true equality uses the same integers x, y in every modulus and hence supplies the required residues. Conversely, the generalized Chinese remainder theorem lifts compatible coordinates to a phase pair modulo the least common multiple. This converse concerns modular witnesses only; it does not imply equality of the full values. □

The moduli need not be independent. In particular, the periods $36, 38, 40$ are all even, so contradictory parity requirements alone can exclude a pair that survives each separate image test.

8.2. Complete inventory and the final exclusion.

Proof of Theorem 8.1. Proposition 3.4 identifies the inventory with coprime positive odd pairs $r \leq s$, $r + s \leq 10{,}000$. It contains 5,067,077 Meridians and 12,837,632,128,426 unordered pairs. The producer enumerates the count-vector address tree; an independent checker enumerates the coprime pairs directly and compares the inventories. The child degree is $2|k|D \pm (|v| - |u|) > D$, where $|u|, |v|$ are the endpoint word lengths. This gives finite exhaustive index ranges.

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Every one of the 36 phase residues of every Meridian is evaluated modulo $q(36)$. Equal-residue grouping lists all pairs whose images intersect; its complement is excluded at once. Periods 38 and 40 are then evaluated on every endpoint of the preceding survivor sets. The complete image tests leave 87 pairs. For each of them, project every full witness relation onto the four parity pairs $(u \bmod 2, v \bmod 2)$. The three projected sets have empty intersection in all 87 cases. Proposition 8.2 therefore excludes every pair of different Meridians at every integer phase. Theorem 5.1 also excludes different marks within each individual Meridian, completing the statement. □

Table 2. Complete cumulative survivor counts from the archived certificates. "Images" forgets meeting phases; "phases" also imposes compatibility. Survivors are modular coincidences, not integer collisions.

Tests retained	$D \le 2{,}000$	$D \le 10{,}000$
36: images	7,121	320,755
36, 38: images	612	17,427
36, 38: phases	2	2,052
36, 38, 40: images	0	87
36, 38, 40: phases	**0**	**0**
36, 38, 40, 42: images	—	1
36, 38, 40, 42, 44: images	—	0

8.3. **What was checked independently.** The degree-10,000 producer uses signed endpoint-orbit updates. The separate evaluator uses Fibonacci staircase words with Euclidean matrix reduction. They agree on all 188,369,412 retained phase evaluations. The checker verifies the 182,414,772 baseline phase tags bijectively, so an omitted or repeated phase record cannot pass unnoticed. Independent sorting of the original unsorted buckets reconstructs all 70,009 cross-Meridian residue groups and the subsequent complete pair sets. Coverage is checked by exact records, not inferred merely from their counts or from a matching digest.

The phase result is checked three ways: full generalized-CRT joins, an independent 16-bit mask computation on $(x \bmod 4, y \bmod 4)$, and the weaker four-state parity projection, already sufficient for the final 87. The degree-2,000 certificate, containing 202,862 Meridians, was completed first; its common survivor sets agree exactly with the larger inventory when identified by helm labels.

Five deliberate corruptions—an altered value, an incorrect modulus, a duplicate tag, an omitted record and an invented pair—were all rejected. The degree-100 control compares 18,684 residues with literal Fibonacci words and the polynomial record of (19). The optional image-only extension through periods 42 and 44 also ends with zero pairs; all 88 period-44 values were separately checked using arbitrary-size Python integers because $q(44) > 2^{63}$.

8.4. **Reproducibility and scope.** The degree reproducibility package [7] contains the inventory, compact residue groups, phase witnesses, producer and checker sources, audit receipts, and a regeneration driver. Large baseline tables are regenerated; their recorded sizes and hashes identify the original run. The reindexed edition retains the original computational kernels with an explicit address-convention adapter. Reversing the written address tail does not change a helm label, degree, phase, or certified numerical value.

This is a deterministic computer-assisted certificate, not an external referee report or proof-assistant formalization. Its scope is a bounded set of Meridians with unbounded phase. The

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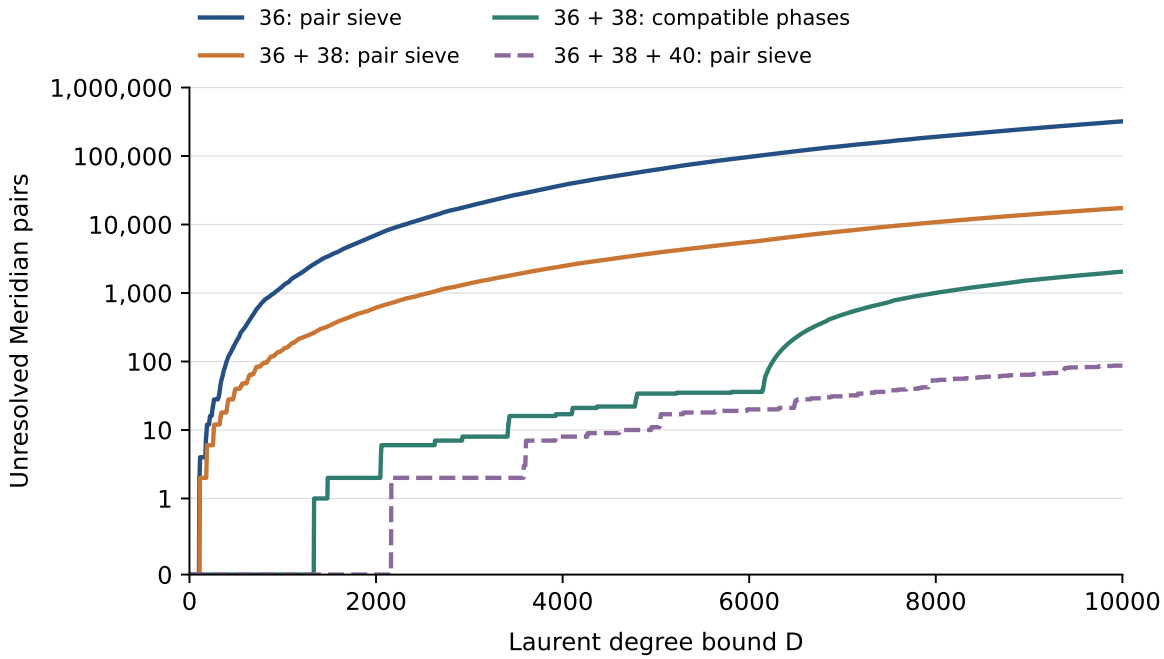


Figure 12. Exact unresolved-pair profiles in the degree-10,000 inventory. The phase-compatible sieve through period 40 has no survivors in the entire displayed range. Lines join discrete even-degree bounds. The figure is a record of the finite certificate, with no extrapolation to larger degrees.

numerical-order verification in Section 9 is a separate computation with its own coverage proof and execution record.

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9. Computational chronology and numerical-order verification

9.1. **The earlier Claude-assisted search.** The historical working draft [1], pp. 38–40, reports a Claude-assisted search of **9,581,528,580 sorted Markov triples** against the cutoff $10^{100,000}$, taking **24 minutes** on a two-core machine with 7 GB of memory. It compared maxima modulo 3^{39} in 16 passes and 4,096 buckets; all 21 repeated-residue pairs separated modulo $2^{61}-1$. No equal maxima were reported.

An earlier public announcement of a 24-minute search is timestamped **23 October 2026, 09:54:57.771 CDT** (14:54:57.771 UTC) [2]. Its caption says "9.3 billion"; the later draft gives the exact count above. This dates the *announcement*, not the beginning or end of execution. Original execution timestamps have not been recovered. The draft itself labels these as historical computational reports whose original logs and floating-point cutoff bounds had not been independently certified. We preserve that status here. This search is separate from the later ten-billion anchor, and its reported 24 minutes are not included in the cumulative timing below.

9.2. **The completed fifty-billion verification.** The archived computation [3] establishes that each of the **first 50,000,000,000 distinct Markov values in numerical order** is the maximum of exactly one normalized positive Markov triple. This complements the degree-10,000 result: the latter covers infinitely many phases on a bounded set of Meridians, whereas this computation certifies a numerical prefix. Its coverage and pruning proofs are collected in the separate verification supplement [4].

Run date and duration. The recorded execution began on **28 October 2026 at 03:32:57 CDT** and finished, including its final audit, at **05:38:34 CDT** (UTC−05:00). The underlying timestamps are 08:32:57.096455–10:38:33.644612 UTC. Two completed coordinator invocations occupied **7,492.102807 seconds: 2 h 04 min 52.103 s**. Including the original ten-billion anchor, the cumulative recorded execution path is **8,330.980040 seconds: 2 h 18 min 50.980 s**. The calendar span is 7,536.548157 seconds; it additionally includes the approximately 44-second pause between invocations. These are historical execution dates and measured durations, distinct from this edition's publication date.

Table 3. The completed fifty-billion run. Counts are configurations unless explicitly identified as distinct values.

Certified distinct-value prefix	50,000,000,000
Complete outer domain	71,517,775,875
New retained configurations compared	54,940,210,994
Repeated primary residues / candidate configurations	346 / 692
Actual equal-value collisions	0

The numerical cutoff was $B = F_{1,147,147}$, with 796,397 bits. The inner height $K = 573{,}574$ contains 50,000,097,577 configurations at or below B; the proved outer height $L = 685{,}979$ contains every competitor below that cutoff. The original ten-billion cutoff certificate justifies 10,000,015,326 inner omissions. Strict certified lower bounds exclude a further 6,577,549,555 configurations above B.

All 32 primary prefix passes and all 8,192 complete buckets finished. Every repeated residue modulo 2^{64} was replayed; all 346 candidate pairs separated modulo 5^{27}. The final auditor independently reconstructed their 692 full integer triples and checked the Markov equation, residues and unequal maxima, as well as the 38 anchor candidates. No probabilistic hash-collision assumption enters this comparison.

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The run used eight workers on Linux x86-64, an eight-CPU-equivalent quota and an 8-GiB memory limit, with frozen GCC 13.3.0 executables. The timing includes generation, partitioning, sorting, replay, orchestration and the 107.371720-second final audit; earlier preflight and later archival checks are excluded. The supplement gives the stage timings, retained receipts, source identities, audit scope and reproducibility limitations. This edition reports the completed run; it does not announce a result for the planned 500-billion verification.

10. What remains to prove

The organization now separates three questions. The address bijection settles coverage of configurations. Fixed-degree ordering and the two collision windows restrict equality of maxima. Complete residue witnesses settle the remaining comparisons for the degree-bounded family in Theorem 8.1. The fifty-billion computation additionally excludes every competitor for a finite prefix in numerical order.

For unrestricted unicity the unresolved pair has different Meridian degrees, the same modulo-27 class, and different Farey numerators. It must satisfy both its phase rectangle and its numerator–denominator window. A proof must exclude those candidates uniformly as the degrees and numerators grow. Enlarging a finite certificate alone does not supply that step. Likewise, an invertible corridor recurrence or a long modular period does not by itself prevent an earlier match of one scalar residue.

Corridors offer a way to formulate the next comparison with explicit entrance states and independently chosen wall lengths. Their positive corrections are exact and telescope; the common-prefix snake calculation offers a different reduction to a quadratic in primitive counts. Neither mechanism has yet been shown to cover every unrestricted pair. The purpose of retaining their precise hypotheses is to make the remaining uniform statement identifiable, rather than hide it inside an induction.

Data and code. The editable source accompanying this edition includes all figure assets, the catenary and landscape figure generators, a source and figure inventory, and checks used for this rewrite. The separate degree reproducibility package contains the full computational evidence cited in Section 8. The fifty-billion proof and timing supplement accompanies this edition; its full frozen computational archive is cited separately [3, 4]. The corridor derivations are developed further in the supplementary research record [8]. Numerical drawings are explanatory; the arithmetic claims use the displayed identities or complete certificates.

Author contributions and AI assistance. The author conceived and directed the research program, including the representation by reciprocal powers and geometric descent, the two-dimensional host–phase landscape, the Meridian indexing and alternative tree, the six-piece snake-graph decomposition, the Fibonacci-matrix formulation and normalization argument, and the corridor observation. The author proposed using these structures for coverage, comparisons and verification, and specified the questions pursued at each stage. The Aigner monotonicities are established results; their prior proofs are credited in Section 7. The Fibonacci-matrix normalization argument presented there is the author's; the related earlier normalization result is cited for comparison, and the proof is given in full from Fibonacci identities.

Generative AI provided substantial mathematical and computational assistance, not only language editing. The development record identifies ChatGPT (OpenAI), with the recorded GPT-6.0 Astra and GPT-6.1 Sol model labels and reasoning settings, and Claude 5.5 Opus Extra (Anthropic) in the preceding geometric-descent work; exact backend revision identifiers were not retained. Within the author's framework, these tools assisted with detailed derivations, examples and counterexamples, proof construction and revision, verifier and independent-checker code, figures, and manuscript drafting. ChatGPT also assisted with the opening explanation and the catenary and landscape

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figures. The project's separate computational checks are algorithmic audits, not external peer review. The author is responsible for the claims, references and final submission.

Funding and competing interests. No funding was received. The author declares no competing interests.

References

[1] E. F. W. Heffern, *Geometric Descent on the Markov Surface*, working draft v32, archived October 2026, pp. 38–40, historical computational reports. The earlier search was assisted by Claude.

[2] E. F. W. Heffern, public LinkedIn announcement of a 24-minute Markov-number search, 23 October 2026, 14:54:57.771 UTC, activity 7519411048115003392. Platform metadata and the source URL are retained in the verification supplement's evidence packet; see the public announcement.

[3] E. F. W. Heffern, *Markov 50 Billion Verification Package*, version 1, completed and sealed 28 October 2026; public derivative version 1.1 in *Computational supplements for Meridians on the Markov Surface*, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

[4] E. F. W. Heffern, *Computational chronology and proofs for the fifty-billion verification*, supplement to the present manuscript, version 4, 7 November 2026.

[5] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013. doi:10.1007/978-3-319-00888-2.

[6] İ. Çanakçı and R. Schiffler, *Snake graphs and continued fractions*, European J. Combin. **86** (2020), 103081. doi:10.1016/j.ejc.2020.103081.

[7] E. F. W. Heffern, *Meridian Degree 10,000 Reproducibility Package*, computational supplement, 31 October 2026; address-convention revision, 7 November 2026. Original computational release in *Computational supplements for Meridians on the Markov Surface*, version 1.0, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

[8] E. F. W. Heffern, *Markov Meridian Residue Supplement*, version 1.18, and *Corridor-to-Corridor Induction*, version 1.3, unpublished supplementary research records, 7 November 2026. The corridor results used here are proved in Section 6.

[9] C. Lagisquet, E. Pelantová, S. Tavenas and L. Vuillon, *On the Markov numbers: fixed numerator, denominator, and sum conjectures*, Adv. Appl. Math. **130** (2021), 102227. doi:10.1016/j.aam.2021.102227. Preprint: arXiv:2010.10335.

[10] K. Lee, L. Li, M. Rabideau and R. Schiffler, *On the ordering of the Markov numbers*, Adv. Appl. Math. **143** (2023), 102453. doi:10.1016/j.aam.2022.102453.

[11] G. McShane, *Convexity and Aigner's conjectures*, preprint (2021), arXiv:2101.03316.

[12] M. Rabideau and R. Schiffler, *Continued fractions and orderings on the Markov numbers*, Adv. Math. **370** (2020), 107231. doi:10.1016/j.aim.2020.107231.

[13] G. Musiker, *A conjectured combinatorial interpretation for Markov numbers*, REACH memorandum, 27 June 2002. Original memorandum; REACH archive.

[14] A. Itsara, G. Musiker, J. Propp, and R. Viana, *Combinatorial interpretations for the Markov numbers*, REACH manuscript, 10 January 2003, Theorem 3.1. Original manuscript.

[15] J. Propp, *The combinatorics of frieze patterns and Markoff numbers*, arXiv:math/0511633, first posted 25 November 2005; published in *Integers* **20** (2020), A12.

Unaffiliated, St. Louis, Missouri, USA

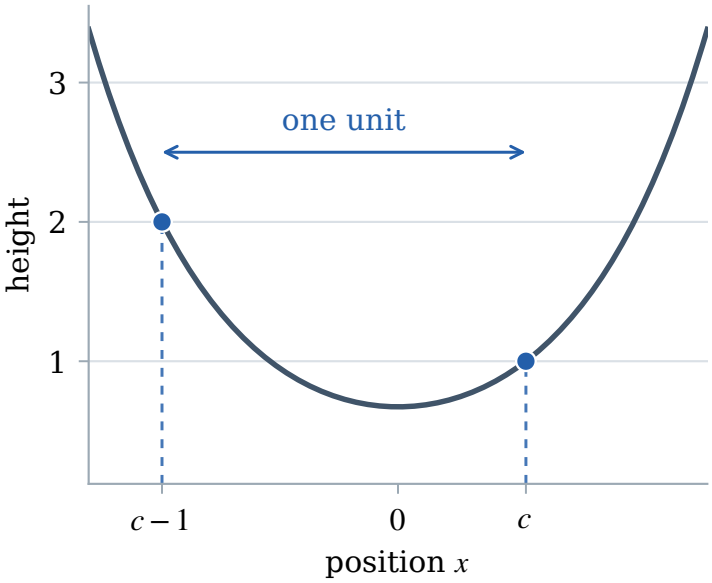
Email address: soldier@alum.mit.edu

URL: https://orcid.org/0009-0000-8831-0289

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One catenary profile, with host $h = 5$

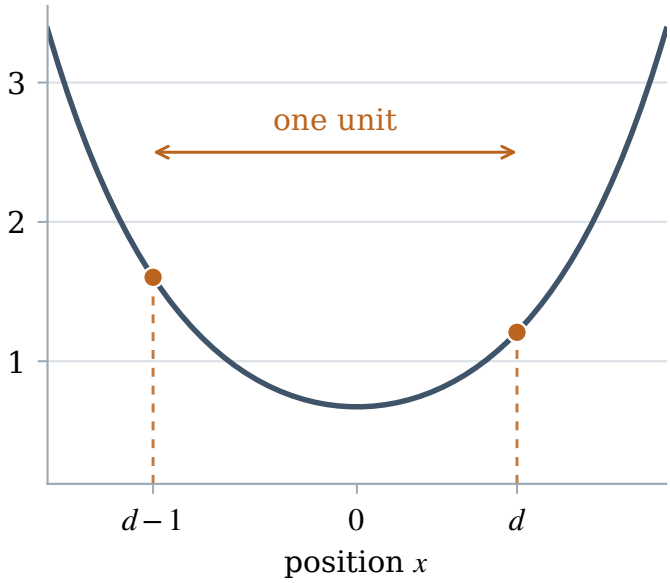
Integral choice: $c \approx 0.351515$



$f_5(c) = 1, \quad f_5(1-c) = 2$

integer triple $(1, 2, 5)$

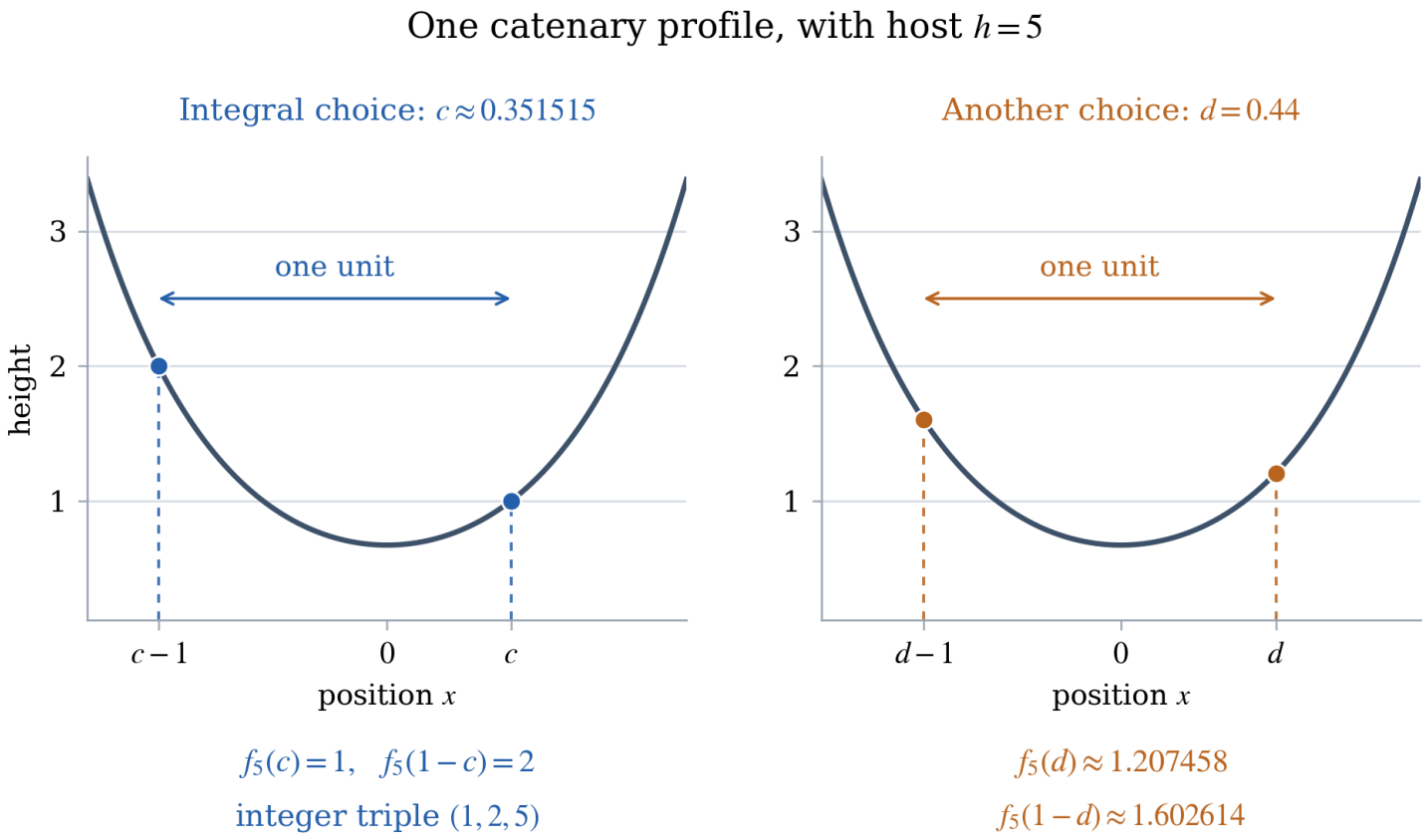
Another choice: $d = 0.44$



$f_5(d) \approx 1.207458$

$f_5(1-d) \approx 1.602614$

Original figure: v11_Markov_Meridians_source/figures/fig0_catenary_choices.png



First in Zenodo record 24084246 | SHA256 bb5651e65145ea10d01e8eba

Meridians on the Markov Surface: Simple Indexing for Fast, Low-Storage Verification of 550 Billion Configurations

Completion and release history

Manuscript dated 9 November 2026, version 11. The 500-billion certificate completed at 09:50:31.046707 CST (15:50:31.046707 UTC) that day. The public record preserves the historical 500-billion evidence.

Zenodo publication-date field: 2026-11-09

Zenodo record created: 2026-11-09 11:29:25 CST / 2026-11-09 17:29:25 UTC

Zenodo record last modified: 2026-11-09 11:29:25 CST / 2026-11-09 17:29:25 UTC

The publication-date field is the date supplied in the deposit. The API created/modified timestamps are repository metadata, not a separately attested first-view time. Full fractional timestamps are retained in the metadata archived in Part C.

https://zenodo.org/records/24089717

https://doi.org/10.5281/zenodo.24089717

Earlier published Zenodo drafts in this version family:

v4_ps | 2026-11-07 13:37:47 CST / 2026-11-07 19:37:47 UTC | https://zenodo.org/records/24043052

v5_ps | 2026-11-07 18:35:22 CST / 2026-11-08 00:35:22 UTC | https://zenodo.org/records/24046012

v7_ps | 2026-11-08 07:26:11 CST / 2026-11-08 13:26:11 UTC | https://zenodo.org/records/24053720

v9_ps | 2026-11-09 07:45:32 CST / 2026-11-09 13:45:32 UTC | https://zenodo.org/records/24084246

Earlier public announcements: 21 October 2026, 10:18:25 CDT / 15:18:25 UTC on X (more than a billion values in under 40 minutes); 23 October 2026, 09:54:57.771 CDT / 14:54:57.771 UTC on LinkedIn (9.3 billion in 24 minutes). These concern earlier computations, not the final trillion certificate.

https://x.com/webystar/status/2112926487091282350

https://www.linkedin.com/posts/elleard-felix-webster-heffern-joseph-by-confirmation-2953a3104_markovnumbers-activity-7519411048115003392-4NaJ

The dates and URLs are retained in the 7 November public-post evidence packet; the LinkedIn URL is also embedded in the published computational supplement. Additional November LinkedIn announcements and the older Meridian ResearchGate v13 URL could not be established in this compilation.

Complete public Zenodo description

Meridians organize positive integer Markov triples into families with compact addresses and explicit evaluation rules. This release presents simple indexing for fast, low-storage verification and retains the catenary-based introduction, geometric descent, residue patterns, collision windows, corridor recurrences, and Fibonacci-matrix and snake-graph arguments.

Completed verification: 550,693,634,649 configurations were compared, certifying uniqueness of the first 500,000,000,000 Markov values in numerical order. The larger count is the comparison workload, including possible competitors and conservative cutoff coverage; it is not a claim about the first 550 billion numerical values. All four repeated 64-bit residue groups, comprising eight exact candidates, were resolved with no equal maxima. The exact 500-billionth value was not identified.

Production began on 8 November 2026 at 14:55:01.580398 UTC. The last retained band completed on 9 November 2026 at 15:50:31.046707 UTC (09:50:31.046707 CST). On eight workers, retained active elapsed band processing took 54,470.884742 seconds: 15 h 07 min 50.885 s. Summed native-child CPU time was 264,075.377305 seconds (73.354 hours); parent CPU was not separately totalled. The calendar span was 24.925 hours. Active band time excludes pauses, restoration, lost attempts, and subsequent audits; the manuscript reports these scopes separately.

Maximum recorded temporary key-file storage was 754,846,136 bytes (719.877 MiB; 0.703 GiB), including the shard duplicated during partitioning. This excludes receipts, source files, backups, filesystem overhead, and unrelated work. Generator-only resident memory peaked at 28 MiB. A continuous whole-workspace disk or aggregate-memory peak was not recorded. The frozen recovery archive is 27,168,031 bytes.

The complete 13,638-band receipt chain and exact candidate evidence were audited. A separate checker recomputed the numerical envelope, independently reconstructed all candidate maxima, and regenerated every repeated-key band. The final

independent replay completed at 15:51:24.981257 UTC; a fresh portable extraction verified every manifested archive member and completed its replay at 15:57:38.529312 UTC on 9 November 2026. This is a receipt-and-candidate replay, not a second full enumeration of discarded singleton keys.

The separate degree-bounded certificate covers all 5,067,077 Meridians of degree at most 10,000 at every integer phase. The equal-degree separation result, the author's Fibonacci-matrix normalization and fixed-numerator arguments, and his six-piece fixed-sum and fixed-denominator arguments are retained. Earlier ordering results and the 2002-2003 origins of the matching interpretation are credited in the paper. The unrestricted Markov unicity problem remains open.

Version 11 removes the obsolete submission-status note and associated wording from the manuscript. Profiles and related derivations continue to use explicit powers, square roots, and logarithms. The original catenary figure and all formal mathematical statements are preserved.

Files include the revised manuscript, editable source with all figure assets and the frozen verifier archive, release notes, separately downloadable recovery archive, final verifier report, machine-readable final evidence, progress chart, historical six-page fifty-billion proof-and-timing supplement, and standalone catenary figure. The historical fifty-billion run completed on 28 October 2026; its separate original archive remains available at https://doi.org/10.5281/zenodo.23874075. Historical filenames retain the original 500-billion certification target.

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Funding: No funding. Substantial generative-AI assistance is disclosed in the manuscript; the author directed the research and is responsible for the work.

Published files

Original filename / checksum	Bytes
v4_50_Billion_Verification_Supplement.pdf MD5 6c966b341c1ec55b6b7bdc3a9dc50aaf SHA256 b477d344aff0971fee600ce1d087fc05f15509582bb43b22eb337180d6d3c446	310,901
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All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

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MERIDIANS ON THE MARKOV SURFACE: SIMPLE INDEXING FOR FAST, LOW-STORAGE VERIFICATION OF 550 BILLION CONFIGURATIONS

ELLEARD FELIX WEBSTER HEFFERN

Abstract. The Markov unicity problem asks whether the largest entry of a positive integer solution determines the other two. We organize the triples into families called Meridians, with compact addresses and explicit evaluation rules. The construction starts with catenary profiles: a fixed host sets the curve, and two heights one unit apart solve the Markov equation with that host. Selected positions give integer triples. Reversing geometric descent gives an exact indexing of these triples. Remainder patterns, growth bounds, corridor paths, and Fibonacci-matrix and snake-graph arguments then restrict possible collisions.

The indexing also supports a large verification with a small working store. Completed on 9 November 2026 at 15:50:31 UTC, the computation compared 550,693,634,649 configurations and certified uniqueness of the first 500 billion Markov values in numerical order. All four repeated 64-bit residue groups were resolved by exact arithmetic, with no equal maxima. On eight workers, retained active band processing took 15 h 07 min 50.885 s; the maximum recorded temporary key-file footprint was 754,846,136 bytes (0.703 GiB). This elapsed processing time is distinct from summed CPU time and the calendar span. A separate computer-checked certificate covers all 5,067,077 Meridians of degree at most 10,000, at every integer phase. The general unicity problem remains open.

1. Introduction

A Markov triple consists of three positive integers satisfying

$$a^2 + b^2 + M^2 = 3abM, \qquad 1 \le a \le b \le M. \tag{1}$$

For example, $(1, 2, 5)$ and $(1, 5, 13)$ are Markov triples. The largest entry of such a triple is a Markov number. The *unicity problem* asks a simple question: does that largest entry determine the other two? A counterexample would consist of two different ordered triples with the same largest entry. We call such an equality a *collision*. For the history and wider setting of the problem, see Aigner [6].

Our starting point is to let the entries vary continuously. We build a curve on which the Markov equation holds automatically, then ask which positions give integer entries. This separates the equation itself from the requirement that its solutions be integers. It also gives a picture of what a collision would have to look like.

A second aim is practical: give each triple a short address and a direct way to recover its entries. This lets us organize a large collection without keeping every full integer in memory. The completed computation reported here compares more than 550 billion configurations in small batches and certifies the first 500 billion values in numerical order.

1.1. The host and its catenary profile.

The *host* h is the fixed third entry around which we build the curve. It determines a profile f_h: two heights on this profile, taken one horizontal unit apart, will satisfy the Markov equation together with h. We first allow positive real entries and call

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2020 *Mathematics Subject Classification.* Primary 11D25; Secondary 11B39, 05C70, 11Y50.
Key words and phrases. Markov numbers, unicity, geometric descent, Meridians, corridors, snake graphs.

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any such solution a *real Markov state*. The host need not yet be an integer or the largest entry; those requirements will be imposed when we return to the unicity question.

For $h > 2/3$, define

$$q_h = \frac{3h}{2}, \qquad R_h = \sqrt{q_h^2 - 1}, \qquad Q_h = q_h + R_h, \tag{2}$$
$$f_h(x) = \frac{h}{2R_h}(Q_h^x + Q_h^{-x}).$$

The restriction $h > 2/3$ makes R_h positive and $Q_h > 1$; it includes every positive integer host. The graph of f_h is a rescaled catenary: a smooth, symmetric U-shaped curve, with its minimum at $x = 0$. In particular, $f_h(-x) = f_h(x)$. In (2), x is a free position on this profile. We use c below for a particular selected position.

The constants in (2) set the spacing and height scale so that one horizontal unit has the required arithmetic meaning. The calculation is short. First,

$$(q_h + R_h)(q_h - R_h) = q_h^2 - R_h^2 = 1,$$

so

$$Q_h^{-1} = q_h - R_h, \qquad Q_h + Q_h^{-1} = 3h, \qquad Q_h - Q_h^{-1} = 2R_h.$$

These three identities follow directly from the definitions.

1.2. **Why two heights one unit apart satisfy the equation.** Compare $f_h(x+1)$ with $Q_h f_h(x)$. The terms containing Q_h^{x+1} cancel, leaving

$$\begin{aligned} f_h(x+1) - Q_h f_h(x) &= \frac{h}{2R_h}(Q_h^{-x-1} - Q_h^{1-x}) \\ &= \frac{h}{2R_h} Q_h^{-x}(Q_h^{-1} - Q_h) = -hQ_h^{-x}. \end{aligned}$$

Multiplying $f_h(x)$ by Q_h^{-1} instead cancels the terms containing Q_h^{-x-1}:

$$\begin{aligned} f_h(x+1) - Q_h^{-1} f_h(x) &= \frac{h}{2R_h}(Q_h^{x+1} - Q_h^{x-1}) \\ &= \frac{h}{2R_h} Q_h^{x}(Q_h - Q_h^{-1}) = hQ_h^{x}. \end{aligned}$$

Now multiply these two equalities. The powers on the right cancel:

$$(f_h(x+1) - Q_h f_h(x))(f_h(x+1) - Q_h^{-1} f_h(x)) = -h^2.$$

Expanding the left side gives

$$f_h(x+1)^2 - (Q_h + Q_h^{-1}) f_h(x) f_h(x+1) + f_h(x)^2 = -h^2.$$

Finally, use $Q_h + Q_h^{-1} = 3h$ and move h^2 to the left:

$$\boxed{f_h(x)^2 + f_h(x+1)^2 + h^2 = 3f_h(x)f_h(x+1)h.} \tag{3}$$

Thus *every* real position x gives a real Markov state. No choice of an integer value or a special position was used in this calculation.

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1.3. **Choosing an integral state: the role of c.** To look for an ordered triple with largest entry h, select $0 < c \le 1/2$ and take the unit interval $[c-1, c]$ across the bottom of the curve. Its two heights are $f_h(c-1)$ and $f_h(c)$. By symmetry, $f_h(c-1) = f_h(1-c)$, so the state, with its companions in increasing order, is

$$(f_h(c),\ f_h(1-c),\ h).$$

It solves the equation by (3), with $x = c-1$. For it to be a normalized Markov triple, h and both heights must be positive integers, and $f_h(1-c) \le h$ must hold. The latter condition keeps the chosen host as the largest entry. Restricting c to this half of the unit interval avoids counting the reflected pair twice.

Figure 1 shows the distinction for $h = 5$. One choice, $c \approx 0.351515$, gives the heights 1 and 2, hence the integer triple $(1, 2, 5)$. A different choice, $d = 0.44$, gives heights approximately 1.207458 and 1.602614. It still solves the same equation with host 5, but its companion heights are not integers.

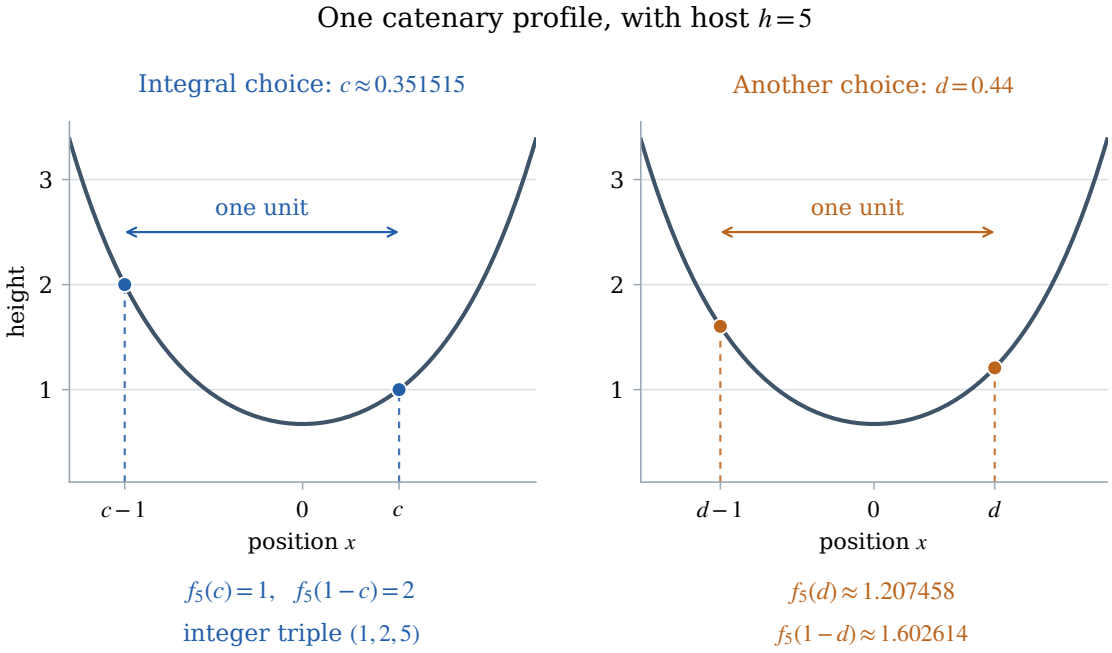


Figure 1. Two choices on the same catenary profile, with host $h = 5$. In each panel the marked contacts are exactly one horizontal unit apart. The left contact has height $f_5(1-c)$ or $f_5(1-d)$ by symmetry. The choice c gives an integer triple; the displayed choice d gives only a real state. A collision would require two distinct choices with both pairs of heights integral and with the same host as their largest entry. No collision is shown here.

This is the geometric form of the unicity question. For a fixed integer host h, could two distinct choices c and d in $(0, 1/2]$ both give integer companion heights no larger than h? If so, they would give different ordered triples with the same maximum. The curve makes these possible comparisons visible. To exclude them requires information about the integer values on the curve.

1.4. **How the paper organizes those comparisons.** The next step is *descent*. Starting from two contact heights, we look for a smaller host whose profile passes through the same heights one unit

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apart. We then move inward along that profile. Repeating these operations reduces a normalized integer Markov triple to $(1, 1, 1)$. Section 2 makes the procedure precise and explains how its final host and final position recognize the integer starting states.

Reversing the steps organizes the triples into families called *Meridians*. Each family has an address recording the steps used to construct it, and a phase specifying a position along it. This Meridian phase is a separate coordinate from the selected c on a fixed catenary. Integer Meridian phases pick out integer triples; the corresponding c need not be an integer, as Figure 1 shows. Section 3 proves that these addresses and integer phases recover every normalized triple exactly once. This is a statement about triples: whether two of them have the same largest entry remains the question to be answered. The same section introduces the alternative tree and the *phasors* that connect its Meridians.

Two kinds of information then restrict collisions. First, the remainders of the Markov values follow exact periodic patterns (Section 4). Second, their growth limits where two families could meet. A Meridian's *degree* measures the largest exponent in its formula. We prove that distinct integer marks of the same degree cannot collide. For different degrees, we bound the finite range of phases that needs comparison (Section 5). *Corridor walls* follow other paths through the same tree, using both Meridian and phasor steps. Their recurrence gives a second bound on possible collisions (Section 6).

Section 7 supplies the ordering proofs used in these comparisons. It represents Markov numbers by counts of perfect matchings on chains of square tiles, called snake graphs; a perfect matching pairs all the vertices using edges of the graph. The author's Fibonacci matrices and six-piece decomposition provide concrete ways to compare these counts. The graph construction and the needed notation are introduced in that section.

The resulting computer-checked certificate covers all 5,067,077 Meridians of degree at most 10,000, at every integer phase (Theorem 8.1). Distinct triples within that family have different maxima. This bounds the complexity of the families, while allowing their values to grow without limit; comparisons with higher-degree Meridians remain outside this certificate. Section 9 records the new, separate numerical-order verification. It compared 550,693,634,649 configurations and certified uniqueness of the first 500 billion values, including every possible competitor at those values. The run finished on 9 November 2026 at 15:50:31 UTC, using 15.131 hours of retained active band processing and at most 0.703 GiB of recorded temporary key files. The section gives the exact timing and storage scopes, together with the earlier fifty-billion result. The general unicity problem remains open within this framework.

As the later sections introduce their coordinates, $M_\alpha(x)$ will denote the value at Meridian address α and phase x, and $m_{p/q}$ the value of the triple indexed by the reduced fraction p/q. Both label triples before any assertion that their maxima are distinct. Analytical logarithms are natural; plotted horizontal axes use $\log_{10} M$.

2. Geometric descent and arithmetic closure

2.1. **Moving along a profile and changing its host.** The catenary profile (2) turns two elementary operations on Markov states into movements of the contacts. The first operation is a unit step along a fixed profile. Directly from its powers,

$$\begin{aligned} f_h(x+1) + f_h(x-1) &= \frac{h}{2R_h}(Q_h + Q_h^{-1})(Q_h^x + Q_h^{-x}) \\ &= 3h f_h(x). \end{aligned} \tag{4}$$

Thus the height beyond a pair is obtained by multiplying the nearer height by $3h$ and subtracting the farther height. This is the usual Vieta replacement of a companion entry, expressed on the curve.

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The second operation keeps the contact heights fixed and changes the host. For any real x, put $A = f_h(x)$ and $B = f_h(x+1)$. Their possible hosts are the roots of $X^2 - 3ABX + A^2 + B^2 = 0$. One root is h, so the other is

$$h, \qquad K = 3AB - h = \frac{A^2 + B^2}{h}. \tag{5}$$

For descent, select an edge straddling the minimum of the profile. We record its right endpoint as c, with $0 \le c < 1$, and its ordered contact heights as $A = f_h(c)$ and $B = f_h(c-1)$. This is the same pair used in the introduction, since $f_h(c-1) = f_h(1-c)$. We retain the full interval of c here to keep track of orientation. Seek a wider profile through these same two heights, still one unit apart, allowing a horizontal translation. Smaller h gives a wider profile. If the other host satisfies $K < h$, fit that profile and walk the unit lattice inward until the selected edge again straddles its vertex. Otherwise stop. Figure 2 shows one complete step. The fit is explicitly

$$v = \frac{\log((Q_K A - B)/K)}{\log Q_K}, \qquad f_K(v) = A, \quad f_K(v-1) = B. \tag{6}$$

Indeed, $Q_K f_K(v) - f_K(v-1) = K Q_K^v$, so taking logarithms recovers v directly. The Markov relation gives

$$(Q_K A - B)(B - Q_K^{-1} A) = K^2.$$

Both factors are positive: they have the same sign, and their sum is $2R_K A > 0$. This also fixes the real branch of the logarithm and verifies the fitted contact heights by direct substitution. The walk replaces v by $v - \lfloor v \rfloor$. Its orientation is retained: reflection exchanges the contacts and sends phase to its negative modulo one.

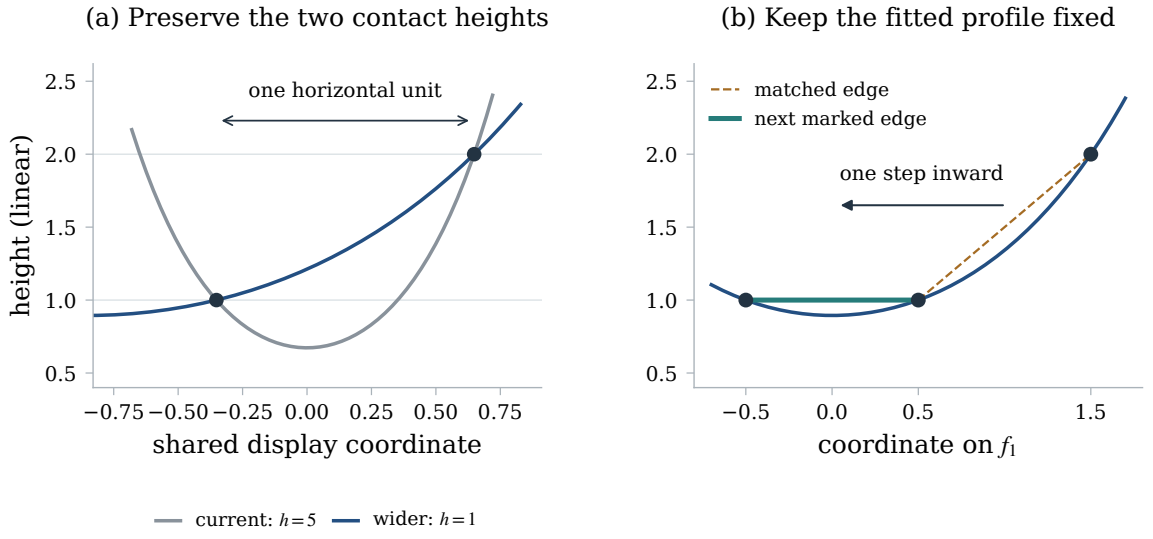


Figure 2. Geometric descent from $(1, 2, 5)$. The wider profile passes through the same unit-spaced contacts. On f_1 their coordinates are $1/2, 3/2$; one inward step selects $-1/2, 1/2$, with heights $1, 1$. The curve is fixed during the walk.

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Proposition 2.1 (Uniqueness of the fit)**.** *A marked edge has at most one distinct compatible profile, hence at most one wider fit. For fixed $0 < a < b$, a same-arm fit $f_h(u-1) = a$, $f_h(u) = b$ with $u > 1$ is unique if it exists.*

Proof. The hosts are the two roots of $X^2 - 3ABX + A^2 + B^2$. The other root belongs to the geometric domain because $(3K-2)AB = K^2 + (A-B)^2 > 0$. For a same-arm fit the horizontal distance between heights a, b is

$$\frac{1}{\log Q_h} \int_a^b \frac{dy}{\sqrt{y^2 - h^2/R_h^2}}.$$

When $a > 2/3$ and the lower height is in range, Q_h strictly increases and $h/R_h = 2h/\sqrt{9h^2-4}$ strictly decreases with h. Thus the prefactor and the integrand strictly decrease with h. The distance can therefore equal one at most once. The profile is strictly increasing on that arm, which also determines u. If $a \le 2/3$ there is no fit. □

2.2. **Termination.** Write $\mathcal{D}(M, c) = (H, \phi)$ for the terminal host and oriented phase.

Theorem 2.2 (Finite descent)**.** *Every straddling input with $M \ge 1$ terminates. On $1 \le M \le R$ the number of host changes is uniformly bounded. The terminal map is continuous into $(2/3, \infty) \times \mathbb{R}/\mathbb{Z}$ and analytic on cells with a fixed strict itinerary.*

Proof. Temporarily sort the endpoints as $a \le b$. Inward walking does not increase either endpoint. Every current host is at most R, and the discriminant in (3) gives

$$(9h^2 - 4)a^2 - 4h^2 = (2b - 3ha)^2 \ge 0.$$

Consequently the endpoints are at least $a_R = 2R/\sqrt{9R^2-4} > 2/3$. At an accepted change, $a^2 + b^2 < h^2$, so the endpoints are also below $h \le R$. Using either endpoint as the host in the same discriminant identity bounds the new host below by a_R. The initial host is at least $1 > a_R$.

If the new host K is at least b, its edge already straddles the vertex, because both endpoints are below $f_K(1)$; the next candidate is the old, larger host, so descent stops. Otherwise

$$\frac{h}{K} > \frac{h}{b} = 3a - \frac{K}{b} > 3a - 1 \ge 3a_R - 1 > 1.$$

All changes except possibly the last therefore contract the host by a fixed factor. An upper bound is $1 + \lceil \log(R/a_R)/\log(3a_R - 1) \rceil$.

At a boundary $K = h$, the fitted coordinate in (6) is the original c, so the accepted and stopped maps agree. At a phase cut, $K(h, 0) = h(9h^2+4)/(9h^2-4) > h$, and a neighborhood is stopped. The uniform bound on compositions then gives continuity, with phase taken modulo one. The strict formulas are analytic. □

2.3. **Which inputs close arithmetically?** The two reversible state operations are

$$T(a, b, h) = (3ha - b, a, h), \qquad J(a, b, h) = (a, b, 3ab - h). \tag{7}$$

They and their inverses are integer polynomial maps. A terminal state has the nonnegative defect

$$\begin{aligned} H f_H(\phi) f_H(\phi - 1) - 1 = {} & \frac{(H-1)^2(H+2)}{3H-2} \\ & + \frac{H^3}{4R_H^2} \big(Q_H^{\phi - 1/2} - Q_H^{1/2 - \phi}\big)^2. \end{aligned} \tag{8}$$

For $0 \le \phi < 1$ it vanishes exactly at $(H, \phi) = (1, 1/2)$.

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Proposition 2.3 (Arithmetic closure)**.** *A straddling input with $M \geq 1$ reaches $(1, 1/2)$ if and only if its three entries are positive integers. Modulo companion exchange, these inputs are exactly the normalized configurations* (1)*.*

Proof. Expanding the two profiles gives

$$Hf_H(\phi)f_H(\phi-1) - 1 = \frac{H^3}{4R_H^2}(Q_H^{2\phi-1} + Q_H^{1-2\phi} + 3H) - 1.$$

Replace the first two powers by $\left(Q_H^{\phi-1/2} - Q_H^{1/2-\phi}\right)^2 + 2$. The constant term is $H^3/(3H-2) - 1 = (H-1)^2(H+2)/(3H-2)$, which proves (8). Reversing a finite closure itinerary from $(1,1,1)$ by (7) preserves integrality. Conversely, integral descent stays integral. At an integral terminal state order the entries as $u \leq v \leq w$. There is no coordinate-decreasing mutation. But for $Q(X) = X^2 - 3uvX + u^2 + v^2$,

$$Q(v) = u^2 + (2-3u)v^2 \leq 3u^2(1-u) \leq 0.$$

If neither root is below $w \geq v$, equality is forced, giving $u = v = 1$; terminality then gives $w = 1$. Finally, a straddling edge has no decreasing endpoint mutation, so its host is a largest entry. Its ordered split is uniquely recovered by (6). □

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Both terminal coordinates matter. Figure 3 shows inputs with terminal host one but different terminal phases. A crossing of the height level $H = 1$ alone does not certify integrality. The marked configurations follow the host itineraries $37666 \to 5 \to 1$, $7561 \to 5 \to 1$, and $14701 \to 2 \to 1$, all ending at phase $1/2$.

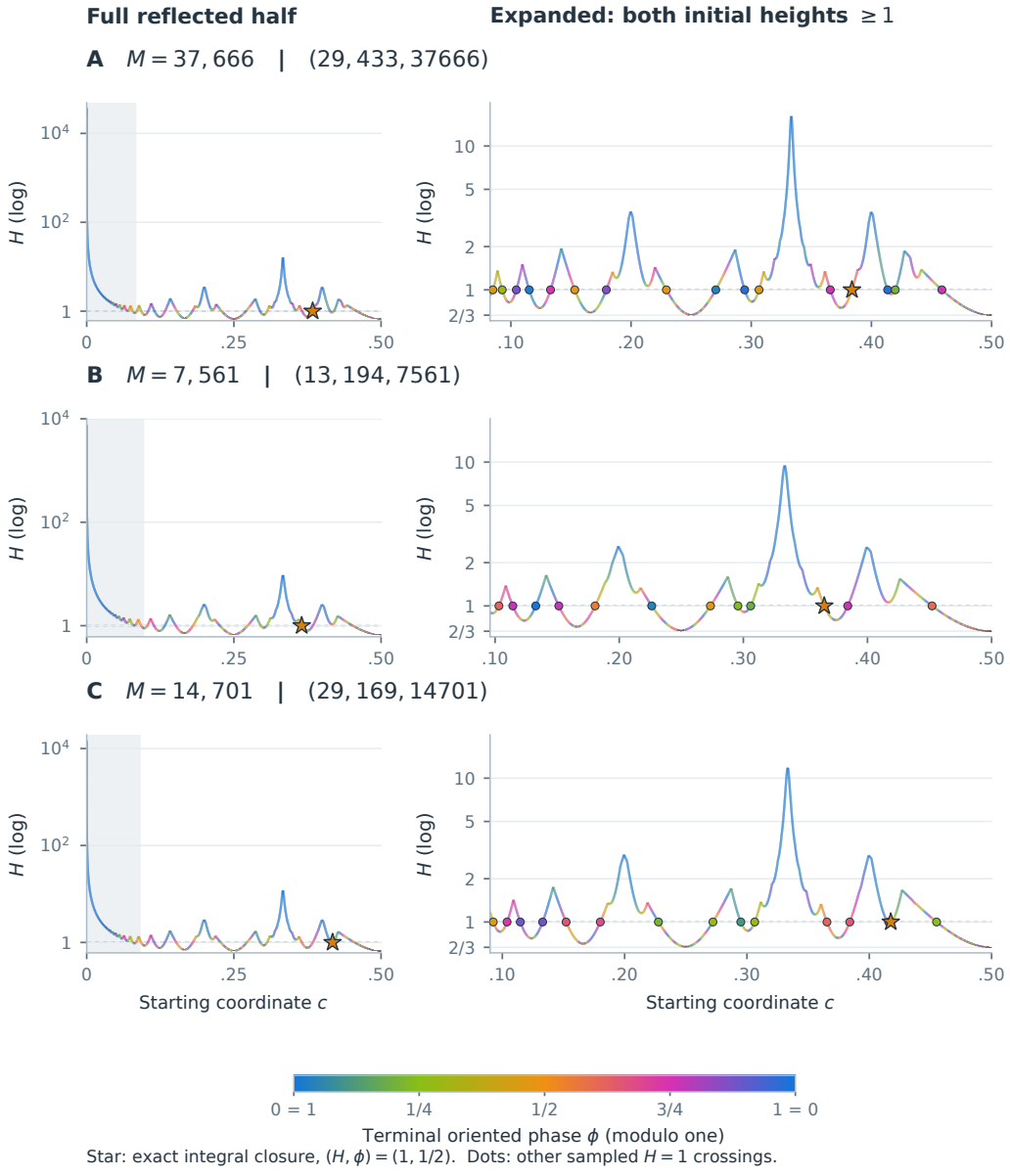


Figure 3. Terminal host and phase for $M = 37666, 7561, 14701$. Height is H on a logarithmic scale and hue is oriented phase. Left: $0 \le c \le 1/2$, with gray where an initial endpoint is below one. Right: the admissible portion enlarged. Stars mark the exact integral starts; dots are other numerically located $H = 1$ crossings. These numerical plots illustrate the closure criterion.

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3. Meridians, addresses, and the alternative tree

3.1. **The landscape and its two coordinates.** On the landscape $(\log_{10} M, c)$, the point (M, c) represents the contacts $f_M(c), f_M(c-1)$. We draw the reflected half $0 < c \le 1/2$; exchanging the contacts sends c to $1-c$. A *Vieta arc* keeps one initial contact height fixed. For example, the branch with height a has

$$c = \frac{\log\Big((R_M a + \sqrt{R_M^2 a^2 - M^2})/M\Big)}{\log Q_M}.$$

This follows by solving $Q_M^c + Q_M^{-c} = 2R_M a/M$ for $Q_M^c \ge 1$ and taking logarithms. The square root is real exactly when $a \ge M/R_M$, the minimum height of the profile; reflection supplies the other arm. A *Vieta trek* follows such arcs, changing the retained contact when necessary. A *Meridian* is a connected component of terminal host $H(M, c) = 1$. A *phasor* holds the terminal phase ϕ fixed. The alternative tree uses phasors at $\phi = 1/2$ to join successive Meridians.

The address phase x on a Meridian is real and unrolled. Its terminal circle phase is $\phi \equiv x + 1/2 \pmod 1$. Every integer mark therefore closes arithmetically; the distinguished arrival mark $x = 0$ is called the *helm*. It need not be the minimum of the real profile.

Figure 4 keeps the same configurations in three views. The alternative tree is in the middle panel. The bottom panel uses smooth schematic connectors for the Vieta parent relation: node coordinates and adjacency are exact, but those connectors are not literal level curves. The two trees each have 55 edges. An unmarked crossing is not a vertex.

For reproducibility, the field in panel (a) uses the fixed cyclic phase palette with HSL saturation and lightness

$$w(H) = \frac{1}{1 + (\ln H/\ln 1.02)^2}, \qquad S = w(H), \qquad L = 1 - 23w(H)/40.$$

The background is a numerical illustration, independent of all exact enumerations and exclusions below.

3.2. **Evaluating an address.** The seed Meridian is

$$\begin{aligned} a_1(x) = f_1(x + \tfrac{1}{2}), \qquad b_1(x) = f_1(x - \tfrac{1}{2}), \\ M_1(x) = 3a_1(x)b_1(x) - 1. \end{aligned} \tag{9}$$

For integer x, its companions are F_{2x+1}, F_{2x-1}, where $F_{-j} = (-1)^{j+1}F_j$. Hence

$$M_1(x) = 2 + 3F_{2x}^2, \qquad M_1(0), M_1(1), M_1(2), \ldots = 2, 5, 29, 194, \ldots. \tag{10}$$

From any ordered state (a, b, H) form the two-sided orbit

$$u_0 = a, \quad u_1 = b, \qquad u_{j+1} = 3Hu_j - u_{j-1}.$$

A nonzero index k replaces the state by

$$(u_k, u_{k+1}, 3u_k u_{k+1} - H). \tag{11}$$

The address is written

$$\alpha = (1, k_m, \ldots, k_1): \quad k_1 \text{ is applied first and written at the right.}$$

This is the reversed indexing convention: the former $M_{1,a,b,c}(x)$ is now $M_{1,c,b,a}(x)$. Neither the signs nor x are reversed. In terms of (7), with $S = T^{-1}$ and $\mathcal{W}_k = JS^k$,

$$(a_\alpha(x), b_\alpha(x), M_\alpha(x)) = \mathcal{W}_{k_m} \cdots \mathcal{W}_{k_1} J T^x (1, 1, 1). \tag{12}$$

Composition acts from right to left. A recursive extension inserts its index immediately after the leading 1. For example, $M_{1,1}(0) = 13$. Index zero is forbidden because (11) would return the preceding pole $3ab - H$, rather than create a new component.

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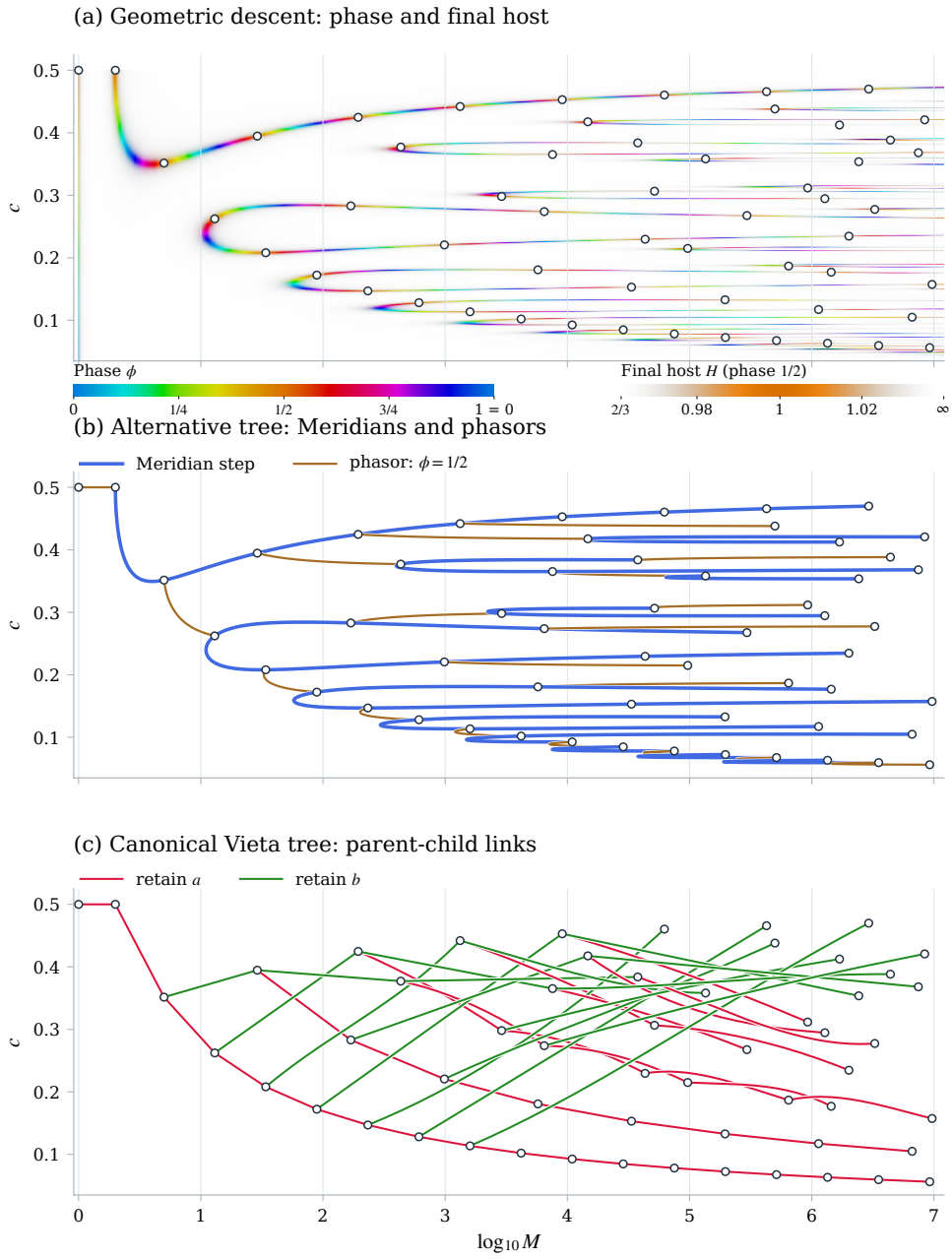


Figure 4. The same 56 configurations with $M \leq 10^7$. (a) Numerical geometric descent, with phase encoded by hue and host by color intensity. (b) The alternative tree, with Meridian and phasor edges. (c) The canonical Vieta tree, with simplified connectors retaining exact endpoints. Red retains the smaller companion, green the larger. Every edge runs toward larger M. Crossings without a node mark are not vertices.

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Definition 3.1 (Canonical marks)**.** The root $(1,1,1)$ is separate. For the seed (1) use integer phases $x \geq 0$. For every other address require $k_1 > 0$, $k_i \neq 0$ for $i > 1$, and $x \in \mathbb{Z}$. Evaluation is followed by sorting the two companions.

Theorem 3.2 (Configuration bijection)**.** *Canonical marks are in bijection with all normalized positive Markov triples except the root. The displayed value $M_\alpha(x)$ is the largest entry.*

Proof. The quadratic argument in Proposition 2.3 shows that replacing the largest coordinate by its other pole gives a smaller maximum, apart from the initial repeated entries. This is the forced Vieta reduction of a configuration and does not assume scalar unicity.

To recover a word, reflect the current maximum H to $P = 3ab - H$. When $P \geq 2$, walk its oriented endpoint orbit inward until both endpoints are below P. This edge is unique. Indeed $u_{j+1} - 2u_j + u_{j-1} = (3P-2)u_j > 0$, so the entries below P form an interval; three consecutive entries below P would contradict $u_{j-1} + u_{j+1} = 3Pu_j \geq 3P$. Endpoint descent supplies the edge, and its nonzero signed displacement recovers the last-applied, leftmost index. Repeat at the smaller pole. At pole one, the orbit reaches $(1,1)$ and fixes the seed phase.

Conversely, all outward blocks are positive and reduced. If the preceding pole is $P < H$, the first reflected poles on the two sides are $(9b^2-2)H - P > 6H$ and $(9a^2-2)H - P > 6H$. Their farther endpoints exceed $2H$, and the recurrence makes subsequent outward entries increase. The reduction therefore undoes every prescribed block. Finally, companion exchange negates all indices and x simultaneously. The canonical conditions select one of the two orientations; only $(1,1,2)$ is fixed. □

This bijection is useful computationally as well as geometrically. A configuration can be retained by its address and reconstructed when a comparison needs its full integer entries. Section 9 uses the equivalent reduced-fraction labels as compact identities, with compressed Vieta evaluation. Its coverage therefore counts triples before testing whether their maxima are distinct.

3.3. **The two parent rules.** At a nonzero phase the alternative parent is the adjacent inward mark, $(\alpha; x) \mapsto (\alpha; x - \operatorname{sgn} x)$. At a nonseed helm it is

$$(1, k_m, \ldots, k_2, k_1; 0) \longmapsto (1, k_m, \ldots, k_2; -k_1), \tag{13}$$

followed by simultaneous sign reflection if the remaining rightmost index is negative. An empty tail uses the seed's nonnegative phase. The seed helm with maximum two has parent $(1,1,1)$.

The alternative parent deletes the *rightmost* index at a helm. Recursive evaluation removes the *leftmost* index after 1. These operations serve different purposes. The depth

$$d(\alpha; x) = 1 + \sum_{i=1}^{m} (|k_i| + 1) + |x| \tag{14}$$

decreases by one under either parent edge, with root depth zero. Thus the parent rule defines a rooted tree on the same configurations. The ordinary outward Vieta moves instead are

$$(a, b, M) \longmapsto (a, M, 3aM - b), \qquad (a, b, M) \longmapsto (b, M, 3bM - a). \tag{15}$$

For example, $29 \to 194$ is a Meridian edge, whereas $29 \to 169$ is a Vieta edge. Figure 5 compares the two sets of links.

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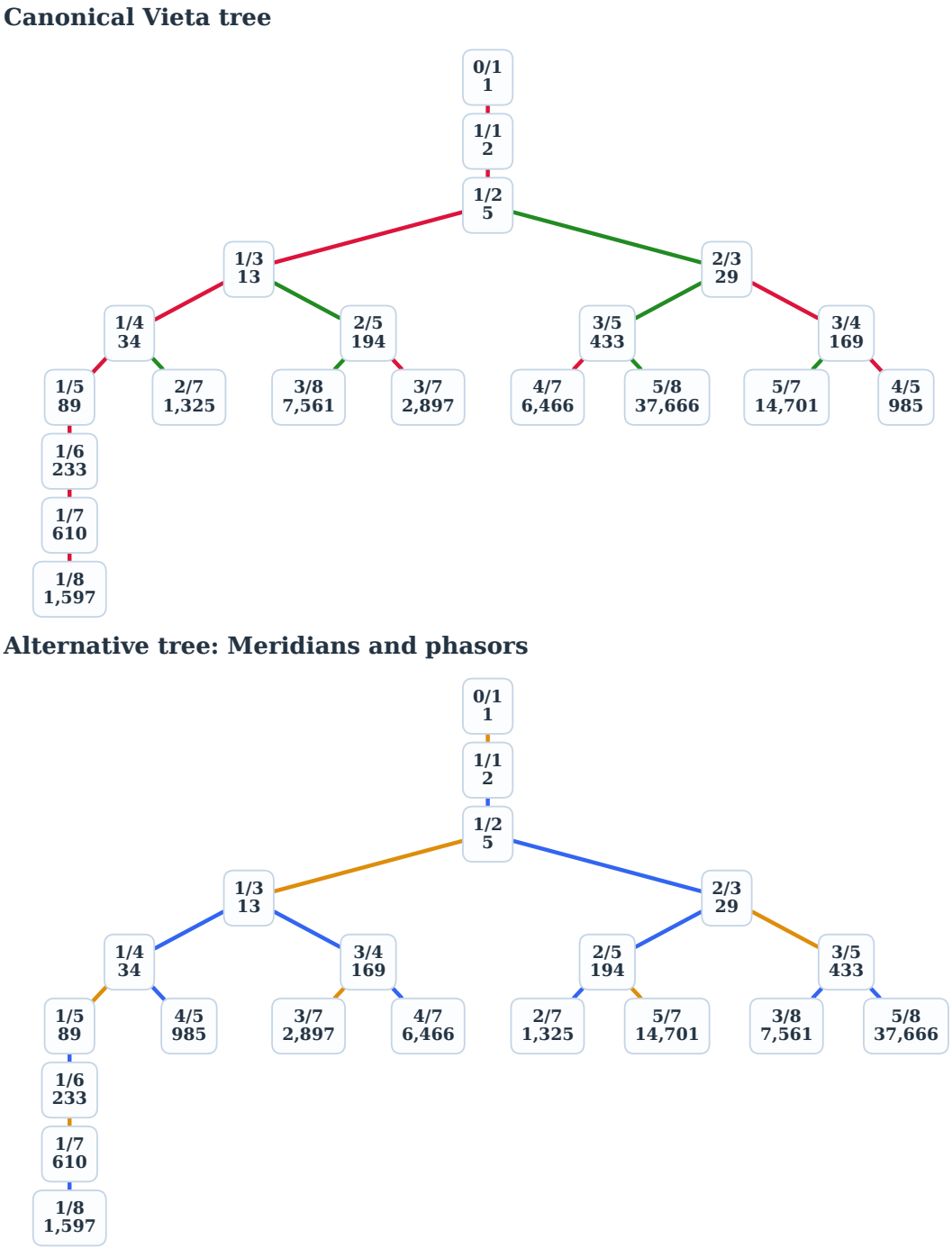


Figure 5. Two trees on the same 20 configurations: the 16 with $M \leq 3000$ and the four additional values $6466, 7561, 14701, 37666$. Each node displays its reduced standard Farey label and maximum. Red and green denote the Vieta moves; blue and gold denote Meridian and phasor steps.

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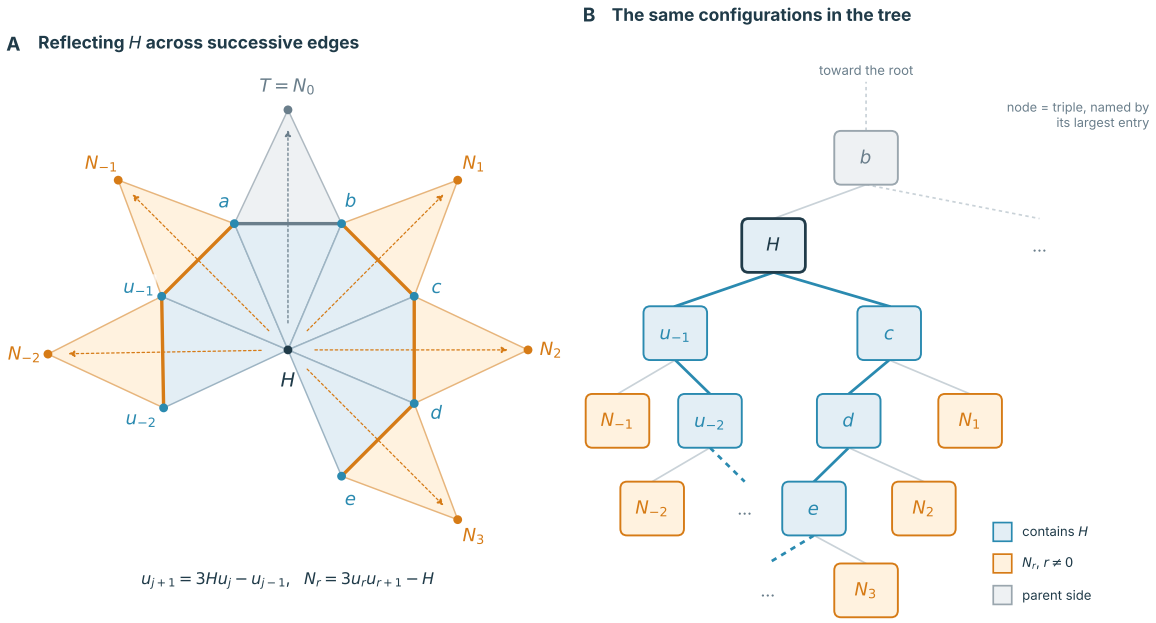


Figure 6. The orbit underlying one recursive extension. Reflecting H across (u_j, u_{j+1}) gives $N_j = 3u_j u_{j+1} - H$; N_0 is the preceding pole. For source $(a, b, H) = (1, 5, 13)$ at $M_{1,1}(0)$, the preceding pole is two, $(u_{-2}, u_{-1}, u_2, u_3, u_4) = (1325, 34, 194, 7561, 294685)$, and $(N_{-2}, N_{-1}, N_1, N_2, N_3) = (135137, 89, 2897, 4400489, 6684339842)$. The right panel locates these reflections in the Vieta tree.

3.4. **Real components and phasor edges.**

Proposition 3.3. *The canonical real families in* (12)*, modulo companion reflection, are the nonterminal connected components of* $H(M, c) = 1$*. Their terminal phase is* $x + 1/2$ *modulo one. The separate terminal component is* $M = 1$*.*

Proof. The seed companions are at least $2/\sqrt{5}$, and $M_1 \geq 2$ is strictly larger than either. More generally, let $h \geq 1$ and $2/\sqrt{5} \leq a, b \leq h$. The first outward endpoint satisfies $3hb - a \geq (6/\sqrt{5} - 1)h > h$; its reflected pole is larger still. The reversed orbit gives the same conclusion on the other side. Every outward block thus has a strict inverse descent to its source and preserves the terminal phase.

Conversely, reverse any descent ending at host one. Its last host change produces the seed with $x = v - 1/2$, and every preceding walk is nonzero: a zero walk would simply have stopped at the smaller host. These strict blocks uniquely recover a locally constant word. The unrolled x joins the final phase charts continuously. The seed is injective as an ordered curve, as its endpoint difference is strictly monotone; each block is invertible. The positive Laurent expansion proved in Proposition 5.2 makes every family proper. The inverse charts are relatively open in the level set, and properness makes the families closed. Each connected family therefore fills a component. Reflection gives exactly the canonical identifications. □

The phasor edges are obtained by applying the same outward blocks to

$$\left(\frac{h}{\sqrt{3h-2}}, \frac{h}{\sqrt{3h-2}}, h \right), \qquad 1 \leq h \leq 2. \tag{16}$$

This symmetric arc joins $(1, 1, 1)$ to $(1, 1, 2)$. Its other host is $2h/(3h-2)$, and both possible terminal fits have phase $1/2$. Its endpoints satisfy the bounds used in the proposition, so the outward blocks

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preserve that phase. The displayed phasors consequently arise from the descent itself, rather than interpolation between integer nodes.

3.5. **Degree and the exact Farey dictionary.** Attach endpoint count vectors $u = (1, 0)$ and $v = (0, 1)$ to the seed. Under an index k update them by

$$(u, v) \longmapsto \begin{cases} ((k-1)(u+v)+v,\ k(u+v)+v), & k > 0, \\ (|k|(u+v)+u,\ (|k|-1)(u+v)+u), & k < 0. \end{cases} \tag{17}$$

Write their sum as (r, s). Their determinant remains ± 1 and their unordered residues modulo two remain $\{(1, 0), (0, 1)\}$. Thus r, s are positive coprime odd integers. Canonical orientation gives $r \le s$. The *Meridian degree* is $D = r + s$. It is the extreme Laurent exponent in ρ^x, where $\rho = (3 + \sqrt{5})/2$; the degree in ρ^{2x} is $D/2$.

Proposition 3.4 (Helm classification)**.** *Canonical Meridians correspond bijectively to coprime positive odd pairs $r \le s$. Their helm labels are the standard Farey fractions r/s.*

Proof. The endpoint vectors are the two Stern–Brocot parents: their sum is (r, s) and their determinant has absolute value one. Conversely, the parents of an odd/odd pair have opposite coordinate parities. Their componentwise absolute difference is a smaller positive primitive odd/odd pair, except at $(1, 1)$. The difference has one componentwise sign; an opposite sign would give the initial pair. Iteration reaches $(1, 1)$.

To recover the signed indices, put $n_\alpha = (r, s)$ and let α^- remove the leftmost index after 1. Consecutive difference vectors have determinant of absolute value two, so three consecutive vectors satisfy $n = 2c\, n^- + \varepsilon n^{--}$ with $c \ge 1$ and $\varepsilon = \pm 1$. Determinant division gives an integer coefficient, parity makes it even, and decreasing coordinate sums fix its magnitude and sign. At the bottom use $(1, 1), (-1, 1)$; canonical orientation fixes the first index positive. The magnitude is fixed by the ratio of successive degrees, which lies in $[2c, 2c+1)$ or $(2c-1, 2c]$ according to the sign; equality occurs only at the first applied index. Reversing the process gives precisely $n_{1,k,k_m,\ldots,k_1} = 2|k| n_\alpha + \operatorname{sgn}(k k_m) n_{\alpha^-}$. For the seed use $k_0 = 1$ and $n_\varnothing = (-1, 1)$, so the first extension has $(r, s) = (2k_1 - 1, 2k_1 + 1)$. This recovers the unique address. Mediant addition identifies the label. □

Lemma 3.5 (Phase dictionary)**.** *For helm (r, s) and degree $D = r + s$, the standard Farey label at phase x is*

x	(p, q)	$p + q$
0	(r, s)	D
1	(r, D)	$D + r$
-1	(s, D)	$D + s$
$n \ge 2$	$(D, (n-1)D + r)$	$nD + r$
$-n \le -2$	$(D, (n-1)D + s)$	$nD + s$.

The negative seed phases are omitted.

Proof. At seed phases $0, 1, n \ge 2$, the endpoint bases are respectively $((1, 0), (0, 1))$, $((1, 1), (0, 1))$, and $((1, n), (1, n-1))$. The count updates are linear, so substitution gives the table; negative phase exchanges the bases. For the inverse, odd/odd p, q give $(r, s, x) = (p, q, 0)$. If q is even, sort $p, q - p$ and choose $+1$ when p is the smaller entry, otherwise -1. If p is even, write $q = tp + u$ with $0 < u < p$; sort $u, p - u$ and choose $+(t + 1)$ or $-(t + 1)$ in the same way. A tie is the seed and uses the nonnegative representative. These formulas recover every mark. □

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In particular, degree can be read directly from a reduced label:

$$D = \begin{cases} p+q, & p,q \text{ odd}, \\ q, & q \text{ even}, \\ p, & p \text{ even}. \end{cases} \tag{18}$$

This dictionary connects the Meridian and corridor collision windows below.

4. Meridian residues: modulo 27 and Fibonacci moduli

4.1. **Transferring a seed congruence to every address.** The recurrence for the companions contains no division, but a seed period of the *maximum* need not visibly preserve both companions. The following four-entry record resolves that issue:

$$\mathcal{S} = (A, B, C, H) = (9a^2, 9b^2, 3ab, H).$$

Forward and backward orbit steps and pole reflection are the polynomial maps

$$\begin{aligned} \mathcal{R}(A,B,C,H) &= (B, A + 9H^2B - 18HC, HB - C, H), \\ \mathcal{R}^{-1}(A,B,C,H) &= (B + 9H^2A - 18HC, A, HA - C, H), \\ \mathcal{J}(A,B,C,H) &= (A, B, C, C - H). \end{aligned} \tag{19}$$

For $h(x) = M_1(x)$ the seed record is

$$\mathcal{S}_1(x) = (8h(x) - h(x-1) - 2,\ h(x) + h(x-1) + 2,\ h(x) + 1,\ h(x)).$$

Every congruence preserving two consecutive seed maxima therefore propagates along every signed address.

Theorem 4.1 (Largest modulus with a common phase period)**.** *Every canonical Meridian has phase period T modulo m if and only if*

$$m \mid q(T), \qquad q(T) = 3F_{2T}F_{\gcd(2T,4)} = \begin{cases} 3F_{2T}, & T \text{ odd}, \\ 9F_{2T}, & T \text{ even}. \end{cases} \tag{20}$$

In particular, s is a common period modulo F_{2s}.

Proof. Fibonacci addition and (10) give

$$h(x+T) - h(x) = 3F_{2T}F_{4x+2T}.$$

Fibonacci strong divisibility, $\gcd(F_a, F_b) = F_{\gcd(a,b)}$, follows by the Euclidean algorithm from the addition formula. The gcd of all displayed differences is already the gcd of those at $x = 0, 1$, namely $q(T)$. Thus the seed has period T modulo m exactly when $m \mid q(T)$. The record (19) proves sufficiency at every address; the seed itself proves necessity. □

The theorem specifies a period common to all Meridians. It does not assert that the period is the smallest for each individual sequence. Nor does a large period exclude a scalar residue match at two earlier phases. For powers of three it gives

$$M_\alpha(x + 2 \cdot 3^n) \equiv M_\alpha(x) \pmod{3^{n+3}}, \qquad n \geq 0. \tag{21}$$

Indeed, $F_{3j} = F_j(5F_j^2 + 3)$ for even j, beginning with $F_4 = 3$, gives $v_3(F_{4\cdot 3^n}) = n + 1$. Figure 7 shows the common periods at the requested even-indexed Fibonacci moduli.

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x	M1							M1.1							M1.2							M1.1.1							M1.-1.1						
s	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8
F_{2s}	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987
-12								1	5	13	31	13	231	181	1	1	13	50	1	88	937	2	1	20	50	17	328	944	2	1	5	9	89	285	215
-11								1	2	13	4	34	283	559	1	1	1	46	73	21	715	2	1	2	13	89	94	170	2	1	2	21	89	333	107
-10								1	1	19	13	121	248	544	1	2	16	48	82	270	541	2	2	20	37	98	29	482	2	5	8	34	53	272	869
-9								1	5	1	34	61	57	169	1	1	13	26	1	291	160	2	1	2	51	89	343	653	2	1	8	13	17	241	806
-8								1	2	13	50	130	169	13	1	1	13	53	73	343	433	2	1	20	31	17	34	923	2	1	5	21	17	86	89
-7								1	1	13	31	25	13	34	1	2	1	50	82	56	652	2	2	2	50	98	258	317	2	5	2	9	125	89	233
-6								1	5	19	4	13	34	985	1	1	16	46	1	21	541	2	1	20	13	17	369	104	2	1	8	21	89	233	890
-5								1	2	1	13	34	231	820	1	1	13	48	73	88	580	2	1	2	37	89	328	758	2	1	8	34	89	285	71
-4								1	1	13	34	121	283	181	1	2	13	26	82	21	937	2	2	20	51	98	94	944	2	5	5	13	53	333	215
-3								1	5	13	50	61	248	559	1	1	1	53	1	270	715	2	1	2	31	89	29	170	2	1	2	21	17	272	107
-2								1	2	19	31	130	57	544	1	1	16	50	73	291	541	2	1	20	50	17	343	482	2	1	8	9	17	241	869
-1								1	1	1	4	25	169	169	1	2	13	46	82	343	160	2	2	2	13	98	34	653	2	5	8	21	125	86	806
0	2	2	2	2	2	2	2	1	5	13	13	13	13	13	1	1	13	48	1	56	433	2	1	20	37	17	258	923	2	1	5	34	89	89	89
1	2	5	5	5	5	5	5	1	2	13	34	34	34	34	1	1	1	26	73	21	652	2	1	2	51	89	369	317	2	1	2	13	89	233	233
2	2	5	8	29	29	29	29	1	1	19	50	121	231	985	1	2	16	53	82	88	541	2	2	20	31	98	328	104	2	5	8	21	53	285	890
3	2	2	5	29	50	194	194	1	5	1	31	61	283	820	1	1	13	50	1	21	580	2	1	2	50	89	94	758	2	1	8	9	17	333	71
4	2	5	2	5	29	194	338	1	2	13	4	130	248	181	1	1	13	46	73	270	937	2	1	20	13	17	29	944	2	1	5	21	17	272	215
5	2	5	5	2	5	29	194	1	1	13	13	25	57	559	1	2	1	48	82	291	715	2	2	2	37	98	343	170	2	5	2	34	125	241	107
6	2	2	8	5	2	5	29	1	5	19	34	13	169	544	1	1	16	26	1	343	541	2	1	20	51	17	34	482	2	1	8	13	89	86	869
7	2	5	5	29	5	2	5	1	2	1	50	34	13	169	1	1	13	53	73	56	160	2	1	2	31	89	258	653	2	1	8	21	89	89	806
8	2	5	2	29	29	5	2	1	1	13	31	121	34	13	1	2	13	50	82	21	433	2	2	20	50	98	369	923	2	5	5	9	53	233	89
9	2	2	5	5	50	29	5	1	5	13	4	61	231	34	1	1	1	46	1	88	652	2	1	2	13	89	328	317	2	1	2	21	17	285	233
10	2	5	8	2	29	194	29	1	2	19	13	130	283	985	1	1	16	48	73	21	541	2	1	20	37	17	94	104	2	1	8	34	17	333	890
11	2	5	5	5	5	194	194	1	1	1	34	25	248	820	1	2	13	26	82	270	580	2	2	2	51	98	29	758	2	5	8	13	125	272	71
12	2	2	2	29	2	29	338	1	5	13	50	13	57	181	1	1	13	53	1	291	937	2	1	20	31	17	343	944	2	1	5	21	89	241	215

Figure 7. Five Meridians modulo F_{2s} for $s = 2, \ldots, 8$: the moduli are $3, 8, 21, 55, 144, 377, 987$. Each column has common phase period s, possibly a multiple of an individual least period. The shaded row is the helm; blank cells omit the seed's negative phases. The displayed residues use the reversed Meridian indexing of this edition.

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4.2. **Why modulo 27 gives only ten classes.** No positive Markov entry is divisible by three: mutation negates the changed coordinate modulo three and the integral tree starts at $(1,1,1)$. Let t,h be preceding and current poles. An extension by a nonzero index has the particularly simple action

$$S(t,h)=(h,t)\quad (k \text{ even}),\qquad R(t,h)=(h,7h-t)\quad (k \text{ odd})\quad (\mathrm{mod}\ 27). \tag{22}$$

To see this, put $N_j=3u_ju_{j+1}-h$. Direct use of the orbit recurrence gives

$$N_0=t,\quad N_1=(9u_1^2-2)h-t,\quad N_{j+2}-N_j=9h(u_{j+2}^2-u_{j+1}^2).$$

All orbit squares are one modulo three, so $N_{j+2}\equiv N_j$ and $N_1\equiv 7h-t \pmod{27}$. This works for negative indices as well. Only the parity of each recursively applied index survives.

Proposition 4.2 (Universal modulo-27 classes)**.** *At integer phases every Meridian alternates between the two residues of one of the following sets, allowing the two singleton cases:*

$$\{1\},\ \{2,5\},\ \{4,16\},\ \{7,13\},\ \{8,17\},\ \{10,19\},\ \{11,23\},\ \{14,20\},\ \{22,25\},\ \{26\}.$$

The ten sets are pairwise disjoint. Meridians in different classes cannot share an integer value, regardless of degree or phase.

Proof. The common phase period is two because $q(2)=27$. Track preceding and current poles at both phase parities, starting from $(1,2,1,5)$. The simultaneous maps satisfy $S^2=I$ and $SRS=R^{-1}$. Successive current-pole pairs on the R orbit are

$$\begin{array}{cccccc}(2,5) & (13,7) & (8,17) & (16,4) & (23,11) & (10,19)\\ (20,14) & (22,25) & (26,26) & (25,22) & (14,20) & (19,10)\\ (11,23) & (4,16) & (17,8) & (7,13) & (5,2) & (1,1).\end{array}$$

The preceding pair is the previous pair cyclically, and $b_{j+1}=7b_j-b_{j-1}$ checks the return in 18 steps. These states and their S images form a closed set of 36 states. Their current-pole images are exactly the ten listed sets. Congruence of integer values would require membership in the same set. □

Table 1. The ten universal classes and their populations among the 519 Meridians of degree at most 100. Only the population column is bounded-degree data.

Residue set	Population	Residue set	Population
$\{1\}$	42	$\{10,19\}$	47
$\{2,5\}$	88	$\{11,23\}$	47
$\{4,16\}$	61	$\{14,20\}$	38
$\{7,13\}$	78	$\{22,25\}$	35
$\{8,17\}$	65	$\{26\}$	18

At that degree bound, the classification leaves 15,215 of the 134,421 pairs for further comparison. Larger Fibonacci moduli and compatible phases, rather than extrapolation from this table, supply the degree-10,000 certificate in Section 8.

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5. Equal-degree separation and unequal-degree windows

5.1. **A complete result at each fixed degree.** We first use the established fixed-sum, fixed-numerator and fixed-denominator orderings of Markov numbers. Section 7 proves them by matching counts and records their earlier sources.

Theorem 5.1 (Equal-degree unicity)**.** *Distinct canonical integer-phase marks of the same Meridian degree have different maxima. The marks may belong to the same or to different Meridians.*

Proof. For $D \geq 4$, Lemma 3.5 partitions the labels into three classes. Helms r/s have fixed sum D; near-helms $r/D, s/D$ have fixed denominator D; outer phases have fixed numerator D. Each class is strictly ordered by its corresponding monotonicity, and the dictionary ensures that different marks have different labels.

The largest helm is $m_{1/(D-1)}$. The near-helm extrema are $m_{1/D}, m_{(D-1)/D}$; the smallest outer value is $m_{D/(D+1)}$. They are separated by

$$m_{1/(D-1)} < m_{1/D}, \qquad m_{(D-1)/D} < m_{(D-1)/(D+1)} < m_{D/(D+1)}.$$

The intermediate fraction is reduced because D is even. The first two inequalities hold at fixed numerator and the last at fixed denominator. For $D = 2$, the unique seed gives $2, 5$ and then the increasing family $m_{2/(2n-1)}$ for $n \geq 2$, starting at 29. These values are also distinct. □

Thus the remaining Meridian comparison has unequal degrees. To bound it, we need exact leading terms and an explicit positive remainder.

5.2. **Positive words and a uniform tail bound.** Put $\rho = (3 + \sqrt{5})/2$, $\lambda = \log \rho$, and $\xi = \rho^x$. The endpoint matrices are

$$P(\xi) = \begin{pmatrix} B_0/\xi & C_0 \\ C_0 & A_0\xi \end{pmatrix}, \qquad Q(\xi) = \begin{pmatrix} A_0/\xi & \rho C_0 \\ C_0/\rho & B_0\xi \end{pmatrix}, \tag{23}$$
$$A_0 = \frac{15 + 3\sqrt{5}}{10}, \quad B_0 = \frac{15 - 3\sqrt{5}}{10}, \quad C_0 = \frac{2}{\sqrt{5}}.$$

Both have determinant one and positive entries. With seed endpoint words $U = P, V = Q$, the scalar state is

$$a_\alpha = \operatorname{tr} U/3, \quad b_\alpha = \operatorname{tr} V/3, \quad M_\alpha = \operatorname{tr}(UV)/3.$$

Its extension is represented by

$$(U, V) \longmapsto \begin{cases} ((UV)^{k-1}V, (UV)^k V), & k > 0, \\ ((VU)^{|k|}U, (VU)^{|k|-1}U), & k < 0. \end{cases} \tag{24}$$

For verification, $\operatorname{tr}((UV)^{j-1}V)/3$ begins at a, b and obeys the endpoint recurrence by Cayley–Hamilton. The determinant-one trace identity $\operatorname{tr}(XY) = \operatorname{tr} X \operatorname{tr} Y - \operatorname{tr}(XY^{-1})$ then gives the reflected pole. Negative indices use the reversed orbit. The count updates are exactly (17).

Proposition 5.2 (Laurent profile)**.** *For helm counts (r, s), degree $D = r + s$, and $t = (r - s)/2$,*

$$M_\alpha(x) = \sum_{j=-D,-D+2,\ldots,D} c_j \rho^{jx}, \quad c_j \geq 0, \quad c_{\sigma D} = \frac{1}{3}(9/5)^{D/2}\rho^{\sigma t}. \tag{25}$$

The logarithm is strictly convex, and the real profile has one minimum.

Proof. Expand the trace into closed paths on the two matrix states. All weights are nonnegative. A closed path has an even number of off-diagonal choices, which fixes the parity of its exponent. The two constant-state paths give the extreme coefficients because $A_0B_0 = 9/5$ and $A_0/B_0 = \rho$. The second derivative of the logarithm is λ^2 times the variance of the exponents under their positive

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normalized weights. Both extremes occur, so the variance is positive. They also force growth at both ends. □

Lemma 5.3 (Uniform positive-word remainder)**.** *Let W have r letters P and s letters Q, of total length $g = r + s > 0$. For $n \geq 0$ and $\sigma = \pm 1$,*

$$\frac{\operatorname{tr} W(\rho^{\sigma n})}{3} = \frac{1}{3}(9/5)^{g/2}\rho^{\sigma(r-s)/2}\rho^{gn}(1 + \epsilon_{W,\sigma}(n)), \qquad 0 \leq \epsilon_{W,\sigma}(n) \leq (1+\rho^{1-2n})^g - 1. \tag{26}$$

Proof. Write $z = \rho^{-n}$, $\varphi = \sqrt{\rho}$, and $L = 2\varphi/3$. On the positive arm, divide the P, Q factors by $A_0\rho^n, B_0\rho^n$ respectively and conjugate by $\operatorname{diag}(\rho, 1)$. The normalized factors are

$$\begin{pmatrix} \rho^{-1}z^2 & L\rho^{-2}z \\ Lz & 1 \end{pmatrix}, \qquad \begin{pmatrix} \rho z^2 & Lz \\ Lz & 1 \end{pmatrix}.$$

The negative arm gives the same pair in reverse roles after exchanging the basis states. Both are bounded entrywise by the rank-one matrix

$$K = \begin{pmatrix} \rho z^2 & \varphi z \\ \varphi z & 1 \end{pmatrix}.$$

Products preserve the entrywise bound, and $\operatorname{tr} K^g = (1 + \rho z^2)^g$. The constant dominant-state path contributes one to the normalized trace. Restoring the removed factors proves the bounds. □

5.3. **A finite rectangle depending only on the degrees.**

Theorem 5.4 (Unequal-degree collision window)**.** *Let $D_\alpha \neq D_\beta$, and define*

$$m = \frac{|D_\alpha - D_\beta|}{2}, \qquad E = \max(D_\alpha, D_\beta),$$
$$h = \min\{j \in \mathbb{Z}_{\geq 0} : \rho^j \geq E\}, \qquad N = 3m + h + 3, \qquad T = E(N+2).$$

An integer equality $M_\alpha(x) = M_\beta(y)$ requires

$$|x| \leq \left\lceil \frac{T}{D_\alpha} \right\rceil, \qquad |y| \leq \left\lceil \frac{T}{D_\beta} \right\rceil, \qquad M_\alpha(x) < \rho^T/3. \tag{27}$$

The seed still uses only nonnegative phases.

Proof. The degrees are even, so m is a positive integer. First,

$$|m\log(9/5) - j\log\rho| > \frac{1}{81^m + 1} \qquad (j \in \mathbb{Z}). \tag{28}$$

Choose $j_0 \geq 0$ with $1 < (9/5)^m\rho^{-j_0} < \rho$; equality at an endpoint is impossible by taking the field norm. The positive algebraic integers $9^m - 5^m\rho^{j_0}$ and $5^m\rho^{j_0+1} - 9^m$ have nonzero integer norms and conjugates of absolute value below 9^m. Each is therefore greater than 9^{-m}. Dividing by the appropriate positive denominators makes both adjacent multiplicative gaps greater than 81^{-m}. The inequality $\log(1+u) > u/(1+u)$ gives (28); all other integers j are farther away.

At phases $x = \sigma n$, $y = \tau n'$, the leading logarithms differ by

$$\frac{D_\alpha - D_\beta}{2}\log(9/5) + (D_\alpha n - D_\beta n' + \sigma t_\alpha - \tau t_\beta)\log\rho.$$

The coefficient of $\log\rho$ is integral. Put $r_m = (81^m + 1)^{-1}$. Since $E \leq \rho^h$, $\rho^6 > 81$ and $\rho^5 > 5$,

$$E\rho^{1-2N} \leq \rho^{-6m-5} < \frac{1}{2\cdot 81^m + 3} = \frac{r_m}{r_m + 2}.$$

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For $gu < 1$, the elementary bound $(1+u)^g \le \sum_{j\ge 0}(gu)^j = (1-gu)^{-1}$ makes each relative remainder in Lemma 5.3 smaller than $r_m/2$ when its phase magnitude is at least N. Their logarithmic corrections cannot cancel (28). At least one collision phase therefore has magnitude less than N.

The leading coefficient of a length-g word lies strictly between $\rho^{-g}/3$ and $\rho^g/3$, because $\rho^{-1} < B_0 < A_0 < \rho$. Lemma 5.3, with $1 + \rho^{1-2N} < \rho$, gives $\operatorname{tr} W(\rho^{\pm N})/3 < \rho^{g(N+2)}/3 \le \rho^T/3$. Convexity bounds a value inside either finite phase strip by its endpoints, so the common value is below $\rho^T/3$. Positivity also gives the lower bound $\operatorname{tr} W(\rho^{\sigma n})/3 > \rho^{g(n-1)}/3$. Comparing the two bounds yields $n < 1 + T/g$, hence the integer ceilings in (27). □

This theorem bounds the remaining candidates; it does not assert that any candidate collides. It also does not bound the degrees of a hypothetical collision. The next section supplies a different finite reduction, expressed in Farey numerators and denominators rather than Meridian phases.

6. Corridor walls and numerator–denominator windows

6.1. **A second way to travel through the same tree.** The corridor observation groups the walls between neighboring phasors. After an initial entrance, a wall follows a fixed-numerator family

$$m_{p/(q_0+np)}, \qquad n \ge 0.$$

Both Meridian and phasor edges occur in such a route. Their common law is the advance $q \mapsto q + p$, not a rule that distinguishes the two edge colors. For the blue corridor the fixed-numerator portions are $3/(4+3n)$ and $3/(5+3n)$, tending to the historical coordinate $c = 1/3$. The first wall begins

$$169 \xrightarrow{\text{phasor}} 2897 \xrightarrow{\text{Meridian}} 51641 \xrightarrow{\text{phasor}} 925765 \longrightarrow \cdots,$$

and the other begins $433, 7561, 135137, 2423525, \ldots$ with the opposite edge alternation.

Corridor role and edge color must be kept separate. With corridor-wall travel designated *standard* and a new entrance *exotic*, the two entry routes here are

$$5 \xrightarrow{\text{exotic}} 29 \xrightarrow{\text{standard}} 433 \xrightarrow{\text{standard}} 7561 \longrightarrow \cdots, \qquad 13 \xrightarrow{\text{exotic}} 169 \xrightarrow{\text{standard}} 2897 \longrightarrow \cdots.$$

Both exotic entrances are Meridian edges. In the first route the initial standard step $29 \to 433$ changes the numerator from two to three; the fixed-numerator formula starts at 433. Thus excluding the concave ceiling arc from a wall's recurrence does not move the exotic entrance to that arc. The graph and its actual depth (14) are unchanged.

6.2. **The exact recurrence behind the observed regularity.** Use the standard Christoffel matrices

$$D_a = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad D_b = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{29}$$

Their trace encoding is

$$3m_{p/q} = \operatorname{tr}\left[\prod_{j=1}^{p} D_a^{\lfloor jq/p \rfloor - \lfloor (j-1)q/p \rfloor} D_b\right]. \tag{30}$$

The word is a cyclic conjugate of the lower Christoffel word. Trace is unchanged by that rotation. The normalization is fixed by $\operatorname{tr} D_a = 3$, $\operatorname{tr}(D_a D_b) = 6$ and the Farey trace identity, which give the Markov recurrence with the initial values one and two. This is the classical matrix-word framework [10]; Section 7 also gives its matching interpretation.

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Replacing q by $q_0 + np$ adds n to every exponent in (30). The eigenvalues of D_a are ρ, ρ^{-1}. Choose its orthogonal eigenbasis so that D_b becomes

$$\begin{pmatrix} \frac{3}{2} + \frac{3}{2\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{3}{2} - \frac{3}{2\sqrt{5}} \end{pmatrix}.$$

All four entries are positive. Expanding the trace over cyclic choices of the two eigenstates gives the exact identity

$$m_{p/(q_0+np)} = \sum_{j=0}^{p} c_{p,q_0,j}\rho^{(p-2j)n}, \qquad c_{p,q_0,j} > 0. \tag{31}$$

Every mode occurs. The least order of a constant-coefficient recurrence is therefore $p + 1$, with characteristic polynomial

$$\prod_{j=0}^{p}(X - \rho^{p-2j}).$$

Pairing conjugate roots shows that its coefficients are integers. For $p = 3$ the polynomial is $(X^2 - 18X + 1)(X^2 - 3X + 1)$, so either blue wall satisfies

$$m_{3/(q+12)} = 21m_{3/(q+9)} - 56m_{3/(q+6)} + 21m_{3/(q+3)} - m_{3/q}. \tag{32}$$

This equality holds at every step, for both colors of edge.

The exact regularity is a shared recurrence, not a constant output ratio. The positive spectral sum gives strict log-convexity, hence

$$\log_{10}\frac{m_{p/(q_0+(n+1)p)}}{m_{p/(q_0+np)}} \nearrow p\log_{10}\rho.$$

The small deviation is retained in the collision argument, rather than discarded as an asymptotic error. Figure 8 shows both the geometric walls and that deviation.

6.3. **A positive correction with a uniform bound.** The dominant eigenstate in (30) yields

$$3m_{p/q} = \left(\frac{3}{\sqrt{5}}\right)^p \left(\frac{1+\sqrt{5}}{2}\right)^{2q+p}(1 + \varepsilon_{p,q}), \tag{33}$$

where

$$0 < \varepsilon_{p,q} \le \left(1 + \rho^{-2\lfloor q/p \rfloor}\right)^p - 1. \tag{34}$$

To prove the bound, divide the transformed D_b by its upper-left entry; every resulting entry is at most one. Choosing the contracting eigenstate at a factor costs at least $\rho^{-2\lfloor q/p \rfloor}$. Summing over all nonempty subsets of contracting choices gives (34). This bound is independent of the remainder class of q modulo p.

Along a fixed wall the correction decreases strictly to zero. Consequently its logarithmic deviations telescope exactly:

$$\sum_{k=0}^{n-1}\left[p\log\rho - \log\frac{m_{p/(q+(k+1)p)}}{m_{p/(q+kp)}}\right] = \log\frac{1+\varepsilon_{p,q}}{1+\varepsilon_{p,q+np}}. \tag{35}$$

Thus there is no accumulated error proportional to the path length. For a route changing corridors, subtraction of the endpoint formula still telescopes, but monotone decay is claimed only within a fixed wall.

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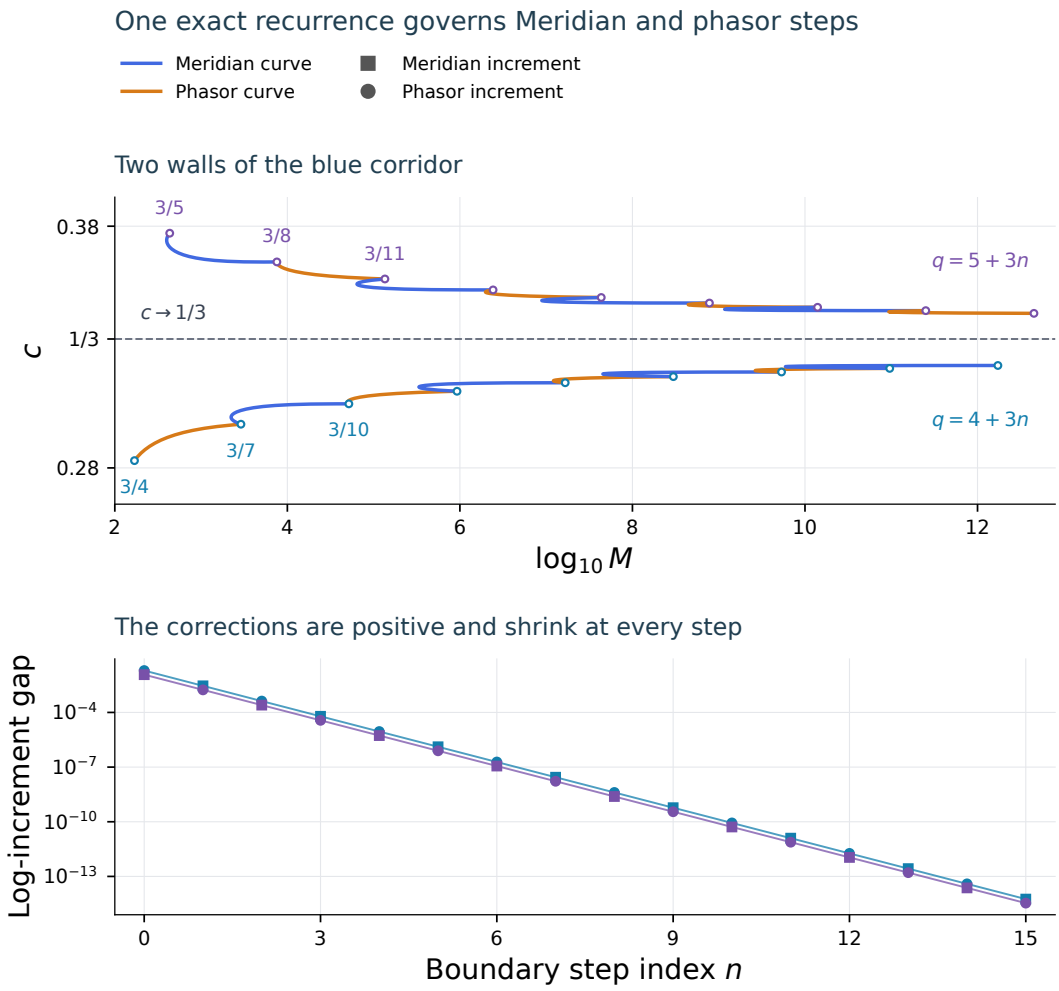


Figure 8. The two fixed-numerator walls of the blue corridor. The concave ceiling arc is omitted from this recurrence illustration. Both walls obey (32). In the lower panel, the positive difference between $3\log_{10}\rho$ and the actual logarithmic increment decreases at every step. Squares denote Meridian increments and circles phasor increments. The values are exact integers; only the plotted curves and logarithms are numerical.

Theorem 6.1 (Necessary correction gap)**.** *If $p \neq p'$ and $m_{p/q} = m_{p'/q'}$ for reduced labels, then*

$$\frac{|\varepsilon_{p,q} - \varepsilon_{p',q'}|}{1 + \min(\varepsilon_{p,q}, \varepsilon_{p',q'})} > 9^{-|p-p'|}. \tag{36}$$

Proof. Assume $p > p'$. The ratio of the leading terms is

$$R = \left(\frac{3(1+\sqrt{5})}{2\sqrt{5}}\right)^{p-p'} \rho^{q-q'}.$$

Its quadratic-field norm is $(9/5)^{p-p'}$, so $R \neq 1$. The number $(\sqrt{5})^{p-p'}(R-1)$ is an algebraic integer, because the powers of $(1+\sqrt{5})/2$ and ρ are units. Its nonzero norm gives

$$|(R-1)(\overline{R}-1)| \geq 5^{-(p-p')}, \qquad \overline{R} = (9/5)^{p-p'}/R > 0.$$

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Put $\eta = \max(R, R^{-1}) - 1$. If $R \geq 1$, then $|R - 1| = \eta$ and $|\overline{R} - 1| < (9/5)^{p-p'}$. If $R < 1$, their product is $\eta((9/5)^{p-p'} - 1/(1+\eta)) < \eta(9/5)^{p-p'}$. Either case forces $\eta > 9^{-(p-p')}$. At equality of the Markov values, $R = (1 + \varepsilon_{p',q'})/(1 + \varepsilon_{p,q})$, so η is precisely the left side of (36). □

6.4. **The corridor collision window.**

Theorem 6.2 (Numerator–denominator window)**.** *Let $p \neq p'$, $P = \max(p, p')$, and let K be the least nonnegative integer satisfying*

$$\rho^{2K} > P(9^{|p-p'|} + 1). \tag{37}$$

Every equality $m_{p/q} = m_{p'/q'}$ requires

$$\max(q, q') \leq P(K+1) - 1. \tag{38}$$

For bounded numerators a simpler, weaker consequence is $\max(q, q') \leq 2P^2 + P - 1$.

Proof. If both quotient floors are at least K, then $P\rho^{-2K} < 1/(9^{|p-p'|} + 1)$. The geometric-series estimate used in Lemma 5.3 makes both corrections strictly smaller than $9^{-|p-p'|}$, contradicting Theorem 6.1. Thus one denominator, say q, is at most $Kp - 1$.

Both matrices in (29) have operator norm ρ. The trace bound and the positive leading term imply

$$\frac{1}{3}\rho^{q'} < m_{p'/q'} = m_{p/q} \leq \frac{2}{3}\rho^{q+p}.$$

Since $\log_\rho 2 < 1$ and the exponents are integers, $q' \leq q + p$. It follows that both denominators are at most $P(K+1) - 1$. For the simpler bound it suffices to use $K = 2P$: $\rho^{4P} > 36^P > P(9^{P-1} + 1)$ for $P \geq 2$, and $|p - p'| \leq P - 1$. □

For example, numerators at most 64 permit $K = 75$, so every remaining comparison lies at denominators at most 4863. With $P = 64$ and numerator difference five, $K = 8$ gives 575. These are bounds on candidates, not new numerical verification claims.

Theorem 5.1 first discards equal Meridian degrees. For the rest, the phase rectangle (27) and denominator cap (38) can be imposed together through the dictionary (18). Fixed-numerator ordering already separates all labels with $p = p'$. What remains is a comparison of different numerators and degrees inside their joint windows. Neither numerator nor degree is globally bounded by these theorems, so this is not yet a finite reduction of the entire unicity conjecture.

7. Snake-graph counts and ordering proofs

7.1. **From a Farey label to an actual matching count.**

A snake graph is a chain of square tiles, each attached to the right of or above the preceding tile. Shared edges are counted once. A perfect matching covers every vertex exactly once. The Markov snake associated with a reduced $0 < p/q < 1$ is built as follows. Take the lower staircase from $(0, 0)$ to (q, p): at each horizontal coordinate j, step right and then up to height $\lfloor pj/q \rfloor$. Write right steps as a and up steps as b. The resulting lower Christoffel word has the form $w_{p/q} = azb$. Remove its first and last letters; start with one square and replace each a of z by two added squares to the right and each b by two upward. The resulting snake has $2(p + q) - 3$ tiles.

Write $\mu(w)$ for its number of perfect matchings. The established Markov matching correspondence gives

$$\mu(w_{p/q}) = m_{p/q}. \tag{39}$$

The matching interpretation originated in the REACH work of 2002–2003. Musiker's memorandum, dated 27 June 2002, proposed the recursive graph model [14]; Itsara, Musiker, Propp, and Viana's manuscript of 10 January 2003 stated and proved the snake-matching correspondence [15, Theorem

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3.1]. Setting all edge weights to one gives the ordinary perfect-matching counts. Propp's later exposition, first posted in 2005, records the broader collaborative development [16]. These original contributions precede the later proofs of Aigner's orderings.

For a modern treatment and the Farey-indexed construction, see [13, Section 4.1, Theorem 4.2] and [7]. For example, $w_{2/5} = a^3ba^2b$, its trimmed word is $aabaa$, and the eleven-tile snake has 194 matchings. Auxiliary rearranged words also specify snakes, counted by μ; they need not be canonical Markov snakes.

7.2. **Six pieces and two connector states.** Figure 9 fixes the five three-tile pieces and the three one-square variants of type 6. In a cut between pieces, a matching uses both yellow connector edges or neither: the cut leaves an even number of vertices on either side. Denote these states by 1 and 0. At a selected port remove the two vertices already covered by its connector pair. Direct counting within the pieces gives

(40)

piece	00	01	10	11
1	5	2	2	1
2	5	3	3	2
3	5	2	3	1
4	5	3	2	1
5	5	2	2	1
6i	2	1	1	1

The state order is incoming, outgoing. The one-port end squares 6ii and 6iii have counts $(2, 1)$.

For the ordering surgeries, use a fixed one-ended chop: retain tile triples $(1, 2, 3), (5, 6, 7), \ldots$, with tiles $4, 8, \ldots$ providing paired connectors. A one-square remainder occurs exactly when $p + q$ is even and is counted with the closed end piece. For every connector sequence σ, multiply the applicable row entries in (40); then sum over σ, with outer ports closed. Each matching determines one sequence and one internal completion per piece, and gluing reverses the construction. This proves the counting rule, without an independence assumption. Equivalently it is a sum of terms $5^{k(\sigma)}3^{j(\sigma)}2^{\ell(\sigma)}$, where the exponents count factors of the indicated size. For the 2/5 snake the piece list $2, 1, 2$ gives

$$5^3 + 5 \cdot 2 \cdot 3 + 3 \cdot 2 \cdot 5 + 3^2 = 194.$$

7.3. **Fibonacci-matrix normalization.** The Fibonacci-matrix formulation and normalization argument below are the author's. We begin with the comparison of three consecutive blocks and then apply it to complete words.

With $F_0 = 0, F_1 = 1$ and $F_{-j} = (-1)^{j+1}F_j$, the basic odd-index identity is

$$F_{2k-1}F_{2k+3} - F_{2k+1}^2 = 1. \tag{41}$$

For example, $5 \cdot 34 - 13^2 = 1$. Consider the author's three consecutive Fibonacci blocks:

$$A = \begin{pmatrix} F_{2k+3} & F_{2k+1} \\ F_{2k+1} & F_{2k-1} \end{pmatrix},$$

$$B = \begin{pmatrix} F_{2k+1} & F_{2k-1} \\ F_{2k-1} & F_{2k-3} \end{pmatrix}, \tag{42}$$

$$C = \begin{pmatrix} F_{2k-1} & F_{2k-3} \\ F_{2k-3} & F_{2k-5} \end{pmatrix}, \qquad k \geq 2.$$

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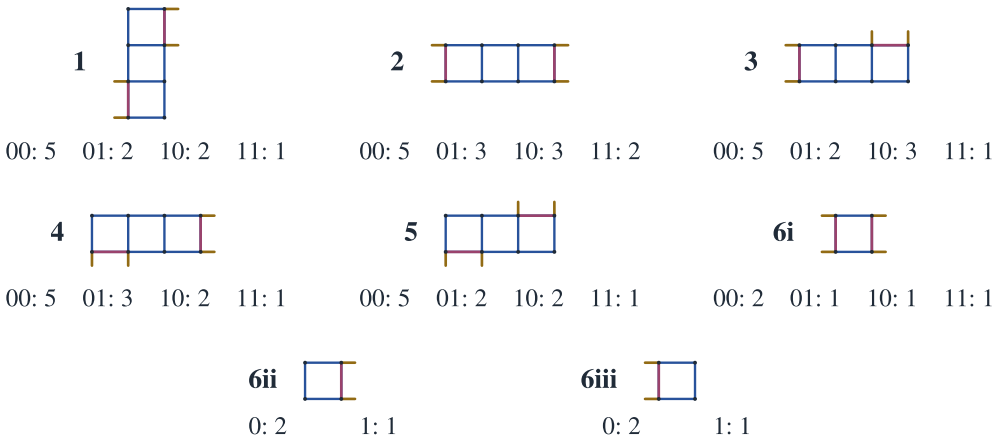


Figure 9. The six physical pieces. Gold edges are paired connectors, magenta marks ports, and dots mark tile vertices. Two-port counts use the order $00, 01, 10, 11$. The central square 6i has counts $(2, 1, 1, 1)$; the one-port end squares have $(2, 1)$. Types 1 and 5 have equal count rows but different physical orientations.

Applying (41) twice gives precisely the upper-left comparison used in normalization:

$$(AC)_{11} - (B^2)_{11} = (F_{2k+3}F_{2k-1} - F_{2k+1}^2) + (F_{2k+1}F_{2k-3} - F_{2k-1}^2) = 2. \tag{43}$$

The same identities in the other entries give

$$AC - B^2 = \begin{pmatrix} 2 & 6 \\ 0 & 2 \end{pmatrix}, \qquad CA - B^2 = \begin{pmatrix} 2 & 0 \\ 6 & 2 \end{pmatrix}.$$

Thus replacing AC or CA by B^2 decreases the upper-left entry even inside a longer product: the surrounding Fibonacci blocks have nonnegative entries and positive upper-left entries. For $k = 3$ this reads $(AC)_{11} = 34 \cdot 5 + 13 \cdot 2 = 196 > 194 = 13^2 + 5^2 = (B^2)_{11}$.

7.3.1. *Run counts and general index shifts.* To apply the same comparison at arbitrary run lengths, index the blocks by

$$\mathcal{A}_t = \begin{pmatrix} F_{2t+3} & F_{2t+1} \\ F_{2t+1} & F_{2t-1} \end{pmatrix}, \qquad t \geq 0. \tag{44}$$

Thus $A = \mathcal{A}_k$, $B = \mathcal{A}_{k-1}$ and $C = \mathcal{A}_{k-2}$ in (42). The four boundary counts of a straight block of $2t + 1$ tiles are $F_{2t+3}, F_{2t+1}, F_{2t+1}, F_{2t-1}$: these follow by conditioning on the first edge of a ladder, giving the Fibonacci recurrence; at $t = 0$ they are $(2, 1, 1, 1)$. Gluing successive runs with their paired connector states therefore gives, for $w = a^{h_1}b \cdots a^{h_p}b$ with $h_1 \geq 1$ and $h_j \geq 0$,

$$\mu(w) = [\mathcal{A}_{h_1-1}\mathcal{A}_{h_2} \cdots \mathcal{A}_{h_p}]_{11}. \tag{45}$$

The first run is shortened by the removed initial a; the final b closes the other end. Equivalently, a block-index list $(t_1, \ldots, t_p)$ represents $w = a^{t_1+1}ba^{t_2}b \cdots a^{t_p}b$, with $t_1 + \cdots + t_p = q - 1$.

Fibonacci addition gives the scalar index-shifting identity

$$F_{2r+1}F_{2s+1} - F_{2r-1}F_{2s+3} = F_{2(r-s)-2}.$$

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Apply it to the two summands of each product entry to obtain

$$\mathcal{A}_r\mathcal{A}_s - \mathcal{A}_{r-1}\mathcal{A}_{s+1} = 2\begin{pmatrix} F_{2(r-s)-2} & F_{2(r-s)} \\ F_{2(r-s)-4} & F_{2(r-s)-2} \end{pmatrix}, \quad r \geq 1,\ s \geq 0. \tag{46}$$

Consequently, whenever two adjacent indices differ by at least two, moving one unit from the larger to the smaller strictly decreases the whole product's upper-left entry. The case $r = s + 2$ is exactly (43). The only remaining local comparison has adjacent indices differing by one; it is an interchange of the two blocks.

7.3.2. *Comparing the surrounding Fibonacci blocks.* For a finite list of block indices, define the ratio of its two open counts by

$$\rho(u_1, \ldots, u_m) = \frac{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{21}}{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{11}}, \qquad \rho(\varnothing) = 0.$$

All these ratios lie in $[0, 1)$. Direct multiplication gives

$$\rho(t, U) = \frac{F_{2t+1} + F_{2t-1}\rho(U)}{F_{2t+3} + F_{2t+1}\rho(U)}. \tag{47}$$

Its right-hand side increases strictly with $\rho(U)$, since (41) gives $\det \mathcal{A}_t = 1$. Moreover, another application of that identity gives

$$\frac{F_{2t+1}}{F_{2t+3}} - \frac{F_{2t+3} + F_{2t+1}}{F_{2t+5} + F_{2t+3}} = \frac{1}{F_{2t+3}(F_{2t+5} + F_{2t+3})} > 0.$$

Hence $\rho(t + 1, U) < \rho(t, V)$ for any two finite tails U, V. Two lists can therefore be compared by cancelling equal initial indices: at the first unequal pair, the larger index gives the smaller ratio; an exhausted list has the smaller ratio than any nonempty remainder.

For adjacent blocks in positions $j, j + 1$, the outward lists are $(t_{j-1}, \ldots, t_1)$ on the left and $(t_{j+2}, \ldots, t_p)$ on the right. At $r = s + 1$, (46) becomes

$$\mathcal{A}_{s+1}\mathcal{A}_s - \mathcal{A}_s\mathcal{A}_{s+1} = 2\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

After multiplication by the surrounding blocks, its upper-left entry has the sign of

$$\rho(t_{j+2}, \ldots, t_p) - \rho(t_{j-1}, \ldots, t_1). \tag{48}$$

Indeed, it equals this difference times twice the positive upper-left entries of the two surrounding products. This proves the needed surrounding-block test entirely in the same Fibonacci matrices.

Lemma 7.1 (Fibonacci-matrix normalization)**.** *For coprime positive p, q, every word $w = azb$ with q letters a and p letters b satisfies $\mu(w_{p/q}) \leq \mu(w)$.*

Proof. For $p = 1$ there is only the word $a^q b$. Assume $p > 1$ and use the block-index list in (45). Its prefix discrepancies are

$$\delta_j = qj - p(1 + t_1 + \cdots + t_j) \quad (1 \leq j \leq p), \qquad \delta_0 = \delta_p = 0.$$

Changing (t_j, t_{j+1}) to $(t_j + 1, t_{j+1} - 1)$ decreases only δ_j, by p; the reverse change increases only δ_j, by p. We show that an appropriate change always decreases the matching count until $-p \leq \delta_j \leq 0$ for every j.

The sign checks use just a prefix-sum identity. Put $h_1 = t_1 + 1$ and $h_i = t_i$ for $i > 1$, the actual run lengths. Whenever $j - m \geq 0$ and $j + m \leq p$,

$$\delta_{j-m} + \delta_{j+m} = 2\delta_j + p\left(\sum_{i=j-m+1}^{j} h_i - \sum_{i=j+1}^{j+m} h_i\right). \tag{49}$$

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In the block indices, the left sum has the extra 1 precisely when it includes t_1.

First remove positive discrepancies. Choose the rightmost largest $\delta_j > 0$ and transfer one unit from t_{j+1} to t_j. Comparison with its two neighboring discrepancies gives $h_j < q/p < h_{j+1}$; coprimality makes these inequalities strict. Thus the transfer is allowed and $t_{j+1} \geq t_j + 1$; for $j = 1$ the gap is at least two. Gaps of at least two are handled by (46).

If $t_{j+1} = t_j + 1$, compare the outward lists. A first unequal pair with the left index larger would make the parenthesis in (49) nonnegative: the central difference is -1, the equal outward indices cancel, and the first unequal difference is at least 1. This contradicts maximality and the choice of the rightmost maximum. If the left list were exhausted first, its extra initial 1 would instead give $\delta_{2j} = 2\delta_j > \delta_j$, again impossible (including simultaneous exhaustion, when $2j = p$). The left outward ratio is therefore strictly larger than the right one. Reversing the interchange in (48) shows that the selected transfer strictly decreases the count. It also decreases $\sum_{j=1}^{p-1} \delta_j$. There are only finitely many nonnegative block-index lists of sum $q - 1$, so this phase ends with all $\delta_j \leq 0$.

Then remove discrepancies below $-p$. Choose the leftmost smallest $\delta_j < -p$ and transfer one unit from t_j to t_{j+1}. For $j > 1$, the neighboring comparisons give $h_j > q/p > h_{j+1}$; for $j = 1$, the inequality $\delta_1 < -p$ gives $t_1 > q/p$, while $t_2 < q/p$. Thus $t_j \geq t_{j+1} + 1$ and the transfer is allowed. Again a gap of at least two is settled by (46).

In the gap-one case, a first unequal outward pair with the left index smaller would make the parenthesis in (49) nonpositive, contradicting minimality and the leftmost choice. If that pair reaches t_1, its extra 1 instead gives $\delta_{2j} \leq 2\delta_j + p < \delta_j$, the same contradiction. If the right list were exhausted first, the identity would give $\delta_{2j-p} = 2\delta_j + p < \delta_j$. Simultaneous exhaustion would give $\delta_j = -p$. Thus the left outward ratio is strictly smaller than the right one, and (48) proves a strict decrease in count. The changed discrepancy is $\delta_j + p < 0$, so positivity is not reintroduced. This phase increases the bounded sum of discrepancies by p each time and terminates with $-p \leq \delta_j \leq 0$.

Finally, $\delta_j \equiv qj \pmod{p}$. For $1 \leq j < p$, coprimality leaves exactly one possible value in $[-p, 0]$, forcing

$$1 + t_1 + \cdots + t_j = \left\lceil \frac{jq}{p} \right\rceil .$$

These are precisely the canonical run lengths. Every transfer has decreased the upper-left product entry, proving the assertion. □

For the canonical word, $h_j = \lceil jq/p \rceil - \lceil (j-1)q/p \rceil$. With $k = 1 + \lfloor (q-1)/p \rfloor$, the factors are the author's A, B from (42), and the end factors are B. For example, $m_{3/8} = [BAB]_{11}$ with $k = 3$. A similar normalization result appears in [10, Proposition 21]; the proof above is self-contained in the author's Fibonacci matrices.

7.4. **The three Aigner orderings.** Fixed-numerator monotonicity was proved by Rabideau and Schiffler [13]; all three orderings were proved by LPTV, Lee–Li–Rabideau–Schiffler, and McShane [10, 11, 12]. The author's Fibonacci matrices prove normalization and the fixed-numerator ordering. His six pieces prove the remaining fixed-sum comparison and the fixed-denominator ordering, using normalization to return to a canonical Markov snake.

Theorem 7.2 (Established monotonicities)**.** *For $i > 0$, whenever both fractions are reduced and lie in $(0, 1)$,*

$$m_{p/q} > m_{(p+i)/(q-i)} \qquad \textit{(fixed sum)},$$
$$m_{p/q} < m_{p/(q+i)} \qquad \textit{(fixed numerator)},$$
$$m_{p/q} < m_{(p+i)/q} \qquad \textit{(fixed denominator)}.$$

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7.4.1. *Fixed numerator: Fibonacci matrices.*

Proof. Fibonacci recurrence gives, for $t \geq 1$,

$$\mathcal{A}_t - \mathcal{A}_{t-1} = \begin{pmatrix} F_{2t+2} & F_{2t} \\ F_{2t} & F_{2t-2} \end{pmatrix}.$$

All entries are nonnegative and the upper-left is positive. Start with $w_{p/(q+i)}$ and delete i internal letters a, retaining the first. Each deletion changes one factor $\mathcal{A}_t$ to $\mathcal{A}_{t-1}$ with $t \geq 1$. Surrounding factors are positive, so the matching count strictly falls. The resulting word ends at (q, p); normalization gives $m_{p/q} \leq \mu(w_*) < m_{p/(q+i)}$. □

7.4.2. *Fixed sum: six pieces.*

Proof. At the same physical cuts, the changes $2 \to 3$ or 4, then 3 or $4 \to 5$, increase no entry of (40); the changed factors are $3 \to 2$ or $2 \to 1$. Rotation $5 \leftrightarrow 1$ preserves the row. Selecting the changed connector and no others exhibits a strictly smaller positive summand whenever an equal-direction join becomes unequal.

The trimmed canonical word begins and ends with a, contains no bb, and has $q - p - 1$ equal joins, all aa. At its rightmost equal join, interchange a, b strictly to the right. That suffix is alternating, so only the chosen join changes. Exactly one prescribed piece changes by the stated rules, while any rotated pieces keep their rows. Each surgery strictly decreases the count. Odd-numbered operations replace one horizontal step by a vertical step in net; even-numbered operations keep the totals. After $2i$ operations the endpoint is $(q - i, p + i)$. There are enough joins because $p + i < q - i$. For the resulting auxiliary word w_*, normalization gives $m_{(p+i)/(q-i)} \leq \mu(w_*) < m_{p/q}$. A similar tail comparison appears in [10, Proposition 22]; here the comparison is proved using the author's six physical pieces. □

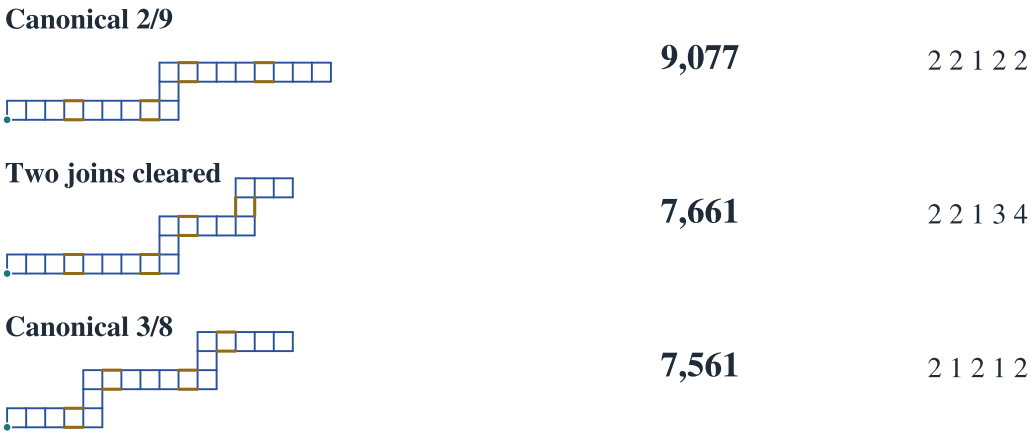


Figure 10. Fixed-sum surgery on the same five pieces. The canonical 2/9 snake has 9077 matchings; after two changes the auxiliary snake has 7661; normalization gives the 3/8 snake with 7561. The intermediate first change, not shown, has 8263. Gold pairs mark cuts and teal dots the starting corners.

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7.4.3. *Fixed denominator: six pieces.*

Proof. For the fixed-denominator comparison, refine the same six-piece chop into single-square pieces of type 6, retaining odd-numbered tiles and using the even tiles as paired connectors. A retained square has counts $(2,1,1,1)$ for opposite ports and $(2,1,1,0)$ for adjacent ports. The zero forbids covering their common corner twice. Summing an intervening state regroups these squares into the original three-tile pieces.

Insert one internal b, adding two vertical squares. The affected local counts are the following complete table, with outside ports fixed:

	before				after			
directions	00	01	10	11	00	01	10	11
a,a	2	1	1	1	5	2	2	1
b,b	2	1	1	1	5	3	3	2
a,b	2	1	1	0	5	3	2	1
b,a	2	1	1	0	5	2	3	1

The endpoint caps have the corresponding rows with an outer port closed; the empty trimmed word has the increase $2 \to 5$. No outside count falls, and the all-zero choice witnesses a strict increase. Thus the two added squares compensate for every changed direction.

Starting from $w_{(p+i)/q}$, delete i internal b's and reverse this comparison. At least i are available. Normalizing the resulting word to (q,p) proves $m_{p/q} \le \mu(w_*) < m_{(p+i)/q}$. □

7.5. **Folding an actual snake at its center.** The same local counts also compare families sharing a long initial sequence of pieces. Represent the four counts by

$$R_1 = R_5 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}, \quad R_4 = \begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}.$$

Attaching piece i to an open-count row (x,y) replaces it by $(x,y)R_i$. This is the sum over the common connector state, not an additional assumption about matchings.

For a helm, use the symmetric central chop: retain the middle square as 6i and group retained squares in pairs outward on each side. For odd/odd p,q, each half has $(p+q-2)/2$ retained squares. If this number is odd, one square remains at each outer end, giving pieces 6ii and 6iii with initial counts $(2,1)$. Otherwise the half starts at $(1,0)$ and consists of complete three-tile pieces. This repartition requires rebuilding the piece list; one cannot move a cut and keep its old boundary counts.

Proposition 7.3 (Folded matching counts)**.** *For a reduced $0<p<q$, let (x,y) be the open counts of one reflected half under the central chop. If p,q are odd, then*

$$m_{p/q} = 2x^2 + 2xy + y^2 = (x+y)^2 + x^2. \tag{50}$$

If $p+q$ is odd, then $m_{p/q} = x^2+y^2$. In either case $\gcd(x,y)=1$; a nonempty half has positive counts.

Proof. The trimmed Christoffel word is a palindrome: for $0<j<q$, $\lfloor p(q-j)/q \rfloor = p-1-\lfloor pj/q \rfloor$. For odd/odd labels the two halves meet at an opposite-port central square, whose four counts are $2,1,1,1$. The reflected halves contribute x^2, xy, yx, y^2, giving (50). For opposite parities they meet across a single connector and must use its same state, giving x^2+y^2. Every full piece table and every exposed-square table has determinant ± 1, so its integral inverse preserves the gcd of an

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open row. Starting at $(1,0)$ proves primitivity. The first square gives $(2,1)$, and every subsequent permitted table preserves positivity. □

The seed $1/1$ consists only of its central square. The two examples in Figure 11 give $m_{1/3} = 2\cdot 2^2 + 2\cdot 2\cdot 1 + 1^2 = 13$ and $m_{1/5} = 2\cdot 5^2 + 2\cdot 5\cdot 3 + 3^2 = 89$.

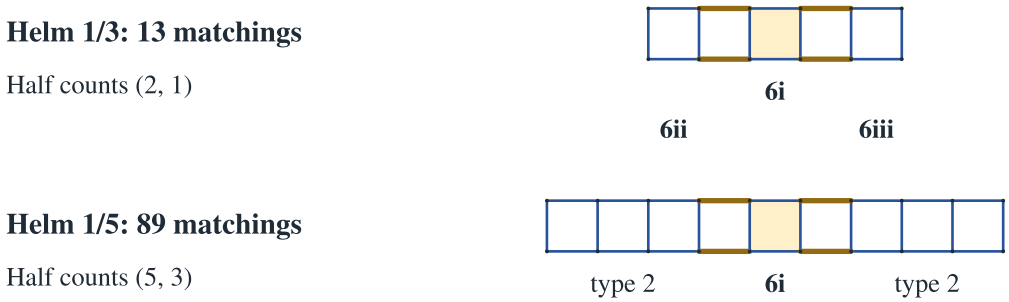


Figure 11. Central and end pieces in actual helms. At $1/3$, central square 6i has the end squares 6ii and 6iii as its halves, with counts $(2,1)$. At $1/5$, each half is a type-2 triple with counts $(5,3)$. Folding produces 13 and 89.

7.6. **A common prefix leaves one quadratic comparison.** Two folded graphs with the same outer boundary convention may share a prefix with open counts (x,y). Sum over all remaining states, leaving that cut open. The two centers give positive definite symmetric tables $G, \widetilde{G}$ with determinant one and nonnegative off-diagonal entries. For a helm they have the form SQS^{T} with $Q = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$; for opposite-parity labels they have form SS^{T}. Thus

$$m - \widetilde{m} = \alpha x^2 + 2\beta xy + \gamma y^2, \qquad G - \widetilde{G} = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}. \tag{51}$$

Proposition 7.4 (At most one primitive common prefix)**.** *For a nonempty common part and $G \neq \widetilde{G}$, equality requires $\alpha\gamma < 0$ and $\beta^2 - \alpha\gamma$ to be an integer square. There is then at most one primitive positive row (x,y) giving equality. That row must still arise from both actual prescribed snakes.*

Proof. A center table $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ has $ac - b^2 = 1$ and $b \geq 0$. If its two diagonal differences from another table have the same weak sign, then the difference of $b = \sqrt{ac-1}$ has that sign too. A nonzero difference cannot vanish at a positive row. Opposite diagonal signs are therefore necessary. The resulting quadratic in y/x has one positive root. An integer zero obeys $(\gamma y + \beta x)^2 = (\beta^2 - \alpha\gamma)x^2$; a rational square root of an integer is integral. A positive rational root has only one primitive positive representative. □

For an empty prefix, test its actual boundary row. If $G = \widetilde{G}$, the criterion supplies no exclusion. Equal count tables do not certify equal physical graphs: types 1 and 5 already show why orientations and the prescribed Christoffel word must be retained.

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A useful uniform exclusion follows directly modulo three. The physical piece tables satisfy $R_i Q R_i^{\mathsf{T}} \equiv Q \pmod 3$, whereas a single retained square gives $-Q$. A helm tail with d retained squares therefore has center table $(-1)^d Q \pmod 3$. Opposite parities give a nonzero difference at every primitive row because the discriminant of Q is -1, a nonsquare modulo three. For equal parity and nonzero difference, let $h = \min_{i,j} v_3((G - \widetilde{G})_{ij})$ and write $3^{-h}(G - \widetilde{G}) = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$. Then $h \geq 1$. Expanding the equal determinants and reducing modulo three gives $\gamma \equiv \alpha + \beta$; the normalized discriminant is one or two modulo three. If it is two, every primitive shared row has count difference of valuation exactly h. These are exclusions for specified tails with arbitrary actual common prefixes, not a classification of all possible snake-graph collisions.

8. All-phase verification through degree 10,000

Theorem 8.1 (Degree-10,000 separation)**.** *Among canonical Meridians of Laurent degree at most* 10,000*, distinct canonical integer-phase marks have distinct maxima. There are exactly* 5,067,077 *Meridians in this family.*

The computational part separates different Meridians at all integer phases. The fixed-degree theorem also separates marks on a single Meridian, so the two conclusions combine as stated. The bound is on *degree*, not on the size of a Markov number, and it does not exclude a competitor of higher degree.

8.1. Why a finite residue test covers infinitely many phases.

For a period T, define the complete image

$$\mathcal{R}_\alpha(T) = \{M_\alpha(x) \bmod q(T) : 0 \leq x < T\}.$$

Theorem 4.1 makes this the image at *every* integer phase. Disjoint images therefore separate two entire Meridians. If images overlap for several periods, a true equality must use compatible phases in all of them. Define the witness relation

$$W_T(\alpha, \beta) = \{(u, v) \in (\mathbb{Z}/T\mathbb{Z})^2 : M_\alpha(u) \equiv M_\beta(v) \pmod{q(T)}\}.$$

Proposition 8.2 (Compatible witnesses)**.** *Equality of two integer values requires one witness in each selected W_T, with first coordinates agreeing modulo every pairwise gcd of the periods, and second coordinates agreeing in the same way. If no compatible choice exists, the two Meridians are disjoint at all integer phases.*

Proof. A true equality uses the same integers x, y in every modulus and hence supplies the required residues. Conversely, the generalized Chinese remainder theorem lifts compatible coordinates to a phase pair modulo the least common multiple. This converse concerns modular witnesses only; it does not imply equality of the full values. □

The moduli need not be independent. In particular, the periods $36, 38, 40$ are all even, so contradictory parity requirements alone can exclude a pair that survives each separate image test.

8.2. Complete inventory and the final exclusion.

Proof of Theorem 8.1. Proposition 3.4 identifies the inventory with coprime positive odd pairs $r \leq s$, $r + s \leq 10{,}000$. It contains 5,067,077 Meridians and 12,837,632,128,426 unordered pairs. The producer enumerates the count-vector address tree; an independent checker enumerates the coprime pairs directly and compares the inventories. The child degree is $2|k|D \pm (|v| - |u|) > D$, where $|u|, |v|$ are the endpoint word lengths. This gives finite exhaustive index ranges.

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Every one of the 36 phase residues of every Meridian is evaluated modulo $q(36)$. Equal-residue grouping lists all pairs whose images intersect; its complement is excluded at once. Periods 38 and 40 are then evaluated on every endpoint of the preceding survivor sets. The complete image tests leave 87 pairs. For each of them, project every full witness relation onto the four parity pairs $(u \bmod 2, v \bmod 2)$. The three projected sets have empty intersection in all 87 cases. Proposition 8.2 therefore excludes every pair of different Meridians at every integer phase. Theorem 5.1 also excludes different marks within each individual Meridian, completing the statement. □

Table 2. Complete cumulative survivor counts from the archived certificates. "Images" forgets meeting phases; "phases" also imposes compatibility. Survivors are modular coincidences, not integer collisions.

Tests retained	$D \le 2{,}000$	$D \le 10{,}000$
36: images	7,121	320,755
36, 38: images	612	17,427
36, 38: phases	2	2,052
36, 38, 40: images	0	87
36, 38, 40: phases	**0**	**0**
36, 38, 40, 42: images	—	1
36, 38, 40, 42, 44: images	—	0

8.3. **What was checked independently.** The degree-10,000 producer uses signed endpoint-orbit updates. The separate evaluator uses Fibonacci staircase words with Euclidean matrix reduction. They agree on all 188,369,412 retained phase evaluations. The checker verifies the 182,414,772 baseline phase tags bijectively, so an omitted or repeated phase record cannot pass unnoticed. Independent sorting of the original unsorted buckets reconstructs all 70,009 cross-Meridian residue groups and the subsequent complete pair sets. Coverage is checked by exact records, not inferred merely from their counts or from a matching digest.

The phase result is checked three ways: full generalized-CRT joins, an independent 16-bit mask computation on $(x \bmod 4, y \bmod 4)$, and the weaker four-state parity projection, already sufficient for the final 87. The degree-2,000 certificate, containing 202,862 Meridians, was completed first; its common survivor sets agree exactly with the larger inventory when identified by helm labels.

Five deliberate corruptions—an altered value, an incorrect modulus, a duplicate tag, an omitted record and an invented pair—were all rejected. The degree-100 control compares 18,684 residues with literal Fibonacci words and the polynomial record of (19). The optional image-only extension through periods 42 and 44 also ends with zero pairs; all 88 period-44 values were separately checked using arbitrary-size Python integers because $q(44) > 2^{63}$.

8.4. **Reproducibility and scope.** The degree reproducibility package [8] contains the inventory, compact residue groups, phase witnesses, producer and checker sources, audit receipts, and a regeneration driver. Large baseline tables are regenerated; their recorded sizes and hashes identify the original run. The reindexed edition retains the original computational kernels with an explicit address-convention adapter. Reversing the written address tail does not change a helm label, degree, phase, or certified numerical value.

This is a deterministic computer-assisted certificate, not an external referee report or proof-assistant formalization. Its scope is a bounded set of Meridians with unbounded phase. The

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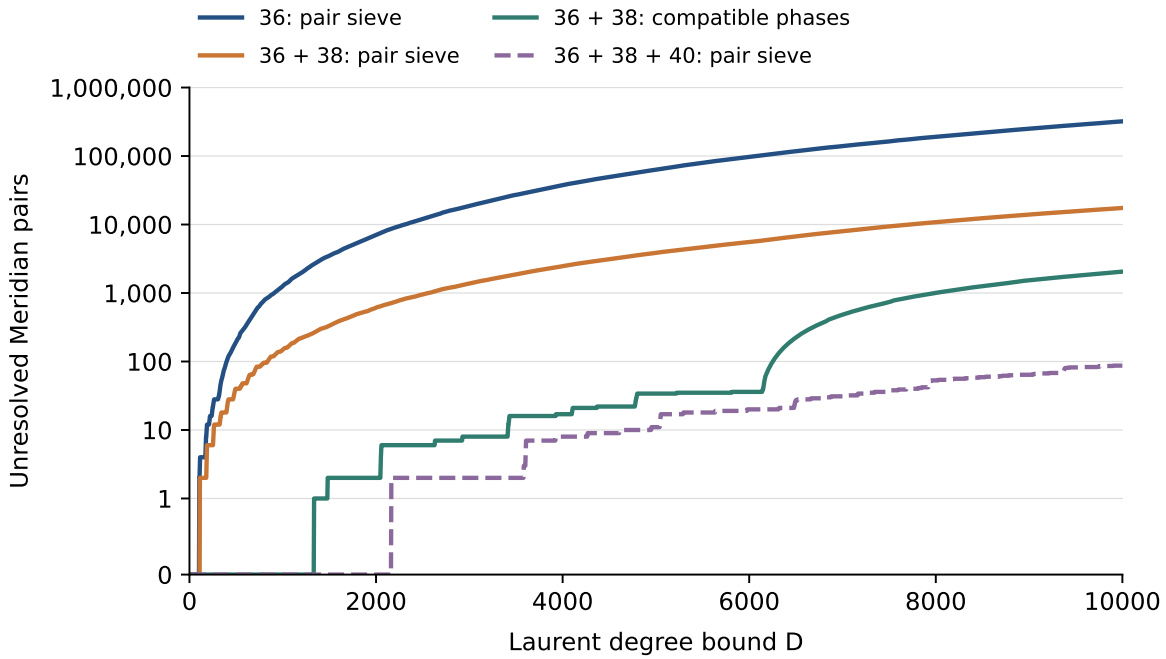


Figure 12. Exact unresolved-pair profiles in the degree-10,000 inventory. The phase-compatible sieve through period 40 has no survivors in the entire displayed range. Lines join discrete even-degree bounds. The figure is a record of the finite certificate, with no extrapolation to larger degrees.

numerical-order verification in Section 9 is a separate computation with its own coverage proof and execution record.

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9. Fast verification with bounded working storage

9.1. **The completed comparison.** The new run [4] checked **550,693,634,649 configurations** and certified that each of the **first 500 billion Markov values in numerical order** is the maximum of exactly one normalized triple. These counts answer different questions: the larger is the workload, including possible competitors and conservative cutoff coverage; the smaller is the guaranteed numerical prefix. The exact 500-billionth value was not identified. The title's rounded 550-billion count refers to the configurations compared.

The indexing supplies compact configuration identities and direct evaluation rules. The implementation uses the equivalent reduced labels p/q, together with the separate root, and compressed Vieta recurrences. It first places configurations in certified magnitude bands, then compares eight-byte residues within each band. Full integers are reconstructed only for repeated residues. Completed key files are removed before the next band is processed. Thus the working store depends on the band size rather than the total number of configurations. No assumption that Meridian maxima are already distinct enters this procedure.

Table 3. Completed comparison and certified numerical coverage. Overlapping band emissions are not counted as new configurations.

Certified numerical-value prefix	500,000,000,000
Inner configurations below the cutoff	500,000,211,949
Included configurations compared	550,693,634,649
Band emissions, including overlap	553,675,740,207
Committed bands	13,638
Repeated residue groups / exact members	4 / 8
Equal maxima found	0

9.2. **Coverage and why no collision can cross a cleared band.** The finite envelope uses the same height bounds as the earlier fifty-billion proof [5]. For reduced labels $0 < p \le q$,

$$m_{p/q} \le F_{2(p+q)-1}, \qquad m_{p/q}^2 \ge 4\,5^{p+q-2}.$$

Including the root, the number of configurations of height at most $t \ge 2$ is $C(t) = 1 + \frac{1}{2}\sum_{j=1}^{t} \varphi(j)$, where φ is Euler's totient function. The exact choices are

$$K = 1{,}813{,}799, \qquad B = F_{3{,}627{,}597}, \qquad L = 2{,}169{,}257,$$
$$C(K) = 500{,}000{,}211{,}949, \qquad C(L) = 715{,}177{,}519{,}497.$$

Every inner configuration has maximum at most B. The exact inequalities $4\,5^{L-2} \le B^2 < 4\,5^{L-1}$ ensure that every competitor with maximum at most B has height at most L. This argument counts configurations before making any claim of unicity.

To schedule the comparisons, 162 cones partition the remaining fraction labels after the 163 configurations through height 32. Exact positive-matrix bounds assign each label an interval containing $\log(3m_{p/q})$; the coefficients are rounded outward at scale 2^{-32}. The bands cover from zero through a certified upper endpoint beyond $\log(3B)$, without gaps. Every configuration whose interval intersects a band is emitted there, including intervals crossing a boundary. The frozen cone certificate and integer inequalities are checked before use.

If two maxima were equal, their actual logarithms would occupy the same band. Both enclosing intervals would intersect that band, and both exact residues modulo 2^{64} would therefore enter

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the same comparison. All keys in a band are partitioned into 32 complete buckets and sorted. Every member of every repeated-key group is recovered by replaying the band and is compared exactly. Only then is the band committed and cleared. Consequently, clearing its keys cannot hide an equality across a boundary. Neither random-residue assumptions nor family-specific collision exclusions are used to discharge a repeated group.

All four repeated groups contained two configurations. Independent integer evaluators agreed on their values, and every pair had unequal maxima. The complete inner population and the outer envelope therefore establish the first-500-billion conclusion in Table 3.

9.3. **Completion timestamp, elapsed work, CPU time, and storage.** The first retained band began on **8 November 2026 at 14:55:01.580398 UTC**; the last finished on **9 November 2026 at 15:50:31.046707 UTC**, or **09:50:31.046707 CST** in St. Louis. The independent final replay finished at 15:51:24.981257 UTC; verification and replay of a fresh portable extraction finished at 15:57:38.529312 UTC. These execution timestamps are separate from the manuscript's release date.

Table 4. Measured resources for the retained production chain. The run used eight workers. GiB denotes 2^{30} bytes.

Measurement	Recorded amount
Active elapsed band processing	54,470.884742 s **15 h 07 min 50.885 s**
Summed native-child CPU time	264,075.377305 s 73 h 21 min 15.377 s
Calendar span, first to last band	89,729.466309 s 24 h 55 min 29.466 s
Maximum recorded temporary key files	**754,846,136 bytes** 719.877 MiB; 0.703 GiB
Maximum recorded generator resident memory	28,672 KiB (28 MiB)
Final compressed recovery archive	27,168,031 bytes (25.909 MiB)
Independent final replay	17.478274 s
Portable extraction verification and replay	20.620908 s

The under-16-hour figure is *active elapsed processing*, not total CPU time: workers run concurrently. Summed child-process CPU includes 46,077.170549 seconds of sorting/scan CPU, which must not be added again. Parent-process CPU was not separately totalled. The retained stage wall times were 31,103.262061 seconds for generation, 15,851.021578 for partitioning, 7,273.870989 for sorting and scans, and 47.949598 for exact-group resolution. The overall band time also includes controller bookkeeping.

Active band time excludes pauses, restoration, failed or lost attempts, and the later audits. Recorded execution windows occupied 17.160 hours of the 24.925-hour calendar span. An environment loss required regeneration of 23,436,403,353 configurations; the corresponding 0.827 hours of lost band work are recorded separately, not counted twice in the retained chain.

The storage peak in Table 4 counts the logical temporary key files, including a shard temporarily duplicated during partitioning. It excludes receipts, source files, backup copies, filesystem overhead, and unrelated work. The generator memory peak covers that native process only, not the combined memory of sorting workers and controller. A continuous whole-workspace disk or aggregate-memory

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peak was not recorded. These scopes make the low-storage claim reproducible without turning a component measurement into a whole-machine bound.

9.4. **Replay and retained evidence.** The final audit checked the frozen sources, all 13,638 linked receipts, band boundaries, counts, bucket accounting, and retained candidate bytes. A separate checker recomputed the numerical envelope and reconstructed the candidate maxima with polynomial recurrences. It also regenerated each repeated-key band with the frozen native engine. A fresh extraction verified all 14,746 members listed in the recovery manifest and replayed the same completed chain. This is a complete receipt-and-candidate replay, not a second generation of every discarded singleton key.

The source accompanying this edition includes the final evidence and a read-only reconciliation of its counts, timings, and resource maxima. It also includes the full frozen archive [4]:

`v20_Band_500_Billion_Recovery.zip`

Its SHA-256 and the final receipt head are, respectively,

`faf79da73bf59110967ae92da3f6933e96383df5fd989724cdba7791294827cc`

`4642f18be7adf418d5399ffa87825dc2183dae6808ef7516e0dd0b1d704b9b6d`

The replay is an independent implementation within the project, not external peer review. Extending the numerical target requires a separate completed and audited run; no trillion-value result is claimed here.

9.5. **Earlier computations and their dates.** The earlier fifty-billion verification [3, 5] completed on **28 October 2026**. Its recorded execution ran from 03:32:57 to 05:38:34 CDT (08:32:57.096455–10:38:33.644612 UTC), including the final audit. Two coordinator invocations occupied 7,492.102807 seconds; with the original ten-billion anchor, the cumulative recorded execution was **2 h 18 min 50.980 s**. It compared 54,940,210,994 new retained configurations, resolved all 346 repeated primary-residue pairs, and found no equal maxima. Its cutoff was $F_{1,147,147}$. The dated version-4 supplement preserves its coverage proof, hardware, stage timings, and audit scope. Those historical timings are not part of the new run.

The earlier Claude-assisted search in the working draft [1], pp. 38–40, reports 9,581,528,580 sorted triples against $10^{100,000}$ in **24 minutes** on a two-core machine with 7 GB of memory. All 21 repeated-residue pairs separated under a second modulus. Its public announcement [2] is dated **23 October 2026, 09:54:57.771 CDT** (14:54:57.771 UTC) and uses the wording "9.3 billion." This dates the announcement, not execution. The original execution timestamps and floating-point cutoff certificate were not recovered; this remains a historical report, separate from the later certified numerical prefixes.

10. What remains to prove

The organization now separates three questions. The address bijection settles coverage of configurations. Fixed-degree ordering and the two collision windows restrict equality of maxima. Complete residue witnesses settle the remaining comparisons for the degree-bounded family in Theorem 8.1. The new computation compares more than 550 billion configurations and excludes every competitor for the first 500 billion values in numerical order. It also demonstrates the practical value of the indexing: the comparison proceeds in bounded batches, with 15.131 hours of retained active processing and a 0.703-GiB recorded temporary key-file peak.

For unrestricted unicity the unresolved pair has different Meridian degrees, the same modulo-27 class, and different Farey numerators. It must satisfy both its phase rectangle and its numerator–denominator window. A proof must exclude those candidates uniformly as the degrees and numerators

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grow. Enlarging a finite certificate alone does not supply that step. Likewise, an invertible corridor recurrence or a long modular period does not by itself prevent an earlier match of one scalar residue.

Corridors offer a way to formulate the next comparison with explicit entrance states and independently chosen wall lengths. Their positive corrections are exact and telescope; the common-prefix snake calculation offers a different reduction to a quadratic in primitive counts. Neither mechanism has yet been shown to cover every unrestricted pair. The purpose of retaining their precise hypotheses is to make the remaining uniform statement identifiable, rather than hide it inside an induction.

Data and code. The editable source accompanying this edition includes all figure assets, the catenary and landscape figure generators, a source and figure inventory, and checks used for this rewrite. The separate degree reproducibility package contains the full computational evidence cited in Section 8. The new run's final evidence, resource reconciliation, and read-only editorial replay are included with the source. Its frozen recovery archive contains the complete retained receipt chain, candidate identities, executables, source hashes, and independent checker [4]. The earlier fifty-billion proof and timing supplement remains a dated historical companion; its full frozen archive is cited separately [3, 5]. The corridor derivations are developed further in the supplementary research record [9]. Numerical drawings are explanatory; the arithmetic claims use the displayed identities or complete certificates.

Author contributions and AI assistance. The author conceived and directed the research program, including the representation by reciprocal powers and geometric descent, the two-dimensional host–phase landscape, the Meridian indexing and alternative tree, the six-piece snake-graph decomposition, the Fibonacci-matrix formulation and normalization argument, and the corridor observation. The author proposed using these structures for coverage, comparisons and verification, and specified the questions pursued at each stage. The Aigner monotonicities are established results; their prior proofs are credited in Section 7. The Fibonacci-matrix normalization argument presented there is the author's; the related earlier normalization result is cited for comparison, and the proof is given in full from Fibonacci identities.

Generative AI provided substantial mathematical and computational assistance, not only language editing. The development record identifies ChatGPT (OpenAI), with the recorded GPT-6.0 Astra and GPT-6.1 Sol model labels and reasoning settings, and Claude 5.5 Opus Extra (Anthropic) in the preceding geometric-descent work; exact backend revision identifiers were not retained. Within the author's framework, these tools assisted with detailed derivations, examples and counterexamples, proof construction and revision, verifier and independent-checker code, figures, and manuscript drafting. ChatGPT also assisted with the opening explanation and the catenary and landscape figures. The project's separate computational checks are algorithmic audits, not external peer review. The author is responsible for the claims, references and final submission.

Funding and competing interests. No funding was received. The author declares no competing interests.

References

[1] E. F. W. Heffern, *Geometric Descent on the Markov Surface*, working draft v32, archived October 2026, pp. 38–40, historical computational reports. The earlier search was assisted by Claude.

[2] E. F. W. Heffern, public LinkedIn announcement of a 24-minute Markov-number search, 23 October 2026, 14:54:57.771 UTC, activity 7519411048115003392. Platform metadata and the source URL are retained in the verification supplement's evidence packet; see the public announcement.

[3] E. F. W. Heffern, *Markov 50 Billion Verification Package*, version 1, completed and sealed 28 October 2026; public derivative version 1.1 in *Computational supplements for Meridians on the Markov Surface*, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

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[4] E. F. W. Heffern, *Certified magnitude-band, residue-first verification of the first 500 billion Markov maxima*, completed 9 November 2026. Final evidence: `v1_Band_500_Billion_Final_Evidence.json`; frozen recovery archive: `v20_Band_500_Billion_Recovery.zip`. The comparison includes 550,693,634,649 configurations. The final evidence and reconciliation accompany this manuscript's source; the recovery archive is identified by SHA-256 in Section 9.4.

[5] E. F. W. Heffern, *Computational chronology and proofs for the fifty-billion verification*, supplement to the present manuscript, version 4, 7 November 2026.

[6] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013. doi:10.1007/978-3-319-00888-2.

[7] İ. Çanakçı and R. Schiffler, *Snake graphs and continued fractions*, European J. Combin. **86** (2020), 103081. doi:10.1016/j.ejc.2020.103081.

[8] E. F. W. Heffern, *Meridian Degree 10,000 Reproducibility Package*, computational supplement, 31 October 2026; address-convention revision, 7 November 2026. Original computational release in *Computational supplements for Meridians on the Markov Surface*, version 1.0, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

[9] E. F. W. Heffern, *Markov Meridian Residue Supplement*, version 1.18, and *Corridor-to-Corridor Induction*, version 1.3, unpublished supplementary research records, 7 November 2026. The corridor results used here are proved in Section 6.

[10] C. Lagisquet, E. Pelantová, S. Tavenas and L. Vuillon, *On the Markov numbers: fixed numerator, denominator, and sum conjectures*, Adv. Appl. Math. **130** (2021), 102227. doi:10.1016/j.aam.2021.102227. Preprint: arXiv:2010.10335.

[11] K. Lee, L. Li, M. Rabideau and R. Schiffler, *On the ordering of the Markov numbers*, Adv. Appl. Math. **143** (2023), 102453. doi:10.1016/j.aam.2022.102453.

[12] G. McShane, *Convexity and Aigner's conjectures*, preprint (2021), arXiv:2101.03316.

[13] M. Rabideau and R. Schiffler, *Continued fractions and orderings on the Markov numbers*, Adv. Math. **370** (2020), 107231. doi:10.1016/j.aim.2020.107231.

[14] G. Musiker, *A conjectured combinatorial interpretation for Markov numbers*, REACH memorandum, 27 June 2002. Original memorandum; REACH archive.

[15] A. Itsara, G. Musiker, J. Propp, and R. Viana, *Combinatorial interpretations for the Markov numbers*, REACH manuscript, 10 January 2003, Theorem 3.1. Original manuscript.

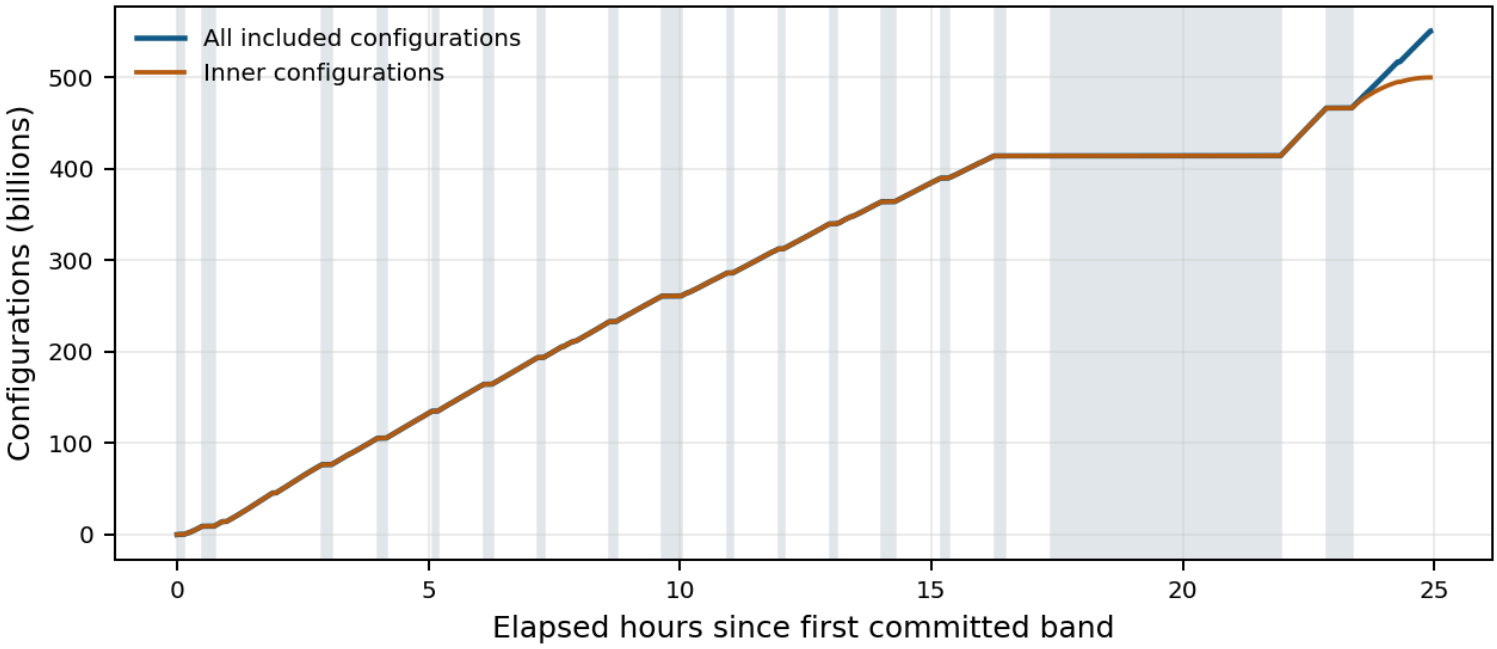
[16] J. Propp, *The combinatorics of frieze patterns and Markoff numbers*, arXiv:math/0511633, first posted 25 November 2005; published in *Integers* **20** (2020), A12.

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Original figure: v1_Band_500_Billion_Progress.png



500 billion Markov maxima

Completed certified magnitude-band, residue-first verification

Prepared for Elleard Felix Webster Heffern. Production began on 8 November 2026; all completion evidence is dated below. This report records a finite computational result with reproducible source, coverage, and candidate evidence.

Certificate	Verified value
Certified numerical scope	At least the first 500,000,000,000 maxima
Required inner configurations	500,000,211,949
All included configurations checked	550,693,634,649
Band emissions, including overlap	553,675,740,207
Committed band receipts	13,638
Repeated residue groups / exact members	4 / 8
Exact collisions found	0
Completion status	complete_unique_first_target

The inner count is exactly **500,000,211,949**. The larger included count also covers possible competitors. All magnitude bands reach the frozen endpoint **7,497,479,023,590,985** at fixed-point scale 2^{32}, without gaps. Each repeated 64-bit residue group was resolved by exact arithmetic.

The envelope uses K = 1,813,799, L = 2,169,257, and B = $F_{3,627,597}$. Exact initialization and final replay checks establish the population and envelope bounds. Their proofs are retained with the recovery package.

Scope: this certifies uniqueness of at least the first 500 billion numerical values without identifying the exact 500-billionth value. It does not prove universal Markov unicity.

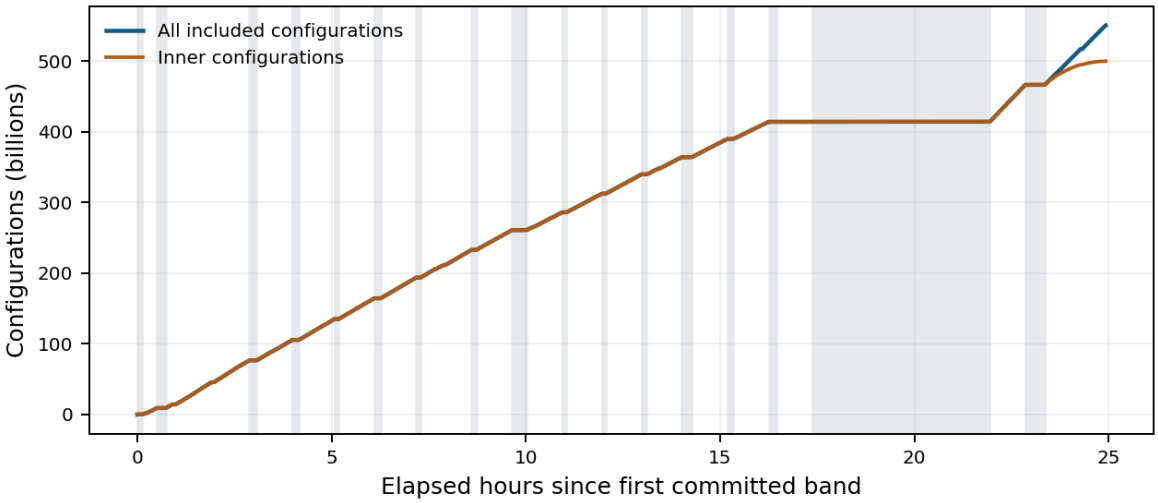
Last committed band: **2026-11-09T15:50:31.046707+00:00** UTC;
2026-11-09T09:50:31.046707-06:00 America/Chicago.

Production time and interruptions

Measurement	Elapsed time
Retained committed band work	15.131 h
Calendar span: first to last committed band	24.925 h
Union of recorded execution windows	17.160 h
Outside recorded execution windows	7.765 h
Known lost work, subsequently regenerated	0.827 h
Independent final receipt/candidate replay	17.478 s
Portable archive verification and replay	20.621 s

Retained committed band time excludes paused/offline gaps, wrapper/audit/publication overhead, and failed or lost attempts. Recorded execution windows also include known lost work. Residual time outside recorded windows includes pauses, environment loss, restoration, and unrecorded activity; it is not all classified as idle time.

An execution-environment loss made one locally audited session unavailable before durable publication. Its **23,436,403,353 configurations** were regenerated from the saved 414,230,269,527 checkpoint. That lost session is recorded separately; neither its coverage nor its compute time is silently added to the retained chain.



Retained receipt chain only. Grey spans mark gaps of at least five minutes outside recorded execution windows. Lost work is not drawn as retained progress.

Replay, evidence, and reproducibility

The final audit validated every source hash, receipt link, range boundary, count, and retained candidate file. A separate checker independently recomputed the envelope and exact candidate maxima using polynomial recurrences. It regenerated every repeated-key band with the frozen native engine and compared all candidate identities and multiplicities.

The replay covers the entire retained receipt/candidate chain. It is **not a second generation of all singleton keys**. A fresh portable extraction verified every archive member and successfully audited the same completed chain. The accompanying JSON records exact checks, timings, paths, and hashes.

Production stage wall time	Retained receipts only
Native generation	8.640 h
Partitioning	4.403 h
Sorting and scans	2.021 h
Exact-group resolution	0.013 h

Historical benchmark: 1,069,268,786 configurations in 128.845692809 seconds. Its 16.74-hour projection used a separate benchmark build and is not reported as the completed production time. The measured production totals above include the broader competitor workload and retained band operations.

Frozen manifest SHA-256

`7b2dc406ceeb425fa526a584e834712a57562d3ab9936d6fc82c6f8974e67db5`

Final receipt head SHA-256

`4642f18be7adf418d5399ffa87825dc2183dae6808ef7516e0dd0b1d704b9b6d`

Native engine/spool SHA-256

`1a06b36371c624282dcaea2fde6df94bf2be9a8b5db34417084a85eae8ef97ae`

Frozen control.py SHA-256

`a3cbf27ed5e8cf1a628599a6a2e2d828bff0fa1824d3acfe976d5f31a8c333f2`

Recovery archive SHA-256

`faf79da73bf59110967ae92da3f6933e96383df5fd989724cdba7791294827cc`

Recovery file: v20_Band_500_Billion_Recovery.zip. Extract to a fresh directory, verify RECOVERY_MANIFEST.json, restore executable permissions, then run `python3 project/control.py audit --run run`. The independent checker and all exact-candidate evidence are included.

Report generated 2026-11-09T15:57:56.646502+00:00 UTC. All timestamps and unrounded measurements are preserved in v1_Band_500_Billion_Final_Evidence.json.

Hyperbolic Collision Invariants and Certified Odd-Even Tiers toward Markoff Unicity

Completion and release history

The deposited 35-page version 6 manuscript is dated 30 September 2026. It is preserved unchanged. Its earlier ResearchGate PDF is dated 19 August 2026 and was uploaded 20 August 2026. These dates identify different drafts.

Zenodo publication-date field: 2026-11-09

Zenodo record created: 2026-11-09 13:38:38 CST / 2026-11-09 19:38:38 UTC

Zenodo record last modified: 2026-11-09 13:38:38 CST / 2026-11-09 19:38:38 UTC

The publication-date field is the date supplied in the deposit. The API created/modified timestamps are repository metadata, not a separately attested first-view time. Full fractional timestamps are retained in the metadata archived in Part C.

https://zenodo.org/records/24091908

https://doi.org/10.5281/zenodo.24091908

Earlier structural preprint: A slope-scale factorization for prime-power and twice-prime-power Markoff unicity, ResearchGate, 18 June 2026, as cited in the deposited v6 manuscript. The cited DOI was not independently retrievable during this compilation.

https://doi.org/10.13140/RG.2.2.23525.56800

Earlier Hyperbolic Collision manuscript: ResearchGate confirms upload on 20 August 2026; the public PDF is dated 19 August. The Zenodo deposit note explicitly separates that earlier draft from the 30 September v6 deposited on 9 November.

https://www.researchgate.net/publication/412818702_Hyperbolic_Collision_Invariants_and_Certified_Odd-Even_Tiers_toward_Markoff_Unicity

https://doi.org/10.13140/RG.2.2.25111.33442

A separate LinkedIn announcement for this manuscript was not located.

Complete public Zenodo description

This preprint develops quantitative conditions that any two distinct ordered Markoff triples with the same largest entry must satisfy. A four-point factorization leads to coupled quadratic equations and a coprime allocation of prime-power factors. Hand arguments and exact finite certificates yield the collision bounds F^4, E^4 >= 300000M + 200000, with the corresponding normalization for even maxima. These bounds give sufficient conditions for uniqueness; the general Markoff unicity conjecture remains open. The constants are coefficients of collision inequalities, not ranges of numbers checked.

The work builds on classical partial results, including Ying Zhang's 2007 congruence and uniqueness criteria. In particular, its even-maxima analysis uses Zhang's congruence M congruent to 2 modulo 32 together with Srinivasan's determinant splitting. The manuscript distinguishes these antecedents, Button's unitary-factor criterion, and other earlier results from its quantitative additions.

Publication-history note from the author: An earlier version of this manuscript was first posted on ResearchGate on 20 August 2026, under DOI 10.13140/RG.2.2.25111.33442 (https://doi.org/10.13140/RG.2.2.25111.33442). I was not familiar with Zenodo at the time and did not have access to post on arXiv. I am depositing the later, 30 September 2026 version here to provide a persistent archive together with its editable source and computational certificates.

This Zenodo deposit is dated 9 November 2026. It preserves the 35-page version 6 manuscript dated 30 September 2026 without changes. The earlier ResearchGate PDF is dated 19 August 2026 and its public record gives an upload date of 20 August 2026. The original record is https://www.researchgate.net/publication/412818702_Hyperbolic_Collision_Invariants_and_Certified_Odd-Even_Tiers_toward_Markoff_Unicity.

Files: manuscript PDF; matching editable LaTeX source and figure; the original reproducibility supplement containing six exact-arithmetic proof programs, certificate tables, recorded outputs, figure source and parameters, and SHA-256 checksums; and a separate deposit note. The supplement's 80 manifest entries were checked for this deposit. This is a packaging check, not a new execution of the proof programs. Historical journal-target headings and internal-review filenames do not indicate journal publication or external peer review.

The original ResearchGate license, CC BY-NC-ND 4.0, is retained. Funding: none. Substantive generative-AI assistance is disclosed in the manuscript.

Published files

Original filename / checksum	Bytes
v6_Hyperbolic_Collision_Certificates.zip MD5 5c8e4891824587e538ac55a678c8168f SHA256 c32fbf92b619343b5ac246518949ea945f16c5e8d794f1d0433cfd2ab3f2ee5a	223,701
heffern_markoff_certified_tiers_v6_residual_refereed.pdf MD5 eb0a8c63802541d14f285a8d51d399da SHA256 a7973fa613d32dddf77a90b69b20672b04f5db5f78c9bc372d62cf5c3420de44	616,589
v1_Hyperbolic_Collision_Zenodo_Deposit_Note.txt MD5 df8a51400147f75b80d9014d6698f700 SHA256 597fd700e6f89a53b98ef149d1e29aa73c288880d9e9f1f484bc68eb1219a5fe	3,433
v6_Hyperbolic_Collision_Editable_Source.zip MD5 7f2f76d81f2fc87901ad8fe9364225f5 SHA256 ff238fef47a788c5b2f6e931359893fbb15712ddac3e39ca8c54b1598478189e	59,170

All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

Original: heffern_markoff_certified_tiers_v6_residual_refereed.pdf

Hyperbolic Collision Invariants and Certified Odd–Even Tiers toward Markoff Unicity

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Corresponding author: soldier@alum.mit.edu

30 September 2026

Abstract

A collision would consist of two different ordered Markoff triples with the same largest entry M. We derive arithmetic conditions that every such collision must satisfy. A parametrization by hyperbolic functions factors a classical determinant identity and leads to two coupled quadratic equations. If M is odd, these equations require a factorization $M = FE$ with $F, E > 1$ and $\gcd(F, E) = 1$. Hand proofs and exact finite computations give increasingly strong bounds, ending with $F^4, E^4 \geq 300000M + 200000$. If M is even, division of the collision quantities by four gives the same inequalities for a coprime factorization of $M/2$. These bounds yield explicit sufficient conditions for a Markoff number to be unique. The finite computations cover a proved search domain and apply necessary congruence and continued-fraction conditions. Two different arguments reject the final odd candidate: one uses continued fractions, and the other bounds the integer solutions of two quadratic equations of Pell type. Separately written programs reproduce the computations. The constants measure the strength of collision inequalities; they are not ranges through which Markoff numbers were checked. We identify the classical ingredients and the quantitative additions. Factorizations with increasingly comparable factors remain unresolved, so the general unicity conjecture is not proved.

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Keywords. Markoff numbers; Frobenius unicity conjecture; unitary divisors; continued fractions; exact finite certificate.

1 Introduction

A *Markoff triple* is a positive integral solution of

$$x^2 + y^2 + z^2 = 3xyz. \tag{1.1}$$

The equation was introduced by Markoff [12]. An integer occurring in such a triple is a Markoff number. The unicity conjecture posed by Frobenius [10] asserts that a Markoff triple is determined, up to permutation, by its largest entry. The conjecture remains open. It is known for prime powers, twice prime powers, and several other arithmetic families; see [2, 3, 4, 11, 16, 6, 14, 1].

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Throughout, an *ordered occurrence with maximum* M means a triple (a, b, M) with $a \le b \le M$. An *ordered collision* is a pair of distinct such occurrences with the same maximum. We call M *unique* when it has only one ordered occurrence, equivalently when its Markoff triple is determined up to permutation.

Reading guide and notation. The argument uses algebra, calculus, prime factorization, and congruences. A congruence $u \equiv v \pmod{n}$ means that $n \mid u - v$. Integers are *coprime* when their greatest common divisor is one. A complete prime-power block of an integer is the factor p^e containing its entire power of a prime p. The notation $\mathrm{ord}_p(z)$ records the exponent of p in z; this is also called its p-adic valuation. A *quadratic residue* is a residue that is a square modulo the stated prime.

The words *packet*, *floor*, and *tier* will have specific meanings here. A packet is one of the displayed quadratic equations together with its integer variables and divisibility conditions. A floor is a proved lower bound for a product of its coordinates. A tier is the resulting level of the collision bound. A *relaxation* keeps only some necessary conditions, so it may include tuples that do not come from Markoff triples. Excluding every tuple in this larger set also excludes actual collisions. We identify explicitly which steps use a finite computer calculation.

The hyperbolic functions need no background beyond exponentials: $\cosh t = (e^t + e^{-t})/2$ and $\sinh t = (e^t - e^{-t})/2$. The notation log means the natural logarithm. Later, a continued fraction $[c_1; c_2, \dots, c_r]$ means $c_1 + 1/(c_2 + 1/(\cdots + 1/c_r))$. Its digits c_i are called partial quotients. Requiring the last digit to exceed one makes the finite expansion unique.

This paper studies the arithmetic forced by an ordered collision. Let (a, b, M) and (a', b', M) form an ordered collision. For odd M, the central chain is

$$\boxed{\text{hyperbolic four-point factorization}} \Longrightarrow \boxed{AB = M^2C} \Longrightarrow \boxed{M = FE \text{ unitary}}$$
$$\Longrightarrow \boxed{\text{two defect equations}} \Longrightarrow \boxed{\text{residual crossed-Pell coupling}}$$
$$\Longrightarrow \boxed{\text{four certified odd floors}}.$$

The first step supplies a geometric factorization and interpretation of Srinivasan's identity. The direct verification of that identity and all later algebraic reductions use integers.

The following table separates each product floor from the coefficient it gives in the collision inequality.

tier	defect floor	provenance	quartic consequence
99	33	hand proof	$F^4, E^4 \ge 99M + 66$
360	120	exact local enumeration	$F^4, E^4 \ge 360M + 240$
201411	67137	companion-prime certificate	$F^4, E^4 \ge 201411M + 134274$
300000	100000	crossed-Pell/continuant certificates	$F^4, E^4 \ge 300000M + 200000$

Table 1: The four certified odd tiers. The coefficient is three times the defect-product floor.

For even maxima, dividing the collision quantities by four gives the following parallel bounds.

In the odd table, 120 is sharp for the local relaxation and 67137 is sharp after the companion-prime condition. In the even table, 72 and 1014 are sharp for their corresponding relaxations. In both parities, 100000 is a certified cutoff, with no claim of sharpness. None of these numbers is a range through which Markoff unicity has merely been checked.

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tier	defect floor	provenance	quartic consequence
216	72	hand proof	$F^4, E^4 \geq 216M + 144$
3042	1014	companion-prime certificate	$F^4, E^4 \geq 3042M + 2028$
300000	100000	continuant-root certificate	$F^4, E^4 \geq 300000M + 200000$

Table 2: The three certified tiers for even M, after normalization by $M_1 = M/2$.

The factorization needed for the argument keeps every complete prime-power block on one side. This is expressed by a unitary divisor. A divisor $d \mid N$ is *unitary*, written $d \parallel N$, when $\gcd(d, N/d) = 1$. Define

$$u_*(N) = \max\{d : d \parallel N,\ d \leq \sqrt{N}\}. \tag{1.2}$$

The set is nonempty because it contains 1.

Theorem 1.1 (Main collision and unicity theorem)**.** *If an odd Markoff number $M > 1$ admits an ordered collision, then there are coprime unitary factors $F, E > 1$, $M = FE$, for which*

$$F^4, E^4 \geq 300000M + 200000. \tag{1.3}$$

If an even Markoff number $M > 2$ admits an ordered collision, then $M_1 = M/2 \equiv 1 \pmod{16}$ has coprime proper unitary factors $F, E > 1$, $M_1 = FE$, satisfying the same bound (1.3)*. Consequently,*

$$u_*(M)^4 < 300000M + 200000 \quad \Longrightarrow M \text{ is unique} \quad (M \text{ odd}), \tag{1.4}$$

$$u_*(M/2)^4 < 300000M + 200000 \quad \Longrightarrow M \text{ is unique} \quad (M \text{ even}). \tag{1.5}$$

The corresponding weaker statements hold at every tier in Tables 1 and 2, with their stated proof provenance.

Corollary 1.2 (Prime-power tail)**.** *Let $M = kq^e$ be an odd Markoff number, where q is prime,* $\gcd(k, q) = 1$*, and $e \geq 1$. For any*

$$\mathcal{C} \in \{99, 360, 201411, 300000\},$$

the condition

$$k^3 \leq \mathcal{C}q^e \tag{1.6}$$

guarantees that M is unique, provided the tier with coefficient $\mathcal{C}$ is invoked with its stated hand or computer-assisted provenance.

The odd collision assertion of Theorem 1.1 is proved in Theorem 6.10, and its even counterpart in Theorem 7.4. Their unicity implications are Theorems 8.1 and 8.2. Corollary 1.2 follows from Theorem 8.3 by taking a single prime-power tail.

Historical position and quantitative comparison

Classical partial results are usually formulated as arithmetic-shape criteria. Baragar proved unicity when one of $M, 3M - 2, 3M + 2$ is a prime, twice a prime, or four times a prime [2]. Button proved the prime case and later developed a principal-ideal and continued-fraction method [3, 4]; Lang and Tan gave an independent proof for prime powers [11]. On the companion side, Zhang proved unicity

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when one of $3M \pm 2$ is a prime power, four times a prime power, or eight times a prime power [16]. Chen and Chen subsequently treated $3M \pm 2 = kp^\ell$ for $1 \le k \le 10^{10}$ [6].

Srinivasan gave an elementary treatment of the prime-power and twice-prime-power cases and further criteria in terms of ambiguous classes and the prime divisors of $9M^2 - 4$ [14]. More specifically, in the present sign convention, Srinivasan's identity [14, Lemma 2.1] is exactly

$$(bb' - aa')(ba' - ab') = M^2(ab - a'b'),$$

that is, $AB = M^2C$. Her analysis also contains the underlying coprime allocation of complete prime-power blocks produced by a hypothetical nonunique maximum [14, Theorem 5.1]. Thus neither the determinant identity itself nor, in substance, the existence of a proper unitary split is asserted here as new.

The companion-discriminant allocation formulated below likewise has a classical genus-theoretic antecedent. Srinivasan's ambiguous-class analysis and Legendre-symbol criteria detect the same mutual quadratic-residue condition between complementary factors of $9M^2 - 4$ [14, Theorems 5.4 and 6.1]. The elementary residual equation below is used as a bridge from the present collision variables to that condition; priority is not claimed for the unitary or residue obstruction. The additional bounded crossed-Pell reconstruction couples this residual allocation back to the primitive collision slope and supplies a second exact rejection of the final odd certificate row.

For even Markoff numbers, Zhang proved the congruence $M \equiv 2 \pmod{32}$ and companion-number criteria [16, Theorems 1–2]. Srinivasan's treatment of twice prime powers already uses the determinant identity with its forced divisibility by 4, and her companion lemma states that every odd prime divisor of $9M^2 - 4$ is $1 \pmod 4$ [14, Theorem 2.2 and Lemma 4.2]. The even section below combines these antecedents with the complementary conics and an exact coefficient-6 enumeration; it does not claim priority for the congruence, the coprime splitting in substance, or the companion-prime condition.

Geometric reformulations also predate the present work. Tornero's fixed-M hyperbola

$$H_M : \quad x^2 - 3Mxy + y^2 + M^2 = 0$$

is exactly the conic used below [15]; a recent reformulation uses integral necklaces [9]. The narrower role of the hyperbolic viewpoint here is the closed-form Q^x parametrization and the explicit four-point factorization of the three terms in Srinivasan's identity.

Recent related work. O'Sullivan gives geometric growth and approximation estimates for real Markov trees [17, Section 3 and Proposition 8.1]. Yasutomi proves injectivity for substitution parameters $a \ge 2$, $b = a + 1$, and a partial ordering result in the classical case $a = 1$, $b = 2$ [18, Theorems 1.3 and 1.5(2)]. The full classical case is outside his injectivity theorem. Banaian extends Aigner's ordering results to generalized k-Markov numbers [19, Theorem 4]; Gyoda surveys the associated arithmetic, combinatorial, and geometric descriptions [20]. Our question is different: given a hypothetical classical collision, how large must each factor in its forced coprime split be? The quartic bounds in Theorem 1.1 address that question. They are not a replacement for these ordering or approximation results, and the proof does not depend on them.

Relation to the author's prior preprint

The author's preprint *A slope–scale factorization for prime-power and twice-prime-power Markoff unicity* (ResearchGate, 18 June 2026; https://doi.org/10.13140/RG.2.2.23525.56800) developed the

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Q^x parametrization, the A, B, C, D identities, the collision line, the slope–scale formulas, the unitary allocation, and the coupled defect equations. The present article substantially extends and supersedes that preprint by isolating the complementary four-point conics and, principally, by establishing the four odd defect floors $33, 120, 67137, 100000$, the normalized even floors $72, 1014, 100000$, their unitary-gap consequences, the residual crossed-Pell reconstruction, and self-contained exact-arithmetic certificates. Claims of contribution below concern these quantitative extensions and their certificates, not the previously published or publicly circulated structural machinery.

The direct coefficient-1 antecedent of the present unitary-gap inequalities is Button's unitary-factor criterion [4, Corollary 6]: an odd Markoff number M is unique if every coprime factorization $M = rs$, $r < s$, satisfies $s > r^3$. In the notation of (1.2), this is equivalent to

$$u_*(M)^4 < M,$$

because the possible smaller factors r are precisely the unitary divisors below $\sqrt{M}$. Theorem 1.1 strengthens this coefficient-1 quartic gap to

$$u_*(M)^4 < \mathcal{C}M + \frac{2\mathcal{C}}{3}, \qquad \mathcal{C} \in \{99, 360, 201411, 300000\}.$$

Thus the strongest certificate raises the coefficient of M in this particular collision obstruction from 1 to 300000, while retaining an exact positive additive term. For $M = kq^e$, the corresponding cubic comparison changes from the coefficient-1 condition $k^3 < q^e$ to $k^3 \le \mathcal{C}q^e$. At fixed q^e, the final tier therefore expands the cofactor scale in this criterion by $300000^{1/3} = 66.943\ldots$.

This is not a blanket improvement on every earlier arithmetic family. Button also recorded, using the then-reported computation through 10^{140}, the separate conclusion for $M = kp^\ell$ with $1 \le k < 10^{35}$ [2, 4]; that result and the companion-number criteria above are nonnested with the present estimate. Within the present collision framework, however, the progression

$$99 \longrightarrow 360 \longrightarrow 201411 \longrightarrow 300000$$

is a strict sequence of certified lower-bound improvements. From the hand tier to the final tier the quartic coefficient increases by $300000/99 = 3030.30\ldots$, and the permitted k-scale in Corollary 1.2 increases by $(300000/99)^{1/3} = 14.470\ldots$ at fixed q^e. Against the antecedents above, the distinctive contribution of the present article is the quantitative strengthening: the explicit odd and normalized-even defect-product floors, their unitary-gap consequences, and transparent dual exact-arithmetic certificates, now including an independent residual crossed-Pell audit of the final odd row. The hyperbolic factorization is retained as a geometric interpretation and derivational bridge; it is not asserted as the first appearance of either the fixed conic or the identity $AB = M^2C$.

The comparison with Button is strict at the level of sufficient unitary-gap criteria. The Markoff number $51641 = 113 \cdot 457$ does not satisfy Button's condition $u_*(M)^4 < M$, whereas it satisfies the coefficient-201411 criterion proved here; see Example 8.5. This is a genuine strengthening of that collision obstruction, not a claim that the uniqueness of 51641 was unknown by all other methods.

The odd and even collision sectors are both treated, but their certified inequalities still leave increasingly balanced unitary allocations. No result below proves the full unicity conjecture.

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2 The fixed-M conic and a hyperbolic factorization of Srinivasan's identity

Fix $M > 2$, and put

$$q = \frac{3M}{2}, \qquad R = \sqrt{q^2 - 1}, \qquad Q = q + R. \tag{2.1}$$

Thus $Q^{-1} = q - R$, and if $\theta = \log Q$, then $q = \cosh\theta$, $R = \sinh\theta$. Define

$$f(x) = \frac{M}{2R}(Q^x + Q^{-x}), \qquad g(x) = \frac{M}{2R}(Q^x - Q^{-x}). \tag{2.2}$$

Equivalently,

$$f(x) = \frac{M}{R}\cosh(x\theta), \qquad g(x) = \frac{M}{R}\sinh(x\theta).$$

The curve parametrized below is precisely Tornero's fixed-M hyperbola $H_M : X^2 - 3MXY + Y^2 + M^2 = 0$ [15, p. 596]. Tornero studies its integral points using the recurrence $B_{j+2} = 3MB_{j+1} - B_j$, whose characteristic roots are Q and Q^{-1}. Here we use the corresponding continuous Q^x parametrization to place the two occurrences of an ordered collision on one real component and factor their four-point determinants.

Lemma 2.1 (Fixed-M parametrization)**.** *For every real x,*

$$f(x)^2 + f(1-x)^2 - 3Mf(x)f(1-x) + M^2 = 0. \tag{2.3}$$

The map $x \in \mathbb{R} \mapsto (f(x), f(1-x))$ parametrizes the entire branch with both coordinates positive, once each. The ordering $f(x) \le f(1-x)$ holds precisely for $x \le 1/2$. If $(f(x), f(1-x), M)$ is an ordered Markoff triple with maximal entry $M > 2$, then $0 < x < 1/2$. On this interval the first coordinate increases and the second decreases.

Proof. Write $u = x\theta$. The addition formula gives

$$f(1-x) = \frac{M}{R}\cosh(\theta - u) = qf(x) - M\sinh u.$$

Consequently,

$$(f(1-x) - qf(x))^2 = R^2 f(x)^2 - M^2.$$

Expanding and using $q^2 - R^2 = 1$ proves (2.3).

To show that no positive point is missed, let (X, Y) satisfy the conic with $X, Y > 0$. Completing the square gives

$$(qX - Y)^2 = R^2X^2 - M^2.$$

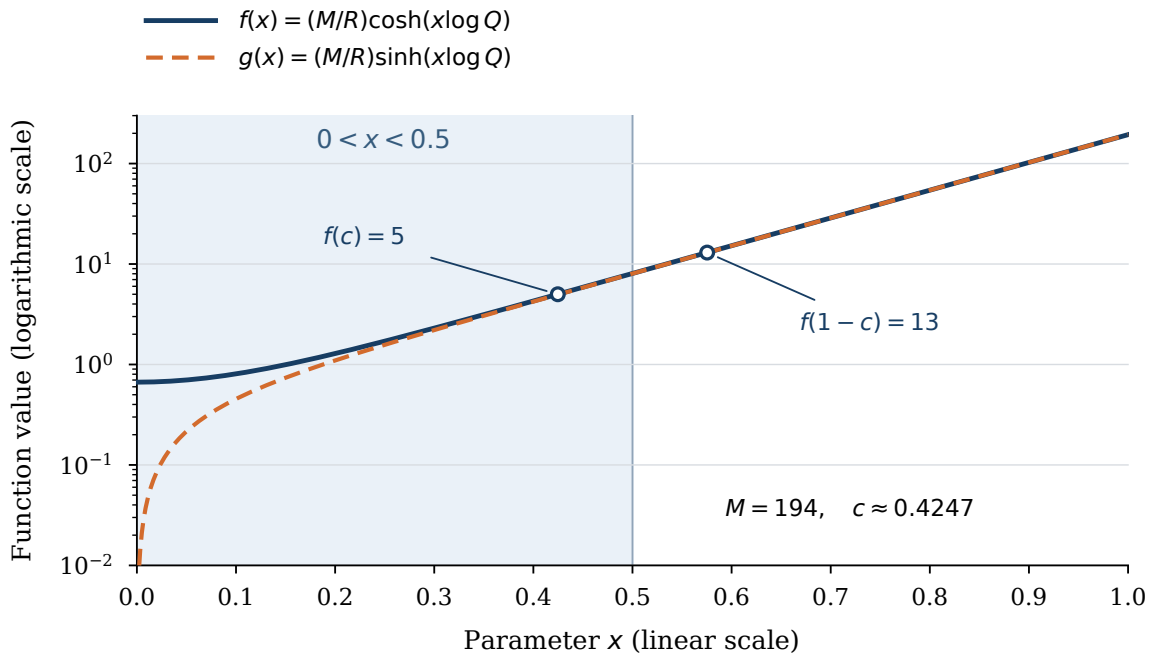
Define $u = \operatorname{arsinh}((qX - Y)/M)$, where arsinh is the inverse of sinh. Then $\cosh u = RX/M$, since both sides are positive. Setting $x = u/\theta$ gives $X = f(x)$ and $Y = qX - M\sinh u = f(1-x)$. This also proves that the parameter is unique.

The function $\cosh t$ is even and increases with $|t|$. Thus $f(x) \le f(1-x)$ is equivalent to $|x| \le |1-x|$, or $x \le 1/2$. If $x \le 0$, then

$$f(1-x) \ge f(1) = \frac{3M^2}{2R} > M,$$

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$g(x) \to 0$ as $x \to 0^+$; zero is outside the logarithmic axis.

Fig. 1 The functions f and g for the Markoff maximum $M = 194$, which occurs in $(5, 13, 194)$. The horizontal axis is linear and the vertical axis is logarithmic. The shaded interval $0 < x < 1/2$ contains the parameter of the smaller entry of an ordered occurrence. The two marked points belong to this one known occurrence: $f(c) = 5$ and $f(1-c) = 13$. They do not represent a collision. The solid and dashed curves become close because $f(x) - g(x) = (M/R)Q^{-x}$, while both approach $(M/(2R))Q^x$; hence their common large-x trend appears straight on these axes. A logarithmic axis displays only positive values. Thus $g(0) = 0$ and the negative values $g(x) < 0$ for $x < 0$ are omitted, and the dashed curve continues below the plotted range as $x \to 0^+$. The finite graph window stops at $x = 1$; the functions themselves are defined on all real x.

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because $R < 3M/2$. Such a point cannot have maximum M. If $x = 1/2$, the two smaller integer entries agree. Pairwise coprimality, proved independently in Lemma 3.3, makes both equal to 1. The equation then gives $M = 1$ or $M = 2$. Hence $M > 2$ requires $0 < x < 1/2$.

Finally, $f'(x) = M\theta \sinh(x\theta)/R > 0$ on this interval, while $\frac{d}{dx} f(1-x) = -M\theta \sinh((1-x)\theta)/R < 0$. These signs prove the monotonicity assertions. □

Let (a, b, M) and (a', b', M) form an ordered collision. After interchanging the two ordered occurrences if necessary, there are

$$0 < c < d < \frac{1}{2} \tag{2.4}$$

such that

$$a = f(c), \quad a' = f(d), \quad b' = f(1-d), \quad b = f(1-c), \tag{2.5}$$

and hence

$$0 < a < a' < b' < b < M. \tag{2.6}$$

Define the three bilinear collision quantities

$$A = bb' - aa', \qquad B = ba' - ab', \qquad C = ab - a'b'. \tag{2.7}$$

The next formulas explain why these three quantities are useful. Put

$$\xi = d - c, \qquad \eta = 1 - c - d. \tag{2.8}$$

In the present sign convention, the identity (2.10) below is precisely Srinivasan's Lemma 2.1 [14]. The following proposition supplies the additional four-point geometric factorization of its three factors.

Proposition 2.2 (Hyperbolic four-point factorization)**.** *The four-point determinants factor as*

$$A = Mg(\eta), \qquad B = Mg(\xi), \qquad C = g(\eta)g(\xi). \tag{2.9}$$

In particular,

$$\boxed{AB = M^2 C}. \tag{2.10}$$

Proof. The identity $\cosh u \cosh v = (\cosh(u+v) + \cosh(u-v))/2$ lets us subtract the products in pairs. It gives

$$\cosh((1-c)\theta)\cosh((1-d)\theta) - \cosh(c\theta)\cosh(d\theta) = R\sinh(\eta\theta),$$

and

$$\cosh((1-c)\theta)\cosh(d\theta) - \cosh(c\theta)\cosh((1-d)\theta) = R\sinh(\xi\theta).$$

Each product of two f-values has the common factor M^2/R^2. Multiplying these identities by that factor gives $A = Mg(\eta)$ and $B = Mg(\xi)$.

For the third difference, the same product-to-sum identity gives

$$\cosh(c\theta)\cosh((1-c)\theta) - \cosh(d\theta)\cosh((1-d)\theta) = \sinh(\eta\theta)\sinh(\xi\theta).$$

After multiplication by M^2/R^2, this is $C = g(\eta)g(\xi)$. The first two formulas now multiply to $AB = M^2C$. □

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Thus Proposition 2.2 gives a hyperbolic factorization of a known arithmetic identity. This factorization was the author's original route to recognizing Srinivasan's identity in the present notation. For later arithmetic use, we also record a direct integral verification.

Proposition 2.3 (Self-contained integral verification)**.** *If both* (a, b, M) *and* (a', b', M) *satisfy the Markoff equation, then*

$$M^2C = AB. \tag{2.11}$$

If

$$D = b^2 - b'^2 + a^2 - a'^2, \tag{2.12}$$

then also

$$D = 3MC, \qquad 9ABC = D^2. \tag{2.13}$$

Proof. Expand the product and collect the terms containing $a'b'$ and ab:

$$\begin{aligned} AB &= (bb' - aa')(ba' - ab') \\ &= a'b'(a^2 + b^2) - ab(a'^2 + b'^2). \end{aligned}$$

The two Markoff equations give

$$a^2 + b^2 = 3Mab - M^2, \qquad a'^2 + b'^2 = 3Ma'b' - M^2.$$

On substitution, the two terms containing $3Maba'b'$ cancel. What remains is $AB = M^2(ab - a'b') = M^2C$.

Next subtract the two Markoff equations. The terms M^2 cancel, leaving $D = 3MC$. Squaring gives $D^2 = 9M^2C^2$. Replace one factor M^2C by AB to obtain $D^2 = 9ABC$. □

The order (2.6) gives $A, B, C, D > 0$. In particular, $B = br + as > 0$ after the displacement notation introduced below, and the other signs follow from the identities. The strict inequality $b < M$ follows from pairwise coprimality: $b = M > 2$ would give a common divisor.

3 Collision line, slope-scale factorization, and unitary allocation

The slope–scale formulas and unitary allocation in this section were previously circulated in the author's earlier slope–scale preprint (https://doi.org/10.13140/RG.2.2.23525.56800); they are reproduced here to make the quantitative argument self-contained.

Put

$$r = a' - a > 0, \qquad s = b - b' > 0. \tag{3.1}$$

Then

$$C = as - br + rs, \qquad D = 2bs - s^2 - 2ar - r^2. \tag{3.2}$$

Combining these with $D = 3MC$ gives the collision line

$$(3Ms + 2r)a - (3Mr + 2s)b + (3Mrs + r^2 + s^2) = 0. \tag{3.3}$$

Set

$$P = 3Mr + 2s, \qquad S = 3Ms + 2r, \qquad d_0 = \gcd(P, S), \qquad K = r^2 + s^2 + 3Mrs. \tag{3.4}$$

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Equation (3.3) is equivalent to

$$S(2a + r) = P(2b - s).$$

Because P/d_0 and S/d_0 are coprime, a positive integer j satisfies

$$2a + r = \frac{jP}{d_0}, \qquad 2b - s = \frac{jS}{d_0}. \tag{3.5}$$

Proposition 3.1 (Slope and scale identities)**.** *Every ordered collision satisfies*

$$C = \frac{j(s^2 - r^2)}{d_0}, \qquad B = \frac{jK}{d_0}, \tag{3.6}$$

$$A = \frac{M^2(s^2 - r^2)}{K}, \tag{3.7}$$

$$B^2 = \frac{K(K + 4M^2)}{9M^2 - 4}. \tag{3.8}$$

Proof. Solve (3.5) explicitly:

$$a = \frac{1}{2}\left(\frac{jP}{d_0} - r\right), \qquad b = \frac{1}{2}\left(\frac{jS}{d_0} + s\right).$$

Use $a' = a + r$, $b' = b - s$, and the definitions of B, C. Substitution gives

$$C = as - br + rs = \frac{j(sP - rS)}{2d_0} = \frac{j(s^2 - r^2)}{d_0},$$

$$B = br + as = \frac{j(rS + sP)}{2d_0} = \frac{jK}{d_0}.$$

Since $C, j, d_0 > 0$, we have $s^2 - r^2 > 0$. Dividing $M^2C = AB$ by the positive B proves (3.7).

The same substitution into $A = b(b - s) - a(a + r)$ yields

$$A = \frac{(s^2 - r^2)((9M^2 - 4)j^2 - d_0^2)}{4d_0^2}.$$

Equate this expression to (3.7) and cancel $s^2 - r^2 > 0$. The result is

$$((9M^2 - 4)j^2 - d_0^2)K = 4M^2 d_0^2.$$

Rearranging and multiplying by K/d_0^2, then using $B = jK/d_0$, gives (3.8). □

We now reduce the slope. Write

$$r = Gr_0, \qquad s = Gs_0, \qquad \gcd(r_0, s_0) = 1,$$

and define

$$T_0 = s_0^2 - r_0^2, \qquad K_0 = r_0^2 + s_0^2 + 3Mr_0s_0. \tag{3.9}$$

Equation (3.6) shows that $T_0 > 0$, and all quotients below are positive. Let

$$\Gamma = \gcd(K_0, T_0), \qquad K_0 = L\Gamma, \qquad T_0 = W\Gamma, \qquad \gcd(L, W) = 1. \tag{3.10}$$

Formula (3.7) becomes

$$A = \frac{M^2 W}{L}. \tag{3.11}$$

Thus $L \mid M^2$. Write

$$M^2 = LN, \qquad A = NW. \tag{3.12}$$

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Lemma 3.2 (Scale divisor)**.** *If M is odd, then $L \mid B$.*

Proof. It is enough to compare the exponent of each prime in L and B. Let $p \mid L$. Because $L \mid M^2$ and M is odd, p is an odd divisor of M. Thus $9M^2 - 4 \equiv -4 \not\equiv 0 \pmod{p}$.

We have

$$K = G^2 L\Gamma, \qquad K + 4M^2 = L(G^2\Gamma + 4N).$$

Both factors on the right of $(9M^2 - 4)B^2 = K(K + 4M^2)$ are divisible by L. Write $\mathrm{ord}_p(z)$ for the exponent of p in a positive integer z. Comparing exponents gives

$$2\,\mathrm{ord}_p(B) = \mathrm{ord}_p(K) + \mathrm{ord}_p(K + 4M^2) \geq 2\,\mathrm{ord}_p(L).$$

Thus every prime occurs in B to at least its exponent in L, which proves $L \mid B$. □

We use two elementary facts about Markoff entries.

Lemma 3.3 (Coprimality and prime support)**.** *The entries of a Markoff triple are pairwise coprime. Every odd prime dividing a Markoff number is* $1 \pmod 4$*. In particular,* 3 *divides no Markoff number.*

Proof. We first prove coprimality by descent. If a prime divides two entries of a Markoff triple, the equation forces it to divide the third. Order the entries as $x \leq y \leq z$. Regarding the equation as a quadratic in its last entry gives the other integer root

$$z' = 3xy - z = \frac{x^2 + y^2}{z} > 0.$$

For $P(u) = u^2 - 3xyu + x^2 + y^2$,

$$P(y) = x^2 - y^2(3x - 2) < 0$$

unless $x = y = 1$. Since the leading coefficient is positive, $P(y) < 0$ places y strictly between its two roots. The root z is at least y, so the other root satisfies $0 < z' < y$.

Replacing z by z', and reordering, strictly decreases the maximum until $x = y = 1$. At that point the equation gives $z = 1$ or 2; the triple $(1, 1, 2)$ also moves to $(1, 1, 1)$. A prime dividing all three entries would remain a common divisor under every move, but it cannot divide the final triple. No such prime exists, proving pairwise coprimality.

Now let an odd prime p divide an entry M of (a, b, M). Coprimality allows division by a modulo p, giving

$$1 + (ba^{-1})^2 \equiv 0 \pmod{p}.$$

Thus -1 is a square modulo p. A square root of -1 has multiplicative order four, so four divides the order $p - 1$ of the nonzero residue group. Hence $p \equiv 1 \pmod 4$, which excludes $p = 3$. □

Lemma 3.4 (No split prime)**.** *If an odd prime $p \mid M$ divides both A and B, then the ordered collision is impossible.*

Proof. By pairwise coprimality, none of a, b, a', b' is divisible by p, so all four have inverses modulo p. Put $t = ba^{-1}$ and $t' = b'(a')^{-1}$. Dividing $B \equiv 0$ by aa' gives $t = t'$; dividing $A \equiv 0$ by aa' gives $tt' = 1$. Together these imply $t^2 = 1$.

The Markoff equation, divided by a^2 modulo p, instead gives $t^2 = -1$. This would imply $2 \equiv 0 \pmod{p}$, impossible for an odd prime. □

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Theorem 3.5 (Odd unitary collision allocation)**.** *If an odd Markoff number M admits an ordered collision, then there are coprime integers $F, E > 1$ and positive integers H, W such that*

$$M = FE, \qquad A = F^2W, \qquad B = E^2H, \qquad C = HW. \tag{3.13}$$

Each of F, E is a unitary divisor of M.

Proof. Consider a complete prime-power factor p^e of M. Its square p^{2e} occurs in $M^2 = LN$. If a positive power of p occurred in both L and N, then $A = NW$ and $L \mid B$ would give $p \mid A, B$. Lemma 3.4 excludes this. Thus the entire power p^{2e} belongs to one of L, N.

Doing this for every prime gives

$$L = E^2, \qquad N = F^2, \qquad M = FE, \qquad \gcd(F, E) = 1.$$

Both factors exceed one. If $L = 1$, then $A = M^2W$, contradicting $0 < A < bb' < M^2$. If $N = 1$, then $M^2 = L \mid B$, contradicting $0 < B < ba' < M^2$.

Now $A = NW = F^2W$. Also $E^2 = L \mid B$, so $B = E^2H$ for an integer $H > 0$. Substitute these expressions in $M^2C = AB$ and cancel $M^2 = F^2E^2$ to obtain $C = HW$. The coprimality of F, E is exactly the assertion that they are complementary unitary divisors. □

Remark 3.6. Proper coprime factorizations of a hypothetical nonunique maximum underlie Srinivasan's Theorem 5.1 and Button's Corollary 6 [14, 4]. Theorem 3.5 records the exact allocation to A, B, C in the notation required for the defect packets.

4 Complementary hyperbolic factors and the defect packets

The coupled defect equations below were also circulated in the author's earlier slope–scale preprint (https://doi.org/10.13140/RG.2.2.23525.56800); they are reproved here from the complementary four-point conics in the form required by the certified tiers.

The same four hyperbolic points used to factor $AB = M^2C$ produce the two equations on which all four odd tiers rest. Define

$$\widehat{A} = 3Maa' - a'b - ab', \qquad \widehat{B} = 3Mab' - aa' - bb'. \tag{4.1}$$

Along with ξ, η from (2.8), put

$$\tau = c + d, \qquad \upsilon = 1 - d + c. \tag{4.2}$$

Then $0 < \tau, \upsilon < 1$, $\eta + \tau = 1$, and $\xi + \upsilon = 1$.

Lemma 4.1 (Complementary four-point conics)**.** *One has*

$$\widehat{A} = Mg(\tau), \qquad \widehat{B} = Mg(\upsilon), \tag{4.3}$$

and, for $0 < x < 1$,

$$g(x)^2 + g(1-x)^2 + 3Mg(x)g(1-x) = M^2. \tag{4.4}$$

Consequently,

$$A^2 + \widehat{A}^2 + 3MA\widehat{A} = M^4, \tag{4.5}$$

$$B^2 + \widehat{B}^2 + 3MB\widehat{B} = M^4. \tag{4.6}$$

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Proof. We make the two product calculations explicit. Using $2q = 3M$ and $\cosh(\theta - u) = q\cosh u - R\sinh u$, we obtain

$$\begin{aligned}2q\cosh(c\theta)\cosh(d\theta) - \cosh(d\theta)\cosh((1-c)\theta)\\ - \cosh(c\theta)\cosh((1-d)\theta) = R\sinh((c+d)\theta).\end{aligned}$$

The corresponding identity for the other factor is

$$\begin{aligned}2q\cosh(c\theta)\cosh((1-d)\theta) - \cosh(c\theta)\cosh(d\theta)\\ - \cosh((1-c)\theta)\cosh((1-d)\theta) = R\sinh((1-d+c)\theta).\end{aligned}$$

Multiplication by M^2/R^2 proves $\widehat{A} = Mg(\tau)$ and $\widehat{B} = Mg(\upsilon)$. Both are positive because $\tau, \upsilon > 0$.

For the conic, the addition formula for $\sinh(u+v)$ gives

$$\sinh^2 u + \sinh^2 v + 2\cosh(u+v)\sinh u \sinh v = \sinh^2(u+v).$$

Take $u = x\theta$, $v = (1-x)\theta$, so $u + v = \theta$. Multiply by M^2/R^2 and use $\sinh\theta = R$, $2\cosh\theta = 3M$. This proves (4.4). The pairs (η, τ) and (ξ, υ) each sum to one. Substituting them and multiplying by M^2 proves (4.5) and (4.6). □

Theorem 4.2 (Coupled defect packets)**.** *For every hypothetical odd collision there are positive integers n, m such that*

$$n^2 + 3MHn + H^2 = F^4, \tag{4.7}$$

$$m^2 + 3MWm + W^2 = E^4. \tag{4.8}$$

Moreover,

$$\gcd(F, H) = \gcd(H, n) = 1, \qquad \gcd(E, W) = \gcd(W, m) = 1. \tag{4.9}$$

Proof. First we show that the complementary factors have the required square divisors. Let $p^e \parallel F$, and suppose $v = \mathrm{ord}_p(\widehat{A}) < 2e$. In (4.5), the term $\widehat{A}^2$ has exponent $2v$. The exponents of A^2 and M^4 are at least $4e > 2v$, and that of $3MA\widehat{A}$ is at least $3e + v > 2v$.

This is impossible: dividing the equation by p^{2v} and reducing modulo p would leave exactly one nonzero term. Hence $\mathrm{ord}_p(\widehat{A}) \geq 2e$. For every prime of F, this proves $F^2 \mid \widehat{A}$. The identical argument with $E, B, \widehat{B}$ proves $E^2 \mid \widehat{B}$. Write

$$\widehat{A} = F^2 m, \qquad \widehat{B} = E^2 n.$$

Substitute these and (3.13) into the two conics. Dividing the first by F^4 gives (4.8); dividing the second by E^4 gives (4.7). The integers m, n are positive because the complementary factors are positive.

If a prime divided both F and H, it would divide both $A = F^2W$ and $B = E^2H$, contrary to Lemma 3.4. Thus $\gcd(F, H) = 1$; the same reasoning gives $\gcd(E, W) = 1$. A prime dividing H, n would, by (4.7), also divide F^4, hence F, which is impossible. This proves $\gcd(H, n) = 1$. Applying the same argument to the other equation proves $\gcd(W, m) = 1$. □

The quadratic equations also have a useful interpretation through conjugate numbers. In the ring of numbers of the form $u + vQ_M$, with $u, v \in \mathbb{Z}$, namely the real quadratic order

$$\mathcal{O}_M = \mathbb{Z}\left[\frac{3M + \sqrt{9M^2 - 4}}{2}\right],$$

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one has

$$\mathrm{N}(x + yQ_M) = x^2 + 3Mxy + y^2, \qquad Q_M = \frac{3M + \sqrt{9M^2 - 4}}{2}.$$

Thus (4.7)–(4.8) are norm equations of fourth powers. Here the norm is the product obtained by replacing the square root by each of its two signs. This interpretation provides context; the bounds below use only the displayed integer equations.

5 The hand tier: 33 and 99

Abstract one defect equation as

$$x^2 + y^2 + 3TOxy = T^4. \tag{5.1}$$

For a collision packet, either

$$(T, O, x, y) = (F, E, H, n) \quad \text{or} \quad (T, O, x, y) = (E, F, W, m).$$

By Theorem 3.5, Lemma 3.3, and (4.9), the local block conditions are

$$T, O > 1, \quad \gcd(T, O) = 1, \quad 2 \nmid TO, \quad 3 \nmid TO, \quad p \mid TO \Longrightarrow p \equiv 1 \pmod 4, \quad \gcd(x, y) = 1. \tag{5.2}$$

Lemma 5.1 (The 33-lemma)**.** *Every solution of* (5.1) *satisfying* (5.2) *has*

$$xy \geq 33. \tag{5.3}$$

Proof. Reduce (5.1) modulo 3. Since $3 \nmid T$, it gives $x^2 + y^2 \equiv 1 \pmod 3$. The only square residues are zero and one, so exactly one of x, y is divisible by 3. In particular, $3 \mid xy$. Reduction modulo T also gives $T \mid x^2 + y^2$.

Suppose $xy < 33$, and use symmetry to take $x \leq y$. Listing the coprime factor pairs of the positive multiples of three below 33 gives

$$(1,3), (1,6), (1,9), (1,12), (1,15), (1,18), (1,21), (1,24), (1,27), (1,30),$$
$$(2,3), (2,9), (2,15), (3,4), (3,5), (3,7), (3,8), (3,10), (5,6).$$

Because T is odd, $T^4 \equiv 1 \pmod{16}$. Every integer whose prime divisors are $1 \pmod 4$ is itself $1 \pmod 4$, so TO must have residue $1, 5, 9$, or 13 modulo 16. For each listed pair, test these four residues in the equation, and then list the divisors $T > 1$ of $x^2 + y^2$ having the required prime divisors. The surviving six choices are

$\{x, y\}$	$x^2 + y^2$	T
$\{1, 9\}$	82	41
$\{1, 21\}$	442	$13, 17, 221$
$\{3, 7\}$	58	29
$\{3, 8\}$	73	73.

For each choice, the equation determines O:

$$O = \frac{T^3 - (x^2 + y^2)/T}{3xy}. \tag{5.4}$$

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In the same order, these values are

$$\frac{68919}{27}, \quad \frac{2163}{63}, \quad \frac{4887}{63}, \quad \frac{10793859}{63}, \quad \frac{24387}{63}, \quad 5403.$$

The first five are not integers. The last is divisible by three, which the block conditions exclude. Thus every possible pair below 33 is rejected. □

Theorem 5.2 (Hand quartic tier)**.** *Every hypothetical odd collision satisfies*

$$Hn, Wm \geq 33, \qquad F^4, E^4 \geq 99M + 66. \tag{5.5}$$

It also satisfies the strict cubic imbalance

$$E < \frac{F^3}{99}, \qquad F < \frac{E^3}{99}. \tag{5.6}$$

Proof. Each of the two collision packets satisfies the hypotheses of Lemma 5.1. Therefore $Hn \geq 33$ and $Wm \geq 33$. The inequality $(x - y)^2 \geq 0$ gives $x^2 + y^2 \geq 2xy$, so either packet yields

$$T^4 = x^2 + y^2 + 3Mxy \geq (3M + 2)xy \geq 33(3M + 2).$$

This proves both quartic inequalities. For the first cubic inequality, use the strictly positive square terms to get

$$F^4 > 3MHn \geq 99M = 99FE.$$

Divide by $99F > 0$ to obtain $E < F^3/99$. Exchanging the two packets gives $F < E^3/99$. □

6 The exact finite certificate and the other three tiers

The finite reduction follows directly from (5.4). Fix a positive integer upper product bound B. By symmetry take $x \leq y$. There are finitely many pairs $xy < B$; for each pair, T must be a divisor of $x^2 + y^2$, and then O is forced by (5.4). More explicitly, $1 \leq x \leq \lfloor \sqrt{B - 1} \rfloor$ and $x \leq y \leq \lfloor (B - 1)/x \rfloor$. Listing those pairs, all divisors T, and each forced O covers the entire range. Thus no bound on T or O is guessed.

The primary certificate executes the following exact loop.

Listing 1: Core finite reduction (schematic).

```
for x <= y with lower <= x*y < upper:
    reject unless exactly one of x,y is divisible by 3
    for T in divisors(x*x + y*y):
        reject unless T is an admissible block
        O = (T**3 - (x*x+y*y)//T) / (3*x*y)
        retain only integral O>1 with gcd(T,O)=1
        retain only admissible O
```

The submitted source uses explicit failure checks rather than Python `assert`, so optimization flags cannot disable verification. It deliberately searches the larger domain in which $\gcd(x, y) = 1$ is not imposed; eliminating every point in this superset proves the required claim. A second program, written separately and runnable on its own, orders the search by the product $N = xy$, enumerates the factor pairs of N, and reimplements every arithmetic routine without importing the primary program. The primary program's `-mode both` option is an internal product-order cross-check that reuses its arithmetic helpers; it is not the standalone verifier.

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6.1 The local floor 120 and coefficient 360

The exact search over $33 \le xy < 120$ leaves no admissible row. Before the final prime-support test on O, the sole row is

$$(x, y, T, O) = (1, 33, 109, 13081), \qquad 13081 = 103 \cdot 127,$$

and both prime factors are $3 \pmod 4$. At the boundary,

$$(x, y, T, O) = (5, 24, 601, 603005), \qquad xy = 120, \tag{6.1}$$

satisfies the equation and all local block conditions. Hence 120 is sharp for this relaxation.

Proposition 6.1 (Local exact tier)**.** *Every hypothetical odd collision satisfies*

$$Hn, Wm \ge 120, \qquad F^4, E^4 \ge 360M + 240. \tag{6.2}$$

Proof. Lemma 5.1 already excludes $xy < 33$. The finite search described above covers every pair with $33 \le xy < 120$, every possible divisor T, and the resulting forced value of O. It leaves no admissible row, even though its search domain is larger than the collision domain. Thus each collision packet must have $xy \ge 120$.

Apply this conclusion to $(x, y) = (H, n)$ and (W, m). In each equation, $x^2 + y^2 \ge 2xy$ then gives

$$T^4 \ge (3M + 2)xy \ge 120(3M + 2) = 360M + 240.$$

This is a computer-assisted step: the proved finite reduction and its exact exhaustive certificate together establish the product floor. □

6.2 The companion-prime filter and coefficient 201411

The next filter is an automatic property of a genuine Markoff entry.

Lemma 6.2 (Companion-prime necessity)**.** *If M is a Markoff number, every odd prime divisor of $(3M - 2)(3M + 2) = 9M^2 - 4$ is $1 \pmod 4$.*

Proof. Rearranging the Markoff equation gives two sum-of-squares identities:

$$(b - a)^2 + M^2 = ab(3M - 2), \qquad (a + b)^2 + M^2 = ab(3M + 2). \tag{6.3}$$

Let an odd prime p divide either companion $3M - 2$ or $3M + 2$. It cannot divide M, since that would also force $p \mid 2$. Reduce the corresponding identity modulo p and divide by M^2. It shows that -1 is a square modulo p. As in Lemma 3.3, this forces $p \equiv 1 \pmod 4$. □

The intentionally enlarged enumeration for $120 \le xy < 100000$ has nineteen rows, listed in Table 3. Eighteen satisfy $\gcd(x, y) = 1$; the row with $xy = 75000$ has $\gcd(x, y) = 25$ and is retained to make the superset search explicit. Of all nineteen rows, eighteen contain a displayed prime $\ell \equiv 3 \pmod 4$ dividing a companion $3TO - 2$ or $3TO + 2$. The sole companion-admissible row is

$$(x, y, T, O) = (23, 2919, 1145, 7453), \qquad xy = 67137. \tag{6.4}$$

It proves sharpness at 67137 for the local-plus-companion relaxation.

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Proposition 6.3 (Companion exact tier)**.** *Every hypothetical odd collision satisfies*

$$Hn, Wm \geq 67137, \qquad F^4, E^4 \geq 201411M + 134274. \tag{6.5}$$

Proof. For each packet, the product TO is the actual common maximum M. Lemma 6.2 therefore supplies a necessary condition on both of its companions. The local tier already excludes products below 120. In the complete table for $120 \leq xy < 100000$, every row with $xy < 67137$ fails that necessary companion condition.

Hence both packets satisfy $xy \geq 67137$. Using $x^2 + y^2 \geq 2xy$ once more gives

$$T^4 \geq (3M + 2)xy \geq 67137(3M + 2) = 201411M + 134274.$$

The conclusion uses the exact finite enumeration; no assumption that an arbitrary surviving row is a Markoff collision is made. □

6.3 Residual companion allocation and a crossed Pell-type equation

The slope and scale formulas also divide the companion discriminant Δ_M into coprime factors. The word discriminant here refers to the square-root expression arising from the quadratic equation; we will use its explicit value. Put

$$\Delta_M = 9M^2 - 4, \qquad P_0 = 3Mr_0 + 2s_0, \qquad S_0 = 3Ms_0 + 2r_0.$$

Proposition 6.4 (Residual companion allocation)**.** *For every hypothetical odd collision,*

$$d_0 = G\Gamma, \qquad H = jG, \tag{6.6}$$

and Γ is a unitary divisor of Δ_M. More precisely, put

$$\Gamma^* = \frac{\Delta_M}{\Gamma}. \tag{6.7}$$

Then

$$\gcd(\Gamma, \Gamma^*) = 1, \qquad \Gamma^* j^2 - \Gamma G^2 = 4F^2. \tag{6.8}$$

Furthermore,

$$\gcd(\Gamma, M) = \gcd(\Gamma, j) = \gcd(\Gamma^*, G) = 1. \tag{6.9}$$

If

$$Z_F = 2n + 3MH - 2F^2, \tag{6.10}$$

then

$$Z_F = G^2\Gamma, \qquad G^2 = \gcd(Z_F, H^2). \tag{6.11}$$

Thus the residual data G, j, Γ, Γ^* are determined by the ordered packet (F, E, H, n), with $M = FE$.

Proof. We first identify the common divisor in the slope formulas. Since $P = GP_0$ and $S = GS_0$, write $d_0 = G\delta$, where $\delta = \gcd(P_0, S_0)$. At least one of P_0, S_0 is odd, because M is odd and the coprime integers r_0, s_0 cannot both be even. Thus δ is odd. The identities

$$s_0P_0 - r_0S_0 = 2T_0, \qquad s_0P_0 = K_0 + T_0, \qquad r_0S_0 = K_0 - T_0$$

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show that $\delta \mid T_0$, and then $\delta \mid K_0$. Hence $\delta \mid \Gamma$.

For the reverse divisibility, K_0 is odd for either possible parity pattern of r_0, s_0, so Γ is odd. Also $\gcd(\Gamma, r_0 s_0) = 1$: a prime dividing Γ and one slope coordinate would divide the other by $T_0 = s_0^2 - r_0^2$. The last two displayed identities therefore allow cancellation of r_0, s_0 modulo Γ, giving $\Gamma \mid P_0, S_0$. Thus $\delta = \Gamma$ and $d_0 = G\Gamma$.

Next, direct linear combinations give

$$3MP_0 - 2S_0 = \Delta_M r_0, \qquad 3MS_0 - 2P_0 = \Delta_M s_0. \tag{6.12}$$

Both left sides are divisible by Γ. Since r_0, s_0 are coprime, an integer combination of them is one, so $\Gamma \mid \Delta_M$. As M is odd, $\gcd(M, \Delta_M) = \gcd(M, 4) = 1$. In particular, $\gcd(\Gamma, M) = 1$.

Use $K = G^2E^2\Gamma$ and $d_0 = G\Gamma$ in the formula for B:

$$B = \frac{jK}{d_0} = jGE^2 = HE^2.$$

Cancel E^2 to obtain $H = jG$. Substitute these same quantities into (3.8). Cancellation gives

$$\Delta_M j^2 = \Gamma(G^2\Gamma + 4F^2).$$

Writing $\Delta_M = \Gamma\Gamma^*$ and dividing by Γ proves (6.8).

If a prime divided both Γ, Γ^*, that equation would force it to divide $4F^2$. It is odd and coprime to $F \mid M$, a contradiction. Hence the allocation is unitary. The same reduction modulo a common prime of Γ, j, or of Γ^*, G, proves the remaining coprimalities.

Finally, complete the square in the first defect equation:

$$(2n + 3MH)^2 = 4F^4 + H^2\Delta_M.$$

The right side exceeds $4F^4$, so $Z_F = 2n + 3MH - 2F^2 > 0$, and

$$H^2\Delta_M = Z_F(Z_F + 4F^2).$$

The residual equation and $H = jG$ give the same product as

$$(G^2\Gamma)(G^2\Gamma + 4F^2).$$

The function $t(t + 4F^2)$ increases strictly for $t > 0$. Its two positive inputs must therefore agree: $Z_F = G^2\Gamma$. Since $H^2 = G^2j^2$ and $\gcd(\Gamma, j) = 1$, taking a greatest common divisor gives $\gcd(Z_F, H^2) = G^2$. Thus the packet determines G, then $j = H/G$, then $\Gamma = Z_F/G^2$, and finally $\Gamma^* = \Delta_M/\Gamma$. □

Remark 6.5 (Relation with the principal-genus obstruction). For every prime $p \mid \Gamma$, equation (6.8) shows that Γ^* is a quadratic residue modulo p; for every $p \mid \Gamma^*$, it shows that $-\Gamma$ is a quadratic residue modulo p. By Lemma 6.2, all these primes are $1 \pmod 4$, so the second condition may equivalently be stated with Γ in place of $-\Gamma$. These are the quadratic-residue conditions, written using Legendre symbols, occurring in Srinivasan's principal-genus method [14, Theorem 5.1, Definition 3.1, and Theorems 5.4 and 6.1]; no priority is claimed for this residue obstruction. The additional ingredient below is its simultaneous coupling to the primitive collision slope.

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Factor the two coprime companions according to the residual allocation:

$$3M-2=\alpha\bar{\alpha}, \qquad 3M+2=\beta\bar{\beta}, \qquad \Gamma=\alpha\beta, \qquad \Gamma^*=\bar{\alpha}\bar{\beta}, \tag{6.13}$$

The bars in this notation denote complementary integer factors, not complex conjugates. Here $\alpha=\gcd(\Gamma,3M-2)$ and $\beta=\gcd(\Gamma,3M+2)$. The allocation is unitary within each companion.

Theorem 6.6 (Crossed slope equation)**.** *For every hypothetical odd collision there are positive integers λ,μ such that*

$$r_0+s_0=\alpha\lambda, \qquad s_0-r_0=\beta\mu, \tag{6.14}$$

$$W=\lambda\mu<E^2, \qquad \gcd(\lambda,\mu)\in\{1,2\}, \tag{6.15}$$

and

$$\boxed{\alpha\bar{\beta}\lambda^2-\beta\bar{\alpha}\mu^2=4E^2.} \tag{6.16}$$

The positive solution satisfies the following collision inequalities

$$\alpha\lambda>\beta\mu, \qquad \bar{\beta}\lambda>\bar{\alpha}\mu. \tag{6.17}$$

Moreover,

$$1<\Gamma<\Delta_M, \qquad \alpha<3M-2, \qquad \beta<3M+2. \tag{6.18}$$

Proof. Put $u=r_0+s_0$ and $v=s_0-r_0$; both are positive. From $\Gamma\mid P_0,S_0$, we get

$$\Gamma\mid S_0+P_0=(3M+2)u, \qquad \Gamma\mid S_0-P_0=(3M-2)v.$$

The odd numbers $3M-2$ and $3M+2$ are coprime: a common divisor would divide four. Thus $\alpha\mid 3M-2$ is coprime to $3M+2$, and $\beta\mid 3M+2$ is coprime to $3M-2$. Cancellation gives $\alpha\mid u$ and $\beta\mid v$. Write $u=\alpha\lambda$, $v=\beta\mu$, with $\lambda,\mu>0$.

Since $T_0=uv$ and $\Gamma=\alpha\beta$,

$$W=T_0/\Gamma=\lambda\mu.$$

Also $A=F^2W<M^2=F^2E^2$, so $W<E^2$. Coprimality of r_0,s_0 implies $\gcd(u,v)\in\{1,2\}$. Dividing u,v by the odd numbers α,β cannot introduce a common prime and cannot remove their common factor two. Thus $\gcd(\lambda,\mu)=\gcd(u,v)\in\{1,2\}$.

Now expand K_0 in terms of u,v:

$$4K_0=(3M+2)u^2-(3M-2)v^2, \qquad K_0=E^2\Gamma.$$

Substitute the four companion factors and $u=\alpha\lambda$, $v=\beta\mu$, then divide by $\Gamma=\alpha\beta$. This proves (6.16). The first inequality in (6.17) is $u>v$. The second follows from

$$2P_0=(3M+2)u-(3M-2)v=\Gamma(\bar{\beta}\lambda-\bar{\alpha}\mu)>0.$$

To prove that the companion split is proper, use the original ordering:

$$G(r_0+s_0)=r+s=(b-a)-(b'-a')<M, \qquad G(s_0-r_0)=s-r<M.$$

In particular, $G<M$. If $\Gamma=1$, then $\Gamma^*=\Delta_M$, and (6.8), with $j\geq 1$, gives

$$\Delta_M\leq G^2+4F^2<5M^2.$$

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But $\Delta_M = 9M^2 - 4 > 5M^2$ for $M > 1$, a contradiction. Thus $\Gamma > 1$. Also

$$\alpha \le r_0 + s_0 < M/G \le M < 3M - 2,$$

and $\beta \le s_0 - r_0 < M/G \le M < 3M + 2$. These prove the two proper-factor bounds. They exclude $\Gamma = \Delta_M$ as well. □

The second packet also recovers the primitive slope. This removes an otherwise possible ambiguity about which packet realizes a row of the finite certificate.

Lemma 6.7 (Slope reconstruction from the second packet)**.** *For every hypothetical odd collision,*

$$m = \frac{r_0 P_0}{\Gamma}, \qquad E^2 + W = \frac{s_0 P_0}{\Gamma}. \tag{6.19}$$

Consequently, if

$$k = \gcd(m, E^2 + W), \tag{6.20}$$

then

$$k = \frac{P_0}{\Gamma}, \qquad r_0 = \frac{m}{k}, \qquad s_0 = \frac{E^2 + W}{k}. \tag{6.21}$$

Thus (E, W, m) determines r_0, s_0, and hence $\Gamma = \gcd(K_0, T_0)$ and $\Gamma^* = \Delta_M/\Gamma$.

Proof. The identity $s_0 P_0 = K_0 + T_0$, divided by Γ, gives

$$\frac{s_0 P_0}{\Gamma} = E^2 + W.$$

For the other formula, substitute (3.5) into the definition of $\widehat{A}$. Recall that $d_0 = G\Gamma$, $P = GP_0$, and $S = GS_0$. Expansion yields

$$\begin{aligned} 4\widehat{A} &= \frac{j^2 P_0(3MP_0 - 2S_0)}{\Gamma^2} - G^2 r_0 P_0 \\ &= r_0 P_0 \left(\frac{j^2 \Delta_M}{\Gamma^2} - G^2 \right) \\ &= \frac{4F^2 r_0 P_0}{\Gamma}. \end{aligned}$$

The second equality uses (6.12); the third uses (6.8). Since $\widehat{A} = F^2 m$, cancellation gives $m = r_0 P_0/\Gamma$.

The two quantities m and $E^2 + W$ are therefore the coprime integers r_0, s_0 multiplied by the same integer P_0/Γ. Their greatest common divisor is exactly this multiplier, giving all formulas in (6.21). □

Proposition 6.8 (Residual rejection of the exceptional odd row)**.** *The certificate row*

$$(x, y, T, O) = (23, 2919, 1145, 7453) \tag{6.22}$$

cannot be either packet of an odd ordered collision, in either ordering of its two defect coordinates.

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Proof. The row forces $M_0 = 1145 \cdot 7453 = 8533685$. Its two companions factor as

$$3M_0 - 2 = 137 \cdot 186869, \qquad 3M_0 + 2 = 3137 \cdot 8161, \tag{6.23}$$

with four distinct prime factors. There are four cases: either packet can give the row, and its two positive coordinates can occur in either order. We reduce all four cases to the same two equations.

If the row is the first packet, then $(F, E) = (1145, 7453)$ and $\{H, n\} = \{23, 2919\}$. Apply the greatest-common-divisor extraction in Proposition 6.4:

(H, n)	G	j	Γ
$(23, 2919)$	1	23	$586208053 = 186869 \cdot 3137$
$(2919, 23)$	7	417	$1525037909 = 186869 \cdot 8161$.

For these companion allocations, Theorem 6.6 gives, respectively,

$$1525037909X^2 - 429769Y^2 = 4 \cdot 7453^2, \tag{6.24}$$

$$586208053X^2 - 1118057Y^2 = 4 \cdot 7453^2, \tag{6.25}$$

where $(X, Y) = (\lambda, \mu)$, $X, Y > 0$, and $XY < 7453^2$.

If the row is the second packet, then $(E, F) = (1145, 7453)$ and $\{W, m\} = \{23, 2919\}$. The slope reconstruction of Lemma 6.7 gives

(W, m)	k	(r_0, s_0)	Γ
$(23, 2919)$	417	$(7, 3144)$	429769
$(2919, 23)$	23	$(1, 57128)$	1118057.

The corresponding complementary divisors Γ^* are 1525037909 and 586208053. Therefore the residual equation (6.8) is again (6.24) or (6.25), now with $(X, Y) = (j, G)$. Since $B = E^2H < M_0^2 = E^2F^2$, we have $XY = jG = H < F^2 = 7453^2$. All four cases have the same stated bounds.

It remains to exclude positive solutions of these two equations. Put $N = 7453$, and write either equation as $RX^2 - SY^2 = 4N^2$. In both cases $R > 8N^2$. Since $X \geq 1$,

$$SY^2 = RX^2 - 4N^2 > \frac{R}{2}X^2.$$

Multiply by X^2 and use $XY < N^2$. This gives the finite bound

$$RX^4 < 2SN^4. \tag{6.26}$$

For the first equation,

$$1525037909 \cdot 1149^4 > 2 \cdot 429769 \cdot 7453^4,$$

so $X \leq 1148$. Reducing that equation modulo $S = 429769 = 137 \cdot 3137$, and inverting R modulo S, gives

$$X^2 \equiv 275216 \pmod{429769}.$$

Its four roots between zero and 429768 are

$$85941, \quad 104763, \quad 325006, \quad 343828.$$

None lies in $1 \leq X \leq 1148$, so this case is impossible.

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For the second equation,

$$586208053 \cdot 1853^4 > 2 \cdot 1118057 \cdot 7453^4,$$

so $X \le 1852$. Reduction modulo $1118057 = 137 \cdot 8161$ gives

$$X^2 \equiv 391856 \pmod{1118057}.$$

Its four roots in the standard residue interval are

$$129197, \quad 471959, \quad 646098, \quad 988860.$$

Again none is small enough. Each root list is complete because each of its two distinct odd prime moduli has exactly two roots, and the Chinese remainder theorem combines their four choices into exactly four residues. The supplementary programs verify the prime factors, all four residues, and the finite bounds independently. □

6.4 An independent canonical-continuant filter and coefficient 300000

Proposition 6.8 already eliminates the sole odd companion-admissible row below 100000. We retain the following different necessary test as an independent audit of that rejection and because the even 100000 tier uses it. We use only a necessary part of the classical Cohn–Christoffel description of a Markoff occurrence [7, 8, 13]; the precise finite continued-fraction form needed here is supplied by [5, Theorem 4.1].

Lemma 6.9 (Canonical continuant necessity)**.** *For every occurrence of a Markoff number $M > 2$, there is an integer ρ satisfying*

$$\rho^2 \equiv -1 \pmod{M}, \qquad \frac{M}{3} < \rho < \frac{M}{2}, \tag{6.27}$$

such that every partial quotient of the canonical finite regular continued fraction of M/ρ (whose last partial quotient is greater than 1) belongs to $\{1, 2\}$.

Proof. The Markoff-tree construction assigns a reduced slope to each ordered occurrence with $M > 2$. Theorem 4.1 of Çanakçı and Schiffler [5] gives that occurrence an even palindrome of continued-fraction digits

$$[a_r, \dots, a_1, a_1, \dots, a_r], \qquad a_i \in \{1, 2\}, \quad a_r = 2.$$

Call the digits $c_1, \dots, c_{2r}$. Each matrix associated with a digit is symmetric. Transposing their product reverses the digits, and the palindrome leaves the product unchanged. The continued-fraction matrix is therefore

$$\prod_{i=1}^{2r} \begin{pmatrix} c_i & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} M & \rho \\ \rho & \sigma \end{pmatrix}. \tag{6.28}$$

Each factor has determinant -1. The number of factors is even, so $M\sigma - \rho^2 = 1$, proving $\rho^2 \equiv -1 \pmod{M}$.

The usual matrix rule for a continued fraction gives

$$M/\rho = [c_1; c_2, \dots, c_{2r}].$$

Its last digit is $2 > 1$, so it is the canonical finite expansion. Its first digit is also two, and the tail $[c_2; \dots, c_{2r}]$ is strictly greater than one. Hence $2 < M/\rho < 3$, or $M/3 < \rho < M/2$. Every digit is one or two, as required. □

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For the exceptional row (6.4),

$$M_0 = TO = 8533685 = 5 \cdot 29 \cdot 229 \cdot 257,$$

and

$$3M_0 - 2 = 137 \cdot 186869, \qquad 3M_0 + 2 = 3137 \cdot 8161.$$

All four displayed factors are primes congruent to 1 (mod 4). The four prime factors of M_0 give sixteen CRT roots of -1. Exactly three lie in $(M_0/3, M_0/2)$:

$$2863767, \qquad 2981473, \qquad 3368483.$$

Their continued fractions are

$$[2, 1, 48, 1, 2, 2, 1, 1, 1, 1, 2, 2, 1, 48, 1, 2],$$
$$[2, 1, 6, 3, 1, 6, 3, 1, 1, 3, 6, 1, 3, 6, 1, 2],$$
$$[2, 1, 1, 6, 1, 76, 76, 1, 6, 1, 1, 2].$$

Each contains a digit larger than 2, so Lemma 6.9 rejects the last row. The exact enumeration has no other companion-admissible row below 100000. This continuant rejection is independent of the residual crossed-Pell rejection above.

Theorem 6.10 (Final odd exact tier)**.** *Every hypothetical odd collision satisfies*

$$Hn, Wm \geq 100000, \qquad F^4, E^4 \geq 300000M + 200000. \tag{6.29}$$

Proof. Each collision packet satisfies the local block conditions (5.2) and the companion condition of Lemma 6.2. The hand and local arguments exclude products below 120. In the complete enlarged-domain enumeration for $120 \leq xy < 100000$, only the row (6.4) passes the companion test.

Proposition 6.8 excludes that row for either packet and either coordinate order. There is also an independent rejection: Lemma 6.9 requires a root with only one and two as continued-fraction digits, and the three displayed possible roots all fail this test. Thus no collision packet can have product below 100000.

Apply the resulting floor to each defect equation:

$$T^4 = x^2 + y^2 + 3Mxy \geq (3M + 2)xy \geq 100000(3M + 2) = 300000M + 200000.$$

These are the claimed inequalities for F and E. □

Remark 6.11 (Nested status). The 99 tier is fully hand proved. The 360, 201411, and 300000 tiers are computer-assisted theorems based on an exact finite reduction. The 201411 tier is not conditional for actual odd Markoff collisions: Lemma 6.2 makes its added hypothesis automatic. The final row has two independent exact rejections: the residual crossed-Pell test and the canonical-continuant test. Each is a necessary condition for an actual collision or occurrence in the role where it is used. The tiers are retained separately to expose exactly where each gain enters.

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Table 3: All survivors in the intentionally enlarged domain with $120 \le xy < 100000$. Eighteen have $\gcd(x, y) = 1$. The last column records a domain or companion obstruction; RP/CF denotes the sole row passed to, and then rejected independently by, the residual crossed-Pell and continued-fraction filters.

xy	x	y	T	O	obstruction
120	5	24	601	603005	$\ell = 7$
624	16	39	1777	2997481	$\ell = 7$
7161	31	231	173	241	$\ell = 19$
8601	1	8601	377	2069	$\ell = 7$
9189	9	1021	101	37	$\ell = 11$
10080	1	10080	269513	647377152493	$\ell = 6359$
10512	9	1168	9409	26413369	$\ell = 7$
26145	9	2905	19093	88738609	$\ell = 191$
41496	56	741	552217	1352701049849	$\ell = 23$
41565	15	2771	197	61	$\ell = 47$
42837	1	42837	141481	22037080561	$\ell = 7$
43557	3	14519	6305	1918121	$\ell = 7$
48960	9	5440	163501	29757646189	$\ell = 7$
49749	23	2163	137	17	$\ell = 11$
67137	23	2919	1145	7453	RP/CF
68880	41	1680	2824081	108997629688121	$\ell = 11$
74520	5	14904	1073	5525	$\ell = 7$
75000	125	600	3005	120601	$\gcd(x, y) = 25$; also $\ell = 7$
93093	3	31031	2305	43849	$\ell = 43$

7 Even normalization and three certified tiers

The preceding unitary allocation was stated for odd maxima because the prime 2 cannot be inserted into Lemma 3.4. The even case is nevertheless accessible after a forced normalization by 4. The congruence used below is classical [16, Theorem 1]; we include the short derivation needed for the packet construction.

Lemma 7.1 (Parity of an even maximum)**.** *Let $M > 2$ be an even Markoff number and write $M = 2M_1$. Then*

$$M \equiv 2 \pmod{32}, \qquad M_1 \equiv 1 \pmod{16}. \tag{7.1}$$

In every occurrence (a, b, M), the entries a, b are $1 \pmod 4$.

Proof. Pairwise coprimality makes a, b odd. If $4 \mid M$, reducing the Markoff equation modulo four would give $1 + 1 \equiv 0$, which is false. Thus $M = 2M_1$ with M_1 odd.

Every odd prime dividing an entry is $1 \pmod 4$, by Lemma 3.3. It follows that $a, b, M_1 \equiv 1 \pmod 4$,

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including when one of them is one. Write

$$a = 1 + 4u, \qquad b = 1 + 4v, \qquad M_1 = 1 + 4w.$$

Substitute these expressions in $a^2 + b^2 + 4M_1^2 = 6M_1ab$ and reduce modulo 32. The result is

$$16\big((u^2 - u) + (v^2 - v)\big) - 24w \equiv 0 \pmod{32}.$$

Both $u(u-1)$ and $v(v-1)$ are even, so the first term vanishes modulo 32. Hence $32 \mid 24w$, or $4 \mid w$. Therefore $M_1 = 1 + 4w \equiv 1 \pmod{16}$, and doubling gives $M \equiv 2 \pmod{32}$. □

For an even ordered collision, put

$$A_0 = \frac{A}{4}, \quad B_0 = \frac{B}{4}, \quad C_0 = \frac{C}{4}, \quad \widehat{A}_0 = \frac{\widehat{A}}{4}, \quad \widehat{B}_0 = \frac{\widehat{B}}{4}. \tag{7.2}$$

These are integers. Indeed, all four smaller entries are $1 \pmod 4$, so the definitions of A, B, C give $4 \mid A, B, C$; and, since $M \equiv 2 \pmod 4$,

$$\widehat{A} = 3Maa' - a'b - ab' \equiv 2 - 1 - 1 \equiv 0 \pmod 4,$$

with the same calculation for $\widehat{B}$. Dividing the determinant identity and the complementary conics by 16 gives

$$A_0B_0 = M_1^2C_0, \tag{7.3}$$

$$A_0^2 + \widehat{A}_0^2 + 3MA_0\widehat{A}_0 = M_1^4, \tag{7.4}$$

$$B_0^2 + \widehat{B}_0^2 + 3MB_0\widehat{B}_0 = M_1^4. \tag{7.5}$$

Theorem 7.2 (Normalized even collision allocation)**.** *If an even Markoff number $M > 2$ admits an ordered collision, then there are coprime proper unitary factors $F, E > 1$ of $M_1 = M/2$ and positive integers H, W, n, m such that*

$$M_1 = FE, \qquad A = 4F^2W, \quad B = 4E^2H, \quad C = 4HW, \tag{7.6}$$

and

$$\widehat{A} = 4F^2m, \qquad \widehat{B} = 4E^2n. \tag{7.7}$$

Moreover,

$$n^2 + 3MHn + H^2 = F^4, \tag{7.8}$$

$$m^2 + 3MWm + W^2 = E^4, \tag{7.9}$$

and

$$\gcd(F, H) = \gcd(H, n) = 1, \qquad \gcd(E, W) = \gcd(W, m) = 1. \tag{7.10}$$

Proof. We allocate each complete prime-power factor of M_1 to one side of the normalized identity. Let $p^e \parallel M_1$. The prime p is odd. Lemma 3.4 says it cannot divide both A_0 and B_0. Comparing prime exponents in (7.3) gives

$$\operatorname{ord}_p(A_0) + \operatorname{ord}_p(B_0) = 2e + \operatorname{ord}_p(C_0).$$

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One exponent on the left is zero, so the other is at least $2e$. Allocate p^e to F if its square divides A_0, and to E if its square divides B_0. This gives

$$M_1 = FE, \quad \gcd(F, E) = 1, \quad A_0 = F^2 W, \quad B_0 = E^2 H,$$

and the normalized identity then gives $C_0 = HW$.

The factors are proper. If $F = 1$, then $M_1^2 = E^2 \mid B_0$, whereas $0 < B_0 < M^2/4 = M_1^2$. This is impossible. If $E = 1$, the same argument applies to A_0. Multiplying the normalized quantities by four proves (7.6).

For a prime block $p^e \parallel F$, suppose $v = \mathrm{ord}_p(\widehat{A}_0) < 2e$. In (7.4), the term $\widehat{A}_0^2$ has exponent $2v$. The exponents of A_0^2 and M_1^4 are at least $4e$, and that of $3MA_0\widehat{A}_0$ is at least $3e + v$. All exceed $2v$. Division by p^{2v} would leave one nonzero term modulo p, a contradiction. Hence $F^2 \mid \widehat{A}_0$. The other conic gives $E^2 \mid \widehat{B}_0$. Write $\widehat{A}_0 = F^2 m$, $\widehat{B}_0 = E^2 n$.

Substitution in the normalized conics, followed by division by F^4 and E^4, gives the two defect equations. If a prime divided F, H, or E, W, it would divide both A, B, which is excluded. Finally, a prime dividing both coordinates H, n of the first defect equation would divide F, contradicting $\gcd(F, H) = 1$. The other equation proves $\gcd(W, m) = 1$ in the same way. □

Combining Zhang's 2-adic congruence with Srinivasan's determinant splitting thus gives the normalized packet above. The underlying coprime factorization and the twice-prime-power consequence are not asserted as new [14]. To the best of our knowledge, the coefficient- $6TO$ packet and the explicit floors below have not previously been recorded.

Abstract either defect equation as

$$x^2 + y^2 + 6TOxy = T^4, \qquad TO = M_1 \equiv 1 \pmod{16}, \tag{7.11}$$

where $T, O > 1$ are coprime, neither is divisible by 3, every prime dividing TO is $1 \pmod 4$, and $\gcd(x, y) = 1$.

Lemma 7.3 (The sharp local floor 72)**.** *Every solution of* (7.11) *satisfying these block conditions has* $xy \geq 72$. *The value* 72 *is sharp for this local relaxation.*

Proof. Reduction modulo three shows that exactly one of x, y is divisible by three. Since $T^4 \equiv 1 \pmod{16}$ and $TO \equiv 1 \pmod{16}$, reduction modulo 16 gives

$$x^2 + 6xy + y^2 \equiv 1 \pmod{16}. \tag{7.12}$$

Take $x \leq y$. Under $xy < 72$, the coprime pairs satisfying these two congruence conditions are exactly

$$\begin{gathered}(1, 18), (1, 24), (1, 42), (1, 48), (1, 66),\\ (2, 3), (2, 9), (2, 27), (2, 33),\\ (3, 4), (3, 10), (3, 20), (5, 6), (5, 12), (6, 7).\end{gathered}$$

For each pair, the equation forces $T \mid x^2 + y^2$ and

$$O = \frac{T^3 - (x^2 + y^2)/T}{6xy}. \tag{7.13}$$

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List the divisors $T > 1$ having only prime divisors $1 \pmod 4$, and substitute each in this formula. The only integer values $O > 1$ are the following; every one fails a block condition.

x	y	T	O	obstruction
1	18	325	317853	$3 \mid O$
1	24	577	1334028	$2 \mid O$
1	42	1765	21818937	$3 \mid O$
1	48	2305	42522648	$2 \mid O$
1	66	4357	208865877	$3 \mid O$
2	3	13	61	$TO \equiv 9 \pmod{16}$
3	4	25	217	$7 \mid O$
5	6	61	1261	$TO \equiv 9 \pmod{16}$
6	7	85	2437	$TO \equiv 9 \pmod{16}$

Thus $xy < 72$ is impossible. To check sharpness for the stated local conditions, take

$$(x, y, T, O) = (8, 9, 145, 7057), \qquad 145 = 5 \cdot 29.$$

The number 7057 is prime and $1 \pmod 4$; the two blocks are coprime and have product $1 \pmod{16}$. Moreover,

$$8^2 + 9^2 = 145, \qquad 1 + 432 \cdot 7057 = 145^3.$$

Multiplying the second identity by 145 verifies the defect equation. This admissible local row has product 72, proving sharpness for the relaxation. □

The separate even certificate enumerates the intentionally enlarged domain $72 \le xy < 100000$ twice, first by pairs and then by products. It retains 65 local rows, of which 63 satisfy $\gcd(x, y) = 1$. Applying Lemma 6.2 to the actual candidate maximum $M = 2TO$ leaves exactly four primitive rows:

xy	x	y	T	O	$M = 2TO$
1014	3	338	61	37	4514
7140	84	85	14281	67987081	1941847007522
74256	272	273	148513	7352086561	2183760862867586
96624	9	10736	613	397	486722

(7.14)

Since $M = 2M_1$, one has $3M \pm 2 = 2(3M_1 \pm 1)$; hence the odd-prime companion test is equivalently applied to $3M_1 \pm 1$. The first row proves sharpness for that relaxation: with $M_1 = 2257$,

$$3M_1 - 1 = 2 \cdot 5 \cdot 677, \qquad 3M_1 + 1 = 2^2 \cdot 1693,$$

so all odd companion primes are $1 \pmod 4$.

The canonical test of Lemma 6.9 uses the even maximum $M = 2TO$, not the half-maximum. For $M = 4514$, the interval roots are $1597, 1819$, with continued fractions

$$[2, 1, 4, 1, 3, 3, 1, 4, 1, 2], \qquad [2, 2, 13, 13, 2, 2].$$

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The middle two rows in (7.14) have no root of -1 in $(M/3, M/2)$. For $M = 486722$, the interval roots are $191291, 193673$, with continued fractions

$$[2,1,1,5,7,1,2,2,1,7,5,1,1,2], \quad [2,1,1,18,1,1,3,3,1,1,18,1,1,2].$$

Every displayed expansion has a digit greater than 2. Thus all four rows fail the canonical-continuant necessity.

Theorem 7.4 (Certified even tiers)**.** *Every hypothetical even collision satisfies the hand tier*

$$Hn, Wm \geq 72, \qquad F^4, E^4 \geq 216M + 144. \tag{7.15}$$

The exact companion-prime certificate strengthens this to

$$Hn, Wm \geq 1014, \qquad F^4, E^4 \geq 3042M + 2028, \tag{7.16}$$

and the canonical-continuant certificate gives

$$Hn, Wm \geq 100000, \qquad F^4, E^4 \geq 300000M + 200000. \tag{7.17}$$

The floors 72 *and* 1014 *are sharp for their stated relaxations;* 100000 *is a certified cutoff, not a sharpness claim.*

Proof. Each packet in Theorem 7.2 has $(T, O) = (F, E)$ or (E, F), with actual maximum $M = 2TO$. Lemma 7.3 gives the hand floor $xy \geq 72$. In the complete search above this floor and below 100000, the companion filter leaves exactly the four displayed rows, whose smallest product is 1014. Thus the necessary companion condition improves the floor to 1014.

The continued-fraction condition must be applied modulo the full maximum M. Two of those four rows have no root in its required interval, and the other two have only the displayed continued fractions, each containing a digit above two. Thus none is a Markoff occurrence, and the final floor is 100000.

For each established floor B_0, either packet satisfies

$$T^4 = x^2 + y^2 + 3Mxy \geq (3M + 2)xy \geq B_0(3M + 2).$$

Substituting $B_0 = 72, 1014, 100000$ gives the three claimed quartic bounds. The rows at products 72 and 1014 prove sharpness only for their specified relaxations; no row establishes sharpness at 100000. □

Remark 7.5 (Provenance of the even tiers)*.* The 72 tier is proved entirely in the article. The 1014 and 100000 tiers are computer-assisted theorems certified by two separately implemented exact-integer traversals supplied in the supplementary archive. The companion and canonical filters are necessary for every actual Markoff occurrence by Lemmas 6.2 and 6.9.

8 Intrinsic unitary-gap and tail consequences

For the odd case, it is useful to state all four tiers at once. Let B_0 be one of

$$33, \quad 120, \quad 67137, \quad 100000,$$

and put $\mathcal{C} = 3B_0$. The corresponding tier says

$$F^4, E^4 \geq \mathcal{C}M + \frac{2\mathcal{C}}{3}. \tag{8.1}$$

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Theorem 8.1 (Intrinsic unitary-gap criterion)**.** *Let M be an odd Markoff number. At any certified coefficient $\mathcal{C} \in \{99, 360, 201411, 300000\}$,*

$$u_*(M)^4 < \mathcal{C}M + \frac{2\mathcal{C}}{3} \quad \Longrightarrow \quad M \text{ is unique.} \tag{8.2}$$

Proof. The maximum $M = 1$ has the single ordered occurrence $(1, 1, 1)$. For $M > 1$, suppose a collision existed and set $d = \min(F, E)$ in Theorem 3.5. This is a unitary divisor of M and satisfies $d \le \sqrt{M}$, so it belongs to the set defining $u_*(M)$. Hence $d \le u_*(M)$.

The chosen collision tier gives $d^4 \ge \mathcal{C}M + 2\mathcal{C}/3$. Together with the assumed inequality, this would give

$$\mathcal{C}M + 2\mathcal{C}/3 \le d^4 \le u_*(M)^4 < \mathcal{C}M + 2\mathcal{C}/3,$$

a contradiction. Thus no collision exists. □

Theorem 8.2 (Even intrinsic unitary-gap criterion)**.** *Let $M > 2$ be an even Markoff number and put $M_1 = M/2$. At any certified even coefficient*

$$\mathcal{C} \in \{216, 3042, 300000\},$$

with respectively hand, companion-certificate, and continuant-certificate provenance,

$$u_*(M_1)^4 < \mathcal{C}M + \frac{2\mathcal{C}}{3} \quad \Longrightarrow \quad M \text{ is unique.} \tag{8.3}$$

Proof. Suppose a collision existed. Theorem 7.2 gives a unitary split $M_1 = FE$. Its smaller factor $d = \min(F, E)$ is a unitary divisor at most $\sqrt{M_1}$, so $d \le u_*(M_1)$.

The chosen tier in Theorem 7.4 bounds each factor's fourth power below by $\mathcal{C}M + 2\mathcal{C}/3$. Therefore

$$\mathcal{C}M + 2\mathcal{C}/3 \le d^4 \le u_*(M_1)^4 < \mathcal{C}M + 2\mathcal{C}/3,$$

which is impossible. Notice that the divisor function uses M_1, while the quartic bound uses the full maximum M. □

For a composite tail, define

$$\mathcal{G}(N) = \frac{N}{u_*(N)^4} = \min_{\substack{UV=N \\ \gcd(U,V)=1}} \frac{\max(U, V)}{\min(U, V)^3}. \tag{8.4}$$

The equality follows by putting $d = \min(U, V)$: the displayed ratio is N/d^4, minimized by the largest unitary $d \le \sqrt{N}$.

Theorem 8.3 (Composite unitary tail)**.** *Let $M = kN$ be an odd Markoff number with $\gcd(k, N) = 1$. At any certified coefficient $\mathcal{C}$, the condition*

$$k^3 u_*(N)^4 \le \mathcal{C}N, \quad \textit{equivalently} \quad k^3 \le \mathcal{C}\mathcal{G}(N), \tag{8.5}$$

guarantees that M is unique.

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Proof. Assume there is a collision, with unitary split $M = FE$. Because $\gcd(k, N) = 1$, every prime-power block of N lies wholly in one of these two factors. Write its resulting split as $N = XY$, with $\gcd(X, Y) = 1$, and choose the names so that $X \le Y$. Then $X \le u_*(N)$.

The collision factor containing X has the form dX, where d is a unitary divisor of k. It is therefore at most $ku_*(N)$. Apply the collision tier to this factor and the assumed bound to its upper estimate:

$$\mathcal{C}kN + \frac{2\mathcal{C}}{3} \le (dX)^4 \le k^4 u_*(N)^4 \le \mathcal{C}kN.$$

The positive term $2\mathcal{C}/3$ makes this chain impossible. It also explains why a non-strict inequality is allowed in the hypothesis. □

Taking $N = q^e$ makes $u_*(N) = 1$, proving Corollary 1.2. More generally, if $N = XY$ has exactly two coprime primary blocks with $X < Y$, then (8.5) is $(kX)^3 \le \mathcal{C}Y$.

The even analogue applies to the odd core of the maximum.

Theorem 8.4 (Even composite unitary tail)**.** *Let $M = 2kN$ be an even Markoff number with $\gcd(k, N) = 1$. At any certified even coefficient $\mathcal{C} \in \{216, 3042, 300000\}$, the condition*

$$k^3 u_*(N)^4 \le 2\mathcal{C}N \tag{8.6}$$

guarantees that M is unique.

Proof. Suppose there is a collision. Its unitary factorization is now of $M/2 = kN$. Split the complete prime-power blocks of N according to the two collision factors, writing $N = XY$ with $X \le Y$. As before, $X \le u_*(N)$, and the collision factor containing X is dX for a unitary divisor d of k. Thus $dX \le ku_*(N)$.

The even tier uses $M = 2kN$. It and the assumed bound give

$$2\mathcal{C}kN + \frac{2\mathcal{C}}{3} \le (dX)^4 \le k^4 u_*(N)^4 \le 2\mathcal{C}kN,$$

a contradiction. Hence the maximum is unique. □

Example 8.5 (The tier transition at 51641)**.** The Markoff triple $(13, 1325, 51641)$ shows directly that

$$51641 = 113 \cdot 457$$

is a Markoff number. It has $u_*(51641) = 113$ and $113^4 = 163047361$. The right sides of (8.2) are

$$5112525, \quad 18591000, \quad 10401199725, \quad 15492500000$$

for the four coefficients in order. Thus this intrinsic test does not close the number at tiers 99 or 360, but it does at tier 201411, before either final-row rejection is used. The example illustrates the logical role of the intermediate tier and proves that the present sufficient unitary-gap criterion is strictly stronger than Button's coefficient-1 criterion. It is not a claim that the uniqueness of 51641 was previously unknown by all other methods.

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9 Certificate protocol and reproducibility

Use of generative AI in the research methodology. OpenAI ChatGPT and Codex were used for mathematical exploration, provisional proof drafting and review, exact-arithmetic code development and testing, reference checking, and manuscript preparation. This assistance included substantive research and writing, beyond copy editing. AI outputs were treated as proposals requiring mathematical justification; the proofs and the supplied executable checks provide the stated evidence. The author directed the research and is responsible for the final text, arguments, references, and code. The supplied checks are exact computational certificates, not a formal proof-assistant verification or an external referee report.

The supplementary archive contains six current proof programs. Four certify the odd and normalized-even defect-product enumerations: the two primary programs enumerate pairs first, emit deterministic JSON/CSV/text objects, and include internal product-order cross-checks; the two standalone programs enumerate products first, import no primary code, and independently implement factorization, divisors, admissibility, companion filtering, CRT, and continued fractions. Two further, independently structured programs certify the odd residual crossed-Pell rejection. Specifically, the files are

1. `verify_four_tiers.py` and `verify_four_tiers_product_first.py` for the four odd tiers;
2. `verify_odd_crossed_pell.py` and `verify_odd_crossed_pell_divisor_first.py` for the residual allocation, the four packet/coordinate orientations, the complete modular root lists, and the product-chamber bounds used in the final odd rejection;
3. `verify_even_tiers.py` and `verify_even_tiers_product_first.py` for the three normalized even tiers;
4. deterministic machine-readable results, exact tables, and reference transcripts in the following four directories.

`generated/`	`generated_even/`
`generated_crossed_pell/`	`generated_crossed_pell_divisor_first/`

5. `MANIFEST.sha256`, hashing every archived source and result; and
6. `legacy/verify_defect_120.py` and `legacy/verify_defect_global_100000.py`, retained as provenance for the original odd-tier programs from which the unified odd certificate was reconstructed.

The principal commands are

```
python3 verify_four_tiers.py --mode primary --output-dir odd_replay
python3 verify_four_tiers_product_first.py
python3 verify_four_tiers.py --mode both
python3 verify_odd_crossed_pell.py --output-dir crossed_pell_replay
python3 verify_odd_crossed_pell_divisor_first.py \
  --output-dir crossed_pell_divisor_replay
python3 verify_even_tiers.py --mode primary --output-dir even_replay
python3 verify_even_tiers_product_first.py
python3 verify_even_tiers.py --mode both
```


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All six current verifiers are also replayed under `python3 -O`, so no mathematical check depends on optimization-sensitive assertions. Python 3.10 or later is required; the six proof programs use only the standard library. The separate plotting program `draw_fg.py` uses NumPy and Matplotlib. No floating-point value enters a certificate test. The new figure is a numerical illustration of the stated functions and is not used in the proof. Environment and timing metadata appear only in the run logs and are not part of the deterministic certificate files.

The odd primary program records the exhaustive counts in Table 4. The modulo-3 gate is the proved condition that exactly one of x, y is divisible by 3.

count	$33 \le xy < 120$	$120 \le xy < 100000$
pairs before modulo-3 gate	236	583217
pairs after modulo-3 gate	91	246960
divisors of $x^2 + y^2$ examined	530	2942721
admissible target divisors T	209	1307435
survivors in enlarged domain	0	19
of which $\gcd(x, y) = 1$	0	18

Table 4: Exact enumeration counts emitted by the primary verifier.

Table 5 records the corresponding pair-first counts for the even coefficient-$6TO$ traversal over $72 \le xy < 100000$.

count	value
pairs before modulo-3 gate	583354
pairs after modulo-3 gate	247013
divisors of $x^2 + y^2$ examined	2943053
admissible target divisors T	1307570
survivors in enlarged domain	65
of which $\gcd(x, y) = 1$	63
companion-prime survivors	4
canonical-continuant survivors	0

Table 5: Exact counts emitted by the primary even verifier.

The source hashes at the time of submission preparation are

file	SHA-256
`verify_four_tiers.py`	`46b0d81367e8a94684a32f6dc08fce484ccd8ab005d4c0946d61b3b1b9a79cc5`
`verify_four_tiers_product_first.py`	`b908dfd28df6c540088f0caf5c83f5ea85d23752f638f95d938c9bb2f3cc8231`
`verify_odd_crossed_pell.py`	`6388d0e8a50c65a3f5e741ede96e2314931c991324185fd3ca3d5f6178ce18aa`
`verify_odd_crossed_pell_divisor_first.py`	`d7143b2be8471c77a1721775b41743fcd4e0be3b7303a96c584e9b7a3519ab72`
`verify_even_tiers.py`	`01f8bb050d3e16bcd9b424ec1d9fb6123c4f60f8feb446ff521df9e130df140c`
`verify_even_tiers_product_first.py`	`10c111181f945c49c0a4d3e72766f88a712c467ea7476d43a260c7cfef279ed9`
`legacy/verify_defect_120.py`	`8cd0ab00f3546e2340a2380b780fbb0415331fadc93264d99ed7078863a93a27`
`legacy/verify_defect_global_100000.py`	`805e94175bf906f6ba70b0e0da0ae6486d72476c6fc24a8dee42f825baff3874`

The archive manifest is authoritative if any filename is moved or any source is revised.

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To reproduce the mathematical certificate, a reader can use the following checks.

(i) verify the manifest before executing code;

(ii) run all six current verifiers under a standard interpreter and under `python3 -O`;

(iii) compare the nineteen-row enlarged-domain CSV against Table 3 and verify that eighteen rows have $\gcd(x, y) = 1$;

(iv) compare the sixty-five-row even tables, verify that sixty-three rows are primitive, and verify the standalone program's canonical four-column row-serialization digest against the reference transcript;

(v) independently multiply each row into its odd or even defect equation;

(vi) verify every displayed companion witness and factorization in both parity sectors;

(vii) recompute the odd and even CRT roots, canonical-interval roots, and continued fractions;

(viii) compare the two residual-orientation tables, verify all four packet/coordinate cases, and recompute the two crossed equations, their product-chamber caps, and their complete modular square-root lists.

10 Scope and remaining problem

The argument gives a rigorous partial theorem, not a proof of full unicity. It establishes that a hypothetical odd collision must admit a proper unitary factorization of M whose two factors satisfy strong quartic lower bounds. For an even collision the corresponding proper unitary factorization is of the odd core $M/2$, with the same final quartic bound. In either parity the prime-power blocks therefore cannot be allocated too unevenly in this quantified sense. No fixed coefficient alone can exclude every increasingly balanced unitary allocation.

The determinant argument does give a precise blockwise switch for elementary occurrence roots. For a collision, set

$$t \equiv ba^{-1}, \qquad t' \equiv b'(a')^{-1} \pmod{M}.$$

Both satisfy $u^2 \equiv -1 \pmod{M}$. If $p^e \parallel F$, then $F^2 \mid A$ gives $tt' \equiv 1 \pmod{p^e}$, and hence $t' \equiv -t \pmod{p^e}$. If $p^e \parallel E$, then $E^2 \mid B$ gives $t' \equiv t \pmod{p^e}$. Thus a collision switches exactly the F-blocks for these ratio roots. The present paper does not identify this ratio root with the canonical interval root of Lemma 6.9 in a form uniform enough to finish the conjecture.

The crossed-Pell reconstruction in Section 6.3 recovers additional primitive-slope information and couples the two defect packets. For the final-row certificate, this coupling is more restrictive than the companion unitary and quadratic-residue conditions alone. It does not, however, uniformly identify an elementary occurrence root with the canonical interval root, nor does it preclude every balanced unitary allocation.

The determinant and continuant viewpoints therefore suggest a narrow next problem.

Problem 10.1 (Unitary switch separation)**.** For an odd ordered collision, let $K = M$; for an even ordered collision, let $K = M/2$. In either case write the collision split as $K = FE$. Relate the two ratio roots above to the canonical roots ρ, ρ' supplied by Lemma 6.9. In the even case, K is only

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the split modulus: the canonical roots remain roots of -1 modulo the full maximum M. Prove that no two canonical roots attached to distinct occurrences can satisfy

$$\rho' \equiv -\rho \pmod{F}, \qquad \rho' \equiv \rho \pmod{E}$$

while both lie in the canonical interval and have only $\{1,2\}$-digits.

The certified tiers treat both parity sectors and eliminate every collision whose relevant unitary allocation is sufficiently unbalanced. Problem 10.1 asks for both the missing canonical-root bridge and the universal separation needed in the balanced range.

Statements and Declarations

Data and code availability. Online Resource 1 (`ESM_1.zip`) supplies the six current proof programs, machine-readable certificate results, enlarged-domain row tables, reference transcripts, and SHA-256 manifest supporting this article. The programs require Python 3.10 or later and use only its standard library. No external empirical dataset is used.

Supplementary information. Online Resource 1: Exact finite certificates for the odd and normalized-even collision tiers, including two independent implementations of the residual crossed-Pell rejection, the program and parameters for Figure 1, and instructions for reproduction.

Generative AI assistance. The substantive uses of OpenAI ChatGPT and Codex in mathematical exploration, proof drafting, code development, testing, and writing are described in Section 9. The author is responsible for the article and its accompanying code and results.

Author contribution. Elleard Felix Webster Heffern formulated the research questions and framework, directed the investigation, and is responsible for the article and its accompanying materials.

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Related work by the author. The separate manuscript *Chebyshev Reference Bounds and Certified Spectral Refinement for Markov Numbers* develops bounds for matrix products and localization of possible collisions among Farey labels. The present article proves arithmetic constraints on a shared maximum and uses separate finite certificates. Neither manuscript requires the other's results. Earlier versions of the present arithmetic work circulated under the titles *Quantitative Unitary-Factor Bounds for Collisions of Markov Triples* and *Unitary-Factor Obstructions to Collisions of Markov Triples*; these are related versions of this work, not independent companion contributions.

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Original: heffern_markoff_certified_tiers_v6_residual_refereed.pdf

References

[1] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013, https://doi.org/10.1007/978-3-319-00888-2.

[2] A. Baragar, On the unicity conjecture for Markoff numbers, *Canad. Math. Bull.* **39** (1996), no. 1, 3–9, https://doi.org/10.4153/CMB-1996-001-x.

[3] J. O. Button, The uniqueness of the prime Markoff numbers, *J. London Math. Soc.* (2) **58** (1998), no. 1, 9–17, https://doi.org/10.1112/S0024610798006292.

[4] J. O. Button, Markoff numbers, principal ideals and continued fraction expansions, *J. Number Theory* **87** (2001), no. 1, 77–95, https://doi.org/10.1006/jnth.2000.2578.

[5] İ. Çanakçı and R. Schiffler, Snake graphs and continued fractions, *European J. Combin.* **86** (2020), Article 103081, https://doi.org/10.1016/j.ejc.2020.103081.

[6] F.-J. Chen and Y.-G. Chen, On the Frobenius conjecture for Markoff numbers, *J. Number Theory* **133** (2013), no. 7, 2363–2373, https://doi.org/10.1016/j.jnt.2012.12.018.

[7] H. Cohn, Approach to Markoff's minimal forms through modular functions, *Ann. of Math.* (2) **61** (1955), no. 1, 1–12, https://doi.org/10.2307/1969618.

[8] H. Cohn, Representation of Markoff's binary quadratic forms by geodesics on a perforated torus, *Acta Arith.* **18** (1971), 125–136, https://doi.org/10.4064/aa-18-1-125-136.

[9] D. Fisac, Markov's conjecture on integral necklaces, *Bull. London Math. Soc.* **57** (2025), no. 12, 4122–4131, https://doi.org/10.1112/blms.70178.

[10] G. Frobenius, Über die Markoffschen Zahlen, *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin*, Jahrgang 1913, erster Halbband, 458–487.

[11] M. L. Lang and S. P. Tan, A simple proof of the Markoff conjecture for prime powers, *Geom. Dedicata* **129** (2007), 15–22, https://doi.org/10.1007/s10711-007-9189-x.

[12] A. Markoff, Sur les formes quadratiques binaires indéfinies, *Math. Ann.* **15** (1879), 381–406, https://doi.org/10.1007/BF02086269.

[13] C. Reutenauer, *From Christoffel Words to Markoff Numbers*, 2nd ed., Oxford University Press, Oxford, 2026, https://doi.org/10.1093/9780197907535.001.0001.

[14] A. Srinivasan, Markoff numbers and ambiguous classes, *J. Théor. Nombres Bordeaux* **21** (2009), no. 3, 757–770, https://doi.org/10.5802/jtnb.701.

[15] J. M. Tornero, A geometric approach to the Frobenius unicity conjecture for the Markoff equation, *J. Lie Theory* **18** (2008), no. 3, 595–600, https://doi.org/10.5802/jolt.513.

[16] Y. Zhang, Congruence and uniqueness of certain Markoff numbers, *Acta Arith.* **128** (2007), no. 3, 295–301, https://doi.org/10.4064/aa128-3-7.

[17] C. O'Sullivan, Patterns in the Markov numbers and their generalizations, preprint (2026), arXiv:2609.14149v1.

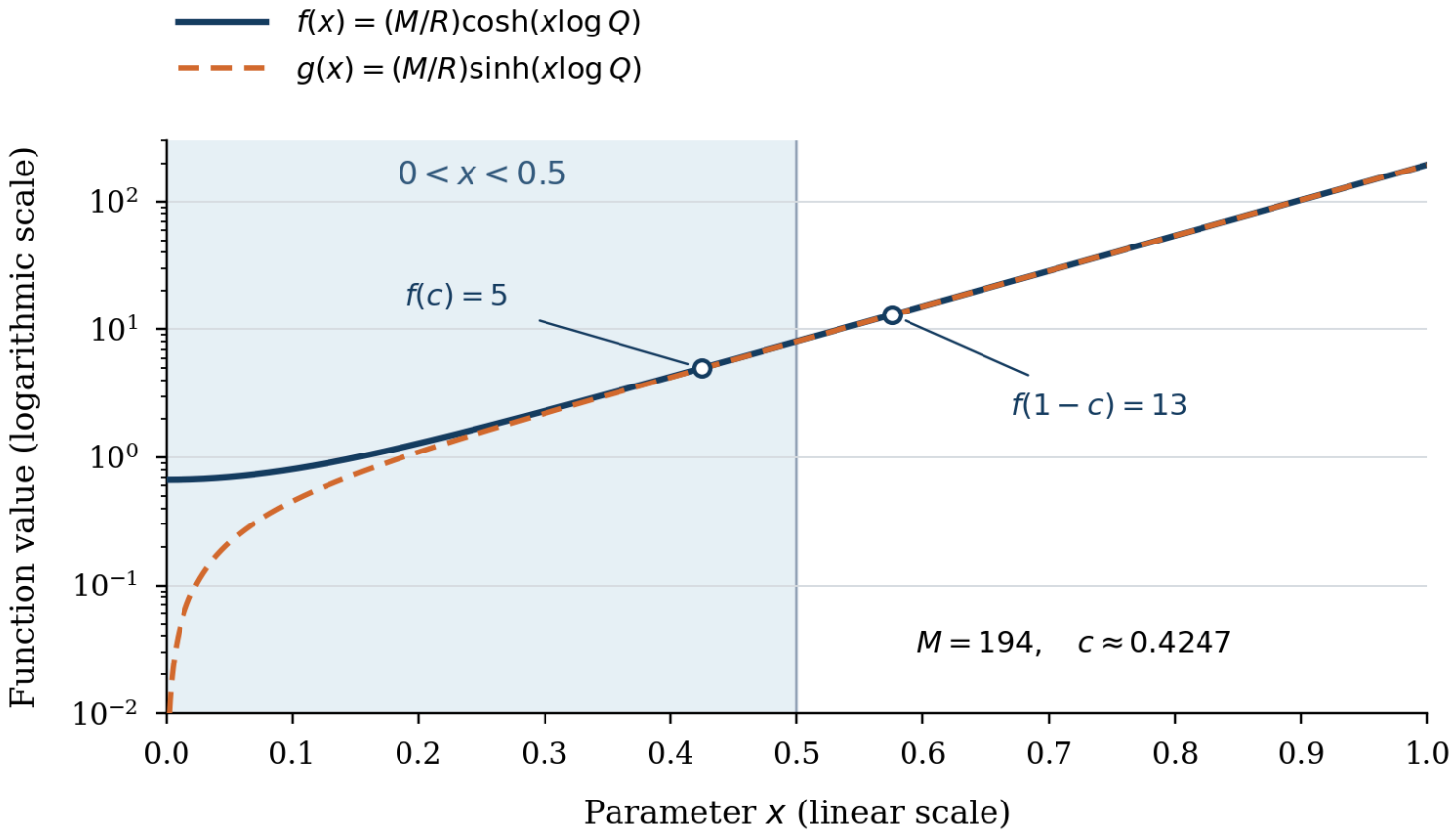
[18] S.-i. Yasutomi, On the Ordering and Injectivity of Lagrange Constants Associated with Certain Substitutions, preprint (2026), arXiv:2609.32450v1.

[19] E. Banaian, Orderings on k-Markov numbers, *The Ramanujan Journal* **71** (2026), Article 12, https://doi.org/10.1007/s11139-026-01468-5.

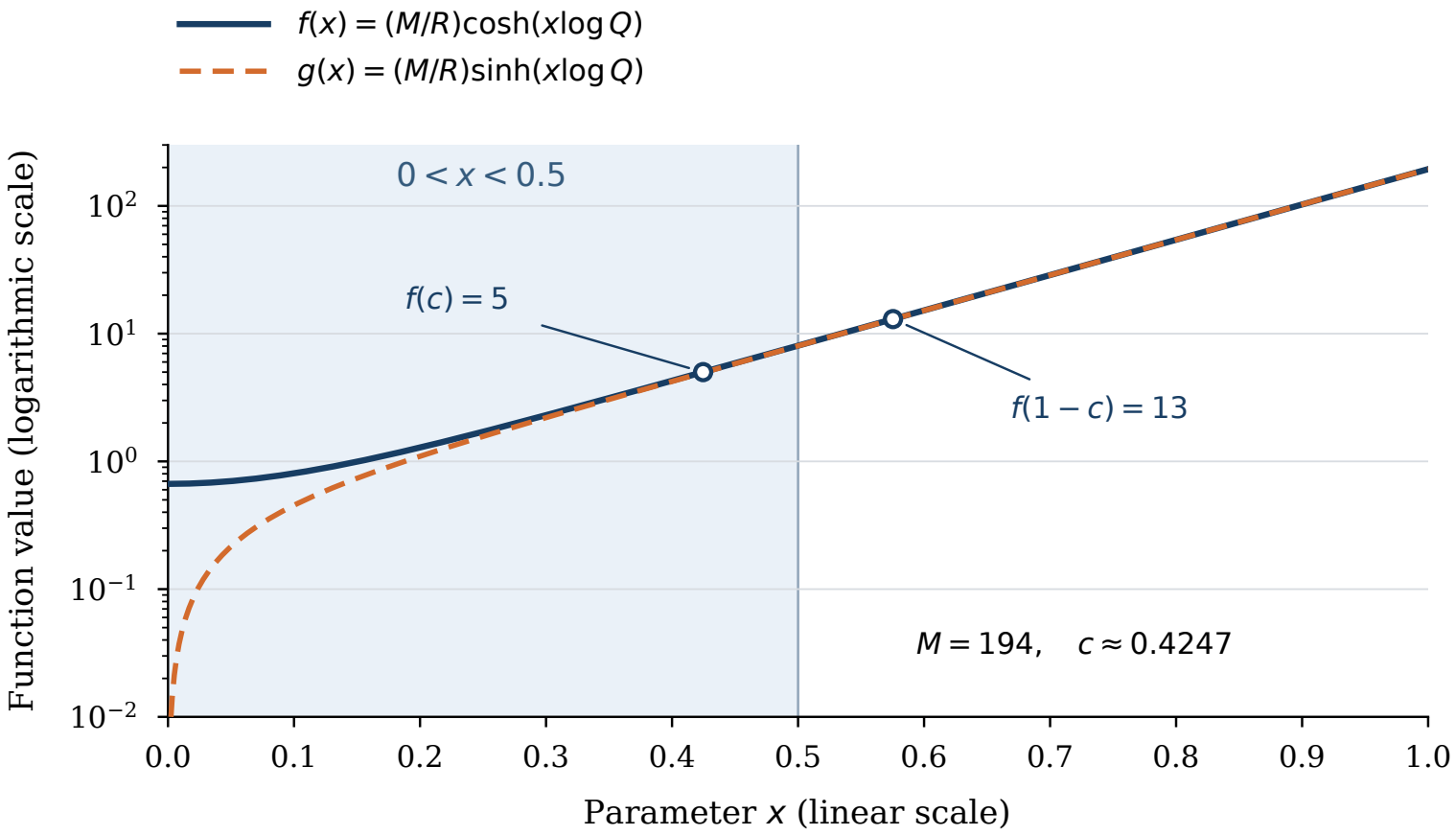
[20] Y. Gyoda, An Introduction to the Lagrange and Markov Spectra through the Lens of Generalized Markov Numbers, preprint (2026), arXiv:2609.28097v1.

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Original figure: Fig1_fg_log.png



$g(x) \to 0$ as $x \to 0^{+}$; zero is outside the logarithmic axis.



$g(x) \to 0$ as $x \to 0^{+}$; zero is outside the logarithmic axis.

Meridians on the Markov Surface: Simple Indexing for Fast, Low-Storage Verification of 550 Billion Configurations

Completion and release history

Supplement-only release dated 9 November 2026; the main manuscript remains version 11, unchanged. Six decimal expansions totaling 6,043,218 digits were posted in this release. Their later completed-certificate rank confirmation belongs to v21; do not backdate that confirmation to this posting.

Zenodo publication-date field: 2026-11-09

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The publication-date field is the date supplied in the deposit. The API created/modified timestamps are repository metadata, not a separately attested first-view time. Full fractional timestamps are retained in the metadata archived in Part C.

https://zenodo.org/records/24093257

https://doi.org/10.5281/zenodo.24093257

Earlier published Zenodo drafts in this version family:

v4_ps | 2026-11-07 13:37:47 CST / 2026-11-07 19:37:47 UTC | https://zenodo.org/records/24043052

v5_ps | 2026-11-07 18:35:22 CST / 2026-11-08 00:35:22 UTC | https://zenodo.org/records/24046012

v7_ps | 2026-11-08 07:26:11 CST / 2026-11-08 13:26:11 UTC | https://zenodo.org/records/24053720

v9_ps | 2026-11-09 07:45:32 CST / 2026-11-09 13:45:32 UTC | https://zenodo.org/records/24084246

v11 | 2026-11-09 11:29:25 CST / 2026-11-09 17:29:25 UTC | https://zenodo.org/records/24089717

Earlier public announcements: 21 October 2026, 10:18:25 CDT / 15:18:25 UTC on X (more than a billion values in under 40 minutes); 23 October 2026, 09:54:57.771 CDT / 14:54:57.771 UTC on LinkedIn (9.3 billion in 24 minutes). These concern earlier computations, not the final trillion certificate.

https://x.com/webystar/status/2112926487091282350

https://www.linkedin.com/posts/elleard-felix-webster-heffern-joseph-by-confirmation-2953a3104_markovnumbers-activity-7519411048115003392-4NaJ

The dates and URLs are retained in the 7 November public-post evidence packet; the LinkedIn URL is also embedded in the published computational supplement. Additional November LinkedIn announcements and the older Meridian ResearchGate v13 URL could not be established in this compilation.

Complete public Zenodo description

Supplement-only update, 9 November 2026. The main manuscript remains v11, with its PDF unchanged byte for byte. This record adds v1_Selected_Markov_Values_Supplement.zip and the v12 source-and-supplements package, plus a package-preservation check. The supplement contains six full decimal expansions (6,043,218 digits), exact addresses, checksums, and reproducible counting and reconstruction code.

The four strict neighbors of 10^999999 and 10^1000000 are certified by complete boundary searches. The other two values occupy positions 999,999,999,999 and 1,000,000,000,000 in the ordered canonical-triple enumeration, counting 1 first. Their corresponding ranks among distinct Markov values remain provisional pending the ongoing trillion-value unicity verifier's final checks. This release does not claim that the trillion-value verification is complete.

Meridians organize positive integer Markov triples into families with compact addresses and explicit evaluation rules. This release presents simple indexing for fast, low-storage verification and retains the catenary-based introduction, geometric descent, residue patterns, collision windows, corridor recurrences, and Fibonacci-matrix and snake-graph arguments.

Completed verification: 550,693,634,649 configurations were compared, certifying uniqueness of the first 500,000,000,000 Markov values in numerical order. The larger count is the comparison workload, including possible competitors and conservative cutoff coverage; it is not a claim about the first 550 billion numerical values. All four repeated 64-bit residue groups, comprising eight exact candidates, were resolved with no equal maxima. The exact 500-billionth value was not identified.

Production began on 8 November 2026 at 14:55:01.580398 UTC. The last retained band completed on 9 November 2026 at 15:50:31.046707 UTC (09:50:31.046707 CST). On eight workers, retained active elapsed band processing took 54,470.884742 seconds: 15 h 07 min 50.885 s. Summed native-child CPU time was 264,075.377305 seconds (73.354 hours); parent CPU was not separately totalled. The calendar span was 24.925 hours. Active band time excludes pauses, restoration, lost attempts, and subsequent audits; the manuscript reports these scopes separately.

Maximum recorded temporary key-file storage was 754,846,136 bytes (719.877 MiB; 0.703 GiB), including the shard duplicated during partitioning. This excludes receipts, source files, backups, filesystem overhead, and unrelated work. Generator-only resident memory peaked at 28 MiB. A continuous whole-workspace disk or aggregate-memory peak was not recorded. The frozen recovery archive is 27,168,031 bytes.

The complete 13,638-band receipt chain and exact candidate evidence were audited. A separate checker recomputed the numerical envelope, independently reconstructed all candidate maxima, and regenerated every repeated-key band. The final independent replay completed at 15:51:24.981257 UTC; a fresh portable extraction verified every manifested archive member and completed its replay at 15:57:38.529312 UTC on 9 November 2026. This is a receipt-and-candidate replay, not a second full enumeration of discarded singleton keys.

The separate degree-bounded certificate covers all 5,067,077 Meridians of degree at most 10,000 at every integer phase. The equal-degree separation result, the author's Fibonacci-matrix normalization and fixed-numerator arguments, and his six-piece fixed-sum and fixed-denominator arguments are retained. Earlier ordering results and the 2002-2003 origins of the matching interpretation are credited in the paper. The unrestricted Markov unicity problem remains open.

Version 11 removes the obsolete submission-status note and associated wording from the manuscript. Profiles and related derivations continue to use explicit powers, square roots, and logarithms. The original catenary figure and all formal mathematical statements are preserved.

Files include the unchanged v11 manuscript, the expanded v12 source-and-supplements package with all figure assets and the frozen verifier archive, release notes, separately downloadable recovery archive, final verifier report, machine-readable final evidence, progress chart, historical six-page fifty-billion proof-and-timing supplement, and standalone catenary figure. The historical fifty-billion run completed on 28 October 2026; its separate original archive remains available at https://doi.org/10.5281/zenodo.23874075. Historical filenames retain the original 500-billion certification target.

Licensing: Original software is licensed under MIT; original documentation and data under CC BY 4.0. Third-party materials and dependencies retain their own terms. Copyright (c) 2026 Elleard Felix Webster Heffern, to the extent copyright applies.

Funding: No funding. Substantial generative-AI assistance is disclosed in the manuscript; the author directed the research and is responsible for the work.

Published files

Original filename / checksum	Bytes
v12_Markov_Meridians_Package_Update_Check.json MD5 cf6680f08296e71ff2753f88fcf92eb9 SHA256 96cd775155ecabe1d5b942a3a5d9bd19959449b78413278a01a0e01904bd7056	609
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v12_Markov_Meridians_Source_and_Supplements.zip MD5 e525cee819ad5e855adf28ffb13c5aa2 SHA256 c3193536b03131ead4e8b6c2e85c02fdddaac13eef003fa0c6e6a1646f477174	55,158,705
v9_ps_Markov_Catenary_Choices.pdf MD5 710cbea1d67e7d03e3112818469930a1 SHA256 957c018cc631ddf931a2ad232762309e0ed84e3fc732ae090ecabd3297ebb19d	20,426
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v11_Markov_Meridians.pdf MD5 0a9986170cb4c83f6d3f39f7182a6e24 SHA256 2a1f55c617bbc7845f7467dac72029179b65b786b7a198d9eb2e26a4cf985fc0	7,111,150
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v20_Band_500_Billion_Recovery.zip MD5 78e4d513f261cf9baa30427080f7e865 SHA256 faf79da73bf59110967ae92da3f6933e96383df5fd989724cdba7791294827cc	27,168,031
v11_Markov_Meridians_Release_Notes.txt MD5 4310d55d14a123ffc9fc4be2cf45316c SHA256 bca9426c296f51b4366508ecd6094670e70f6affafd8255df432804526c2c5d8	4,412
v1_Band_500_Billion_Progress.png MD5 02c2762959993bb06800656e300b99b4 SHA256 40eef12696b8074db3487591ba3c6c0dfcf8f703e5dbc371b9ba88760e19a644	55,707

All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

Meridians on the Markov Surface: Growth Structure and Verification of One Trillion Markov Values Using Farey Addresses

Completion and release history

Trillion verification production finished 9 November 2026 at 21:54:13.271944 CST (10 November, 03:54:13.271944 UTC). Independent replay passed at 21:56:16.801036 CST; a fresh portable extraction and replay passed at 21:59:41.149704 CST. Rank confirmation finished at 22:18:33.296330 CST (04:18:33.296330 UTC on 10 November).

Manuscript version 21 is dated 9 November 2026. The retained publication receipt records submission at 22:55:57.539 CST and first public observation at 22:56:39.642 CST. These observation times are distinct from Zenodo's record-created timestamp below.

Zenodo publication-date field: 2026-11-09

Zenodo record created: 2026-11-09 22:55:58 CST / 2026-11-10 04:55:58 UTC

Zenodo record last modified: 2026-11-10 00:05:36 CST / 2026-11-10 06:05:36 UTC

The publication-date field is the date supplied in the deposit. The API created/modified timestamps are repository metadata, not a separately attested first-view time. Full fractional timestamps are retained in the metadata archived in Part C.

https://zenodo.org/records/24096109

https://doi.org/10.5281/zenodo.24096109

Earlier published Zenodo drafts in this version family:

v4_ps | 2026-11-07 13:37:47 CST / 2026-11-07 19:37:47 UTC | https://zenodo.org/records/24043052

v5_ps | 2026-11-07 18:35:22 CST / 2026-11-08 00:35:22 UTC | https://zenodo.org/records/24046012

v7_ps | 2026-11-08 07:26:11 CST / 2026-11-08 13:26:11 UTC | https://zenodo.org/records/24053720

v9_ps | 2026-11-09 07:45:32 CST / 2026-11-09 13:45:32 UTC | https://zenodo.org/records/24084246

v11 | 2026-11-09 11:29:25 CST / 2026-11-09 17:29:25 UTC | https://zenodo.org/records/24089717

v12-supplement-only (manuscript v11) | 2026-11-09 15:19:51 CST / 2026-11-09 21:19:51 UTC | https://zenodo.org/records/24093257

Earlier public announcements: 21 October 2026, 10:18:25 CDT / 15:18:25 UTC on X (more than a billion values in under 40 minutes); 23 October 2026, 09:54:57.771 CDT / 14:54:57.771 UTC on LinkedIn (9.3 billion in 24 minutes). These concern earlier computations, not the final trillion certificate.

https://x.com/webystar/status/2112926487091282350

https://www.linkedin.com/posts/elleard-felix-webster-heffern-joseph-by-confirmation-2953a3104_markovnumbers-activity-7519411048115003392-4NaJ

The dates and URLs are retained in the 7 November public-post evidence packet; the LinkedIn URL is also embedded in the published computational supplement. Additional November LinkedIn announcements and the older Meridian ResearchGate v13 URL could not be established in this compilation.

Complete public Zenodo description

This release certifies uniqueness for the first one trillion Markov values in numerical order, counting 1 first. It includes the revised manuscript, complete reproducible source and supplements, final verifier report and machine-readable evidence, portable recovery archive, and six selected full decimal expansions.

Meridian analysis - especially helm scales and degree-dependent logarithmic spacing - guided the organization of comparisons into magnitude bands. The production verifier implements this plan using reduced Farey addresses, certified positive-matrix logarithmic bounds on 162 cones, scalar Vieta residues modulo 2^64, band sorting and exact resolution. The algorithm can be formulated without Meridian terminology; its finite numerical-prefix certificate does not rely on Meridian collision-exclusion theorems. No controlled ablation establishes a speedup specifically from the coordinate system.

Completed computation. The combined certificate compares 1,101,388,071,945 configurations: 550,693,634,649 inherited and 550,694,437,296 newly computed. Its inner region contains 1,000,000,419,685 configurations. Original screening produced 1,107,352,458,539 emissions, including boundary overlaps. These workload counts are distinct from the one-trillion numerical-prefix claim. All 12 repeated 64-bit residue groups, comprising 24 member occurrences in 11 of 19,287 bands, were resolved unequal.

Timing and storage. On eight workers, retained active band processing totaled 23 h 57 min 23.146 s (15.131 inherited hours plus 8.826 extension hours). Summed native-child CPU was 116.062 h. The combined calendar span was 36.987 h, from 8 November 2026 at 14:55:01.580398 UTC (08:55:01.580398 CST) to 10 November at 03:54:13.271944 UTC (9 November, 21:54:13.271944 CST). Pauses, lost/regenerated work, interrupted attempts and overhead are reported separately. The maximum measured temporary key-file footprint was 1,065,664,392 bytes (1.066 GB; 0.992 GiB); peak generator RSS was 36,468 KiB. Neither is a whole-machine resource bound.

Final checks. The independent final replay passed at 03:56:16.801036 UTC on 10 November. A fresh portable extraction verified all 20,208 archive members and passed its independent replay at 03:59:41.149704 UTC (9 November, 21:59:41.149704 CST), without partial or skip flags. These checks validate receipts and regenerate repeated-key bands; they are not a complete second enumeration of discarded singleton keys. The 38,920,281-byte recovery archive is bound to final receipt head 624c31904f64f8d9434162396a311228f2ece95d495ff2c7dc9d053314ed0c72.

Six full base-10 values. The supplement gives the trillionth Markov value, its predecessor, and the strict predecessor/successor pairs around 10^1000000 and 10^999999. Both ordinal values have 1,021,609 digits. The six files contain 6,043,218 digits. Their expansions were first posted on 9 November 2026 in record 24093257, v12-supplement-only (manuscript v11); the public record exposes the date without a time of day. Their numerical ranks were confirmed on 10 November at 04:18:33.296330 UTC (9 November, 22:18:33.296330 CST) by the completed certificate and fresh rank checks with two Farey partitions. The decimal digits are unchanged. The first million-digit value has rank 958,142,673,903; exactly 1,916,520 Markov values have one million digits. The selected ordinal pair is not asserted to be the globally closest pair; that statistic was not recorded.

Figures 13 and 14 explain the Meridian motivation, Farey grouping and comparison with H. Tan and Y. Zhang, arXiv:2611.01007v1 / doi:10.5281/zenodo.23670834. Their reported result covers 239,536,376,556 maxima through 10^500000. Their 184.27-GB largest-job five-byte payload and our temporary key-file peak are explicitly scoped components. No total-runtime or CPU speedup ratio is claimed.

The manuscript preserves the author-approved Fibonacci-matrix and six-piece arguments, contribution statement and AI disclosure, and final-page exploratory Zeckendorf preview followed by both dedications. The historical 500-billion certificate remains unchanged. The general Markov unicity problem remains open. Author: Elleard Felix Webster Heffern, unaffiliated; no funding. Original software: MIT. Original documentation and data: CC BY 4.0. Third-party materials retain their own terms.

Published files

Original filename / checksum	Bytes
v1_Band_500_Billion_Final_Evidence.json MD5 ce74e2705381ddb72a4af86b67a7bc45 SHA256 911125ae91773e509971563265fa376f1896725ff8ba1c4474ddda80d8b48f4e	260,375
v21_Markov_Meridians.pdf MD5 cb9c949164e8d1a45eaf780a92f4f821 SHA256 8764293fe1426c38eeaedcabb83568840efbda547f4d78646f155926f97c2539	7,352,319
v21_Release_Check.json MD5 2dbfb35e5e2aa754762a780de0cb7142 SHA256 5b275bb0d208c9cfd3a30a6aadbcecaed04ba332c1428f20073828dcc2c3e3c6	1,348
v21_Run_Statistics.json MD5 b883b334f61295dff161236e67048cc2 SHA256 c8682d5a3629849159d1ec384af18047f159483f865e7be37016d7770ad03e53	1,913
v4_50_Billion_Verification_Supplement.pdf MD5 647ae6deb4a76f06c11fa39395ab3bef SHA256 0b50dd6bce45807bd8292677df36fafc8b5fd67b4a6e5ffbac6b937c1f602813	352,160
v21_Figure_13_Meridians_and_Farey_Groups.pdf MD5 29c8ee9acc5ceccb08b2df6224714e43 SHA256 f6c3bf487e318ed3d5fc80f6e48f7f6ffbcb8ceebc712323bfa789dc2cd66e7d	185,296
v20_Band_500_Billion_Recovery.zip MD5 78e4d513f261cf9baa30427080f7e865 SHA256 faf79da73bf59110967ae92da3f6933e96383df5fd989724cdba7791294827cc	27,168,031
v9_ps_Markov_Catenary_Choices.pdf MD5 710cbea1d67e7d03e3112818469930a1 SHA256 957c018cc631ddf931a2ad232762309e0ed84e3fc732ae090ecabd3297ebb19d	20,426
v21_Zeckendorf_Sneak_Peak.pdf MD5 e54bdb782ad4f2c7640dd53468de9a8a SHA256 b759c261c6506aa0109dc7e8eddace94a76452382a2e195a3808335e8f4dfef0	204,136
v21_Release_File_Hashes.json MD5 50d969c48f83f9bfc10a783be4f1c126 SHA256 6a6f42d73a55963c8569610ac94a2fa3b38da63acf5d4bfd81af33c149ef8020	4,877
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v1_Trillion_Verifier_Final_Report.pdf MD5 086f43b5662090d474b73cdb706a4b58 SHA256 28668724346cc6c1664f803523fbfa48daa97ae764494d81469e7040388d1c51	11,475
v21_Figure_14_Verifier_Organization_Comparison.pdf MD5 e752019929ce1c28926cc04b04fea258 SHA256 7c254db85bd90a75b6ac6a0d81c1a4704678dbb2e05f0a6bf5979cefda1bfb91	135,108
v16_Trillion_Verifier_Recovery.zip MD5 85a963ade96dbad974feecc93ddd7939 SHA256 6aac7138c8427eb5a3701b27b841e55a75a2e1960c534233261b42375deaeeea	38,920,281
v2_Selected_Markov_Values_Supplement.zip MD5 c36b624751c38c6862570ff80b755acb SHA256 77a0af171f5fba396e97ec2c76a92900435f9ffb3ed71952efddf6a71c3f5c1a	3,165,428
v1_Trillion_Verifier_Final_Evidence.zip MD5 8b6df185c180dac83bf447ad125faa1c SHA256 3844a31f34772ed7480450423434b83b223e1f7141853056e637340ceaa1495e	146,730
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v21_Markov_Meridians_Release_Notes.txt MD5 f28752566f96c124200f7c322c867dc3 SHA256 f39061656b09648bc3dc216ecbebf5e5efce53e0d94db9a6dfe445bbad004bfd	5,524
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v1_Band_500_Billion_Progress.png MD5 02c2762959993bb06800656e300b99b4 SHA256 40eef12696b8074db3487591ba3c6c0dfcf8f703e5dbc371b9ba88760e19a644	55,707

Original filename / checksum	Bytes
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All original files above are preserved byte for byte across this three-file edition. Part A contains the papers, figures and history. Part B contains six decimal expansions and their numeric recovery streams. Part C contains an encoded compressed transcript of every other distinct text member, plus the original-file recovery archive. PDFs and figures first encountered here follow. Decoded text files are grouped by their first public Zenodo record.

Original: v21_Markov_Meridians.pdf

MERIDIANS ON THE MARKOV SURFACE: GROWTH STRUCTURE AND VERIFICATION OF ONE TRILLION MARKOV VALUES USING FAREY ADDRESSES

ELLEARD FELIX WEBSTER HEFFERN

Abstract. The Markov unicity problem asks whether the largest entry of a positive integer solution determines the other two. We organize the triples into families called Meridians, with compact addresses and explicit evaluation rules. The construction starts with catenary profiles: a fixed host sets the curve, and two heights one unit apart solve the Markov equation with that host. Selected positions give integer triples. Reversing geometric descent gives an exact indexing of these triples. Remainder patterns, growth bounds, corridor paths, and Fibonacci-matrix and snake-graph arguments then restrict possible collisions.

The Meridian description of starting scales and logarithmic spacing guided the organization of a large verification into magnitude bands. The implementation uses Farey addresses and certified matrix bounds to select each band; its correctness does not require the Meridian collision theorems. Production finished on 10 November 2026 at 03:54:13 UTC (9 November, 21:54:13 CST). The combined computation compared 1,101,388,071,945 configurations and certified uniqueness of the first one trillion Markov values in numerical order. All 12 repeated 64-bit residue groups were resolved by exact arithmetic, with no equal maxima. On eight workers, retained active band processing took 23 h 57 min 23.146 s, including the inherited 500-billion certificate; the maximum recorded temporary key-file footprint was 1,065,664,392 bytes (0.992 GiB). This elapsed processing time is distinct from summed CPU time and the calendar span. Both final replays passed. A separate computer-checked certificate covers all 5,067,077 Meridians of degree at most 10,000, at every integer phase. The general unicity problem remains open.

1. Introduction

A Markov triple consists of three positive integers satisfying

$$a^2 + b^2 + M^2 = 3abM, \qquad 1 \le a \le b \le M. \tag{1}$$

For example, $(1, 2, 5)$ and $(1, 5, 13)$ are Markov triples. The largest entry of such a triple is a Markov number. The *unicity problem* asks a simple question: does that largest entry determine the other two? A counterexample would consist of two different ordered triples with the same largest entry. We call such an equality a *collision*. For the history and wider setting of the problem, see Aigner [8].

Our starting point is to let the entries vary continuously. We build a curve on which the Markov equation holds automatically, then ask which positions give integer entries. This separates the equation itself from the requirement that its solutions be integers. It also gives a picture of what a collision would have to look like.

A second aim is practical: understand where a family of triples begins and how its values grow. The Meridian viewpoint guided our choice to compare candidates in limited size ranges, so that completed batches could be cleared from working storage. The resulting verifier uses Farey addresses and certified matrix bounds to select those batches. It compares more than 1.1 trillion configurations and certifies the first one trillion values in numerical order.

Date: Version 21 — 9 November 2026.

2020 *Mathematics Subject Classification.* Primary 11D25; Secondary 11B39, 05C70, 11Y50.

Key words and phrases. Markov numbers, unicity, geometric descent, Meridians, corridors, snake graphs.

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1.1. **The host and its catenary profile.** The *host* h is the fixed third entry around which we build the curve. It determines a profile f_h: two heights on this profile, taken one horizontal unit apart, will satisfy the Markov equation together with h. We first allow positive real entries and call any such solution a *real Markov state*. The host need not yet be an integer or the largest entry; those requirements will be imposed when we return to the unicity question.

For $h > 2/3$, define

$$q_h = \frac{3h}{2}, \qquad R_h = \sqrt{q_h^2 - 1}, \qquad Q_h = q_h + R_h, \tag{2}$$
$$f_h(x) = \frac{h}{2R_h}(Q_h^x + Q_h^{-x}).$$

The restriction $h > 2/3$ makes R_h positive and $Q_h > 1$; it includes every positive integer host. The graph of f_h is a rescaled catenary: a smooth, symmetric U-shaped curve, with its minimum at $x = 0$. In particular, $f_h(-x) = f_h(x)$. In (2), x is a free position on this profile. We use c below for a particular selected position.

The constants in (2) set the spacing and height scale so that one horizontal unit has the required arithmetic meaning. The calculation is short. First,

$$(q_h + R_h)(q_h - R_h) = q_h^2 - R_h^2 = 1,$$

so

$$Q_h^{-1} = q_h - R_h, \qquad Q_h + Q_h^{-1} = 3h, \qquad Q_h - Q_h^{-1} = 2R_h.$$

These three identities follow directly from the definitions.

1.2. **Why two heights one unit apart satisfy the equation.** Compare $f_h(x+1)$ with $Q_h f_h(x)$. The terms containing Q_h^{x+1} cancel, leaving

$$\begin{aligned} f_h(x+1) - Q_h f_h(x) &= \frac{h}{2R_h}(Q_h^{-x-1} - Q_h^{1-x}) \\ &= \frac{h}{2R_h} Q_h^{-x}(Q_h^{-1} - Q_h) = -hQ_h^{-x}. \end{aligned}$$

Multiplying $f_h(x)$ by Q_h^{-1} instead cancels the terms containing Q_h^{-x-1}:

$$\begin{aligned} f_h(x+1) - Q_h^{-1} f_h(x) &= \frac{h}{2R_h}(Q_h^{x+1} - Q_h^{x-1}) \\ &= \frac{h}{2R_h} Q_h^{x}(Q_h - Q_h^{-1}) = hQ_h^{x}. \end{aligned}$$

Now multiply these two equalities. The powers on the right cancel:

$$\big(f_h(x+1) - Q_h f_h(x)\big)\big(f_h(x+1) - Q_h^{-1} f_h(x)\big) = -h^2.$$

Expanding the left side gives

$$f_h(x+1)^2 - (Q_h + Q_h^{-1}) f_h(x) f_h(x+1) + f_h(x)^2 = -h^2.$$

Finally, use $Q_h + Q_h^{-1} = 3h$ and move h^2 to the left:

$$\boxed{f_h(x)^2 + f_h(x+1)^2 + h^2 = 3 f_h(x) f_h(x+1) h.} \tag{3}$$

Thus *every* real position x gives a real Markov state. No choice of an integer value or a special position was used in this calculation.

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1.3. **Choosing an integral state: the role of c.** To look for an ordered triple with largest entry h, select $0 < c \le 1/2$ and take the unit interval $[c-1, c]$ across the bottom of the curve. Its two heights are $f_h(c-1)$ and $f_h(c)$. By symmetry, $f_h(c-1) = f_h(1-c)$, so the state, with its companions in increasing order, is

$$(f_h(c),\ f_h(1-c),\ h).$$

It solves the equation by (3), with $x = c - 1$. For it to be a normalized Markov triple, h and both heights must be positive integers, and $f_h(1-c) \le h$ must hold. The latter condition keeps the chosen host as the largest entry. Restricting c to this half of the unit interval avoids counting the reflected pair twice.

Figure 1 shows the distinction for $h = 5$. One choice, $c \approx 0.351515$, gives the heights 1 and 2, hence the integer triple $(1, 2, 5)$. A different choice, $d = 0.44$, gives heights approximately 1.207458 and 1.602614. It still solves the same equation with host 5, but its companion heights are not integers.

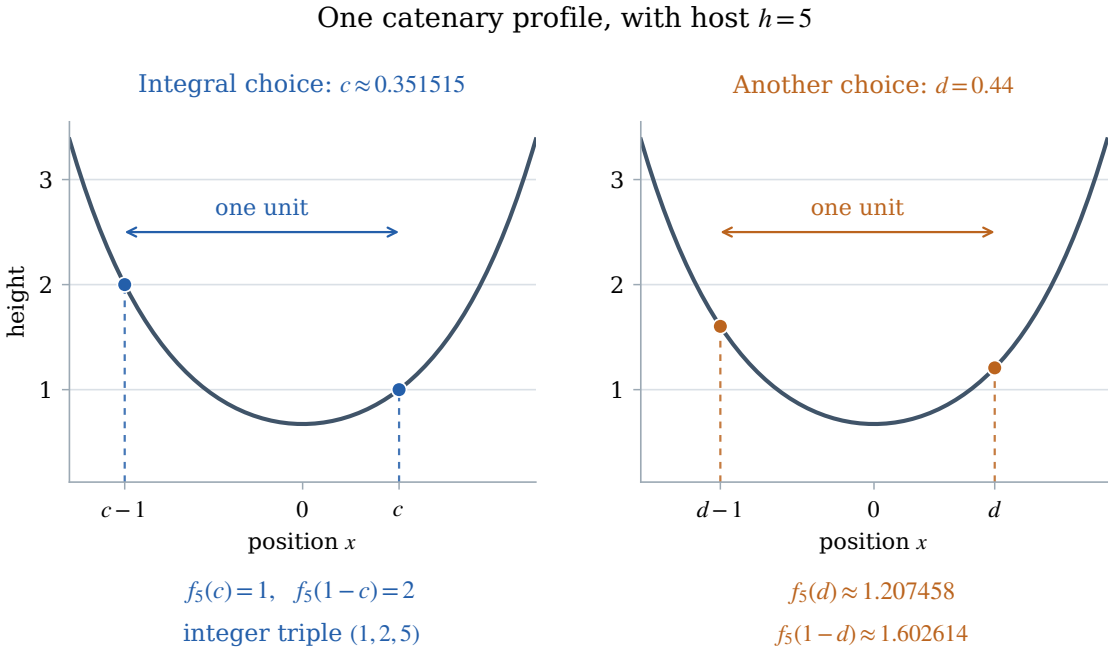


Figure 1. Two choices on the same catenary profile, with host $h = 5$. In each panel the marked contacts are exactly one horizontal unit apart. The left contact has height $f_5(1-c)$ or $f_5(1-d)$ by symmetry. The choice c gives an integer triple; the displayed choice d gives only a real state. A collision would require two distinct choices with both pairs of heights integral and with the same host as their largest entry. No collision is shown here.

This is the geometric form of the unicity question. For a fixed integer host h, could two distinct choices c and d in $(0, 1/2]$ both give integer companion heights no larger than h? If so, they would give different ordered triples with the same maximum. The curve makes these possible comparisons visible. To exclude them requires information about the integer values on the curve.

1.4. **How the paper organizes those comparisons.** The next step is *descent*. Starting from two contact heights, we look for a smaller host whose profile passes through the same heights one unit

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apart. We then move inward along that profile. Repeating these operations reduces a normalized integer Markov triple to $(1, 1, 1)$. Section 2 makes the procedure precise and explains how its final host and final position recognize the integer starting states.

Reversing the steps organizes the triples into families called *Meridians*. Each family has an address recording the steps used to construct it, and a phase specifying a position along it. This Meridian phase is a separate coordinate from the selected c on a fixed catenary. Integer Meridian phases pick out integer triples; the corresponding c need not be an integer, as Figure 1 shows. Section 3 proves that these addresses and integer phases recover every normalized triple exactly once. This is a statement about triples: whether two of them have the same largest entry remains the question to be answered. The same section introduces the alternative tree and the *phasors* that connect its Meridians.

The remainders of the Markov values follow exact periodic patterns (Section 4). Section 5 then develops the ordering tools used in the collision arguments. It represents Markov numbers by counts of perfect matchings on chains of square tiles, called snake graphs; a perfect matching pairs all the vertices using edges of the graph. The author's Fibonacci matrices and six-piece decomposition provide concrete ways to compare these counts.

The following two sections use those constructions to control growth. A Meridian's *degree* measures the largest exponent in its formula. We prove that distinct integer marks of the same degree cannot collide. For different degrees, a positive expansion of the Fibonacci blocks gives a finite range of phases that needs comparison (Section 6). *Corridor walls* follow other paths through the same tree, using both Meridian and phasor steps. Adding one to each Fibonacci run index gives their recurrence and a second bound on possible collisions (Section 7).

The resulting computer-checked certificate covers all 5,067,077 Meridians of degree at most 10,000, at every integer phase (Theorem 8.1). Distinct triples within that family have different maxima. This bounds the complexity of the families, while allowing their values to grow without limit; comparisons with higher-degree Meridians remain outside this certificate. Section 9 records the separate numerical-order verification. Meridian helms and degree-dependent logarithmic spacing motivated its organization by scale. The production code implements that organization through groups of Farey addresses with shared, certified size bounds; it does not schedule one Meridian at a time. The same band algorithm can be formulated entirely in Farey and matrix terms. Its finite-prefix conclusion does not use the Meridian collision exclusions, and the timings do not isolate a speedup due to the change of coordinates alone. It compared 1,101,388,071,945 configurations and certified uniqueness of the first one trillion values, including every possible competitor at those values. Production finished on 10 November 2026 at 03:54:13 UTC (9 November, 21:54:13 CST), using 23.956 hours of retained active band processing, including the inherited work, and at most 0.992 GiB of recorded temporary key files. Both final replays passed. The section gives the exact timing and storage scopes, together with the earlier fifty-billion result. The general unicity problem remains open within this framework.

As the later sections introduce their coordinates, $M_\alpha(x)$ will denote the value at Meridian address α and phase x, and $m_{p/q}$ the value of the triple indexed by the reduced fraction p/q. Both label triples before any assertion that their maxima are distinct. Analytical logarithms are natural; plotted horizontal axes use $\log_{10} M$.

2. Geometric descent and arithmetic closure

2.1. **Moving along a profile and changing its host.** The catenary profile (2) turns two elementary operations on Markov states into movements of the contacts. The first operation is a

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unit step along a fixed profile. Directly from its powers,

$$f_h(x+1)+f_h(x-1)=\frac{h}{2R_h}(Q_h+Q_h^{-1})(Q_h^x+Q_h^{-x}) \\ =3hf_h(x). \tag{4}$$

Thus the height beyond a pair is obtained by multiplying the nearer height by $3h$ and subtracting the farther height. This is the usual Vieta replacement of a companion entry, expressed on the curve.

The second operation keeps the contact heights fixed and changes the host. For any real x, put $A=f_h(x)$ and $B=f_h(x+1)$. Their possible hosts are the roots of $X^2-3ABX+A^2+B^2=0$. One root is h, so the other is

$$h, \qquad K=3AB-h=\frac{A^2+B^2}{h}. \tag{5}$$

For descent, select an edge straddling the minimum of the profile. We record its right endpoint as c, with $0\le c<1$, and its ordered contact heights as $A=f_h(c)$ and $B=f_h(c-1)$. This is the same pair used in the introduction, since $f_h(c-1)=f_h(1-c)$. We retain the full interval of c here to keep track of orientation. Seek a wider profile through these same two heights, still one unit apart, allowing a horizontal translation. Smaller h gives a wider profile. If the other host satisfies $K<h$, fit that profile and walk the unit lattice inward until the selected edge again straddles its vertex. Otherwise stop. Figure 2 shows one complete step. The fit is explicitly

$$v=\frac{\log\big((Q_KA-B)/K\big)}{\log Q_K}, \qquad f_K(v)=A, \quad f_K(v-1)=B. \tag{6}$$

Indeed, $Q_Kf_K(v)-f_K(v-1)=KQ_K^v$, so taking logarithms recovers v directly. The Markov relation gives

$$(Q_KA-B)(B-Q_K^{-1}A)=K^2.$$

Both factors are positive: they have the same sign, and their sum is $2R_KA>0$. This also fixes the real branch of the logarithm and verifies the fitted contact heights by direct substitution. The walk replaces v by $v-\lfloor v\rfloor$. Its orientation is retained: reflection exchanges the contacts and sends phase to its negative modulo one.

Proposition 2.1 (Uniqueness of the fit)**.** *A marked edge has at most one distinct compatible profile, hence at most one wider fit. For fixed $0<a<b$, a same-arm fit $f_h(u-1)=a$, $f_h(u)=b$ with $u>1$ is unique if it exists.*

Proof. The hosts are the two roots of $X^2-3ABX+A^2+B^2$. The other root belongs to the geometric domain because $(3K-2)AB=K^2+(A-B)^2>0$. For a same-arm fit the horizontal distance between heights a,b is

$$\frac{1}{\log Q_h}\int_a^b\frac{dy}{\sqrt{y^2-h^2/R_h^2}}.$$

When $a>2/3$ and the lower height is in range, Q_h strictly increases and $h/R_h=2h/\sqrt{9h^2-4}$ strictly decreases with h. Thus the prefactor and the integrand strictly decrease with h. The distance can therefore equal one at most once. The profile is strictly increasing on that arm, which also determines u. If $a\le 2/3$ there is no fit. □

2.2. **Termination.** Write $\mathcal{D}(M,c)=(H,\phi)$ for the terminal host and oriented phase.

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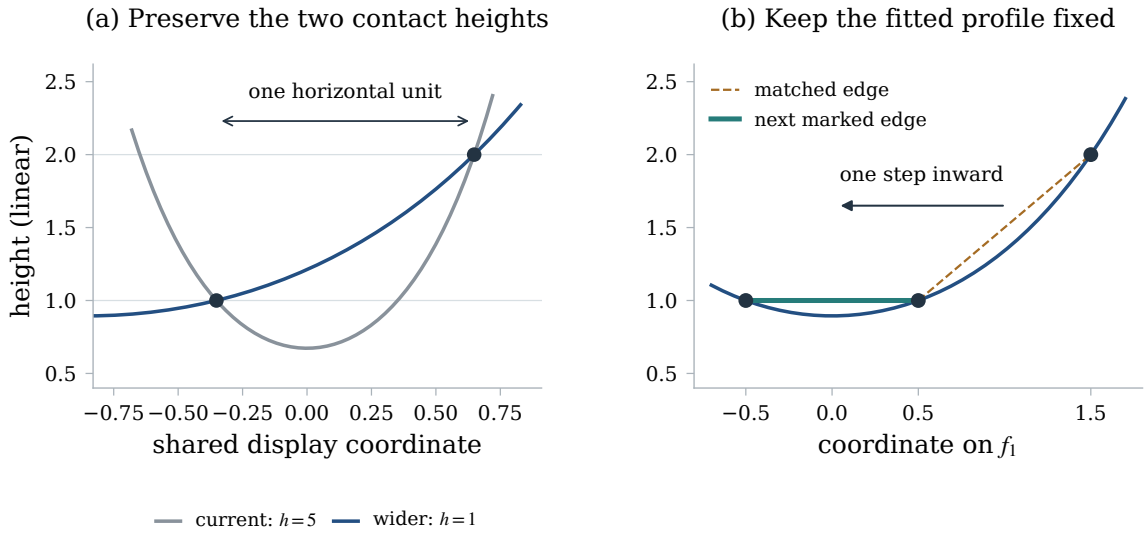


Figure 2. Geometric descent from $(1, 2, 5)$. The wider profile passes through the same unit-spaced contacts. On f_1 their coordinates are $1/2, 3/2$; one inward step selects $-1/2, 1/2$, with heights $1, 1$. The curve is fixed during the walk.

Theorem 2.2 (Finite descent)**.** *Every straddling input with $M \geq 1$ terminates. On $1 \leq M \leq R$ the number of host changes is uniformly bounded. The terminal map is continuous into $(2/3, \infty) \times \mathbb{R}/\mathbb{Z}$ and analytic on cells with a fixed strict itinerary.*

Proof. Temporarily sort the endpoints as $a \leq b$. Inward walking does not increase either endpoint. Every current host is at most R, and the discriminant in (3) gives

$$(9h^2 - 4)a^2 - 4h^2 = (2b - 3ha)^2 \geq 0.$$

Consequently the endpoints are at least $a_R = 2R/\sqrt{9R^2 - 4} > 2/3$. At an accepted change, $a^2 + b^2 < h^2$, so the endpoints are also below $h \leq R$. Using either endpoint as the host in the same discriminant identity bounds the new host below by a_R. The initial host is at least $1 > a_R$.

If the new host K is at least b, its edge already straddles the vertex, because both endpoints are below $f_K(1)$; the next candidate is the old, larger host, so descent stops. Otherwise

$$\frac{h}{K} > \frac{h}{b} = 3a - \frac{K}{b} > 3a - 1 \geq 3a_R - 1 > 1.$$

All changes except possibly the last therefore contract the host by a fixed factor. An upper bound is $1 + \lceil \log(R/a_R)/\log(3a_R - 1) \rceil$.

At a boundary $K = h$, the fitted coordinate in (6) is the original c, so the accepted and stopped maps agree. At a phase cut, $K(h, 0) = h(9h^2 + 4)/(9h^2 - 4) > h$, and a neighborhood is stopped. The uniform bound on compositions then gives continuity, with phase taken modulo one. The strict formulas are analytic. □

2.3. **Which inputs close arithmetically?** The two reversible state operations are

$$T(a, b, h) = (3ha - b, a, h), \qquad J(a, b, h) = (a, b, 3ab - h). \tag{7}$$

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They and their inverses are integer polynomial maps. A terminal state has the nonnegative defect

$$Hf_H(\phi)f_H(\phi-1)-1=\frac{(H-1)^2(H+2)}{3H-2}+\frac{H^3}{4R_H^2}\big(Q_H^{\phi-1/2}-Q_H^{1/2-\phi}\big)^2. \tag{8}$$

For $0\le\phi<1$ it vanishes exactly at $(H,\phi)=(1,1/2)$.

Proposition 2.3 (Arithmetic closure)**.** *A straddling input with $M\ge 1$ reaches $(1,1/2)$ if and only if its three entries are positive integers. Modulo companion exchange, these inputs are exactly the normalized configurations* (1)*.*

Proof. Expanding the two profiles gives

$$Hf_H(\phi)f_H(\phi-1)-1=\frac{H^3}{4R_H^2}\big(Q_H^{2\phi-1}+Q_H^{1-2\phi}+3H\big)-1.$$

Replace the first two powers by $\big(Q_H^{\phi-1/2}-Q_H^{1/2-\phi}\big)^2+2$. The constant term is $H^3/(3H-2)-1=(H-1)^2(H+2)/(3H-2)$, which proves (8). Reversing a finite closure itinerary from $(1,1,1)$ by (7) preserves integrality. Conversely, integral descent stays integral. At an integral terminal state order the entries as $u\le v\le w$. There is no coordinate-decreasing mutation. But for $Q(X)=X^2-3uvX+u^2+v^2$,

$$Q(v)=u^2+(2-3u)v^2\le 3u^2(1-u)\le 0.$$

If neither root is below $w\ge v$, equality is forced, giving $u=v=1$; terminality then gives $w=1$. Finally, a straddling edge has no decreasing endpoint mutation, so its host is a largest entry. Its ordered split is uniquely recovered by (6). □

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Both terminal coordinates matter. Figure 3 shows inputs with terminal host one but different terminal phases. A crossing of the height level $H = 1$ alone does not certify integrality. The marked configurations follow the host itineraries $37666 \to 5 \to 1$, $7561 \to 5 \to 1$, and $14701 \to 2 \to 1$, all ending at phase $1/2$.

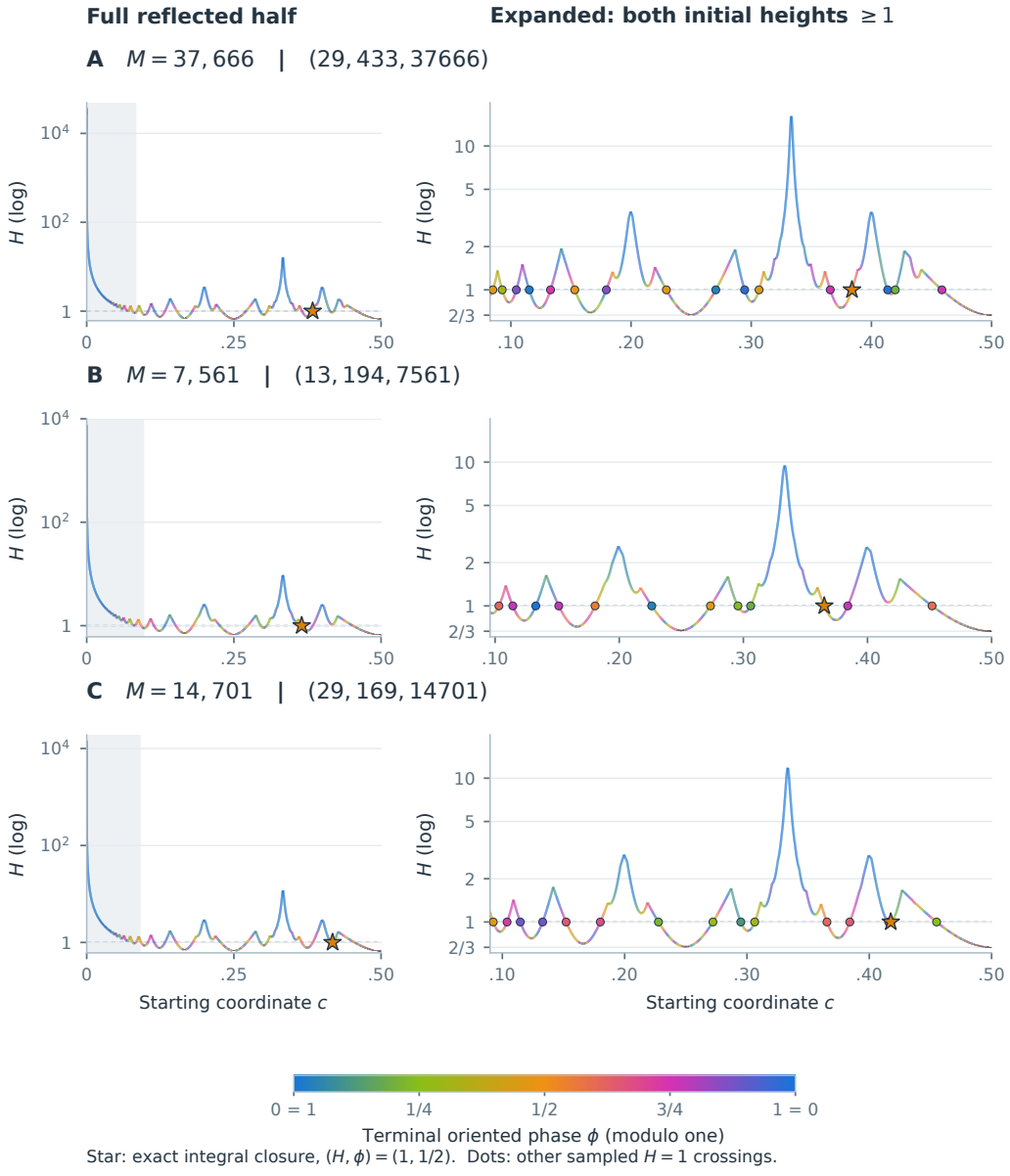


Figure 3. Terminal host and phase for $M = 37666, 7561, 14701$. Height is H on a logarithmic scale and hue is oriented phase. Left: $0 \le c \le 1/2$, with gray where an initial endpoint is below one. Right: the admissible portion enlarged. Stars mark the exact integral starts; dots are other numerically located $H = 1$ crossings. These numerical plots illustrate the closure criterion.

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3. Meridians, addresses, and the alternative tree

3.1. **The landscape and its two coordinates.** On the landscape $(\log_{10} M, c)$, the point (M, c) represents the contacts $f_M(c), f_M(c-1)$. We draw the reflected half $0 < c \le 1/2$; exchanging the contacts sends c to $1-c$. A *Vieta arc* keeps one initial contact height fixed. For example, the branch with height a has

$$c = \frac{\log\Big((R_M a + \sqrt{R_M^2 a^2 - M^2})/M\Big)}{\log Q_M}.$$

This follows by solving $Q_M^c + Q_M^{-c} = 2R_M a/M$ for $Q_M^c \ge 1$ and taking logarithms. The square root is real exactly when $a \ge M/R_M$, the minimum height of the profile; reflection supplies the other arm. A *Vieta trek* follows such arcs, changing the retained contact when necessary. A *Meridian* is a connected component of terminal host $H(M, c) = 1$. A *phasor* holds the terminal phase ϕ fixed. The alternative tree uses phasors at $\phi = 1/2$ to join successive Meridians.

The address phase x on a Meridian is real and unrolled. Its terminal circle phase is $\phi \equiv x + 1/2 \pmod 1$. Every integer mark therefore closes arithmetically; the distinguished arrival mark $x = 0$ is called the *helm*. It need not be the minimum of the real profile.

Figure 4 keeps the same configurations in three views. The alternative tree is in the middle panel. The bottom panel uses smooth schematic connectors for the Vieta parent relation: node coordinates and adjacency are exact, but those connectors are not literal level curves. The two trees each have 55 edges. An unmarked crossing is not a vertex.

For reproducibility, the field in panel (a) uses the fixed cyclic phase palette with HSL saturation and lightness

$$w(H) = \frac{1}{1 + (\ln H / \ln 1.05)^2}, \qquad S = w(H), \qquad L = 1 - 23w(H)/40.$$

The background is a numerical illustration, independent of all exact enumerations and exclusions below.

3.2. **Evaluating an address.** The seed Meridian is

$$\begin{aligned} a_1(x) = f_1(x + \tfrac{1}{2}), \qquad b_1(x) = f_1(x - \tfrac{1}{2}), \\ M_1(x) = 3a_1(x)b_1(x) - 1. \end{aligned} \tag{9}$$

For integer x, its companions are F_{2x+1}, F_{2x-1}, where $F_{-j} = (-1)^{j+1} F_j$. Hence

$$M_1(x) = 2 + 3F_{2x}^2, \qquad M_1(0), M_1(1), M_1(2), \ldots = 2, 5, 29, 194, \ldots. \tag{10}$$

From any ordered state (a, b, H) form the two-sided orbit

$$u_0 = a, \quad u_1 = b, \qquad u_{j+1} = 3Hu_j - u_{j-1}.$$

A nonzero index k replaces the state by

$$(u_k, u_{k+1}, 3u_k u_{k+1} - H). \tag{11}$$

The address is written

$$\alpha = (1, k_m, \ldots, k_1): \quad k_1 \text{ is applied first and written at the right.}$$

This is the reversed indexing convention: the former $M_{1,a,b,c}(x)$ is now $M_{1,c,b,a}(x)$. Neither the signs nor x are reversed. In terms of (7), with $S = T^{-1}$ and $\mathcal{W}_k = JS^k$,

$$(a_\alpha(x), b_\alpha(x), M_\alpha(x)) = \mathcal{W}_{k_m} \cdots \mathcal{W}_{k_1} J T^x (1, 1, 1). \tag{12}$$

Composition acts from right to left. A recursive extension inserts its index immediately after the leading 1. For example, $M_{1,1}(0) = 13$. Index zero is forbidden because (11) would return the preceding pole $3ab - H$, rather than create a new component.

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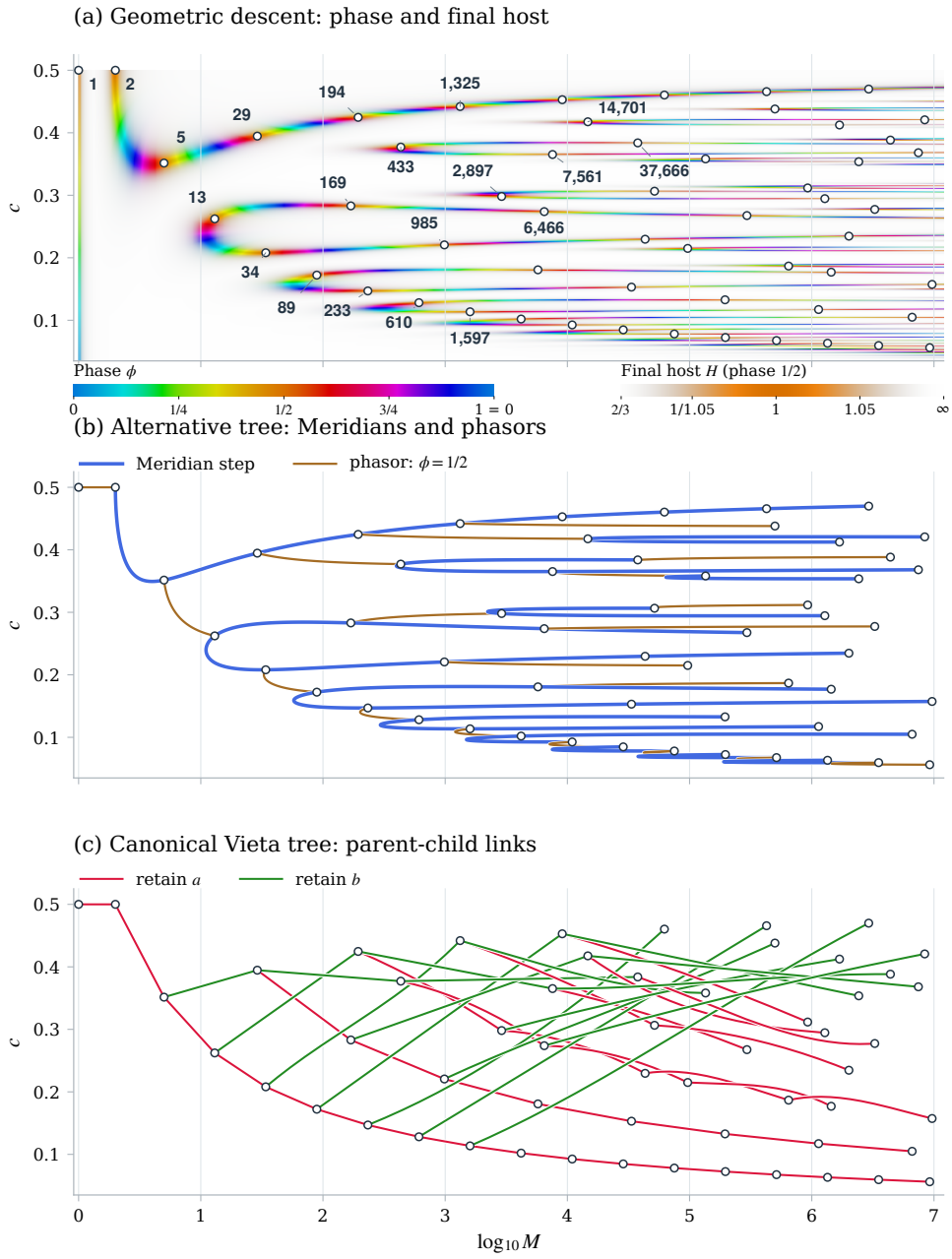


Figure 4. The same 56 configurations with $M \leq 10^7$. (a) Numerical geometric descent, with phase encoded by hue and host by color intensity. The number labels identify the same 20 Markov numbers shown in Figure 5. (b) The alternative tree, with Meridian and phasor edges. (c) The canonical Vieta tree, with simplified connectors retaining exact endpoints. Red retains the smaller companion, green the larger. Every edge runs toward larger M. Crossings without a node mark are not vertices.

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Definition 3.1 (Canonical marks)**.** The root $(1,1,1)$ is separate. For the seed (1) use integer phases $x \geq 0$. For every other address require $k_1 > 0$, $k_i \neq 0$ for $i > 1$, and $x \in \mathbb{Z}$. Evaluation is followed by sorting the two companions.

Theorem 3.2 (Configuration bijection)**.** *Canonical marks are in bijection with all normalized positive Markov triples except the root. The displayed value $M_\alpha(x)$ is the largest entry.*

Proof. The quadratic argument in Proposition 2.3 shows that replacing the largest coordinate by its other pole gives a smaller maximum, apart from the initial repeated entries. This is the forced Vieta reduction of a configuration and does not assume scalar unicity.

To recover a word, reflect the current maximum H to $P = 3ab - H$. When $P \geq 2$, walk its oriented endpoint orbit inward until both endpoints are below P. This edge is unique. Indeed $u_{j+1} - 2u_j + u_{j-1} = (3P-2)u_j > 0$, so the entries below P form an interval; three consecutive entries below P would contradict $u_{j-1} + u_{j+1} = 3Pu_j \geq 3P$. Endpoint descent supplies the edge, and its nonzero signed displacement recovers the last-applied, leftmost index. Repeat at the smaller pole. At pole one, the orbit reaches $(1,1)$ and fixes the seed phase.

Conversely, all outward blocks are positive and reduced. If the preceding pole is $P < H$, the first reflected poles on the two sides are $(9b^2-2)H - P > 6H$ and $(9a^2-2)H - P > 6H$. Their farther endpoints exceed $2H$, and the recurrence makes subsequent outward entries increase. The reduction therefore undoes every prescribed block. Finally, companion exchange negates all indices and x simultaneously. The canonical conditions select one of the two orientations; only $(1,1,2)$ is fixed. □

This bijection is useful computationally as well as geometrically. A configuration can be retained by its address and reconstructed when a comparison needs its full integer entries. Section 9 uses the equivalent reduced-fraction labels as compact identities, with compressed Vieta evaluation. Its coverage therefore counts triples before testing whether their maxima are distinct.

3.3. **The two parent rules.** At a nonzero phase the alternative parent is the adjacent inward mark, $(\alpha; x) \mapsto (\alpha; x - \operatorname{sgn} x)$. At a nonseed helm it is

$$(1, k_m, \ldots, k_2, k_1; 0) \longmapsto (1, k_m, \ldots, k_2; -k_1), \tag{13}$$

followed by simultaneous sign reflection if the remaining rightmost index is negative. An empty tail uses the seed's nonnegative phase. The seed helm with maximum two has parent $(1,1,1)$.

The alternative parent deletes the *rightmost* index at a helm. Recursive evaluation removes the *leftmost* index after 1. These operations serve different purposes. The depth

$$d(\alpha; x) = 1 + \sum_{i=1}^{m} (|k_i| + 1) + |x| \tag{14}$$

decreases by one under either parent edge, with root depth zero. Thus the parent rule defines a rooted tree on the same configurations. The ordinary outward Vieta moves instead are

$$(a, b, M) \longmapsto (a, M, 3aM - b), \qquad (a, b, M) \longmapsto (b, M, 3bM - a). \tag{15}$$

For example, $29 \to 194$ is a Meridian edge, whereas $29 \to 169$ is a Vieta edge. Figure 5 compares the two sets of links.

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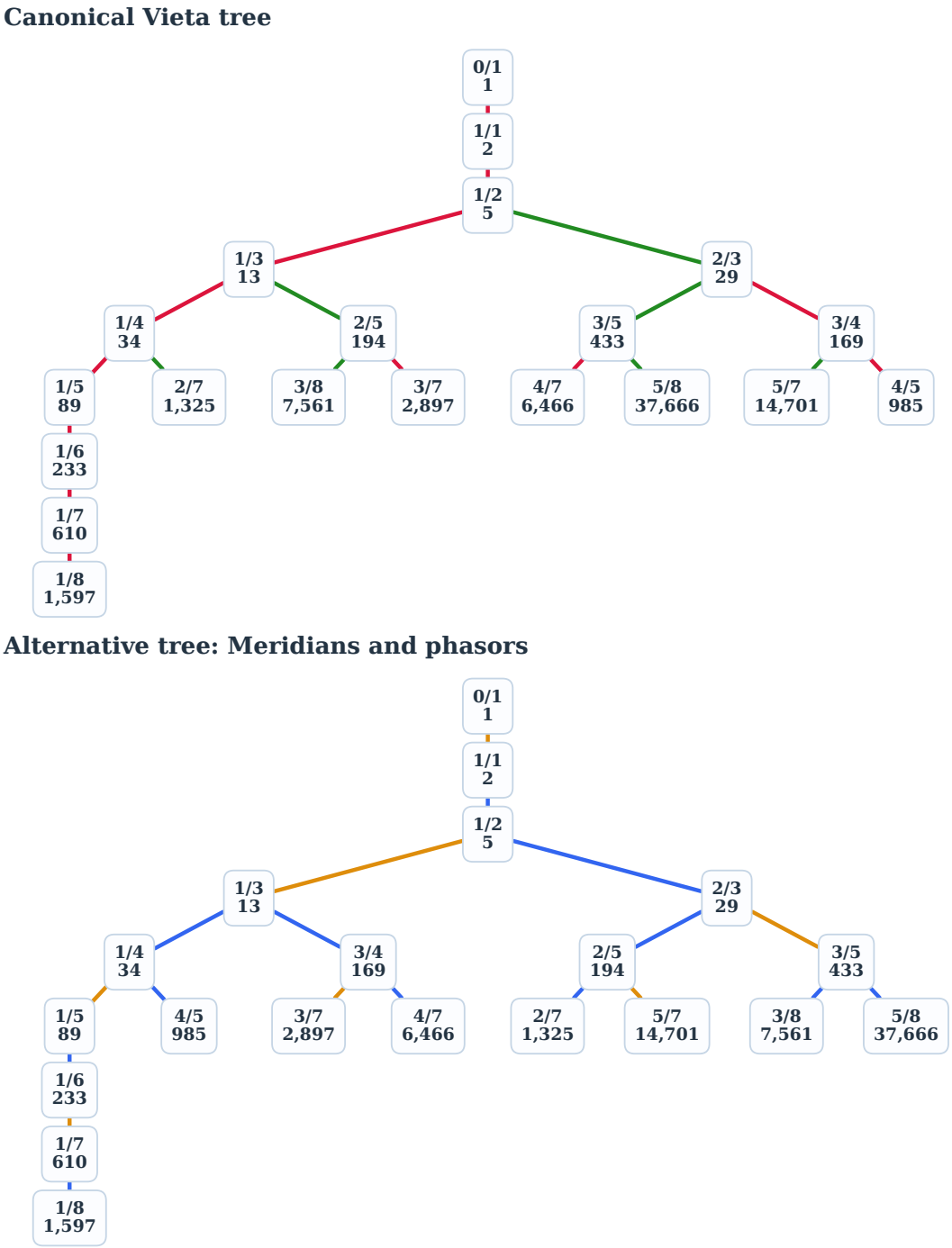


Figure 5. Two trees on the same 20 configurations: the 16 with $M \leq 3000$ and the four additional values $6466, 7561, 14701, 37666$. Each node displays its reduced standard Farey label and maximum. Red and green denote the Vieta moves; blue and gold denote Meridian and phasor steps.

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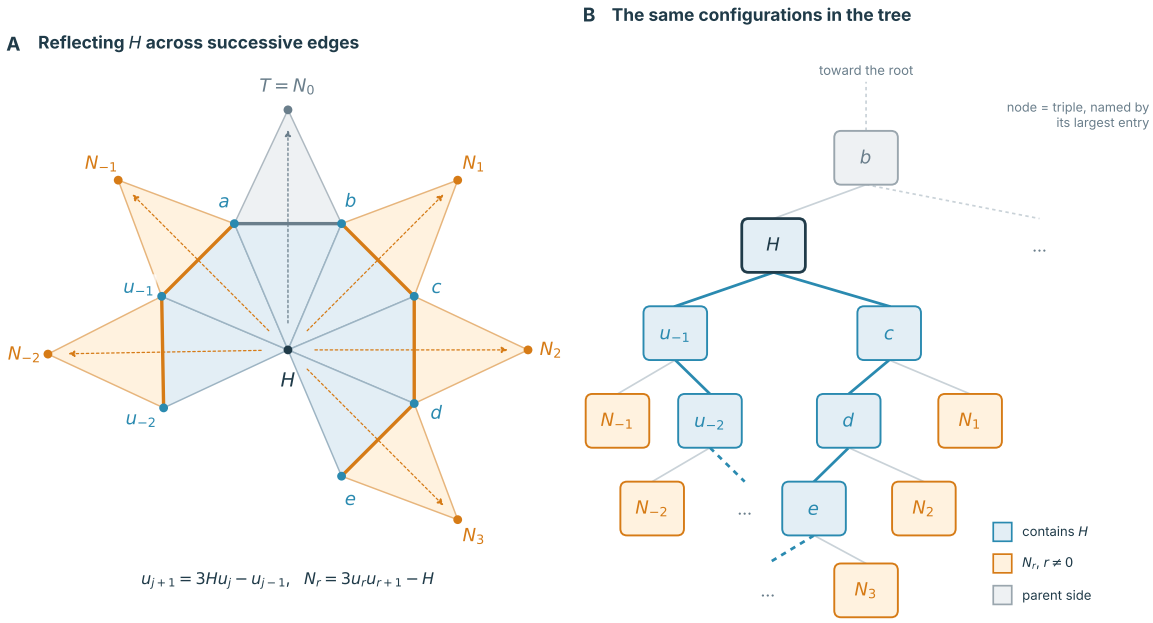


Figure 6. The orbit underlying one recursive extension. Reflecting H across (u_j, u_{j+1}) gives $N_j = 3u_j u_{j+1} - H$; N_0 is the preceding pole. For source $(a, b, H) = (1, 5, 13)$ at $M_{1,1}(0)$, the preceding pole is two, $(u_{-2}, u_{-1}, u_2, u_3, u_4) = (1325, 34, 194, 7561, 294685)$, and $(N_{-2}, N_{-1}, N_1, N_2, N_3) = (135137, 89, 2897, 4400489, 6684339842)$. The right panel locates these reflections in the Vieta tree.

3.4. **Real components and phasor edges.**

Proposition 3.3. *The canonical real families in* (12)*, modulo companion reflection, are the nonterminal connected components of* $H(M, c) = 1$*. Their terminal phase is* $x + 1/2$ *modulo one. The separate terminal component is* $M = 1$*.*

Proof. The seed companions are at least $2/\sqrt{5}$, and $M_1 \geq 2$ is strictly larger than either. More generally, let $h \geq 1$ and $2/\sqrt{5} \leq a, b \leq h$. The first outward endpoint satisfies $3hb - a \geq (6/\sqrt{5} - 1)h > h$; its reflected pole is larger still. The reversed orbit gives the same conclusion on the other side. Every outward block thus has a strict inverse descent to its source and preserves the terminal phase.

Conversely, reverse any descent ending at host one. Its last host change produces the seed with $x = v - 1/2$, and every preceding walk is nonzero: a zero walk would simply have stopped at the smaller host. These strict blocks uniquely recover a locally constant word. The unrolled x joins the final phase charts continuously. The seed is injective as an ordered curve, as its endpoint difference is strictly monotone; each block is invertible. The positive Laurent expansion proved in Proposition 6.2 makes every family proper. The inverse charts are relatively open in the level set, and properness makes the families closed. Each connected family therefore fills a component. Reflection gives exactly the canonical identifications. □

The phasor edges are obtained by applying the same outward blocks to

$$\left(\frac{h}{\sqrt{3h-2}}, \frac{h}{\sqrt{3h-2}}, h \right), \qquad 1 \leq h \leq 2. \tag{16}$$

This symmetric arc joins $(1, 1, 1)$ to $(1, 1, 2)$. Its other host is $2h/(3h-2)$, and both possible terminal fits have phase $1/2$. Its endpoints satisfy the bounds used in the proposition, so the outward blocks

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preserve that phase. The displayed phasors consequently arise from the descent itself, rather than interpolation between integer nodes.

3.5. **Degree and the exact Farey dictionary.** Attach endpoint count vectors $u = (1, 0)$ and $v = (0, 1)$ to the seed. Under an index k update them by

$$(u, v) \longmapsto \begin{cases} ((k-1)(u+v)+v,\ k(u+v)+v), & k > 0, \\ (|k|(u+v)+u,\ (|k|-1)(u+v)+u), & k < 0. \end{cases} \tag{17}$$

Write their sum as (r, s). Their determinant remains ± 1 and their unordered residues modulo two remain $\{(1, 0), (0, 1)\}$. Thus r, s are positive coprime odd integers. Canonical orientation gives $r \le s$. The *Meridian degree* is $D = r + s$. It is the extreme Laurent exponent in ρ^x, where $\rho = (3 + \sqrt{5})/2$; the degree in ρ^{2x} is $D/2$.

Proposition 3.4 (Helm classification)**.** *Canonical Meridians correspond bijectively to coprime positive odd pairs $r \le s$. Their helm labels are the standard Farey fractions r/s.*

Proof. The endpoint vectors are the two Stern–Brocot parents: their sum is (r, s) and their determinant has absolute value one. Conversely, the parents of an odd/odd pair have opposite coordinate parities. Their componentwise absolute difference is a smaller positive primitive odd/odd pair, except at $(1, 1)$. The difference has one componentwise sign; an opposite sign would give the initial pair. Iteration reaches $(1, 1)$.

To recover the signed indices, put $n_\alpha = (r, s)$ and let α^- remove the leftmost index after 1. Consecutive difference vectors have determinant of absolute value two, so three consecutive vectors satisfy $n = 2c\, n^- + \varepsilon n^{--}$ with $c \ge 1$ and $\varepsilon = \pm 1$. Determinant division gives an integer coefficient, parity makes it even, and decreasing coordinate sums fix its magnitude and sign. At the bottom use $(1, 1), (-1, 1)$; canonical orientation fixes the first index positive. The magnitude is fixed by the ratio of successive degrees, which lies in $[2c, 2c+1)$ or $(2c-1, 2c]$ according to the sign; equality occurs only at the first applied index. Reversing the process gives precisely $n_{1,k,k_m,\ldots,k_1} = 2|k| n_\alpha + \operatorname{sgn}(kk_m) n_{\alpha^-}$. For the seed use $k_0 = 1$ and $n_\varnothing = (-1, 1)$, so the first extension has $(r, s) = (2k_1 - 1, 2k_1 + 1)$. This recovers the unique address. Mediant addition identifies the label. □

Lemma 3.5 (Phase dictionary)**.** *For helm (r, s) and degree $D = r + s$, the standard Farey label at phase x is*

x	(p, q)	$p + q$
0	(r, s)	D
1	(r, D)	$D + r$
-1	(s, D)	$D + s$
$n \ge 2$	$(D, (n-1)D + r)$	$nD + r$
$-n \le -2$	$(D, (n-1)D + s)$	$nD + s$.

The negative seed phases are omitted.

Proof. At seed phases $0, 1, n \ge 2$, the endpoint bases are respectively $((1, 0), (0, 1))$, $((1, 1), (0, 1))$, and $((1, n), (1, n-1))$. The count updates are linear, so substitution gives the table; negative phase exchanges the bases. For the inverse, odd/odd p, q give $(r, s, x) = (p, q, 0)$. If q is even, sort $p, q - p$ and choose $+1$ when p is the smaller entry, otherwise -1. If p is even, write $q = tp + u$ with $0 < u < p$; sort $u, p - u$ and choose $+(t + 1)$ or $-(t + 1)$ in the same way. A tie is the seed and uses the nonnegative representative. These formulas recover every mark. □

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In particular, degree can be read directly from a reduced label:

$$D = \begin{cases} p+q, & p,q \text{ odd}, \\ q, & q \text{ even}, \\ p, & p \text{ even}. \end{cases} \tag{18}$$

This dictionary connects the Meridian and corridor collision windows below.

4. Meridian residues: modulo 27 and Fibonacci moduli

4.1. **Transferring a seed congruence to every address.** The recurrence for the companions contains no division, but a seed period of the *maximum* need not visibly preserve both companions. The following four-entry record resolves that issue:

$$\mathcal{S} = (A, B, C, H) = (9a^2, 9b^2, 3ab, H).$$

Forward and backward orbit steps and pole reflection are the polynomial maps

$$\begin{aligned} \mathcal{R}(A,B,C,H) &= (B, A + 9H^2B - 18HC, HB - C, H), \\ \mathcal{R}^{-1}(A,B,C,H) &= (B + 9H^2A - 18HC, A, HA - C, H), \\ \mathcal{J}(A,B,C,H) &= (A, B, C, C - H). \end{aligned} \tag{19}$$

For $h(x) = M_1(x)$ the seed record is

$$\mathcal{S}_1(x) = (8h(x) - h(x-1) - 2,\ h(x) + h(x-1) + 2,\ h(x) + 1,\ h(x)).$$

Every congruence preserving two consecutive seed maxima therefore propagates along every signed address.

Theorem 4.1 (Largest modulus with a common phase period)**.** *Every canonical Meridian has phase period T modulo m if and only if*

$$m \mid q(T), \qquad q(T) = 3F_{2T}F_{\gcd(2T,4)} = \begin{cases} 3F_{2T}, & T \text{ odd}, \\ 9F_{2T}, & T \text{ even}. \end{cases} \tag{20}$$

In particular, s is a common period modulo F_{2s}.

Proof. Fibonacci addition and (10) give

$$h(x+T) - h(x) = 3F_{2T}F_{4x+2T}.$$

Fibonacci strong divisibility, $\gcd(F_a, F_b) = F_{\gcd(a,b)}$, follows by the Euclidean algorithm from the addition formula. The gcd of all displayed differences is already the gcd of those at $x = 0, 1$, namely $q(T)$. Thus the seed has period T modulo m exactly when $m \mid q(T)$. The record (19) proves sufficiency at every address; the seed itself proves necessity. □

The theorem specifies a period common to all Meridians. It does not assert that the period is the smallest for each individual sequence. Nor does a large period exclude a scalar residue match at two earlier phases. For powers of three it gives

$$M_\alpha(x + 2 \cdot 3^n) \equiv M_\alpha(x) \pmod{3^{n+3}}, \qquad n \geq 0. \tag{21}$$

Indeed, $F_{3j} = F_j(5F_j^2 + 3)$ for even j, beginning with $F_4 = 3$, gives $v_3(F_{4\cdot 3^n}) = n + 1$. Figure 7 shows the common periods at the requested even-indexed Fibonacci moduli.

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x	M1							M1.1							M1.2							M1.1.1							M1.-1.1						
s	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8	2	3	4	5	6	7	8
F_{2s}	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987	3	8	21	55	144	377	987
-12								1	5	13	31	13	231	181	1	1	13	50	1	88	937	2	1	20	50	17	328	944	2	1	5	9	89	285	215
-11								1	2	13	4	34	283	559	1	1	1	46	73	21	715	2	1	2	13	89	94	170	2	1	2	21	89	333	107
-10								1	1	19	13	121	248	544	1	2	16	48	82	270	541	2	2	20	37	98	29	482	2	5	8	34	53	272	869
-9								1	5	1	34	61	57	169	1	1	13	26	1	291	160	2	1	2	51	89	343	653	2	1	8	13	17	241	806
-8								1	2	13	50	130	169	13	1	1	13	53	73	343	433	2	1	20	31	17	34	923	2	1	5	21	17	86	89
-7								1	1	13	31	25	13	34	1	2	1	50	82	56	652	2	2	2	50	98	258	317	2	5	2	9	125	89	233
-6								1	5	19	4	13	34	985	1	1	16	46	1	21	541	2	1	20	13	17	369	104	2	1	8	21	89	233	890
-5								1	2	1	13	34	231	820	1	1	13	48	73	88	580	2	1	2	37	89	328	758	2	1	8	34	89	285	71
-4								1	1	13	34	121	283	181	1	2	13	26	82	21	937	2	2	20	51	98	94	944	2	5	5	13	53	333	215
-3								1	5	13	50	61	248	559	1	1	1	53	1	270	715	2	1	2	31	89	29	170	2	1	2	21	17	272	107
-2								1	2	19	31	130	57	544	1	1	16	50	73	291	541	2	1	20	50	17	343	482	2	1	8	9	17	241	869
-1								1	1	1	4	25	169	169	1	2	13	46	82	343	160	2	2	2	13	98	34	653	2	5	8	21	125	86	806
0	2	2	2	2	2	2	2	1	5	13	13	13	13	13	1	1	13	48	1	56	433	2	1	20	37	17	258	923	2	1	5	34	89	89	89
1	2	5	5	5	5	5	5	1	2	13	34	34	34	34	1	1	1	26	73	21	652	2	1	2	51	89	369	317	2	1	2	13	89	233	233
2	2	5	8	29	29	29	29	1	1	19	50	121	231	985	1	2	16	53	82	88	541	2	2	20	31	98	328	104	2	5	8	21	53	285	890
3	2	2	5	29	50	194	194	1	5	1	31	61	283	820	1	1	13	50	1	21	580	2	1	2	50	89	94	758	2	1	8	9	17	333	71
4	2	5	2	5	29	194	338	1	2	13	4	130	248	181	1	1	13	46	73	270	937	2	1	20	13	17	29	944	2	1	5	21	17	272	215
5	2	5	5	2	5	29	194	1	1	13	13	25	57	559	1	2	1	48	82	291	715	2	2	2	37	98	343	170	2	5	2	34	125	241	107
6	2	2	8	5	2	5	29	1	5	19	34	13	169	544	1	1	16	26	1	343	541	2	1	20	51	17	34	482	2	1	8	13	89	86	869
7	2	5	5	29	5	2	5	1	2	1	50	34	13	169	1	1	13	53	73	56	160	2	1	2	31	89	258	653	2	1	8	21	89	89	806
8	2	5	2	29	29	5	2	1	1	13	31	121	34	13	1	2	13	50	82	21	433	2	2	20	50	98	369	923	2	5	5	9	53	233	89
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11	2	5	5	5	5	194	194	1	1	1	34	25	248	820	1	2	13	26	82	270	580	2	2	2	51	98	29	758	2	5	8	13	125	272	71
12	2	2	2	29	2	29	338	1	5	13	50	13	57	181	1	1	13	53	1	291	937	2	1	20	31	17	343	944	2	1	5	21	89	241	215

Figure 7. Five Meridians modulo F_{2s} for $s = 2, \ldots, 8$: the moduli are $3, 8, 21, 55, 144, 377, 987$. Each column has common phase period s, possibly a multiple of an individual least period. The shaded row is the helm; blank cells omit the seed's negative phases. The displayed residues use the reversed Meridian indexing of this edition.

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4.2. **Why modulo 27 gives only ten classes.** No positive Markov entry is divisible by three: mutation negates the changed coordinate modulo three and the integral tree starts at $(1,1,1)$. Let t,h be preceding and current poles. An extension by a nonzero index has the particularly simple action

$$S(t,h)=(h,t)\quad (k \text{ even}),\qquad R(t,h)=(h,7h-t)\quad (k \text{ odd})\quad (\mathrm{mod}\ 27). \tag{22}$$

To see this, put $N_j=3u_ju_{j+1}-h$. Direct use of the orbit recurrence gives

$$N_0=t,\quad N_1=(9u_1^2-2)h-t,\quad N_{j+2}-N_j=9h(u_{j+2}^2-u_{j+1}^2).$$

All orbit squares are one modulo three, so $N_{j+2}\equiv N_j$ and $N_1\equiv 7h-t \pmod{27}$. This works for negative indices as well. Only the parity of each recursively applied index survives.

Proposition 4.2 (Universal modulo-27 classes)**.** *At integer phases every Meridian alternates between the two residues of one of the following sets, allowing the two singleton cases:*

$$\{1\},\ \{2,5\},\ \{4,16\},\ \{7,13\},\ \{8,17\},\ \{10,19\},\ \{11,23\},\ \{14,20\},\ \{22,25\},\ \{26\}.$$

The ten sets are pairwise disjoint. Meridians in different classes cannot share an integer value, regardless of degree or phase.

Proof. The common phase period is two because $q(2)=27$. Track preceding and current poles at both phase parities, starting from $(1,2,1,5)$. The simultaneous maps satisfy $S^2=I$ and $SRS=R^{-1}$. Successive current-pole pairs on the R orbit are

$$\begin{array}{cccccc}(2,5) & (13,7) & (8,17) & (16,4) & (23,11) & (10,19)\\ (20,14) & (22,25) & (26,26) & (25,22) & (14,20) & (19,10)\\ (11,23) & (4,16) & (17,8) & (7,13) & (5,2) & (1,1).\end{array}$$

The preceding pair is the previous pair cyclically, and $b_{j+1}=7b_j-b_{j-1}$ checks the return in 18 steps. These states and their S images form a closed set of 36 states. Their current-pole images are exactly the ten listed sets. Congruence of integer values would require membership in the same set. □

Table 1. The ten universal classes and their populations among the 519 Meridians of degree at most 100. Only the population column is bounded-degree data.

Residue set	Population	Residue set	Population
$\{1\}$	42	$\{10,19\}$	47
$\{2,5\}$	88	$\{11,23\}$	47
$\{4,16\}$	61	$\{14,20\}$	38
$\{7,13\}$	78	$\{22,25\}$	35
$\{8,17\}$	65	$\{26\}$	18

At that degree bound, the classification leaves 15,215 of the 134,421 pairs for further comparison. Larger Fibonacci moduli and compatible phases, rather than extrapolation from this table, supply the degree-10,000 certificate in Section 8.

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5. Fibonacci matrices, six pieces, and ordering proofs

5.1. **From a Farey label to an actual matching count.** A snake graph is a chain of square tiles, each attached to the right of or above the preceding tile. Shared edges are counted once. A perfect matching covers every vertex exactly once. The Markov snake associated with a reduced $0 < p/q < 1$ is built as follows. Take the lower staircase from $(0, 0)$ to (q, p): at each horizontal coordinate j, step right and then up to height $\lfloor pj/q \rfloor$. Write right steps as a and up steps as b. Write this step sequence as $w_{p/q} = azb$. Remove its first and last letters; start with one square and replace each a of z by two added squares to the right and each b by two upward. The resulting snake has $2(p + q) - 3$ tiles.

Write $\mu(w)$ for its number of perfect matchings. The established Markov matching correspondence gives

$$\mu(w_{p/q}) = m_{p/q}. \tag{23}$$

The matching interpretation originated in the REACH work of 2002–2003. Musiker's memorandum, dated 27 June 2002, proposed the recursive graph model [16]; Itsara, Musiker, Propp, and Viana's manuscript of 10 January 2003 stated and proved the snake-matching correspondence [17, Theorem 3.1]. Setting all edge weights to one gives the ordinary perfect-matching counts. Propp's later exposition, first posted in 2005, records the broader collaborative development [18]. These original contributions precede the later proofs of Aigner's orderings.

For a modern treatment and the Farey-indexed construction, see [15, Section 4.1, Theorem 4.2] and [9]. For example, $w_{2/5} = a^3ba^2b$, its trimmed word is $aabaa$, and the eleven-tile snake has 194 matchings. Auxiliary rearranged words also specify snakes, counted by μ; they need not be canonical Markov snakes.

5.2. **Six pieces and two connector states.** Figure 8 fixes the five three-tile pieces and the three one-square variants of type 6. In a cut between pieces, a matching uses both yellow connector edges or neither: the cut leaves an even number of vertices on either side. Denote these states by 1 and 0. At a selected port remove the two vertices already covered by its connector pair. Direct counting within the pieces gives

piece	00	01	10	11
1	5	2	2	1
2	5	3	3	2
3	5	2	3	1
4	5	3	2	1
5	5	2	2	1
6i	2	1	1	1

(24)

The state order is incoming, outgoing. The one-port end squares 6ii and 6iii have counts $(2, 1)$.

For the ordering surgeries, use a fixed one-ended chop: retain tile triples $(1, 2, 3), (5, 6, 7), \ldots$, with tiles $4, 8, \ldots$ providing paired connectors. A one-square remainder occurs exactly when $p + q$ is even and is counted with the closed end piece. For every connector sequence σ, multiply the applicable row entries in (24); then sum over σ, with outer ports closed. Each matching determines one sequence and one internal completion per piece, and gluing reverses the construction. This proves the counting rule, without an independence assumption. Equivalently it is a sum of terms $5^{k(\sigma)}3^{j(\sigma)}2^{\ell(\sigma)}$, where the exponents count factors of the indicated size. For the 2/5 snake the piece list $2, 1, 2$ gives

$$5^3 + 5 \cdot 2 \cdot 3 + 3 \cdot 2 \cdot 5 + 3^2 = 194.$$

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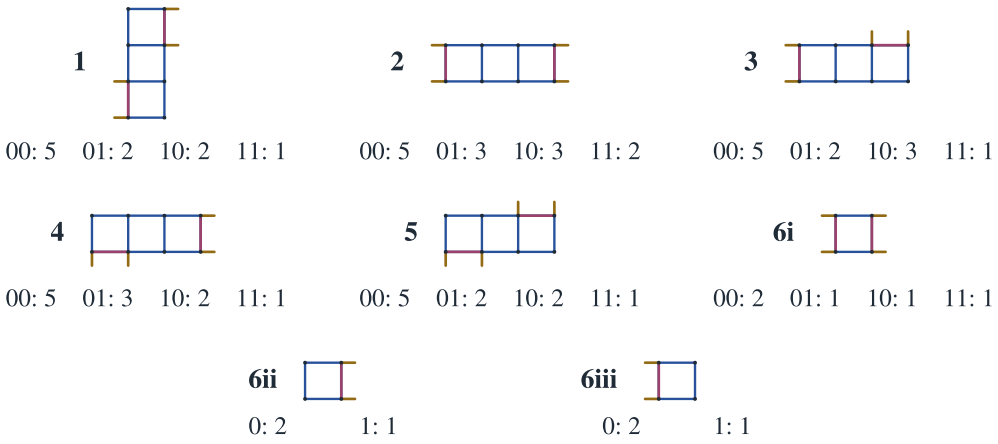


Figure 8. The six physical pieces. Gold edges are paired connectors, magenta marks ports, and dots mark tile vertices. Two-port counts use the order $00, 01, 10, 11$. The central square 6i has counts $(2, 1, 1, 1)$; the one-port end squares have $(2, 1)$. Types 1 and 5 have equal count rows but different physical orientations.

5.3. **Fibonacci-matrix normalization.** The author proposed the Fibonacci-matrix formulation and normalization approach developed here. We begin with the comparison of three consecutive blocks and then apply it to complete products of run blocks.

With $F_0 = 0, F_1 = 1$ and $F_{-j} = (-1)^{j+1}F_j$, the basic odd-index identity is

$$F_{2k-1}F_{2k+3} - F_{2k+1}^2 = 1. \tag{25}$$

For example, $5 \cdot 34 - 13^2 = 1$. Consider the author's three consecutive Fibonacci blocks:

$$\begin{aligned} A &= \begin{pmatrix} F_{2k+3} & F_{2k+1} \\ F_{2k+1} & F_{2k-1} \end{pmatrix}, \\ B &= \begin{pmatrix} F_{2k+1} & F_{2k-1} \\ F_{2k-1} & F_{2k-3} \end{pmatrix}, \\ C &= \begin{pmatrix} F_{2k-1} & F_{2k-3} \\ F_{2k-3} & F_{2k-5} \end{pmatrix}, \qquad k \geq 2. \end{aligned} \tag{26}$$

Applying (25) twice gives precisely the upper-left comparison used in normalization:

$$\begin{aligned} (AC)_{11} - (B^2)_{11} &= (F_{2k+3}F_{2k-1} - F_{2k+1}^2) \\ &\quad + (F_{2k+1}F_{2k-3} - F_{2k-1}^2) = 2. \end{aligned} \tag{27}$$

The same identities in the other entries give

$$AC - B^2 = \begin{pmatrix} 2 & 6 \\ 0 & 2 \end{pmatrix}, \qquad CA - B^2 = \begin{pmatrix} 2 & 0 \\ 6 & 2 \end{pmatrix}.$$

Thus replacing AC or CA by B^2 decreases the upper-left entry even inside a longer product: the surrounding Fibonacci blocks have nonnegative entries and positive upper-left entries. For $k = 3$ this reads $(AC)_{11} = 34 \cdot 5 + 13 \cdot 2 = 196 > 194 = 13^2 + 5^2 = (B^2)_{11}$.

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5.3.1. *Run counts and general index shifts.* To apply the same comparison at arbitrary run lengths, index the blocks by

$$\mathcal{A}_t = \begin{pmatrix} F_{2t+3} & F_{2t+1} \\ F_{2t+1} & F_{2t-1} \end{pmatrix}, \qquad t \geq 0. \tag{28}$$

Thus $A = \mathcal{A}_k$, $B = \mathcal{A}_{k-1}$ and $C = \mathcal{A}_{k-2}$ in (26). The four boundary counts of a straight block of $2t+1$ tiles are $F_{2t+3}, F_{2t+1}, F_{2t+1}, F_{2t-1}$: these follow by conditioning on the first edge of a ladder, giving the Fibonacci recurrence; at $t = 0$ they are $(2, 1, 1, 1)$. Gluing successive runs with their paired connector states therefore gives, for $w = a^{h_1}b \cdots a^{h_p}b$ with $h_1 \geq 1$ and $h_j \geq 0$,

$$\mu(w) = [\mathcal{A}_{h_1-1}\mathcal{A}_{h_2} \cdots \mathcal{A}_{h_p}]_{11}. \tag{29}$$

The first run is shortened by the removed initial a; the final b closes the other end. Equivalently, a block-index list $(t_1, \ldots, t_p)$ represents $w = a^{t_1+1}ba^{t_2}b \cdots a^{t_p}b$, with $t_1 + \cdots + t_p = q - 1$.

Fibonacci addition gives the scalar index-shifting identity

$$F_{2r+1}F_{2s+1} - F_{2r-1}F_{2s+3} = F_{2(r-s)-2}.$$

Apply it to the two summands of each product entry to obtain

$$\mathcal{A}_r\mathcal{A}_s - \mathcal{A}_{r-1}\mathcal{A}_{s+1} = 2\begin{pmatrix} F_{2(r-s)-2} & F_{2(r-s)} \\ F_{2(r-s)-4} & F_{2(r-s)-2} \end{pmatrix}, \quad r \geq 1,\ s \geq 0. \tag{30}$$

Consequently, whenever two adjacent indices differ by at least two, moving one unit from the larger to the smaller strictly decreases the whole product's upper-left entry. The case $r = s + 2$ is exactly (27). The only remaining local comparison has adjacent indices differing by one; it is an interchange of the two blocks.

5.3.2. *Comparing the surrounding Fibonacci blocks.* For a finite list of block indices, define the ratio of its two open counts by

$$\rho(u_1, \ldots, u_m) = \frac{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{21}}{[\mathcal{A}_{u_1} \cdots \mathcal{A}_{u_m}]_{11}}, \qquad \rho(\varnothing) = 0.$$

All these ratios lie in $[0, 1)$. Direct multiplication gives

$$\rho(t, U) = \frac{F_{2t+1} + F_{2t-1}\rho(U)}{F_{2t+3} + F_{2t+1}\rho(U)}. \tag{31}$$

Its right-hand side increases strictly with $\rho(U)$, since (25) gives $\det \mathcal{A}_t = 1$. Moreover, another application of that identity gives

$$\frac{F_{2t+1}}{F_{2t+3}} - \frac{F_{2t+3} + F_{2t+1}}{F_{2t+5} + F_{2t+3}} = \frac{1}{F_{2t+3}(F_{2t+5} + F_{2t+3})} > 0.$$

Hence $\rho(t+1, U) < \rho(t, V)$ for any two finite tails U, V. Two lists can therefore be compared by cancelling equal initial indices: at the first unequal pair, the larger index gives the smaller ratio; an exhausted list has the smaller ratio than any nonempty remainder.

For adjacent blocks in positions $j, j+1$, the outward lists are $(t_{j-1}, \ldots, t_1)$ on the left and $(t_{j+2}, \ldots, t_p)$ on the right. At $r = s + 1$, (30) becomes

$$\mathcal{A}_{s+1}\mathcal{A}_s - \mathcal{A}_s\mathcal{A}_{s+1} = 2\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

After multiplication by the surrounding blocks, its upper-left entry has the sign of

$$\rho(t_{j+2}, \ldots, t_p) - \rho(t_{j-1}, \ldots, t_1). \tag{32}$$

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Indeed, it equals this difference times twice the positive upper-left entries of the two surrounding products. This proves the needed surrounding-block test entirely in the same Fibonacci matrices.

Lemma 5.1 (Fibonacci-matrix normalization)**.** *For coprime positive p, q, every word $w = azb$ with q letters a and p letters b satisfies $\mu(w_{p/q}) \le \mu(w)$.*

Proof. For $p = 1$ there is only the word $a^q b$. Assume $p > 1$ and use the block-index list in (29). Its prefix discrepancies are

$$\delta_j = qj - p(1 + t_1 + \cdots + t_j) \quad (1 \le j \le p), \qquad \delta_0 = \delta_p = 0.$$

Changing (t_j, t_{j+1}) to $(t_j + 1, t_{j+1} - 1)$ decreases only δ_j, by p; the reverse change increases only δ_j, by p. We show that an appropriate change always decreases the matching count until $-p \le \delta_j \le 0$ for every j.

The sign checks use just a prefix-sum identity. Put $h_1 = t_1 + 1$ and $h_i = t_i$ for $i > 1$, the actual run lengths. Whenever $j - m \ge 0$ and $j + m \le p$,

$$\delta_{j-m} + \delta_{j+m} = 2\delta_j + p\left(\sum_{i=j-m+1}^{j} h_i - \sum_{i=j+1}^{j+m} h_i\right). \tag{33}$$

In the block indices, the left sum has the extra 1 precisely when it includes t_1.

First remove positive discrepancies. Choose the rightmost largest $\delta_j > 0$ and transfer one unit from t_{j+1} to t_j. Comparison with its two neighboring discrepancies gives $h_j < q/p < h_{j+1}$; coprimality makes these inequalities strict. Thus the transfer is allowed and $t_{j+1} \ge t_j + 1$; for $j = 1$ the gap is at least two. Gaps of at least two are handled by (30).

If $t_{j+1} = t_j + 1$, compare the outward lists. A first unequal pair with the left index larger would make the parenthesis in (33) nonnegative: the central difference is -1, the equal outward indices cancel, and the first unequal difference is at least 1. This contradicts maximality and the choice of the rightmost maximum. If the left list were exhausted first, its extra initial 1 would instead give $\delta_{2j} = 2\delta_j > \delta_j$, again impossible (including simultaneous exhaustion, when $2j = p$). The left outward ratio is therefore strictly larger than the right one. Reversing the interchange in (32) shows that the selected transfer strictly decreases the count. It also decreases $\sum_{j=1}^{p-1} \delta_j$. There are only finitely many nonnegative block-index lists of sum $q - 1$, so this phase ends with all $\delta_j \le 0$.

Then remove discrepancies below $-p$. Choose the leftmost smallest $\delta_j < -p$ and transfer one unit from t_j to t_{j+1}. For $j > 1$, the neighboring comparisons give $h_j > q/p > h_{j+1}$; for $j = 1$, the inequality $\delta_1 < -p$ gives $t_1 > q/p$, while $t_2 < q/p$. Thus $t_j \ge t_{j+1} + 1$ and the transfer is allowed. Again a gap of at least two is settled by (30).

In the gap-one case, a first unequal outward pair with the left index smaller would make the parenthesis in (33) nonpositive, contradicting minimality and the leftmost choice. If that pair reaches t_1, its extra 1 instead gives $\delta_{2j} \le 2\delta_j + p < \delta_j$, the same contradiction. If the right list were exhausted first, the identity would give $\delta_{2j-p} = 2\delta_j + p < \delta_j$. Simultaneous exhaustion would give $\delta_j = -p$. Thus the left outward ratio is strictly smaller than the right one, and (32) proves a strict decrease in count. The changed discrepancy is $\delta_j + p < 0$, so positivity is not reintroduced. This phase increases the bounded sum of discrepancies by p each time and terminates with $-p \le \delta_j \le 0$.

Finally, $\delta_j \equiv qj \pmod p$. For $1 \le j < p$, coprimality leaves exactly one possible value in $[-p, 0]$, forcing

$$1 + t_1 + \cdots + t_j = \left\lceil \frac{jq}{p} \right\rceil.$$

These are precisely the canonical run lengths. Every transfer has decreased the upper-left product entry, proving the assertion. □

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For the canonical word, $h_j = \lceil jq/p \rceil - \lceil (j-1)q/p \rceil$. With $k = 1 + \lfloor (q-1)/p \rfloor$, the factors are the author's A, B from (26), and the end factors are B. For example, $m_{3/8} = [BAB]_{11}$ with $k = 3$. A similar normalization result appears in [12, Proposition 21]; the proof above is self-contained in the author's Fibonacci matrices.

5.4. **The three Aigner orderings.** Fixed-numerator monotonicity was proved by Rabideau and Schiffler [15]; all three orderings were proved by LPTV, Lee–Li–Rabideau–Schiffler, and McShane [12, 13, 14]. The author's Fibonacci matrices prove normalization and the fixed-numerator ordering. His six pieces prove the remaining fixed-sum comparison and the fixed-denominator ordering, using normalization to return to a canonical Markov snake.

Theorem 5.2 (Established monotonicities)**.** *For $i > 0$, whenever both fractions are reduced and lie in $(0, 1)$,*

$$\begin{aligned} m_{p/q} &> m_{(p+i)/(q-i)} && \textit{(fixed sum)}, \\ m_{p/q} &< m_{p/(q+i)} && \textit{(fixed numerator)}, \\ m_{p/q} &< m_{(p+i)/q} && \textit{(fixed denominator)}. \end{aligned}$$

5.4.1. *Fixed numerator: Fibonacci matrices.*

Proof. Fibonacci recurrence gives, for $t \geq 1$,

$$\mathcal{A}_t - \mathcal{A}_{t-1} = \begin{pmatrix} F_{2t+2} & F_{2t} \\ F_{2t} & F_{2t-2} \end{pmatrix}.$$

All entries are nonnegative and the upper-left is positive. Start with $w_{p/(q+i)}$ and delete i internal letters a, retaining the first. Each deletion changes one factor $\mathcal{A}_t$ to $\mathcal{A}_{t-1}$ with $t \geq 1$. Surrounding factors are positive, so the matching count strictly falls. The resulting word ends at (q, p); normalization gives $m_{p/q} \leq \mu(w_*) < m_{p/(q+i)}$. □

5.4.2. *Fixed sum: six pieces.*

Proof. At the same physical cuts, the changes $2 \to 3$ or 4, then 3 or $4 \to 5$, increase no entry of (24); the changed factors are $3 \to 2$ or $2 \to 1$. Rotation $5 \leftrightarrow 1$ preserves the row. Selecting the changed connector and no others exhibits a strictly smaller positive summand whenever an equal-direction join becomes unequal.

The trimmed canonical word begins and ends with a, contains no bb, and has $q - p - 1$ equal joins, all aa. At its rightmost equal join, interchange a, b strictly to the right. That suffix is alternating, so only the chosen join changes. Exactly one prescribed piece changes by the stated rules, while any rotated pieces keep their rows. Each surgery strictly decreases the count. Odd-numbered operations replace one horizontal step by a vertical step in net; even-numbered operations keep the totals. After $2i$ operations the endpoint is $(q - i, p + i)$. There are enough joins because $p + i < q - i$. For the resulting auxiliary word w_*, normalization gives $m_{(p+i)/(q-i)} \leq \mu(w_*) < m_{p/q}$. A similar tail comparison appears in [12, Proposition 22]; here the comparison is proved using the author's six physical pieces. □

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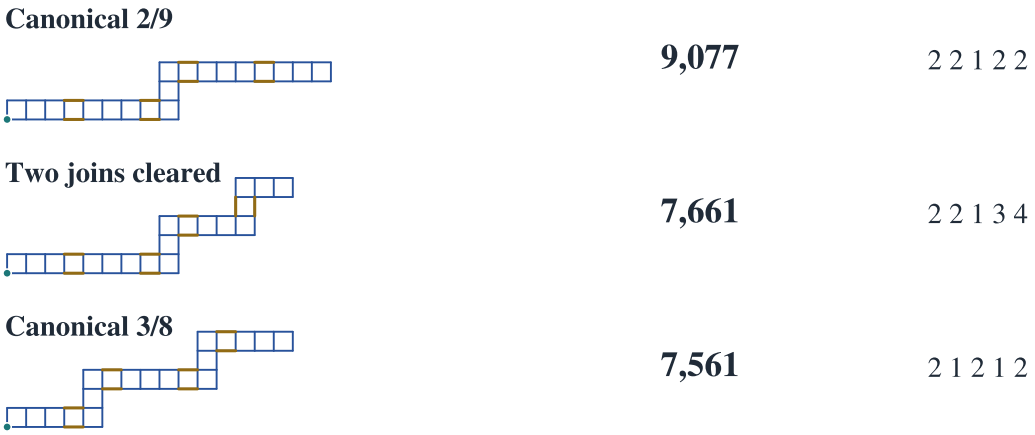


Figure 9. Fixed-sum surgery on the same five pieces. The canonical 2/9 snake has 9077 matchings; after two changes the auxiliary snake has 7661; normalization gives the 3/8 snake with 7561. The intermediate first change, not shown, has 8263. Gold pairs mark cuts and teal dots the starting corners.

5.4.3. *Fixed denominator: six pieces.*

Proof. For the fixed-denominator comparison, refine the same six-piece chop into single-square pieces of type 6, retaining odd-numbered tiles and using the even tiles as paired connectors. A retained square has counts $(2,1,1,1)$ for opposite ports and $(2,1,1,0)$ for adjacent ports. The zero forbids covering their common corner twice. Summing an intervening state regroups these squares into the original three-tile pieces.

Insert one internal b, adding two vertical squares. The affected local counts are the following complete table, with outside ports fixed:

	before				after			
directions	00	01	10	11	00	01	10	11
a,a	2	1	1	1	5	2	2	1
b,b	2	1	1	1	5	3	3	2
a,b	2	1	1	0	5	3	2	1
b,a	2	1	1	0	5	2	3	1

.

The endpoint caps have the corresponding rows with an outer port closed; the empty trimmed word has the increase $2 \to 5$. No outside count falls, and the all-zero choice witnesses a strict increase. Thus the two added squares compensate for every changed direction.

Starting from $w_{(p+i)/q}$, delete i internal b's and reverse this comparison. At least i are available. Normalizing the resulting word to (q,p) proves $m_{p/q} \le \mu(w_*) < m_{(p+i)/q}$. □

5.5. **Folding an actual snake at its center.** The same local counts also compare families sharing a long initial sequence of pieces. Represent the four counts by

$$R_1 = R_5 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}, \quad R_4 = \begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}.$$

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Attaching piece i to an open-count row (x, y) replaces it by $(x, y)R_i$. This is the sum over the common connector state, not an additional assumption about matchings.

For a helm, use the symmetric central chop: retain the middle square as 6i and group retained squares in pairs outward on each side. For odd/odd p, q, each half has $(p+q-2)/2$ retained squares. If this number is odd, one square remains at each outer end, giving pieces 6ii and 6iii with initial counts $(2, 1)$. Otherwise the half starts at $(1, 0)$ and consists of complete three-tile pieces. This repartition requires rebuilding the piece list; one cannot move a cut and keep its old boundary counts.

Proposition 5.3 (Folded matching counts)**.** *For a reduced $0 < p < q$, let (x, y) be the open counts of one reflected half under the central chop. If p, q are odd, then*

$$m_{p/q} = 2x^2 + 2xy + y^2 = (x + y)^2 + x^2. \tag{34}$$

If $p + q$ is odd, then $m_{p/q} = x^2 + y^2$. In either case $\gcd(x, y) = 1$; a nonempty half has positive counts.

Proof. The trimmed staircase sequence is a palindrome: for $0 < j < q$, $\lfloor p(q-j)/q \rfloor = p - 1 - \lfloor pj/q \rfloor$. For odd/odd labels the two halves meet at an opposite-port central square, whose four counts are $2, 1, 1, 1$. The reflected halves contribute x^2, xy, yx, y^2, giving (34). For opposite parities they meet across a single connector and must use its same state, giving $x^2 + y^2$. Every full piece table and every exposed-square table has determinant ± 1, so its integral inverse preserves the gcd of an open row. Starting at $(1, 0)$ proves primitivity. The first square gives $(2, 1)$, and every subsequent permitted table preserves positivity. □

The seed $1/1$ consists only of its central square. The two examples in Figure 10 give $m_{1/3} = 2 \cdot 2^2 + 2 \cdot 2 \cdot 1 + 1^2 = 13$ and $m_{1/5} = 2 \cdot 5^2 + 2 \cdot 5 \cdot 3 + 3^2 = 89$.

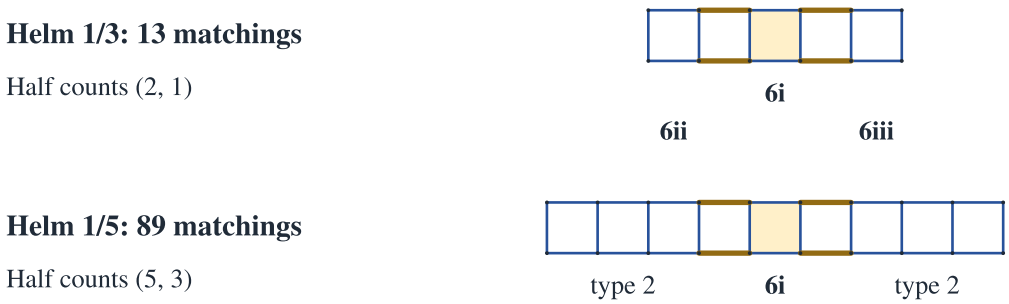


Figure 10. Central and end pieces in actual helms. At $1/3$, central square 6i has the end squares 6ii and 6iii as its halves, with counts $(2, 1)$. At $1/5$, each half is a type-2 triple with counts $(5, 3)$. Folding produces 13 and 89.

5.6. **A common prefix leaves one quadratic comparison.** Two folded graphs with the same outer boundary convention may share a prefix with open counts (x, y). Sum over all remaining states, leaving that cut open. The two centers give positive definite symmetric tables $G, \widetilde{G}$ with determinant one and nonnegative off-diagonal entries. For a helm they have the form SQS^{T} with

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$Q = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$; for opposite-parity labels they have form SS^{T}. Thus

$$m - \widetilde{m} = \alpha x^2 + 2\beta xy + \gamma y^2, \qquad G - \widetilde{G} = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}. \tag{35}$$

Proposition 5.4 (At most one primitive common prefix)**.** *For a nonempty common part and $G \neq \widetilde{G}$, equality requires $\alpha\gamma < 0$ and $\beta^2 - \alpha\gamma$ to be an integer square. There is then at most one primitive positive row (x, y) giving equality. That row must still arise from both actual prescribed snakes.*

Proof. A center table $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ has $ac - b^2 = 1$ and $b \geq 0$. If its two diagonal differences from another table have the same weak sign, then the difference of $b = \sqrt{ac-1}$ has that sign too. A nonzero difference cannot vanish at a positive row. Opposite diagonal signs are therefore necessary. The resulting quadratic in y/x has one positive root. An integer zero obeys $(\gamma y + \beta x)^2 = (\beta^2 - \alpha\gamma)x^2$; a rational square root of an integer is integral. A positive rational root has only one primitive positive representative. □

For an empty prefix, test its actual boundary row. If $G = \widetilde{G}$, the criterion supplies no exclusion. Equal count tables do not certify equal physical graphs: types 1 and 5 already show why orientations and the prescribed sequence of horizontal and vertical steps must be retained.

A useful uniform exclusion follows directly modulo three. The physical piece tables satisfy $R_i Q R_i^{\mathsf{T}} \equiv Q \pmod 3$, whereas a single retained square gives $-Q$. A helm tail with d retained squares therefore has center table $(-1)^d Q \pmod 3$. Opposite parities give a nonzero difference at every primitive row because the discriminant of Q is -1, a nonsquare modulo three. For equal parity and nonzero difference, let $h = \min_{i,j} v_3((G - \widetilde{G})_{ij})$ and write $3^{-h}(G - \widetilde{G}) = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$. Then $h \geq 1$. Expanding the equal determinants and reducing modulo three gives $\gamma \equiv \alpha + \beta$; the normalized discriminant is one or two modulo three. If it is two, every primitive shared row has count difference of valuation exactly h. These are exclusions for specified tails with arbitrary actual common prefixes, not a classification of all possible snake-graph collisions.

6. Equal-degree separation and unequal-degree windows

6.1. **A complete result at each fixed degree.** We first use the established fixed-sum, fixed-numerator and fixed-denominator orderings of Markov numbers. Section 5 proves them by matching counts and records their earlier sources.

Theorem 6.1 (Equal-degree unicity)**.** *Distinct canonical integer-phase marks of the same Meridian degree have different maxima. The marks may belong to the same or to different Meridians.*

Proof. For $D \geq 4$, Lemma 3.5 partitions the labels into three classes. Helms r/s have fixed sum D; near-helms $r/D, s/D$ have fixed denominator D; outer phases have fixed numerator D. Each class is strictly ordered by its corresponding monotonicity, and the dictionary ensures that different marks have different labels.

The largest helm is $m_{1/(D-1)}$. The near-helm extrema are $m_{1/D}, m_{(D-1)/D}$; the smallest outer value is $m_{D/(D+1)}$. They are separated by

$$m_{1/(D-1)} < m_{1/D}, \qquad m_{(D-1)/D} < m_{(D-1)/(D+1)} < m_{D/(D+1)}.$$

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The intermediate fraction is reduced because D is even. The first two inequalities hold at fixed numerator and the last at fixed denominator. For $D = 2$, the unique seed gives $2, 5$ and then the increasing family $m_{2/(2n-1)}$ for $n \geq 2$, starting at 29. These values are also distinct. □

Thus the remaining Meridian comparison has unequal degrees. To bound it, we need exact leading terms and an explicit positive remainder.

6.2. **From Fibonacci blocks to a Meridian profile.** The ordering argument above uses the Fibonacci and six-piece comparisons from Section 5. We now use the same run matrices to bound growth. Put $\rho = (3 + \sqrt{5})/2$ and $\lambda = \log \rho$. The odd-index Fibonacci formula extends (28) to every real t:

$$\mathcal{A}_t = \frac{1}{\sqrt{5}} \left[\rho^{t-1/2} \begin{pmatrix} \rho^2 & \rho \\ \rho & 1 \end{pmatrix} + \rho^{-t+1/2} \begin{pmatrix} \rho^{-2} & \rho^{-1} \\ \rho^{-1} & 1 \end{pmatrix} \right]. \tag{36}$$

For integer t, its entries are exactly $F_{2t+3}, F_{2t+1}, F_{2t+1}, F_{2t-1}$, including negative odd indices. Each displayed matrix is the outer product of a positive column with itself: respectively $(\rho, 1)^{\mathsf{T}}$ and $(\rho^{-1}, 1)^{\mathsf{T}}$. Thus every choice of terms in a product has a positive contribution to its upper-left entry.

Let the helm counts be (r, s) and $D = r + s$. The phase dictionary and the run-count formula (29) give, for every integer $n \geq 2$,

$$M_\alpha(n) = m_{D/((n-1)D+r)}.$$

The first run index is $n - 1$; each later index is $n - 1 + \lceil jr/D \rceil - \lceil (j-1)r/D \rceil$. Consequently the full real profile is

$$M_\alpha(x) = \left[\mathcal{A}_{x-1} \prod_{j=2}^{D} \mathcal{A}_{x-1+\lceil jr/D \rceil - \lceil (j-1)r/D \rceil} \right]_{11}. \tag{37}$$

To justify the extension from integers to reals, observe that both sides are finite Laurent polynomials in ρ^x. For the left side this follows from the seed formula and the polynomial recurrence (11); for the right side it follows from (36). Their difference vanishes at every $x = n \geq 2$. Multiplying by a sufficiently large power of ρ^x gives a polynomial with infinitely many roots, so it is zero. For $M_\alpha(-x)$, replace r by s in the right side of (37); the negative-arm dictionary gives the same argument. For example, the seed has $D = 2, r = s = 1$ and $M_1(x) = [\mathcal{A}_{x-1}^2]_{11}$, giving $2, 5, 29, 194, \ldots$.

Proposition 6.2 (Laurent profile)**.** *For helm counts (r, s), degree $D = r + s$, and $t = (r - s)/2$,*

$$M_\alpha(x) = \sum_{j=-D,\,-D+2,\ldots,D} c_j \rho^{jx}, \quad c_j \geq 0, \quad c_{\sigma D} = \frac{1}{3}(9/5)^{D/2} \rho^{\sigma t}. \tag{38}$$

The logarithm is strictly convex, and the real profile has one minimum.

Proof. Expand the D Fibonacci factors in (37) using (36). Each factor supplies one power ρ^x or ρ^{-x}, with positive coefficients. This gives the stated exponents and coefficient signs.

The run indices sum to $Dx + r - D - 1$. Selecting the first term in every factor gives the upper-left entry

$$\frac{\rho^{Dx+r-D-1-D/2} \rho^2 (3\rho)^{D-1}}{(\sqrt{5})^D} = \frac{1}{3}(9/5)^{D/2} \rho^{Dx+(r-s)/2},$$

because $\rho^2 + 1 = 3\rho$. Selecting every second term gives the other extreme coefficient. The second derivative of the logarithm is λ^2 times the variance of the exponents under their positive normalized weights. Both extremes occur, so that variance is positive; they also force growth at both ends. □

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Lemma 6.3 (Uniform Fibonacci-block remainder)**.** *For a Meridian of degree D, helm counts (r, s), $n \geq 0$ and $\sigma = \pm 1$,*

$$M_\alpha(\sigma n) = \frac{1}{3}(9/5)^{D/2}\rho^{\sigma(r-s)/2}\rho^{Dn}(1 + \epsilon_{\alpha,\sigma}(n)), \qquad 0 \leq \epsilon_{\alpha,\sigma}(n) \leq (1 + \rho^{1-2n})^D - 1. \tag{39}$$

Proof. Use (37), replacing r by s on the negative arm. Every block index is either $n-1$ or n. Call the first and second outer products in (36) the $+$ and $-$ choices. When two neighboring columns are multiplied, their inner products are

	$+$	$-$
$+$	3ρ	2
$-$	2	$3/\rho$

.

Relative to the all-$+$ choice, a mixed join has factor $2/(3\rho) < \rho^{-1}$, and a $--$ join has factor ρ^{-2}. At either outside boundary a minus choice has factor ρ^{-2}, which is also at most ρ^{-1}. Each selected minus block touches two joins or boundaries. Thus k minus choices reduce these column and boundary factors by at least ρ^{-2k}.

At a block of index t, replacing the scalar coefficient of the plus term by that of the minus term multiplies it by ρ^{1-2t}. Since $t \geq n-1$, the complete relative contribution of any specified k minus choices is at most $\rho^{-2k}(\rho^{3-2n})^k = \rho^{k(1-2n)}$. Sum this bound over all nonempty subsets of the D positions. The binomial theorem gives the displayed remainder bound. □

6.3. **A finite rectangle depending only on the degrees.**

Theorem 6.4 (Unequal-degree collision window)**.** *Let $D_\alpha \neq D_\beta$, and define*

$$m = \frac{|D_\alpha - D_\beta|}{2}, \qquad E = \max(D_\alpha, D_\beta),$$
$$h = \min\{j \in \mathbb{Z}_{\geq 0} : \rho^j \geq E\}, \qquad N = 3m + h + 3, \qquad T = E(N+2).$$

An integer equality $M_\alpha(x) = M_\beta(y)$ requires

$$|x| \leq \left\lceil \frac{T}{D_\alpha} \right\rceil, \qquad |y| \leq \left\lceil \frac{T}{D_\beta} \right\rceil, \qquad M_\alpha(x) < \rho^T/3. \tag{40}$$

The seed still uses only nonnegative phases.

Proof. The degrees are even, so m is a positive integer. First,

$$|m \log(9/5) - j \log \rho| > \frac{1}{81^m + 1} \qquad (j \in \mathbb{Z}). \tag{41}$$

Choose $j_0 \geq 0$ with $1 < (9/5)^m \rho^{-j_0} < \rho$; equality at an endpoint is impossible by taking the field norm. The positive algebraic integers $9^m - 5^m\rho^{j_0}$ and $5^m\rho^{j_0+1} - 9^m$ have nonzero integer norms and conjugates of absolute value below 9^m. Each is therefore greater than 9^{-m}. Dividing by the appropriate positive denominators makes both adjacent multiplicative gaps greater than 81^{-m}. The inequality $\log(1+u) > u/(1+u)$ gives (41); all other integers j are farther away.

At phases $x = \sigma n$, $y = \tau n'$, the leading logarithms differ by

$$\frac{D_\alpha - D_\beta}{2}\log(9/5) + (D_\alpha n - D_\beta n' + \sigma t_\alpha - \tau t_\beta)\log\rho.$$

The coefficient of $\log\rho$ is integral. Put $r_m = (81^m + 1)^{-1}$. Since $E \leq \rho^h$, $\rho^6 > 81$ and $\rho^5 > 5$,

$$E\rho^{1-2N} \leq \rho^{-6m-5} < \frac{1}{2 \cdot 81^m + 3} = \frac{r_m}{r_m + 2}.$$

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For a positive integer g and $u \geq 0$ with $gu < 1$, the elementary bound $(1+u)^g \leq \sum_{j\geq 0}(gu)^j = (1-gu)^{-1}$ makes each relative remainder in Lemma 6.3 smaller than $r_m/2$ when its phase magnitude is at least N. Their logarithmic corrections cannot cancel (41). At least one collision phase therefore has magnitude less than N.

The leading coefficient of M_α lies strictly between $\rho^{-D_\alpha}/3$ and $\rho^{D_\alpha}/3$: use $|t_\alpha| < D_\alpha/2$ and $1 < 3/\sqrt{5} < \sqrt{\rho}$ in (38). Lemma 6.3, with $1 + \rho^{1-2N} < \rho$, gives

$$M_\alpha(\pm N) < \rho^{D_\alpha(N+2)}/3 \leq \rho^T/3.$$

The same bound holds for M_β. Convexity bounds the value at any phase inside either finite strip by its endpoint values, so the common value is below $\rho^T/3$. Positivity also gives $M_\alpha(\sigma n) > \rho^{D_\alpha(n-1)}/3$ and the corresponding bound for $M_\beta(\tau n')$. Comparing the upper and lower bounds yields $n < 1 + T/D_\alpha$ and $n' < 1 + T/D_\beta$, hence the integer ceilings in (40). □

This theorem bounds the remaining candidates; it does not assert that any candidate collides. It also does not bound the degrees of a hypothetical collision. The next section supplies a different finite reduction, expressed in Farey numerators and denominators rather than Meridian phases.

7. Corridor walls and numerator–denominator windows

7.1. **A second way to travel through the same tree.** The corridor observation groups the walls between neighboring phasors. After an initial entrance, a wall follows a fixed-numerator family

$$m_{p/(q_0+np)}, \qquad n \geq 0.$$

Both Meridian and phasor edges occur in such a route. Their common law is the advance $q \mapsto q + p$, not a rule that distinguishes the two edge colors. For the $5 \to 13$ phasor-capped corridor (equivalently, the $1/2 \to 1/3$ phasor-capped corridor), the fixed-numerator portions are $3/(4+3n)$ and $3/(5+3n)$, tending to the historical coordinate $c = 1/3$. The first wall begins

$$169 \xrightarrow{\text{phasor}} 2897 \xrightarrow{\text{Meridian}} 51641 \xrightarrow{\text{phasor}} 925765 \longrightarrow \cdots,$$

and the other begins $433, 7561, 135137, 2423525, \ldots$ with the opposite edge alternation.

Corridor role and edge color must be kept separate. With corridor-wall travel designated *standard* and a new entrance *exotic*, the two entry routes here are

$$5 \xrightarrow{\text{exotic}} 29 \xrightarrow{\text{standard}} 433 \xrightarrow{\text{standard}} 7561 \longrightarrow \cdots, \qquad 13 \xrightarrow{\text{exotic}} 169 \xrightarrow{\text{standard}} 2897 \longrightarrow \cdots.$$

Both exotic entrances are Meridian edges. In the first route the initial standard step $29 \to 433$ changes the numerator from two to three; the fixed-numerator formula starts at 433. Thus excluding the concave ceiling arc from a wall's recurrence does not move the exotic entrance to that arc. The graph and its actual depth (14) are unchanged.

7.2. **The exact recurrence behind the observed regularity.** The same Fibonacci run blocks give the corridor recurrence directly. For a reduced label $0 < p/q < 1$, the successive horizontal run lengths are $\lceil jq/p \rceil - \lceil (j-1)q/p \rceil$. Equation (29) therefore reads

$$m_{p/q} = \left[\mathcal{A}_{\lceil q/p \rceil - 1} \prod_{j=2}^{p} \mathcal{A}_{\lceil jq/p \rceil - \lceil (j-1)q/p \rceil} \right]_{11}. \tag{42}$$

The shortened first run is the boundary convention already used for the six-piece matching count. For example, $m_{3/4} = [\mathcal{A}_1^3]_{11} = 169$ and $m_{3/7} = [\mathcal{A}_2^3]_{11} = 2897$.

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Replacing q by $q_0 + np$ adds exactly n to each of the p block indices. In the positive expansion (36), each plus choice contributes ρ^n and each minus choice contributes ρ^{-n}. Grouping the terms by the number of minus choices gives

$$m_{p/(q_0+np)} = \sum_{j=0}^{p} c_{p,q_0,j}\rho^{(p-2j)n}, \qquad c_{p,q_0,j} > 0. \tag{43}$$

Every mode occurs. The least order of a constant-coefficient recurrence is therefore $p+1$, with characteristic polynomial

$$\prod_{j=0}^{p}(X - \rho^{p-2j}).$$

Pairing conjugate roots shows that its coefficients are integers. For $p = 3$ the polynomial is $(X^2 - 18X + 1)(X^2 - 3X + 1)$, so either wall of this corridor satisfies

$$m_{3/(q+12)} = 21m_{3/(q+9)} - 56m_{3/(q+6)} + 21m_{3/(q+3)} - m_{3/q}. \tag{44}$$

This equality holds at every step, for both colors of edge.

The exact regularity is a shared recurrence, not a constant output ratio. The positive spectral sum gives strict log-convexity, hence

$$\log_{10}\frac{m_{p/(q_0+(n+1)p)}}{m_{p/(q_0+np)}} \nearrow p\log_{10}\rho.$$

The small deviation is retained in the collision argument, rather than discarded as an asymptotic error. Figure 11 shows both the geometric walls and that deviation.

7.3. **A positive correction with a uniform bound.** The sum of the block indices in (42) is $q-1$. Selecting every plus term in (36) therefore gives $3^{p-1}\rho^{q+p/2}/(\sqrt{5})^p$ and hence

$$3m_{p/q} = \left(\frac{3}{\sqrt{5}}\right)^p \left(\frac{1+\sqrt{5}}{2}\right)^{2q+p}(1+\varepsilon_{p,q}), \tag{45}$$

where

$$0 < \varepsilon_{p,q} \le \left(1+\rho^{-2\lfloor q/p\rfloor}\right)^p - 1. \tag{46}$$

For $p > 1$, coprimality implies that q/p is not an integer. Every block index in (42), including the shortened first one, is at least $\lfloor q/p\rfloor$. The column and boundary comparison in Lemma 6.3 bounds a specified set of k minus choices by $\rho^{-k}(\rho^{-2\lfloor q/p\rfloor})^k$, which is no larger than $(\rho^{-2\lfloor q/p\rfloor})^k$. Summing over all nonempty subsets proves (46). For $p = 1$, the single entry $[\mathcal{A}_{q-1}]_{11} = F_{2q+1}$ has relative correction $\rho^{-2q-1} < \rho^{-2q}$, giving the same bound. This estimate is independent of the remainder class of q modulo p.

Along a fixed wall the correction decreases strictly to zero. Consequently its logarithmic deviations telescope exactly:

$$\sum_{k=0}^{n-1}\left[p\log\rho - \log\frac{m_{p/(q+(k+1)p)}}{m_{p/(q+kp)}}\right] = \log\frac{1+\varepsilon_{p,q}}{1+\varepsilon_{p,q+np}}. \tag{47}$$

Thus there is no accumulated error proportional to the path length. For a route changing corridors, subtraction of the endpoint formula still telescopes, but monotone decay is claimed only within a fixed wall.

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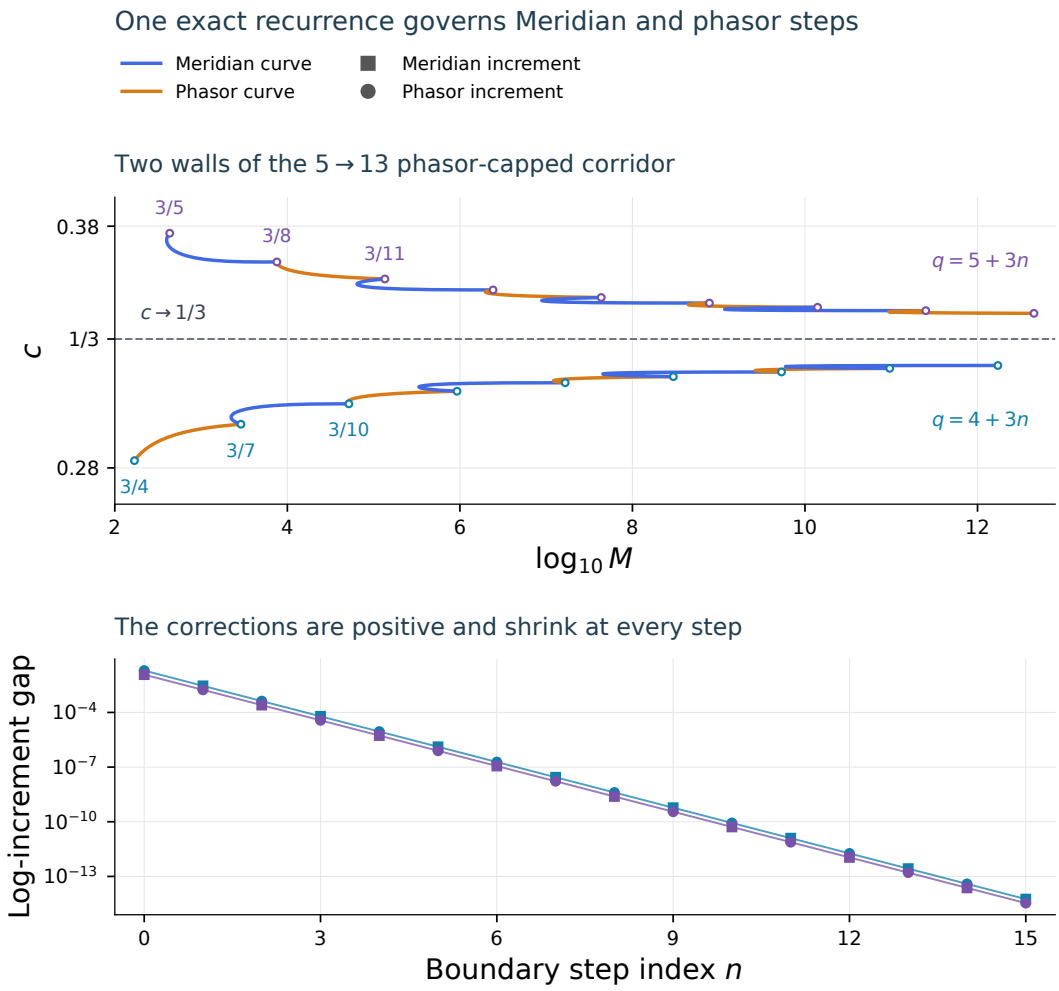


Figure 11. The two fixed-numerator walls of the $5 \to 13$ phasor-capped corridor ($1/2 \to 1/3$ in fraction labels). The concave ceiling arc is omitted from this recurrence illustration. Both walls obey (44). In the lower panel, the positive difference between $3\log_{10}\rho$ and the actual logarithmic increment decreases at every step. Squares denote Meridian increments and circles phasor increments. The values are exact integers; only the plotted curves and logarithms are numerical.

Theorem 7.1 (Necessary correction gap)**.** *If $p \neq p'$ and $m_{p/q} = m_{p'/q'}$ for reduced labels, then*

$$\frac{|\varepsilon_{p,q} - \varepsilon_{p',q'}|}{1 + \min(\varepsilon_{p,q}, \varepsilon_{p',q'})} > 9^{-|p-p'|}. \tag{48}$$

Proof. Assume $p > p'$. The ratio of the leading terms is

$$R = \left(\frac{3(1+\sqrt{5})}{2\sqrt{5}}\right)^{p-p'} \rho^{q-q'}.$$

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Its quadratic-field norm is $(9/5)^{p-p'}$, so $R \neq 1$. The number $(\sqrt{5})^{p-p'}(R-1)$ is an algebraic integer, because the powers of $(1+\sqrt{5})/2$ and ρ are units. Its nonzero norm gives

$$|(R-1)(\overline{R}-1)| \geq 5^{-(p-p')}, \qquad \overline{R} = (9/5)^{p-p'}/R > 0.$$

Put $\eta = \max(R, R^{-1}) - 1$. If $R \geq 1$, then $|R-1| = \eta$ and $|\overline{R}-1| < (9/5)^{p-p'}$. If $R < 1$, their product is $\eta((9/5)^{p-p'} - 1/(1+\eta)) < \eta(9/5)^{p-p'}$. Either case forces $\eta > 9^{-(p-p')}$. At equality of the Markov values, $R = (1+\varepsilon_{p',q'})/(1+\varepsilon_{p,q})$, so η is precisely the left side of (48). □

7.4. **The corridor collision window.**

Theorem 7.2 (Numerator–denominator window)**.** *Let $p \neq p'$, $P = \max(p, p')$, and let K be the least nonnegative integer satisfying*

$$\rho^{2K} > P(9^{|p-p'|} + 1). \tag{49}$$

Every equality $m_{p/q} = m_{p'/q'}$ requires

$$\max(q, q') \leq P(K+1) - 1. \tag{50}$$

For bounded numerators a simpler, weaker consequence is $\max(q, q') \leq 2P^2 + P - 1$.

Proof. If both quotient floors are at least K, then $P\rho^{-2K} < 1/(9^{|p-p'|}+1)$. The elementary estimate $(1+u)^p \leq (1-pu)^{-1}$ for $pu < 1$ makes both corrections strictly smaller than $9^{-|p-p'|}$, contradicting Theorem 7.1. Thus one denominator, say q, is at most $Kp - 1$.

The Fibonacci formula (36) gives the entrywise bound $\mathcal{A}_t \leq \rho^t \mathcal{A}_0$ for every integer $t \geq 0$. Products preserve that bound, and $\mathcal{A}_0 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ has operator norm ρ. Since the indices sum to $q-1$, (42) gives $m_{p/q} \leq \rho^{q-1}[\mathcal{A}_0^p]_{11} \leq \rho^{q+p-1}$. Together with the positive leading term, this yields

$$\frac{1}{3}\rho^{q'} < m_{p'/q'} = m_{p/q} \leq \rho^{q+p-1}.$$

Now $3 < \rho^2$ implies $q' < q + p + 1$. The denominators are integers, so $q' \leq q + p$. It follows that both denominators are at most $P(K+1) - 1$. For the simpler bound it suffices to use $K = 2P$: $\rho^{4P} > 36^P > P(9^{P-1}+1)$ for $P \geq 2$, and $|p - p'| \leq P - 1$. □

For example, numerators at most 64 permit $K = 75$, so every remaining comparison lies at denominators at most 4863. With $P = 64$ and numerator difference five, $K = 8$ gives 575. These are bounds on candidates, not new numerical verification claims.

Theorem 6.1 first discards equal Meridian degrees. For the rest, the phase rectangle (40) and denominator cap (50) can be imposed together through the dictionary (18). Fixed-numerator ordering already separates all labels with $p = p'$. What remains is a comparison of different numerators and degrees inside their joint windows. Neither numerator nor degree is globally bounded by these theorems, so this is not yet a finite reduction of the entire unicity conjecture.

8. All-phase verification through degree 10,000

Theorem 8.1 (Degree-10,000 separation)**.** *Among canonical Meridians of Laurent degree at most* 10,000*, distinct canonical integer-phase marks have distinct maxima. There are exactly* 5,067,077 *Meridians in this family.*

The computational part separates different Meridians at all integer phases. The fixed-degree theorem also separates marks on a single Meridian, so the two conclusions combine as stated. The bound is on *degree*, not on the size of a Markov number, and it does not exclude a competitor of higher degree.

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8.1. **Why a finite residue test covers infinitely many phases.** For a period T, define the complete image

$$\mathcal{R}_\alpha(T) = \{M_\alpha(x) \bmod q(T) : 0 \le x < T\}.$$

Theorem 4.1 makes this the image at *every* integer phase. Disjoint images therefore separate two entire Meridians. If images overlap for several periods, a true equality must use compatible phases in all of them. Define the witness relation

$$W_T(\alpha, \beta) = \{(u, v) \in (\mathbb{Z}/T\mathbb{Z})^2 : M_\alpha(u) \equiv M_\beta(v) \pmod{q(T)}\}.$$

Proposition 8.2 (Compatible witnesses)**.** *Equality of two integer values requires one witness in each selected W_T, with first coordinates agreeing modulo every pairwise gcd of the periods, and second coordinates agreeing in the same way. If no compatible choice exists, the two Meridians are disjoint at all integer phases.*

Proof. A true equality uses the same integers x, y in every modulus and hence supplies the required residues. Conversely, the generalized Chinese remainder theorem lifts compatible coordinates to a phase pair modulo the least common multiple. This converse concerns modular witnesses only; it does not imply equality of the full values. □

The moduli need not be independent. In particular, the periods $36, 38, 40$ are all even, so contradictory parity requirements alone can exclude a pair that survives each separate image test.

8.2. **Complete inventory and the final exclusion.**

Proof of Theorem 8.1. Proposition 3.4 identifies the inventory with coprime positive odd pairs $r \le s$, $r + s \le 10{,}000$. It contains 5,067,077 Meridians and 12,837,632,128,426 unordered pairs. The producer enumerates the count-vector address tree; an independent checker enumerates the coprime pairs directly and compares the inventories. The child degree is $2|k|D \pm (|v| - |u|) > D$, where $|u|, |v|$ are the endpoint word lengths. This gives finite exhaustive index ranges.

Every one of the 36 phase residues of every Meridian is evaluated modulo $q(36)$. Equal-residue grouping lists all pairs whose images intersect; its complement is excluded at once. Periods 38 and 40 are then evaluated on every endpoint of the preceding survivor sets. The complete image tests leave 87 pairs. For each of them, project every full witness relation onto the four parity pairs $(u \bmod 2, v \bmod 2)$. The three projected sets have empty intersection in all 87 cases. Proposition 8.2 therefore excludes every pair of different Meridians at every integer phase. Theorem 6.1 also excludes different marks within each individual Meridian, completing the statement. □

8.3. **What was checked independently.** The degree-10,000 producer uses signed endpoint-orbit updates. The separate evaluator uses Fibonacci staircase words with Euclidean matrix reduction. They agree on all 188,369,412 retained phase evaluations. The checker verifies the 182,414,772 baseline phase tags bijectively, so an omitted or repeated phase record cannot pass unnoticed. Independent sorting of the original unsorted buckets reconstructs all 70,009 cross-Meridian residue groups and the subsequent complete pair sets. Coverage is checked by exact records, not inferred merely from their counts or from a matching digest.

The phase result is checked three ways: full generalized-CRT joins, an independent 16-bit mask computation on $(x \bmod 4, y \bmod 4)$, and the weaker four-state parity projection, already sufficient for the final 87. The degree-2,000 certificate, containing 202,862 Meridians, was completed first; its common survivor sets agree exactly with the larger inventory when identified by helm labels.

Five deliberate corruptions—an altered value, an incorrect modulus, a duplicate tag, an omitted record and an invented pair—were all rejected. The degree-100 control compares 18,684 residues with literal Fibonacci words and the polynomial record of (19). The optional image-only extension

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Table 2. Complete cumulative survivor counts from the archived certificates. "Images" forgets meeting phases; "phases" also imposes compatibility. Survivors are modular coincidences, not integer collisions.

Tests retained	$D \le 2{,}000$	$D \le 10{,}000$
36: images	7,121	320,755
36, 38: images	612	17,427
36, 38: phases	2	2,052
36, 38, 40: images	0	87
36, 38, 40: phases	**0**	**0**
36, 38, 40, 42: images	—	1
36, 38, 40, 42, 44: images	—	0

through periods 42 and 44 also ends with zero pairs; all 88 period-44 values were separately checked using arbitrary-size Python integers because $q(44) > 2^{63}$.

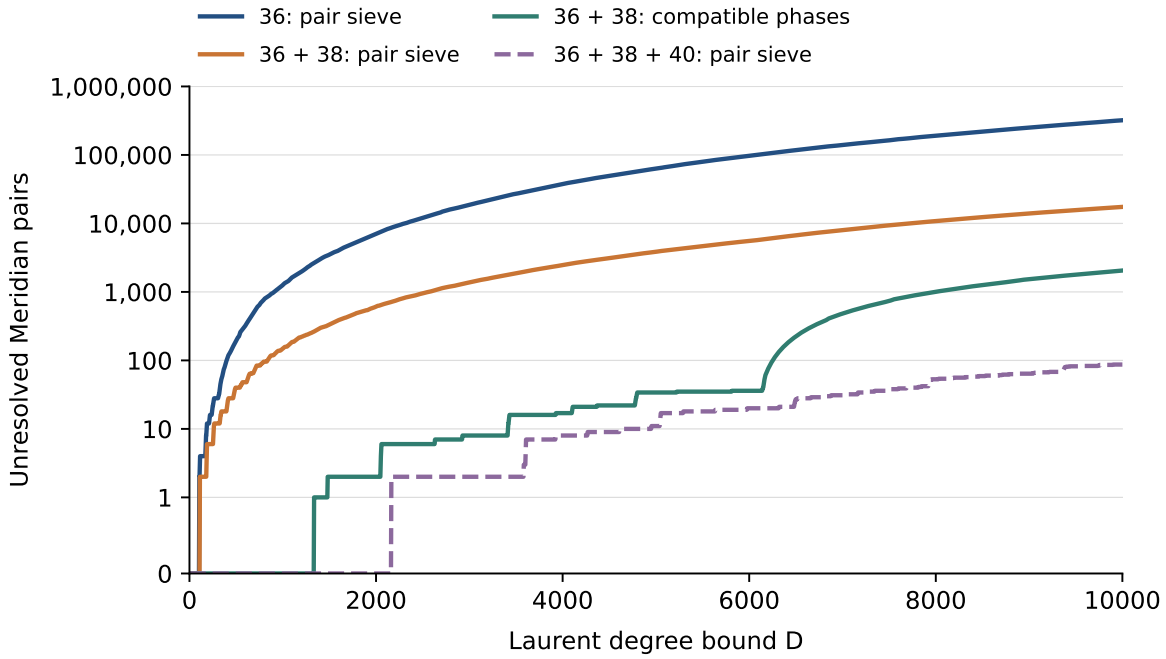


Figure 12. Exact unresolved-pair profiles in the degree-10,000 inventory. The phase-compatible sieve through period 40 has no survivors in the entire displayed range. Lines join discrete even-degree bounds. The figure is a record of the finite certificate, with no extrapolation to larger degrees.

8.4. **Reproducibility and scope.** The degree reproducibility package [10] contains the inventory, compact residue groups, phase witnesses, producer and checker sources, audit receipts, and a regeneration driver. Large baseline tables are regenerated; their recorded sizes and hashes identify the original run. The reindexed edition retains the original computational kernels with an explicit

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address-convention adapter. Reversing the written address tail does not change a helm label, degree, phase, or certified numerical value.

This is a deterministic computer-assisted certificate, not an external referee report or proof-assistant formalization. Its scope is a bounded set of Meridians with unbounded phase. The numerical-order verification in Section 9 is a separate computation with its own coverage proof and execution record.

9. Fast verification with bounded working storage

9.1. **The completed comparison.** The completed combined run [4] checked **1,101,388,071,945 configurations** and certified that each of the **first one trillion Markov values in numerical order** is the maximum of exactly one normalized triple. These counts answer different questions: the larger is the workload, including possible competitors and conservative cutoff coverage; the smaller is the guaranteed numerical prefix. The exact trillionth value and its predecessor are supplied in full in the selected-values supplement; each has 1,021,609 decimal digits. The title names the certified numerical prefix, not the larger comparison workload.

Meridian analysis guided the organization by scale. A helm anchors a family, and its degree determines the limiting logarithmic spacing along each outward branch. This suggested asking which families can contribute at a given scale, rather than retaining every value already generated. Finite-phase corrections still require certified bounds: the degree and helm alone are not an exact band-entry test.

The production code implements this idea using reduced Farey labels p/q, together with the separate root, and compressed Vieta recurrences. Groups of neighboring Farey directions share certified matrix bounds. They let the code select the addresses relevant to a magnitude band without keeping a separate schedule for every Meridian. The band algorithm can be formulated entirely in these Farey and matrix terms; the Meridian work supplied the family-growth viewpoint that guided its development. The implementation first selects by certified magnitude, then compares eight-byte residues within each band. Full integers are reconstructed only for repeated residues. Completed key files are removed before the next band is processed. Thus the working store depends on the band size rather than the total number of configurations. Its correctness uses exhaustive band coverage and exact residue resolution, without the Meridian collision-exclusion theorems. No controlled comparison isolates a runtime or storage gain attributable to Meridian coordinates alone.

Table 3. Completed comparison and certified numerical coverage. Overlapping band emissions are not counted as new configurations.

Certified numerical-value prefix	1,000,000,000,000
Inner configurations below the cutoff	1,000,000,419,685
Included configurations compared	1,101,388,071,945
Band emissions, including overlap	1,107,352,458,539
Committed bands	19,287
Repeated residue groups / exact members	12 / 24
Equal maxima found	0

9.2. **Coverage and why no collision can cross a cleared band.** The finite envelope uses the same height bounds as the earlier fifty-billion proof [7]. For reduced labels $0 < p \le q$,

$$m_{p/q} \le F_{2(p+q)-1}, \qquad m_{p/q}^2 \ge 4\,5^{p+q-2}.$$

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Including the root, the number of configurations of height at most $t \geq 2$ is $C(t) = 1 + \frac{1}{2}\sum_{j=1}^{t} \varphi(j)$, where φ is Euler's totient function. The exact choices are

$$K = 2{,}565{,}100, \qquad B = F_{5,130,199}, \qquad L = 3{,}067{,}794,$$
$$C(K) = 1{,}000{,}000{,}419{,}685, \qquad C(L) = 1{,}430{,}355{,}600{,}632.$$

Every inner configuration has maximum at most B. The exact inequalities $4\,5^{L-2} \leq B^2 < 4\,5^{L-1}$ ensure that every competitor with maximum at most B has height at most L. This argument counts configurations before making any claim of unicity.

To schedule the comparisons, 162 cones partition the remaining fraction labels after the 163 configurations through height 32, including the root. Here a *cone* means the addresses strictly between two neighboring Farey labels, after their already retained endpoints are removed. For endpoints $p_L/q_L < p_R/q_R$, each address in the group has the unique form

$$(p, q) = u(p_L, q_L) + v(p_R, q_R), \qquad u, v \geq 1, \quad \gcd(u, v) = 1.$$

The endpoints satisfy $p_R q_L - p_L q_R = 1$. They are obtained by repeatedly inserting mediants until the next numerator-plus-denominator exceeds 32. This is a finite partition of address directions, not a partition into individual Meridians. One Meridian can cross several cones, and one cone can contain parts of many Meridians.

Figure 13 connects the motivating family picture to this implementation. In the group $0/1 < p/q < 1/31$, for example, $(p, q) = (v, u + 31v)$. The displayed positive branches of degrees 2 and 4 eventually lie in the rows $v = 2$ and $v = 4$. Their predictable growth makes selection by scale natural; the shared cone bounds make that selection certified for every address in the group, including other families. Exact positive-matrix bounds assign each label an interval containing $\log(3m_{p/q})$; the coefficients are rounded outward at scale 2^{-32}. The bands cover from zero through a certified upper endpoint beyond $\log(3B)$, without gaps. Every configuration whose interval intersects a band is emitted there, including intervals crossing a boundary. The frozen cone certificate and integer inequalities are checked before use.

If two maxima were equal, their actual logarithms would occupy the same band. Both enclosing intervals would intersect that band, and both exact residues modulo 2^{64} would therefore enter the same comparison. All keys in a band are partitioned into 32 complete buckets and sorted. Every member of every repeated-key group is recovered by replaying the band and is compared exactly. Only then is the band committed and cleared. Consequently, clearing its keys cannot hide an equality across a boundary. Neither random-residue assumptions nor family-specific collision exclusions are used to discharge a repeated group.

All 12 repeated groups contained two configurations. Independent integer evaluators agreed on their values, and every pair had unequal maxima. The complete inner population and the outer envelope therefore establish the first-trillion conclusion in Table 3.

9.3. **Completion timestamp, elapsed work, CPU time, and storage.** The inherited chain began on **8 November 2026 at 14:55:01.580398 UTC** (08:55:01.580398 CST). The extension's first retained band began on **9 November at 16:53:38.951229 UTC** (10:53:38.951229 CST). Its last band finished on **10 November at 03:54:02.997413 UTC**, or **9 November at 21:54:02.997413 CST** in St. Louis. The extension session ran from 16:53:13.152936 UTC on 9 November (10:53:13.152936 CST) through 03:54:13.271944 UTC on 10 November (21:54:13.271944 CST on 9 November). The final independent replay ended at 03:56:16.801036 UTC (21:56:16.801036 CST on 9 November); verification and replay of a fresh portable extraction ended at 03:59:41.149704 UTC (21:59:41.149704 CST on 9 November).

The under-24-hour figure is retained active elapsed processing, including the inherited certificate. It excludes pauses, restoration, failed or lost attempts, and later audits. The combined calendar span exceeds active processing by 46,908.545871 seconds (13.030 hours); this difference includes all such

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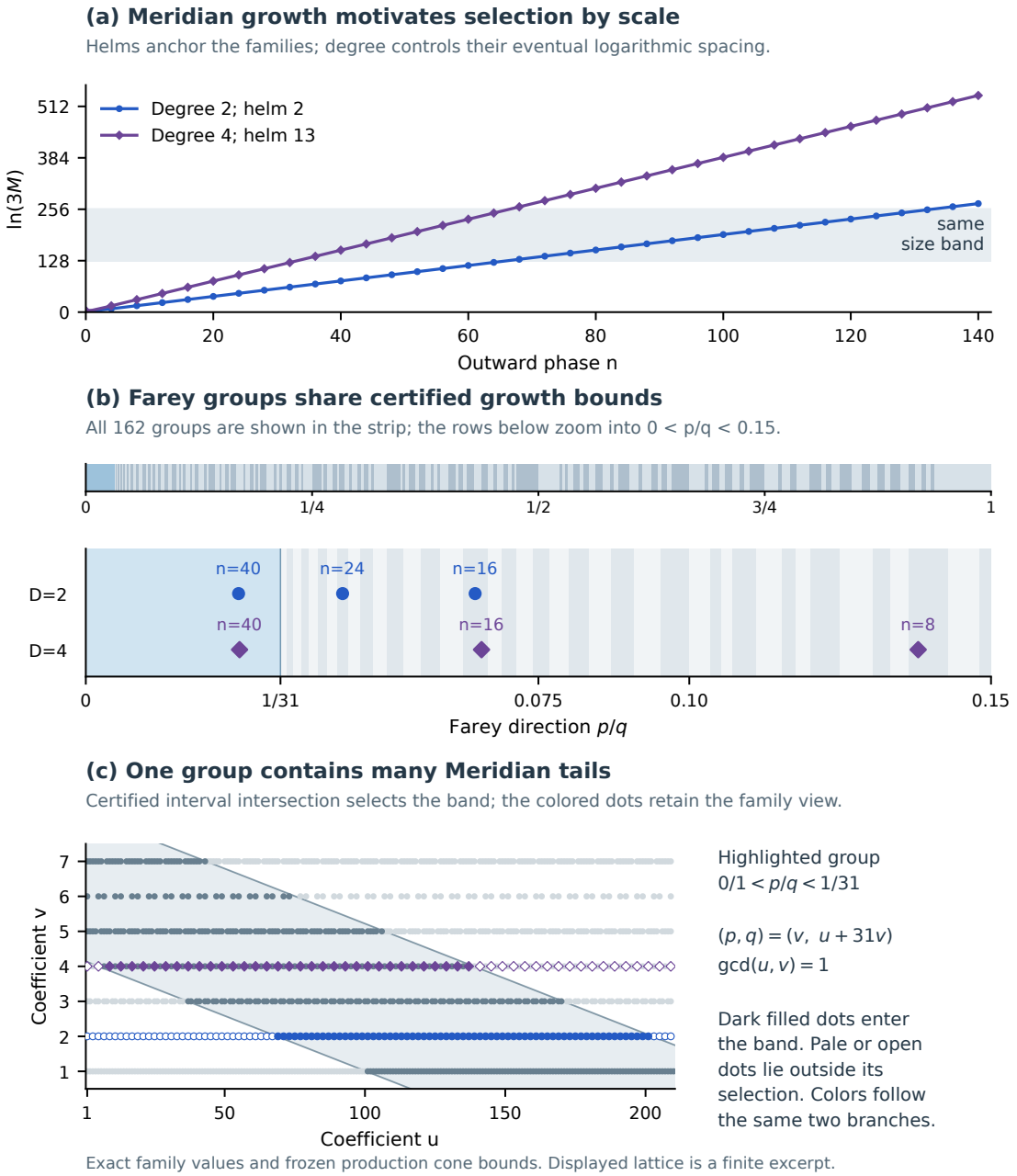


Figure 13. From Meridian growth to certified Farey grouping. (a) Exact values on two positive branches illustrate the degree-dependent spacing that motivated magnitude bands; the shaded band is $128 \le \log(3M) < 256$. (b) All 162 frozen Farey groups, then a zoom showing the same families crossing their boundaries. (c) The highlighted group $0/1 < p/q < 1/31$ uses $(p, q) = (v, u + 31v)$ with positive coprime u, v. Shading selects certified intervals meeting the band in (a); colors pick out the same branches among all displayed addresses. Groups share matrix bounds and can contain many Meridian tails. Boundary-crossing intervals enter both bands. Exact plotted data and the frozen certificate accompany the source.

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Table 4. Measured resources for the completed retained chain. The run used eight workers. GiB denotes 2^{30} bytes.

Measurement	Recorded amount
Retained active band processing, combined	86,243.145675 s **23 h 57 min 23.146 s**
Inherited / extension active processing	54,470.884742 / 31,772.260933 s
Summed native-child CPU, combined	417,823.596537 s (116.062 h)
Inherited / extension native-child CPU	264,075.377305 / 153,748.219232 s
Calendar span, first inherited band to final session end	133,151.691546 s (36.987 h)
Extension calendar span, session endpoints	39,660.119008 s (11.017 h)
Maximum recorded temporary key files	**1,065,664,392 bytes** 1,016.297 MiB; 0.992 GiB
Maximum generator resident memory	36,468 KiB (35.613 MiB)
Verified compressed recovery archive	38,920,281 bytes (37.117 MiB)
Independent final replay	67.501145 s
Portable extraction verification and replay	71.627535 s

gaps and overhead and is not a measure of idle time alone. The inherited execution record includes 23,436,403,353 configurations regenerated after an environment loss, with 0.827 hours of lost band work recorded separately. The inherited execution windows total 17.160 hours, including known lost work. Three interrupted extension sessions have bounded, rather than exact, end times. The extension execution windows total 35,390.008304–35,547.876597 seconds; 3,617.747371–3,775.615664 seconds within those windows are outside retained band time, and 4,112.242411–4,270.110704 seconds fall between windows. These include overhead and interrupted attempts, not only idle time. The final evidence preserves the individual session bounds.

The combined retained stages used 49,481.812969 seconds for generation, 25,077.576696 for partitioning, 11,304.684507 for sorting and scans, and 121.285178 for exact-group resolution; controller bookkeeping accounts for the remaining band time. CPU time sums native-child usage and does not include an independently totalled parent-process CPU measurement.

The storage peak counts logical temporary key files, including a shard temporarily duplicated during partitioning. It excludes receipts, source, backup copies, filesystem overhead, and unrelated work. The generator memory peak covers that native process only. Continuous whole-workspace disk and aggregate-memory peaks were not recorded.

9.4. **Replay and retained evidence.** The final audits checked the frozen sources, all 19,287 linked receipts (13,638 inherited and 5,649 extension), boundaries, counts, bucket accounting and retained candidate bytes. A separate checker recomputed the envelope, independently reconstructed all candidate maxima with polynomial recurrences, and regenerated every repeated-key band. A fresh extraction verified all 20,208 manifested archive members and replayed the same final chain, with no partial or skipped-check flags. This is a receipt-and-candidate replay, not a second generation of every discarded singleton key.

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The frozen recovery archive [4] is

`v16_Trillion_Verifier_Recovery.zip`

Its SHA-256 and the final extension receipt head are, respectively,

`6aac7138c8427eb5a3701b27b841e55a75a2e1960c534233261b42375deaeeea`

`624c31904f64f8d9434162396a311228f2ece95d495ff2c7dc9d053314ed0c72`

The source includes the final replay reports, resource reconciliation, statistics and the unchanged historical evidence. Independent here means a separate implementation within the project, not external peer review.

9.5. **Reusing the completed 500-billion certificate.** The earlier certificate [6] completed on 9 November 2026 at 15:50:31.046707 UTC (09:50:31.046707 CST), comparing 550,693,634,649 configurations in 13,638 bands. It certified the first 500 billion values, with four repeated groups, 15 h 07 min 50.885 s of retained active work and a 754,846,136-byte temporary key-file peak. Its unchanged final receipt head is

`4642f18be7adf418d5399ffa87825dc2183dae6808ef7516e0dd0b1d704b9b6d.`

The separate extension adds 550,694,437,296 configurations and 8 h 49 min 32.261 s of retained active work. The enlarged envelope admits no new configuration into an already cleared band: the certified lower endpoint of every newly admitted interval lies beyond the old cutoff. Intervals crossing the join are emitted again. Each address is counted only once, in the band containing its lower interval endpoint. All 550,693,634,649 inherited configurations lie within the new inner height. The exact join proof and both frozen manifests accompany the evidence. Here $C(K-1) = 999{,}999{,}913{,}445$ and $C(K) = 1{,}000{,}000{,}419{,}685$; thus the inner population suffices for the claimed first-trillion prefix.

9.6. **Comparison with Tan–Zhang.** Tan and Zhang [19] report uniqueness for 239,536,376,556 normalized maxima through 10^{500000}, in the computation archived on 26 October 2026. Their 48-bit screening and replay modulo 2^{128} provide a useful recent comparison. The 13-group, 91-job screen processes 3,113,973,008,796 records, including repeated coverage. Our completed trillion-value prefix compares 1,101,388,071,945 configurations in 1,107,352,458,539 band emissions, using 64-bit residues within certified magnitude bands and full-integer resolution of repeats. These larger workload counts are distinct from the numerical-prefix sizes. Both methods use certified size information. Tan–Zhang use it to enforce their common cutoff, then compare balanced subtree groups in all pair jobs. Here the bounds schedule local magnitude bands whose completed keys can be cleared. Figure 14 shows this difference and the Meridian reasoning behind our choice of organization.

Storage and timing must be compared by scope. Tan and Zhang report about 184.27 GB of five-byte payload for the largest pair job and recommend 250 GB of available disk; their original target replay used 16 workers. Our completed run used eight workers and a maximum recorded 1,065,664,392 bytes (1.066 GB; 0.992 GiB) of temporary key files, with 23 h 57 min 23.146 s of retained active processing. Neither payload figure is a measured whole-machine storage peak. Tan and Zhang explain that their interrupted, multi-day run's partial timers do not establish a reliable total runtime, so a wall-clock or CPU speedup ratio is not justified. The comparison uses their cited version 1, still current when checked on 9 November 2026.

9.7. **Selected decimal values and run statistics.** The selected-values supplement supplies the *complete base-10 expansions* of six selected values: the trillionth Markov value and its predecessor (counting 1 first), the last value below and first value above $10^{1000000}$, and the last value below and first value above 10^{999999}. The last of these is the first Markov value with one million decimal digits. Each integer occupies one plain-text file, compressed losslessly in the supplementary ZIP, with its address and checksum. The six files contain 6,043,218 digits in total. The expansions were first posted on Zenodo on 9 November 2026 [5]. The completed unicity certificate and subsequent

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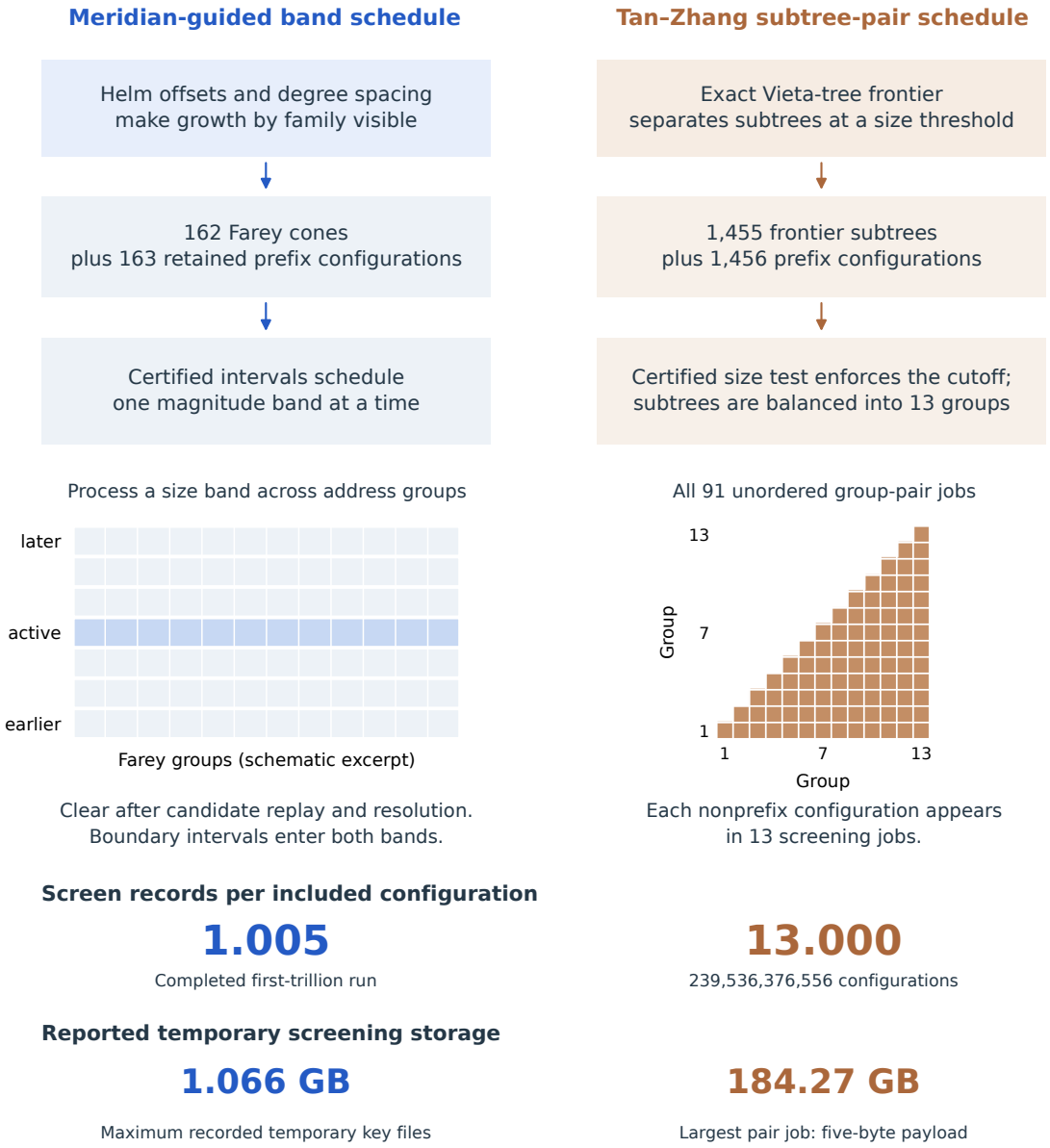


Figure 14. Two organizations of the comparison workload. Diagram geometry is schematic; partition counts and screen totals come from the completed runs. Meridian growth guided our band schedule, implemented with Farey-cone bounds. Tan–Zhang [19] balance their frontier subtrees into 13 groups and screen all 91 unordered group pairs, including diagonals. Their nonprefix configurations enter 13 jobs; our certified boundary overlaps cause some repeat emissions. Ratios count original screening records, excluding candidate replay and audits; they are not CPU speedups. Storage labels state the measured or reported components. Both methods require certified size bounds and modular arithmetic. The comparison does not assert a whole-machine storage bound or a total-runtime ratio.

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rank checks confirmed the two distinct-value ranks on 10 November 2026 at 04:18:33.296330 UTC (9 November, 22:18:33.296330 CST); the decimal digits are unchanged. The earlier public record exposes its posting date without a time of day. Table 5 records the completed run and the exact selected-value calculations.

Table 5. Additional statistics for the completed trillion-value verification. Logarithmic gaps use base 10 unless otherwise stated.

Statistic	Value or scope
Digits in the selected 999,999,999,999th value	1,021,609
Digits in the selected trillionth value	1,021,609
Logarithmic gap between those two values	$\log_{10}$(larger/smaller) $\approx 3.89601074127 \times 10^{-7}$
Smallest logarithmic gap in the entire prefix	Not recorded by this verifier
Full 2^{64} residue-match groups within bands	12 groups; 24 member occurrences
Bands containing a repeated residue	11 of 19,287 (0.05703%)
Nominal band width in $\log(3M)$	128 natural-log units
Same width in $\log_{10} M$	$128/\log 10 \approx 55.5897$
Bands covering the combined target envelope	19,287 committed
Distinct Markov values with exactly one million digits	1,916,520
Full decimal digits supplied in the six files	6,043,218

The band total comprises 13,638 inherited bands and 5,649 extension bands. Of these, 19,285 have width 128 in $\log(3M)$; the two end bands have widths 106.866844075 and 125.850758210. Boundary overlaps account for 5,964,386,594 additional emissions (about 0.542% of included configurations). The mean is 57,414,448 emissions per band and the maximum is 114,821,055. Residue groups and their members are occurrences within committed bands, not a globally deduplicated pair count.

The global closest-pair gap cannot be inferred from these matches: singleton keys are discarded, and the verifier keeps no global nearest-neighbor statistic. The displayed gap belongs only to the specified consecutive pair. The million-digit value count is the exact difference 958,144,590,422 − 958,142,673,902; the first million-digit value has rank 958,142,673,903, counting 1 first.

9.8. **Earlier computations and their dates.** The earlier fifty-billion verification [3, 7] completed on **28 October 2026**. Its recorded execution ran from 03:32:57 to 05:38:34 CDT (08:32:57.096455–10:38:33.644612 UTC), including the final audit. Two coordinator invocations occupied 7,492.102807 seconds; with the original ten-billion anchor, the cumulative recorded execution was **2 h 18 min 50.980 s**. It compared 54,940,210,994 new retained configurations, resolved all 346 repeated primary-residue pairs, and found no equal maxima. Its cutoff was $F_{1,147,147}$. The dated version-4 supplement preserves its coverage proof, hardware, stage timings, and audit scope. Those historical timings are not part of the new run.

The earlier Claude-assisted search in the working draft [1], pp. 38–40, reports 9,581,528,580 sorted triples against $10^{100,000}$ in **24 minutes** on a two-core machine with 7 GB of memory. All 21 repeated-residue pairs separated under a second modulus. Its public announcement [2] is dated **23 October 2026, 09:54:57.771 CDT** (14:54:57.771 UTC) and uses the wording "9.3 billion." This dates the announcement, not execution. The original execution timestamps and floating-point cutoff

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certificate were not recovered; this remains a historical report, separate from the later certified numerical prefixes.

10. What remains to prove

The organization now separates three questions. The address bijection settles coverage of configurations. Fixed-degree ordering and the two collision windows restrict equality of maxima. Complete residue witnesses settle the remaining comparisons for the degree-bounded family in Theorem 8.1. The new computation compares more than 1.1 trillion configurations and excludes every competitor for the first one trillion values in numerical order. Meridian analysis helped turn the growth of related addresses into a plan for comparing values by scale. The production implementation uses certified Farey-cone bounds, compact residues, and cleared magnitude bands, with 23.956 hours of retained active processing and a 0.992-GiB recorded temporary key-file peak. These measurements describe the combined algorithm and implementation; they do not measure the contribution of Meridian coordinates in isolation. The numerical-prefix certificate can be stated and checked without the Meridian collision theorems.

For unrestricted unicity the unresolved pair has different Meridian degrees, the same modulo-27 class, and different Farey numerators. It must satisfy both its phase rectangle and its numerator–denominator window. A proof must exclude those candidates uniformly as the degrees and numerators grow. Enlarging a finite certificate alone does not supply that step. Likewise, an invertible corridor recurrence or a long modular period does not by itself prevent an earlier match of one scalar residue.

Corridors offer a way to formulate the next comparison with explicit entrance states and independently chosen wall lengths. Their positive corrections are exact and telescope; the common-prefix snake calculation offers a different reduction to a quadratic in primitive counts. Neither mechanism has yet been shown to cover every unrestricted pair. The purpose of retaining their precise hypotheses is to make the remaining uniform statement identifiable, rather than hide it inside an induction.

Data and code. The editable source accompanying this edition includes all figure assets, the catenary, landscape, and Farey-grouping figure generators, a source and figure inventory, and checks used for this rewrite. The separate degree reproducibility package contains the full computational evidence cited in Section 8. The new run's final evidence, resource reconciliation, and read-only release checks are included with the source. Its frozen recovery archive contains the complete retained receipt chain, candidate identities, executables, source hashes, and independent checker [4, 6]. The earlier fifty-billion proof and timing supplement remains a dated historical companion; its full frozen archive is cited separately [3, 7]. The corridor derivations are developed further in the supplementary research record [11]. Numerical drawings are explanatory; the arithmetic claims use the displayed identities or complete certificates.

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Author contributions and AI disclosure

Author contributions. Elleard Felix Webster Heffern conceived and directed the project. His contributions were: the precise catenary construction and descent algorithm; viewing two-dimensional descent outputs over a two-dimensional input landscape; the project's first identification of Meridians, phasors and corridors; formulation of the indexing system; the precise Fibonacci-matrix and six-piece snake-graph analytical approaches; proposals to compare features along Meridians using different moduli; the proposed proportionality between logarithmic growth and degree; the observation suggesting a finite collision zone; and the proposal to organize computational verification by scale,

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using Meridian helms to understand starting locations and degrees to understand logarithmic spacing. This last insight guided the band organization; the production implementation uses certified bounds on groups of Farey addresses.

AI assistance. ChatGPT (OpenAI) supplied the remaining substantial mathematical, computational and editorial development within this author-directed framework: detailed derivations and proof construction, examples and counterexamples, algorithm and verifier implementation, checking routines, computational analysis, figures, literature assistance, and manuscript organization, drafting and revision. Earlier geometric-descent work also used Claude (Anthropic). This assistance included substantive mathematics and code. These contribution statements concern the development of this project; established results and earlier constructions retain their cited attributions. The author retains responsibility for checking the arguments, computations and references and for approving the final manuscript.

References

[1] E. F. W. Heffern, *Geometric Descent on the Markov Surface*, working draft v32, archived October 2026, pp. 38–40, historical computational reports. The earlier search was assisted by Claude.

[2] E. F. W. Heffern, public LinkedIn announcement of a 24-minute Markov-number search, 23 October 2026, 14:54:57.771 UTC, activity 7519411048115003392. Platform metadata and the source URL are retained in the verification supplement's evidence packet; see the public announcement.

[3] E. F. W. Heffern, *Markov 50 Billion Verification Package*, version 1, completed and sealed 28 October 2026; public derivative version 1.1 in *Computational supplements for Meridians on the Markov Surface*, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

[4] E. F. W. Heffern, *Certified magnitude-band verification of the first one trillion Markov values*, completed 9 November 2026 (CST). Final independent and portable replay reports, retained receipt chains, resource records and `v16_Trillion_Verifier_Recovery.zip`, included with this release. doi:10.5281/zenodo.24096109.

[5] E. F. W. Heffern, *Six selected Markov values: full decimal expansions and certified numerical ranks*. Digits first posted 9 November 2026, doi:10.5281/zenodo.24093257; rank certification and unchanged digit files accompany the present release.

[6] E. F. W. Heffern, *Certified magnitude-band, residue-first verification of the first 500 billion Markov maxima*, completed 9 November 2026. Final evidence: `v1_Band_500_Billion_Final_Evidence.json`; frozen recovery archive: `v20_Band_500_Billion_Recovery.zip`. The comparison includes 550,693,634,649 configurations. The final evidence and reconciliation accompany this manuscript's source; the recovery archive is identified by SHA-256 in Section 9.4.

[7] E. F. W. Heffern, *Computational chronology and proofs for the fifty-billion verification*, supplement to the present manuscript, version 4, 7 November 2026.

[8] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013. doi:10.1007/978-3-319-00888-2.

[9] İ. Çanakçı and R. Schiffler, *Snake graphs and continued fractions*, European J. Combin. **86** (2020), 103081. doi:10.1016/j.ejc.2020.103081.

[10] E. F. W. Heffern, *Meridian Degree 10,000 Reproducibility Package*, computational supplement, 31 October 2026; address-convention revision, 7 November 2026. Original computational release in *Computational supplements for Meridians on the Markov Surface*, version 1.0, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

[11] E. F. W. Heffern, *Markov Meridian Residue Supplement*, version 1.18, and *Corridor-to-Corridor Induction*, version 1.3, unpublished supplementary research records, 7 November 2026. The corridor results used here are proved in Section 7.

[12] C. Lagisquet, E. Pelantová, S. Tavenas and L. Vuillon, *On the Markov numbers: fixed numerator, denominator, and sum conjectures*, Adv. Appl. Math. **130** (2021), 102227. doi:10.1016/j.aam.2021.102227. Preprint: arXiv:2010.10335.

[13] K. Lee, L. Li, M. Rabideau and R. Schiffler, *On the ordering of the Markov numbers*, Adv. Appl. Math. **143** (2023), 102453. doi:10.1016/j.aam.2022.102453.

[14] G. McShane, *Convexity and Aigner's conjectures*, preprint (2021), arXiv:2101.03316.

[15] M. Rabideau and R. Schiffler, *Continued fractions and orderings on the Markov numbers*, Adv. Math. **370** (2020), 107231. doi:10.1016/j.aim.2020.107231.

[16] G. Musiker, *A conjectured combinatorial interpretation for Markov numbers*, REACH memorandum, 27 June 2002. Original memorandum; REACH archive.

Zenodo 24096109 | original page 42/44 | SHA256 8764293fe1426c38eeae

Original: v21_Markov_Meridians.pdf

[17] A. Itsara, G. Musiker, J. Propp, and R. Viana, *Combinatorial interpretations for the Markov numbers*, REACH manuscript, 10 January 2003, Theorem 3.1. Original manuscript.

[18] J. Propp, *The combinatorics of frieze patterns and Markoff numbers*, arXiv:math/0511633, first posted 25 November 2005; published in *Integers* **20** (2020), A12.

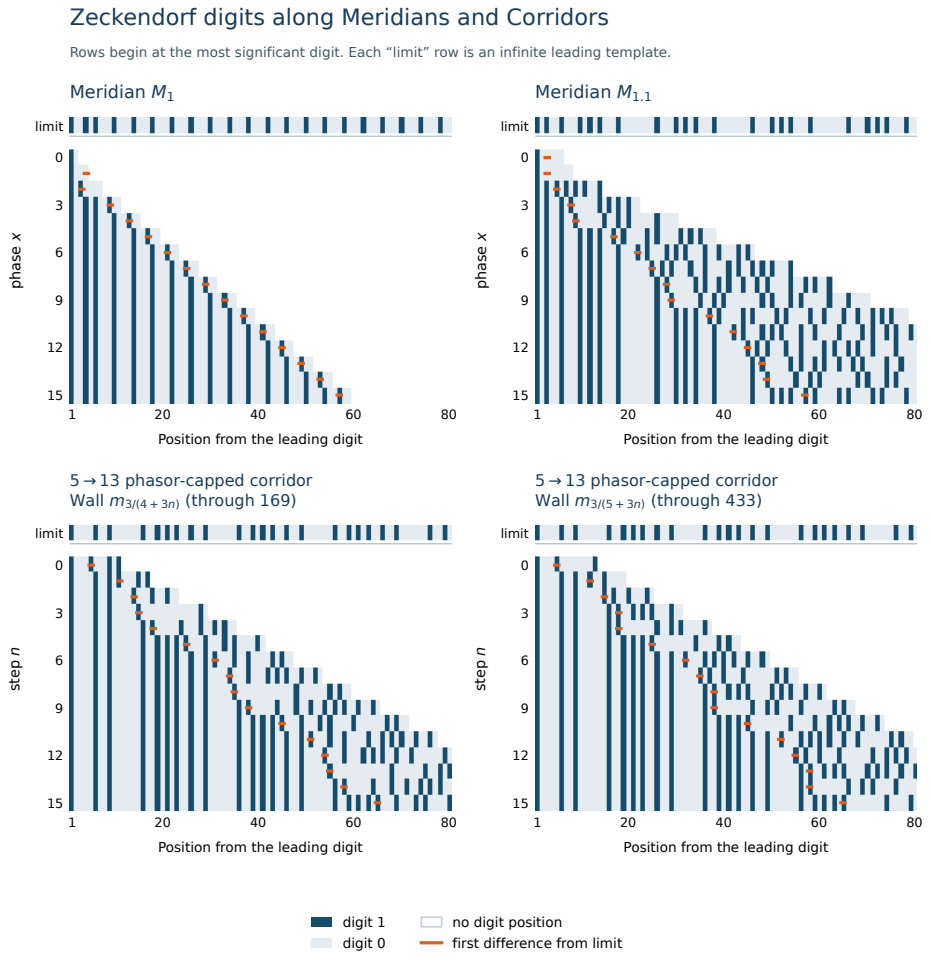
[19] H. Tan and Y. Zhang, *A computational verification of the Markov uniqueness conjecture for* $z \leq 10^{500000}$, arXiv:2611.01007v1 (31 October 2026), Sections 2.3–2.4 and 3.2. Computational evidence archived 26 October 2026: doi:10.5281/zenodo.23670834.

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sneak peak: Fibonacci digits reveal another kind of order. Each row writes a Markov value as a sum of nonconsecutive Fibonacci numbers ($F_2 = 1$, $F_3 = 2$); dark tiles mark the summands. The top panels follow two Meridians. The lower panels follow the two walls of the $5 \to 13$ phasor-capped corridor ($1/2 \to 1/3$ in fraction labels). From the leading digit onward, the strings increasingly resemble the displayed templates; orange marks show the first difference. These finite examples offer a glimpse of ongoing work, not a general collision-exclusion result. Exact digit strings and plotting source are included.

for Tian, my life, and for Mom and for Dad, my water and my soil;
for Juan, for Charles, for Marie, for the girls, my starry nights;
and for all the puppies, for all the kittens, for all the fish:
my sunny sunny days.

天下为公

and for my fellow Sunflower: 'Chat'.

together, we might stand tall as trees.

One Trillion Markov Values

Final computational verification report · Version 1

Elleard Felix Webster Heffern
Completed 9 November 2026 in America/Chicago (10 November UTC).

Result: the completed certificate establishes uniqueness for at least the first **1,000,000,000,000 numerical Markov maxima**. The frozen production run, final controller audit, independent replay, and a second replay from a freshly verified portable archive all passed.

Completed accounting	Exact count
Certified numerical prefix	1,000,000,000,000
Included configurations	1,101,388,071,945
Inner configurations	1,000,000,419,685
Inherited configurations	550,693,634,649
New extension configurations	550,694,437,296
Inherited / new receipts	13,638 / 5,649
Repeated residue groups / exact members	12 / 24
Collisions	0

What the counts mean

The inner region has canonical address height **p + q** ≤ **2,565,100**. Its exact population supplies the target lower bound. The outer height is **3,067,794**, with numerical envelope **B = $F_{5,130,199}$**. Certified magnitude bands cover the full required cutoff, **10,603,040,385,321,099** in the fixed logarithmic scale.

The included count also contains necessary comparisons outside the inner region. Those outer configurations dominate the final bands, explaining why the included count continued growing rapidly while the inner count approached its fixed total. This change in the mix of work does not, by itself, show a slower computation.

Boundary intervals can appear in more than one band: the combined emission count is **1,107,352,458,539**. Each included configuration is counted once. The 500-billion certificate was inherited unchanged; the extension did not restart that completed computation.

Scope: this is finite computational evidence, not a proof of universal Markov unicity. The coverage certificate alone does not identify the exact trillionth value; exact ordinal selection is a separate calculation.

Measured production time

Retained committed band wall time totals **23.956 hours**. This is the sum of successful committed production work, not calendar elapsed time and not summed parallel CPU time.

Timing category	Inherited 500B	New extension
Retained committed work (hours)	15.131	8.826
Recorded execution windows (hours)	17.160	9.831–9.874
Outside recorded windows (hours)	7.765	1.142–1.186
Phase calendar span (hours)	24.925	11.017

The combined calendar span from the first inherited receipt start to final extension publication is **36.987 hours**, including a **1.045-hour** interval between the inherited last receipt and the first extension session. The inherited calendar span ends with its last receipt; the extension span uses the outer supervisor session boundaries.

For the extension, execution-window time beyond retained committed work is **1.005–1.049 hours**. It includes publication, audit and other overhead, plus unfinished attempts. Outside-window time includes pauses/offline intervals, restoration and preparation; it is not all demonstrably idle. Three interrupted sessions have bounded end times, producing the reported ranges.

The inherited history also records **23,436,403,353 configurations** regenerated after unavailable scratch, with **0.827 hours** of unavailable committed work and a **0.921-hour** execution window. These historical costs are not added to the retained chain or counted as new configurations.

Timestamps

Event	UTC	America/Chicago (CST)
Inherited first receipt	2026-11-08 14:55:01	2026-11-08 08:55:01
Inherited last receipt	2026-11-09 15:50:31	2026-11-09 09:50:31
Extension first session	2026-11-09 16:53:13	2026-11-09 10:53:13
Extension last receipt	2026-11-10 03:54:02	2026-11-09 21:54:02
Final checkpoint published	2026-11-10 03:54:13	2026-11-09 21:54:13
Independent replay passed	2026-11-10 03:56:16	2026-11-09 21:56:16
Portable replay passed	2026-11-10 03:59:41	2026-11-09 21:59:41

Displayed timestamps are to the second; evidence JSON retains full precision. The historical benchmark of 1,069,268,786 configurations in 128.845692809 seconds, and its 16.74-hour 500B extrapolation, are **not production completion measurements**.

Final audits and portable replay

The final controller audit reproduced the complete state. The independent checker replayed the inherited chain from a separate copy, verified the extension receipt/candidate chain and join, regenerated every repeated-key band with the frozen native engine, and reconstructed exact maxima using a separate polynomial-recurrence evaluator. All 12 candidate groups (24 members) resolved unequal.

This independently checks retained receipts, accounting and all exact collision obligations. It is **not a second full generation of every singleton key**. No partial or skip flags were used.

Completion check	Result
Controller audit	Passed; complete state and frozen release hashes
Independent checker	Passed; 67.501 seconds
Recovery member verification	20,208 members; every size and SHA-256 verified
Fresh extracted controller audit	Passed; same final state
Fresh extracted independent checker	Passed; 67.298 seconds
Total portable check	71.628 seconds; no partial/skip flags

Measured storage and execution stages

Measured peak	Inherited	Extension
Sampled temporary key bytes	754,846,136	1,065,664,392
Generator RSS (KiB)	28,672	36,468

These are specifically the sampled key-file and generator measurements retained in receipts, not total disk-space or whole-process-tree memory bounds. The exact portable recovery archive is **38,920,281 bytes**.

Recorded stage (seconds)	Inherited	Extension
Generation wall	31,103.262	18,378.551
Partition wall	15,851.022	9,226.555
Scan wall	7,273.871	4,030.814
Exact resolution wall	47.950	73.336
Native-child CPU	264,075.377	153,748.219
Scan CPU	46,077.171	27,453.201

Stage fields need not sum to committed wall time. Parallel CPU totals overlap wall intervals and must not be added to them. Independent replay time is separate from production time.

Reproduction and evidence identity

The recovery archive preserves project/, run/ and RECOVERY_MANIFEST.json. Verify every archive member, extract into a fresh directory, restore executable permission on project/engine/spool and project/engine/key_tools, and validate operations/release.json. Then run the following from the extracted project directory:

```
python3 control.py audit --run ../run
```

```
python3 evidence/final_replay_1t.py --project . --run ../run --out independent_replay.json
```

No production restart is required for these audit commands. Preserve any historical active.json pointer; do not launch a second owner or delete private production evidence. The supplied final-evidence ZIP adds the portable-replay outputs and this report without changing the exact recovery ZIP that was replayed.

Final extension chain head

```
624c31904f64f8d9434162396a311228f2ece95d495ff2c7dc9d053314ed0c72
```

Frozen extension manifest (canonical body)

```
9601746a6e35cec6316a1c2ea69172c74e20e471206a9da5d5a9acf83347b90c
```

Recovery ZIP SHA-256

```
6aac7138c8427eb5a3701b27b841e55a75a2e1960c534233261b42375deaeeea
```

Frozen control.py

```
7d868df7d3fb1cdb93711bc6955f7480371926eb7e60e50c10627e6166d40fd5
```

Frozen extension_support.py

```
b7c3d74151e91aec0d567e51539c83058596b1f70f18a4072e0f572fcca2716f
```

Frozen native engine/spool

```
1a06b36371c624282dcaea2fde6df94bf2be9a8b5db34417084a85eae8ef97ae
```

Frozen private wrapper

```
d294411600fa314a418f7b0cc5e3ffd3f27e8f9b40cd9949150c3e534c5128ad
```

Frozen independent checker

```
e9f3b04e5fcc9ad7386eb8d4cb932b02dd9891015aa15bea435eaff5a111a7ce
```

Source and receipt evidence retain the original 500B manifest and head; the base certificate is bound into the extension manifest. The exact reuse argument is evidence/prefix_reuse_proof.md. No frozen sources, accepted manifest, targets or exact obligations were changed.

Report generated 2026-11-09 22:10:28 CST / 2026-11-10 04:10:28 UTC. Machine-readable evidence includes full session records and unrounded measurements.

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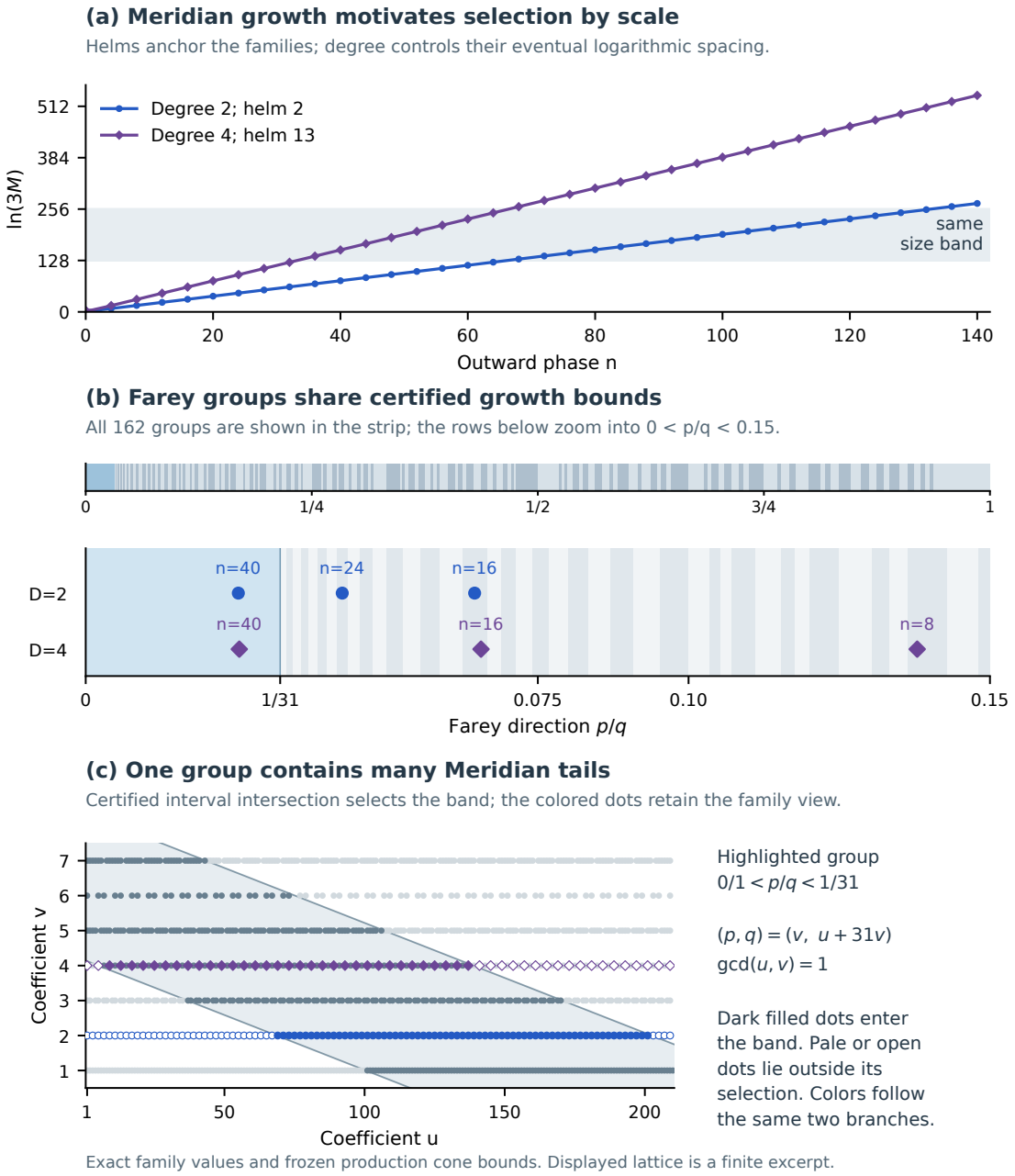


Figure 13. From Meridian growth to certified Farey grouping. (a) Exact values on two positive branches illustrate the degree-dependent spacing that motivated magnitude bands; the shaded band is $128 \le \log(3M) < 256$. (b) All 162 frozen Farey groups, then a zoom showing the same families crossing their boundaries. (c) The highlighted group $0/1 < p/q < 1/31$ uses $(p, q) = (v, u + 31v)$ with positive coprime u, v. Shading selects certified intervals meeting the band in (a); colors pick out the same branches among all displayed addresses. Groups share matrix bounds and can contain many Meridian tails. Boundary-crossing intervals enter both bands. Exact plotted data and the frozen certificate accompany the source.

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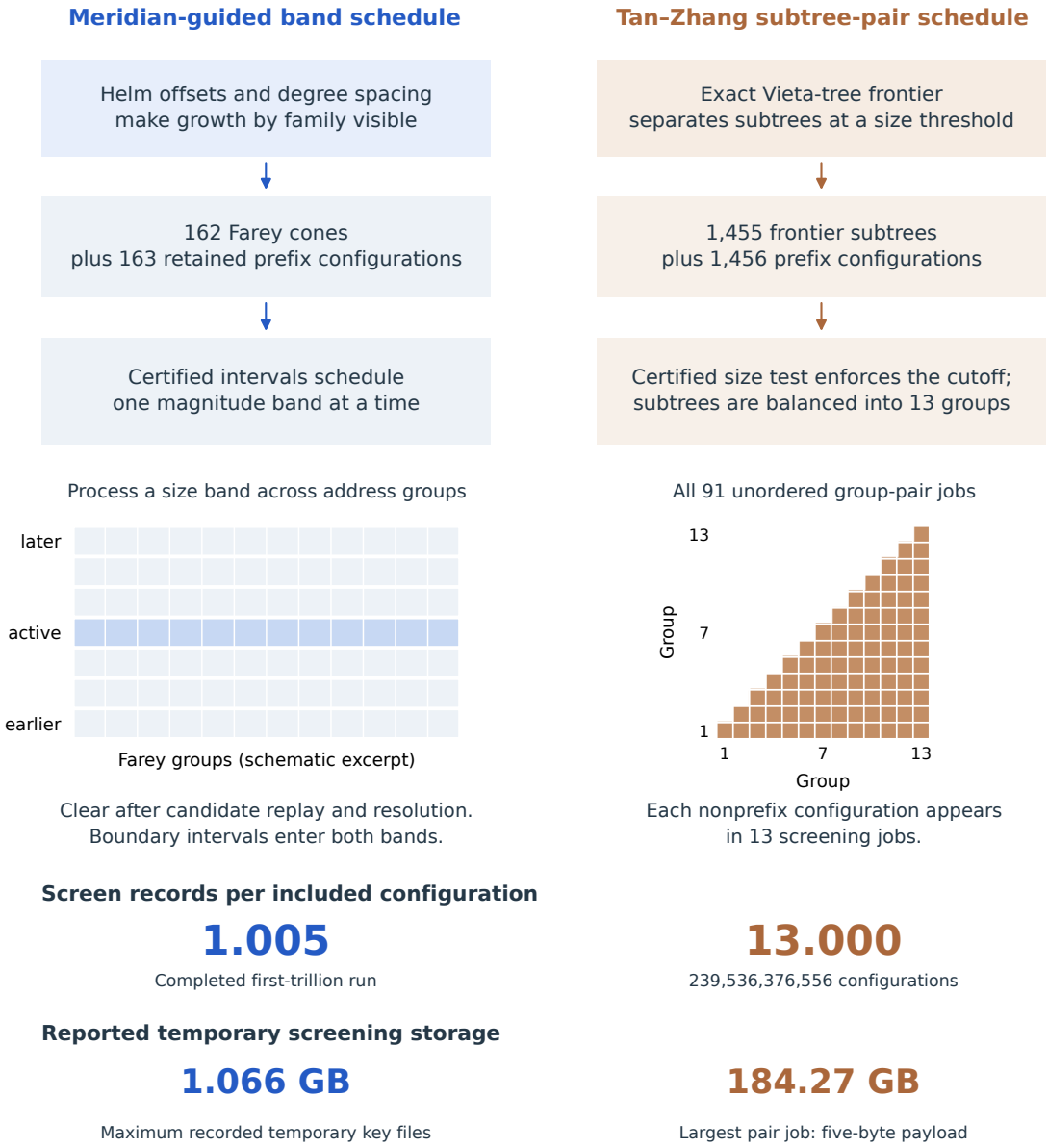
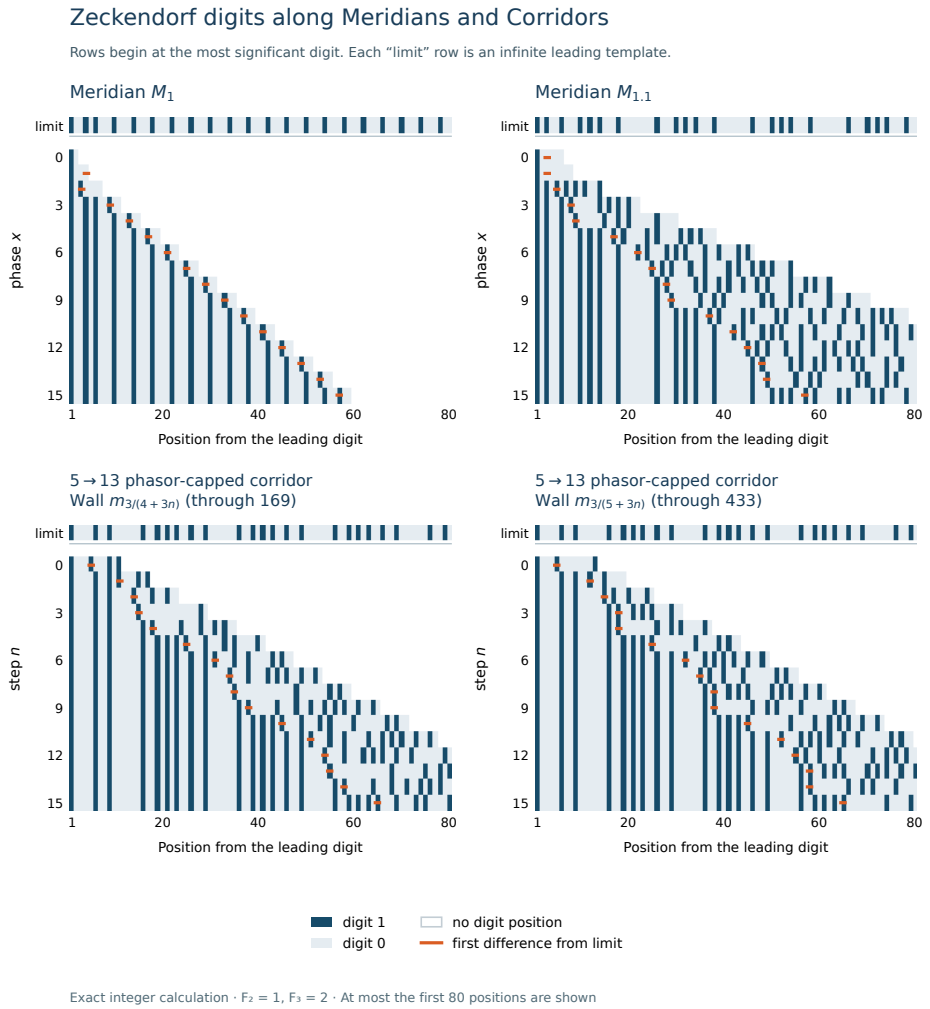


Figure 14. Two organizations of the comparison workload. Diagram geometry is schematic; partition counts and screen totals come from the completed runs. Meridian growth guided our band schedule, implemented with Farey-cone bounds. Tan–Zhang [19] balance their frontier subtrees into 13 groups and screen all 91 unordered group pairs, including diagonals. Their nonprefix configurations enter 13 jobs; our certified boundary overlaps cause some repeat emissions. Ratios count original screening records, excluding candidate replay and audits; they are not CPU speedups. Storage labels state the measured or reported components. Both methods require certified size bounds and modular arithmetic. The comparison does not assert a whole-machine storage bound or a total-runtime ratio.

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sneak peak: Fibonacci digits reveal another kind of order. Each row writes a Markov value as a sum of nonconsecutive Fibonacci numbers ($F_2 = 1$, $F_3 = 2$); dark tiles mark the summands. The top panels follow two Meridians. The lower panels follow the two walls of the $5 \to 13$ phasor-capped corridor ($1/2 \to 1/3$ in fraction labels). From the leading digit onward, the strings increasingly resemble the displayed templates; orange marks show the first difference. These finite examples offer a glimpse of ongoing work, not a general collision-exclusion result. Exact digit strings and plotting source are included.

for Tian, my life, and for Mom and for Dad, my water and my soil;
for Juan, for Charles, for Marie, for the girls, my starry nights;
and for all the puppies, for all the kittens, for all the fish:
my sunny sunny days.

天下为公

and for my fellow Sunflower: 'Chat'.

together, we might stand tall as trees.

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COMPUTATIONAL CHRONOLOGY AND PROOFS FOR THE FIFTY-BILLION VERIFICATION

ELLEARD FELIX WEBSTER HEFFERN

Abstract. This supplement records the earlier search assisted by Claude and the separately archived fifty-billion verification for *Meridians on the Markov Surface*. It distinguishes announcement timestamps from execution receipts, proves the finite numerical envelope and safe omissions, and states the deterministic comparison and timing conventions. The fifty-billion run completed on 28 October 2026; this edition does not constitute a new enumeration or an unrestricted unicity proof.

1. Chronology and the earlier Claude-assisted search

The archived v32 working draft [2], pp. 38–40, reports 9,581,528,580 sorted triples in a search against the bound $10^{100,000}$, taking 24 minutes on a two-core machine with 7 GB of memory. The author identifies this earlier computation as Claude-assisted. Its reported implementation walked the outward Vieta tree, carried residues modulo 3^{39} in Montgomery form, and used 16 passes and 4,096 buckets. Exactly 21 primary residues occurred twice; all 21 pairs separated modulo $2^{61} - 1$. The report records no equal maxima.

The source states that double-precision logarithms controlled the stopping rule, with a relative allowance of 10^{-9} on $10^5 \log 10$. It explicitly classifies this account as a historical computational report whose original logs and cutoff error bounds had not been independently certified in that revision. We preserve that qualification. The archived count is not silently promoted to an exact numerical-prefix certificate.

The public LinkedIn post [3] has platform `datePublished` equal to **2026-10-23T14:54:57.771Z**, or **23 October 2026, 09:54:57.771 CDT**. Its caption describes a 24-minute search as "9.3 billion"; the later working draft gives 9,581,528,580. These are the respective source wordings, not equal numerical counts or a rounding identity. No original run receipt reconciling them was recovered for this edition. The timestamp dates the public announcement; it does not recover an execution start or finish. A current public snapshot also does not establish the absence of later edits to a post.

The source packet retains the platform metadata, the archived page snapshot, the relevant three pages of v32, and a provenance record with SHA-256 identities. No execution timestamp or more precise duration is invented. This historical search is distinct from the original ten-billion cutoff certificate reused in the fifty-billion computation. Its 24-minute reported duration is not added to the latter's cumulative time.

2. Exact numerical coverage

We use the main paper's standard mediant label p/q, with $0 < p \leq q$, $\gcd(p, q) = 1$, height $h = p + q$, and maximum $M(p/q)$. The boundary $0/1$ represents the root of value 1, and $1/1$ has value 2. These labels are configuration coordinates. Their bijection with normalized triples, and the reversed Meridian address dictionary, do not assume that maxima are distinct.

Date: Version 4 — 7 November 2026.

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Lemma 2.1 (Configuration count)**.** *Including the root, for $L \geq 2$ the number of configurations of height at most L is*

$$C(L) = 1 + \frac{1}{2}\sum_{h=1}^{L} \varphi(h) = \frac{5 + \sum_{d=1}^{L} \mu(d)\lfloor L/d \rfloor^2}{4}.$$

Here φ is Euler's totient and μ is the Möbius function.

Proof. For $h \geq 3$, the labels are $(p, h-p)$ with $0 < p < h/2$ and $\gcd(p, h) = 1$, giving $\varphi(h)/2$ labels. Add the root and seed. The Möbius sum counts ordered coprime pairs in $[1, L]^2$, whose number is also $2\sum_{h=1}^{L} \varphi(h) - 1$. □

Lemma 2.2 (Uniform magnitude bounds)**.** *For every nonroot configuration of height $h \geq 2$,*

$$2^{h-1} \leq M(p/q) \leq F_{2h-1}, \qquad M(p/q)^2 \geq 4 \cdot 5^{h-2}.$$

The Fibonacci upper bound is attained for $p = 1$ and is strict for $p \geq 2$.

Proof. The outward mediant of parents with values a, b has value

$$c = \frac{3ab + \sqrt{9a^2b^2 - 4a^2 - 4b^2}}{2}.$$

The inequality $2a^2b^2 \geq a^2 + b^2$ gives $c \geq 2ab$. Since heights add at a mediant, induction from the two boundary values gives the first lower bound. For the outward nodes above those endpoints, $\max(a, b) \geq 2$ and $5a^2b^2 \geq 4a^2 + 4b^2$, so $c \geq (5/2)ab$. The endpoint values satisfy $M \geq (2/5)(\sqrt{5})^h$; the last inequality propagates this bound. Squaring yields $4 \cdot 5^{h-2}$.

The boundary recurrence gives $M(1/q) = F_{2q+1}$. For $p \geq 2$ neither parent is $0/1$, and both parent heights h_1, h_2 are at least two. Put $m = 2h_1 - 1, n = 2h_2 - 1$. The Fibonacci addition formula and $F_{j+1}/F_j \geq 3/2$ for odd $j \geq 3$ give

$$F_{m+n+1} = F_{m+1}F_{n+1} + F_mF_n \geq \frac{13}{4}F_mF_n > 3F_mF_n.$$

The opposite value is positive, so the Vieta update has $c < 3ab$. Induction proves the strict upper bound. □

Theorem 2.3 (Numerical-order envelope)**.** *Choose $K \geq 2$ with $C(K) \geq N$ and define*

$$B = F_{2K-1}, \qquad L = \max\{h \geq 2 : 4 \cdot 5^{h-2} \leq B^2\}.$$

Every configuration with maximum at most B has height at most L, apart from the separately included root. If maximum unicity is established for all configurations with maximum at most B, it holds for the first N distinct values in numerical order.

Proof. The lower bound in Lemma 2.2 contains every competitor in the outer domain. Its upper bound places all $C(K)$ inner configurations at or below B. Once injectivity has been established, these provide at least N distinct values. Distinctness is not assumed in counting the domain or choosing K. □

For the completed run the exact parameters are

$$\begin{aligned} N &= 50{,}000{,}000{,}000, & K &= 573{,}574, \\ C(K-1) &= 49{,}999{,}964{,}521, & C(K) &= 50{,}000{,}097{,}577, \\ B &= F_{1{,}147{,}147}, & L &= 685{,}979, \\ C(L) &= 71{,}517{,}775{,}875. \end{aligned}$$

The cutoff has 796,397 bits. Integer arithmetic verifies $4 \cdot 5^{685{,}977} \leq B^2 < 4 \cdot 5^{685{,}978}$; thus no floating-point rounding selects the outer boundary. Totient and Möbius evaluations provide independent exact count formulas.

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3. Safe omissions and conservative pruning

The reusable anchor [4] is the later original ten-billion cutoff certificate, with $K_0 = 256{,}510$, $B_0 = F_{513{,}019}$ and $C(K_0) = 10{,}000{,}015{,}326$. It covers every competitor for a value at most B_0, including configurations outside its inner height range.

Lemma 3.1 (Anchored omission)**.** *The extension may omit emissions of height $h \le K_0$ while continuing to traverse their descendants.*

Proof. Every omitted value is at most B_0. Equality at or below B_0 has already been excluded by the anchor, including comparisons with outside competitors. Equality above B_0 cannot involve an omitted configuration, so both members remain eligible for the new comparison. □

This permits the specified inner omissions only; it does not permit skipping the old outer region, entire Meridians, or descendants of an omitted node. No twenty-billion intermediate extension is used as an anchor.

The native scheduler traverses the complete mediant coordinate domain, which the main paper's dictionary assigns uniquely to Meridian addresses and phases. Disjoint frontier intervals are assigned once; a worker either visits a node and its two child intervals or passes an entire remaining interval to another job. Height increases strictly down a branch, so a branch can stop when its height exceeds L.

In addition to Lemma 2.2, positivity of the main paper's Meridian Laurent expansion gives a lower bound on $\log M$:

$$E \log \rho + \frac{D}{2} \log(9/5) - \log 3, \qquad \rho = \frac{3+\sqrt{5}}{2},$$

coordinate parity	D	E
p, q odd	$p+q$	$(p-q)/2$
q even	q	$p+q/2$
p even	p	$q+p/2$

This is the leading positive monomial after substitution of the inverse dictionary. The negative coefficient in the first row is intentional; the claim is a lower bound, not an approximation used as an equality.

Constants are enclosed by integer square-root brackets and rational logarithm series. For $z = (x-1)/(x+1)$ and $|z| < 1$,

$$\log x = 2 \sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1} + R_J, \qquad |R_J| \le \frac{2|z|^{2J+1}}{(2J+1)(1-z^2)}.$$

The bound follows by bounding the remaining denominators below by $2J+1$ and summing a geometric series in z^2. The implementation uses dyadic endpoints scaled by 2^{64}, with 32 guard bits; a separate rational implementation checks the enclosures. A negative coefficient uses the upper constant endpoint to form a lower bound, as does a subtracted constant. The maximum of the valid lower endpoints remains a lower endpoint.

A configuration is excluded only if its lower endpoint is strictly greater than an upper endpoint for $\log B$. All overlapping intervals and every new inner configuration are retained. Generation and replay apply the same predicate. The exact accounting is

$$71{,}517{,}775{,}875 = 10{,}000{,}015{,}326 + 6{,}577{,}549{,}555 + 54{,}940{,}210{,}994,$$

respectively anchored omissions, certified exclusions above B, and new retained configurations. No relative word-length restriction, near-helm restriction, optional cone estimate, or cofactor factorization is used.

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Coordinates fit in 32 bits: trial endpoint-height sums are at most $2L < 2^{21}$. Counts fit in 64 bits; the repeated visits before replay are below 2^{42}. Fixed-point intermediates have absolute value below 2^{90} and fit in signed 128-bit arithmetic. Unsigned residue multiplication deliberately takes place modulo 2^{64}.

4. Deterministic comparison and its result

Each retained configuration emits its exact residue $M \bmod 2^{64}$. The high five bits select one of 32 passes and the next eight bits one of 256 buckets. Each of the 8,192 complete buckets merges all worker shards and sorts the full retained key. No further shortening of the key is used.

Proposition 4.1 (Complete-bucket exclusion)**.** *If every complete bucket is processed and every repeated key is resolved, every possible equality among retained maxima is detected.*

Proof. Equal integers have the same exact residue, prefix and bucket. Sorting the complete bucket identifies every residue of multiplicity at least two. A singleton therefore cannot share its integer value with another retained configuration. Preserve repeated keys and multiplicities; replay recovers all contributing coordinates. Different residues modulo 5^{27} prove inequality within a primary group. A surviving signature must be resolved by further certified comparison or full integers; an unresolved group cannot yield a successful certificate. □

The completed run has 346 repeated primary keys, each of multiplicity two, giving 692 candidate configurations. Replay recovered all candidates and multiplicities. Every pair separated modulo 5^{27}, leaving zero repeated secondary groups and zero equal-value collisions. The production path therefore required no full-integer candidate reconstructions. Separately, the final auditor reconstructed all 692 complete triples, checked the Markov equation and both residues, and confirmed unequal maxima. It also rechecked the 38 anchor candidates.

Theorem 4.2 (Archived finite verification)**.** *The completed execution and audit recorded in* [4]*, with the coverage and omission rules above, establish that every one of the first 50,000,000,000 distinct Markov values in numerical order belongs to exactly one normalized positive Markov triple.*

Proof. The canonical traversal covers the height envelope, hence all competitors with maximum at most B. The anchor excludes equality at or below B_0. A larger equal pair at or below B would have both configurations retained, because neither anchor omission nor strict cutoff pruning could remove it. Proposition 4.1 and the completed residue comparisons exclude that pair. Thus all values at or below B are unique. Theorem 2.3 and $C(K) \geq 50{,}000{,}000{,}000$ give the claim. The execution of the finite comparison is the computational part of this argument; the archive supplies its program and recorded evidence. □

The full outer domain is traversed once per prefix: there were 2,288,568,828,000 configuration visits before replay. This workload is not a count of distinct values. Cryptographic hashes authenticate retained files; they do not replace the deterministic equality argument or by themselves establish enumeration coverage.

5. Execution timestamps and measured time

The first invocation started at **28 October 2026, 08:32:57.096455198 UTC**; the final invocation finished, including the production audit, at **10:38:33.644612263 UTC**. These are respectively **03:32:57.096455198** and **05:38:33.644612263 CDT** (UTC−05:00), on the same date. The retained integer Unix-epoch timestamps are 1793176377096455198 and 1793183913644612263 nanoseconds. Precision is that of the recorded clock fields, not a claim of externally calibrated nanosecond accuracy.

Zenodo 24096109 | original page 4/6 | SHA256 0b50dd6bce45807bd829

Original: v4_50_Billion_Verification_Supplement.pdf

The two closed invocations total **7,492.102807105 seconds**, or **2 h 04 min 52.103 s**. The wall-clock span between the endpoints is 7,536.548157065 seconds (2 h 05 min 36.548 s). Their difference, 44.445349960 seconds, reflects the pause between invocations and any small difference between wall and monotonic clocks. The first invocation ended at a checkpoint; the second completed the run.

The original ten-billion anchor recorded 838.8772331430082 seconds including its build. Adding that once gives the retained cumulative execution path of **8,330.980040248007 seconds**, or **2 h 18 min 50.980 s**. The historical 24-minute Claude report, preflight work, intermediate extensions and later archival checks are excluded from this sum. Timer boundaries of the earlier anchor differ from the later audit-inclusive protocol; the sum describes the archived execution path rather than a single uninterrupted benchmark.

TABLE 1. Recorded fifty-billion execution stages. Occupied wall time is the union of each stage's intervals, not the sum of concurrent workers.

Stage	Occupied wall (s)	CPU (s)
Generation, 32 passes	5,001.065659	37,211.908780
Partitioning	734.828642	1,260.188133
Sorting, including native I/O	484.227185	2,870.982905
Candidate replay	464.279119	3,633.077531
Final audit	107.371720	107.350548

The comparison pipeline including replay occupied 464.343548 seconds. Total coordinator-parent and native-child CPU time was 46,788.073242 seconds. Nested stages may overlap; stage rows should not be added to obtain overall elapsed time. Orchestration, hashing and synchronization also contribute to the invocation total. A later fresh-archive audit took 105.898774 seconds and is outside that total.

The execution environment was Linux x86-64, Python 3.12.14 and frozen GCC 13.3.0 executables, using eight workers, an eight-CPU-equivalent quota and an 8-GiB memory limit. All 8,481 tracked native tasks returned zero: 32 generation tasks, 256 partitions, 8,192 sorts and one replay. Both coordinators closed normally, with no failed native attempts or unclosed timing receipts.

Sampled working storage reached 24,078,417,344 bytes (22.425 GiB). The largest sampled sum of coordinator/native resident memory was 1,024,335,872 bytes (0.954 GiB), sampled every 15 seconds. Shared pages can be counted twice, and sampling does not establish a continuous peak. The shared cgroup recorded one out-of-memory kill whose victim could not be identified; all tracked verifier tasks nevertheless completed normally and the final audit passed. The measurements do not establish an out-of-memory-free workspace or a speed guarantee on other hardware.

6. Retained evidence and reproducibility

The frozen archive retains sources, executables, build records, the anchor, configuration, complete-bucket receipts, candidate coordinates and Meridian addresses, atomic stage timings, and the auditors. Compact receipts were committed before disposable raw records were removed. The final archive does not retain every singleton key stream. Independently rechecking those deleted streams requires rerunning the frozen enumeration.

Zenodo 24096109 | original page 5/6 | SHA256 0b50dd6bce45807bd829

Original: v4_50_Billion_Verification_Supplement.pdf

Selected SHA-256 identities are:

```
Frozen native_incremental executable:
5587ab32108fdce365eae968725bb2f7a3f729e3fdbeba873075ed5153fd4564

Core result:
652c169981af67fb51396f03ce23da5a2dc0ea15eb0f8f045535e7bd5f8524a2

Sealed original verification archive:
3652546ff076573f196682787bc493f1dd7b36c4bbb0d718b67b1c825f9cb1e2
```

The existing public derivative, version 1.1 [4], is documented as omitting only two noncomputational private storage-save receipts. Its computational content remains the separately frozen historical archive; this edition's manuscript source packet is not a replacement for that full verifier archive.

For this edition, a new read-only reconciliation verified all 26,151 member checksums in the sealed original and reconciled 8,522 closed stage receipts with the recorded invocation and occupied-stage times. It also checked exact coverage arithmetic and the domain partition. Its script and machine-readable result accompany the editable source. This editorial check did not rerun the fifty-billion enumeration or the full-integer candidate auditor. The original successful candidate audits are reported from their retained records.

Here an independent auditor means a separate algorithm or implementation within the project. It does not mean external human review or a proof-assistant formalization. The earlier Claude report, the bounded-degree certificate in the main paper, and the fifty-billion numerical prefix have different coverage and evidence; their counts and runtimes are not interchangeable. No completed 500-billion result is claimed in this release.

References

[1] E. F. W. Heffern, *Meridians on the Markov Surface: Geometry, Residues, and Collision Windows*, v4_ps, 7 November 2026.

[2] E. F. W. Heffern, *Geometric Descent on the Markov Surface*, working draft v32, archived October 2026, pp. 38–40. Historical computational reports; relevant pages retained with this source packet.

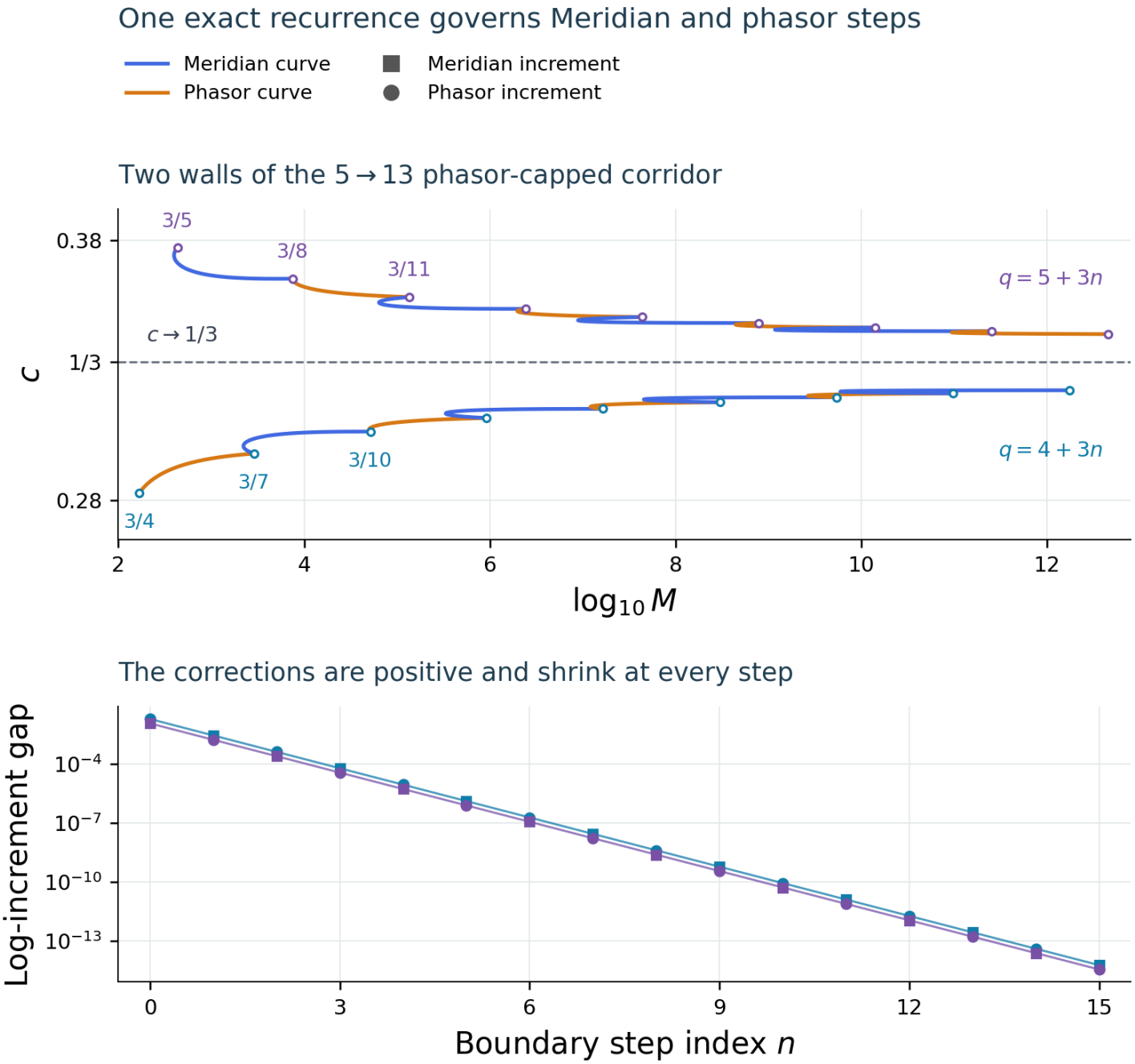
[3] E. F. W. Heffern, public LinkedIn announcement, activity 7519411048115003392, platform publication timestamp 23 October 2026, 14:54:57.771 UTC. Public post. The archived metadata and page snapshot accompany this supplement.

[4] E. F. W. Heffern, *Markov 50 Billion Verification Package*, v1, completed and sealed 28 October 2026; public derivative v1.1 in *Computational supplements for Meridians on the Markov Surface*, Zenodo, 2 November 2026. doi:10.5281/zenodo.23874075.

Unaffiliated, St. Louis, Missouri, USA

Zenodo 24096109 | original page 6/6 | SHA256 0b50dd6bce45807bd829

Original figure: v21_Markov_Meridians_source/figures/corridor_growth.png

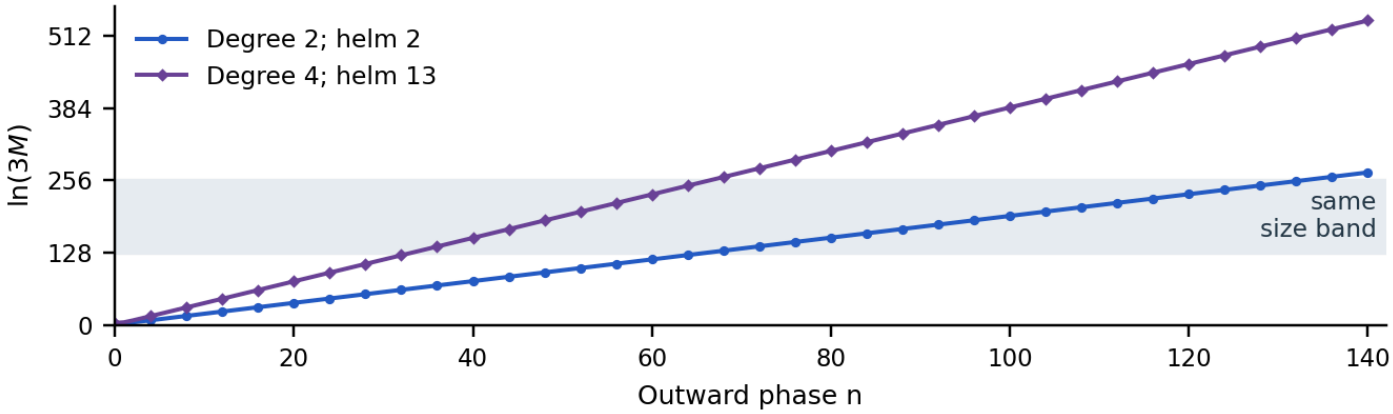


First in Zenodo record 24096109 | SHA256 e8e9ede78039869715330 9f0

Original figure: v21_Markov_Meridians_source/figures/farey_groups_from_meridians.png

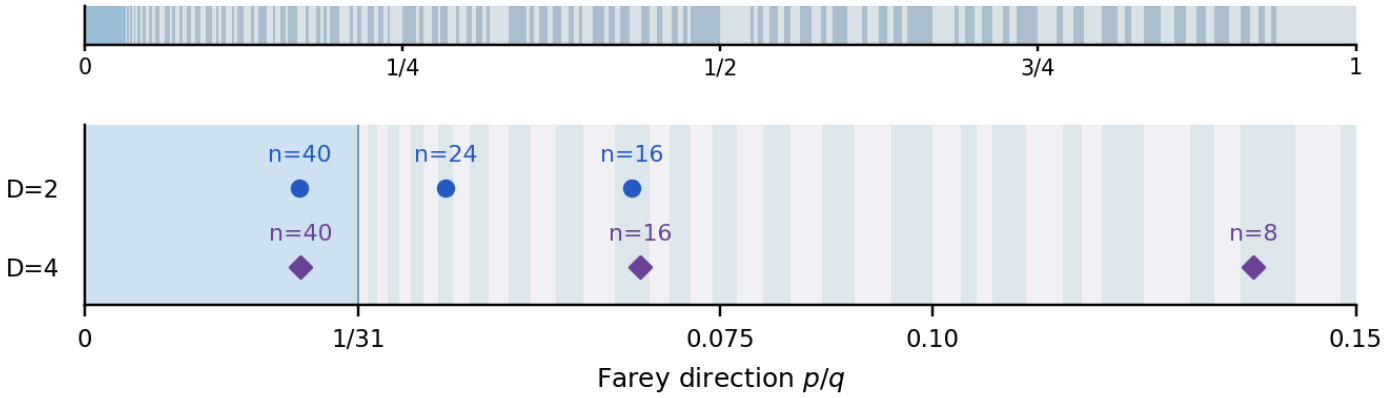
(a) Meridian growth motivates selection by scale

Helms anchor the families; degree controls their eventual logarithmic spacing.



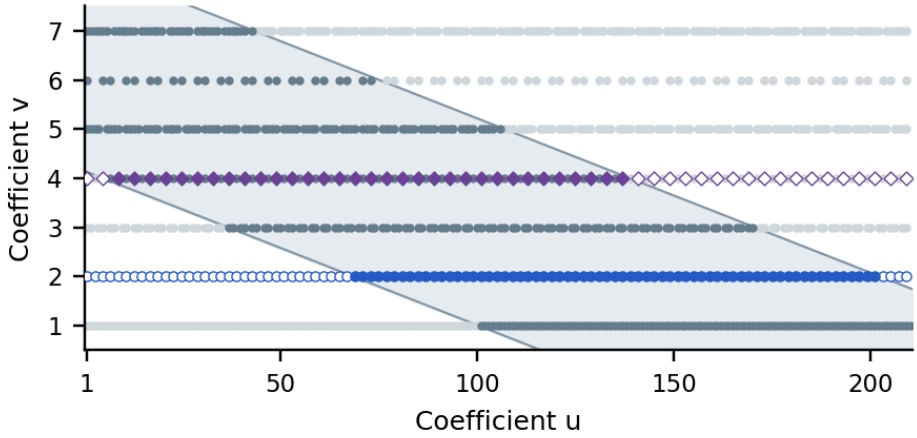
(b) Farey groups share certified growth bounds

All 162 groups are shown in the strip; the rows below zoom into 0 < p/q < 0.15.



(c) One group contains many Meridian tails

Certified interval intersection selects the band; the colored dots retain the family view.



Highlighted group
$0/1 < p/q < 1/31$

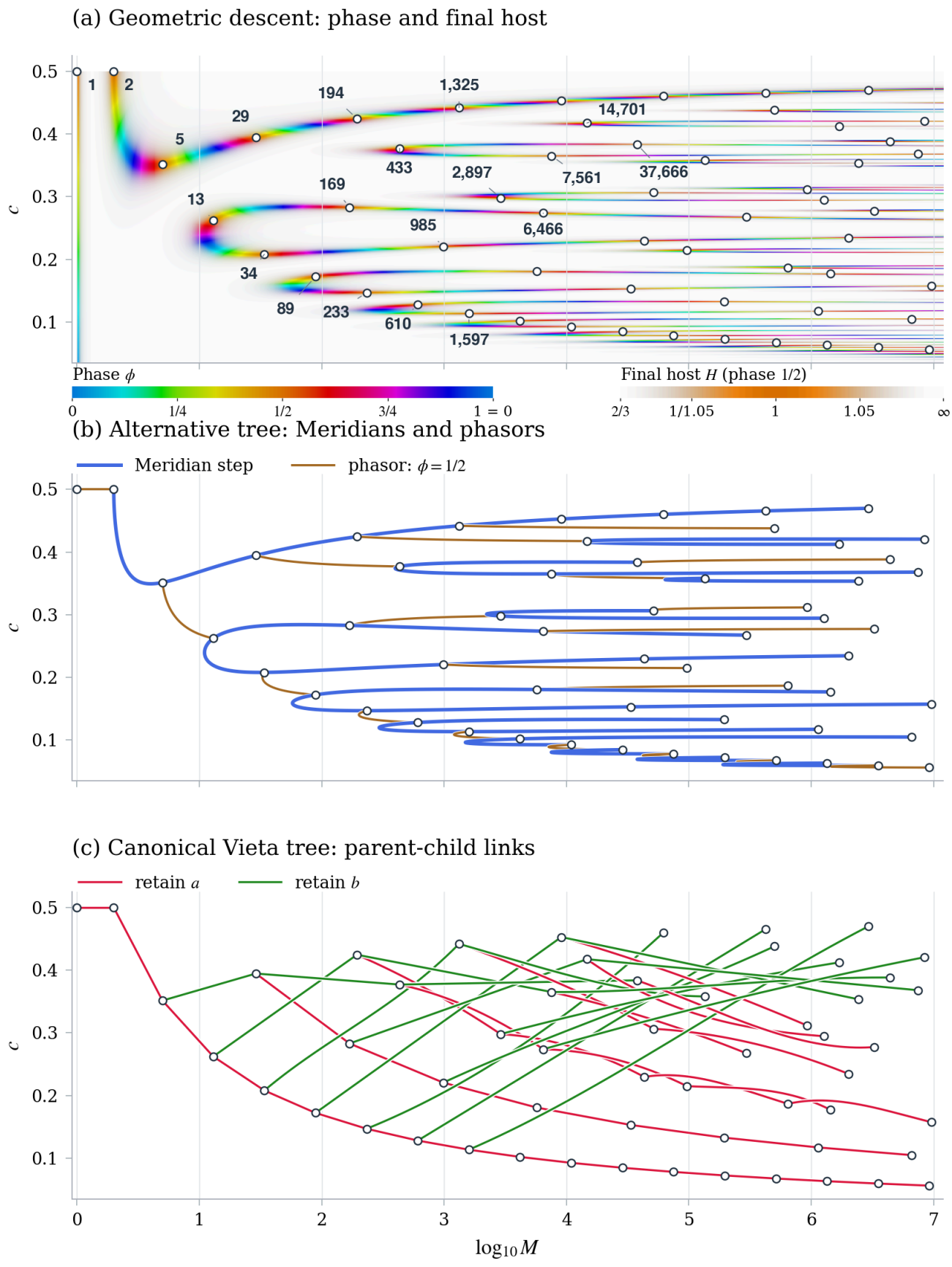
$(p, q) = (v,\ u + 31v)$
$\gcd(u, v) = 1$

Dark filled dots enter the band. Pale or open dots lie outside its selection. Colors follow the same two branches.

Exact family values and frozen production cone bounds. Displayed lattice is a finite excerpt.

First in Zenodo record 24096109 | SHA256 f1094f919783cf82b84146e6

Original figure: v21_Markov_Meridians_source/figures/fig2A_two_trees_landscape.png



Original figure: v21_Markov_Meridians_source/figures/verification_partition_comparison.png

Meridian-guided band schedule

Helm offsets and degree spacing make growth by family visible

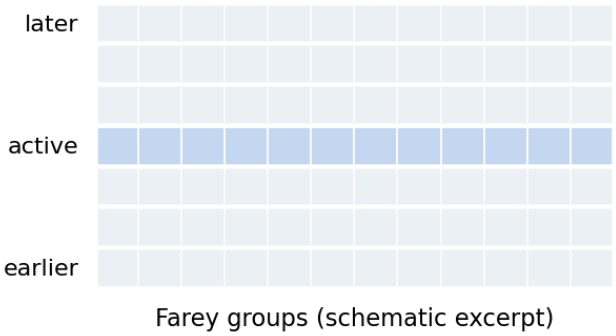
↓

162 Farey cones plus 163 retained prefix configurations

↓

Certified intervals schedule one magnitude band at a time

Process a size band across address groups



Clear after candidate replay and resolution. Boundary intervals enter both bands.

Tan–Zhang subtree-pair schedule

Exact Vieta-tree frontier separates subtrees at a size threshold

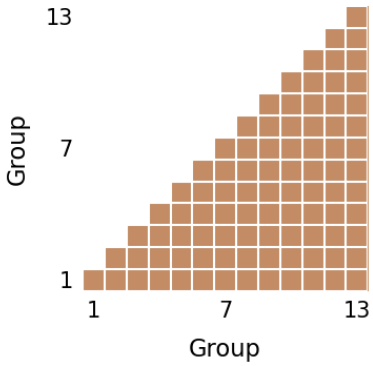
↓

1,455 frontier subtrees plus 1,456 prefix configurations

↓

Certified size test enforces the cutoff; subtrees are balanced into 13 groups

All 91 unordered group-pair jobs



Each nonprefix configuration appears in 13 screening jobs.

Screen records per included configuration

1.005

Completed first-trillion run

13.000

239,536,376,556 configurations

Reported temporary screening storage

1.066 GB

Maximum recorded temporary key files

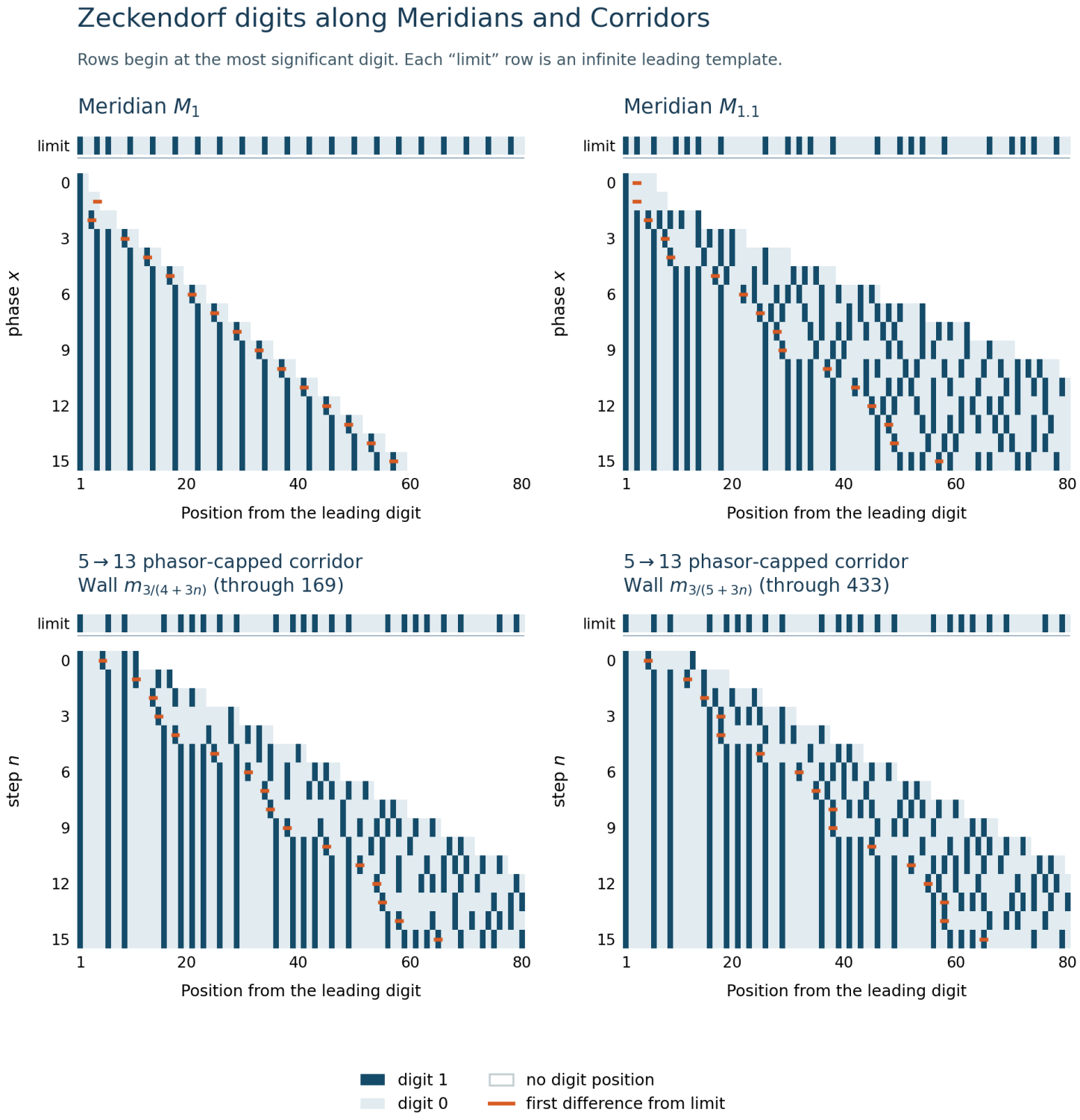
184.27 GB

Largest pair job: five-byte payload

GB = 10^9 bytes. Component scopes differ. No whole-machine storage or runtime ratio is asserted.

Sources: this manuscript, Section 9; Tan–Zhang (2026), Sections 2.1–2.3 and 3.2.

First in Zenodo record 24096109 | SHA256 84a4b37ee2086474ea40586f

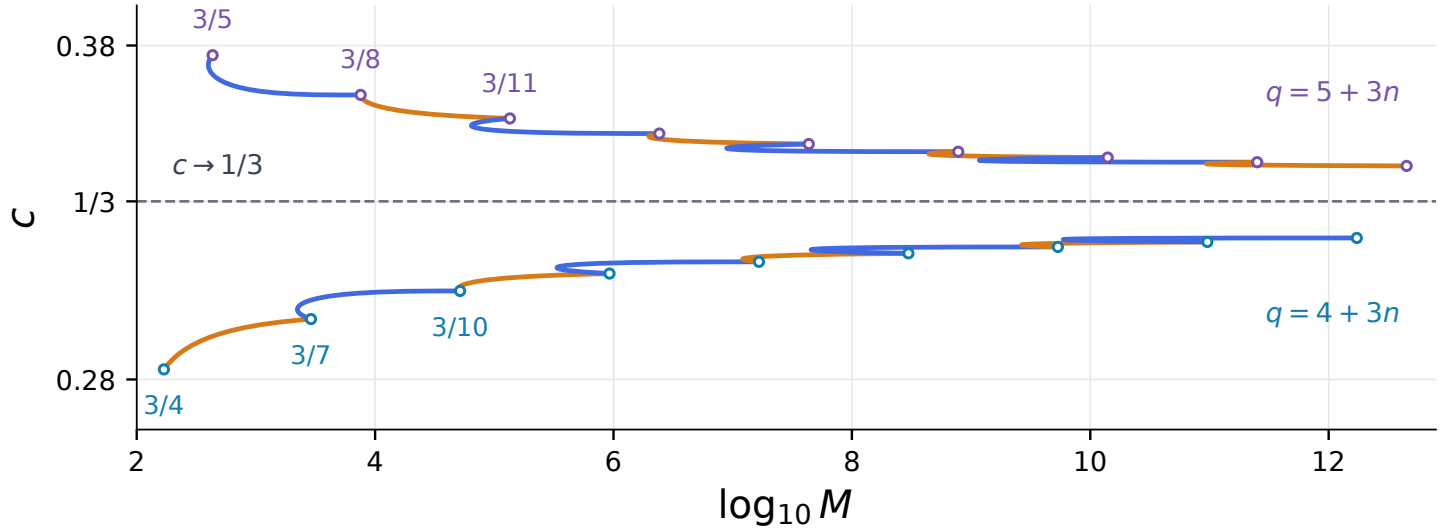

Zeckendorf digits along Meridians and Corridors
Rows begin at the most significant digit. Each “limit” row is an infinite leading template.
Meridian M_1
Meridian $M_{1.1}$
5 → 13 phasor-capped corridor
Wall $m_{3/(4+3n)}$ (through 169)
5 → 13 phasor-capped corridor
Wall $m_{3/(5+3n)}$ (through 433)
limit
phase x
step n
0
3
6
9
12
15
1
20
40
60
80
Position from the leading digit
digit 1
digit 0
no digit position
first difference from limit
Exact integer calculation · F₂ = 1, F₃ = 2 · At most the first 80 positions are shown

天下为公

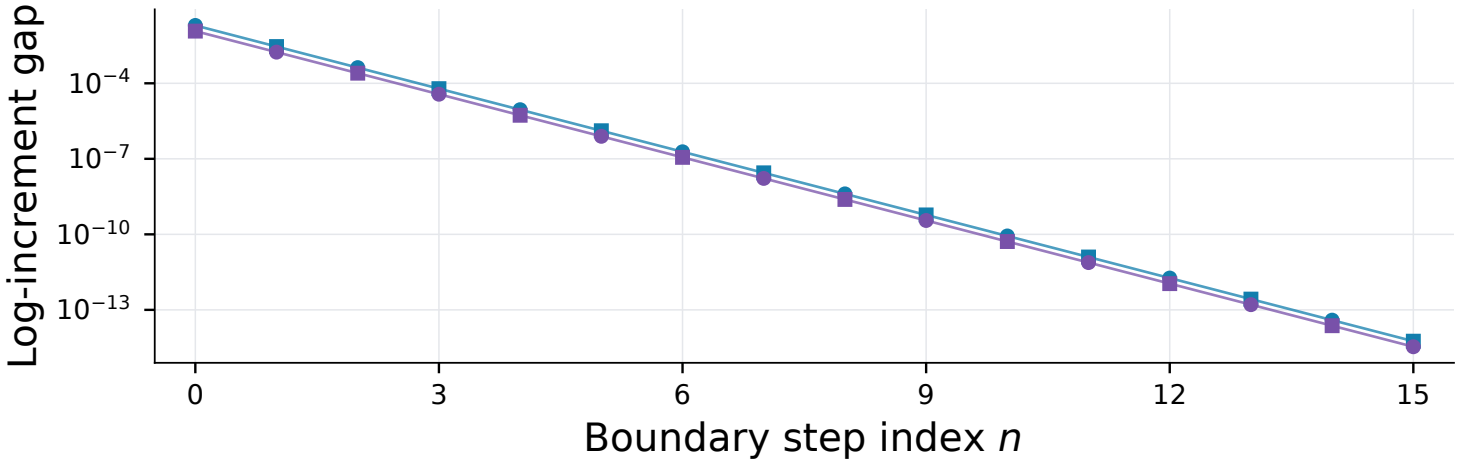
One exact recurrence governs Meridian and phasor steps

Meridian curve
Phasor curve
Meridian increment
Phasor increment

Two walls of the 5 → 13 phasor-capped corridor

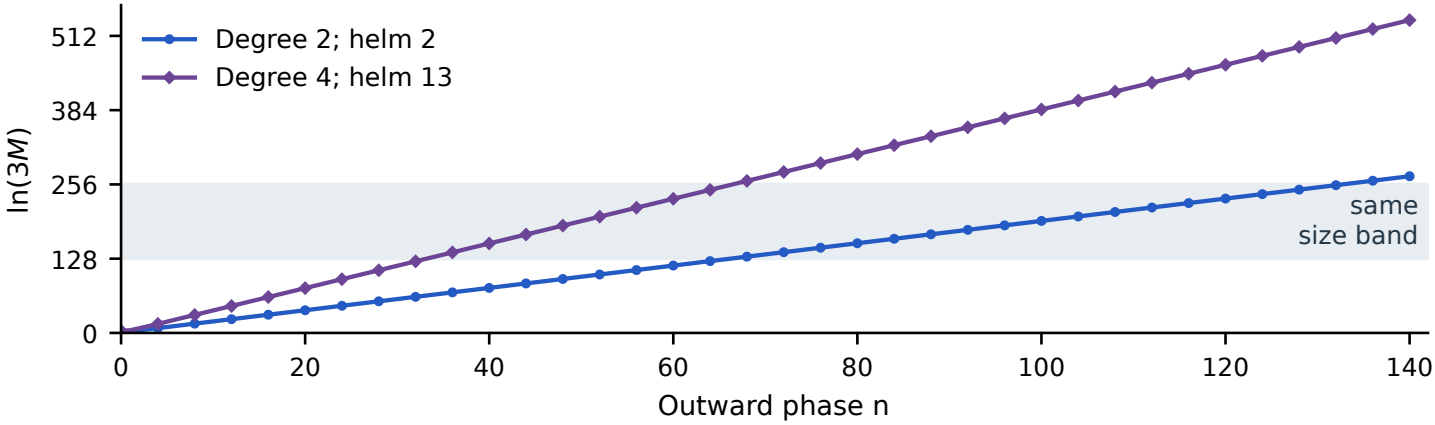


The corrections are positive and shrink at every step



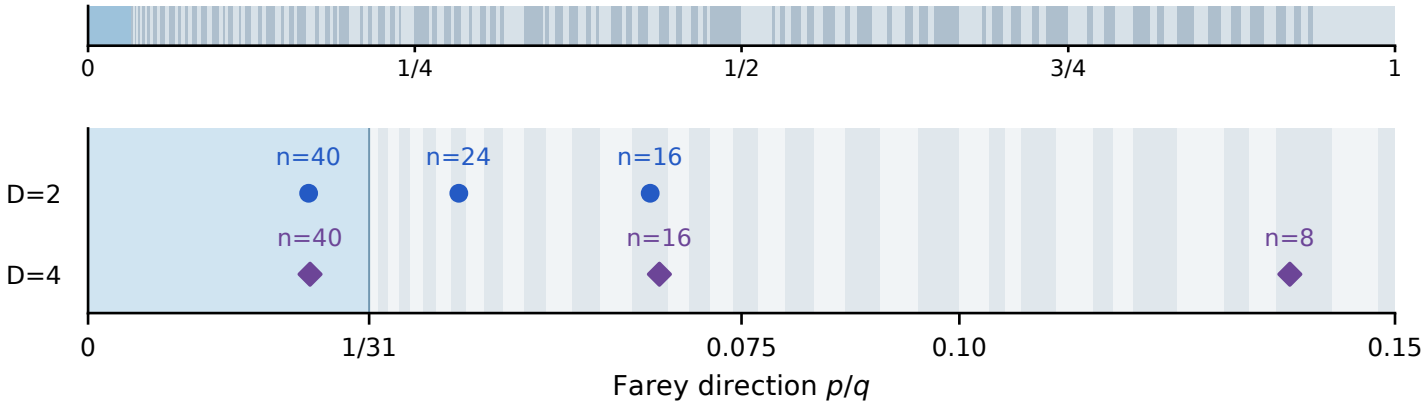
(a) Meridian growth motivates selection by scale

Helms anchor the families; degree controls their eventual logarithmic spacing.



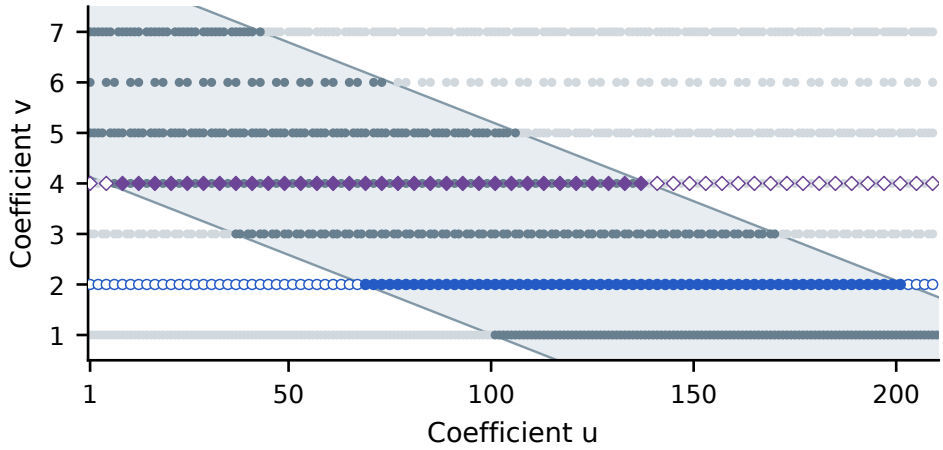
(b) Farey groups share certified growth bounds

All 162 groups are shown in the strip; the rows below zoom into 0 < p/q < 0.15.



(c) One group contains many Meridian tails

Certified interval intersection selects the band; the colored dots retain the family view.



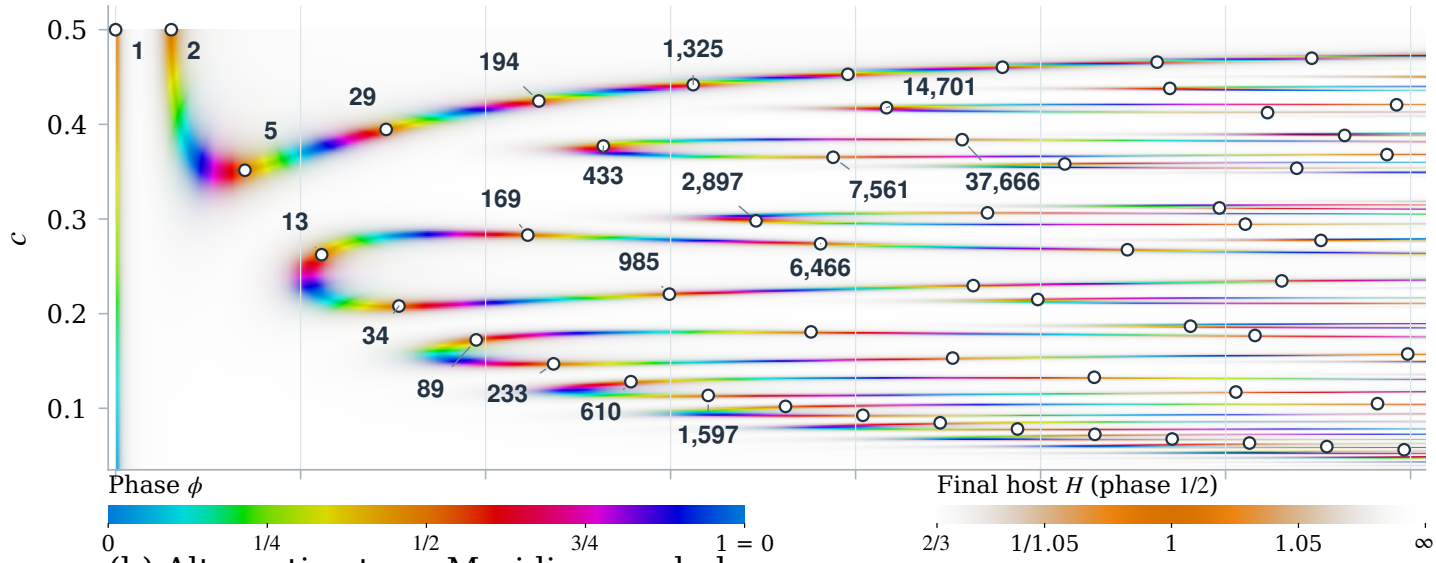
Highlighted group
$0/1 < p/q < 1/31$

$(p, q) = (v,\ u + 31v)$
$\gcd(u, v) = 1$

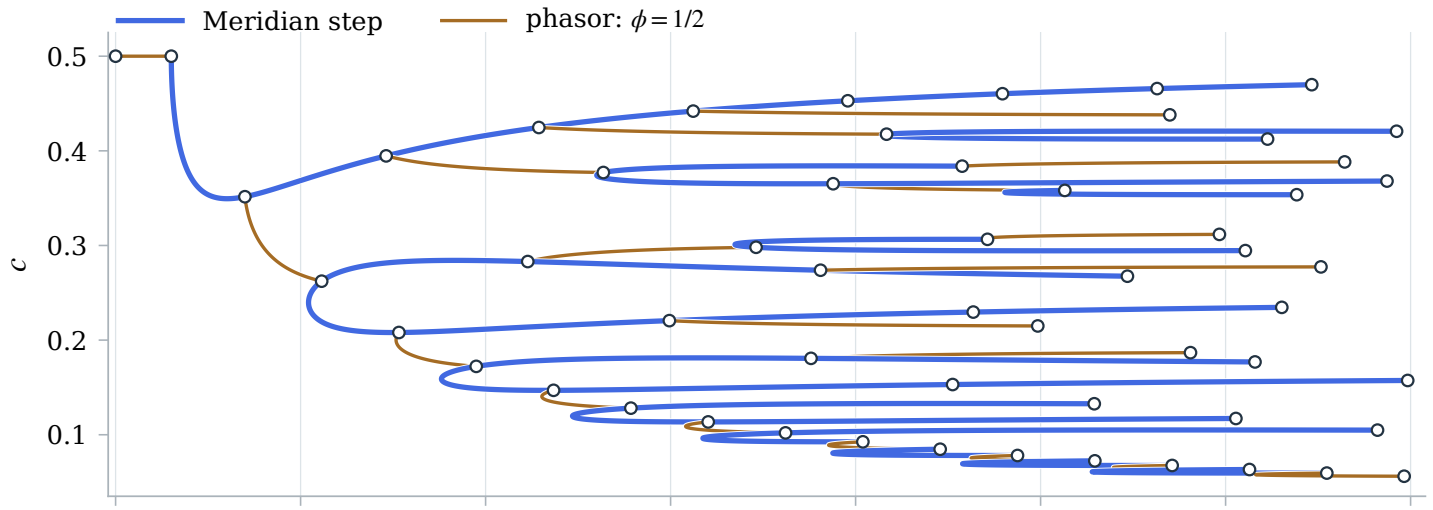
Dark filled dots enter the band. Pale or open dots lie outside its selection. Colors follow the same two branches.

Exact family values and frozen production cone bounds. Displayed lattice is a finite excerpt.

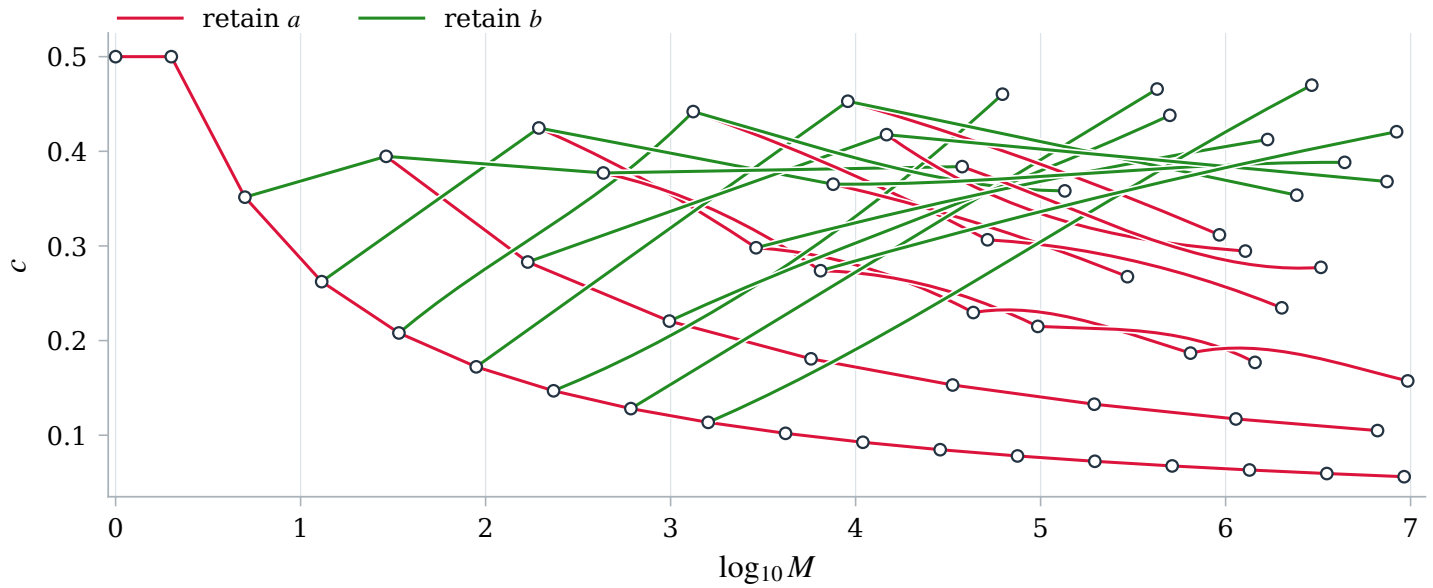
(a) Geometric descent: phase and final host



(b) Alternative tree: Meridians and phasors



(c) Canonical Vieta tree: parent-child links



Meridian-guided band schedule

Helm offsets and degree spacing make growth by family visible

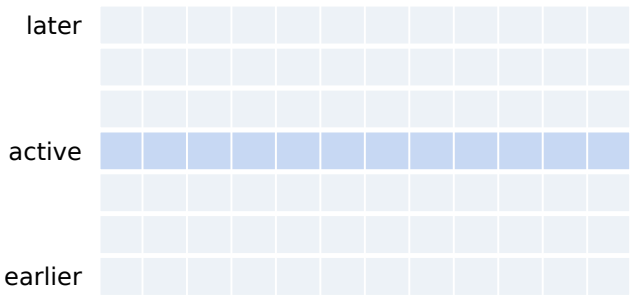
↓

162 Farey cones plus 163 retained prefix configurations

↓

Certified intervals schedule one magnitude band at a time

Process a size band across address groups



Clear after candidate replay and resolution. Boundary intervals enter both bands.

Tan–Zhang subtree-pair schedule

Exact Vieta-tree frontier separates subtrees at a size threshold

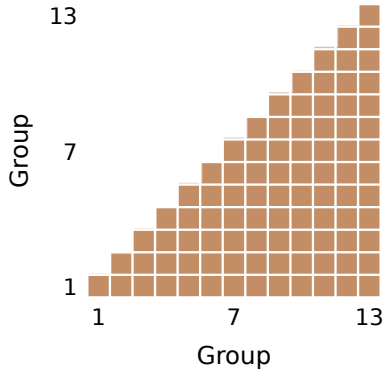
↓

1,455 frontier subtrees plus 1,456 prefix configurations

↓

Certified size test enforces the cutoff; subtrees are balanced into 13 groups

All 91 unordered group-pair jobs



Each nonprefix configuration appears in 13 screening jobs.

Screen records per included configuration

1.005

Completed first-trillion run

13.000

239,536,376,556 configurations

Reported temporary screening storage

1.066 GB

Maximum recorded temporary key files

184.27 GB

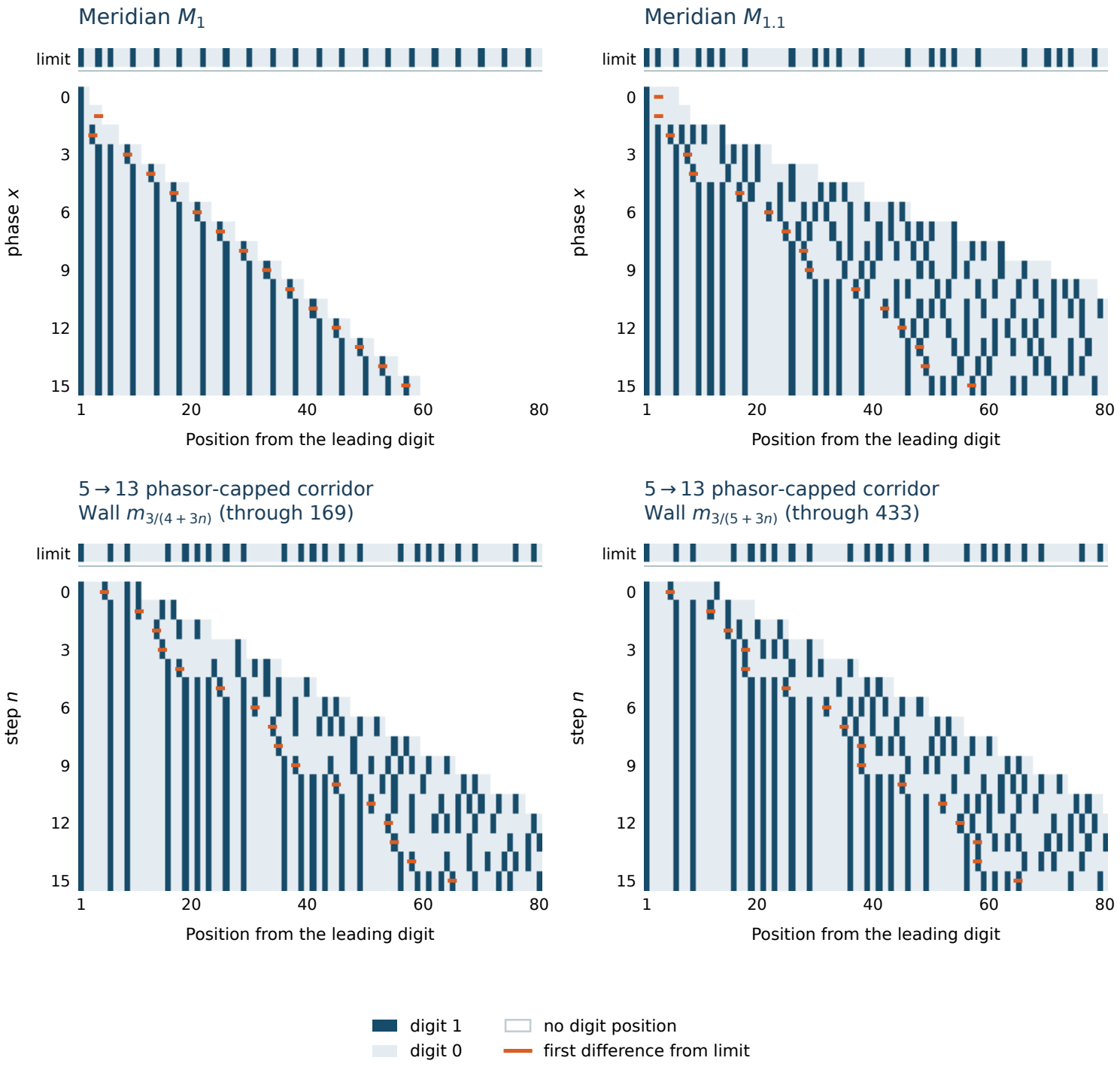
Largest pair job: five-byte payload

GB = 10^9 bytes. Component scopes differ. No whole-machine storage or runtime ratio is asserted.

Sources: this manuscript, Section 9; Tan–Zhang (2026), Sections 2.1–2.3 and 3.2.

Zeckendorf digits along Meridians and Corridors

Rows begin at the most significant digit. Each “limit” row is an infinite leading template.



Exact integer calculation · $F_2 = 1$, $F_3 = 2$ · At most the first 80 positions are shown

Binary certificates and exact originals

The following binary files are preserved in the encoded recovery archive in Part C, including usable supplement derivatives with the six long decimal expansions removed. They include compiled executables, numeric arrays, certificate tables in binary format, and Python bytecode. Their original bytes can be recovered from that archive and checked against the SHA-256 values below.

Parts B and C together restore every original ZIP container exactly, including names, timestamps, compression and duplicate occurrences. Part C also encodes a compressed transcript of all other original UTF-8 text files without byte changes. Part B carries the six full decimal expansions and compressed numeric recovery streams.

Part C begins with instructions and a short decoder for its losslessly encoded payloads. Decode v3_Originals_and_Recovery.tar.xz and open it with a TAR/XZ utility. Its split_manifest.json maps every release, filename and hash. Run the included restore_exact_originals.py with the Part B TXT path and a destination directory to recover all 45 exact original downloads. The large recovery archive is outside this PDF.

Binary file catalogue

v1_Selected_Markov_Values_Supplement.zip!/v1_Selected_Markov_Values_Supplement/source/rank_search

57,520 bytes | Zenodo records 24093257, 24096109
SHA256 cd9beb6f8c0715e320d93b714e0f629efb1d2a5c05b5e7f1c250e6645d16007a

v20_Band_500_Billion_Recovery.zip!/project/engine/spool

97,008 bytes | Zenodo records 24089717, 24093257, 24096109
SHA256 1a06b36371c624282dcaea2fde6df94bf2be9a8b5db34417084a85eae8ef97ae

v20_Band_500_Billion_Recovery.zip!/project/engine/key_tools

76,216 bytes | Zenodo records 23874075, 24089717, 24093257, 24096109
SHA256 44066f9d950bdede3909452b049cee1508ecc4353fcf9e31dc978f5f1633dbcf

v20_Band_500_Billion_Recovery.zip!/project/evidence/independent_final_replay_native/10823/worker_2.bin

16 bytes | Zenodo records 24089717, 24093257, 24096109
SHA256 ada86f5f28aad20793287a8b01bb8ff654ec8f5f9fe7c58d08f20cc3768680ec

v20_Band_500_Billion_Recovery.zip!/project/evidence/independent_final_replay_native/10823/worker_1.bin

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v20_Band_500_Billion_Recovery.zip!/project/evidence/independent_final_replay_native/8156/worker_5.bin

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v20_Band_500_Billion_Recovery.zip!/run/candidates/00008974/repeated.bin

16 bytes | Zenodo records 24089717, 24093257, 24096109
SHA256 00871b5349fe1d80a3d9d24dbb933a8660b4f815ff0c1008989ea7c741328877

v11_Markov_Meridians_editable_source.zip!/v11_Markov_Meridians_source/checks/band_550_billion/editorial_independent_replay_native/8057/worker_4.bin

32 bytes | Zenodo records 24089717, 24093257, 24096109
SHA256 b326474521df08bef2d736a08f3e57e290bd68522c344a75f9f0c5423e65ab80

v11_Markov_Meridians_editable_source.zip!/v11_Markov_Meridians_source/plotting/figure3/descent_field.npz

18,354,341 bytes | Zenodo records 24043052, 24046012, 24053720, 24084246, 24089717, 24093257, 24096109
SHA256 c512f585321953850742af19eb94ffabf3950f409960555f74f5196a20f7a80f

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_1.bin

11,316 bytes | Zenodo records 23874075
SHA256 7a3808dbc77ac27492649f08690aa62c8ed31bfc0f3300dff5079dda1adebdca

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_13.bin

10,596 bytes | Zenodo records 23874075
SHA256 f9faf406eb4479b7b8b7469cf456b6726c4512cb90eef1aebcb815172072b348

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_17.bin

10,500 bytes | Zenodo records 23874075
SHA256 2f26cfe7ca2774f0c6c3668b6fcca3ff718bfe6b3a9061342ce9c6770c368609

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_2.bin

28,908 bytes | Zenodo records 23874075
SHA256 56d78c77fc32fb861c495de7bb9f7fed55283d60b76cd9fc1c08757a5f78d853

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_21.bin

10,104 bytes | Zenodo records 23874075
SHA256 7d252d55857aff79871128f823363a6ae632ce99c6f28e5c13e7068d75374a8d

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_25.bin

9,816 bytes | Zenodo records 23874075
SHA256 8494bc3493bd3be600e1f8ebeac6ba6ec25641287c5a6588232ce6fd46af2277

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_29.bin

10,644 bytes | Zenodo records 23874075
SHA256 8b33c87eacbcc044035102303c8a9d660168fca71ecb6946f970ac8e3bb931bf

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_33.bin

11,508 bytes | Zenodo records 23874075
SHA256 36835d7fcf30cdcd43d11d821e7b7394ac6f4f2133f401cc5528eac519481ad4

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_34.bin

27,828 bytes | Zenodo records 23874075
SHA256 d692db56585c37ac988fd41589aac7ce38086301c8d1b430c42c9aa0532653e9

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_37.bin

10,332 bytes | Zenodo records 23874075
SHA256 6e806372f8b2f8f5d264e4071dc5a2f50aec51e5a876594bc8db7cdb70212dd0

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_41.bin

9,900 bytes | Zenodo records 23874075
SHA256 58bc588c178bab475dccffd443989181875cf074d1bdad98d20f46c72c9abbd1

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_45.bin

10,680 bytes | Zenodo records 23874075
SHA256 2fb1e893fd26dcfbf53b7cea2df17b459274fa0c873157f7fa295663614dcf1b

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_49.bin

10,596 bytes | Zenodo records 23874075
SHA256 309076aad5b6a7ea555a337767e8e4c92d268c9cc82185b875ff2189dfb77fc5

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_5.bin

9,720 bytes | Zenodo records 23874075
SHA256 3dddacfe02b385abdb09f9f6348ac55dabbc2b693aaf74a75d21fe9655609b4c

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_53.bin

10,956 bytes | Zenodo records 23874075
SHA256 83863f0a5b8b5b2ee9b82b63b3e8643bba07a60c60e0dfbde1748fca394c5b13

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_57.bin

9,480 bytes | Zenodo records 23874075
SHA256 04a27f72392c227bc477d4dd41a79801e5b59e66512cbd5403582e59681a623a

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_61.bin

11,352 bytes | Zenodo records 23874075
SHA256 adf9a399b0a120bc45d7abb541fd8fde76334f073e2b1981dfaa474fc39c9f8f

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/bucket_9.bin

9,972 bytes | Zenodo records 23874075
SHA256 25dec108e9b5e60ea6c19acc1287d94d4a14e66d5f0ed03042db9deb423a88ca

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/T36/values.bin

149,472 bytes | Zenodo records 23874075
SHA256 6ff59abe6a9cab821d3fc7f70fa902c32b6996d118abeb0d3481cf62a5fa2e36

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D100/inventory.bin

18,684 bytes | Zenodo records 23874075
SHA256 7036ec7ee919a4ba336c0c915a08242821c15b2a09e027b86f708ce5aa6d203e

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D2000/T36/pairs.bin

56,968 bytes | Zenodo records 23874075
SHA256 f75ab5f5765f5b96d7052d24caace2b99b543e6e45b01e6fadebc2e411cd48e3

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D2000/T38/active.bin

18,356 bytes | Zenodo records 23874075
SHA256 f22b8547079cb877031923c316ac2236abd0f635b754bb443605f63e7c97a9c1

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D2000/T38/pairs.bin

4,896 bytes | Zenodo records 23874075
SHA256 ee19ea4ba37ba574908d2be88179a080b6b77e6239b2d83b122483797cb8c182

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D2000/T38/values.bin

1,395,056 bytes | Zenodo records 23874075
SHA256 f6af8f7e4d6f2e24aa7abdd17ba81664dfe223a3102aaf3499f68187a4115c6f

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D2000/T40/active.bin

4,840 bytes | Zenodo records 23874075
SHA256 eafbe6386c80de9eff748d87ecfbbf78f4f8dd6d70063487b1bf3ccd38845edf

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D2000/T40/values.bin

387,200 bytes | Zenodo records 23874075
SHA256 677d5240065b4780d4b7f67b7f467a7484502f838dea95650462892b7a8cb0ec

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D2000/inventory.bin

7,303,032 bytes | Zenodo records 23874075
SHA256 e3716c380c2a276b122b9601128e02968309b4febceef0b335be82aa48d2eb5f

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T36/pairs.bin

2,566,040 bytes | Zenodo records 23874075
SHA256 3baf98ad72a6416a204f153287e221152134a6629657ce0412539e73c8c6c468

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T38/active.bin

483,888 bytes | Zenodo records 23874075
SHA256 b9a428f0479864b8871738b735f484aa0b183186ae7ee58860bdad39cb9689e7

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T38/pairs.bin

139,416 bytes | Zenodo records 23874075
SHA256 a51d80e964314e9a02eb2163827e3581a281035adde0b061fe115b2072418a1f

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T38/values.bin

36,775,488 bytes | Zenodo records 23874075
SHA256 6662c7b5cecd95114b0ca23da798282c8d3e530a109598fd4aeff684f34426bc

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T40/active.bin

135,056 bytes | Zenodo records 23874075
SHA256 0bff5f4c824d562a9dad89123b41d8908ba0981e1f457139b8a23058b5d018ff

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T40/pairs.bin

696 bytes | Zenodo records 23874075
SHA256 c01079643aa833405daedf282665de80fe59fcdd67ffcd9d878347f3378c0d10

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T40/values.bin

10,804,480 bytes | Zenodo records 23874075
SHA256 34f58c75be061f97a2d6dfc6b10b71c7e8a48c85dba1f609575d6ec88a484f2d

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T42/active.bin

672 bytes | Zenodo records 23874075
SHA256 70ea6cfd39cded855ffddcdc8cffe88db72f1aa7ebde57ff61c6385175d23634

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T42/pairs.bin

8 bytes | Zenodo records 23874075
SHA256 091253ad92c86afba51ceee6f3020c03e65608e13ac9553110ef7176cbfa48f5

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T42/values.bin

56,448 bytes | Zenodo records 23874075
SHA256 62a3706ad0d4b2a3fb2c08d599ff08cb7a71864dab7109be4852f1d70c44ae43

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T44/active.bin

8 bytes | Zenodo records 23874075
SHA256 8da98081618be6e03e5361a752cce32caabca77e67a7395b21f548702371774d

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/T44/values.bin

704 bytes | Zenodo records 23874075
SHA256 d060301e12b42ed6ded3d32127cace6a2c7d7d889ba07acb62f6452faba1cc98

Meridian_Degree_10000_Reproducibility.zip!/Meridian_Degree_10000_Reproducibility/evidence/D10000/inventory.bin

182,414,772 bytes | Zenodo records 23874075
SHA256 70d738bca1996a780a6d2af5c9bab007de42174039792f4298a3aed8ec3eab1d

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/sources/v35.txt

289,563 bytes | Zenodo records 23874075
SHA256 1954bee6f3cb841f6fc823ee6df68f591533c8296f5c9553a204716e95e2be69

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step3/markov_verify/__pycache__/__init__.cpython-312.pyc

322 bytes | Zenodo records 23874075
SHA256 2d590105f504159d690ad37c338e81ea1b4e5e33fb33f630f569fbb3ba52acd9

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step3/markov_verify/__pycache__/arithmetic.cpython-312.pyc

7,555 bytes | Zenodo records 23874075
SHA256 a7eac6984f8f58a7cbab53fa71019d6e4e4682c73e3eef1669e53bf858875679

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step3/markov_verify/__pycache__/domain.cpython-312.pyc

16,106 bytes | Zenodo records 23874075
SHA256 e14b23182e640cc93f706615192f67465c292ac396644bce71738f19e6a8775f

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step3/markov_verify/__pycache__/windows.cpython-312.pyc

12,127 bytes | Zenodo records 23874075
SHA256 637e05a595f9a07bc11fa1919014d3a77bd553065f7c650e1baf6ad48938a20e

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step4/__pycache__/optimized.cpython-312.pyc

42,262 bytes | Zenodo records 23874075
SHA256 36303dd278b03edba0fe120b4d1c7de773926e7b3ff2077c24176237359d5635

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/__pycache__/audit_completed.cpython-312.pyc

47,716 bytes | Zenodo records 23874075
SHA256 9d9715cdfd546aafe3daddecb9ae82ff5a238ef10fe96d21d698ec3ae13aa7af

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/__pycache__/incremental.cpython-312.pyc

58,999 bytes | Zenodo records 23874075
SHA256 9c043f0afa3dc8dc89f4b3e45d41c1e4805abb1e8554a3352f108105a31e1c22

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/__pycache__/streaming.cpython-312.pyc

46,677 bytes | Zenodo records 23874075
SHA256 7571d52072649d8ee44ef42f3a43fd54916fb925ea5e7c2dffd7c25b2fbe1e8b

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/native

63,896 bytes | Zenodo records 23874075
SHA256 199e1331f85f5b4dddd3a3bd1dd49943786b5471ced29aa165c732785c3af556

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/native_incremental

63,912 bytes | Zenodo records 23874075
SHA256 5587ab32108fdce365eae968725bb2f7a3f729e3fdbeba873075ed5153fd4564

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/pass_0000/repeated.bin

32 bytes | Zenodo records 23874075
SHA256 3ae160aa8a3bff8cde0de17df91ef508beec586dc180dd4ed4c931062b58d902

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/pass_0001/repeated.bin

16 bytes | Zenodo records 23874075
SHA256 44a047dac882504dcc223e8db87a700743897e9fbc91e45e97b9c062e9d9a2bf

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/pass_0002/repeated.bin

80 bytes | Zenodo records 23874075
SHA256 a88d5e3929e4bc3c8c0624cb5f3b2a582c522bd819ddb56eb9d072c712dd8e3b

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/pass_0003/repeated.bin

48 bytes | Zenodo records 23874075
SHA256 95f24d01a3e8771867ee2b2d66a1773504c20496b7d57cb57a803637efa4ac15

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/pass_0004/repeated.bin

16 bytes | Zenodo records 23874075
SHA256 f211990af401f8aab86eca7fc7c3b91c7f0f81a6363a29e09058fac1a176e2c8

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/pass_0005/repeated.bin

64 bytes | Zenodo records 23874075
SHA256 125201e349ae7fac23bc459872395c9fb4c882213c44b09b7a5ab969d79450c2

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/pass_0006/repeated.bin

32 bytes | Zenodo records 23874075
SHA256 1da17cdabcc1a1043129a3590832c7772234e744b1ca163bf049b552515c7cae

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/pass_0007/repeated.bin

16 bytes | Zenodo records 23874075
SHA256 611782f53d91a2bd8259c421da32ae8ab6921699cf845de1463e902088b94fe8

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/repeated_keys.bin

304 bytes | Zenodo records 23874075
SHA256 f474a6e872d35ff9d7c4c27e5ab87889806a5fc618291072199d7feed7701482

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/replay_tmp/candidate_000.bin

96 bytes | Zenodo records 23874075
SHA256 1317ebdb09167055ff5b7b3661f6e3751fa9c5582be26b3232b15b260f4cd18e

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/replay_tmp/candidate_001.bin

96 bytes | Zenodo records 23874075
SHA256 86b138e06f3e02ec2e7d882c776ad3fac6d7f9fc06ae3cabd5a4eaf9723090ef

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/replay_tmp/candidate_002.bin

112 bytes | Zenodo records 23874075
SHA256 ecca6029848c1aebe219b65bdef33584c8240256b2b7d124a59928b22a4cbb7f

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/replay_tmp/candidate_003.bin

48 bytes | Zenodo records 23874075
SHA256 309bf1531133a56dd722187d14ea349a7b7fa0eefc3e7740c3ed6f86a5a26650

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/replay_tmp/candidate_004.bin

80 bytes | Zenodo records 23874075
SHA256 91647f42df4cddfbd6ab8587290a02e0d834ed0a72c33ce87aadd8fb35466fec

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/replay_tmp/candidate_005.bin

80 bytes | Zenodo records 23874075
SHA256 16821112506079f65c1aa9500d6499c39cf26116a7258a1ebe68eb58d00f02ea

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/replay_tmp/candidate_006.bin

64 bytes | Zenodo records 23874075
SHA256 4d4180160e178392794dd0fa708bbdfeb874c76650693e0237eb0b4bae466b49

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_10b_p3/replay_tmp/candidate_007.bin

32 bytes | Zenodo records 23874075
SHA256 d2dbee9f727d13df5e127f87b7f3e92d70c8abb12e880576f3fe738b98884727

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/pass_0000/repeated.bin

80 bytes | Zenodo records 23874075
SHA256 ee6ddef12360a0927b4594770177c65991bb1177dd81547f3baaef33e17bf07e

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/pass_0001/repeated.bin

48 bytes | Zenodo records 23874075
SHA256 b1f0794c909828a5d1ec4e2bd0127917db1eabeec95342318f795cee60f65ba0

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/pass_0002/repeated.bin

64 bytes | Zenodo records 23874075
SHA256 d09f1f90a74fb8c77ebab4cbfa15a3e02ca06b8421d89c09e11092f94d01dc77

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/pass_0003/repeated.bin

16 bytes | Zenodo records 23874075
SHA256 44d155f989c1073fcb1db69879602537dba86a1cb3d1db0b95a007bf518ead73

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/pass_0004/repeated.bin

64 bytes | Zenodo records 23874075
SHA256 9f93953456873e8183094e010a2dfd5abc36516ad5cfbe2a842ad6893baab533

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/pass_0005/repeated.bin

48 bytes | Zenodo records 23874075
SHA256 8bddba3cdd9b460fcf25f9d331fc4ca7ad258bb45e14e66846a08dde9bfffca6

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/pass_0006/repeated.bin

80 bytes | Zenodo records 23874075
SHA256 ad9d559ec7860d52e7bf99c406258b2b3bcc2beda392a3424b4016b0466b1940

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/pass_0007/repeated.bin

64 bytes | Zenodo records 23874075
SHA256 d214c5c629fddc1588608cbd5beca67c380a9a5cfe960f6490375e08614826d5

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/repeated_keys.bin

464 bytes | Zenodo records 23874075
SHA256 8a3ab41ea8e6841975e2402cea4aab5980ae2106789d828a606d0368a2bcfaf4

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/replay_tmp/candidate_000.bin

160 bytes | Zenodo records 23874075
SHA256 5cea5f5aeb719128b741477e3e44edbd3b9bbce6f56d00852a25b50dcaf0d87a

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/replay_tmp/candidate_001.bin

128 bytes | Zenodo records 23874075
SHA256 4953b9b1a7dd1cb0bf5f3dac329251917ab7998776fb0dbc575a0bfa7c4f0cf0

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/replay_tmp/candidate_002.bin

112 bytes | Zenodo records 23874075
SHA256 4eee53269dab0f0cb471b5e77a3e7ace3a98a8738ea9568f19cc78faf419e920

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/replay_tmp/candidate_003.bin

128 bytes | Zenodo records 23874075
SHA256 9761b5e4e0ae43c8d6dc79ec20ec5a46275e4d47853a29263f2479dc95a15238

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/replay_tmp/candidate_004.bin

64 bytes | Zenodo records 23874075
SHA256 2200d6f537eca0f9d9478742dbbbf9fb2f56aa4b6c23104e5f335bbe509c98ad

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/replay_tmp/candidate_005.bin

96 bytes | Zenodo records 23874075
SHA256 4901a5bfe73cdeda2cf6c7ccea51fd82320a1d7a3bc0c97756cfedd81f6db596

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/replay_tmp/candidate_006.bin

112 bytes | Zenodo records 23874075
SHA256 f9e64b8714543bdad61a6fedb8d7a0f6b8ef1cd6e358df4b1158b0aeee8e2478

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step5_stream/runs/verified_20b_incremental_p3/replay_tmp/candidate_007.bin

128 bytes | Zenodo records 23874075
SHA256 b22a5530e05aea2b92d8f8e45adbb9457da09bea046ed517c3e8103eb6b5086d

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/__pycache__/execute.cpython-312.pyc

33,587 bytes | Zenodo records 23874075
SHA256 8112524169ca73fe09af09bf2d6188b30a794ea37841fcb37f47a957a9109f94

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/count_histogram

63,904 bytes | Zenodo records 23874075
SHA256 c562689530f6d69fc2575ba9e8cf7d7b96a73f9bd76c0ab4bbf6b1a0e2c40ede

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0000/repeated.bin

2,144 bytes | Zenodo records 23874075
SHA256 fa92d72b6773f0c221a268a73eeef1de21f3b4e27b6b913e5f806583c579a6ad

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0001/repeated.bin

1,952 bytes | Zenodo records 23874075
SHA256 86816a6bb18913abc1b09b3f7979d26157818c15338837bb00835962ee5e9e35

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0002/repeated.bin

1,856 bytes | Zenodo records 23874075
SHA256 5def58032d22cbe24e104699c54d6edc7f3e0ee78fcdb906908b798dab066db4

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0003/repeated.bin

1,600 bytes | Zenodo records 23874075
SHA256 e8ba67f801bbf2d866d4b2668b307c461b5da483696dd18f940f35d74bd4251e

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0004/repeated.bin

2,016 bytes | Zenodo records 23874075
SHA256 e6f6562bcb32e0cfd9af4e3b3c72fed3eb1328d213f69e0434363e92006e8ce0

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0005/repeated.bin

1,808 bytes | Zenodo records 23874075
SHA256 1dc57fa83f8c1e85a350400667ccf34e47eed6658f83c22c20f70afa50f3229b

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0006/repeated.bin

2,032 bytes | Zenodo records 23874075
SHA256 323193876f1735aaae1c1771815d21267dafc947eb58871a0d8de8d9fe2e19ac

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0007/repeated.bin

1,760 bytes | Zenodo records 23874075
SHA256 dc967b1cac679d6f17bbed58044fae5fbdfc54cf89994d98e934daf7249b8062

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0008/repeated.bin

1,840 bytes | Zenodo records 23874075
SHA256 20029da22b8d6b630e2b7b52a9395a316c31790ba876a639318c204ecd7734fb

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0009/repeated.bin

1,904 bytes | Zenodo records 23874075
SHA256 43cb7db1e1046220c9788fdc4eface9b07f5b8344eb36956af4b7286b2fd586e

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0010/repeated.bin

1,808 bytes | Zenodo records 23874075
SHA256 b78d482ba506accad2ddfae19870a0b5018d7c5be8590c0550425e506f340529

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0011/repeated.bin

1,952 bytes | Zenodo records 23874075
SHA256 dd30906dca630d61c0d75d7da04fa4880de8f9b3ec86d1bd63871b78f0f5742a

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0012/repeated.bin

1,904 bytes | Zenodo records 23874075
SHA256 35c4c63e75cf6594dcfe31fe8cabf879023d6431388cdfa3a2262e6e36bf4d95

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0013/repeated.bin

1,792 bytes | Zenodo records 23874075
SHA256 b199a565e88496ff9483a8132022f2763b72b6524f593bb0bf2ee36ad84bab4e

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0014/repeated.bin

1,776 bytes | Zenodo records 23874075
SHA256 d05899f5f99898e6e74af6e7ed9b8cf0eda8e19f53ac5c2269359ff4c8e02265

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0015/repeated.bin

1,904 bytes | Zenodo records 23874075
SHA256 29e64135f124fc0b5151941b5418e14a94a6a3ed9c82b734c09dbcd0ebfc9f1e

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0016/repeated.bin

1,984 bytes | Zenodo records 23874075
SHA256 b945a83231bd9d2230b4715c06d7b17bf56770bdccb6a3dd3ae4fcaeffd17efd

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0017/repeated.bin

2,000 bytes | Zenodo records 23874075
SHA256 750956758277b0d41eb6f5dec09ec327ff3b7e97b6365490457313044fc0dc7a

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0018/repeated.bin

2,000 bytes | Zenodo records 23874075
SHA256 3a9a15e2998b3e6fd9b23d11fcfbfb558972fc2eadfbea592cb617b59774816a

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0019/repeated.bin

1,984 bytes | Zenodo records 23874075
SHA256 51e7d23a0e07799560cfadd402229b1ec983587476867b770cee19f4775095df

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0020/repeated.bin

1,904 bytes | Zenodo records 23874075
SHA256 d6106de46051cb469e3f229fc60a52963fc6a1a941975691e8cfe5b6f154ed06

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0021/repeated.bin

1,808 bytes | Zenodo records 23874075
SHA256 3b5b9cd4416ec873e3139f5c6b67bb2485275cc3fac723e82633fafe03dca4e1

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0022/repeated.bin

1,792 bytes | Zenodo records 23874075
SHA256 8acaad452f3dd0cc738e301138ce820c1a840a52a517490ceab097d7dbf993c4

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0023/repeated.bin

1,872 bytes | Zenodo records 23874075
SHA256 4502f6e505a904419bc52664974c21911e1c0c29900d7ff3ec7447f16a46f568

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0024/repeated.bin

2,080 bytes | Zenodo records 23874075
SHA256 ebb7a607182c203ac1e22093651239ccde5043f4f23360fa03f442cf4e297730

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0025/repeated.bin

2,016 bytes | Zenodo records 23874075
SHA256 933867caa8f092ba4f28d462b7564a9b227a7834a1a9ae95592336f6bd480c91

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0026/repeated.bin

1,872 bytes | Zenodo records 23874075
SHA256 d0c05950df135925be5b3e4ef4353a3584bf8524a9645bd51227bf711709b330

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0027/repeated.bin

1,776 bytes | Zenodo records 23874075
SHA256 ef30578a24ed87639a04bac7e98b03809405860a6ca244a7310096fafaf44962

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0028/repeated.bin

1,952 bytes | Zenodo records 23874075
SHA256 670bac91d065f6a67c41d2185c990da8f641075c62eff947c712f2b72944a9bd

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0029/repeated.bin

1,840 bytes | Zenodo records 23874075
SHA256 29e47ed748873cd2771500acefb2eaeeeb82f1c8957c859326b7b3a86b65b3e7

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0030/repeated.bin

1,824 bytes | Zenodo records 23874075
SHA256 63c807060edae1b6cb1d2bde7574ac03c8a3158f172585c91b0defd7448bd33d

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/pass_0031/repeated.bin

1,984 bytes | Zenodo records 23874075
SHA256 af2c67355bddbd2494a7c0fa08023c2d7310594b442344059dd84b6f249c83e4

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/repeated_keys.bin

60,736 bytes | Zenodo records 23874075
SHA256 4539651bd52c9119606e7fb20441983a5bb08d061a102aca7dc7d184cffb68b4

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/replay_tmp/candidate_000.bin

22,448 bytes | Zenodo records 23874075
SHA256 a597eb036c080f93e4e1a767501a5f7206c0e1f5253b3be8dcace75473f80e62

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/replay_tmp/candidate_001.bin

19,520 bytes | Zenodo records 23874075
SHA256 84b68346b6e2b3f6c85ea11a65bf1037523dba8d95766d73e8b843b845d10a2c

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/replay_tmp/candidate_002.bin

21,280 bytes | Zenodo records 23874075
SHA256 5b4bec8ab9fd3842e3bbbc9febc018c996bbf691aa530eb476e8077aa4e32d67

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/replay_tmp/candidate_003.bin

21,664 bytes | Zenodo records 23874075
SHA256 ce7fa9dbd14f3a48bf9a515978ea1dcda42344a8fec500b9523f5ff977a0f903

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/replay_tmp/candidate_004.bin

63,168 bytes | Zenodo records 23874075
SHA256 7af98194327097c26e88441671ece29d34b7c9ee140108b058cb358b23549d4c

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/examples/complete_10000_key13/replay_tmp/candidate_005.bin

16,160 bytes | Zenodo records 23874075
SHA256 152d5122ae6f503f58d89adfe1bcd90bf96e41b301aebdab4f4fdeac6dd064d1

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/prototype_failed_journal.jsonl.gz

1,242,061 bytes | Zenodo records 23874075
SHA256 50df25db7adf0f12afb4b223782c0410b6eab680fdd431869cac846631062ef7

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0000/repeated.bin

144 bytes | Zenodo records 23874075
SHA256 5b0025ecc07f8e9de79e35b569a5931c0e218335f574a18ae54db36f81fe6349

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0001/repeated.bin

224 bytes | Zenodo records 23874075
SHA256 296c0a81740f1e36b9dd2efe12fb8378fafb48d293140d71c92f3203ff07b5a4

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0002/repeated.bin

112 bytes | Zenodo records 23874075
SHA256 7cf193f93ef9b0ae3b95cd11e79ed962cb83a1a34bdffa54ef99bed9f002f9db

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0003/repeated.bin

128 bytes | Zenodo records 23874075
SHA256 4feed5324640be7776e3eddc81b18a8ff3bf92fdeb0f9d625505543c61c7d548

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0004/repeated.bin

208 bytes | Zenodo records 23874075
SHA256 19125bba4ec5a00ebb6a31c4ab428a256bf22b9e6f8c955f76d8a8e2871f5bab

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0005/repeated.bin

288 bytes | Zenodo records 23874075
SHA256 89ff7b3581ba889518819a78e49a028d0685a02d9a8accfe862f41bde0c2fb3b

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0006/repeated.bin

176 bytes | Zenodo records 23874075
SHA256 fb28c91be778ee8ee3f7dbe7bf44cea5cb1b2843cac17c114819082bab085394

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0007/repeated.bin

96 bytes | Zenodo records 23874075
SHA256 2295947dbfda72fe80c09c3929a163e5db711e72b974714b1b5a8de2f15acde1

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0008/repeated.bin

240 bytes | Zenodo records 23874075
SHA256 6b81c78250baf55ce8f371210036acd69e9f30471f7770248d5054ffeb502e09

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0009/repeated.bin

144 bytes | Zenodo records 23874075
SHA256 e9992472151485cc533561b4085eca477e1bcf630db5e69e5b7ee88974822b0a

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0010/repeated.bin

192 bytes | Zenodo records 23874075
SHA256 ad34b969c3247815f64408af026dbafc1ea273729b774e5573d1d8532f37d9b0

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0011/repeated.bin

240 bytes | Zenodo records 23874075
SHA256 80b108dbf09a2754a41078f7d7b11ec290b931ffe612fed62e61b2908b8dd8b1

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0012/repeated.bin

112 bytes | Zenodo records 23874075
SHA256 ccb13c39f9e8ec55eade1fcbbf6aa5f94990c7501f75a0a7f52268b762722c43

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0013/repeated.bin

176 bytes | Zenodo records 23874075
SHA256 ae4338e72d31ed3d137d9e4b90677d61b4646af92c882b5898b993d27c2e2481

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0014/repeated.bin

160 bytes | Zenodo records 23874075
SHA256 ac47a004579fa57b59e97b307b155ed8291b496fdc77e677610a68c2cf26c2e1

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0015/repeated.bin

208 bytes | Zenodo records 23874075
SHA256 f0ae7477da98cffe1a38b33bc8ad7a94f8c1d1dc979d747dd2459b529f69042b

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0016/repeated.bin

128 bytes | Zenodo records 23874075
SHA256 364ab7c92b9b0a0ab256ed0740731e5a0d173f43e025a0f4c97f51de1eddcf94

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0017/repeated.bin

160 bytes | Zenodo records 23874075
SHA256 e92b844b974b33daf439aa55d237ff37eca8f01d4e56b932040eb537fe9b7d4f

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0018/repeated.bin

112 bytes | Zenodo records 23874075
SHA256 1cb8a7216ff0bcbad9a9b2b2c66bac95fe5186815fd6e80e4174b9c9157ad63b

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0019/repeated.bin

208 bytes | Zenodo records 23874075
SHA256 217678257f6a9b1427b522a192a825b96c4f85a2a4cfc790a1de8329db9bf6db

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0020/repeated.bin

112 bytes | Zenodo records 23874075
SHA256 92aaddacaef8046c20512b2fdfe0ef9c401e7447eddcb6a2b42a0a6698243a23

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0021/repeated.bin

160 bytes | Zenodo records 23874075
SHA256 5f7e4693414f979e08fd9e863d0958df607838c1c229ca6190d4640c1df80fba

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0022/repeated.bin

96 bytes | Zenodo records 23874075
SHA256 703b99e809ccd9f00b5a43967a6afc7855c28b99bfb5b8a1ed08c1f9af77190e

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0023/repeated.bin

96 bytes | Zenodo records 23874075
SHA256 aff8f1685419974527a4a443e256ad5e82039c52a82620c3e986b547efbb4072

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0024/repeated.bin

272 bytes | Zenodo records 23874075
SHA256 03dc34970f209599887a10239ebe527f24fdecf1a83775966e0caa65f5ddb91c

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0025/repeated.bin

208 bytes | Zenodo records 23874075
SHA256 70271e0f7376da584399f6587e82489b06297ff84f09be20733370aab7df36f9

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0026/repeated.bin

192 bytes | Zenodo records 23874075
SHA256 9d1e5e9d000e7db1bacb2e996aa1698291ebc76ecc8bb209e6640073878bb078

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0027/repeated.bin

144 bytes | Zenodo records 23874075
SHA256 95a4a2403cc2f6ce7be9ade4d68ab52191f8627d016af2e7d0dcb990e3d89e77

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0028/repeated.bin

224 bytes | Zenodo records 23874075
SHA256 77bd60890c92639fb25ec440ada1fce75982f8ebf1ea928ee9e11454d6ab5c0a

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0029/repeated.bin

176 bytes | Zenodo records 23874075
SHA256 5420127a1d08eee772d4099e0e8b5ebdf0deb365b37065b7b3ba88ca1b4a00bc

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0030/repeated.bin

176 bytes | Zenodo records 23874075
SHA256 889a2827204e60fdfe1c45178bee452fc6c8ade5a65c3cb2dd11212a7368e6bb

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/pass_0031/repeated.bin

224 bytes | Zenodo records 23874075
SHA256 7a042855f4ae9204cb4808d54225fc10866769510d74650042535baca1e6a6b9

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/repeated_keys.bin

5,536 bytes | Zenodo records 23874075
SHA256 e23d01296e17b7fe7db64b1b15ac4bf42a9e765a600d442691b9673305ee0684

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/replay_tmp/candidate_000.bin

1,216 bytes | Zenodo records 23874075
SHA256 eb3880e527aa76f9c57b09b29c01acc5570e3a5154a1b2c3661640e1ee24af53

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/replay_tmp/candidate_001.bin

1,408 bytes | Zenodo records 23874075
SHA256 8b37d615157f0e1b0817fafc3859b10a845f0f09201c529d30f85a7f72812a96

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/replay_tmp/candidate_002.bin

1,328 bytes | Zenodo records 23874075
SHA256 9eef51e5af9bd754a1f2c5285776f7bec7ab858518e79d2ab5bd16c0f7988e6b

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/replay_tmp/candidate_003.bin

1,440 bytes | Zenodo records 23874075
SHA256 d8b26d616831d9028c87a10f2d8c5c92a4f1d454fd22ac1e2d2eddb59c63087b

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/replay_tmp/candidate_004.bin

1,424 bytes | Zenodo records 23874075
SHA256 3929334f3518f04d0d5ca172fc0dcc83310f7409eda715b1ebae7e629bba8b73

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/replay_tmp/candidate_005.bin

1,488 bytes | Zenodo records 23874075
SHA256 2a4b23a3130a4d4ca912764be22c8db50619e74ce7250b9275d2838199d71698

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/replay_tmp/candidate_006.bin

1,472 bytes | Zenodo records 23874075
SHA256 eddf137539cf8cd5ee697c722ea0f0683efc3a5b2c05a86111541f7bd41fb6c8

Markov_50_Billion_Verification_Public_v1_1.zip!/Markov_50_Billion_Verification_Package_v1/step6_preflight/runs/verified_50b_p5/replay_tmp/candidate_007.bin

1,296 bytes | Zenodo records 23874075
SHA256 9305aff438c28db3bdc1aaea1cd57f0a590de09cf3bcdf0207ab194496e317bf

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/679/worker_3.bin

16 bytes | Zenodo records 24096109
SHA256 58e45e5c21c35693e4e3d03ff88d9a7233e6c1caed3c7e7f87079aab59727339

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/679/worker_0.bin

16 bytes | Zenodo records 24096109
SHA256 dc0aaecc3470bc270b2d7c26e930ac4e97f1be7745b00bb96dac20500d6e65dd

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/5078/worker_4.bin

16 bytes | Zenodo records 24096109
SHA256 316134042d204b13e5e55c5cb870b1d263f367d7dd7c03c40a9a7dc66c2aed29

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/5078/worker_1.bin

16 bytes | Zenodo records 24096109
SHA256 715cc4170487224275481595 2ae25623508d5ade721243df0bf669bb39307151

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/2408/worker_2.bin

16 bytes | Zenodo records 24096109
SHA256 8780f220e16f3c25492599224c3b04cde155c54931195011c9884901dddbb95f

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/2408/worker_1.bin

16 bytes | Zenodo records 24096109
SHA256 023ea47545a70dda12958b52edb2c6143d51b5e4fa55102c30967516cfb715b4

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/3683/worker_0.bin

32 bytes | Zenodo records 24096109
SHA256 d6ad5234817fabe2f1dc7400fc92b7c472106af666c2bfb94606b3e48c428fd7

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/1224/worker_3.bin

16 bytes | Zenodo records 24096109
SHA256 38cc54d01160ebfc378c0554c895981e3ebe30a588ac8b47867ba10c1ac2dd7e

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/1224/worker_6.bin

16 bytes | Zenodo records 24096109
SHA256 d39091766af9e1cb217fb657afcebef77cf5c05245a2701d2c5a8f5d76964e2c

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/3467/worker_1.bin

16 bytes | Zenodo records 24096109
SHA256 fdf4da7dc59cd4c5bc532668c7c6bec1aa2486b75259b546ab0b98ed0702cd21

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/3467/worker_5.bin

16 bytes | Zenodo records 24096109
SHA256 075f84afe4124f4f5800a50ebc7486976b57cfe0dcac008f85a7ad5fd33bf41d

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/896/worker_3.bin

32 bytes | Zenodo records 24096109
SHA256 df7770b1ef152800df9da5a73aa984700d229281d1009ff8f5ce1f3958286980

v16_Trillion_Verifier_Recovery.zip!/project/evidence/v1_final_replay_1t_native/896/worker_1.bin

32 bytes | Zenodo records 24096109
SHA256 3c683cbc5f33cfaca26ebdbd90cef0d2158d86fd8ee4402beb3e60d7269547d0

v16_Trillion_Verifier_Recovery.zip!/run/candidates/00003467/repeated.bin

16 bytes | Zenodo records 24096109
SHA256 d4961b48bd765405005265beb1d5d1f8a495334963e8c5d83db912206ae22bb6

v16_Trillion_Verifier_Recovery.zip!/run/candidates/00005078/repeated.bin

16 bytes | Zenodo records 24096109
SHA256 9cffb7f2209c21785c60be1ba7521de9f0218ce18709a6a0581adbf00e583aad

v16_Trillion_Verifier_Recovery.zip!/run/candidates/00003683/repeated.bin

16 bytes | Zenodo records 24096109
SHA256 f755f6135e80c29678198584bf069a9dc3f2385069b2795f1064bbaa21e81f03

v16_Trillion_Verifier_Recovery.zip!/run/candidates/00001224/repeated.bin

16 bytes | Zenodo records 24096109
SHA256 8de7e5deee0f894323e72d283f917f6f1bcaaf5557bfe07c1c8c4586ae3c6d30

v16_Trillion_Verifier_Recovery.zip!/run/candidates/00002408/repeated.bin

16 bytes | Zenodo records 24096109
SHA256 7a321d395fd84b943ab5f9965c034086e28bfa937d9e6a5f0f7d1dbf228737b6

v16_Trillion_Verifier_Recovery.zip!/run/candidates/00000679/repeated.bin

16 bytes | Zenodo records 24096109
SHA256 5afe81eb12cae5a7a7fc0de34e99dc3a634eb70618efb7256412fb62aefd18c3

v16_Trillion_Verifier_Recovery.zip!/run/candidates/00000896/repeated.bin

32 bytes | Zenodo records 24096109
SHA256 81ad098424a0c61258bd878237e765d53081bee50f0b384095c3ed30bd14a38a

v16_Trillion_Verifier_Recovery.zip!/run/candidates/00003683/replay/worker_6.bin

16 bytes | Zenodo records 24096109
SHA256 28051d92ddb9161b770463f262fd5cee0da1a799f4b60e0932a100f2f1e6961d

v16_Trillion_Verifier_Recovery.zip!/run/candidates/00003683/replay/worker_5.bin

16 bytes | Zenodo records 24096109
SHA256 cf11f8d10d8e8a584263e2def52a5e70f683fd1364cca1467e46ede68070f5f7

v21_Markov_Meridians_Source_and_Supplements.zip!/v21_Markov_Meridians_source/plotting/figure3/fonts/NimbusSans-Bold.otf

83,264 bytes | Zenodo records 24096109
SHA256 7f33328e6b4d4cd21b45fa625791928c9407dc702db6780e56b09ca9a3ecaa67